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数学物理方法

习题解答



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斯颂乐 徐世良 高永椿 编
张官南 张立志

天津科学技术出版社

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前 言

“数学物理方法”是物理类专业的重要基础课程，它不仅为后继课程研究有关的数学物理问题作准备，也为实际工作中遇到的数学物理问题的求解提供基础。为了掌握这门课程中解决问题的方法，在学习过程中解算一定数量的习题是至关重要的。

斯颂乐、徐世良、高永椿、张官南、张立志等同志将我编写的《数学物理方法》（第二版）的习题一一解答出来，有的习题还有几种解法，以资比较，并对整个题解进行了反复的修订。我认为这样一份题解可以起如下几方面的作用：

担任这门课程的老师，在给学生布置习题作业之前，需要先解算大量的习题，然后从中挑选适当的习题布置给学生，而《数学物理方法》习题的解算往往是很费时间的。《题解》可以节约任课老师挑选习题的时间，让他们把精力用于更好地提高教学质量。

学习这门课程的大学生或自修这门课程的读者，在独立思考 and 独立解算基础上，可以与《题解》进行比较，以总结自己解法的优缺点。如果某些习题虽经反复思考犹有困雄，那么，从《题解》可以找出困雄的症结所在，这就前进了一步。但是，这里需要强调的是独立思考，切不可依赖《题解》，依赖《题解》对于学习是有害无益的。

实际工作者遇到有关数学物理问题时也可能从《题解》中取得某些借鉴。

原书由于编写时间十分仓促，习题答案有某些不妥之处，

解题时已作了订正。

在《数学物理方法习题解答》行将出版之际，天津科学技术出版社的编辑同志要我写个简短的前言，我就把上面的想法写了出来，以就教于各方人士。

梁 昆 森

一九八一年元月

内 容 提 要

本书对梁昆森教授所编《数学物理方法》(第二版)中的全部习题作出了解答。内容分复变函数论、傅里叶级数和积分、数学物理方程三个部份,共十七章包括习题约四百条,有些习题列出了多种解法。

本书是配合综合大学、高等师范院校物理类各专业数学物理方法课程的教学用书,也可为工科院校有关专业的工程数学课程所选用,对于有关科学技术工作者也有一定的参考价值。

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第一篇 复变函数论

第一章 复变函数

§1. 复数与复数运算

1. 下列式子在复数平面上各具有怎样的意义?

$$(1) |z| \leq 2.$$

$$\text{解一: } |z| = |x + iy| = \sqrt{x^2 + y^2} \leq 2,$$

$$\text{或 } x^2 + y^2 \leq 4.$$

这是以原点为圆心而半径为2的圆及其内部.

解二: 按照模的几何意义, $|z|$ 是复数 $z = x + iy$ 与原点间的距离, 若此距离总是 ≤ 2 , 则即表示以原点为圆心而半径为2的圆及其内部.

$$(2) |z - a| = |z - b| \quad (a, b \text{ 为复常数}).$$

$$\text{解一: 设 } z = x + iy, \quad a = a_1 + ia_2, \quad b = b_1 + ib_2;$$

$$|z - a| = \sqrt{(x - a_1)^2 + (y - a_2)^2},$$

$$|z - b| = \sqrt{(x - b_1)^2 + (y - b_2)^2},$$

于是

$$(x - a_1)^2 + (y - a_2)^2 = (x - b_1)^2 + (y - b_2)^2,$$

$$\text{即 } (2y - a_2 - b_2)(b_2 - a_2) = (2x - a_1 - b_1)(a_1 - b_1)$$

亦即

$$\frac{y - \frac{a_2 + b_2}{2}}{x - \frac{a_1 + b_1}{2}} = \frac{a_1 - b_1}{b_2 - a_2}.$$

这是一条直线,是一条过点 a 和点 b 连线的中点 $(\frac{a_1 + b_1}{2}, \frac{a_2 + b_2}{2})$ 且与该连线垂直的直线.

解二: 等式的几何意义是, 点 z 到定点 a 和点 b 的距离相等的各点的轨迹, 即表示点 a 和点 b 的连线的垂直平分线.

$$(3) \operatorname{Re} z > \frac{1}{2}.$$

解: 设 $z = x + iy$, 则 $\operatorname{Re} z = x$, 故原式为 $x > \frac{1}{2}$, 它表示 $x > \frac{1}{2}$ 的半平面, 即直线 $x = \frac{1}{2}$ 右边的区域 (不包括该直线).

$$(4) |z| + \operatorname{Re} z \leq 1.$$

解: 设 $z = x + iy$, 则原式即 $x^2 + y^2 \leq (1 - x)^2$, 亦即 $y^2 \leq 1 - 2x$, 它表示抛物线 $y^2 = 1 - 2x$ 及其内部.

$$(5) \alpha < \arg z < \beta, a < \operatorname{Re} z < b \quad (\alpha, \beta, a \text{ 和 } b \text{ 为实常数}).$$

解: 注意到 $\arg z = \varphi$, $\operatorname{Re} z = x$, 则原二式

$$\text{即} \quad \begin{cases} \alpha < \varphi < \beta, \\ a < x < b. \end{cases}$$

为两直线 $x = a$ 、 $x = b$ 和两射线 $\varphi = \alpha$ 、 $\varphi = \beta$ 所围成的区域 (不包括边界).

$$(6) 0 < \arg \frac{z-i}{z+i} < \frac{\pi}{4}.$$

$$\text{解: 因为 } \frac{z-i}{z+i} = \frac{x+i(y-1)}{x+i(y+1)}$$

$$\begin{aligned}
&= \frac{[x+i(y-1)][x-i(y+1)]}{[x+i(y+1)][x-i(y+1)]} \\
&= \frac{x^2+y^2-1}{x^2+(y+1)^2} + i \frac{-2x}{x^2+(y+1)^2} \\
&\equiv X+iY=Z.
\end{aligned}$$

所以，原式即 $0 < \arg z < \frac{\pi}{4}$ 。如以 X 轴为实轴， Y 轴为虚

轴，上式在复平面 Z 上表示由射线 $\Phi = 0$ 和 $\Phi = \frac{\pi}{4}$ 所围成的区域（不包括射线本身），这就意味着要求 $X > 0$ 和 $Y > 0$ ，

即要求 $\frac{x^2+y^2-1}{x^2+(y+1)^2} > 0$ 和 $\frac{-2x}{x^2+(y+1)^2} > 0$ ，

亦即

$$\begin{cases} x < 0, \\ x^2 + y^2 - 1 > 0. \end{cases} \quad (1)$$

又由 $0 < \arg Z < \frac{\pi}{4}$ 得 $0 < \arctg(Y/X) < \frac{\pi}{4}$ ，即

$$0 < \arctg\left(\frac{-2x}{x^2+y^2-1}\right) < \frac{\pi}{4},$$

亦即 $0 < \frac{-2x}{x^2+y^2-1} < 1$ ，注意到(1)式，

则

$$\begin{cases} -2x > 0, \\ -2x < x^2 + y^2 - 1. \end{cases} \quad \text{即} \quad \begin{cases} x < 0, \\ x^2 + y^2 + 2x - 1 > 0. \end{cases} \quad (2)$$

在 $x < 0$ 的条件下，凡满足 $x^2 + y^2 + 2x - 1 > 0$ 的点必定也满足 $x^2 + y^2 - 1 > 0$ 。所以，(1)式无需单独提出，而(2)式表示复平面上的左半平面 $x < 0$ ，但除去圆周 $(x+1)^2 + y^2 = 2$ 及其内部（图1-1）。

注意：应排除

$$\begin{cases} x > 0, \\ x^2 + y^2 - 1 < 0; \end{cases}$$

$$\text{及 } (x+1)^2 + y^2 < 2$$

(这相当于 $X < 0, Y < 0$;

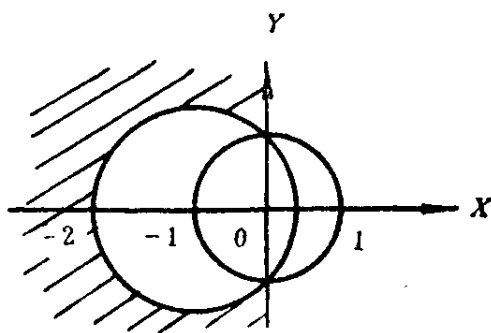


图 1-1

即 $\pi < \Phi < \frac{5}{4}\pi$, $\pi < \arg \frac{z-i}{z+i} < \frac{5}{4}\pi$) 这个解。

$$(7) \left| \frac{z-1}{z+1} \right| \leq 1.$$

$$\begin{aligned} \text{解: } \left| \frac{z-1}{z+1} \right| &= \left| \frac{(x-1) + iy}{(x+1) + iy} \right| \\ &= \frac{\sqrt{(x-1)^2 + y^2}}{\sqrt{(x+1)^2 + y^2}} \leq 1, \end{aligned}$$

$$\text{即 } (x-1)^2 + y^2 \leq (x+1)^2 + y^2,$$

亦即 $0 \leq x$, 这表示连同 Y 轴在内的右半平面。

$$(8) \operatorname{Re}\left(\frac{1}{z}\right) = 2.$$

$$\text{解: } \frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2},$$

$$\text{故 } \operatorname{Re}\left(\frac{1}{z}\right) = \frac{x}{x^2+y^2} = 2, 2x^2 + 2y^2 = x,$$

$$\text{即 } \left(x - \frac{1}{4}\right)^2 + y^2 = \frac{1}{16}.$$

这是中心在 $\left(\frac{1}{4}, 0\right)$ 而半径为 $\frac{1}{4}$ 的圆周。

$$(9) \operatorname{Re} z^2 = a^2 \quad (a \text{ 是实常数}).$$

$$\text{解: } z^2 = (x+iy)^2 = (x^2 - y^2) + i \cdot 2xy,$$

故 $\operatorname{Re} z^2 = x^2 - y^2$, 则原式即为

$$x^2 - y^2 = a^2.$$

此轨迹为双曲线 $x^2 - y^2 = a^2$.

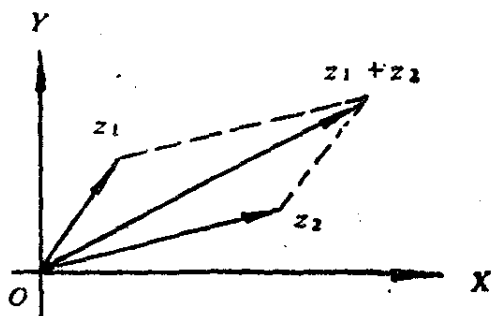
$$(10) |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2.$$

解：这是一个恒等式，对于复平面上任意的 z_1 和 z_2 都成立，因为

$$\begin{aligned} |z_1 + z_2|^2 + |z_1 - z_2|^2 &= (x_1 + x_2)^2 + (y_1 + y_2)^2 \\ &\quad + (x_1 - x_2)^2 + (y_1 - y_2)^2 \\ &= 2x_1^2 + 2x_2^2 + 2y_1^2 + 2y_2^2 \\ &= 2|z_1|^2 + 2|z_2|^2. \end{aligned}$$

它表示平行四边形对角线的平方和等于两邻边平方和的两倍。

此外，如把 z_1 和 z_2 表示成复平面上的矢量，那么 z_1 和 z_2 的加减运算与相应的矢量的加减运算（平行四边形法则）是相同的，



这可由图1-2清楚地看出。

图 1-2

2. 把下列复数用代数式、三角式和指数式几种形式表示出来。

(1) i .

解： i 本身即为代数式，此时在 $z = x + iy$ 中， $x = 0$ 、 $y = 1$ ；

三角式： $\rho = \sqrt{x^2 + y^2} = 1$,

$$\varphi = \operatorname{arctg}\left(\frac{y}{x}\right) = \operatorname{arctg}\left(\frac{1}{0}\right) = \frac{\pi}{2},$$

$$\text{所以 } z = i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2};$$

$$\text{指数式: } z = i = e^{i \frac{\pi}{2}}.$$

(2) -1 .

解: -1 本身即为代数式;

三角式: $z = \cos\pi + i\sin\pi$;

指数式: $z = e^{i\pi}$.

(3) $1 + i\sqrt{3}$.

解: $z = 1 + i\sqrt{3}$ 本身即为代数式;

三角式: $\rho = \sqrt{1^2 + (\sqrt{3})^2} = 2$, $\varphi = \operatorname{arctg} \frac{\sqrt{3}}{1} = \frac{\pi}{3}$,

所以 $z = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$;

指数式: $z = 2e^{i\frac{\pi}{3}}$.

(4) $1 - \cos\alpha + i\sin\alpha$ (α 是实常数).

解: $z = (1 - \cos\alpha) + i\sin\alpha$ 本身即为代数式;

三角式: $\rho = \sqrt{(1 - \cos\alpha)^2 + \sin^2\alpha} = \sqrt{2(1 - \cos\alpha)}$

$$= 2\sin \frac{\alpha}{2},$$

$$\varphi = \operatorname{arctg} \frac{\sin\alpha}{1 - \cos\alpha}, \quad \operatorname{tg}\varphi = \frac{\sin\alpha}{1 - \cos\alpha} = \operatorname{ctg} \frac{\alpha}{2},$$

$$\varphi = \left(n + \frac{1}{2} \right) \pi - \frac{\alpha}{2},$$

在主值范围内 $\varphi = \frac{1}{2}(\pi - \alpha)$ ($0 \leq \alpha \leq \pi$), 所以

$$z = 2\sin \frac{\alpha}{2} \left[\cos \left(\operatorname{arctg} \operatorname{ctg} \frac{\alpha}{2} \right) + i \sin \left(\operatorname{arctg} \operatorname{ctg} \frac{\alpha}{2} \right) \right],$$

或
$$z = 2\sin\frac{\alpha}{2}\left(\cos\frac{\pi-\alpha}{2} + i\sin\frac{\pi-\alpha}{2}\right)$$

$$(0 \leq \alpha \leq \pi);$$

指数式:
$$z = 2\sin\frac{\alpha}{2}e^{i\operatorname{arctg}\operatorname{ctg}\frac{\alpha}{2}},$$

或
$$z = 2\sin\frac{\alpha}{2}e^{i\left(\frac{\pi-\alpha}{2}\right)}.$$

(5) z^3 .

解: 代数式:
$$z^3 = (x+iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$$

三角式:
$$z^3 = \rho^3(\cos 3\varphi + i\sin 3\varphi),$$

其中 $\rho = \sqrt{x^2 + y^2}$, $\varphi = \operatorname{arctg}\left(\frac{y}{x}\right);$

指数式:
$$z^3 = \rho^3 e^{i3\varphi}.$$

(6) e^{1+i} .

解: 指数式即为 $z = e^{1+i} = e \cdot e^i$, 显然, 其中 $\rho = e$, $\varphi = 1$;

三角式:
$$z = e(\cos 1 + i\sin 1);$$

代数式:
$$z = e\cos 1 + ie\sin 1.$$

(7) $\frac{1-i}{1+i}$.

解: 代数式:
$$z = \frac{1-i}{1+i} = \frac{1}{2}(1-i)^2 = -i.$$

三角式: 因 $\rho = 1$, $\varphi = \operatorname{arctg}\left(\frac{-1}{0}\right) = \frac{3}{2}\pi$, 所以

$$z = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2};$$

指数式:
$$z = e^{i\frac{3\pi}{2}}.$$

3. 计算下列数值 (a, b 和 φ 为实常数) .

$$(1) \sqrt{a+ib}.$$

解：先化 $a+ib$ 为三角式

$$a+ib = \sqrt{a^2+b^2} (\cos\varphi + i\sin\varphi),$$

其中 $\cos\varphi = \frac{a}{\sqrt{a^2+b^2}}, \sin\varphi = \frac{b}{\sqrt{a^2+b^2}}$, 于是

$$\begin{aligned} \sqrt{a+ib} &= \sqrt[4]{a^2+b^2} \left(\cos \frac{\varphi}{2} + i \sin \frac{\varphi}{2} \right) \\ &= \sqrt[4]{a^2+b^2} \left[\sqrt{\frac{1}{2}(1+\cos\varphi)} \right. \\ &\quad \left. + i \sqrt{\frac{1}{2}(1-\cos\varphi)} \right] \\ &= \sqrt[4]{a^2+b^2} \left[\sqrt{\frac{1}{2} \left(1 + \frac{a}{\sqrt{a^2+b^2}} \right)} \right. \\ &\quad \left. + i \sqrt{\frac{1}{2} \left(1 - \frac{a}{\sqrt{a^2+b^2}} \right)} \right] \\ &= \frac{\sqrt{2}}{2} \left(\sqrt{\sqrt{a^2+b^2} + a} \right. \\ &\quad \left. + i \sqrt{\sqrt{a^2+b^2} - a} \right). \end{aligned}$$

$$(2) \sqrt[3]{i}.$$

解：因 $i = 1 \left[\cos \left(\frac{\pi}{2} + 2n\pi \right) + i \sin \left(\frac{\pi}{2} + 2n\pi \right) \right]$,

所以

$$\begin{aligned} \sqrt[3]{i} &= \sqrt[3]{1} \left[\cos \left(\frac{\pi}{6} + \frac{2}{3} n\pi \right) + i \sin \left(\frac{\pi}{6} \right. \right. \\ &\quad \left. \left. + \frac{2}{3} n\pi \right) \right], \end{aligned}$$

或 $\sqrt[3]{i} = e^{i \left(\frac{\pi}{6} + \frac{2}{3} n\pi \right)} \quad (n = 0, 1, 2).$

(3) i^i .

解: 因 $i = e^{i(\frac{\pi}{2} + 2n\pi)}$, 所以

$$i^i = \left[e^{i(\frac{\pi}{2} + 2n\pi)} \right]^i = e^{-\frac{\pi}{2} - 2n\pi} \quad (n = 0, \pm 1, \pm 2, \dots).$$

(4) $\sqrt[4]{i}$.

解: 仿上题,

$$\sqrt[4]{i} = \left[e^{i(\frac{\pi}{2} + 2n\pi)} \right]^{\frac{1}{4}} = e^{\frac{\pi}{8} + \frac{1}{2}n\pi} \quad (n = 0, \pm 1, \pm 2, \dots).$$

(5) $\cos 5\varphi$.

(6) $\sin 5\varphi$.

解: 由乘幂的公式

$$(\cos \varphi + i \sin \varphi)^n = \cos n\varphi + i \sin n\varphi,$$

及二项式定理

$$\begin{aligned} (a+b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \dots \\ &\quad + \frac{n!}{(n-k)!k!}a^{n-k}b^k + \dots \end{aligned}$$

可知

$$\begin{aligned} \cos 5\varphi + i \sin 5\varphi &= (\cos \varphi + i \sin \varphi)^5 \\ &= \cos^5 \varphi + i 5 \cos^4 \varphi \sin \varphi \\ &\quad - 10 \cos^3 \varphi \sin^2 \varphi \\ &\quad - i 10 \cos^2 \varphi \sin^3 \varphi \\ &\quad + 5 \cos \varphi \sin^4 \varphi + i \sin^5 \varphi, \end{aligned}$$

比较等式两边的实部和虚部得

$$\cos 5\varphi = \cos^5 \varphi - 10 \cos^3 \varphi \sin^2 \varphi + 5 \cos \varphi \sin^4 \varphi,$$

$$\sin 5\varphi = 5 \cos^4 \varphi \sin \varphi - 10 \cos^2 \varphi \sin^3 \varphi + \sin^5 \varphi.$$

$$(7) \cos \varphi + \cos 2\varphi + \cos 3\varphi + \dots + \cos n\varphi.$$

$$(8) \sin \varphi + \sin 2\varphi + \sin 3\varphi + \dots + \sin n\varphi.$$

解一：从初等代数知道， n 项的等比级数 $x + x^2 + \cdots + x^n$ 的和为 $x \frac{1-x^n}{1-x}$ 。

现在所求为

$$\begin{aligned}
 & \cos\varphi + \cos 2\varphi + \cos 3\varphi + \cdots + \cos n\varphi \\
 & + i(\sin\varphi + \sin 2\varphi + \sin 3\varphi + \cdots + \sin n\varphi) \\
 & = (\cos\varphi + i\sin\varphi) + (\cos 2\varphi + i\sin 2\varphi) + \cdots \\
 & + (\cos n\varphi + i\sin n\varphi) \\
 & = e^{i\varphi} + e^{2i\varphi} + \cdots + e^{ni\varphi} \\
 & = e^{i\varphi} \frac{1 - e^{in\varphi}}{1 - e^{i\varphi}} \\
 & = \frac{e^{i\varphi}(1 - e^{-i\varphi})(1 - e^{in\varphi})}{(1 - e^{-i\varphi})(1 - e^{i\varphi})} \\
 & = \frac{(e^{i\varphi} - 1)(1 - e^{in\varphi})}{2 - 2\cos\varphi} \\
 & = \frac{e^{i\varphi/2}(e^{i\varphi/2} - e^{-i\varphi/2})e^{in\varphi/2}(e^{-in\varphi/2} - e^{in\varphi/2})}{4\sin^2\frac{\varphi}{2}} \\
 & = \frac{e^{i\varphi/2}\left(2i\sin\frac{\varphi}{2}\right)e^{in\varphi/2}\left(-2i\sin\frac{n\varphi}{2}\right)}{4\sin^2\frac{\varphi}{2}} \\
 & = \frac{e^{i(n+1)\varphi/2}\sin\frac{n\varphi}{2}}{\sin\frac{\varphi}{2}} \\
 & = \frac{\sin\frac{n\varphi}{2}\left(\cos\frac{n+1}{2}\varphi + i\sin\frac{n+1}{2}\varphi\right)}{\sin\frac{\varphi}{2}},
 \end{aligned}$$

比较等式两边的实部和虚部得

$$\begin{aligned}
& \cos\varphi + \cos 2\varphi + \cos 3\varphi + \cdots + \cos n\varphi \\
&= \frac{1}{\sin \frac{\varphi}{2}} \sin \frac{n\varphi}{2} \cos \frac{(n+1)\varphi}{2} \\
&= \frac{1}{2\sin \frac{\varphi}{2}} \left[\sin \left(n + \frac{1}{2} \right) \varphi - \sin \frac{\varphi}{2} \right], \\
& \sin\varphi + \sin 2\varphi + \sin 3\varphi + \cdots + \sin n\varphi \\
&= \frac{1}{\sin \frac{\varphi}{2}} \sin \frac{n\varphi}{2} \sin \frac{(n+1)\varphi}{2} \\
&= \frac{1}{2\sin \frac{\varphi}{2}} \left[\cos \frac{\varphi}{2} - \cos \left(n + \frac{1}{2} \right) \varphi \right].
\end{aligned}$$

解二:

$$\begin{aligned}
& \cos\varphi + \cos 2\varphi + \cdots + \cos n\varphi + i\sin\varphi + i\sin 2\varphi \\
& \quad + \cdots + i\sin n\varphi \\
&= (\cos\varphi + i\sin\varphi) + (\cos 2\varphi + i\sin 2\varphi) + \cdots \\
& \quad + (\cos n\varphi + i\sin n\varphi) \\
&= (\cos\varphi + i\sin\varphi) + (\cos\varphi + i\sin\varphi)^2 + \cdots \\
& \quad + (\cos\varphi + i\sin\varphi)^n \\
&= \frac{(\cos\varphi + i\sin\varphi)[1 - (\cos\varphi + i\sin\varphi)^{n+1}]}{1 - (\cos\varphi + i\sin\varphi)} \\
&= \frac{(\cos\varphi + i\sin\varphi)[(1 - \cos(n+1)\varphi) - i\sin(n+1)\varphi]}{(1 - \cos\varphi) + i\sin\varphi} \\
&= \frac{(\cos\varphi + i\sin\varphi)[(1 - \cos n\varphi) - i\sin n\varphi]}{[(1 - \cos\varphi) - i\sin\varphi] \cdot [(1 - \cos\varphi) + i\sin\varphi]} \\
&= \frac{1}{4\sin^2 \frac{\varphi}{2}} \left\{ \left[4\sin^2 \frac{\varphi}{2} \sin \frac{n\varphi}{2} \cos \varphi \right. \right. \\
& \quad \left. \left. - 2\sin^2 \frac{n\varphi}{2} \sin^2 \varphi \right] \right\}
\end{aligned}$$

$$\begin{aligned}
& + 2\sin^2 \frac{\varphi}{2} \sin \varphi \sin n\varphi + \sin \varphi \cos \varphi \sin n\varphi \Big] \\
& + i \left[4\sin^2 \frac{\varphi}{2} \sin^2 \frac{n\varphi}{2} \sin \varphi \right. \\
& + 2\sin^2 \frac{n\varphi}{2} \sin \varphi \cos \varphi \\
& \left. - 2\sin^2 \frac{\varphi}{2} \cos \varphi \sin n\varphi + \sin^2 \varphi \sin n\varphi \right] \Big\} \\
& = -\frac{1}{4\sin^2 \frac{\varphi}{2}} \left\{ \left[\sin \left(n + \frac{1}{2} \right) \varphi - \sin \frac{\varphi}{2} \right] 2\sin \frac{\varphi}{2} \right. \\
& \left. + i \left[\cos \frac{\varphi}{2} - \cos \left(n + \frac{1}{2} \right) \varphi \right] 2\sin \frac{\varphi}{2} \right\} \\
& = -\frac{1}{2\sin \frac{\varphi}{2}} \left\{ \left[\sin \left(n + \frac{1}{2} \right) \varphi - \sin \frac{\varphi}{2} \right] \right. \\
& \left. + i \left[\cos \frac{\varphi}{2} - \cos \left(n + \frac{1}{2} \right) \varphi \right] \right\},
\end{aligned}$$

比较等式两边的实部和虚部也得到解①中的答案。

§2. 复变函数

1. 试验证(2.11) — (2.14) 几个式子.

$$\begin{aligned}
(1) \quad (2.11) \text{ 式: } \sin(z + 2\pi) &= \sin z, \cos(z + 2\pi) \\
&= \cos z.
\end{aligned}$$

$$\begin{aligned}
\text{验证: } \sin(z + 2\pi) &= \frac{1}{2i} \left[e^{i(z+2\pi)} - e^{-i(z+2\pi)} \right] \\
&= \frac{1}{2i} \left(e^{iz} - e^{-iz} \right) = \sin z,
\end{aligned}$$

$$\begin{aligned}\cos(z+2\pi) &= \frac{1}{2} \left[e^{i(z+2\pi)} + e^{-i(z+2\pi)} \right] \\ &= \frac{1}{2} (e^{iz} + e^{-iz}) = \cos z.\end{aligned}$$

由此可见, 三角函数有实周期 2π .

(2) (2.12) 式:

$$|\sin z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + (2\sin^2 x - \cos^2 x)}.$$

$$\begin{aligned}\text{验证: 因 } \sin z &= \frac{1}{2i} (e^{iz} - e^{-iz}) = -\frac{i}{2} [e^{i(x+iy)} \\ &\quad - e^{-i(x+iy)}] \\ &= -\frac{i}{2} (e^{-y}e^{ix} - e^ye^{-ix}) \\ &= -\frac{i}{2} [e^{-y}(\cos x + i\sin x) \\ &\quad - e^{-y}(\cos x - i\sin x)] \\ &= \frac{1}{2} [(e^y + e^{-y})\sin x + i(e^y - e^{-y})\cos x],\end{aligned}$$

$$\begin{aligned}\text{所以 } |\sin z| &= \frac{1}{2} \sqrt{(e^y + e^{-y})^2 \sin^2 x + (e^y - e^{-y})^2 \cos^2 x} \\ &= \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\sin^2 x - \cos^2 x)}.\end{aligned}$$

(3) (2.13) 式:

$$|\cos z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\cos^2 x - \sin^2 x)}.$$

验证一: 其步骤全同于(2).

验证二: 由 $\cos z = \sin\left(\frac{\pi}{2} - z\right)$ 再利用(2)的答案,

$$\text{则 } |\cos z| = \left| \sin\left(\frac{\pi}{2} - z\right) \right|$$

$$= \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2 \left[\sin^2 \left(\frac{\pi}{2} - x \right) - \cos^2 \left(\frac{\pi}{2} - x \right) \right]}$$

$$= \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\cos^2 x - \sin^2 x)}.$$

(4) (2.14) 式: $e^{z+2\pi i} = e^z$, $\operatorname{sh}(z+2\pi i) = \operatorname{sh} z$,
 $\operatorname{ch}(z+2\pi i) = \operatorname{ch} z$.

验证: $e^{z+2\pi i} = e^z \cdot e^{i2\pi} = e^z$,

$$\operatorname{sh}(z+2\pi i) = \frac{1}{2} [e^{z+2\pi i} - e^{-z-2\pi i}]$$

$$= \frac{1}{2} (e^z - e^{-z}) = \operatorname{sh} z.$$

$$\operatorname{ch}(z+2\pi i) = \frac{1}{2} [e^{z+2\pi i} + e^{-z-2\pi i}]$$

$$= \frac{1}{2} (e^z + e^{-z}) = \operatorname{ch} z.$$

显然, 双曲函数有纯虚周期 $2\pi i$.

2. 计算下列数值 (a 和 b 为实常数, x 为实变数).

(1) $\sin(a+ib)$.

解: $\sin(a+ib) = \frac{1}{2i} [e^{i(a+ib)} - e^{-(a+ib)i}]$

$$= \frac{1}{2i} [e^{-b}(\cos a + i \sin a) - e^{+b}(\cos a - i \sin a)]$$

$$= \frac{1}{2} [e^{-b} \sin a + e^b \sin a + i(e^b \cos a - e^{-b} \cos a)]$$

$$= \frac{1}{2} [(e^b + e^{-b}) \sin a + i(e^b - e^{-b}) \cos a].$$

(2) $\cos(a+ib)$.

$$\begin{aligned}
 \text{解: } \cos(a+ib) &= \frac{1}{2}(e^{i(a+ib)} + e^{-i(a+ib)}) \\
 &= \frac{1}{2}[e^{-b}(\cos a + i \sin a) + e^b(\cos a \\
 &\quad - i \sin a)] \\
 &= \frac{1}{2}[(e^{-b} + e^b)\cos a + i(e^{-b} - e^b)\sin a].
 \end{aligned}$$

$$(3) \ln(-1).$$

$$\text{解一: } \ln(-1) = \ln|-1| + i \arg(-1) = i(2n+1)\pi;$$

$$\begin{aligned}
 \text{解二: } \ln(-1) &= \ln e^{i(x+2n\pi)} = \ln e^{i(2n+1)\pi} \\
 &= i(2n+1)\pi \quad (n = 0, \pm 1, \dots).
 \end{aligned}$$

$$(4) \operatorname{ch}^2 z - \operatorname{sh}^2 z.$$

$$\text{解: } \operatorname{ch}^2 z - \operatorname{sh}^2 z = \frac{e^{2z} + 2 + e^{-2z}}{4} - \frac{e^{2z} - 2 + e^{-2z}}{4} = 1.$$

$$(5) \cos ix.$$

$$\text{解: } \cos ix = \frac{e^{i(ix)} + e^{-i(ix)}}{2} = \frac{e^x + e^{-x}}{2} = \operatorname{ch} x.$$

$$(6) \sin ix.$$

$$\begin{aligned}
 \text{解: } \sin ix &= \frac{e^{i(ix)} - e^{-i(ix)}}{2i} = \frac{e^x - e^{-x}}{2}i \\
 &= i \operatorname{sh} x.
 \end{aligned}$$

$$(7) \operatorname{ch} ix.$$

$$\text{解: } \operatorname{ch} ix = \frac{e^{ix} + e^{-ix}}{2} = \cos x.$$

$$(8) \operatorname{sh} ix.$$

$$\text{解: } \operatorname{sh} ix = \frac{e^{ix} - e^{-ix}}{2} = i \sin x.$$

$$(9) |e^{iax - ib \sin x}|.$$

$$\text{解: } \text{因 } \sin z = \sin(x+iy)$$

$$= \frac{1}{2} \left[(e^y + e^{-y}) \sin x + i(e^y - e^{-y}) \cos x \right],$$

所以

$$\begin{aligned} \text{原式} &= \left| e^{ia(x+iy) - ib \cdot \frac{1}{2} [(e^y + e^{-y}) \sin x + i(e^y - e^{-y}) \cos x]} \right| \\ &= \left| e^{-ay} \cdot e^{i[ax - \frac{b}{2}(e^y + e^{-y}) \sin x - \frac{b}{2}(e^y - e^{-y}) \cos x]} \right| \\ &= \left| e^{-ay + \frac{b}{2}(e^y - e^{-y}) \cos x} \cdot e^{i[ax - \frac{b}{2}(e^y + e^{-y}) \sin x]} \right| \\ &= e^{-ay + \frac{b}{2}(e^y - e^{-y}) \cos x} = e^{-ay + b \operatorname{sh} y \cos x}. \end{aligned}$$

3. 求解方程 $\sin z = 2$.

解一：原方程即 $\frac{1}{2i}(e^{iz} - e^{-iz}) = 2$, 即 $e^{iz} - e^{-iz} = 4i$,

亦即

$$(e^{iz})^2 - 4i(e^{iz}) - 1 = 0.$$

由一元二次代数方程的根的公式得

$$e^{iz} = 2i \pm \sqrt{(2i)^2 + 1} = (2 \pm \sqrt{3})i,$$

于是

$$\begin{aligned} iz &= \ln \left[(2 \pm \sqrt{3})i \right] = \ln(2 \pm \sqrt{3}) + \ln i \\ &= \ln(2 \pm \sqrt{3}) + \ln \left(e^{i(\frac{\pi}{2} + 2n\pi)} \right) \\ &= \ln(2 \pm \sqrt{3}) + i \left(\frac{\pi}{2} + 2n\pi \right), \end{aligned}$$

所以

$$\begin{aligned} z &= \frac{1}{i} \left[\ln(2 \pm \sqrt{3}) + i \left(\frac{\pi}{2} + 2n\pi \right) \right] \\ &= \frac{\pi}{2} + 2n\pi - i \ln(2 \pm \sqrt{3}). \end{aligned}$$

因 $-\ln(2 \pm \sqrt{3}) = \ln(2 \mp \sqrt{3})$, 故上式又可表为

$$z = \frac{\pi}{2} + 2n\pi + i \ln(2 \pm \sqrt{3}).$$

$$\text{解二: } \sin z = \frac{1}{2} \left[(e^y + e^{-y}) \sin x + i(e^y - e^{-y}) \cos x \right] = 2,$$

比较等式两边的实部和虚部得

$$\begin{cases} (e^y + e^{-y}) \sin x = 4, & (1) \end{cases}$$

$$\begin{cases} (e^y - e^{-y}) \cos x = 0. & (2) \end{cases}$$

在(2)式中, 如果 $e^y - e^{-y} = 0$, 则 $y = 0$, 以 $y = 0$ 代入(1)式中则得出 $\sin x = 2$ 的错误结果; 所以 y 不能为零, 即 $e^y - e^{-y} \neq 0$. 只有 $\cos x = 0$, 即

$$x = \frac{\pi}{2} + n\pi (n = 0, 1, 2, \dots).$$

但以 $x = (2k+1)\pi + \frac{\pi}{2}$ 代入(1)式, 则得 $-(e^y + e^{-y}) = 4$,

显然是不合理的, 必须在 $x = \frac{\pi}{2} + n\pi$ 的解中舍去 $x = (2k+1)\pi + \frac{\pi}{2}$ 的部分解; 只保留 $x = \left(2k + \frac{1}{2}\right)\pi$ 的部分解, 以 $x =$

$\left(2k + \frac{1}{2}\right)\pi$ 代入(1)式得

$$e^y + e^{-y} = 4,$$

即

$$(e^y)^2 - 4e^y + 1 = 0,$$

由此解出

$$e^y = 2 \pm \sqrt{3},$$

即

$$y = \ln(2 \pm \sqrt{3}),$$

所以

$$z = \left(2k + \frac{1}{2}\right)\pi + i \ln(2 \pm \sqrt{3}).$$

§3. 多值函数

指出下列多值函数的支点及其阶，并作出里曼面。

(1) $\sqrt{z-a}$.

解：(i) 根式 $w = \sqrt{z-a}$ 的定义是 $w^2 = z-a$ ，今用指数式表示出 $w = \rho e^{i\varphi}$ ， $z-a = re^{i\theta}$ ($r, \rho \geq 0$)。以此代入 $w^2 = z-a$ 中得 $\rho^2 e^{i2\varphi} = re^{i\theta}$ ，所以 $\rho^2 = r$ ， $e^{i2\varphi} = e^{i\theta}$ ， $w = \sqrt{r} e^{i\frac{\theta}{2}}$ ，即

$$\begin{cases} \rho = \sqrt{r}, \\ 2\varphi = \theta + 2n\pi, \quad (n = 0, \pm 1, \pm 2, \dots), \end{cases}$$

由此可见， w 的模与 $z-a$ 的模 r 的对应关系是唯一确定的，但辐角不是如此，而是对应于每一个 θ 值，有两个不同的 φ 值，如：

$\varphi_1 = \frac{\theta}{2} (n=0)$ ， $\varphi_2 = \frac{\theta}{2} + \pi (n=1)$ 。相应的 w 值是： $w_1 =$

$\sqrt{r} e^{i\frac{\theta}{2}}$ ， $w_2 = \sqrt{r} e^{i(\frac{\theta}{2} + \pi)} = -\sqrt{r} e^{i\frac{\theta}{2}}$ ，其它 n 值给出

的只是这两个 w 值的重复。

(ii) 对于 $w = \sqrt{z-a}$ 来说， a 点具有这样的特性，而 z 绕 a 点转一圈回到原处时，相应的函数值 w 不还原，改变了正负号；而当 z 不绕 a 点转一圈回到原处时，函数值还原，所以 a 点是该多值函数的支点。当 z 绕 a 点转两圈回到原处时，对应的函数值还原，所以 a 点是该多值函数的一阶支点。

(iii) 如令 $z = \frac{1}{t}$ ，则 $w = \frac{\sqrt{1-at}}{\sqrt{t}}$ ；当 t 绕 $t=0$ 转一圈回

到原处时， w 值不能还原；绕两圈回到原处时， w 值还原，所以 $z = \infty$ 也是一阶支点。

作出里曼面如图1-3.

$$(2) \quad \sqrt{(z-a)(z-b)}.$$

解: (i) 如令 $z-a=r_1e^{i\theta_1}$,
 $z-b=r_2e^{i\theta_2}$, $w=\rho e^{i\varphi}$, 则

$$\begin{aligned} w &= \sqrt{(z-a)(z-b)} \\ &= \rho e^{i\varphi} = \sqrt{r_1 r_2} e^{i \frac{\theta_1 + \theta_2}{2}}, \end{aligned}$$

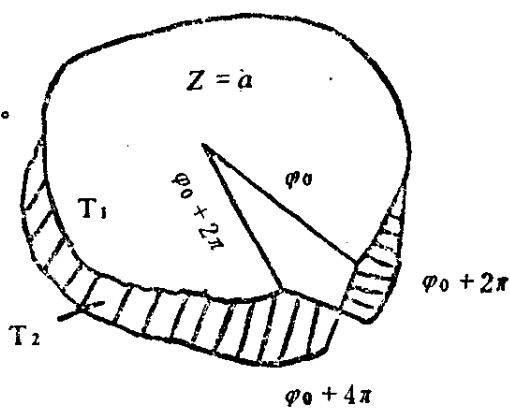


图 1-3

即

$$\begin{cases} \rho = \sqrt{r_1 r_2}, \\ 2\varphi = \theta_1 + \theta_2 + 2n\pi (n = 0, \pm 1, \pm 2, \dots). \end{cases}$$

(ii) 同上题分析, $z=a$ 和 $z=b$ 是多值函数 w 的一阶支点.

(iii) 里曼面有两叶, 在 T_1 上从 $z=a$ 到 $z=b$ 作切割, T_1 的切割下岸连结于 T_2 的上岸, T_2 的下岸连结于 T_1 的上岸. 事实上, 沿着不包围点 a 和 b 的闭路 1 环行一周, 辐角 θ_1

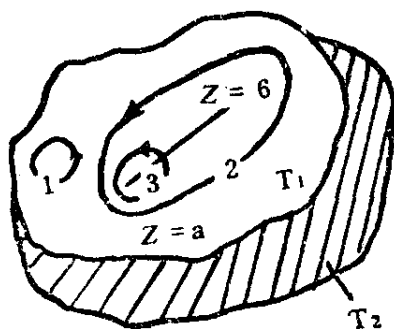


图 1-4

和 θ_2 又返回原来的值. 沿着包围两个点 a 和 b 的闭路 2 环行一周, 此二辐角各增加 2π , 所以 $\frac{1}{2}(\theta_1 + \theta_2)$ 也增加 2π , 而函数值 w 还原. 如果在同一叶上沿着只包围 a 点 (或 b 点) 的闭路 3 环行一周, 函数值 w 并不还原, 所作切割就是为了截断此种闭路.

(3) $\ln z$

解: (i) 对数函数 $w = \ln z$ 的定义是: $e^w = z$, 令 $w = u + iv$ 和 $z = re^{i\theta}$ 代入上式得 $e^u \cdot e^{iv} = re^{i\theta}$, 比较两边的模和辐角得

$$e^u = r, \text{ 即 } u = \ln r = \ln |z|,$$

$$v = \arg z = \theta + 2n\pi (n = 0, \pm 1, \pm 2, \dots).$$

(ii) 由上可见,对数函数的多值性表现在函数值 w 的虚部 v 与自变量 z 的辐角的对应关系上,对于每一个 z 值,有无穷多个 w 值,这些不同的 w 值只是虚部不同而已,相差为 2π 的整数倍,即 $w_n(z) = \ln|z| + i(\theta + 2n\pi)$,其支点是 $z = 0$,而且是无限阶支点.

(iii) 里曼面如图1-5所示,它有无穷多叶,在第一叶上从 $z = 0$ 到 $z = \infty$ 作切割,每一叶的切割下岸连接于下一叶的上岸($z = \infty$ 亦为无限阶支点).

(4) $\ln(z - a)$.

解:除了以 $z = a$ 代替上题中的 $z = 0$ 以外,其它的分析完全和上题相同.

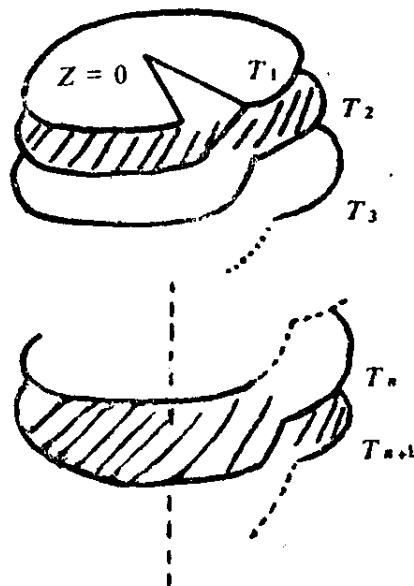


图 1-5

§4. 导数 (微商)

试推导极坐标系中的科希-里曼方程

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi}, \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho}. \end{cases}$$

解一: 从直角坐标系中的科希-里曼方程

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \end{cases}$$

出发,按照变换公式: $\rho = \sqrt{x^2 + y^2}$ 和 $\varphi = \operatorname{arctg} \left(\frac{y}{x} \right)$,即

$x = \rho \cos \varphi$ 和 $y = \rho \sin \varphi$ 变换到极坐标. 计算如下:

从变换公式可得

$$\left\{ \begin{aligned} \frac{\partial \rho}{\partial x} &= \frac{1}{2} \frac{2x}{\sqrt{x^2 + y^2}} = \frac{x}{\rho} = \cos \varphi, \\ \frac{\partial \rho}{\partial y} &= \frac{1}{2} \frac{2y}{\sqrt{x^2 + y^2}} = \frac{y}{\rho} = \sin \varphi, \\ \frac{\partial \varphi}{\partial x} &= \frac{y \left(-\frac{1}{x^2} \right)}{1 + \left(\frac{y}{x} \right)^2} = \frac{-y}{x^2 + y^2} = -\frac{\sin \varphi}{\rho}, \\ \frac{\partial \varphi}{\partial y} &= \frac{\frac{1}{x}}{1 + \left(\frac{y}{x} \right)^2} = \frac{x}{x^2 + y^2} = \frac{\cos \varphi}{\rho}, \end{aligned} \right.$$

又

$$\left\{ \begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial \rho} \frac{\partial \rho}{\partial x} + \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x} = \cos \varphi \frac{\partial u}{\partial \rho} - \frac{1}{\rho} \sin \varphi \frac{\partial u}{\partial \varphi}, \\ \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial \rho} \frac{\partial \rho}{\partial y} + \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial y} = \sin \varphi \frac{\partial u}{\partial \rho} + \frac{1}{\rho} \cos \varphi \frac{\partial u}{\partial \varphi}, \\ \frac{\partial v}{\partial x} &= \frac{\partial v}{\partial \rho} \frac{\partial \rho}{\partial x} + \frac{\partial v}{\partial \varphi} \frac{\partial \varphi}{\partial x} = \cos \varphi \frac{\partial v}{\partial \rho} - \frac{1}{\rho} \sin \varphi \frac{\partial v}{\partial \varphi}, \\ \frac{\partial v}{\partial y} &= \frac{\partial v}{\partial \rho} \frac{\partial \rho}{\partial y} + \frac{\partial v}{\partial \varphi} \frac{\partial \varphi}{\partial y} = \sin \varphi \frac{\partial v}{\partial \rho} + \frac{1}{\rho} \cos \varphi \frac{\partial v}{\partial \varphi}. \end{aligned} \right.$$

把以上四式代入直角坐标系中的科希-里曼方程得

$$\left\{ \begin{aligned} \cos \varphi \frac{\partial u}{\partial \rho} - \frac{1}{\rho} \sin \varphi \frac{\partial u}{\partial \varphi} &= \sin \varphi \frac{\partial v}{\partial \rho} + \frac{1}{\rho} \cos \varphi \frac{\partial v}{\partial \varphi}, \end{aligned} \right. \quad (1)$$

$$\left\{ \begin{aligned} \sin \varphi \frac{\partial u}{\partial \rho} + \frac{1}{\rho} \cos \varphi \frac{\partial u}{\partial \varphi} &= -\cos \varphi \frac{\partial v}{\partial \rho} + \frac{1}{\rho} \sin \varphi \frac{\partial v}{\partial \varphi}. \end{aligned} \right. \quad (2)$$

(1) 式 $\times \sin \varphi$ - (2) 式 $\times \cos \varphi$ 给出

$$-\frac{1}{\rho} \frac{\partial u}{\partial \varphi} = \frac{\partial v}{\partial \rho}, \quad (3)$$

(1)式 $\times \cos\varphi$ + (2)式 $\times \sin\varphi$ 给出

$$\frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi}, \quad (4)$$

(3)与(4)即为极坐标系中的科希-里曼方程。

解二：从定义出发进行推导。

$$w = u(z) + iv(z) = u(\rho, \varphi) + iv(\rho, \varphi),$$

在极坐标系中，先令 Δz 沿径向逼近零，即 $\Delta z = e^{i\varphi} \Delta\rho \rightarrow 0$ ，则

$$\begin{aligned} \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} &= \lim_{\Delta \rho \rightarrow 0} \frac{\Delta w}{\Delta \rho} \frac{\Delta \rho}{\Delta z} = \lim_{\Delta \rho \rightarrow 0} \frac{\Delta w}{\Delta \rho} e^{i\varphi} \\ &= \lim_{\Delta \rho \rightarrow 0} \frac{\Delta u + i\Delta v}{\Delta \rho} e^{-i\varphi} \\ &= \left(\frac{\partial u}{\partial \rho} + i \frac{\partial v}{\partial \rho} \right) e^{-i\varphi}; \end{aligned}$$

再令 Δz 沿横向逼近零，即 $\Delta z = \rho \Delta(e^{i\varphi}) = i\rho e^{i\varphi} \Delta\varphi \rightarrow 0$ ，则

$$\begin{aligned} \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} &= \lim_{\Delta \varphi \rightarrow 0} \frac{\Delta w}{\Delta \varphi} \frac{\Delta \varphi}{\Delta z} = \lim_{\Delta \varphi \rightarrow 0} \frac{\Delta w}{\Delta \varphi} \frac{1}{i\rho} e^{-i\varphi} \\ &= -\frac{i}{\rho} e^{-i\varphi} \lim_{\Delta \varphi \rightarrow 0} \frac{\Delta u + i\Delta v}{\Delta \varphi} \\ &= -\frac{i}{\rho} e^{-i\varphi} \left(\frac{\partial u}{\partial \varphi} + i \frac{\partial v}{\partial \varphi} \right) \\ &= \left(\frac{1}{\rho} \frac{\partial v}{\partial \varphi} - i \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \right) e^{-i\varphi}. \end{aligned}$$

如果函数 $w(z)$ 在点 z 可导，则上述二极限必须都存在而且彼此相等，即

$$\left(\frac{\partial u}{\partial \rho} + i \frac{\partial v}{\partial \rho} \right) e^{-i\varphi} = \left(\frac{1}{\rho} \frac{\partial v}{\partial \varphi} - i \frac{1}{\rho} \frac{\partial u}{\partial \varphi} \right) e^{-i\varphi},$$

比较上式中的实部和虚部即得

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi}, \\ \frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi}. \end{cases}$$

§5. 解析函数

1. 某个区域上的解析函数如为实函数, 试证它必为常数.

解: 设这个解析函数为 $w(z) = u(x, y) + iv(x, y)$, 因为它是实数, 所以 $v(x, y) \equiv 0$; 因为它是解析函数, 所以它满足科希-里曼方程

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

注意到 $v(x, y) \equiv 0$, 则

$$\frac{\partial u}{\partial x} = 0, \quad (1)$$

$$\frac{\partial u}{\partial y} = 0. \quad (2)$$

由(1)知 $u = f_1(y)$, 由(2)知 $u = f_2(x)$; 因为 x, y 在该区域中皆为独立变数, 要 $f_1(y) = f_2(x) = u$, 则只有 $f_1(y) = f_2(x) = \text{常数}$, 即 u 必为常数, 亦即该解析函数必为常数.

2. 已知解析函数 $f(z)$ 的实部 $u(x, y)$ 或虚部 $v(x, y)$, 求该解析函数.

$$(1) \quad u = e^x \sin y.$$

解一: $\frac{\partial u}{\partial x} = e^x \sin y, \quad -\frac{\partial u}{\partial y} = -e^x \cos y$. 根据科希-里

曼方程, 则

$$\frac{\partial v}{\partial y} = e^x \sin y, \quad \frac{\partial v}{\partial x} = -e^x \cos y. \text{ 于是}$$

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy = -e^x \cos y dx + e^x \sin y dy \\ = d(-e^x \cos y).$$

所以

$$v(x, y) = -e^x \cos y + C. \\ f(z) = e^x \sin y + i(-e^x \cos y + C) \\ = -ie^x (\cos y + i \sin y) + iC = -ie^x \cdot e^{iy} + iC \\ = -ie^{x+iy} + iC = -ie^z + iC.$$

解二：因为

$$\frac{\partial v}{\partial x} = -e^x \cos y, \quad (1)$$

$$\frac{\partial v}{\partial y} = e^x \sin y. \quad (2)$$

所以，由 (1) 式，暂且把 y 当作参数，对 x 积分，

$$v(x, y) = \int^{(x)} -e^x \cos y dx = -e^x \cos y + \varphi(y). \quad (3)$$

把 (3) 式对 y 求偏导数，

$$\frac{\partial v}{\partial y} = e^x \sin y + \varphi'(y) \quad (4)$$

比较 (2) 式和 (4) 式得 $\varphi'(y) = 0$ ，即 $\varphi(y) = C$ 。所以

$$v(x, y) = -e^x \cos y + C, \\ f(z) = e^x \sin y + i(-e^x \cos y + C) = -ie^z + iC.$$

必须指出：下面各题都可用这两种方法求解，限于篇幅，我们将只任给出一种。

$$(2) \quad n = e^x (x \cos y - y \sin y), \quad f(0) = 0,$$

解：

$$\begin{cases} \frac{\partial u}{\partial x} = e^x (x \cos y + \cos y - y \sin y) = \frac{\partial v}{\partial y}, \\ -\frac{\partial u}{\partial y} = e^x (x \sin y + \sin y + y \cos y) = \frac{\partial v}{\partial x}. \end{cases}$$

$$\begin{aligned}
dv &= e^x(x\cos y + \cos y - y\sin y)dy + e^x(x\sin y \\
&\quad + \sin y + y\cos y)dx \\
&= e^x d(x\sin y + \sin y + y\cos y - \sin y) + e^x d(x\sin y \\
&\quad - \sin y + \sin y + \cos y) \\
&= d(e^x x\sin y + e^x y\cos y),
\end{aligned}$$

所以 $v = e^x x\sin y + e^x y\cos y + C$.

$$\begin{aligned}
f(z) &= e^x(x\cos y - y\sin y) + ie^x(x\sin y + y\cos y) + iC \\
&= xe^x(\cos y + isiny) - e^xy(\sin y - i\cos y) + iC \\
&= xe^xe^{iy} + iye^xe^{iy} + iC = e^{x+iy}(x+iy) + iC \\
&= ze^z + iC.
\end{aligned}$$

因为 $f(0) = 0 \cdot e^{i0} + iC = 0$, 故 $C = 0$, 于是

$$f(z) = ze^z.$$

$$(3) \quad u = \frac{2\sin x}{e^{2y} + e^{-2y} - 2\cos 2x}, \quad f\left(\frac{\pi}{2}\right) = 0,$$

$$\text{解: } \begin{cases} \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = \frac{4\sin 2x(e^{2y} - e^{-2y})}{(e^{2y} + e^{-2y} - 2\cos 2x)^2} \\ \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = \frac{4\cos 2x(e^{2y} + e^{-2y} - 2\cos 2x) - 8\sin^2 2x}{(e^{2y} + e^{-2y} - 2\cos 2x)^2} \end{cases}$$

$$dv = \frac{4\sin 2x(e^{2y} + e^{-2y})dx + 4[\cos 2x(e^{2y} + e^{-2y}) - 2]dy}{[e^{2y} + e^{-2y} - 2\cos 2x]^2}$$

同 (1) 题, 把 $\frac{\partial v}{\partial x}$ 对 x 积分, 把 y 暂且当作参数,

$$v = -\frac{e^{2y} - e^{-2y}}{e^{2y} + e^{-2y} - 2\cos 2x} + \varphi(y).$$

于是,

$$\begin{aligned}
\frac{\partial v}{\partial y} &= \frac{2(e^{2y} - e^{-2y})^2 - 2(e^{2y} + e^{-2y})(e^{2y} + e^{-2y} - 2\cos 2x)}{(e^{2y} + e^{-2y} - 2\cos 2x)^2} \\
&\quad + \varphi'(y)
\end{aligned}$$

$$= \frac{4[\cos 2x(e^{2y} + e^{-2y}) - 2]}{(e^{2y} + e^{-2y} - 2\cos 2x)^2} + \varphi'(y).$$

把上式与前式比较知 $\varphi(y) = C$ ；又由于 $f\left(\frac{\pi}{2}\right) = 0$,

$$\therefore C = 0$$

则
$$v = -\frac{e^{2y} - e^{-2y}}{e^{2y} + e^{-2y} - 2\cos 2x}.$$

所以 $f(z) = u + iv = \frac{2\sin 2x - i(e^{2y}e^{-2y})}{e^{2y} + e^{-2y} - 2\cos 2x} = \operatorname{ctg} z.$

读者可以自己验证

$$\begin{aligned} \operatorname{ctg} z &= i \frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}} = \frac{(e^y - e^{-y})\sin x + i(e^y + e^{-y})\cos x}{(e^{-y} - e^y)\cos x - i(e^{-y} + e^y)\sin x} \\ &= \frac{2\sin 2x - i(e^{2y} - e^{-2y})}{e^{2y} + e^{-2y} - 2\cos 2x}. \end{aligned}$$

(4) $v = \frac{y}{x^2 + y^2}, f(2) = 0.$

解. 因为在 $v = \frac{y}{x^2 + y^2}$ 中的分母是 $x^2 + y^2$, 这种情况下改用极坐标处理比较方便, 这时

$$v = \frac{1}{\rho} \sin \varphi.$$

注意到极坐标系中的科希-里曼方程, 则

$$\begin{cases} \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = \frac{1}{\rho^2} \cos \varphi = \frac{\partial u}{\partial \rho}, \\ -\frac{\partial v}{\partial \rho} = \frac{1}{\rho^2} \sin \varphi = \frac{1}{\rho} \frac{\partial u}{\partial \varphi}. \end{cases}$$

即

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho^2} \cos \varphi, \\ \frac{\partial u}{\partial \varphi} = \frac{1}{\rho} \sin \varphi. \end{cases}$$

$$\begin{aligned}
 du &= \left(\frac{1}{\rho^2} \cos \varphi \right) d\rho + \left(\frac{1}{\rho} \sin \varphi \right) d\varphi \\
 &= \cos \varphi d\left(-\frac{1}{\rho}\right) + \frac{1}{\rho} d(-\cos \varphi) \\
 &= d\left(-\frac{1}{\rho} \cos \varphi\right),
 \end{aligned}$$

所以 $u = -\frac{1}{\rho} \cos \varphi + C,$

$$\begin{aligned}
 f(z) &= \frac{1}{\rho} (-\cos \varphi + i \sin \varphi) + C \\
 &= \frac{1}{\rho} e^{-i\varphi} + C = -\frac{1}{z} + C.
 \end{aligned}$$

又因 $f(2) = -\frac{1}{2} + C = 0$, 则 $C = \frac{1}{2}$, 从而

$$f(z) = \frac{1}{2} - \frac{1}{z}.$$

(5) $u = \frac{x^2 - y^2}{(x^2 + y^2)^2}, f(\infty) = 0.$

解: u 的表达式的分母与上题相似, 也含有因子 $x^2 + y^2$,

改用极坐标后 $u = \frac{1}{\rho^2} \cos 2\varphi$. 则

$$\begin{cases} \frac{\partial u}{\partial \rho} = -\frac{2}{\rho^2} \cos 2\varphi = \frac{1}{\rho} \frac{\partial v}{\partial \varphi}, \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{2}{\rho^3} \sin 2\varphi = -\frac{\partial v}{\partial \rho}. \end{cases}$$

即

$$\begin{cases} \frac{\partial v}{\partial \rho} = -\frac{2}{\rho^2} \cos 2\varphi, \\ \frac{\partial v}{\partial \varphi} = \frac{2}{\rho^3} \sin 2\varphi. \end{cases}$$

$$dv = \left(-\frac{2}{\rho^2} \cos 2\varphi\right) d\rho + \left(\frac{2}{\rho^3} \sin 2\varphi\right) d\varphi$$

$$\begin{aligned}
&= \frac{1}{\rho^2} d(-\sin 2\varphi) + \sin 2\varphi d\left(-\frac{1}{\rho^2}\right) \\
&= d\left(-\frac{1}{\rho^2} \sin 2\varphi\right),
\end{aligned}$$

所以
$$v = -\frac{1}{\rho^2} \sin 2\varphi + C.$$

$$\begin{aligned}
f(z) &= \frac{1}{\rho^2} \cos 2\varphi - i \frac{1}{\rho^2} \sin 2\varphi + iC \\
&= \frac{1}{\rho^2} e^{-i2\varphi} + iC = \frac{1}{z^2} + iC.
\end{aligned}$$

又因 $f(\infty) = 0 + iC = 0$, 则 $C = 0$, 从而

$$f(z) = \frac{1}{z^2}.$$

(6) $u = x^2 - y^2 + xy$, $f(0) = 0$.

解:

$$\text{因 } \begin{cases} \frac{\partial u}{\partial x} = 2x + y = \frac{\partial v}{\partial y}, \\ -\frac{\partial u}{\partial y} = 2y - x = \frac{\partial v}{\partial x}. \end{cases}$$

则 $dv = (2x + y)dy + (2y - x)dx$

$$= d\left(2xy + \frac{1}{2}y^2\right) + d\left(2xy - \frac{1}{2}x^2\right)$$

$$= d\left(2xy + \frac{1}{2}y^2 - \frac{1}{2}x^2\right),$$

$$v = 2xy + \frac{1}{2}(y^2 - x^2) + C.$$

所以
$$\begin{aligned}
f(z) &= x^2 - y^2 + xy + i\left[2xy + \frac{1}{2}(y^2 - x^2)\right] + iC \\
&= x^2 - y^2 + i2xy - \left[\frac{1}{2}i(x^2 - y^2) - xy\right] + iC
\end{aligned}$$

$$\begin{aligned}
 &= (x+iy)^2 - i\frac{1}{2}\left[(x^2-y^2) + i2xy\right] + iC \\
 &= z^2 - i\frac{1}{2}z^2 + iC.
 \end{aligned}$$

又因 $f(0) = 0 + iC = 0$, 则 $C = 0$, 从而

$$f(z) = z^2\left(1 - \frac{i}{2}\right).$$

$$(7) \quad u = x^3 - 3xy^2, f(0) = 0.$$

解:

$$\text{因} \begin{cases} \frac{\partial u}{\partial x} = 3x^2 - 3y^2 = \frac{\partial v}{\partial y}, \\ -\frac{\partial u}{\partial y} = 6xy = \frac{\partial v}{\partial x}. \end{cases}$$

$$\begin{aligned}
 \text{则} \quad dv &= (3x^2 - 3y^2)dy + 6xydx \\
 &= d(3x^2y - y^3) + d(3x^2y) \\
 &= d(3x^2y - y^3), \\
 v &= 3x^2y - y^3 + C.
 \end{aligned}$$

$$\begin{aligned}
 \text{所以} \quad f(z) &= x^3 - 3xy^2 + i(3x^2y - y^3 + C) \\
 &= (x+iy)^3 + iC = z^3 + iC.
 \end{aligned}$$

又因 $f(0) = 0 + iC = 0$, 则 $C = 0$, 从而

$$f(z) = z^3.$$

$$(8) \quad u = x^3 + 6x^2y - 3xy^2 - 2y^3, f(0) = 0.$$

$$\text{解:} \begin{cases} \frac{\partial u}{\partial x} = 3x^2 + 12xy - 3y^2 = \frac{\partial v}{\partial y}, \\ -\frac{\partial u}{\partial y} = -6x^2 + 6xy + 6y^2 = \frac{\partial v}{\partial x}. \end{cases}$$

$$\begin{aligned}
 \text{则} \quad dv &= (3x^2 + 12xy - 3y^2)dy + (-6x^2 + 6xy + 6y^2)dx \\
 &= d(3x^2y + 6xy^2 - y^3) + d(-2x^3 + 3x^2y + 6xy^2). \\
 v &= -2x^3 + 3x^2y + 6xy^2 - y^3 + C.
 \end{aligned}$$

$$\text{所以} \quad f(z) = x^3 + 6x^2y - 3xy^2 - 2y^3 + i$$

$$\begin{aligned}
 & (-2x^3 + 3x^2y + 6xy^2 - y^3) + iC \\
 & = (x + iy)^3 - 2i(x + iy)^3 + iC = z^3(1 - 2i) + iC.
 \end{aligned}$$

又因 $f(0) = 0 + iC = 0$, 则 $C = 0$, 从而

$$f(z) = z^3(1 - 2i).$$

$$(9) \quad u = x^4 - 6x^2y^2 + y^4, f(0) = 0.$$

$$\text{解: } \begin{cases} \frac{\partial v}{\partial x} = 4x^3 - 12xy^2 = \frac{\partial v}{\partial y}, \\ -\frac{\partial u}{\partial y} = 12x^2y - 4y^3 = -\frac{\partial v}{\partial x}. \end{cases}$$

$$\begin{aligned}
 dv &= (4x^3 - 12xy^2)dy + (12x^2y - 4y^3)dx \\
 &= d(4x^3y - 4xy^3) + d(4x^3y - 4xy^3).
 \end{aligned}$$

$$v = 4x^3y - 4xy^3 + C.$$

$$\begin{aligned}
 \text{于是 } f(z) &= x^4 - 6x^2y^2 + y^4 + i(4x^3y - 4xy^3 + C) \\
 &= (x + iy)^4 + iC = Z^4 + iC.
 \end{aligned}$$

因 $f(0) = 0 + iC = 0$, 则 $C = 0$, 所以

$$f(z) = z^4,$$

$$(10) \quad u = \ln \rho, f(1) = 0.$$

$$\text{解: } \begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = \frac{1}{\rho}, \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} = 0. \end{cases}$$

即

$$\begin{cases} \frac{\partial v}{\partial \varphi} = 1, \\ \frac{\partial v}{\partial \rho} = 0. \end{cases}$$

则

$$dv = d\varphi,$$

$$v = \varphi + C.$$

$$\begin{aligned}
 \text{所以 } f(z) &= \ln \rho + i\varphi + iC = \ln |z| + i \arg z + iC \\
 &= \ln z + iC.
 \end{aligned}$$

又因 $f(1) = 0 + iC = 0$, 则 $C = 0$, 从而

$$f(z) = \ln z,$$

$$(11) \quad u = \varphi, \quad f(1) = 0.$$

解: 因

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = 0, \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} = \frac{1}{\rho}, \end{cases}$$

即

$$\begin{cases} \frac{\partial v}{\partial \varphi} = 0, \\ \frac{\partial v}{\partial \rho} = -\frac{1}{\rho}. \end{cases}$$

则

$$dv = -\frac{1}{\rho} d\rho = d(-\ln \rho),$$

$$v = -\ln \rho + C.$$

所以

$$\begin{aligned} f(z) &= \varphi - i \ln \rho + iC \\ &= -i(\ln \rho + i\varphi) + iC = -i \ln z + iC. \end{aligned}$$

又因 $f(1) = 0 + iC = 0$, 则 $C = 0$, 从而

$$f(z) = -i \ln z + iC.$$

3. 试从极坐标系中的科希-里曼方程

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \\ \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = -\frac{\partial v}{\partial \rho} \end{cases}$$

中消去 u 或 v 。

解: 该方程可改写为

$$\begin{cases} \rho \frac{\partial u}{\partial \rho} = \frac{\partial v}{\partial \varphi}, & (1) \end{cases}$$

$$\begin{cases} -\frac{1}{\rho} \frac{\partial u}{\partial \varphi} = \frac{\partial v}{\partial \rho}. & (2) \end{cases}$$

(1) 式对 ρ 微分一次, (2) 式对 φ 微分一次,

$$\left\{ \begin{aligned} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) &= \frac{\partial^2 v}{\partial \rho \partial \varphi}, \end{aligned} \right. \quad (3)$$

$$\left\{ \begin{aligned} -\frac{1}{\rho} \frac{\partial^2 u}{\partial \varphi^2} &= \frac{\partial^2 v}{\partial \rho \partial \varphi}. \end{aligned} \right. \quad (4)$$

(3) - (4) 得

$$\frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho} \frac{\partial^2 u}{\partial \varphi^2} = 0. \quad (5)$$

科希-里曼方程还可改写为

$$\left\{ \begin{aligned} \frac{\partial u}{\partial \rho} &= \frac{1}{\rho} \frac{\partial v}{\partial \varphi}, \end{aligned} \right. \quad (6)$$

$$\left\{ \begin{aligned} \frac{\partial u}{\partial \varphi} &= -\rho \frac{\partial v}{\partial \rho}. \end{aligned} \right. \quad (7)$$

(6) 式对 φ 微分一次, (7) 式对 ρ 微分一次,

$$\left\{ \begin{aligned} \frac{\partial^2 v}{\partial \rho \partial \varphi} &= \frac{\partial}{\partial \varphi} \left(\frac{1}{\rho} \frac{\partial v}{\partial \varphi} \right), \end{aligned} \right. \quad (8)$$

$$\left\{ \begin{aligned} \frac{\partial^2 u}{\partial \rho \partial \varphi} &= \frac{\partial}{\partial \rho} \left(-\rho \frac{\partial v}{\partial \rho} \right). \end{aligned} \right. \quad (9)$$

$$(8) - (9) \text{ 得 } \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{1}{\rho} \frac{\partial^2 v}{\partial \varphi^2} = 0 \quad (10)$$

显然, 消去 v (或 u) 后的方程 (9) (或 (10)) 即极坐标系中的拉普拉斯方程 (5.2) 或 (5.3) .

§6. 平面标量场

1. 已知复势 $f(z) = \frac{1}{z-2+i}$, 试描画等温网。

$$\begin{aligned} \text{解: 由 } f(z) &= \frac{1}{z-2+i} = \frac{1}{(x-2)+i(y+1)} \\ &= \frac{x-2}{(x-2)^2+(y+1)^2} + i \frac{-(y+1)}{(x-2)^2+(y+1)^2} \end{aligned}$$

得到等温网的两族曲线方程

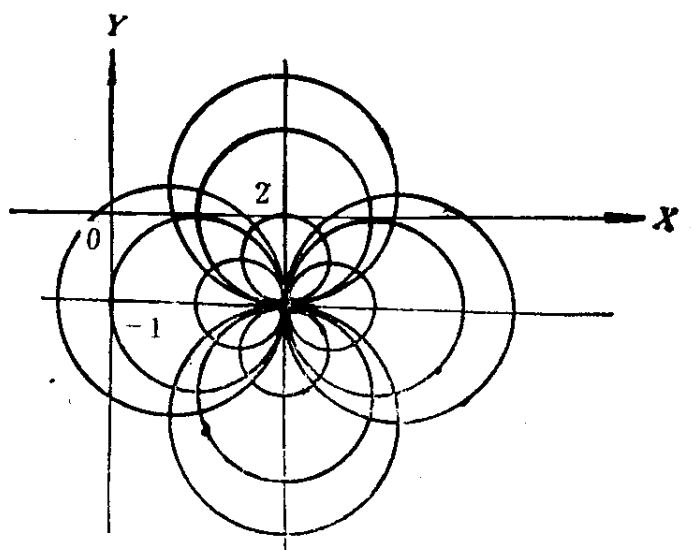


图 1-6

$$\begin{cases} \frac{x-2}{(x-2)^2 + (y+1)^2} = C'_1, \\ \frac{y+1}{(x-2)^2 + (y+1)^2} = C'_2, \end{cases}$$

或
$$\begin{cases} (x-2-C_1)^2 + (y+1)^2 = C_1^2, \\ (x-2)^2 + (y+1-C_2)^2 = C_2^2. \end{cases}$$

故等温网为：在点 $(2, -1)$ 跟直线 $x = 2$, $y = -1$ 相切的圆族。

2. 已知流线族的方程为 “ $\frac{y}{x} = \text{常数}$ ”，求复势。

解：(i) 如令 $v = \frac{y}{x}$, 则 $v_{xx} = \frac{2y}{x^3}$, $v_{yy} = 0$,

从而 $v_{xx} + v_{yy} \neq 0$, $v = \frac{y}{x}$ 不是调和函数。

(ii) 改令 $v = F(t)$, $\left(t = \frac{y}{x}\right)$,

则
$$v_x = F' \left[-\frac{y}{x^2} \right], \quad v_{xx} = F'' \left[\frac{y^2}{x^4} \right] + F' \left[\frac{2y}{x^3} \right];$$

$$v_y = F' \left[\frac{1}{x} \right], \quad v_{yy} = F'' \left[\frac{1}{x^2} \right];$$

应指出：这里必须有 $v_{xx} + v_{yy} = 0$ ，

即
$$F'' \left[\frac{x^2 + y^2}{x^4} \right] + F' \left[\frac{2y}{x^3} \right] = 0.$$

$$\frac{F''}{F'} = -\frac{2y}{x^3} \cdot \frac{x^4}{x^2 + y^2} = \frac{2xy}{x^2 + y^2} = \frac{-2}{\frac{x}{y} + \frac{y}{x}}$$

$$= \frac{-2}{t + \frac{1}{t}} = -\frac{2t}{1 + t^2},$$

$$\ln F'(t) = - \int \frac{2t}{1 + t^2} dt = -\ln(1 + t^2) + \ln C_1,$$

$$F'(t) = \frac{C_1}{1 + t^2};$$

$$F(t) = C_1 \int \frac{dt}{1 + t^2} = C_1 \arctg t + C_2 = C_1 \arctg \frac{y}{x} + C_2,$$

所以
$$v = C_1 \arctg \frac{y}{x} + C_2.$$

这里的记号 v_x 和 v_y 分别代表 $\frac{\partial v}{\partial x}$ 和 $\frac{\partial v}{\partial y}$ ， v_{xx} 和 v_{yy} 分别代表

$\frac{\partial^2 v}{\partial x^2}$ 和 $\frac{\partial^2 v}{\partial y^2}$ (下同)。

(iii) 根据科希-里曼方程 $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$ 知

$$u_y = -v_x = C_1 \frac{y}{x^2 + y^2},$$

因而

$$u = C_1 \int \frac{y}{x^2 + y^2} dy = C_1 \cdot \frac{1}{2} \ln(x^2 + y^2) + C_4(x).$$

现在要确定 $C_4(x)$, 注意到

$$u_x = \frac{C_1 x}{x^2 + y^2} + C_4'(x)$$

根据柯希-里曼方程 $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, 这应等于 v_y , 即 $\frac{C_1 x}{x^2 + y^2}$,

所以 $C_4'(x) = 0, C_4(x) = C_3$, 于是

$$u = C_1 \cdot \frac{1}{2} \ln(x^2 + y^2) + C_3;$$

$$f(z) = C_1 \cdot \frac{1}{2} \ln(x^2 + y^2) + C_3 + iC_1 \operatorname{arctg} \frac{y}{x} + iC_2$$

$$= C_1 \frac{1}{2} \ln(x^2 + y^2) + C_3$$

$$+ C_1 i \left[-\frac{1}{2} i \ln \frac{1 + i(y/x)}{1 - i(y/x)} \right] + iC_2$$

$$= C_1 \left\{ \frac{1}{2} \ln(x^2 + y^2) + \frac{1}{2} \ln \frac{(x + iy)^2}{x^2 + y^2} \right\} + C_3 + iC_2$$

$$= C_1 \ln(x + iy) + C_3 + iC_2$$

$$= C_1 \ln z + C_3 + iC_2,$$

这就是所要求的复势。

3. 已知等势线族的方程为“ $x^2 + y^2 = \text{常数}$ ”, 求复势。

解: (i) 令 $u = F(t)$, ($t = x^2 + y^2$)

则

$$\begin{cases} u_x = 2xF', & u_{xx} = 2F' + 4x^2F'', \\ u_y = 2yF', & u_{yy} = 2F' + 4y^2F''. \end{cases}$$

$$(4x^2 + 4y^2)F'' + 4F' = 0,$$

$$\frac{F''}{F'} = -\frac{1}{x^2 + y^2} = -\frac{1}{t}, F' = \frac{C_1}{t}.$$

求出 $F = C_1 \ln t + C_2 = C_1 \ln(x^2 + y^2) + C_2.$

即 $u = C_1 \ln(x^2 + y^2) + C_2.$

(ii) $u_x = C_1 \frac{2x}{x^2 + y^2}, u_y = C_1 \frac{2y}{x^2 + y^2},$ 根据科希—里曼方程

$$v_y = u_x = C_1 \frac{2x}{x^2 + y^2},$$

因而 $v = C_1 \int \frac{2x}{x^2 + y^2} dy = 2C_1 \operatorname{arctg} \frac{y}{x} + C_4(x).$

又, $v_x = 2C_1 \cdot \frac{-y}{x^2 + y^2} + C_4'(x) = -u_y = -2C_1 \frac{y}{x^2 + y^2}.$

则 $C_4'(x) = 0, \quad C_4(x) = C_3.$

所以 $v = 2C_1 \operatorname{arctg} \left(\frac{y}{x} \right) + C_3 = -iC_1 \ln \frac{(x+iy)^2}{x^2 + y^2} + C_3.$

$$\begin{aligned} \text{(iii)} \quad f(z) &= C_1 \ln(x^2 + y^2) + C_2 + i \left[-iC_1 \ln \frac{(x+iy)^2}{x^2 + y^2} \right. \\ &\quad \left. + C_3 \right] \end{aligned}$$

$$= C_1 \ln(x^2 + y^2) + C_2 + C_1 \ln \frac{(x+iy)^2}{x^2 + y^2} + iC_3$$

$$= C \ln z^2 + C_2 + iC_3 = 2C_1 \ln z + C_2 + iC_3.$$

这就是所要求的复势。

4. 已知电力线为跟实轴相切于原点的圆族, 求复势。

解: 如图1-7所示, 该圆族的方程是

$$x^2 + (y - C_4)^2 = C_4^2,$$

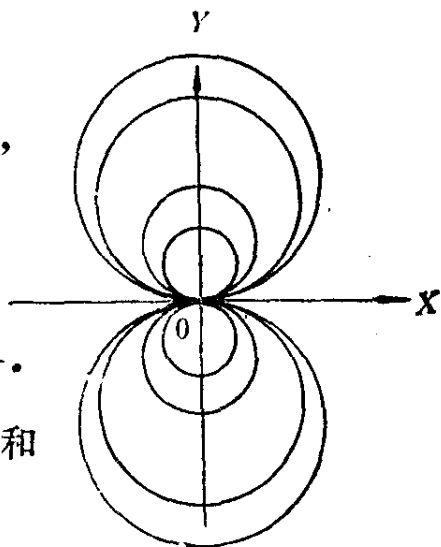
或 $\frac{-y}{x^2 + y^2} = C_4^*, (C_4^* \text{ 亦为常数}),$

如令 $v = \frac{-y}{x^2 + y^2},$

则

$$\begin{aligned} v_x &= \frac{2xy}{(x^2 + y^2)^2}, \\ v_{xx} &= \frac{2y}{(x^2 + y^2)^2} - \frac{8x^2y}{(x^2 + y^2)^3}, \\ v_y &= \frac{2y^2}{(x^2 + y^2)^2} - \frac{1}{x^2 + y^2}, \\ v_{yy} &= \frac{6y}{(x^2 + y^2)^2} - \frac{8y^3}{(x^2 + y^2)^3}. \end{aligned}$$

由此得 $v_{xx} + v_{yy} = 0$, 故这里的 v 是调和函数.



应指出: 既然 $v = -\frac{y}{x^2 + y^2}$ 是调和函数, 图 1-7

所以我们可令复势的虚部 $v(x, y)$ 就等于这个 v , 下面再求 u .

因 v_x, v_y 已在上面写出, 由科希-里曼方程,

$$u_y = -v_x = -\frac{2xy}{(x^2 + y^2)^2},$$

$$u = -2x \int \frac{y dy}{(x^2 + y^2)^2} = \frac{x}{x^2 + y^2} + C_3(x).$$

则

$$\begin{aligned} u_x &= \frac{1}{x^2 + y^2} - \frac{2x^2}{(x^2 + y^2)^2} + C'_3(x) \\ &= v_y = \frac{2y^2}{(x^2 + y^2)^2} - \frac{1}{x^2 + y^2}, \end{aligned}$$

给出 $C'_3(x) = 0$, $C_3(x) = C_2$, 故 $u = \frac{x}{x^2 + y^2} + C_2$.

于是求出复势 $f(z) = \frac{x}{x^2 + y^2} + C_2 + i \frac{-y}{x^2 + y^2} = \frac{x - iy}{x^2 + y^2} + C_2$

$$= \frac{1}{x + iy} + C_2 = \frac{1}{z} + C_2.$$

5. 在圆柱 $|z| = R$ 的外部的平面静电场的复势为 $f(z) =$

$i2\sigma \ln\left(\frac{R}{z}\right)$ 求柱面上的电荷面密度.

$$\begin{aligned}\text{解: } f(z) &= i2\sigma \ln \frac{R}{z} = i2\sigma \ln \frac{R}{\rho e^{i\varphi}} \\ &= 2i\sigma \left[\ln \frac{R}{\rho} - i\varphi \right] = 2\sigma\varphi + 2i\sigma \ln \frac{R}{\rho}\end{aligned}$$

这里, 取电势 $u = 2\sigma \ln \frac{R}{\rho}$, 则圆柱表面外的法向场强

$$\begin{aligned}E \Big|_R &= - \frac{\partial u}{\partial \rho} \Big|_R = - \frac{\partial}{\partial \rho} (2\sigma \ln R - 2\sigma \ln \rho) \\ &= \frac{2\sigma}{\rho} \Big|_R = \frac{2\sigma}{R}.\end{aligned}$$

设电势以高斯单位表示, 以高斯单位表示的高斯定理为

$$\oint \vec{E} \cdot d\vec{S} = 4\pi q.$$

设面密度为 σ_e , 面积为 S , 则

$$\frac{2\sigma}{R} S = 4\pi \sigma_e S, \sigma_e = \frac{\sigma}{2\pi R}.$$

其实, 电势 $u = 2\sigma \ln \frac{R}{\rho}$ 的共轭调和函数 $2\sigma\varphi$ 就是通量函数, 而按照高斯定理

$$2\sigma\varphi_2 - 2\sigma\varphi_1 = 4\pi\sigma_e R(\varphi_2 - \varphi_1),$$

$$2\sigma(\varphi_2 - \varphi_1) = 4\pi\sigma_e R, \text{ 所以 } \sigma_e = \frac{\sigma}{2\pi R}.$$

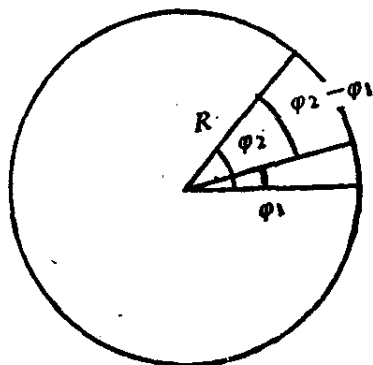


图 1-8

6. 有二个平行而均匀带电的线电荷, 每单位长度所带电量分别是 $+q$ 和 $-q$, 两线相距为 $2a$, 求这个平面静电场的复势、电力线和等势线.

解: 考虑一线电荷在原点、单位长度所带电量为 Q , 显然可取通量函数为 $v = 2Q\varphi$ (高斯单位制), u 为电势, 则

$$\frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = \frac{2Q}{\rho}, \quad \frac{\partial u}{\partial \varphi} = -\rho \frac{\partial v}{\partial \rho} = 0.$$

于是

$$u = 2Q \ln \rho + C,$$

所以复势 $f(z) = C + 2Q(\ln \rho + i\varphi) = C + 2Q \ln z$, 由此可知令 $Q = +q$, 并将线电荷移至 $(a, 0)$, 复势为 $f_1(z) = C_1 + 2q \ln(z - a)$, 令 $Q = -q$, 并将线电荷移至 $(-a, 0)$, 复势 $f_2(z) = C_2 - 2q \ln(z + a)$, 所要求的复势即为 $f_1(z) + f_2(z)$ (依电势迭加原理以及和的通量等于通量的和)。

$$f(z) = 2q \ln \frac{z-a}{z+a} + C, \quad (C = C_1 + C_2),$$

或者置 $+q$ 于 $(-a, 0)$, 置 $-q$ 于 $(a, 0)$, 则

$$f(z) = -2q \ln \frac{z-a}{z+a} = 2q \ln \frac{z+a}{z-a},$$

电力线族为 $I_m \ln \frac{z-a}{z+a} = \text{常数}$,

等势线族为 $R_e \ln \frac{z-a}{z+a} = \text{常数}$,

$$\begin{aligned} \ln \frac{z-a}{z+a} &= \ln \frac{x+iy-a}{x+iy+a} = \ln \frac{x^2+y^2-a^2+2ia y}{(x+a)^2+y^2} \\ &= \ln \left[\frac{\sqrt{(x^2+y^2-a^2)^2+4a^2 y^2}}{(x+a)^2+y^2} \right. \\ &\quad \left. e^{i \arctg \frac{2ay}{x^2+y^2-a^2}} \right] \\ &= \frac{1}{2} \ln \frac{(x^2+y^2-a^2)^2-4a^2 y^2}{[(x+a)^2+y^2]^2} \\ &\quad + \arctg \frac{2ay}{x^2+y^2-a^2} \\ &= \frac{1}{2} \ln \frac{(x-a)^2+y^2}{(x+a)^2+y^2} + \arctg \frac{2ay}{x^2+y^2-a^2}, \end{aligned}$$

电力线族为 $x^2 + y^2 - a^2 = 2ac_1 y$,

即 $x^2 + y^2 - 2ac_1 y + a^2 c_1^2 = a^2 + a^2 c_1^2$,

$$x^2 + (y - ac_1)^2 = a^2(1 + c_1^2),$$

等势线族为 $c_2[(x-a)^2 + y^2] = (x+a)^2 + y^2$,

$$(c_2 - 1)x^2 - 2(c_2 + 1)ax + (c_2 - 1)y^2 = (1 - c_2)a^2,$$

$$x^2 - 2\frac{c_2 + 1}{c_2 - 1}ax + \left(\frac{c_2 + 1}{c_2 - 1}\right)^2 a^2 + y^2$$

$$= -a^2 + \left(\frac{c_2 + 1}{c_2 - 1}\right)^2 a^2,$$

$$\left(x - \frac{c_2 + 1}{c_2 - 1}a\right)^2 + y^2 = \frac{(c_2 + 1)^2 - (c_2 - 1)^2}{(c_2 - 1)^2} a^2,$$

$$\left(x - \frac{c_2 + 1}{c_2 - 1}a\right)^2 + y^2 = \frac{4c_2}{(c_2 - 1)^2} a^2.$$

第二章 复变函数的积分

§9. 科希公式

1. 已知函数 $\psi(t, x) = e^{2tx - t^2}$, 把 x 当作参数, 把 t 认作是复变数, 试应用科希公式把 $\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0}$ 表为回路积分.

对回路积分进行积分变数的代换 $t = x - z$, 并借以证明 $\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$.

解: (i) 把 $\frac{\partial^n \psi}{\partial t^n}$ 表为回路积分如下:

$$\begin{aligned} \left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} &= \frac{n!}{2\pi i} \oint \frac{e^{2\xi x - \xi^2}}{(\xi - t)^{n+1}} d\xi \\ &= \frac{n!}{2\pi i} \oint \frac{e^{2\xi x - \xi^2}}{\xi^{n+1}} d\xi. \end{aligned}$$

(ii) 证明: 以 $\xi = x - z$ 代入上式

$$\begin{aligned} \left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} &= \frac{n!}{2\pi i} \oint \frac{e^{x^2 - z^2}}{(x - z)^{n+1}} d(-z) \\ &= \frac{n!}{2\pi i} \oint \frac{e^{x^2} \cdot e^{-z^2}}{(-1)^n (z - x)^{n+1}} dz \\ &= e^{x^2} \frac{n!}{2\pi i} \oint \frac{(-1)^n e^{-z^2} dz}{(z - x)^{n+1}} \\ &= (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}, \text{ 得证.} \end{aligned}$$

2. 已知函数 $\psi(x, t) = \frac{e^{-xt/(1-t)}}{1-t}$, 试把 x 当作参数, 把 t 认

为是复变数, 试应用科希公式把 $\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0}$ 表为回路积分.

对回路积分进行积分变数的代换, $t = (z-x)/z$, 并借以证明 $\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = e^x \frac{d^n}{dx^n} (x^n e^{-x})$.

解: (i) 把 $\frac{\partial^n \psi}{\partial t^n}$ 表为回路积分如下:

$$\frac{\partial^n \psi}{\partial t^n} = \frac{n!}{2\pi i} \oint_1 \frac{e^{-\frac{x\xi}{1-\xi}} / (1-\xi) d\xi}{(\xi-t)^{n+1}},$$

$$\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = \frac{n!}{2\pi i} \oint_1 \frac{e^{-\frac{x\xi}{1-\xi}} / (1-\xi)}{\xi^{n+1} (1-\xi)} d\xi.$$

(ii) 证明: 以 $\xi = (z-x)/z$ 代入上式,

$$\begin{aligned} \frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} &= \frac{n!}{2\pi i} \oint_1 \frac{e^{-x\left(\frac{z-x}{z}\right)} / \left(1 - \frac{z-x}{z}\right)}{\left(\frac{z-x}{z}\right)^{n+1} \left(1 - \frac{z-x}{z}\right)} \\ &\quad \left(\frac{x}{z^2}\right) dz \end{aligned}$$

$$= \frac{n!}{2\pi i} \oint_1 \frac{z^{n+1} \cdot e^{-(z-x)} \cdot \frac{z}{x} \left(\frac{x}{z^2}\right) dz}{(z-x)^{n+1}}$$

$$= e^x \frac{n!}{2\pi i} \oint_1 \frac{z^n e^{-z}}{(z-x)^{n+1}} dz$$

$$= e^x \frac{d^n}{dx^n} (x^n e^{-x}), \text{ 得证.}$$

第三章 幂级数展开

§11. 幂级数

1. 把幂级数 $\sum_{k=0}^{\infty} a_k (z - z_0)^k = a_0 + a_1 (z - z_0) + a_2 (z - z_0)^2 + \cdots + a_k (z - z_0)^k + \cdots$ 逐项求导，求所得级数的收敛半径，以此验证逐项求导，并不改变收敛半径。

解：该幂级数的收敛半径是 $R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right|$ 。

对该级数逐项求导后得：

$$\frac{d}{dz_1} \sum_{k=0}^{\infty} a_k (z - z_0)^k = a_1 + 2a_2 (z_1 - z_0) + \cdots + K a_k (z - z_0)^{k-1} + (K+1) a_{k+1} (z - z_0)^k + \cdots$$

$$\begin{aligned} \text{其收敛半径为 } R &= \lim_{k \rightarrow \infty} \left| \frac{K a_k}{(K+1) a_{k+1}} \right| \lim_{k \rightarrow \infty} \left| \frac{a_k}{\left(1 + \frac{1}{K}\right) a_{k+1}} \right| \\ &= \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right|, \end{aligned}$$

所以逐项求导后，并不改变其收敛半径。

2. 把上题的幂级数逐项积分，求所得级数的收敛半径，以此验证逐项积分并不改变收敛半径。

解：对该级数逐项积分后得：

$$\int \sum_{k=0}^{\infty} a_k (z - z_0)^k d(z - z_0) = a_0 (z - z_0) + \frac{1}{2} a_1 (z - z_0)^2 + \cdots$$

$$+ \frac{1}{3} a_2 (z - z_0)^3 + \cdots + \frac{1}{K+1} a_K (z - z_0)^{K+1} + \frac{1}{K+2} a_{K+1} (z - z_0)^{K+2} + \cdots,$$

其收敛半径为:

$$\begin{aligned} R &= \lim_{k \rightarrow \infty} \left| \frac{\frac{1}{K+1} a_K}{\frac{1}{K+2} a_{K+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{(K+2) a_K}{(K+1) a_{K+1}} \right| \\ &= \lim_{k \rightarrow \infty} \left| \frac{\left(1 + \frac{2}{K}\right) a_K}{\left(1 + \frac{1}{K}\right) a_{K+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_K}{a_{K+1}} \right|, \end{aligned}$$

故逐项积分后并不改变收敛半径.

3. 求下列幂级数的收敛圆.

$$(1) \sum_{k=1}^{\infty} \frac{1}{K} (z-i)_k$$

$$\begin{aligned} \text{解: 其收敛半径 } R &= \lim_{k \rightarrow \infty} \left| \frac{1/K}{1/(K+1)} \right| = \lim_{k \rightarrow \infty} \left| \frac{K+1}{K} \right| \\ &= \lim_{k \rightarrow \infty} \left| 1 + \frac{1}{K} \right| = 1 \end{aligned}$$

$\therefore$ 收敛圆为 $|z-i| = 1$.

$$(2) \sum_{k=1}^{\infty} K^{\ln K} (z_1 - 2)^K.$$

$$\text{解: 收敛半径 } R = \lim_{k \rightarrow \infty} \left| \frac{K^{\ln K}}{(K+1)^{\ln(K+1)}} \right|,$$

$$\text{令 } (K+1)^{\ln(K+1)} = (K+1)^{\ln \left[K \left(1 + \frac{1}{K} \right) \right]}$$

$$= (K+1)^{\ln K} \cdot (K+1)^{\ln \left(1 + \frac{1}{K} \right)},$$

故

$$R = \lim_{k \rightarrow \infty} \left[\frac{K \ln K}{(K+1) \ln K} \cdot \frac{1}{(K+1) \ln \left(1 + \frac{1}{K}\right)} \right]$$

$$= \frac{1}{\lim_{k \rightarrow \infty} \left(1 + \frac{1}{K}\right)^{\ln K}} \cdot \frac{1}{\lim_{k \rightarrow \infty} (K+1) \ln \left(1 + \frac{1}{K}\right)},$$

$$\text{记 } l_1 = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{K}\right)^{\ln K},$$

$$l_2 = \lim_{k \rightarrow \infty} (K+1) \ln \left(1 + \frac{1}{K}\right),$$

则

$$R = \frac{1}{l_1 l_2}.$$

现计算 l_1 .

$$\ln l_1 = \lim_{k \rightarrow \infty} \left[\ln K \cdot \ln \left(1 + \frac{1}{K}\right) \right]$$

$$= \lim_{k \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{K}\right)}{\frac{1}{\ln K}},$$

这是 $\frac{0}{0}$ 型的不定式, 可用罗毕达法则确定极限,

$$\ln l_1 = \lim_{k \rightarrow \infty} \frac{\frac{1}{1+yK} \left(-\frac{1}{K^2}\right)}{-\frac{1}{(\ln K)^2} \cdot \frac{1}{K}} = \lim_{k \rightarrow \infty} \frac{(\ln K)^2}{K+1},$$

这是 $\frac{\infty}{\infty}$ 型的不定式, 再用罗毕达法则,

$$\ln l_1 = \lim_{k \rightarrow \infty} \frac{(2 \ln K) \cdot \frac{1}{K}}{1} = \lim_{k \rightarrow \infty} \frac{2 \ln K}{K},$$

再用罗毕达法则,

$$\ln l_1 = \lim_{k \rightarrow \infty} \frac{2 \cdot \frac{1}{K}}{1} = 0,$$

因而

$$l_1 = 1,$$

同理,

$$\begin{aligned} \ln l_2 &= \lim_{k \rightarrow \infty} \left[\ln \left(1 + \frac{1}{K} \right) \cdot \ln (K+1) \right] \\ &= \lim_{k \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{K} \right)}{\frac{1}{\ln (K+1)}}, \end{aligned}$$

用罗毕达法则,

$$\begin{aligned} \ln l_2 &= \lim_{k \rightarrow \infty} - \frac{\frac{1}{1 + 1/K} \left(-\frac{1}{K^2} \right)}{\frac{1}{[\ln (K+1)]^2} \cdot \frac{1}{K+1}} \\ &= \lim_{k \rightarrow \infty} \frac{[\ln (K+1)]^2}{K} \cdot \lim_{k \rightarrow \infty} \left(1 + \frac{1}{K} \right) \\ &= \lim_{k \rightarrow \infty} \frac{[\ln (K+1)]^2}{K}, \end{aligned}$$

用罗毕达法则,

$$\begin{aligned} \ln l_2 &= \lim_{k \rightarrow \infty} \frac{\left[2 \ln (K+1) \right] \frac{1}{K+1}}{1} \\ &= \lim_{k \rightarrow \infty} \frac{2 \ln (K+1)}{K+1}, \end{aligned}$$

再用罗毕达法则,

$$\ln l_2 = \lim_{k \rightarrow \infty} \frac{2 \cdot \frac{1}{K+1}}{1} = 0,$$

因而

$$l_2 = 1,$$

结果, 收敛半径

$$R = \frac{1}{l_1 l_2} = 1,$$

所以收敛圆为 $|z_1 - 2| = 1$.

$$(3) \sum_{k=1}^{\infty} \left(\frac{z}{K} \right)^k.$$

解一: 收敛半径

$$\begin{aligned} R &= \lim_{k \rightarrow \infty} \frac{1}{k \sqrt[k]{|a_k|}} = \lim_{k \rightarrow \infty} \frac{1}{k \sqrt[k]{\frac{1}{K^k}}} \\ &= \lim_{k \rightarrow \infty} k \sqrt[k]{K^k} \\ &= \lim_{k \rightarrow \infty} K = \infty. \end{aligned}$$

解二: 收敛半径为

$$\begin{aligned} R &= \lim_{k \rightarrow \infty} \left| \frac{K^{-k}}{(K+1)^{-(K+1)}} \right| = \lim_{k \rightarrow \infty} \left| \frac{(K+1)^{k+1}}{K^k} \right| \\ &= \lim_{k \rightarrow \infty} \left| \frac{1}{K^k} [K^{k+1} + (K+1)K^k + \dots] \right| \\ &= \lim_{k \rightarrow \infty} |K + (K+1) + \dots| = \infty, \end{aligned}$$

或

$$R = \lim_{k \rightarrow \infty} \frac{1}{k \sqrt[k]{\left(\frac{1}{K}\right)^k}} = \lim_{k \rightarrow \infty} K = \infty,$$

所以只要 z 是有限的, 此幂级数就收敛, 收敛圆 $|z| = R < \infty$.

$$(4) \sum_{k=1}^{\infty} K! \left(\frac{z}{K} \right)^k.$$

解: 收敛半径

$$R = \lim_{k \rightarrow \infty} \left[\frac{K!}{(K+1)!} \cdot \frac{(K+1)^{k+1}}{K^k} \right] = \lim_{k \rightarrow \infty} \left[\frac{1}{K+1} \cdot \right]$$

$$= \frac{(K+1)^{k+1}}{K^k} \Bigg] = \lim_{k \rightarrow \infty} \frac{(K+1)^k}{K^k}$$

$$= \lim_{k \rightarrow \infty} \left(1 + \frac{1}{K}\right)^k = e,$$

所以收敛圆是 $|z| = e$.

$$(5) \sum_{k=1}^{\infty} K^k (z-3)^k.$$

解一：收敛半径

$$R = \lim_{k \rightarrow \infty} \frac{1}{k\sqrt[k]{|a_k|}} = \lim_{k \rightarrow \infty} \frac{1}{k\sqrt[k]{K^k}} = \lim_{k \rightarrow \infty} \frac{1}{K} = 0.$$

解二：收敛半径

$$R = \lim_{k \rightarrow \infty} \left| \frac{K^k}{(K+1)^{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \left[K + (K+1) + \dots \right]^{-1} \right|$$

$$= 0,$$

所以收敛圆为 $|z-3| = 0$ ，只要 $z \neq 3$ ，此幂级数就发散。

4. 已知幂级数 $\sum_{k=0}^{\infty} a_k z^k$ 和 $\sum_{k=0}^{\infty} b_k z^k$ 的收敛半径分别为 $R_1 =$

$$\lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| \left(\text{或 } R_1 = \lim_{k \rightarrow \infty} \frac{1}{k\sqrt[k]{|a_k|}} \right) \text{ 和 } R_2 = \lim_{k \rightarrow \infty} \left| \frac{b_k}{b_{k+1}} \right|$$

$\left(\text{或 } R_2 = \lim_{k \rightarrow \infty} \frac{1}{k\sqrt[k]{|b_k|}} \right)$ ，求下列幂级数的收敛半径。

$$(1) \sum_{k=0}^{\infty} (a_k + b_k) z^k.$$

解一：如果 $R_1 \leq R_2$ ，则在圆 $|z| = R_1$ 的内部，幂级数 $\sum_{k=0}^{\infty}$

$a_k z^k$ 和 $\sum_{k=0}^{\infty} b_k z^k$ 都绝对收敛，从而 $\sum_{k=0}^{\infty} (a_k + b_k) z^k$ 必是绝对

收敛的。所以该幂级数的收敛半径不小于 R_1 和 R_2 中的较小者。

解二：记 $|a_k|$ 和 $|b_k|$ 中的较大者为 A_k ，则 $\sum_{k=0}^{\infty} (a_k + b_k)z^k$ 的

收敛半径

$$\begin{aligned} R &= \lim_{k \rightarrow \infty} \frac{1}{k \sqrt{|a_k + b_k|}} = \frac{1}{\lim_{k \rightarrow \infty} k \sqrt{|a_k + b_k|}} \\ &\geq \frac{1}{\lim_{k \rightarrow \infty} k \sqrt{|a_k| + |b_k|}} \geq \frac{1}{\lim_{k \rightarrow \infty} k \sqrt{A_k + A_k}} \\ &= \frac{1}{\lim_{k \rightarrow \infty} k \sqrt{2} \sqrt{A_k}} = \lim_{k \rightarrow \infty} \frac{1}{k \sqrt{A_k}} = \lim_{k \rightarrow \infty} \frac{1}{k \sqrt{A_k}} \\ &= \min(R_1, R_2). \end{aligned}$$

$$(2) \sum_{k=0}^{\infty} (a_k - b_k) z^k.$$

解：方法及结论同于上题。

$$(3) \sum_{k=0}^{\infty} a_k b_k z^k.$$

$$\begin{aligned} \text{解：} R &= \lim_{k \rightarrow \infty} \left| \frac{a_k b_k}{a_{k+1} b_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \cdot \frac{b_k}{b_{k+1}} \right| \\ &= R_1 R_2. \end{aligned}$$

$$(4) \sum_{k=0}^{\infty} \frac{a_k}{b_k} z^k \quad (b_k \neq 0).$$

解：

$$R = \lim_{k \rightarrow \infty} \left| \frac{a_k/b_k}{a_{k+1}/b_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_k/a_{k+1}}{b_k/b_{k+1}} \right| = \frac{R_1}{R_2}.$$

§12. 泰勒级数

在指定的点 z_0 的邻域上把下列函数展开为泰勒级数。

(1) $\operatorname{arctg} z$ 在 $z_0 = 0$.

解一: 按照公式 $f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{K!} (z - z_0)^k$ 求解, 令

$f(z) = \operatorname{arctg} z$, 则

$$f(z) = \operatorname{arctg} z, \quad f(0) \text{ 的主值} = 0;$$

$$f'(z) = \frac{1}{1+z^2}, \quad f'(0) = 1;$$

$$f''(z) = \frac{-2z}{(1+z^2)^2}, \quad f''(0) = 0;$$

$$f'''(z) = \frac{6z^2-2}{(1+z^2)^3}, \quad f'''(0) = -2;$$

$$f^{(4)}(z) = \frac{24(z-z^3)}{(1+z^2)^4}, \quad f^{(4)}(0) = 0;$$

$$\dots\dots, \quad \dots\dots.$$

$$\text{所以 } f(z) = z - \frac{1}{3}z^3 + \frac{1}{5}z^5 - \frac{1}{7}z^7 + \dots\dots, (|z| < 1).$$

解二: 已知函数 $\frac{1}{1+z^2}$ 的泰勒级数是

$$\frac{1}{1+z^2} = \sum_{k=0}^{\infty} (-1)^k z^{2k}, \quad (|z| < 1),$$

对该级数逐项积分并不改变收敛半径, 所以

$$\begin{aligned} \operatorname{arctg} z &= \int \frac{1}{1+z^2} dz = \sum_{k=0}^{\infty} (-1)^k \int z^{2k} dz \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{1}{2K+1} z^{2k+1} = z - \frac{1}{3} z^3 \\ &\quad + \frac{1}{5} z^5 - \frac{1}{7} z^7 + \dots\dots (|z| < 1). \end{aligned}$$

(2) $\sqrt[3]{z}$ 在 $z_0 = i$.

解一：因为

$$\begin{aligned} f(z) &= z^{1/3}, & f(i) &= i^{1/3}, \\ f'(z) &= \frac{1}{3z} z^{1/3}, & f'(i) &= \frac{1}{3i} i^{1/3}, \\ f''(z) &= -\frac{2}{3^2 z^2} z^{1/3}, & f''(i) &= -\frac{1 \cdot 2}{3^2 i^2} i^{1/3}, \\ f'''(z) &= \frac{2 \cdot 5}{3^3 z^3} z^{1/3}, & f'''(i) &= \frac{2 \cdot 5}{3^3 i^3} i^{1/3}, \\ &\dots\dots, & &\dots\dots. \end{aligned}$$

故其泰勒级数为

$$\begin{aligned} f(z) = \sqrt[3]{i} \left\{ 1 + \frac{1}{1! i} \frac{1}{3} (z-i) \right. \\ \left. - \frac{1}{2! i^2} \frac{1 \cdot 2}{3^2} (z-i)^2 \right. \\ \left. + \frac{1}{3! i^3} \frac{2 \cdot 5}{3^3} (z-i)^3 - \dots\dots \right\} \\ (|z| < 1). \end{aligned}$$

解二：根据二项式定理，对于非整数 K ，有

$$\begin{aligned} (a+z)^K &= a^K \left\{ 1 + \frac{K}{1! a} z + \frac{K(K-1)}{2! a^2} z^2 + \dots\dots \right. \\ &\quad \left. + \frac{K(K-1)\dots\dots(K-m+1)}{m! a^m} z^m + \dots\dots \right\}. \end{aligned}$$

所以 $\sqrt[3]{z} = [i + (z-i)]^{1/3}$ 可展开为泰勒级数

$$\begin{aligned} f(z) &= [i + (z-i)]^{1/3} \\ &= \sqrt[3]{i} \left\{ 1 + \frac{1}{1! i} \frac{1}{3} (z-i) \right. \\ &\quad \left. - \frac{1}{2! i^2} \frac{1 \cdot 2}{3^2} (z-i)^2 \right. \end{aligned}$$

$$+ \frac{1}{3!i^3} - \frac{2 \cdot 5}{3^3} (z-i)^3 - \dots \left\} (|z| < 1) .$$

(3) $\ln z$ 在 $z_0 = i$.

解: 因为

$$\begin{aligned} f(z) &= \ln z, & f(i) &= \ln i; \\ f'(z) &= \frac{1}{z}, & f'(i) &= \frac{1}{i}; \\ f''(z) &= -\frac{1}{z^2}, & f''(i) &= -\frac{1}{i^2}; \\ f'''(z) &= \frac{2!}{z^3}, & f'''(i) &= \frac{2!}{i^3}; \\ &\dots, & &\dots. \end{aligned}$$

故其泰勒级数为

$$\begin{aligned} f(z) &= \ln i + \frac{1}{i}(z-i) - \frac{1}{2i^2}(z-i)^2 \\ &\quad + \frac{1}{3i^3}(z-i)^3 + \dots. \end{aligned}$$

(4) $\sqrt[m]{z}$ 在 $z_0 = 1$.

解一: 因为

$$\begin{aligned} f(z) &= z^{1/m}, & f(1) \text{的主值} &= 1; \\ f'(z) &= \frac{1}{m} z^{\frac{1}{m}-1}, & f'(1) &= \frac{1}{m}; \\ f''(z) &= \frac{1-m}{m^2} z^{\frac{1}{m}-2}, & f''(1) &= \frac{1-m}{m^2}; \\ f'''(z) &= \frac{(1-m)(1-2m)}{m^3} z^{\frac{1}{m}-3}, \\ f'''(1) &= \frac{(1-m)(1-2m)}{m^3}; \\ &\dots, & &\dots. \end{aligned}$$

故其泰勒级数为

$$f(z) = 1 + \frac{1}{m}(z-1) + \frac{1-m}{2!m^2}(z-1)^2 + \frac{(1-m)(1-2m)}{3!m^3}(z-1)^3 + \dots$$

解二：注意到 $\sqrt[m]{z} = [1 + (z - 1)]^{1/m}$ ，则根据二项式定理也可求出上述的答案。

(5) $e^{1/(1-z)}$ 在 $z_0 = 0$.

解一：因为

$$f(z) = e^{\frac{1}{1-z}}, \quad f(0) = e;$$

$$f'(z) = e^{\frac{1}{1-z}}(1-z)^{-2}, \quad f'(0) = e;$$

$$f''(z) = e^{\frac{1}{1-z}} [(1-z)^{-2} \cdot (1-z)^{-2} + 2(1-z)^{-3}],$$

$$f''(0) = 3e;$$

$$f'''(z) = e^{\frac{1}{1-z}} [1 - z)^{-6} + 2 (1 - z)^{-5} + 4 (1 - z)^{-4} + 6 (1 - z)^{-3}], \quad f'''(0) = 13e;$$

故其泰勒级数为

$$f(z) = e\left(1 + z + \frac{3}{2!}z^2 + \frac{13}{3!}z^3 + \dots\right).$$

解二：注意到几何级数 $\frac{1}{1-z} = \sum_{k=0}^{\infty} z^k$ ($|z| < 1$)，则

$$e^{\frac{1}{1-z}} = e^1 + \frac{z}{1-z} = e \cdot e^{\frac{z}{1-z}}$$

$$= e \left[1 + \frac{z}{1-z} + \frac{1}{2!} \left(\frac{z}{1-z} \right)^2 + \dots \right]$$

$$\begin{aligned}
&= e [1 + (z + z^2 + z^3 + \dots) \\
&\quad + \frac{1}{2!} (z + z^2 + z^3 + \dots)^2 + \dots] \\
&= e \left[1 + z + \left(1 + \frac{1}{2}\right) z^2 + \left(1 + \frac{2}{2!} \right. \right. \\
&\quad \left. \left. + \frac{1}{3!}\right) z^3 + \dots \right] \\
&= e \left(1 + z + \frac{3}{2} z^2 + \frac{13}{6} z^3 + \dots \right), \\
&\qquad\qquad\qquad (|z| < 1).
\end{aligned}$$

(6) $\ln(1 + e^z)$ 在 $z_0 = 0$.

解: 因为

$$\begin{aligned}
f(z) &= \ln(1 + e^z), & f(0) &= \ln 2 \\
f'(z) &= e^z / (1 + e^z), & f'(0) &= \frac{1}{2}, \\
f''(z) &= e^z / (1 + e^z)^2, & f''(0) &= \frac{1}{4}, \\
f'''(z) &= \frac{-2e^{2z}}{(1 + e^z)^3} + \frac{e^z}{(1 + e^z)^2}, & f'''(0) &= 0, \\
&\dots, & & \dots.
\end{aligned}$$

故其泰勒级数为

$$f(z) = \ln 2 + \frac{1}{1!2} z + \frac{1}{2!4} z^2 - \frac{1}{4!8} z^4 + \dots.$$

(7) $(1 + z)^{1/2}$ 在 $z_0 = 0$.

解一: 因为

$$\begin{aligned}
f(z) &= (1 + z)^{1/2}, & f(0) &= 1, \\
f'(z) &= \frac{z/(1+z) - \ln(1+z)}{z^2} e^{\frac{1}{2}\ln(1+z)},
\end{aligned}$$

$$f'(0) = -\frac{e}{2} \quad (\text{用罗毕达法则}),$$

$$f''(z) = \left\{ \left[\frac{z/(1+z) - \ln(1+z)}{z^2} \right]^2 + \frac{z^2/(1+z^2) - 2z/(1+z) + 2\ln(1+z)}{z^3} \right\} e^{\frac{1}{z}\ln(1+z)}.$$

也用罗毕达法则求出 $f''(0) = \frac{11}{12}e$,

.....,

所以其泰勒级数为

$$f(z) = e \left(1 - \frac{z}{2} + \frac{11}{24}z^2 + \dots \right).$$

解二：注意到 $\ln(1+z)$ 的泰勒展式是

$$\ln(1+z) = z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \frac{1}{4}z^4 + \dots \quad (|z| < 1),$$

以及 e^z 的泰勒级数是

$$e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \dots \quad (|z| < \infty).$$

则 $f(z) = (1+z)^{1/z} = e^{\frac{1}{z}\ln(1+z)}$

$$= e^{\frac{1}{z} \left(z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \frac{1}{4}z^4 + \dots \right)}$$

$$= e^{1 - \frac{1}{2}z + \frac{1}{3}z^2 - \frac{1}{4}z^3 + \dots}$$

$$= e \cdot e^{-\frac{1}{2}z + \frac{1}{3}z^2 - \frac{1}{4}z^3 + \dots}$$

$$\begin{aligned}
&= e \left[1 + \left(-\frac{1}{2}z + \frac{1}{3}z^2 - \frac{1}{4}z^3 + \dots \right) \right. \\
&\quad + \frac{1}{2!} \left(-\frac{1}{2}z + \frac{1}{3}z^2 - \frac{1}{4}z^3 + \dots \right) \\
&\quad \left. + \frac{1}{3!} \left(-\frac{1}{2}z + \frac{1}{3}z^2 + \frac{1}{4}z^3 + \dots \right)^3 + \dots \right] \\
&= e \left(1 - \frac{z}{2} + \frac{11}{24}z^2 + \dots \right).
\end{aligned}$$

显然，其收敛半径 $R = 1$ 。值得注意的是这个级数在函数 $(1+z)^{1/z}$ 的奇点 $z = 0$ 处也收敛；在这种情况下，我们不妨重新定义一个函数

$$f(z) = \begin{cases} (1+z)^{1/z}, & (z \neq 0), \\ \lim_{z \rightarrow 0} (1+z)^{1/z} = e, & (z = 0). \end{cases}$$

它在整个开平面上是解析的，所以函数 $f(z)$ 可在 $z = 0$ 处展开为泰勒级数。显然， $z = 0$ 作为奇点是可去奇点。

(8) $\sin^2 z$ 和 $\cos^2 z$ 在 $z_0 = 0$ 。

解一：因为

$$\begin{aligned}
f(z) &= \sin^2 z, & f(0) &= 0; \\
f'(z) &= 2\sin z \cos z = \sin 2z, & f'(0) &= 0; \\
f''(z) &= 2\cos 2z, & f''(0) &= 2^1; \\
f'''(z) &= -4\sin 2z, & f'''(0) &= 0; \\
f^{(4)}(z) &= -8\cos 2z, & f^{(4)}(0) &= -2^3; \\
&\dots, & &\dots.
\end{aligned}$$

故其泰勒级数为

$$\begin{aligned}
f(z) &= \frac{2}{2!}z^2 - \frac{2^3}{4!}z^4 + \frac{2^5}{6!}z^6 - \dots \\
&= \frac{1}{2} \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(2z)^{2k}}{(2k)!}.
\end{aligned}$$

解二: 若已知 $\sin z = z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \dots$ 且在收敛域内绝对收敛, 则可逐项相乘, 即

$$\begin{aligned}\sin^2 z &= z^2 - \frac{2}{3!}z^4 + \frac{1}{(3!)^2}z^6 + \frac{2}{5!}z^6 - \dots \\ &= z^2 - \frac{1}{3}z^4 + \frac{2}{45}z^6 - \dots \\ &= \frac{1}{2} \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(2z)^{2k}}{(2K)!}.\end{aligned}$$

可用类似于上述的两种解法把 $\cos^2 z$ 展开, 此外, 还可把 $\cos^2 z$ 用下法展开为泰勒级数

$$\begin{aligned}f(z) = \cos^2 z &= 1 - \sin^2 z \\ &= 1 + \frac{1}{2} \sum_{k=1}^{\infty} (-1)^k \frac{(2z)^{2k}}{(2K)!}.\end{aligned}$$

§14. 罗朗级数

在挖去奇点 z_0 的环域上或指定的环域上把下列函数展开为罗朗级数.

(1) $z^5 e^{1/z}$ 在 $z_0 = 0$.

解: 由 $e^t = 1 + t + \frac{1}{2!}t^2 + \dots + \frac{1}{n!}t^n + \dots (|t| < \infty)$ 知

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \dots + \frac{1}{n!} \left(\frac{1}{z}\right)^n + \dots (0 < |z|), \text{ 所以}$$

$$\begin{aligned}f(z) = z^5 e^{1/z} &= z^5 + z^4 + \frac{1}{2!}z^3 + \frac{1}{3!}z^2 + \dots + \frac{1}{n!}z^{5-n} \\ &\quad + \dots (0 < |z|).\end{aligned}$$

(2) $\frac{1}{z^2(z-1)}$ 在 $z_0 = 1$.

解一：因为 $\frac{1}{z^2(z-1)} = \frac{1}{z-1} \frac{1}{[1-(1-z)]^2}$,

并注意到当 $|t| < 1$ 时,

$$\frac{1}{(1-t)^2} = \frac{d}{dt} \frac{1}{1-t} = \frac{d}{dt} \sum_{k=0}^{\infty} t^k = \sum_{k=1}^{\infty} K t^{K-1},$$

所以, 当 $0 < |z-1| < 1$ 时, 有

$$\begin{aligned} \frac{1}{z^2(z-1)} &= \frac{1}{z-1} \sum_{K=1}^{\infty} K (1-z)^{K-1} \\ &= \sum_{K=1}^{\infty} (-1)^{K-1} K (z-1)^{K-2} \end{aligned}$$

亦即

$$\begin{aligned} \frac{1}{z^2(z-1)} &= \sum_{K=-1}^{\infty} (-1)^{K+1} (K+2) (z-1)^K, \\ (0 < |z-1| < 1). \end{aligned}$$

解二：还可把原式表为

$$\frac{1}{z^2(z-1)} = \frac{1}{z-1} - \frac{z+1}{z^2} = \frac{1}{z-1} - \left(\frac{1}{z} + \frac{1}{z^2} \right),$$

注意到 $\frac{1}{z} = \frac{1}{1+(z-1)} = \sum_{n=0}^{\infty} (-1)^n (z-1)^n, (|z-1| < 1),$

$$\begin{aligned} \text{及 } -\frac{1}{z^2} &= \left(\frac{1}{z} \right)' = \sum_{K=1}^{\infty} (-1)^K n (z-1)^{K-1} \\ &= \sum_{K=0}^{\infty} (-1)^{K+1} (K+1) (z-1)^K, \end{aligned}$$

$$\text{则 } -\frac{1}{z} - \frac{1}{z^2} = \sum_{K=0}^{\infty} (-1)^{K+1} (K+2) (z-1)^K,$$

$$\text{所以, } \frac{1}{z^2(z-1)} = (z-1)^{-1} + \sum_{K=0}^{\infty} (-1)^{K+1} (K+2) (z-1)^K$$

$$= \sum_{K=-1}^{\infty} (-1)^{K+1} (K+2) (z-1)^K \quad (0 < |z-1| < 1).$$

解三：注意到函数 $\frac{1}{z^2}$ 在 $z_0 = 1$ 处解析，故可把 $\frac{1}{z^2}$ 在 $z_0 = 1$ 处作泰勒展开，

$$\frac{1}{z^2} = \sum_{K=0}^{\infty} (-1)^K (K+1) (z-1)^K, \quad (|z-1| < 1),$$

所以

$$\begin{aligned} \frac{1}{z^2(z-1)} &= \sum_{K=0}^{\infty} (-1)^K (K+1) (z-1)^{K-1} \\ &= \sum_{K=-1}^{\infty} (-1)^{K+1} (K+2) (z-1)^K, \\ &\quad (0 < |z-1| < 1), \end{aligned}$$

还有其它的解法，不再一一列举。以下各题我们也将只写出一种解法。

三 (3) $\frac{1}{z(z-1)}$ 在 $z_0 = 0$ ，在 $z_0 = 1$ 。

解：因为 $\frac{1}{z(z-1)} = \frac{1}{z-1} - \frac{1}{z}$ 。

(i) 注意到在 $z_0 = 0$ 处 $\frac{1}{z-1}$ 解析，可展开为泰勒级数， $\frac{1}{z-1}$

$$= -\frac{1}{1-z} = -\sum_{k=0}^{\infty} z^k, \text{ 所以}$$

$$\text{在 } z_0 = 0: \frac{1}{z(z-1)} = -\frac{1}{z} + \frac{1}{z-1} = -\frac{1}{z} - \sum_{k=0}^{\infty} z^k = -\sum_{k=-1}^{\infty} z^k,$$

$$(0 < |z| < 1).$$

(ii) 注意到在 $z_0 = 1$ 处 $\frac{1}{z}$ 解析，可展开为泰勒级数，

$$\frac{1}{z} = \frac{1}{(z-1)+1} = \sum_{k=0}^{\infty} (-1)^k (z-1)^k, \text{ 所以}$$

$$\begin{aligned} \text{在 } z_0 = 1: \quad \frac{1}{z(z-1)} &= \frac{1}{z-1} - \sum_{k=0}^{\infty} (-1)^k (z-1)^k \\ &= \sum_{k=-1}^{\infty} (-1)^{k+1} (z-1)^k, (0 < |z-1| < 1). \end{aligned}$$

(4) $e^{1/(1-z)}$ 在 $|z| > 1$.

解: 因为 $|z| > 1$, 所以 $\left|\frac{1}{z}\right| < 1$, 则

$$\begin{aligned} \frac{1}{1-z} &= \frac{-1}{z\left(1-\frac{1}{z}\right)} = -\frac{1}{z}\left(1 + \frac{1}{z} + \frac{1}{z^2} + \cdots\right) \\ &= -\left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right). \end{aligned}$$

从而可得

$$\begin{aligned} e^{\frac{1}{1-z}} &= 1 - \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right) + \frac{1}{2!}\left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right)^2 \\ &\quad - \frac{1}{3!}\left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right)^3 \\ &\quad + \frac{1}{4!}\left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right)^4 - \frac{1}{5!}\left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right)^5 \\ &\quad + \cdots \\ &= 1 - \frac{1}{z} - \frac{1}{2z^2} - \frac{1}{6z^3} + \frac{1}{24z^4} - \frac{19}{120z^5} + \cdots, \\ &\quad (|z| > 1). \end{aligned}$$

(5) $\frac{1}{(z-2)(z-3)}$ 在 $|z| > 3$.

$$\text{解: 因为 } \frac{1}{(z-2)(z-3)} = \frac{z-2-(z-3)}{(z-2)(z-3)} = \frac{1}{z-3} - \frac{1}{z-2}$$

$$= \frac{1}{z} \frac{1}{1 - \frac{3}{z}} - \frac{1}{z} \frac{1}{1 - \frac{2}{z}},$$

并注意到当 $|z| > 3$ 时, 有

$$\frac{1}{z} \cdot \frac{1}{1 - \frac{3}{z}} = \sum_{k=0}^{\infty} \frac{3^k}{z^{k+1}} = \sum_{k=-\infty}^{-1} 3^{-(k+1)} z^k,$$

$$\text{以及 } \frac{1}{z} \left(\frac{1}{1 - \frac{2}{z}} \right) = \sum_{k=-\infty}^{-1} 2^{-(k+1)} z^k,$$

所以

$$\frac{1}{(z-2)(z-3)} = \sum_{k=-\infty}^{-1} \left[3^{-(k+1)} - 2^{-(k+1)} \right] z^k \quad (|z| > 3).$$

$$(6) \quad \frac{(z-1)(z-2)}{(z-3)(z-4)} \text{ 在 } R < |z| < \infty \text{ (} R \text{ 很大)}.$$

$$\text{解: 原式} = \frac{\left(1 - \frac{1}{z}\right)\left(1 - 2\frac{1}{z}\right)}{\left(1 - 3\frac{1}{z}\right)\left(1 - 4\frac{1}{z}\right)} = 1 + \frac{6\frac{1}{z}}{1 - 4\frac{1}{z}} - \frac{2\frac{1}{z}}{1 - 3\frac{1}{z}},$$

$$\text{注意到 } \frac{6\frac{1}{z}}{1 - 4\frac{1}{z}} = 6 \cdot \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{4}{z}\right)^k = 6 \sum_{k=-\infty}^{-1} 4^{-(k+1)} z^k$$

$$\text{及 } 2\frac{1}{z} \cdot \frac{1}{1 - \frac{3}{z}} = 2 \sum_{k=-\infty}^{-1} 3^{-(k+1)} z^k = 2 \sum_{k=-\infty}^{-1} 3^{-(k+1)} z^k,$$

$$\text{所以 } \frac{(z-1)(z-2)}{(z-3)(z-4)} = 1 + \sum_{k=-\infty}^{-1} \left[6 \cdot 4^{-(k+1)} - 2 \cdot 3^{-(k+1)} \right] z^k, \\ (|z| > 4).$$

$$(7) \quad \frac{1}{z^2 - 3z + 2} \text{ 在 } 1 < |z| < 2.$$

解: 原式又 $= \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1},$

$$\text{而 } \frac{1}{z-2} = -\frac{\frac{1}{2}}{1-\frac{z}{2}} = -\frac{1}{2} \left[1 + \frac{z}{2} + \left(\frac{z}{2}\right)^2 + \left(\frac{z}{2}\right)^3 + \dots \right],$$

$$\left(\left| \frac{z}{2} \right| < 1, |z| < 2 \right),$$

$$\begin{aligned} \frac{-1}{z-1} &= -\frac{\frac{1}{z}}{1-\frac{1}{z}} = -\frac{1}{z} \left[1 + \frac{1}{z} + \left(\frac{1}{z}\right)^2 + \left(\frac{1}{z}\right)^3 + \dots \right] \\ &= -\left[\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \dots \right], \quad \left(\left| \frac{1}{z} \right| < 1, |z| > 1. \right) \end{aligned}$$

所以 $\frac{1}{z^2-3z+2} = -\frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k - \sum_{k=-\infty}^{-1} z^k, (1 < |z| < 2),$

即 $\frac{1}{z^2-3z+2} = -\sum_{k=0}^{\infty} \frac{z^k}{2^{k+1}} - \sum_{k=-\infty}^{-1} z^k, \quad (1 < |z| < 2).$

(8) $\frac{1}{z^2-3z+2}$ 在 $2 < |z| < \infty$.

解: 原式 $= \frac{1}{(z-2)(z-1)} = \frac{1}{z-2} - \frac{1}{z-1} = \frac{\frac{1}{z}}{1-\frac{2}{z}} - \frac{\frac{1}{z}}{1-\frac{1}{z}}$

$$\text{而 } \frac{1}{z} \frac{1}{1-\frac{2}{z}} = \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{2}{z}\right)^k = \sum_{k=0}^{\infty} 2^k z^{-(k+1)}$$

$$= \sum_{k=-\infty}^{-1} 2^{-(k+1)} z^k, (|z| > 2),$$

$$-\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^{k+1} = -\sum_{k=-\infty}^{-1} z^k,$$

所以 $\frac{1}{z^2 - 3z + 2} = \sum_{k=-\infty}^{-1} (2^{-(k+1)} - 1)z^k, (2 < |z| < \infty).$

(9) e^z/z 在奇点.

解: 奇点为 $z = 0$, 而 e^z 在 $z = 0$ 解析, 故可作泰勒展开, 所以

$$\begin{aligned}\frac{1}{z}e^z &= \frac{1}{z} \sum_{k=0}^{\infty} \frac{1}{K!} z^K = \sum_{k=-1}^{\infty} \frac{1}{(K+1)!} z^k \\ &= \frac{1}{z} + \sum_{k=1}^{\infty} \frac{1}{K!} z^{k-1}, (0 < |z| < \infty).\end{aligned}$$

(10) $(1 - \cos z)/z$ 在奇点.

解: 奇点为 $z = 0$, 因为 $\lim_{z \rightarrow 0} \frac{1 - \cos z}{z} = 0$, 故该奇点为可去奇点. 所以

$$\begin{aligned}\frac{1 - \cos z}{z} &= \frac{1}{z} - \frac{\cos z}{z} = \frac{1}{z} - \frac{1}{z} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2K)!} z^{2K} \\ &= \sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{(2K)!} z^{2k-1} (|z| < \infty).\end{aligned}$$

(11) $\sin \frac{1}{z}$ 在奇点.

解: $z = 0$ 为函数的奇点, 所以

$$\sin \frac{1}{z} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2K+1)!} \left(\frac{1}{z}\right)^{2k+1}, (0 < |z| < \infty).$$

(12) $\text{ctg} z$ 在奇点.

解: 在半径可以任意小的内圆中只有一个奇点 $z = 0$. 离 $z = 0$ 最近的另一个奇点是 $z = \pi$, 故可在 $0 < |z| < \pi$ 上展开.

解一: $f(z) = \text{ctg} z = \frac{1}{\text{tg} z}$. 先求 $\text{tg} z$, 用待定系数法求

$\operatorname{tg} z_A$ 在 $z_A = 0$ 的邻域里的泰勒级数.

$$\text{设 } \operatorname{tg} z_A = \sum_{l=0}^{\infty} b_l z_A^{2l+1}$$

$$\begin{aligned} \sin z_A &= z_A - \frac{z_A^3}{3!} + \frac{z_A^5}{5!} - \frac{z_A^7}{7!} + \cdots + (-1)^n \frac{z_A^{2n+1}}{(2n+1)!} + \cdots \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{z_A^{2n+1}}{(2n+1)!} \end{aligned}$$

$$\begin{aligned} \cos z_A &= 1 - \frac{z_A^2}{2!} + \frac{z_A^4}{4!} - \frac{z_A^6}{6!} + \cdots + (-1)^k \frac{z_A^{2k}}{(2k)!} + \cdots \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{z_A^{2k}}{(2k)!}, \end{aligned}$$

$$\begin{aligned} \text{则 } \sum_{n=0}^{\infty} \frac{(-1)^n z_A^{2n+1}}{(2n+1)!} &= \sum_{k=0}^{\infty} \frac{(-1)^k z_A^{2k}}{(2k)!} \cdot \sum_{l=0}^{\infty} b_l z_A^{2l+1} \\ &= \sum_{n=0}^{\infty} z_A^{2n+1} \sum_{l=0}^n \frac{(-1)^{n-l} b_l}{(2n-2l)!}. \end{aligned}$$

根据展开的唯一性 (这里是 $\sin z_A$)，两边级数中 z_A^{2n+1} ($n = 0, 1, 2, \dots$) 的系数应相等，

$$\therefore \sum_{l=0}^n \frac{(-1)^l b_l}{(2n-2l)!} = \frac{1}{(2n+1)!}$$

这是系数 b_l 之间的递推关系，可以据此推出这些系数，前几个是：

$$n = 0, \quad b_0 = 1;$$

$$n = 1, \quad \frac{1}{2!} b_0 - b_1 = \frac{1}{3!}, \quad b_1 = \frac{1}{3};$$

$$n = 2, \quad \frac{1}{4!} b_0 - \frac{1}{2!} b_1 + b_2 = \frac{1}{5!}, \quad b_2 = \frac{2}{15};$$

$$n = 3, \quad \frac{1}{6!} b_0 - \frac{1}{4!} b_1 + \frac{1}{2!} b_2 - b_3 = \frac{1}{7!}, \quad b_3 = \frac{17}{315};$$

$$\dots\dots, \quad \dots\dots.$$

$$\therefore \operatorname{tg} z_A = z_A + \frac{1}{3} z_A^3 + \frac{2}{15} z_A^5 + \frac{17}{315} z_A^7 + \dots, \left(|z_A| < \frac{\pi}{2} \right).$$

下面再回到求 $\operatorname{ctg} z_A$:

$$\begin{aligned} \operatorname{ctg} z_A &= \frac{1}{\operatorname{tg} z_A} = \left(z_A + \frac{1}{3} z_A^3 + \frac{2}{15} z_A^5 + \frac{17}{315} z_A^7 + \dots \right)^{-1} \\ &= \frac{1}{z_A} \left(1 + \frac{1}{3} z_A^2 + \frac{2}{15} z_A^4 + \frac{17}{315} z_A^6 + \dots \right)^{-1}, \end{aligned}$$

注意到 $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots, |x| < 1$,

$$\begin{aligned} \operatorname{ctg} z_A &= \frac{1}{z_A} \left\{ 1 - \left(\frac{1}{3} z_A^2 + \frac{2}{15} z_A^4 + \frac{17}{315} z_A^6 + \dots \right) \right. \\ &\quad + \left(\frac{1}{3} z_A^2 + \frac{2}{15} z_A^4 + \frac{17}{315} z_A^6 + \dots \right)^2 \\ &\quad \left. - \left(\frac{1}{3} z_A^2 + \frac{2}{15} z_A^4 + \frac{17}{315} z_A^6 + \dots \right)^3 + \dots \right\} \\ &= \frac{1}{z_A} - \frac{1}{3} z_A - \frac{1}{45} z_A^3 - \frac{2}{945} z_A^5 - \frac{1}{4725} z_A^7 - \dots, \\ &\quad (0 < |z| < \pi). \end{aligned}$$

解法二: 直接用待定系数法求 $\operatorname{tg} z_A$ 在 $z_A = 0$ 的邻域内的罗朗级数.

设 $\operatorname{ctg} z = \frac{1}{z_A} \sum_{l=0}^{\infty} b_l z_A^{2l}$ 再结合 $\sin z$ 和 $\cos z$ 的展开式得;

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(-1)^n z_A^{2n}}{(2n)!} &= \frac{1}{z_A} \sum_{k=0}^{\infty} \frac{(-1)^k z_A^{2k+1}}{(2k+1)!} \cdot \sum_{l=0}^{\infty} b_l z_A^{2l} \\ &= \frac{1}{z} \sum_{n=0}^{\infty} z_A^{2n} \sum_{l=0}^n \frac{(-1)^{n-l} b_l}{(2n-2l+1)!}, \end{aligned}$$

根据展开的唯一性, 得系数 b_l 之间的递推关系式:

$$\sum_{l=0}^n \frac{(-1)^l b_l}{(2n-2l+1)!} = \frac{1}{(2n)!} \text{ 前几个系数是:}$$

$$n = 0, b_0 = 1;$$

$$n = 1, \quad \frac{b_0}{3!} - b_1 = \frac{1}{2!}, \quad b_1 = -\frac{1}{3};$$

$$n = 2, \quad \frac{b_0}{5!} - \frac{b_1}{3!} + b_2 = \frac{1}{4!}, \quad b_2 = -\frac{1}{45};$$

$$n = 3, \quad \frac{b_0}{7!} - \frac{b_1}{5!} + \frac{b_2}{3!} - b_3 = \frac{1}{6!}, \quad b_3 = -\frac{2}{945};$$

.....,

$$\therefore \operatorname{ctg} z = \frac{1}{z} \sum_{n=0}^{\infty} b_n z^{2n}$$

$$= \frac{1}{z} \left(1 - \frac{1}{3} z^2 - \frac{1}{45} z^4 - \frac{2}{945} z^6 - \dots \right)$$

$$= \frac{1}{z} - \frac{1}{3} z - \frac{1}{45} z^3 - \frac{2}{945} z^5 - \dots, \quad (0 < |z| < \pi).$$

$$(13) \quad \frac{z}{(z-1)(z-2)^2} \text{ 在 } |z| < 1, \text{ 在 } 1 < |z| < 2,$$

在 $2 < |z|$.

解：把原式分解为三项，并在不同的区域作泰勒展开，

$$\begin{aligned} \text{即 } \frac{z}{(z-1)(z-2)^2} &= \frac{2(-1) - (z-2)}{(z-1)(z-2)^2} \\ &= \frac{2}{(z-2)^2} - \frac{1}{(z-1)(z-2)} \\ &= \frac{2}{(z-2)^2} + \frac{1}{z-1} - \frac{1}{z-2}, \end{aligned}$$

各自展开为：

$$\left\{ \begin{aligned} \frac{2}{(z-2)^2} &= \frac{\frac{1}{2}}{\left(1 - \frac{z}{2}\right)^2} = \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k (K+1), \\ \left(\left|\frac{z}{2}\right| < 1 \text{ 即 } |z| < 2\right), \end{aligned} \right. \quad (1)$$

$$\left\{ \begin{aligned} \frac{2}{(z-2)^2} &= \frac{2 \frac{1}{z^2}}{\left(1 - \frac{2}{z}\right)^2} = 2 \sum_{k=0}^{\infty} (K+1) \left(\frac{2}{z}\right)^k \frac{1}{z^2} \\ &= \sum_{k=-\infty}^{-2} -(K+1) 2^{-(k+1)} z^k, \\ \left(\left|\frac{2}{z}\right| < 1 \text{ 即 } |z| > 2\right); \end{aligned} \right. \quad (2)$$

$$\left\{ \begin{aligned} \frac{1}{z-1} &= - \sum_{k=0}^{\infty} z^k \quad (|z| < 1), \end{aligned} \right. \quad (3)$$

$$\left\{ \begin{aligned} \frac{1}{z-1} &= \frac{1}{z} \cdot \frac{1}{1 - \frac{1}{z}} = \sum_{k=0}^{\infty} \frac{1}{z^{k+1}} = \sum_{k=-\infty}^{-1} z^k, \quad (|z| > 1), \end{aligned} \right. \quad (4)$$

$$\left\{ \begin{aligned} -\frac{1}{z-2} &= \frac{1}{2} \frac{1}{1 - \frac{z}{2}} = \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k, \quad (|z| < 2), \end{aligned} \right. \quad (5)$$

$$\left\{ \begin{aligned} -\frac{1}{z-2} &= -\frac{1}{z} \frac{1}{1 - \frac{2}{z}} = - \sum_{k=0}^{\infty} \frac{2^k}{z^{k+1}} \\ &= - \sum_{k=-\infty}^{-1} 2^{-(1+k)} z^k, \quad (|z| > 2). \end{aligned} \right. \quad (6)$$

所以, (i) 在 $|z| < 1$ 时, 由(1)(3)(5)可得罗朗级数

$$\frac{z}{(z-1)(z-2)^2} = \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k (K+1) - \sum_{k=0}^{\infty} z^k$$

$$\begin{aligned}
& + \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \\
& = \sum_{k=0}^{\infty} \left[\left(\frac{1}{2}\right)^k \left(\frac{K}{2} + 1\right) - 1 \right] z^k \\
& = \sum_{k=0}^{\infty} \left(\frac{K+2}{2^{k+1}} - 1 \right) z^k.
\end{aligned}$$

其实这是泰勒级数.

(ii) 在 $1 < |z| < 2$ 时, 由 (1) (4) (5) 可得罗朗级数

$$\begin{aligned}
\frac{z}{(z-1)(z-2)^2} &= \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k (K+1) + \sum_{k=-\infty}^{-1} z^k \\
&+ \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \\
&= \sum_{k=-\infty}^{-1} z^k + \sum_{k=0}^{\infty} \left[\left(\frac{1}{2}\right)^k \left(\frac{K}{2} + 1\right) \right] z^k \\
&= \sum_{k=-\infty}^{-1} z^k + \sum_{k=0}^{\infty} \frac{K+2}{2^{k+1}} z^k.
\end{aligned}$$

(iii) 在 $2 < |z|$ 时, 由 (2) (4) (6) 可得罗朗级数

$$\begin{aligned}
\frac{z}{(z-1)(z-2)^2} &= \sum_{k=-\infty}^{-2} - (K+1) 2^{-(k+1)} z^k \\
&+ \sum_{k=-\infty}^{-1} z^k - \sum_{k=-\infty}^{-1} 2^{-(k+1)} z^k \\
&= \sum_{k=-\infty}^{-2} \left(1 - \frac{K+2}{2^{k+1}} \right) z^k.
\end{aligned}$$

(14) $z/(z-1)(z-2)$ 在 $|z| < 1$, 在 $1 < |z| < 2$, 在 $2 < |z|$.

解: 与上题类似, 把原式分解为

$$\frac{z}{(z-1)(z-2)} = \frac{2(z-1) - (z-2)}{(z-1)(z-2)} = \frac{2}{z-2} - \frac{1}{z-1}.$$

再把上式右边各项在不同的区域内泰勒展开为

$$\begin{cases} \frac{2}{z-2} = -\frac{1}{1-\frac{z}{2}} = -\sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \quad (|z| < 2), \end{cases} \quad (1)$$

$$\begin{cases} \frac{2}{z-2} = \frac{2}{z} \cdot \frac{1}{1-\frac{2}{z}} = \sum_{k=0}^{\infty} \left(\frac{2}{z}\right)^{k+1} \\ = \sum_{k=-\infty}^{-1} \left(\frac{z}{2}\right)^k \quad (|z| > 2); \end{cases} \quad (2)$$

$$\begin{cases} -\frac{1}{z-1} = \frac{1}{1-z} = \sum_{k=0}^{\infty} z^k \quad (|z| < 1), \end{cases} \quad (3)$$

$$\begin{cases} -\frac{1}{z-1} = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{k=0}^{\infty} \frac{1}{z^{k+1}} \\ = -\sum_{k=-\infty}^{-1} z^k \quad (|z| > 1); \end{cases} \quad (4)$$

∴ (i) 在 $|z| < 1$ 时, 由(1)(3)式可得罗朗级数

$$\begin{aligned} \frac{z}{(z-1)(z-2)} &= -\sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k + \sum_{k=0}^{\infty} z^k \\ &= \sum_{k=0}^{\infty} \left(1 - \frac{1}{2^k}\right) z^k. \end{aligned}$$

其实这是泰勒级数.

(ii) 在 $1 < |z| < 2$ 时, 由(1)(4)可得罗朗级数

$$\frac{z}{(z-1)(z-2)} = -\sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k - \sum_{k=-\infty}^{-1} z^k.$$

(iii) 在 $2 < |z|$ 时, 由(2)(4)可得罗朗级数

$$\frac{z}{(z-1)(z-2)} = \sum_{k=-\infty}^{-1} \left(\frac{z}{2}\right)^k - \sum_{k=-\infty}^{-1} z^k$$

$$= \sum_{k=-\infty}^{-1} \left(\frac{1}{2^k} - 1 \right) z^k.$$

(15) $\frac{1}{z^2(z^2-1)^2}$ 在 $0 < |z| < 1$ 在 $1 < |z| < \infty$.

解：可仿前两题的解法求解，这里我们用另法求解如下：

(i) 在 $0 < |z| < 1$ 时，

$$\begin{aligned} \frac{1}{z^2(z^2-1)^2} &= \frac{1}{z^2} \cdot \frac{1}{2z} \frac{d}{dz} \left(\frac{1}{1-z^2} \right) = \frac{1}{2z^3} \frac{d}{dz} \sum_{k=0}^{\infty} z^{2k} \\ &= \frac{1}{2z^3} \sum_{k=0}^{\infty} 2K z^{2k-1} = \sum_{k=-1}^{\infty} (K+2) z^{2k}. \end{aligned}$$

(ii) 在 $1 < |z| < \infty$ 时，

$$\begin{aligned} \frac{1}{z^2(z^2-1)^2} &= \frac{1}{z^6 \left(1 - \frac{1}{z^2} \right)^2} = \frac{1}{z^6} \left(\frac{-z^3}{2} \right) \frac{d}{dz} \left(\frac{1}{1 - \frac{1}{z^2}} \right) \\ &= - \frac{1}{2z^3} \frac{d}{dz} \sum_{k=0}^{\infty} \left(\frac{1}{z^2} \right)^k \\ &= - \frac{1}{2z^3} \sum_{k=0}^{\infty} (-2K) \frac{1}{z^{2k+1}} \\ &= - \sum_{k=-\infty}^{-1} (K+2) z^{2k}. \end{aligned}$$

§15. 奇点分类

设函数 $f(z)$ 和 $g(z)$ 分别以点 z_0 为 m 阶和 n 阶极点，同对于下列函数而言， z_0 是何种性质的点。

(1) $f(z)g(z)$ 。

解： $f(z)$ 和 $g(z)$ 可分别表为

$$f(z) = \frac{\phi(z)}{(z-z_0)^m}, \quad g(z) = \frac{\psi(z)}{(z-z_0)^n}.$$

其中 $\phi(z)$ 和 $\psi(z)$ 在 $z = z_0$ 的邻域上是解析的, 且 $\phi(z_0) \neq 0$, $\psi(z_0) \neq 0$. 于是

$$f(z)g(z) = \frac{\phi(z)\psi(z)}{(z-z_0)^m(z-z_0)^n} = \frac{\phi(z)\psi(z)}{(z-z_0)^{m+n}},$$

$\therefore z_0$ 是 $f(z)g(z)$ 的 $(m+n)$ 阶极点.

(2) $f(z)/g(z)$.

解: 分析同上题, 这时有

$$\frac{f(z)}{g(z)} = \frac{\phi(z)/\psi(z)}{(z-z_0)^{m-n}}.$$

如 $m > n$, 则 z_0 是 $f(z)/g(z)$ 的 $(m-n)$ 阶极点;

如 $m < n$, 则 z_0 不是 $f(z)/g(z)$ 的奇点.

(3) $f(z) + g(z)$.

解: 分析同(1)题, 这时有

$$f(z) + g(z) = \frac{\phi(z)}{(z-z_0)^m} + \frac{\psi(z)}{(z-z_0)^n}.$$

z_0 是 $f(z) + g(z)$ 的极点, 其阶数为 m 和 n 中较大的一个, 如 $m = n$, 则极点的阶数可能 $< m$.

第四章 留数定理

§16. 留数定理

1. 确定下列函数的奇点, 求出函数在各奇点的留数.

$$(1) \frac{e^z}{1+z}.$$

解: (i) 因为 $\lim_{z \rightarrow -1} \left(\frac{e^z}{1+z} \right) = \infty$, 所以 $z_0 = -1$ 是函数的极点. 又因 $\lim_{z \rightarrow -1} \left[(1+z) \left(\frac{e^z}{1+z} \right) \right] = \lim_{z \rightarrow -1} e^z = \frac{1}{e}$, 这是非零有限值, 所以 $z_0 = -1$ 是函数的一阶极点 (或称单极点), 其留数就是 $\frac{1}{e}$, 即

$$\operatorname{Res} f(-1) = \frac{1}{e},$$

(ii) 因为 $\lim_{z \rightarrow \infty} \left(\frac{e^z}{1+z} \right)$ 不存在, 所以 $z_0 = \infty$ 是函数的本性奇点. 函数在全平面上只有这两个奇点, 根据 (16.7) {全平面各留数之和} = 0, 可求出函数在本性奇点 $z_0 = \infty$ 的留数.

$$\operatorname{Res} f(\infty) = -\{f(z) \text{ 在所有 (有限个) 有限远奇点的留数之和}\} = -\operatorname{Res} f(-1) = -\frac{1}{e}.$$

以下各题皆应如此分析, 但限于篇幅, 我们只给出简捷的步骤.

$$(2) \frac{z}{(z-1)(z-2)^2}.$$

解: (i) 单极点 $z_0 = 1$,

$$\operatorname{Res} f(1) = \lim_{z \rightarrow 1} \frac{z}{(z-2)^2} = 1.$$

(ii) 又二阶极点 $z_0 = 2$,

$$\begin{aligned} \operatorname{Res} f(2) &= \lim_{z \rightarrow 2} \frac{d}{dz} \left(\frac{z}{z-1} \right) \\ &= \lim_{z \rightarrow 2} \left[\frac{1}{z-1} - \frac{z}{(z-2)^2} \right] = -1. \end{aligned}$$

(3) $e^z/z^2 + a^2$.

解: (i) 单极点 $z_0 = ia$,

$$\operatorname{Res} f(ia) = \lim_{z \rightarrow ia} \left(\frac{e^z}{z+ia} \right) = \frac{e^{ia}}{2ia}.$$

(ii) 单极点 $z_0 = -ia$,

$$\begin{aligned} \operatorname{Res} f(-ia) &= \lim_{z \rightarrow -ia} \left(\frac{e^z}{z-ia} \right) = \frac{e^{-ia}}{-2ia} \\ &= -\frac{e^{-ia}}{2ia}. \end{aligned}$$

(iii) 本性奇点 $z_0 = \infty$,

$$\begin{aligned} \operatorname{Res} f(\infty) &= -[\operatorname{Res} f(ia) + \operatorname{Res} f(-ia)] \\ &= \frac{e^{-ia} - e^{ia}}{2ia} = -\frac{\sin a}{a}. \end{aligned}$$

(4) $e^{iz}/(z^2 + a^2)$.

解: (i) 单极点 $z_0 = ia$,

$$\operatorname{Res} f(ia) = \lim_{z \rightarrow ia} \left(\frac{e^{iz}}{z+ia} \right) = \frac{e^{-a}}{2ia}.$$

(ii) 单极点 $z_0 = -ia$,

$$\operatorname{Res} f(-ia) = \lim_{z \rightarrow -ia} \left(\frac{e^{iz}}{z-ia} \right) = -\frac{e^a}{2ia}.$$

(iii) 本性奇点 $z_0 = \infty$,

$$\begin{aligned}\operatorname{Res} f(\infty) &= -[\operatorname{Res} f(ia) + \operatorname{Res} f(-ia)] \\ &= \frac{e^a - e^{-a}}{2ia} = \frac{\operatorname{sh} a}{ia}.\end{aligned}$$

(5) $ze^z/(z-a)^3$.

解: (i) 三阶极点 $z_0 = a$,

$$\operatorname{Res} f(a) = \lim_{z \rightarrow a} \frac{1}{2!} \frac{d^2}{dz^2} (ze^z) = \left(1 + \frac{a}{2}\right)e^a.$$

(ii) 本性奇点 $z_0 = \infty$,

$$\operatorname{Res} f(\infty) = -\operatorname{Res} f(a) = -\left(1 + \frac{a}{2}\right)e^a.$$

(6) $\frac{1}{z^3 - z^5}$.

解: $f(z) = \frac{1}{z^3 - z^5} = \frac{1}{z^3(1 - z^2)}$.

(i) 单极点 $z_0 = 1$,

$$\operatorname{Res} f(1) = \lim_{z \rightarrow 1} \left[-\frac{1}{z^3(z+1)} \right] = -\frac{1}{2}.$$

(ii) 单极点 $z_0 = -1$,

$$\operatorname{Res} f(-1) = \lim_{z \rightarrow -1} \left[\frac{1}{z^3(1-z)} \right] = -\frac{1}{2}.$$

(iii) 三阶极点 $z_0 = 0$,

$$\begin{aligned}\operatorname{Res} f(0) &= \lim_{z \rightarrow 0} \frac{1}{2!} \frac{d^2}{dz^2} \left(\frac{1}{1 - z^2} \right) \\ &= \lim_{z \rightarrow 0} \frac{1}{2!} \left[\frac{2}{(1 - z^2)^2} + \frac{8z^2}{(1 - z^2)^3} \right] = 1\end{aligned}$$

或由(16.7)得

$$\operatorname{Res} f(0) = -[\operatorname{Res} f(1) + \operatorname{Res} f(-1)] = 1.$$

(7) $\frac{z^2}{(z^2 + 1)^2}$.

解: (i) 二阶极点 $z_0 = i$,

$$\operatorname{Res} f(i) = \lim_{z \rightarrow i} \frac{d}{dz} \left[\frac{z^2}{(z+i)^2} \right] = -\frac{i}{4}.$$

(ii) 二阶极点 $z_0 = -i$,

$$\operatorname{Res} f(-i) = -\operatorname{Res} f(i) = \frac{i}{4}.$$

(8) $z^{2n}/(z+1)^n$.

解: (i) n 阶极点 $z_0 = -1$,

$$\begin{aligned} \operatorname{Res} f(-1) &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} (z+1)^n f(z) \\ &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} z^{2n} \\ &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} [2n(2n-1) \cdots (2n-n+2) \\ &\quad \times z^{2n-n+1}] \\ &= (-1)^{n+1} \frac{2n(2n-1) \cdots (n+2)}{(n-1)!} \\ &= (-1)^{n+1} \frac{(2n)!}{(n-1)!(n+1)!}. \end{aligned}$$

(ii) n 阶极点 $z_0 = \infty$,

$$\operatorname{Res} f(\infty) = -\operatorname{Res} f(-1) = (-1)^n \frac{(2n)!}{(n-1)!(n+1)!}.$$

(9) $e^{\frac{1}{1-z}}$.

解: 本性奇点 $z_0 = 1$, 要求 $f(z) = e^{\frac{1}{1-z}}$ 的留数, 必须把 $f(z)$ 进行罗朗展开 (见 §14 习题(4)),

$$f(z) = 1 - \frac{1}{z} - \frac{1}{2z^2} - \frac{1}{6z^3} + \frac{1}{24z^4} + \cdots,$$

所以

$$\operatorname{Res} f(1) = -1.$$

$$(10) \frac{1}{1+z^{2n}}.$$

解：令原式分母 $1+z^{2n}=0, z^{2n}=-1,$

$$z^n = \pm i = e^{i(2k+1)\pi/2},$$

所以 $z_0 = e^{i(2k+1)\pi/2n} (k=0, 1, 2, \dots, 2n-1)$

为函数 $f(z)$ 的单极点,

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} [(z - e^{i(2k+1)\pi/2n}) / (1 + z^{2n})],$$

应用罗毕达法则, 则

$$\begin{aligned} \operatorname{Res} f(z_0) &= \lim_{z \rightarrow z_0} [1/2nz^{2n-1}] = \frac{1}{2n} e^{-i \frac{(2n-1)(2k+1)}{2n}\pi} \\ &= \frac{1}{2n} \cdot \frac{e^{i(2k+1)\pi/2n}}{e^{i(2k+1)\pi}} = -\frac{1}{2n} e^{i(2k+1)\pi/2n}. \end{aligned}$$

2. 计算下列回路积分.

$$(1) \oint_l \frac{dz}{(z^2+1)(z-1)^2} \quad (l \text{ 的方程是 } x^2+y^2-2x-2y=0).$$

解: l 的方程可化为: $(x-1)^2 + (y-1)^2 = (\sqrt{2})^2$ 如图4-1, 在复平面上, 它是一个以 $(1, i)$ 为圆心, $\sqrt{2}$ 为半径的圆.

被积函数 $f(z) = 1/(z^2+1)(z-1)^2$, 它有两个单极点 $z_0 = \pm i$, 和一个二阶极点 $z_0 = 1$, 在这三个极点中, $z_0 = -i$ 不在积分回路之内, 只有极点 $z_0 = i$ 和 $z_0 = 1$ 在积分回路之内, 它们的留数分别为:

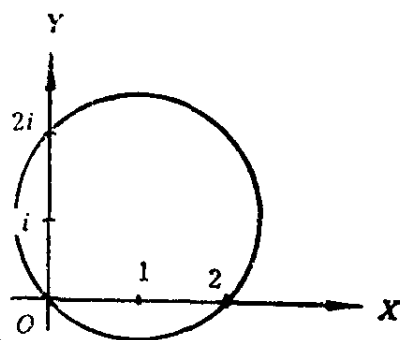


图 4-1

$$\operatorname{Res} f(i) = \lim_{z \rightarrow i} [1/(z+i)(1-z)^2] = \frac{1}{4},$$

$$\begin{aligned}\operatorname{Res} f(1) &= \lim_{z \rightarrow 1} \frac{d}{dz} (1/(1+z^2)) = \lim_{z \rightarrow 1} [-2z/(1+z^2)^2] \\ &= -1/2.\end{aligned}$$

应用留数定理:

$$\begin{aligned}\oint_C \frac{dz}{(z^2+1)(z-1)^2} &= 2\pi i [\operatorname{Res} f(i) + \operatorname{Res} f(1)] \\ &= 2\pi i \left[\frac{1}{4} - \frac{1}{2} \right] = -\frac{\pi i}{2}.\end{aligned}$$

$$(2) \oint_{|z|=1} \cos z dz / z^3.$$

解: 被积 $f(z) = \cos z / z^3$ 的三阶极点 $z_0 = 0$ 在单位圆内, 其留数.

$$\operatorname{Res} f(0) = \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} (\cos z) = -\frac{1}{2},$$

$$\therefore \oint_{|z|=1} \cos z dz / z^3 = 2\pi i \operatorname{Res} f(0) = -\pi i.$$

$$(3) \oint_{|z|=2} e^{1/z^2} dz.$$

解: 被积函数的本性奇点 $z_0 = 0$ 在积分回路之内, $\operatorname{Res} f(0) = 0$, 所以

$$\oint_{|z|=2} e^{1/z^2} dz = 0.$$

$$(4) \oint_{|z|=2} \frac{z dz}{\frac{1}{2} - \sin^2 z}.$$

$$\text{解: 被积函数 } f(z) = \frac{z}{\frac{1}{2} - \sin^2 z} = \frac{2z}{\cos 2z};$$

令 $\cos 2z = 0$, 即 $e^{i2z} + e^{-i2z} = 0$, 由此解出

$$z = \frac{(2k+1)\pi}{4} \quad (k=0, \pm 1, \pm 2, \dots).$$

这些都是 $f(z)$ 的单极点,但其中只有 $z_0 = \pm \frac{\pi}{4}$ 这个单极点在积分回路之内,而

$$\begin{aligned}\operatorname{Res}f\left(-\frac{\pi}{4}\right) &= \lim_{z \rightarrow -\frac{\pi}{4}} \frac{2z\left(z + \frac{\pi}{4}\right)}{\cos 2z} = \lim_{z \rightarrow -\frac{\pi}{4}} \frac{4z + \frac{\pi}{2}}{-2\sin 2z} \\ &= -\frac{\pi}{4},\end{aligned}$$

$$\begin{aligned}\operatorname{Res}f\left(\frac{\pi}{4}\right) &= \lim_{z \rightarrow \frac{\pi}{4}} \frac{2z(z - \pi/4)}{\cos 2z} = \lim_{z \rightarrow \frac{\pi}{4}} \frac{4z - \frac{\pi}{2}}{-2\sin 2z} \\ &= -\frac{\pi}{4}.\end{aligned}$$

$$\begin{aligned}\therefore \oint_{|z|=2} \frac{zdz}{\frac{1}{2} - \sin^2 z} &= 2\pi i \left[\operatorname{Res}f\left(\frac{\pi}{4}\right) \right. \\ &\quad \left. + \operatorname{Res}f\left(-\frac{\pi}{4}\right) \right] \\ &= -\pi^2 i.\end{aligned}$$

3. 应用留数定理计算回路积分 $\frac{1}{2\pi i} \oint_l \frac{f(z)}{z-\alpha} dz$, 函数 $f(z)$ 在 l 所围区域上是解析的, α 是区域的一个内点.

解: 设被积函数 $g(z) = \frac{f(z)}{z-\alpha}$, 因为 $f(z)$ 在 l 所围区域上是解析的, 所以 $g(z)$ 在积分回路(即 l 所围区域)内只有一个单极点 $z_0 = \alpha$, 而

$$\operatorname{Res}f(\alpha) = \lim_{z \rightarrow \alpha} \left[\frac{f(z)}{z-\alpha} (z-\alpha) \right] = f(\alpha),$$

$$\therefore \oint_l \frac{f(z)}{z-\alpha} dz = 2\pi i \operatorname{Res}f(\alpha) = 2\pi i f(\alpha), \text{ 于是}$$

$$\frac{1}{2\pi i} \oint \frac{f(z)}{z-a} dz = f(a).$$

这正是科希公式.

§17. 应用留数定理计算实变函数定积分

1. 计算下列实变函数定积分

$$(1) \int_0^{2\pi} \frac{dx}{2 + \cos x}.$$

解: 这是属于类型一的积分, 为此, 作变换 $z = e^{ix}$ 使原积分化为单位圆内的回路积分

$$\begin{aligned} I &= \oint_{|z|=1} \frac{dz/iz}{2 + \frac{z+z^{-1}}{2}} = \oint_{|z|=1} \frac{2}{i} \frac{dz}{z^2 + 4z + 1} \\ &= \frac{2}{i} \oint_{|z|=1} \frac{dz}{(z+2-\sqrt{3})(z+2+\sqrt{3})} \\ &= \frac{2}{i} \oint_{|z|=1} f(z) dz. \end{aligned}$$

$f(z)$ 有两个单极点 $z_0 = -2 \pm \sqrt{3}$, 其中 $z_0 = -2 + \sqrt{3}$ 在单位圆内, 且

$$\operatorname{Res} f(\sqrt{3} - 2) = \lim_{z \rightarrow \sqrt{3} - 2} \left[\frac{1}{z + 2 + \sqrt{3}} \right] = \frac{1}{2\sqrt{3}}.$$

$$\therefore I = 2\pi i \cdot \frac{2}{i} \operatorname{Res} f(\sqrt{3} - 2) = \frac{2\pi}{\sqrt{3}}.$$

和本题一样, 下面的几小题都是属于类型一的积分, 处理方法和本题类似, 因此, 我们将只给出简捷步骤.

$$(2) \int_0^{2\pi} \frac{dx}{(1 + \varepsilon \cos x)^2} \quad (0 < \varepsilon < 1).$$

解：作变换 $z = e^{ix}$ ，则

$$\begin{aligned} I &= \oint_{|z|=1} \frac{dz/iz}{\left[1 + \frac{\varepsilon}{2}(z + z^{-1})\right]^2} \\ &= \frac{4}{i\varepsilon^2} \oint_{|z|=1} \frac{zdz}{\left(z^2 + \frac{2}{\varepsilon}z + 1\right)^2} \\ &= \frac{4}{i\varepsilon^2} \oint_{|z|=1} f(z)dz. \end{aligned}$$

$f(z)$ 有两个二阶极点 $z_0 = \frac{1}{\varepsilon}(-1 \pm \sqrt{1 - \varepsilon^2})$ ，其中 $z_0 = \frac{1}{\varepsilon}(-1 + \sqrt{1 - \varepsilon^2})$ 在单位圆内，且

$$\operatorname{Res} f\left[\frac{1}{\varepsilon}(-1 + \sqrt{1 - \varepsilon^2})\right] = \frac{\varepsilon^2}{4(1 - \varepsilon^2)^{3/2}}.$$

$$\begin{aligned} \therefore I &= 2\pi i \cdot \frac{4}{i\varepsilon^2} \operatorname{Res} f\left[\frac{1}{\varepsilon}(-1 + \sqrt{1 - \varepsilon^2})\right] \\ &= \frac{2\pi}{(1 - \varepsilon^2)^{3/2}}. \end{aligned}$$

$$(3) \int_0^{2\pi} \frac{\cos^2 2x dx}{1 - 2\varepsilon \cos x + \varepsilon^2} \quad (|\varepsilon| < 1).$$

解：令 $z = e^{ix}$ ，则 $dx = \frac{dz}{iz}$ ， $\cos x = \frac{1 + z^2}{2z}$ ， $\cos^2 2x =$

$\frac{1 + z^4}{2z^2}$ ，以此代入原式得：

$$\begin{aligned} I &= \oint_{|z|=1} \frac{\left[\frac{1 + z^4}{2z^2}\right]^2 \frac{dz}{iz}}{1 - 2\varepsilon \frac{1 + z^2}{2z} + \varepsilon^2} \\ &= \oint_{|z|=1} \frac{(1 + z^4)^2 dz}{4iz^4[-\varepsilon z^2 + (1 + \varepsilon^2)z - \varepsilon]} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4i} \oint_{|z|=1} \frac{(1+z^4)^2 dz}{z^4(1-\varepsilon z)(z-\varepsilon)} \\
&= \frac{1}{4i} \oint_{|z|=1} f(z) dz.
\end{aligned}$$

被积函数的极点是：四阶极点 $z_0 = 0$ ，单极点 $z_0 = \varepsilon, \frac{1}{\varepsilon}$ 。因 $|\varepsilon| < 1$ ，则 $|1/\varepsilon| > 1$ ，故只有 $z_0 = 0$ 和 $z_0 = \varepsilon$ 两个极点在单位圆内，其留数分别为：

$$\begin{aligned}
\text{Res}f(0) &= \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d^3}{dz^3} \left[\frac{(1+z^4)^2}{(1-\varepsilon z)(z-\varepsilon)} \right] \\
&= \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left[\frac{(1+z^4)^2 [2\varepsilon z - (1+\varepsilon^2)]}{[(1-\varepsilon z)(z-\varepsilon)]^2} \right. \\
&\quad \left. + \frac{8z^3(1+z^4)}{(1-\varepsilon z)(z-\varepsilon)} \right] \\
&= \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d}{dz} \left[\frac{2(1+z^4)^2 [2\varepsilon z - (1+\varepsilon^2)]^2}{[(1-\varepsilon z)(z-\varepsilon)]^3} \right. \\
&\quad \left. + \frac{2(1+z^4)^2 \varepsilon + 8z^3(1+z^4) [2\varepsilon z - (1+\varepsilon^2)]}{[(1-\varepsilon z)(z-\varepsilon)]^2} \right. \\
&\quad \left. + \frac{24z^2(1+z^4) + 32z^6}{(1-\varepsilon z)(z-\varepsilon)} \right] \\
&= \frac{1}{3!} \lim_{z \rightarrow 0} \left\{ \frac{6(1+z^4)^2 [2\varepsilon z - (1+\varepsilon^2)]^3}{[(1-\varepsilon z)(z-\varepsilon)]^4} \right. \\
&\quad \left. + \frac{2\{(1+z^4)^2 \cdot 2[2\varepsilon z - (1+\varepsilon^2)]2\varepsilon + 2z^3 \cdot (1+z^4)[2\varepsilon z - (1+\varepsilon^2)]\}}{[(1-\varepsilon z)(z-\varepsilon)]^3} \right. \\
&\quad \left. + \frac{2\{(1+z^4)^2 \cdot 2\varepsilon[2\varepsilon z - (1+\varepsilon^2)] + 16z^3 \cdot (1+z^4)[2\varepsilon z - (1+\varepsilon^2)]^2\}}{[(1-\varepsilon z)(z-\varepsilon)]^3} \right. \\
&\quad \left. + \frac{16\varepsilon z^3(1+z^4)}{[(1-\varepsilon z)(z-\varepsilon)]^2} \right\}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\frac{d}{dz} \{16z^3(1+z^4)[2\epsilon z - (1+\epsilon^2)]\}}{[(1-\epsilon z)(z-\epsilon)]^2} \\
& + \frac{d}{dz} \left[\frac{24z^2(1+z^4) + 32z^6}{(1-\epsilon z)(z-\epsilon)} \right] \Bigg\} \\
& = \frac{1}{3!} \left[-\frac{6}{\epsilon^4} (1+\epsilon^2)^3 + \frac{8\epsilon}{\epsilon^2} (1+\epsilon^2) \right. \\
& \quad \left. + \frac{4\epsilon}{\epsilon^3} (1+\epsilon^2) \right] \\
& = -\frac{(1+\epsilon^2)(1+\epsilon^4)}{\epsilon^4},
\end{aligned}$$

$$\operatorname{Res} f(\epsilon) = \lim_{z \rightarrow \epsilon} \left[\frac{(1+z^4)^2}{z^4(1-\epsilon z)} \right] = \frac{(1+\epsilon^4)^2}{\epsilon^4(1-\epsilon^2)}.$$

$$\begin{aligned}
\therefore I &= 2\pi i \cdot \frac{1}{4i} \left[\frac{(1+\epsilon^4)^2}{\epsilon^4(1-\epsilon^2)} - \frac{(1+\epsilon^2)(1+\epsilon^4)}{\epsilon^4} \right] \\
&= \frac{(1+\epsilon^4)\pi}{1-\epsilon^2}.
\end{aligned}$$

$$(4) \int_0^{2\pi} \frac{\sin^2 x}{a+b\cos x} dx \quad (a>b>0).$$

$$\begin{aligned}
\text{解: 作变换后原式} &= \oint_{|z|=1} \frac{[(z^2-1)/2iz]^2 \cdot dz/iz}{a+b[z^2+1]/2z} \\
&= - \oint_{|z|=1} \frac{(z^2-1)^2 dz}{4iz^3 \left[a + \frac{b}{2z} (z^2+1) \right]} \\
&= - \frac{1}{2bi} \oint_{|z|=1} \frac{(z^2-1)^2 dz}{z^2 \left[z^2 + \frac{2a}{b} z + 1 \right]} \\
&= - \frac{1}{2bi} \oint_{|z|=1} \frac{(z^2-1)^2 dz}{z^2 \left[z + \frac{1}{b} (a + \sqrt{a^2-b^2}) \right] \left[z + \frac{1}{b} (a - \sqrt{a^2-b^2}) \right]}
\end{aligned}$$

$$= -\frac{1}{2bi} \oint_{|z|=1} f(z) dz.$$

上式的被积函数的极点是：二阶极点 $z_0 = 0$ ，单极点 $z_0 = -\frac{1}{b}$

$(a + \sqrt{a^2 - b^2})$ 和单极点 $z_0 = -\frac{1}{b} (a - \sqrt{a^2 - b^2})$ ，其中单极点 $z_0 = -\frac{1}{b} (a + \sqrt{a^2 - b^2})$ 在单位圆外（即 $|z_0| > 1$ ，亦即 $a +$

$\sqrt{a^2 - b^2} > b$ ），其余的极点在单位圆内，其留数分别是：

$$\text{Res}f(0) = \lim_{z \rightarrow 0} \frac{d}{dz} \left[\frac{(z^2 - 1)^2}{z^2 + \frac{2a}{b}z + 1} \right] = -\frac{2a}{b},$$

$$\text{Res}f\left(-\frac{a - \sqrt{a^2 - b^2}}{b}\right)$$

$$= \lim_{z \rightarrow -\frac{a - \sqrt{a^2 - b^2}}{b}} \left[\frac{(z^2 - 1)^2}{z^2 \left[z + \frac{1}{b} (a + \sqrt{a^2 - b^2}) \right]} \right]$$

$$= \frac{\left[\frac{(\sqrt{a^2 - b^2} - a)^2}{b^2} - 1 \right]^2}{\left(\frac{\sqrt{a^2 - b^2} - a}{b} \right)^2 \left[\frac{\sqrt{a^2 - b^2} - a}{b} + \frac{\sqrt{a^2 - b^2} + a}{b} \right]}$$

$$= \frac{(2a^2 - 2b^2 - 2a\sqrt{a^2 - b^2})^2}{2b(2a^2 - b^2 - 2a\sqrt{a^2 - b^2})\sqrt{a^2 - b^2}} = \frac{2\sqrt{a^2 - b^2}}{b}.$$

$$\therefore I = 2\pi i \cdot \left(-\frac{1}{2bi} \right) \left[\frac{2\sqrt{a^2 - b^2}}{b} - \frac{2a}{b} \right]$$

$$= \frac{(a - \sqrt{a^2 - b^2}) 2\pi}{b^2}.$$

$$(5) \int_0^{\pi} \frac{a dx}{a^2 + \sin^2 x} \quad (a > 0).$$

解: 把原式化为 $I = \frac{1}{2} \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} + \frac{1}{2} \int_0^\pi \frac{a dy}{a^2 + \sin^2 y}$.

在后一个积分中令 $y = x - \pi$, 则上式又

$$\begin{aligned} &= \frac{1}{2} \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} + \frac{1}{2} \int_\pi^{2\pi} \frac{a dx}{a^2 + \sin^2 x} = \frac{a}{2} \int_0^{2\pi} \frac{dx}{a^2 + \sin^2 x} \\ &= \frac{a}{2} \oint_{|z|=1} \frac{dz}{iz[a^2 + (z + z^{-1})^2 / (2i)^2]} \\ &= \frac{a}{2} \oint_{|z|=1} \frac{dz}{iz \left(a + \frac{z + z^{-1}}{2} \right) \left(a - \frac{z - z^{-1}}{2} \right)} \\ &= -\frac{2a}{i} \oint_{|z|=1} \frac{z dz}{(z^2 + 2az - 1)(z^2 - 2az - 1)} = -\frac{2a}{i} \oint_{|z|=1} \frac{z dz}{(z + a + \sqrt{a^2 + 1})(z + a - \sqrt{a^2 + 1})(z - a + \sqrt{a^2 + 1})(z - a - \sqrt{a^2 + 1})} \\ &= -\frac{2a}{i} \oint_{|z|=1} f(z) dz. \end{aligned}$$

$f(z)$ 在单位圆内有单极点 $z_0 = -a + \sqrt{a^2 + 1}$ 及 $z_0 = a - \sqrt{a^2 + 1}$, 且

$$\begin{aligned} \operatorname{Res} f(-a + \sqrt{a^2 + 1}) &= \frac{-a + \sqrt{a^2 + 1}}{2\sqrt{a^2 + 1} \cdot 2 \cdot (-a + \sqrt{a^2 + 1}) \cdot (-2a)} \\ &= \frac{-1}{8a\sqrt{a^2 + 1}}, \end{aligned}$$

$$\begin{aligned} \operatorname{Res} f(a - \sqrt{a^2 + 1}) &= \frac{a - \sqrt{a^2 + 1}}{2a \cdot 2(a - \sqrt{a^2 + 1}) \cdot 2(-\sqrt{a^2 + 1})} \\ &= \frac{-1}{8a\sqrt{a^2 + 1}}. \end{aligned}$$

$$\therefore \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} = \frac{2a}{i} 2\pi i \frac{1}{-4a\sqrt{a^2 + 1}} = \frac{\pi}{\sqrt{a^2 + 1}}.$$

$$(6) \int_0^{2\pi} \frac{\cos x dx}{1 - 2\varepsilon \cos x + \varepsilon^2} (|\varepsilon| < 1).$$

$$\begin{aligned}
 \text{解: 作变换后原式} &= \oint_{|z|=1} \frac{\frac{z^2+1}{2z} \cdot \frac{dz}{iz}}{1 - 2\varepsilon \frac{z^2+1}{2z} + \varepsilon^2} \\
 &= \oint_{|z|=1} \frac{(z^2+1)dz}{2iz^2 \left(1 - \varepsilon \frac{z^2+1}{z} + \varepsilon^2\right)} \\
 &= \oint_{|z|=1} \frac{(z^2+1)dz}{[(1+\varepsilon^2)z - \varepsilon z^2 - \varepsilon]} \\
 &= \frac{1}{2i} \oint_{|z|=1} \frac{(z^2+1)dz}{z(1-\varepsilon z)(z-\varepsilon)}.
 \end{aligned}$$

被积函数有三个单极点 $z_0 = 0, \varepsilon, 1/\varepsilon$; 因 $|\varepsilon| < 1$, 则 $\left| \frac{1}{\varepsilon} \right| > 1$, 故只有单极点 $z_0 = 0, \varepsilon$ 在积分回路之内, 其留数分别是:

$$\begin{aligned}
 \text{Res}f(0) &= \lim_{z \rightarrow 0} \left[\frac{z^2+1}{(1-\varepsilon z)(z-\varepsilon)} \right] = -\frac{1}{\varepsilon}, \\
 \text{Res}f(\varepsilon) &= \lim_{z \rightarrow \varepsilon} \left[\frac{z^2+1}{z(1-\varepsilon z)} \right] = \frac{1+\varepsilon^2}{\varepsilon(1-\varepsilon^2)}, \\
 \therefore I &= 2\pi i \cdot \frac{1}{2i} \left[\frac{1+\varepsilon^2}{\varepsilon(1-\varepsilon^2)} - \frac{1}{\varepsilon} \right] = \frac{2\pi\varepsilon}{1-\varepsilon^2}.
 \end{aligned}$$

$$(7) \int_0^{\pi/2} \frac{dx}{1+\cos^2 x}.$$

解: 因被积函数是偶函数, 故可作下列的延拓

$$\begin{aligned}
 I &= \frac{1}{4} \int_0^{2\pi} \frac{dx}{1+\cos^2 x} = \frac{1}{4} \oint_{|z|=1} \frac{\frac{dz}{iz}}{1 + \left[\frac{z^2+1}{2z} \right]^2} \\
 &= \frac{1}{i} \oint_{|z|=1} \frac{zdz}{z^4 + 6z^2 + 1} \\
 &= \frac{1}{i} \oint_{|z|=1} \frac{zdz}{[z^2 + 3 + 2\sqrt{2}][z^2 + 3 - 2\sqrt{2}]}
 \end{aligned}$$

$$= \frac{1}{i} \oint_{|z|=1} \frac{z dz}{[z^2 + (3 + 2\sqrt{2})][z + \sqrt{3 - 2\sqrt{2}}i][z - \sqrt{3 - 2\sqrt{2}}i]},$$

被积函数的四个单极点中, 只是 $z_0 = \pm \sqrt{3 - 2\sqrt{2}}i$, 即 $z_0 = (\sqrt{2} - 1)i$ 和 $z_0 = (1 - \sqrt{2})i$ 在积分回路之内, 其留数分别是

$$\begin{aligned} \operatorname{Res} f(\sqrt{3 - 2\sqrt{2}}i) &= \lim_{z \rightarrow z_0} \left\{ \frac{z}{[z^2 + 3 + 2\sqrt{2}][z + \sqrt{3 - 2\sqrt{2}}i]} \right\} \\ &= \frac{1}{8\sqrt{2}}, \end{aligned}$$

$$\begin{aligned} \operatorname{Res} f(-\sqrt{3 - 2\sqrt{2}}i) &= \lim_{z \rightarrow z_0} \left\{ \frac{z}{[z^2 + 3 + 2\sqrt{2}][z - \sqrt{3 - 2\sqrt{2}}i]} \right\} \\ &= \frac{1}{8\sqrt{2}}, \end{aligned}$$

$$\therefore I = 2\pi i \cdot \frac{1}{i} \cdot \frac{1}{4\sqrt{2}} = \frac{\pi}{2\sqrt{2}}.$$

$$(8) \int_0^{2\pi} \cos^{2n} x dx.$$

$$\begin{aligned} \text{解: 作变换后, 原式} &= \oint_{|z|=1} \left[\frac{z^2 + 1}{2z} \right]^{2n} \frac{dz}{iz} \\ &= \frac{1}{2^{2n}i} \oint_{|z|=1} \frac{(1 + z^2)^{2n} dz}{z^{2n+1}}, \end{aligned}$$

被积函数有一个 $(2n+1)$ 阶极点 $z=0$, 且

$$\operatorname{Res} f(0) = \frac{1}{(2n)!} \lim_{z \rightarrow 0} \frac{d^{2n}}{dz^{2n}} (1 + z^2)^{2n};$$

根据二项式公式: $(a+b)^n = \dots + \frac{n! a^{n-k} b^k}{(n-k)! k!} + \dots$ 知

$$(1 + z^2)^{2n} = \dots + \frac{(2n)! z^{2K}}{(2n-K)! K!} + \dots$$

还要对 z 微分 $2n$ 次, 故凡是 $2k < 2n$ 的 z^{2K} 项, 在微分 $2n$ 次后都为零; 而 $2K > 2n$ 项中, 在微分 $2n$ 次后仍含有变数 z , 当 $z \rightarrow z_0 = 0$

时, 这些项全部为零; 只有当 $2K = 2n$ 的项在微分 $2n$ 次并以 $z_0 = 0$ 代入后的结果才不为零, 即

$$\begin{aligned} \operatorname{Res} f(0) &= \frac{1}{(2n)!} \lim_{z \rightarrow 0} \frac{d^{2n}}{dz^{2n}} \left[\frac{(2n)! z^{2n}}{(2n-n)! n!} \right] = \frac{(2n)!}{(n!)^2}, \\ \therefore I &= \frac{1}{2^{2n} i} \cdot 2\pi i \cdot \frac{(2n)!}{(n!)^2} = \frac{2\pi \cdot 2^n (n!) [1 \cdot 3 \cdot 5 \cdots (2n-1)]}{2^n (n!) 2^n (n!)} \\ &= \frac{2\pi [1 \cdot 3 \cdot 5 \cdots (2n-1)]}{2 \cdot 4 \cdot 6 \cdots 2n}. \end{aligned}$$

2. 计算下列实变函数定积分.

$$(1) \int_{-\infty}^{\infty} \frac{x^2 + 1}{x^4 + 1} dx.$$

$$\begin{aligned} \text{解: } f(z) &= \frac{z^2 + 1}{z^4 + 1} = \frac{z^2 + 1}{(z^2 - i)(z^2 + i)} \\ &= \frac{z^2 + 1}{\left[z - \frac{\sqrt{2}}{2}(1-i) \right] \left[z + \frac{\sqrt{2}}{2}(1-i) \right] \left[z - \frac{\sqrt{2}}{2}(1+i) \right] \left[z + \frac{\sqrt{2}}{2}(1+i) \right]} \end{aligned}$$

它具有四个单极点, 其中只有 $z_0 = -\frac{\sqrt{2}}{2}(1-i), \frac{\sqrt{2}}{2}(1+i)$

在上半平面, 其留数分别为:

$$\operatorname{Res} f\left[\frac{\sqrt{2}}{2}(i-1)\right] = \lim_{z \rightarrow z_0} \left[\frac{z^2 + 1}{(z^2 + i) \left[z - \frac{\sqrt{2}}{2}(1-i) \right]} \right] = \frac{1}{2\sqrt{2}i},$$

$$\operatorname{Res} f\left[\frac{\sqrt{2}}{2}(i+1)\right] = \lim_{z \rightarrow z_0} \left[\frac{z^2 + 1}{(z^2 + i) \left[z + \frac{\sqrt{2}}{2}(1-i) \right]} \right] = \frac{1}{2\sqrt{2}i},$$

$$\therefore I = 2\pi i \cdot \left[\frac{1}{2\sqrt{2}i} + \frac{1}{2\sqrt{2}i} \right] = \sqrt{2}\pi.$$

本题和下面几小题都属于类型二.

$$(2) \int_0^{\infty} \frac{x^2 dx}{(x^2 + 9)(x^2 + 4)^2}.$$

解: 由于被积函数是偶函数, 所以

$$\text{原式} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2 + 9)(x^2 + 4)^2},$$

$$\begin{aligned} \text{被积函数 } f(z) &= \frac{z^2}{(z^2 + 9)(z^2 + 4)^2} \\ &= \frac{z^2}{(z + 3i)(z - 3i)(z + 2i)^2(z - 2i)^2}, \end{aligned}$$

它在上半平面的奇点是两个，一个极点 $z_0 = 3i$ ，一个二阶极点 $z_0 = 2i$ ，其留数分别是：

$$\text{Res}f(3i) = \lim_{z \rightarrow 3i} \left[\frac{z^2}{(z + 3i)(z^2 + 4)^2} \right] = \frac{3}{50}i,$$

$$\begin{aligned} \text{Res}f(2i) &= \lim_{z \rightarrow 2i} \frac{d}{dz} \left[\frac{z^2}{(z^2 + 9)(z + 2i)^2} \right] \\ &= \lim_{z \rightarrow 2i} \left\{ \frac{2z}{(z^2 + 9)(z + 2i)^2} \right. \\ &\quad \left. - \frac{2z^3(z + 2i)^2 + 2z^2(z^2 + 9)(z + 2i)}{[(z^2 + 9)(z + 2i)^2]^2} \right\} \\ &= -\frac{13}{200}i, \end{aligned}$$

$$\therefore I = 2\pi i \cdot \frac{1}{2} \left[\frac{3i}{50} - \frac{13i}{200} \right] = \frac{\pi}{200}.$$

$$(3) \quad \int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2(x^2 + b^2)}.$$

$$\begin{aligned} \text{解: 被积函数 } f(z) &= \frac{1}{(z^2 + a^2)^2(z^2 + b^2)} \\ &= \frac{1}{(z + ai)^2(z - ai)^2(z + bi)(z - bi)}. \end{aligned}$$

(i) 若 $a > b, b > 0$, 则其在上半平面的奇点是：单极点 $z_0 = bi$ ，二阶极点 $z_0 = ai$ ，其留数分别为：

$$\text{Res}f(bi) = \lim_{z \rightarrow bi} \left[\frac{1}{(z^2 + a^2)^2(z + bi)} \right] = \frac{-i}{2b(b^2 - a^2)^2},$$

$$\begin{aligned}
 \operatorname{Res} f(ai) &= \lim_{z \rightarrow ai} \frac{d}{dz} \left[\frac{1}{(z^2 + b^2)(z + ai)^2} \right] \\
 &= \lim_{z \rightarrow ai} \left[\frac{-2z(z + ai)^2 - 2(z^2 + b^2)(z + ai)}{[(z^2 + b^2)(z + ai)^2]^2} \right] \\
 &= \frac{(3a^2 - b^2)i}{4a^3(b^2 - a^2)^2};
 \end{aligned}$$

$$\therefore I = 2\pi i \left[\frac{(3a^2 - b^2)i}{4a^3(b^2 - a^2)^2} - \frac{i}{2b(b^2 - a^2)^2} \right] = \frac{(2a + b)\pi}{2a^3b(a + b)^2}.$$

(ii) 对于 $a < 0$, $b < 0$ 或 $a > 0$, $b < 0$ 或 $a < 0$, $b > 0$ 等三种情况均可作类似的计算.

$$(4) \quad \int_0^{\infty} \frac{dx}{x^4 + a^4}.$$

解一: 因被积函数是偶函数, 故原式 $= \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{x^4 + a^4}$, 其

$$\text{中被积函数 } f(z) = \frac{1}{z^4 + a^4} = \frac{1}{[z^2 + a^2i][z^2 - a^2i]} =$$

$$\frac{1}{\left[z - \frac{\sqrt{2}}{2}a(1-i) \right] \left[z + \frac{\sqrt{2}}{2}a(1-i) \right] \left[z - \frac{\sqrt{2}}{2}a(1+i) \right] \left[z + \frac{\sqrt{2}}{2}a(1+i) \right]}$$

设 $a > 0$, 它在上半平面有两个单极点 $z_0 = \frac{\sqrt{2}}{2}a(i-1)$, $z_0 = \frac{\sqrt{2}}{2}a(i+1)$, 其留数分别是:

$$\begin{aligned}
 \operatorname{Res} f \left(\frac{\sqrt{2}}{2}a(i-1) \right) &= \lim_{z \rightarrow z_0} \left[\frac{1}{\left[z - \frac{\sqrt{2}}{2}a(1-i) \right] [z^2 - a^2i]} \right] \\
 &= \frac{1}{2\sqrt{2}a^3(1+i)},
 \end{aligned}$$

$$\operatorname{Res} f \left[\frac{\sqrt{2}}{2}a(1+i) \right] = \lim_{z \rightarrow z_0} \left[\frac{1}{[z^2 + a^2i] \left[z + \frac{\sqrt{2}}{2}a(1+i) \right]} \right]$$

$$= \frac{1}{2\sqrt{2} a^3(i-1)},$$

$$\therefore I = 2\pi i \cdot \frac{1}{2} \frac{1}{2\sqrt{2} a^3} \left[\frac{1}{i+1} + \frac{1}{i-1} \right] = \frac{\pi}{2\sqrt{2} a^3}.$$

解二：被积函数 $f(z)$ 有四个单极点 $z_0 = ae^{i\frac{\pi}{4}}$ 、 $z_0 = ae^{i\frac{3\pi}{4}}$ 、 $z_0 = ae^{i\frac{5\pi}{4}}$ 、 $z_0 = ae^{i\frac{7\pi}{4}}$ ，其中只有单极点 $z_0 = ae^{i\frac{\pi}{4}}$ 和 $z_0 = ae^{i\frac{3\pi}{4}}$ 在上半平面，其留数分别是（应用罗毕达法则）：

$$\begin{aligned} \operatorname{Res}f(ae^{i\frac{\pi}{4}}) &= \lim_{z \rightarrow z_0} \left[(z - ae^{i\frac{\pi}{4}}) \frac{1}{z^4 + a^4} \right] = \lim_{z \rightarrow z_0} \frac{1}{4z^3} \\ &= \frac{1}{4a^3} e^{-i\frac{3\pi}{4}}, \end{aligned}$$

$$\begin{aligned} \operatorname{Res}f(ae^{i\frac{3\pi}{4}}) &= \lim_{z \rightarrow z_0} \left[(z - ae^{i\frac{3\pi}{4}}) \frac{1}{z^4 + a^4} \right] \\ &= \lim_{z \rightarrow z_0} \frac{1}{4z^3} = \frac{1}{4a^3} e^{-i\frac{9\pi}{4}}. \end{aligned}$$

$$\begin{aligned} \therefore I &= \pi i \left[\operatorname{Res}f(ae^{i\frac{\pi}{4}}) + \operatorname{Res}f(ae^{i\frac{3\pi}{4}}) \right] \\ &= \frac{\pi i}{4a^3} \left[e^{-i\frac{3\pi}{4}} + e^{-i\frac{9\pi}{4}} \right] = \frac{\pi}{2\sqrt{2} a^3}. \end{aligned}$$

显然，解二比解一的计算要简单些。

$$(5) \int_0^{\infty} \frac{(x^2+1)dx}{x^6+1}.$$

解：因被积函数是偶函数，故原式 $= \frac{1}{2} \int_{-\infty}^{\infty} \frac{(x^2+1)dx}{x^6+1}$ ，

$$\begin{aligned}
 \text{被积函数 } f(z) &= \frac{z^2 + 1}{z^6 + 1} = \frac{1}{z^4 - z^2 + 1} \\
 &= \frac{1}{\left[z^2 - \frac{1}{2}(1 + \sqrt{3}i) \right] \left[z^2 - \frac{1}{2}(1 - \sqrt{3}i) \right]} = \\
 &= \frac{1}{\left[z + \sqrt{\frac{1}{2}(1 + \sqrt{3}i)} \right] \left[z - \sqrt{\frac{1}{2}(1 + \sqrt{3}i)} \right] \left[z + \sqrt{\frac{1}{2}(1 - \sqrt{3}i)} \right] \left[z - \sqrt{\frac{1}{2}(1 - \sqrt{3}i)} \right]}
 \end{aligned}$$

$$\begin{aligned}
 \text{注意到: } \sqrt{\frac{1}{2}(1 + \sqrt{3}i)} &= \sqrt{\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}} = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \\
 &= \frac{1}{2}(1 + \sqrt{3}i),
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{\frac{1}{2}(1 - \sqrt{3}i)} &= \sqrt{\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}} = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \\
 &= \frac{1}{2}(i - \sqrt{3}),
 \end{aligned}$$

故 $f(z) =$

$$\frac{1}{\left[z + \frac{1}{2}(i + \sqrt{3}) \right] \left[z - \frac{1}{2}(i + \sqrt{3}) \right] \left[z + \frac{1}{2}(i - \sqrt{3}) \right] \left[z - \frac{1}{2}(i - \sqrt{3}) \right]}.$$

被积函数在上半平面有两个单极点 $z_0 = \frac{1}{2}(i + \sqrt{3})$,

$z_0 = \frac{1}{2}(i - \sqrt{3})$, 其留数为:

$$\begin{aligned}
 &\text{Res}f\left[\frac{1}{2}(i + \sqrt{3})\right] \\
 &= \lim_{z \rightarrow z_0} \frac{1}{\left[z^2 - \frac{1}{2}(1 - \sqrt{3}i) \right] \left[z + \frac{1}{2}(i + \sqrt{3}) \right]} \\
 &= \frac{1}{\left\{ \left[\frac{1}{2}(i - \sqrt{3}) \right]^2 - \frac{1}{2}(i - \sqrt{3}i) \right\} \left\{ \frac{1}{2}(i + \sqrt{3}) + \frac{1}{2}(i + \sqrt{3}) \right\}}
 \end{aligned}$$

$$= \frac{1}{\sqrt{3}(\sqrt{3}i-1)},$$

$$\operatorname{Res} f\left[\frac{1}{2}(i-\sqrt{3})\right] = \lim_{z \rightarrow z_0} \left[\frac{1}{\left[z^2 - \frac{1}{2}(1+\sqrt{3}i)\right]\left[z + \frac{1}{2}(i-\sqrt{3})\right]} \right]$$

$$= \frac{1}{\left\{\left[\frac{1}{2}(i-\sqrt{3})\right]^2 - \frac{1}{2}(1+\sqrt{3}i)\right\}\left\{\frac{1}{2}(i-\sqrt{3}) + \frac{1}{2}(i-\sqrt{3})\right\}}$$

$$= \frac{1}{\sqrt{3}(\sqrt{3}i+1)}.$$

$$\therefore I = 2\pi i \cdot \frac{1}{2} \left[\frac{1}{\sqrt{3}(\sqrt{3}i+1)} - \frac{1}{\sqrt{3}(\sqrt{3}i-1)} \right] = \frac{\pi}{2}.$$

必须指出：本题也可用上题解二的方法求解。

$$(6) \int_0^{\infty} \frac{x^2}{(x^2+a^2)^2} dx.$$

解：因被积函数是偶函数，所以

$$\text{原式} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)^2} dx.$$

被积函数 $f(z) = \frac{z^2}{(z^2+a^2)^2} = \frac{z^2}{(z+ai)^2(z-ai)^2}$ 在上半平面有一个二阶极点 $z_0 = ai$ ，且

$$\begin{aligned} \operatorname{Res} f(ai) &= \lim_{z \rightarrow ai} \frac{d}{dz} \left[\frac{z^2}{(z+ai)^2} \right] = \lim_{z \rightarrow ai} \left[\frac{2z}{(z+ai)^2} - \frac{2z^2}{(z+ai)^3} \right] \\ &= \frac{2ai}{(2ai)^2} - \frac{2(ai)^2}{(2ai)^3} = -\frac{i}{4a}. \end{aligned}$$

$$\therefore I = 2\pi i \cdot \frac{1}{2} \left(-\frac{i}{4a} \right) = \frac{\pi}{4a}.$$

$$(7) \int_{-\infty}^{\infty} \frac{x^{2m}}{1+x^{2n}} dx \quad (m < n).$$

解：被积函数 $f(z) = \frac{z^{2m}}{1+z^{2n}}$ 在上半平面有 n 个单极点

($z^{2n} + 1 = 0$, $z^{2n} = -1$) $z_0 = e^{(2K+1)\pi i/2n}$ ($K = 0, 1, 2, \dots, n-1$)。现在计算留数

$$\operatorname{Res} f(e^{(2K+1)\pi i/2n}) = \lim_{z \rightarrow z_0} \left[(z - e^{(2K+1)\pi i/2n}) \frac{z^{2m}}{1+z^{2n}} \right],$$

用罗毕达法则,

$$\begin{aligned} \text{上式} &= \lim_{z \rightarrow z_0} \frac{2mz^{2m-1}(z - e^{(2K+1)\pi i/2n}) + z^{2m}}{2nz^{2n-1}} \\ &= \frac{1}{2ne^{(2K+1)(2n-2m-1)\pi i/2n}}; \end{aligned}$$

故上半平面各留数之和为

$$\begin{aligned} &\frac{1}{2ne^{(2n-2m-1)\pi i/2n}} \sum_{K=0}^{n-1} \frac{1}{e^{K(2n-2m-1)\pi i/n}} \\ &= \frac{-e^{(2m+1)\pi i/2n}}{2n} \cdot \frac{1 - e^{-(2n-2m-1)\pi i}}{1 - e^{-(2n-2m-1)\pi i/n}} \\ &= \frac{1}{2n} \cdot \frac{2}{e^{(2m+1)\pi i/2n} - e^{-(2m+1)\pi i/2n}} \\ &= \frac{1}{2ni \sin \frac{2m+1}{2n} \pi} \cdot \\ \therefore I &= 2\pi i \frac{1}{2ni \sin \frac{2m+1}{2n} \pi} = \frac{\pi}{n \sin \frac{2m+1}{2n} \pi}. \end{aligned}$$

3. 计算下列实变函数定积分。

$$(1) \int_0^{\infty} \frac{\cos mx}{1+x^4} dx \quad (m > 0).$$

解: 本题和下面几小题都属于类型三。

$$\because F(z) e^{imz} = \frac{e^{imz}}{1+z^4}$$

$$= \frac{e^{miz}}{\left[z - \frac{\sqrt{2}}{2}(1-i) \right] \left[z + \frac{\sqrt{2}}{2}(1-i) \right] \left[z - \frac{\sqrt{2}}{2}(1+i) \right] \left[z + \frac{\sqrt{2}}{2}(1+i) \right]}.$$

在上半平面有两个单极点 $z_0 = \frac{\sqrt{2}}{2}(i-1)$, $z_0 = \frac{\sqrt{2}}{2}(i+1)$,

其留数分别为:

$$\begin{aligned} \text{Res}f(z_0) &= \lim_{z \rightarrow \frac{\sqrt{2}}{2}(i-1)} \left\{ \frac{e^{imz}}{\left[z - \frac{\sqrt{2}}{2}(1-i) \right] (z^2 - i)} \right\} \\ &= \frac{e^{-im \left[\frac{\sqrt{2}}{2}(1-i) \right]}}{(-2i)(i-1)\sqrt{2}} = \frac{e^{-im \left[\frac{\sqrt{2}}{2}(1-i) \right]}}{2\sqrt{2}(i+1)}, \end{aligned}$$

$$\begin{aligned} \text{Res}f(z_0) &= \lim_{z \rightarrow \frac{\sqrt{2}}{2}(i+1)} \left\{ \frac{e^{imz}}{\left[z^2 + i \right] \left[z + \frac{\sqrt{2}}{2}(1+i) \right]} \right\} \\ &= \frac{e^{im \left[\frac{\sqrt{2}}{2}(1+i) \right]}}{2i \cdot \sqrt{2}(1+i)} = \frac{e^{im \left[\frac{\sqrt{2}}{2}(1+i) \right]}}{2\sqrt{2}(i-1)}. \end{aligned}$$

$$\begin{aligned} \therefore I &= \pi i \left[\frac{e^{-im \left[\frac{\sqrt{2}}{2}(1-i) \right]}}{2\sqrt{2}(i-1)} + \frac{e^{im \left[\frac{\sqrt{2}}{2}(1+i) \right]}}{2\sqrt{2}(i+1)} \right] \\ &= \frac{(i-1)e^{-im \left[\frac{\sqrt{2}}{2}(1-i) \right]} - (i+1)e^{im \left[\frac{\sqrt{2}}{2}(1+i) \right]}}{4\sqrt{2}} \pi i \end{aligned}$$

$$= \frac{2e^{-\frac{m}{\sqrt{2}}} \left[-i \cos \frac{m}{\sqrt{2}} - i \sin \frac{m}{\sqrt{2}} \right]}{4\sqrt{2}} \pi i$$

$$= \frac{\sqrt{2}\pi e^{-\frac{m}{\sqrt{2}}} \left(\cos \frac{m}{\sqrt{2}} + \sin \frac{m}{\sqrt{2}} \right)}{4}.$$

本题也可用指数来表示被积函数在上半平面的极点, 即 $z_0 = e^{i\frac{\pi}{4}}$ 和 $z_0 = e^{i\frac{3\pi}{4}}$. 注意应用罗毕达法则计算被积函数在这两个极点的留数, 也可同样求出上述答案.

$$(2) \int_0^{\infty} \frac{\sin mx}{x(x^2+a^2)} dx \quad (m>0, a>0).$$

$$\text{解一: } \int_0^{\infty} \frac{\sin mx}{x(x^2+a^2)} dx = \frac{1}{2i} \int_0^{\infty} \frac{e^{imx}}{x(x^2+a^2)} dx = I$$

考虑积分

$$\oint_l \frac{e^{imz}}{z(z^2+a^2)} dz = \int_{C_R} \frac{e^{imz}}{z(z^2+a^2)} dz + \int_{C_\varepsilon} \frac{e^{imz}}{z(z^2+a^2)} dz + \left(\int_{-R}^{-\varepsilon} + \int_{\varepsilon}^R \right) \frac{e^{imx}}{x(x^2+a^2)} dx. \quad (1)$$

如图4-2, l 内有一单极点 ia ,

留数是 $\frac{-e^{-ma}}{2a^2}$, 所以, (1) 式

$$\text{左端} = 2\pi i \frac{-e^{-ma}}{2a^2} = \frac{-\pi e^{-ma}}{a^2} i,$$

又在 (1) 式两端令 $\varepsilon \rightarrow 0$,

$R \rightarrow \infty$, 则右端第一项依约当引理为零, 右端最后两项 $= 2iI$, 于是,

$$-\frac{\pi e^{-ma}}{a^2} \lim_{\varepsilon \rightarrow 0} \int_{C_\varepsilon} \frac{e^{imz}}{z(z^2+a^2)} dz + 2iI.$$

而

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \int_{C_\varepsilon} \frac{e^{imz}}{z(z^2+a^2)} dz &= \lim_{\varepsilon \rightarrow 0} \int_{C_\varepsilon} \left[\frac{1}{a^2 z} + \text{解析部分 } P(z) \right] dz \\ &= \lim_{\varepsilon \rightarrow 0} \int_0^{2\pi} \frac{i\varepsilon e^{i\varphi}}{a^2 \varepsilon e^{i\varphi}} d\varphi = -\frac{i}{a^2} \pi. \end{aligned}$$

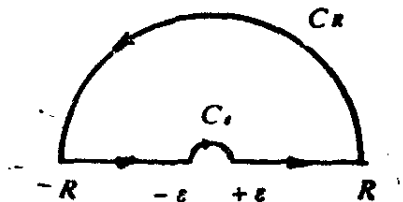


图 4-2

$$\therefore 2iI = \frac{i}{a^2} \pi - \frac{i\pi}{a^2} e^{-ma}, \text{ 即 } I = (1 - e^{-ma}) \frac{\pi}{2a^2}.$$

解二：注意到 $\frac{1}{x(x^2 + a^2)} = \frac{1}{a^2 x} - \frac{x}{a^2(x^2 + a^2)}$ 以及

$$\int_0^\infty \frac{\sin mx}{x} dx = \frac{\pi}{2}, \text{ 则}$$

$$\begin{aligned} \int_0^\infty \frac{\sin mx}{x(x^2 + a^2)} dx &= \frac{1}{a^2} \int_0^\infty \frac{\sin mx}{x} dx - \frac{1}{a^2} \int_0^\infty \frac{x \sin mx}{x^2 + a^2} dx \\ &= \frac{1}{a^2} \left(\frac{\pi}{2} - \int_0^\infty \frac{x \sin mx}{x^2 + a^2} dx \right), \end{aligned}$$

而 $\int_0^\infty \frac{x \sin mx}{x^2 + a^2} dx = \pi \left\{ \frac{ze^{imz}}{z^2 + a^2} \text{ 在上半平面所有奇点留数之和} \right\}$

$$= \pi \left\{ \operatorname{Res} f(ia) \right\} = \pi \left\{ \lim_{z \rightarrow ia} \left[(z - ia) \frac{ze^{imz}}{z^2 + a^2} \right] \right\} =$$

$$\frac{\pi e^{-ma}}{2}, \text{ 所以}$$

$$\begin{aligned} \int_0^\infty \frac{\sin mx}{x(x^2 + a^2)} dx &= \frac{1}{a^2} \left(\frac{\pi}{2} - \frac{\pi}{2} e^{-ma} \right) \\ &= (1 - e^{-ma}) \frac{\pi}{2a^2}. \end{aligned}$$

$$(3) \int_{-\infty}^\infty \frac{x \sin x}{1 + x^2} dx.$$

解：因被积函数是偶函数，

$$\therefore \text{原式} = 2 \int_0^\infty \frac{x \sin x}{1 + x^2} dx.$$

上式中的被积函数 $G(z)e^{iz} = \frac{z}{1+z^2} e^{iz} = \frac{ze^{iz}}{(z+i)(z-i)}$ 在上半平面有一个单极点 $z_0 = i$ ，且

$$\operatorname{Res} f(i) = \lim_{z \rightarrow i} \left(\frac{z}{z+i} \right) e^{iz} = \frac{1}{2e}.$$

$$\therefore I = \pi \cdot 2 \left(\frac{1}{2e} \right) = \frac{\pi}{e}.$$

$$(4) \int_{-\infty}^{\infty} \frac{x \sin mx}{2x^2 + a^2} dx, \quad (m > 0, a > 0).$$

解: 因为被积函数是偶函数,

$$\therefore \text{原式} = 2 \int_0^{\infty} \frac{x \sin mx}{2x^2 + a^2} dx,$$

$$\begin{aligned} \text{上式中的被积函数 } G(z) e^{imz} &= \frac{z}{2z^2 + a^2} e^{imz} \\ &= \frac{z e^{imz}}{2 \left[z + \frac{a}{\sqrt{2}} i \right] \left[z - \frac{a}{\sqrt{2}} i \right]} \end{aligned}$$

在上半平面有一个单极点 $z_0 = \frac{a}{\sqrt{2}} i$, 且

$$\text{Res}f(z_0) = \lim_{z \rightarrow ai/\sqrt{2}} \left[\frac{z e^{imz}}{2 \left(z + \frac{a}{\sqrt{2}} i \right)} \right] = \frac{1}{4} e^{-ma/\sqrt{2}}.$$

$$\therefore I = \pi \cdot 2 \cdot \frac{1}{4} e^{-ma/\sqrt{2}} = \frac{\pi}{2} e^{-ma/\sqrt{2}}.$$

$$(5) \int_0^{\infty} \frac{\cos mx}{(x^2 + a^2)^2} dx,$$

解: $F(z) e^{imz} = \frac{e^{imz}}{(z^2 + a^2)^2} = \frac{e^{imz}}{(z + ai)^2 (z - ai)^2}$ 在上半平面只有一个二阶极点 $z_0 = ai$, 其留数

$$\begin{aligned} \text{Res}f(z_0) &= \lim_{z \rightarrow ai} \frac{d}{dz} \left[\frac{e^{imz}}{(z + ai)^2} \right] \\ &= \lim_{z \rightarrow ai} \left[\frac{ime^{imz}}{(z + ai)^2} - \frac{2e^{imz}}{(z + ai)^3} \right] \\ &= -\frac{(am + 1)e^{-ma}}{4a^3}. \end{aligned}$$

$$\therefore I = \pi i \left[-\frac{(am+1)e^{-ma}}{4a^3} i \right] = \frac{\pi(am+1)e^{-ma}}{4a^3}.$$

$$(6) \int_0^{\infty} \frac{\cos x}{(x^2+a^2)(x^2+b^2)} dx.$$

$$\begin{aligned} \text{解: } F(z) e^{iz} &= \frac{e^{iz}}{(z^2+a^2)(z^2+b^2)} \\ &= \frac{e^{iz}}{(z+ia)(z-ia)(z+ib)(z-ib)} \end{aligned}$$

在上半平面有两个单极点 $z_0 = ai$, $z_0 = bi$, 其留数分别是:

$$\text{Res}f(z_0) = \lim_{z \rightarrow a} \left[\frac{e^{iz}}{(z+ia)(z^2+b^2)} \right] = \frac{ie^{-a}}{2a(a^2-b^2)},$$

$$\text{Res}f(z_0) = \lim_{z \rightarrow b} \left[\frac{e^{iz}}{(z^2+a^2)(z+ib)} \right] = \frac{-ie^{-b}}{2b(a^2-b^2)}.$$

$$\therefore I = \pi i \left[\frac{ie^{-a}}{2a(a^2-b^2)} - \frac{ie^{-b}}{2b(a^2-b^2)} \right] = \frac{\pi(ae^{-b} - be^{-a})}{2ab(a^2-b^2)}$$

$$= \frac{\pi \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right)}{2(a^2-b^2)}.$$

$$(7) \int_0^{\infty} \frac{\sin^2 x}{x^2} dx,$$

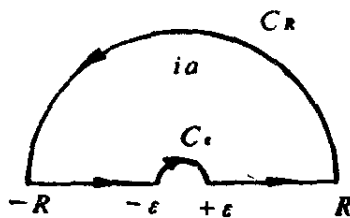


图 4-3

$$\text{解: } \int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{1}{2i} \int_{-\infty}^{\infty} \frac{e^{ix} \sin x}{x^2} dx = I$$

我们考虑积分 $\oint_l \frac{e^{iz} \sin z dz}{z^2}$

$$= \left[\int_{C_R} + \int_{C_\epsilon} \right] \frac{e^{iz} \sin z}{z^2} dz + \left(\int_{-\infty}^{-\epsilon} + \int_{\epsilon}^{\infty} \right) \frac{e^{ix} \sin x}{x^2} dx.$$

如图4-3, l 中无奇点, 所以上式左端为零, 令 $\epsilon \rightarrow 0$, $R \rightarrow \infty$, 右端第一项为

$$\int_{C_R} \frac{e^{iz}(e^{iz} - e^{-iz})dz}{2iz^2} = \frac{1}{2i} \int_{C_R} \left[\frac{e^{i2z}}{z^2} - \frac{1}{z^2} \right] dz.$$

在上式中, 第一项依约当引理 $\rightarrow 0$, 第二项 $\frac{1}{z^2}$ 因 z 一致趋于 0

也 $\rightarrow 0$, 所以 $\lim_{R \rightarrow \infty} \int_{C_R} = 0$,

$$\begin{aligned} \therefore 2iI &= \lim_{\varepsilon \rightarrow 0} - \int_{C_\varepsilon} \frac{e^{iz} \sin z}{z^2} dz \\ &= \lim_{\varepsilon \rightarrow 0} \int_{C_\varepsilon} - \left[\frac{1}{z} + \text{解析部分 } P(z) \right] dz \\ &= \int_{\varphi}^0 - \frac{i\varepsilon e^{i\varphi}}{\varepsilon e^{i\varphi}} d\varphi = i\pi, \quad I = \frac{\pi}{2}. \\ \text{即 } \int_0^\infty \frac{\sin^2 x}{x^2} dx &= \frac{\pi}{2}. \end{aligned}$$

解本题的方法不仅这一种, 其它的方法留给读者自己练习.

$$(8) \int_{-\infty}^{\infty} \frac{e^{imx}}{x - i\alpha} dx, \int_{-\infty}^{\infty} \frac{e^{imx}}{x + i\alpha} dx \quad (m > 0, \operatorname{Re} \alpha > 0).$$

解: 在上半平面 $\frac{e^{imz}}{z - i\alpha}$ 有单极点 $i\alpha$, $\frac{e^{imz}}{z + i\alpha}$ 在上半平面无

奇点

$$\therefore \int_{-\infty}^{\infty} \frac{e^{imx}}{x - i\alpha} dx = 2\pi i \left[\lim_{z \rightarrow i\alpha} e^{imz} \right] = 2\pi i e^{-ma},$$

而 $\int_{-\infty}^{\infty} \frac{e^{imx}}{x + i\alpha} dx = 0.$

第五章 拉普拉斯变换

§ 21. 拉普拉斯变换

1. 求下列函数的拉普拉斯变换函数.

(1) $\text{sh}\omega t$, $\text{ch}\omega t$.

$$\begin{aligned}\text{解一: } \varphi(t) &= \text{sh}\omega t = \frac{1}{2}(e^{\omega t} - e^{-\omega t}) \\ &\doteq \frac{1}{2} \left[\frac{1}{p - \omega} - \frac{1}{p + \omega} \right] = \frac{\omega}{p^2 - \omega^2}.\end{aligned}$$

$$\begin{aligned}\text{解二: } \varphi(t) &= \text{ch}\omega t = \frac{1}{2}(e^{\omega t} + e^{-\omega t}) \\ &\doteq \frac{1}{2} \left[\frac{1}{p - \omega} + \frac{1}{p + \omega} \right] = \frac{p}{p^2 - \omega^2}.\end{aligned}$$

(2) $e^{-\lambda t} \sin \omega t$, $e^{-\lambda t} \cos \omega t$;

$$\begin{aligned}\text{解一: } \varphi(t) &= e^{-\lambda t} \sin \omega t = \frac{1}{2i} e^{-\lambda t} (e^{i\omega t} - e^{-i\omega t}) \\ &\doteq \frac{1}{2i} \left[\frac{1}{(p + \lambda) - i\omega} - \frac{1}{(p + \lambda) + i\omega} \right] \\ &= \frac{\omega}{(p + \lambda)^2 + \omega^2};\end{aligned}$$

$$\begin{aligned}\text{解二: } \varphi(t) &= e^{-\lambda t} \cos \omega t = \frac{1}{2} e^{-\lambda t} (e^{i\omega t} + e^{-i\omega t}), \\ &\doteq \frac{1}{2} \left[\frac{1}{(p + \lambda) - i\omega} + \frac{1}{(p + \lambda) + i\omega} \right] \\ &= \frac{p + \lambda}{(p + \lambda)^2 + \omega^2}.\end{aligned}$$

$$(3) \frac{1}{\sqrt{\pi t}}.$$

$$\text{解: } \varphi(t) = \frac{1}{\sqrt{\pi t}},$$

$$\bar{\varphi}(p) = \int_0^{\infty} \frac{1}{\sqrt{\pi t}} e^{-pt} dt,$$

$$\text{若令 } t = x^2, dt = 2x dx,$$

$$\begin{aligned} \text{则 } \bar{\varphi}(p) &= \int_0^{\infty} \frac{1}{\sqrt{\pi}} \frac{1}{x} e^{-px^2} \cdot 2x dx \\ &= \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-px^2} dx \\ &= \frac{2}{\sqrt{\pi}} \int_0^{\infty} \frac{1}{\sqrt{p}} e^{-px^2} d(\sqrt{p} x) \\ &= \frac{2}{\sqrt{\pi p}} \int_0^{\infty} e^{-y^2} dy \\ &= \frac{2}{\sqrt{\pi p}} \cdot \frac{\sqrt{\pi}}{2} = \frac{1}{\sqrt{p}}. \end{aligned}$$

2. 对下列常微分方程施行拉普拉斯变换

$$(1) \frac{d^3 y}{dt^3} + 3 \frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + y = 6e^{-t},$$

$$y(0) = \left. \frac{dy}{dt} \right|_{t=0} = \left. \frac{d^2 y}{dt^2} \right|_{t=0} = 0.$$

$$\text{解: } p^3 \bar{y}(p) + 3p^2 \bar{y}(p) + 3p \bar{y}(p) + \bar{y}(p) = 6 \cdot \frac{1}{p+1},$$

$$(p+1)^3 \bar{y}(p) = \frac{6}{p+1}, \quad (p+1)^4 \bar{y}(p) = 6.$$

$$\text{亦即 } \bar{y}(p) = \frac{6}{(p+1)^4}.$$

$$(2) \frac{d^2 y}{dt^2} + 9y = 30 \operatorname{ch} t, \quad y(0) = 3,$$

$$\left. \frac{dy}{dt} \right|_{t=0} = 0.$$

$$\text{解: } p^2 \bar{y}(p) - 3p + 9\bar{y}(p) = 30 \cdot \frac{p}{p^2 - 1},$$

$$\begin{aligned} (p^2 + 9) \bar{y}(p) &= \frac{30p}{p^2 - 1} + 3p \\ &= \frac{3p(p^2 + 9)}{p^2 - 1}, \end{aligned}$$

$$\bar{y}(p) = \frac{3p}{p^2 - 1}.$$

$$(3) \begin{cases} \frac{dy}{dt} + 2y + 2z = 10e^{2t}, \\ \frac{dz}{dt} - 2y + z = 7e^{2t}; \end{cases} \quad \begin{cases} y(0) = 1, \\ z(0) = 3. \end{cases}$$

$$\text{解: } \begin{cases} p\bar{y}(p) - 1 + 2\bar{y}(p) + 2\bar{z}(p) = 10 \cdot \frac{1}{p-2}, \\ p\bar{z}(p) - 3 - 2\bar{y}(p) + \bar{z}(p) = 7 \cdot \frac{1}{p-2}, \end{cases}$$

$$\begin{cases} (p+2)\bar{y}(p) + 2\bar{z}(p) = \frac{1}{p-2} + 1, \\ (p+1)\bar{z}(p) - 2\bar{y}(p) = \frac{7}{p-2} + 3. \end{cases}$$

$$(4) \frac{d^2 y}{dt^2} - 2 \frac{dy}{dt} + y = t^2 e^t, y(0) = \left. \frac{dy}{dt} \right|_{t=0} = 0.$$

$$\text{解: } t^2 e^t \doteq \frac{d^2}{dp^2} \frac{1}{p-1} = \frac{2}{(p-1)^3}, \text{ 对原方程进行拉普}$$

拉斯变换,

$$\text{得 } p^2 \bar{y}(p) - 2p\bar{y}(p) + \bar{y}(p) = \frac{2}{(p-1)^3},$$

$$(p-1)^2 \bar{y}(p) = \frac{2}{(p-1)^3}, \quad (p-1)^5 \bar{y}(p) = 2.$$

$$\bar{y}(p) = \frac{2}{(p-1)^5}.$$

$$(5) \quad \frac{dy_1}{dt} = -c_1 y_1, \quad \frac{dy_2}{dt} = c_1 y_1 - c_2 y_2,$$

$$\frac{dy_3}{dt} = c_2 y_2 - c_3 y_3, \quad \frac{dy_4}{dt} = c_3 y_3.$$

$$y_1(0) = N_0, \quad y_2(0) = y_3(0) = y_4(0) = 0.$$

解:
$$\begin{cases} p\bar{y}_1(p) - N_0 = -c_1\bar{y}_1(p), \\ p\bar{y}_2(p) = c_1\bar{y}_1(p) - c_2\bar{y}_2(p), \\ p\bar{y}_3(p) = c_2\bar{y}_2(p) - c_3\bar{y}_3(p), \\ p\bar{y}_4(p) = c_3\bar{y}_3(p); \end{cases}$$

即

$$\begin{cases} (p + c_1)\bar{y}_1(p) = N_0, \\ (p + c_2)\bar{y}_2(p) = c_1\bar{y}_1(p), \\ (p + c_3)\bar{y}_3(p) = c_2\bar{y}_2(p), \\ p\bar{y}_4(p) = c_3\bar{y}_3(p). \end{cases}$$

$$(6) \quad \text{厄米方程} \quad \frac{d^2 y}{dt^2} - 2t \frac{dy}{dt} + \lambda y = 0.$$

解:
$$p^2 \bar{y}(p) - p y(0) - y'(0) + 2 \frac{d}{dp}$$

$$\times [p \bar{y}(p) - y(0)] + \lambda \bar{y} = 0.$$

$$p^2 \bar{y}(p) - p y(0) - y'(0) + 2 \bar{y}(p) + 2p \frac{d \bar{y}(p)}{dp}$$

$$+ \lambda \bar{y}(p) = 0,$$

$$2p \frac{d \bar{y}(p)}{dp} + (p^2 + \lambda + 2) \bar{y}(p) = p y(0) + y'(0).$$

$$(7) \quad \text{拉盖尔方程} \quad t \frac{d^2 y}{dt^2} + (1-t) \frac{dy}{dt} + \lambda y = 0.$$

解:
$$-\frac{d}{dp} [p^2 \bar{y}(p) - p y(0) - y'(0)] + p \bar{y}(p) - y(0)$$

$$+ \frac{d}{dp} [p \bar{y}(p) - y(0)] + \lambda \bar{y}(p) = 0,$$

$$\begin{aligned}
& -p^2 \frac{d\bar{y}(p)}{dp} - 2p\bar{y}(p) + y(0) + p\bar{y}(p) - y(0) \\
& + p \frac{d\bar{y}(p)}{dp} + \bar{y}(p) + \lambda \bar{y}(p) = 0, \\
& (p^2 - p) \frac{d\bar{y}(p)}{dp} + (p - \lambda - 1) \bar{y}(p) = 0, \\
& p(p-1) \frac{d\bar{y}(p)}{dp} + (p - \lambda - 1) \bar{y}(p) = 0.
\end{aligned}$$

§22. 拉普拉斯变换的反演

1. 把下列像函数反演:

$$(1) \quad \bar{y}(p) = \frac{6}{(p+1)^4}.$$

解: 由位移定律 $\frac{3!}{(p+1)^{3+1}} \doteq t^3 e^{-t}$.

$$(2) \quad \bar{y}(p) = \frac{3p}{p^2 - 1}.$$

解: $\frac{3p}{p^2 - 1} = \frac{3}{2} \left(\frac{1}{p+1} + \frac{1}{p-1} \right) \doteq \frac{3}{2} (e^{-t} + e^t) = 3 \cosh t.$

$$(3) \quad \bar{y}(p) = \frac{1}{p-2}, \quad \bar{z}(p) = \frac{3}{p-2}.$$

解: $\frac{1}{p-2} \doteq e^{2t} = y^{(t)},$

$$\frac{3}{p-2} \doteq 3e^{2t} = z(t).$$

$$(4) \quad \bar{y}(p) = \frac{2}{(p-1)^5}.$$

解: $\frac{2}{(p-1)^{4+1}} \doteq \frac{2}{4!} t^4 e^{-t}.$

2. 求 $\bar{j}(P) = \frac{E}{LP^2 + RP + \frac{1}{C}}$ 的原函数.

解: $\bar{j}(P) =$

$$\frac{E}{L\left(P + \frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}\right)\left(P + \frac{R}{2L} - \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}\right)}.$$

(1) 如果 $R^2 - \frac{4L}{C} = 0$, 则

$$\bar{j}(P) = \frac{E}{L} \frac{1}{\left(P + \frac{R}{2L}\right)^2} = \frac{E}{L} t e^{-\frac{R}{2L}t} = j(t).$$

(2) 如果 $R^2 - \frac{4L}{C} > 0$, 则

$$\begin{aligned} \bar{j}(P) &= \frac{E}{L} \frac{1}{\left(P + \frac{R}{2L}\right)^2 - \left(\frac{R^2}{4L^2} - \frac{1}{LC}\right)} \\ &= \frac{E}{L\sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}} e^{-\frac{R}{2L}t} \operatorname{sh} \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} t. \end{aligned}$$

(3) 如 $R^2 - \frac{4L}{C} < 0$, 则

$$\begin{aligned} \bar{j}(P) &= \frac{E}{L} \frac{1}{\left(P + \frac{R}{2L}\right)^2 + \left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)} \\ &= \frac{E}{L\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} e^{-\frac{R}{2L}t} \sin \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} t. \end{aligned}$$

3. 求 $N_4(P) = \frac{N_0 C_1 C_2 C_3}{P(P + C_1)(P + C_2)(P + C_3)}$ 的原函数.

$$\begin{aligned}\text{解: 令 } \bar{N}_4(P) &= \frac{N_0 C_1 C_2 C_3}{P(P+C_1)(P+C_2)(P+C_3)} \\ &= \frac{A}{P} + \frac{B}{P+C_1} + \frac{C}{P+C_2} + \frac{D}{P+C_3},\end{aligned}$$

求出: $A = N_0$,

$$D = \frac{C_1 C_2 N_0}{(C_3 - C_1)(C_2 - C_3)},$$

$$C = \frac{C_3 - C_1}{C_1 - C_2} - \frac{C_1 N_0}{C_1 - C_2} = \frac{C_1 C_3 N_0}{(C_1 - C_2)(C_2 - C_3)},$$

$$B = -(C + D + N_0) = \frac{C_2 C_3 N_0}{(C_1 - C_2)(C_3 - C_1)}.$$

$$\begin{aligned}\therefore \bar{N}_4(P) &= \frac{N_0}{P} + \frac{C_2 C_3 N_0}{(C_1 - C_2)(C_3 - C_1)} \cdot \frac{1}{(P+C_1)} \\ &\quad + \frac{C_1 C_3 N_0}{(C_1 - C_2)(C_2 - C_3)} \cdot \frac{1}{(P+C_2)} \\ &\quad + \frac{C_1 C_2 N_0}{(C_2 - C_3)(C_3 - C_1)(P+C_3)}.\end{aligned}$$

进而求得:

$$\begin{aligned}N_4(t) &= N_0 + \frac{C_2 C_3 N_0}{(C_1 - C_2)(C_3 - C_1)} e^{-C_1 t} \\ &\quad + \frac{C_1 C_3 N_0}{(C_1 - C_2)(C_2 - C_3)} e^{-C_2 t} \\ &\quad + \frac{C_1 C_2 N_0}{(C_2 - C_3)(C_3 - C_1)} e^{-C_3 t}.\end{aligned}$$

4. 求 $\bar{y}(P) = \lambda\mu \frac{P}{(P+C)^4}$ 的原函数.

$$\begin{aligned}\text{解: } \bar{y}(P) &= \lambda\mu \left[\frac{P+C}{(P+C)^4} - \frac{C}{(P+C)^4} \right] \\ &= \lambda\mu \left[\frac{1}{(P+C)^3} - \frac{C}{(P+C)^4} \right],\end{aligned}$$

$$y(t) = \lambda \mu \left[-\frac{1}{2!} t^2 e^{-ct} - \frac{C}{3!} t^3 e^{-ct} \right]$$

$$= \frac{1}{2} \lambda \mu e^{-ct} \left[t^2 - \frac{C}{3} t^3 \right].$$

5. 求 $\bar{j}(P) = \frac{E_0 \omega}{\left(P + \frac{1}{RC}\right)(P^2 + \omega^2)}$ 的原函数.

解: 令 $\bar{j}(P) = \frac{E_0 \omega P}{R \left(P + \frac{1}{RC}\right)(P^2 + \omega^2)}$

$$= \frac{AP}{P^2 + \omega^2} + \frac{B}{P^2 + \omega^2} + \frac{D}{P + \frac{1}{RC}},$$

求出: $A = \frac{E_0}{R^2 \omega C + \frac{1}{C\omega}},$

$$B = \frac{E_0}{R} \left(\frac{\omega}{1 + \frac{1}{R^2 C^2 \omega^2}} \right),$$

$$D = -\frac{E_0}{R^2 C \omega + \frac{1}{C\omega}}.$$

$$\therefore \bar{j}(P) = \frac{E_0}{R^2 C \omega + \frac{1}{C\omega}} \cdot \frac{P}{P^2 + \omega^2}$$

$$+ \frac{E_0}{R} \left(\frac{1}{1 + \frac{1}{R^2 C^2 \omega^2}} \right) \frac{\omega}{P^2 + \omega^2}$$

$$- \frac{E_0}{R^2 C \omega + \frac{1}{C\omega}} \cdot \frac{1}{P + \frac{1}{CR}}$$

$$\begin{aligned}
&= \frac{E_0}{R^2 + \frac{1}{C^2\omega^2}} \left[\left(R \frac{\omega}{P^2 + \omega^2} \right) \right. \\
&\quad \left. + \frac{1}{C\omega} \left(\frac{P}{P^2 + \omega^2} \right) \right] - \frac{E_0}{R^2 + \frac{1}{C^2\omega^2}} \\
&\quad \times \frac{1}{C\omega} \cdot \frac{1}{P + \frac{1}{RC}}, \\
j(t) &= \frac{E_0}{R^2 + \frac{1}{C^2\omega^2}} \left[R \sin \omega t + \frac{1}{C\omega} \cos \omega t \right] \\
&\quad - \frac{E_0/C\omega}{R^2 + \frac{1}{C^2\omega^2}} e^{-\frac{t}{RC}}.
\end{aligned}$$

6. 求 $\bar{T}(P) = A \frac{\omega}{P^2 + \omega^2} - \frac{1}{P^2 + \pi^2 a^2/l^2}$ 的原函数.

$$\begin{aligned}
\text{解: 令 } \bar{T}(P) &= A \frac{\omega}{P^2 + \omega^2} - \frac{1}{P^2 + \pi^2 a^2/l^2} \\
&= \frac{E}{P^2 + \omega^2} + \frac{F}{P^2 + \pi^2 a^2/l^2},
\end{aligned}$$

$$\text{求出 } E = \frac{A\omega}{\pi^2 a^2/l^2 - \omega^2}, \quad F = -\frac{\omega A}{\pi^2 a^2/l^2 - \omega^2}.$$

$$\begin{aligned}
\therefore \bar{T}(P) &= \frac{A}{\pi^2 a^2/l^2 - \omega^2} \cdot \frac{\omega}{P^2 + \omega^2} \\
&\quad - \frac{\omega A}{\pi^2 a^2/l^2 - \omega^2} \cdot \frac{1}{P^2 + \pi^2 a^2/l^2},
\end{aligned}$$

$$\begin{aligned}
T(t) &= \frac{A}{\pi^2 a^2/l^2 - \omega^2} \left[\sin \omega t - \omega \frac{l}{\pi a} \sin \frac{\pi a}{l} t \right] \\
&= \frac{lA}{\pi a} \cdot \frac{1}{\omega^2 - \frac{\pi^2 a^2}{l^2}} \left(\omega \sin \frac{\pi a t}{l} \right.
\end{aligned}$$

$$-\frac{\pi a}{l} \sin \omega t \Big).$$

7. 求 $\bar{T}(P) = \frac{1}{P^2 + \omega^2 a^2} \bar{g}(P)$ 的原函数, $\bar{g}(P)$ 是某个已知的 $g(t)$ 的像函数.

$$\text{解: 设 } \bar{f}(P) = \frac{1}{P^2 + \omega^2 a^2},$$

$$\text{则 } f(t) = \frac{1}{\omega a} \sin \omega a t$$

$$= \frac{1}{\omega a} \cdot \frac{1}{2i} (e^{i\omega a t} - e^{-i\omega a t}).$$

根据卷积定理: 因为 $\bar{f}(P) \doteq f(t)$, $\bar{g}(P) \doteq g(t)$.

$$\begin{aligned} \therefore T(t) &\doteq \bar{f}(P) \bar{g}(P) \doteq \int_0^t g(\tau) f(t-\tau) d\tau \\ &= \frac{1}{\omega a} \cdot \frac{1}{2i} \int_0^t g(\tau) [e^{i\omega a(t-\tau)} \\ &\quad - e^{-i\omega a(t-\tau)}] d\tau. \end{aligned}$$

8. 求 $\bar{T}(P) = \frac{1}{P + \omega^2 a^2} \bar{g}(P)$ 的原函数, $\bar{g}(P)$ 是某个已知的 $g(t)$ 的像函数.

$$\text{解: 设 } \bar{f}(P) = \frac{1}{P + \omega^2 a^2}, \text{ 则 } f(t) = e^{-\omega^2 a^2 t}.$$

根据卷积定理, 因为 $\bar{f}(P) \doteq f(t)$, $\bar{g}(P) \doteq g(t)$.

$$\therefore T(t) \doteq \bar{f}(P) \bar{g}(P) \doteq \int_0^t g(\tau) e^{-\omega^2 a^2(t-\tau)} d\tau.$$

$$\begin{aligned} 9. \text{ 已知像函数 } \bar{y}(P) &= e^{-P^2/4} P - \left(\frac{\lambda}{2} + 1\right) \\ &\times \int e^{P^2/4} P^{-\left(\frac{\lambda}{2} + 1\right)} \left(C_1 + \frac{C_2}{P}\right) dP, \end{aligned}$$

其中 C_1 和 C_2 是两个任意常数, 问 λ 应取怎样的数值才有可能选

定 C_1 和 C_2 使原函数 $y(t)$ 为多项式?

$$\text{解: } \bar{y}(P) = e^{-P^2/4} P^{-\left(\frac{\lambda}{2} + 1\right)}$$

$$\times \int e^{P^2/4} P^{\left(\frac{\lambda}{2} + 1\right)} \left(C_1 + \frac{C_2}{P}\right) dP$$

$$= e^{-P^2/4} P^{-\left(\frac{\lambda}{2} + 1\right)}$$

$$\times \left[C_1 \int e^{P^2/4} P^{\left(\frac{\lambda}{2} + 1\right)} dP + C_2 \right.$$

$$\left. \times \int e^{P^2/4} P^{\frac{\lambda}{2}} dP \right]$$

$$= e^{-P^2/4} P^{-\left(\frac{\lambda}{2} + 1\right)}$$

$$\times \left[2C_1 P^{\frac{\lambda}{2}} e^{P^2/4} - 2C_1 \left(\frac{\lambda}{2}\right) \int e^{P^2/4} P^{\left(\frac{\lambda}{2} - 1\right)} dP \right.$$

$$+ 2C_2 P^{\left(\frac{\lambda}{2} - 1\right)} e^{P^2/4} - 2C_2 \left(\frac{\lambda}{2} - 1\right)$$

$$\left. \times \int e^{P^2/4} P^{\left(\frac{\lambda}{2} - 2\right)} dP \right]$$

$$= e^{-P^2/4} P^{-\left(\frac{\lambda}{2} + 1\right)} \left\{ 2C_1 P^{\frac{\lambda}{2}} e^{P^2/4} \right.$$

$$- 2C_1 \left(\frac{\lambda}{2}\right) \left[2e^{P^2/4} P^{\left(\frac{\lambda}{2} - 2\right)} \right.$$

$$- 2\left(\frac{\lambda}{2} - 2\right) \int e^{P^2/4} P^{\left(\frac{\lambda}{2} - 2\right)} dP \left. \right]$$

$$+ 2C_2 P^{\left(\frac{\lambda}{2} - 1\right)} e^{P^2/4}$$

$$- 2C_2 \left(\frac{\lambda}{2} - 1\right) \left[2e^{P^2/4} P^{\left(\frac{\lambda}{2} - 3\right)} \right.$$

$$\left. - 2 \left(\frac{\lambda}{2} - 3 \right) \int e^{P^2/4} P^{\left(\frac{\lambda}{2} - 3 \right)} dP \right\} \\ = \dots\dots.$$

(i) 如 $\frac{\lambda}{2}$ 为偶数, 可选 $C_1 \neq 0$, $C_2 = 0$, 一次又一次的分部积分, 可得 $\bar{y}(P)$ 为 $\frac{1}{P}$ 的多项式, 相应的原函数必亦为多项式.

(ii) 如 $\frac{\lambda}{2}$ 为奇数, 可选 $C_2 \neq 0$, $C_1 = 0$, 亦可得多项式.

(iii) 如 $\frac{\lambda}{2}$ 不是整数, 则不可能得到多项式.

10. 已知 $\bar{y}(P) = \frac{(P-1)^\lambda}{P^{\lambda+1}}$, 问 λ 应取怎样的数值, 原函数才是多项式?

解: 当 λ 为正整数时,

$$\begin{aligned} \bar{y}(P) &= \frac{(P-1)^\lambda}{P^{\lambda+1}} = \frac{1}{P^{\lambda+1}} \left[P^\lambda - \lambda P^{(\lambda-1)} \right. \\ &\quad + \frac{\lambda(\lambda-1)}{2!} P^{(\lambda-2)} - \dots\dots \\ &\quad + (-1)^k \frac{\lambda(\lambda-1)\dots\dots(\lambda-K+1)}{K!} P^{(\lambda-k)} \\ &\quad \left. + \dots\dots + (-1)^\lambda \right] \\ &= \frac{1}{P} - \frac{\lambda}{P^2} + \frac{\lambda(\lambda-1)}{2!} \cdot \frac{1}{P^3} - \dots\dots \\ &\quad + \frac{\lambda(\lambda-1)\dots\dots(\lambda-K+1)}{K!} \frac{(-1)^k}{P^{k+1}} \\ &\quad + \dots\dots + \frac{(-1)^\lambda}{P^{\lambda+1}}. \end{aligned}$$

$\bar{y}(P)$ 为 $\frac{1}{P}$ 的多项式, 相应的原函数亦必为多项式.

11. 已知 $\bar{X}(P) = F_0 \cdot \frac{\omega}{P^2 + \omega^2} \frac{mP^2 + R}{D(P)}$, 其中 $D(P) = (MP^2 + RP + K + k) \cdot (mP^2 + k) - k^2$, 而 $F_0, \omega, m, k, K, M, R$ 都是正的常数. 试论证 $D(P)$ 没有正的根, 也没有纯虚数根, 在什么条件下, 原函数 $X(t)$ 不含有稳定振荡的部分而只含指数式衰减的部分, 或衰减振荡部分.

$$\text{解: (1) } D(P) = (MP^2 + RP + K + k)(mP^2 + k) - k^2 \\ = 0,$$

$$\text{即 } MmP^4 + RmP^3 + (kM + km + Km)P^2 + kRP + kK = 0.$$

(i) 若 P_1 为正数, 则

$$(MP_1^2 + RP_1 + K + k)(mP_1^2 + k) > k^2 \text{ 即 } D(P_1) > 0,$$

所以 $D(P)$ 没有正根, 从而 $X(t)$ 没有指数式增长项, 即 $X(t)$ 不包含 $e^{st} (S > 0)$.

(ii) 设方程 $D(P) = 0$ 有某个纯虚数根 iy , 则

$$\begin{cases} \operatorname{Re} D(iy) = 0, \\ \operatorname{Im} D(iy) = 0; \end{cases}$$

$$\text{即 } \begin{cases} (-My^2 + K)(-my^2 + k) - kmy^2 = 0, & (1) \\ R(-my^2 + k) = 0. & (2) \end{cases}$$

但 (1)、(2) 两式有矛盾, 所以方程 $D(P) = 0$ 没有纯虚数根, 所以 $X(t)$ 不包含 $e^{\pm i\omega t}$ (ω 为实数), 即不包含有 $\cos \omega t$ 和 $\sin \omega t$, 没有稳定振荡部份.

(iii) 设方程 $D(P) = 0$ 有 $x + iy (x > 0)$ 的根,

$$\text{则 } D(x + iy) = (Mx^2 - My^2 + 2iMxy + Rx \\ + iRy + K + k)(mx^2 - my^2 + i2mxy \\ + k) - k^2 \\ = [(Mx^2 - My^2 + Rx + K + k)$$

$$\begin{aligned}
& \times (mx^2 - my^2 + k) - 2mxy^2 \\
& \times (2Mx + R) - k^2] + i[(Mx^2 - My^2 \\
& + Rx + K + k)2mxy \\
& + (mx^2 - my^2 + k)(2Mx + R)y] \\
& = 0.
\end{aligned}$$

$$\text{即} \quad \begin{cases} (Mx^2 - My^2 + Rx + K + k)(mx^2 - my^2 + k) \\ - 2mxy^2(2Mx + R) - k^2 = 0, & (3) \\ (Mx^2 - My^2 + Rx + K + k)2mx + (mx^2 - my^2 \\ + k)(2Mx + R) = 0, & (4) \end{cases}$$

由(4)式

$$Mx^2 - My^2 + Rx + K + k = -\frac{2Mx + R}{2mx}(mx^2 - my^2 + k),$$

以此代入(3)式,

$$-\frac{2Mx + R}{2mx}(mx^2 - my^2 + k) - 2mxy^2(2Mx + R) - k^2 = 0.$$

上式左边三项都是负的, 其和不可能为零, 所以原假设不成立, 方程 $D(P) = 0$ 没有 $x + iy (x > 0)$ 的根.

由上述可见, $X(t)$ 只可能有指数式衰减 $e^{-\sigma t}$ 部分和衰减振荡 $e^{-\sigma t} \cos \omega t, e^{-\sigma t} \sin \omega t$.

(2) 但 $(P^2 + \omega^2)D(P)$ 有纯虚数根 $\pm i\omega$, 所以 $\bar{X}(P)$ 的分项分式有 $(AP + B)/(P^2 + \omega^2)$ 项, 反演后给出 $X(t)$ 的稳定振荡项. 要消除 $X(t)$ 的稳定振荡项, 必须 $\bar{X}(P)$ 的分母里 $P^2 + \omega^2$ 与分子里 $P^2 + k/m$ 互相约去, 即

$$P^2 + \omega^2 = P^2 + \frac{k}{m},$$

亦即在条件

$$\omega^2 = \frac{k}{m}$$

之下, 原函数 $X(t)$ 不包含有稳定振荡部分而只含指数式衰减

的部分或衰减振荡部分.

12. 求下列像函数的原函数.

$$(1) \quad \bar{I}(P) = \frac{\pi}{2a} \cdot \frac{1}{P+a}.$$

$$\text{解: } I(t) = \frac{\pi}{2a} e^{-at}.$$

$$(2) \quad \bar{I}(P) = \frac{\pi}{2P}.$$

$$\text{解: } I(t) = \frac{\pi}{2}.$$

$$(3) \quad \bar{I}(P) = \frac{\pi}{2} \cdot \frac{1}{P(P+1)}.$$

$$\text{解: } \bar{I}(P) = \frac{\pi}{2} \left(\frac{1}{P} - \frac{1}{P+1} \right),$$

$$\text{所以 } I(t) = \frac{\pi}{2} (1 - e^{-t}).$$

$$(4) \quad \bar{I}(P) = \frac{\pi}{2P^2}.$$

$$\text{解: } I(t) = \frac{\pi}{2} t.$$

§23. 运算微积应用例

1. 求解下列常微分方程

$$(1) \quad \frac{d^3 y}{dt^3} + 3 \frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + y = 6e^{-t},$$

$$y(0) = \left. \frac{dy}{dt} \right|_{t=0} = \left. \frac{d^2 y}{dt^2} \right|_{t=0} = 0.$$

解: 对该方程施行拉普拉斯变换(见 § 21 习题2(1)后),

得:
$$\bar{y}(P) = \frac{6}{(P+1)^4},$$

然后再求出 $\bar{y}(P)$ 的原函数 (见 § 22 习题 1(1)) 为

$$y(t) = t^3 e^{-t}, \text{ 此即该常微分方程的解.}$$

(2) $\frac{d^2 y}{dt^2} + 9y = 30 \operatorname{cht}, y(0) = 3, y'(0) = 0.$

解: 对该方程施行拉普拉斯变换后 (见 § 21 习题 2(2)) 得

$$\bar{y}(P) = \frac{3P}{P^2 - 1},$$

然后再求出 $\bar{y}(P)$ 的原函数 (见 § 22 习题 1(2)) 为

$$y(t) = 3 \operatorname{cht}, \text{ 此即该常微分方程的解.}$$

(3)
$$\begin{cases} \frac{dy}{dt} + 2y + 2z = 10e^{2t}, \\ \frac{dz}{dt} - 2y + z = 7e^{2t}, \end{cases} \quad \begin{cases} y(0) = 1, \\ z(0) = 3. \end{cases}$$

解: 对该方程施行拉普拉斯变换后 (见 § 21 习题 2(3)) 得

$$\begin{cases} (P+2)\bar{y}(P) + 2\bar{z}(P) = \frac{10}{P-2} + 1 = \frac{P+8}{P-2}, \\ (P+1)\bar{z}(P) - 2\bar{y}(P) = \frac{7}{P-2} + 3 = \frac{3P+1}{P-2}. \end{cases}$$

$$\bar{y}(P) = \frac{\begin{vmatrix} (P+8)/(P-2) & 2 \\ (3P+1)/(P-2) & P+1 \end{vmatrix}}{\begin{vmatrix} P+2 & 2 \\ -2 & P+1 \end{vmatrix}} = \frac{1}{P-2},$$

$$\bar{z}(P) = \frac{\begin{vmatrix} P+2 & (P+8)/(P-2) \\ -2 & (3P+1)/(P-2) \end{vmatrix}}{\begin{vmatrix} P+2 & 2 \\ -2 & P+1 \end{vmatrix}} = \frac{3}{P-2}.$$

然后再求出 $\bar{y}(P)$ 和 $\bar{z}(P)$ 的原函数 (见 § 22 习题 1(3)) 为

$$y(t) = e^{2t}, \quad z(t) = 3e^{2t} \text{ 此即该常微分方程的解.}$$

$$(4) \frac{d^2 y}{dt^2} - 2 \frac{dy}{dt} + y = t^2 e^t, \quad y(0) = \left. \frac{dy}{dt} \right|_{t=0} = 0.$$

解: 对该方程施行拉普拉斯变换后 (见 § 21 习题 2(4)) 得

$$\bar{y}(p) = \frac{2}{(p-1)^3},$$

然后再求出 $\bar{y}(p)$ 的原函数 (见 § 22 习题 1(4)) 为 $y(t) = \frac{1}{12} t^4 e^t$, 此即该常微分方程的解.

2. 电压为 E_0 的直流电源通过电感 L 和电阻 R 对电容 C 充电.

求解充电电流 j 的变化情况.

解: 设电键 K 关闭前电路中没有电流,

$$\text{即 } j(0) = 0.$$

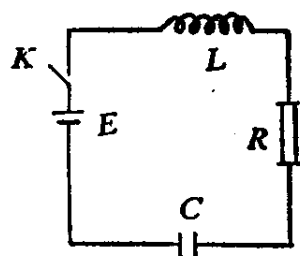


图 5-1

电键 K 关闭后电流 j 所满足的微分方程是

$$L \frac{dj}{dt} + Rj + \frac{1}{C} \int_0^t j dt = E.$$

结合初始条件 $j(0) = 0$ 对上述方程施行拉普拉斯变换后得

$$LP\bar{j}(P) + R\bar{j}(P) + \frac{1}{C} \cdot \frac{1}{P} \bar{j}(P) = \frac{E}{P},$$

$$LP^2\bar{j}(P) + RP\bar{j}(P) + \frac{1}{C} \bar{j}(P) = E,$$

$$\bar{j}(P) = \frac{E}{LP^2 + RP + \frac{1}{C}}.$$

然后再求出 $\bar{j}(P)$ 的原函数 (见 § 22 习题 2) 为

(i) 如 $R^2 - \frac{4L}{C} = 0$,

则 $j(t) = \frac{E}{L} t e^{-\frac{R}{2L}t}$.

(ii) 如 $R^2 - \frac{4L}{C} > 0$,

则 $j(t) = \frac{E}{L\sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}} e^{-\frac{R}{2L}t} \operatorname{sh} \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} t$.

(iii) 如 $R^2 - \frac{4L}{C} < 0$,

则 $j(t) = \frac{E}{L\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} e^{-\frac{R}{2L}t} \sin \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} t$.

3. 放射性元素 E_1 蜕变为 E_2 , 元素 E_1 的原子数 N_1 变化规律为 $\frac{dN_1}{dt} = -C_1 N_1$. 元素 E_2 又蜕变为 E_3 , 元素 E_2 的原子数 N_2 变化规律为 $\frac{dN_2}{dt} = C_1 N_1 - C_2 N_2$, 元素 E_3 又蜕变为 E_4 , 元素 E_3 的原子数 N_3 变化规律 $\frac{dN_3}{dt} = C_2 N_2 - C_3 N_3$, 元素 E_4 是稳定的, 不再蜕变, 它的原子数 N_4 的变化规律为 $\frac{dN_4}{dt} = C_3 N_3$, 以上 C_1, C_2, C_3 和 C_4 都是常数, 设开始时只有元素 E_1 的 N_0 个原子, 求解 N_4 的变化情况 $N_4(t)$.

解: $\frac{dN_1}{dt} = -C_1 N_1, \frac{dN_2}{dt} = C_1 N_1 - C_2 N_2,$

$\frac{dN_3}{dt} = C_2 N_2 - C_3 N_3, \frac{dN_4}{dt} = C_3 N_3,$

$$N_1(0) = N_0, \quad N_2(0) = N_3(0) = N_4(0) = 0,$$

对上述方程施行拉普拉斯变换后 (见 § 21 习题2(5)) 得:

$$(P + C_1)\bar{N}_1(P) = N_0, \quad (P + C_2)\bar{N}_2(P) = C_1\bar{N}_1(P),$$

$$(P + C_3)\bar{N}_3(P) = C_2\bar{N}_2(P), \quad P\bar{N}_4(P) = C_3\bar{N}_3(P), \quad \text{进一步求出:}$$

$$\bar{N}_1(P) = \frac{N_0}{P + C_1}, \quad \bar{N}_2(P) = \frac{C_1 N_0}{(P + C_1)(P + C_2)},$$

$$\bar{N}_3(P) = \frac{C_1 C_2 N_0}{(P + C_1)(P + C_2)(P + C_3)},$$

$$\bar{N}_4(P) = \frac{C_1 C_2 C_3 N_0}{P(P + C_1)(P + C_2)(P + C_3)},$$

然后再求出 $\bar{N}_4(P)$ 的原函数 (见 § 22 习题3) 为:

$$\begin{aligned} N_4(t) = N_0 + & \frac{C_2 C_3 N_0}{(C_1 - C_2)(C_3 - C_1)} e^{-c_1 t} \\ & + \frac{C_1 C_3 N_0}{(C_1 - C_2)(C_2 - C_3)} e^{-c_2 t} \\ & + \frac{C_1 C_2 N_0}{(C_2 - C_3)(C_3 - C_1)} e^{-c_3 t}, \end{aligned}$$

4. 设地面有一震动, 其速度 $v = H(t)$, 地震仪中的感生电流 j 遵守规律 $\frac{dj}{dt} + 2cj + c^2 \int_0^t j dt = \lambda \frac{dv}{dt}$, 这电流通过检

流计, 使检流计发生偏转。偏转 y 遵守规律 $\frac{d^2 y}{dt^2} + 2c \frac{dy}{dt}$

$+ c^2 y = \mu j$, 求解偏转 y 的变化情况 $y(t)$ 。

解:

$$\begin{cases} \frac{dj}{dt} + 2Cj + C^2 \int_0^t j dt = \lambda \frac{dH(t)}{dt}, \\ \frac{d^2 y}{dt^2} + 2c \frac{dy}{dt} + c^2 y = \mu j, \end{cases}$$

$$\begin{cases} j(0) = 0, \\ y(0) = \frac{dy}{dt} \Big|_{t=0} = 0. \end{cases}$$

由于 $H(t) \doteq \frac{1}{P}$ 所以 $\frac{dH}{dt} \doteq P \frac{1}{P} = 1$.

再对方程组施行拉普拉斯变换后得:

$$\begin{cases} \left(P + 2C + \frac{C^2}{P} \right) \bar{j} = \lambda, & \bar{j}(P) = \frac{\lambda P}{P^2 + 2CP + C^2}, \\ (P^2 + 2CP + C^2) \bar{y}(P) = \mu \bar{j}(P), \end{cases}$$

$$\bar{y}(P) = \frac{\mu \bar{j}(P)}{P^2 + 2CP + C^2} = \frac{\mu \lambda P}{(P^2 + 2CP + C^2)^2} = \frac{\lambda \mu P}{(P + C)^4}.$$

然后再求出 $\bar{y}(P)$ 的原函数 (见 § 22 习题4) 为:

$$y(t) = \frac{1}{2} \lambda \mu e^{-ct} \left(t^2 - \frac{C}{3} t^3 \right).$$

5. 求解交流RC电路的方程

$$\begin{cases} Rj + \frac{1}{C} \int_0^t j dt = E_0 \sin \omega t, \\ j(0) = 0. \end{cases}$$

解: 对上述方程施行拉普拉斯变换后得:

$$R\bar{j}(P) + \frac{1}{CP} \bar{j}(P) = E_0 \frac{\omega}{P^2 + \omega^2},$$

$$\bar{j}(P) = \frac{E_0 \omega P}{(P^2 + \omega^2) \left(RP + \frac{1}{C} \right)},$$

然后再求出 $\bar{j}(P)$ 的原函数 (见 § 22 习题5) 为:

$$\begin{aligned} j(t) &= \frac{E_0}{R^2 + \frac{1}{C^2 \omega^2}} \left[R \sin \omega t + \frac{1}{C \omega} \cos \omega t \right] \\ &\times \frac{E_0 / C \omega}{R^2 + \frac{1}{C^2 \omega^2}} e^{-\frac{t}{RC}}. \end{aligned}$$

6. 求解 $T'' + \frac{\pi^2 a^2}{l^2} T = A \sin \omega t$, $T(0) = 0$, $T'(0) = 0$.

解: 对该方程施行拉普拉斯变换后得:

$$P^2 \bar{T}(P) + \frac{\pi^2 a^2}{l^2} \bar{T}(P) = A \frac{\omega}{P^2 + \omega^2},$$

$$\bar{T}(P) = A \frac{\omega}{p^2 + \omega^2} \cdot \frac{1}{p^2 + \frac{\pi^2 a^2}{l^2}},$$

然后再求出 $\bar{T}(P)$ 的原函数 (见 § 22 习题6) 为

$$T(t) = \frac{lA}{\pi a} \cdot \frac{1}{\omega^2 - \frac{\pi^2 a^2}{l^2}} \left(\omega \sin \frac{\pi a t}{l} - \frac{\pi a}{l} \sin \omega t \right).$$

7. 求解 $T'' + \omega^2 a^2 T = g(t)$, $T(0) = 0$, $T'(0) = 0$, $g(t)$ 是某个已知函数.

解: 对该方程施行拉普拉斯变换后得:

$$P^2 \bar{T}(P) + \omega^2 a^2 \bar{T}(P) = \bar{g}(p),$$

$$\bar{T}(P) = \frac{1}{p^2 + \omega^2 a^2} \bar{g}(p),$$

然后再求出 $\bar{T}(P)$ 的原函数 (见 § 22 习题7) 为:

$$T(t) = \frac{1}{\omega a} \cdot \frac{1}{2i} \int_0^t g(\tau) [e^{i\omega a(t-\tau)} - e^{-i\omega a(t-\tau)}] d\tau.$$

8. 求解 $T' + \omega^2 a^2 T = g(t)$, $T(0) = 0$, $g(t)$ 是某个已知函数.

解: 对该方程施行拉普拉斯变换后得:

$$P \bar{T}(P) + \omega^2 a^2 \bar{T}(P) = \bar{g}(P),$$

$$\bar{T}(P) = \frac{1}{p + \omega^2 a^2} \bar{g}(P),$$

然后再求出 $\bar{T}(P)$ 的原函数 (见 § 22 习题8) 为:

$$T(t) = \int_0^t g(\tau) e^{-\omega^2 a^2(t-\tau)} d\tau.$$

9. 厄米方程 $\frac{d^2 y}{dt^2} - 2t \frac{dy}{dt} + \lambda y = 0$ 里的 λ 值 应取怎样的数值才有可能使方程的解为多项式?

解: 对厄米方程施行拉普拉斯变换后 (见 § 21 习题2(6)) 得:

$$\begin{aligned} 2P \frac{d\bar{y}(P)}{dP} + (P^2 + 2 + \lambda) \bar{y}(P) &= P y(0) + y'(0), \\ \frac{d\bar{y}}{dP} + \frac{P^2 + 2 + \lambda}{2P} \bar{y}(P) &= \frac{1}{2} y(0) + \frac{1}{2P} y'(0), \\ \bar{y}(P) &= e^{-\int \frac{P^2 + 2 + \lambda}{2P} dP'} \left\{ \int \left[\frac{1}{2} y(0) + \frac{1}{2P} y'(0) \right] e^{\int \frac{P^2 + 2 + \lambda}{2P} dP} dP \right\} \\ &= e^{-P^2/4} \cdot e^{-(\frac{\lambda}{2} + 1) \ln P} \left\{ \int \left[\frac{1}{2} y(0) + \frac{1}{2P} y'(0) \right] e^{-P^2/4} \cdot e^{-(\frac{\lambda}{2} + 1) \ln P} dP \right\} \\ &= e^{-P^2/4} P^{-(\frac{\lambda}{2} + 1)} \int e^{-P^2/4} P^{-(\frac{\lambda}{2} - 1)} \\ &\quad \times \left[\frac{1}{2} y(0) + \frac{1}{2P} y'(0) \right] dP, \end{aligned}$$

记 $\frac{y(0)}{2} = C_1, \frac{y'(0)}{2} = C_2,$

$$\begin{aligned} \text{则 } \bar{y}(P) &= e^{-P^2/4} P^{-(\frac{\lambda}{2} + 1)} \int e^{-P^2/4} P^{-(\frac{\lambda}{2} + 1)} \\ &\quad \times \left(C_1 + \frac{C_2}{P} \right) dP. \end{aligned}$$

以下的讨论见 § 22 习题9.

10. 拉盖尔方程 $t \frac{d^2 y}{dt^2} + (1-t) \frac{dy}{dt} + \lambda y = 0$ 的 λ 应取怎样的数值才有可能使方程的解为多项式?

解: 对拉盖尔方程进行拉普拉斯变换后 (见 § 21 习题 2(7)) 得

$$P(P-1) \frac{d\bar{y}(P)}{dP} + (P-\lambda-1)\bar{y}(P) = 0,$$

$$\frac{d\bar{y}(P)}{dP} + \frac{P-\lambda-1}{P(P-1)} \bar{y}(P) = 0,$$

$$\frac{d\bar{y}(P)}{\bar{y}(P)} = -\frac{P-\lambda-1}{P(P-1)} dP,$$

$$\begin{aligned} \ln \bar{y}(P) &= \int \frac{(P-\lambda-1)dP}{P(P-1)} \\ &= \ln(P-1)^{\lambda+1} - \ln P^{\lambda+1} + \ln C, \end{aligned}$$

$$\bar{y}(P) = C \frac{(P-1)^{\lambda+1}}{P^{\lambda+1}}.$$

以下的讨论见 § 22 习题 10.

11. 有一种船舶减震器利用的是耦合振动原理. 在水面上颠簸的船体不妨看作是一个阻尼振子, 其质量为 M , 倔强系数为 K , 阻尼系数为 R . 减震器则是附着在船体上的振子, 其质量为 m , 倔强系数为 k , 因此, 船体的位移 $X(t)$ 和减震器的位移 $x(t)$ 的运动方程是:

$$\begin{cases} M\ddot{X} = F_0 \sin \omega t - KX - R\dot{X} - k(X-x), \\ m\ddot{x} = -k(x-X). \end{cases}$$

其中 $F_0 \sin \omega t$ 是使船体颠簸的外力. 在什么条件下, 船体的运动不含有稳定振荡而只含有指数式衰减或衰减振荡?

解: 先对方程 $m\ddot{x} = -k(x-X)$ 施行拉普拉斯变换后得:

$$m[P^2 \bar{x}(P) - Px(0) - \dot{x}(0)] = -k[\bar{x}(P) - \bar{X}(P)],$$

$$\bar{X}(P) = \frac{m p x(0) + m \dot{X}(0) + k \bar{X}(P)}{m p^2 + k} \quad (1)$$

再对另一个运动方程施行拉普拉斯变换后得:

$$\begin{aligned} & M[P^2 \bar{X}(P) - P X(0) - \dot{X}(0)] \\ &= F_0 \frac{\omega}{P^2 + \omega} \\ &\quad - K \bar{X}(P) - R[P \bar{X}(P) - X(0)] \\ &\quad - k[\bar{X}(P) - \bar{X}(P)], \\ & (M P^2 + R P + K + k) \bar{X}(P) \\ &= F_0 \frac{\omega}{P^2 + \omega^2} \\ &\quad + M P X(0) + M \dot{X}(0) - R X(0) + k \bar{X}(P), \end{aligned}$$

将 (1) 式代入上式并整理即得:

若 $t = 0$ 时, $X(0) = \dot{X}(0) = X(0) = \dot{X}(0) = 0$, 就有

$$\begin{aligned} \bar{X}(P) &= F_0 \frac{\omega}{P^2 + \omega^2} \frac{m P^2 + k}{(M P^2 + R P + K + k)(m P^2 + k) - k^2} \\ &= F_0 - \frac{\omega}{P^2 + \omega^2} \cdot \frac{m P^2 + k}{D(P)}. \end{aligned}$$

以下的讨论见 § 22 习题 11.

12. 用运算微积方法求出下列积分

$$(1) I(t) = \int_0^\infty \frac{\cos t x}{x^2 + a^2} dx.$$

解: 先进行拉普拉斯变换, 再调换积分秩序,

$$\begin{aligned} \bar{I}(P) &= \int_0^\infty \frac{P dx}{(x^2 + a^2)(x^2 + p^2)} \\ &= P \int_0^\infty \frac{[(x^2 + a^2) - (x^2 + P^2)] dx}{(a^2 - P^2)(x^2 + a^2)(x^2 + P^2)} \\ &= \frac{P}{a^2 - P^2} \int_0^\infty \frac{1/P^2}{x^2/P^2 + 1} - \frac{1/a^2}{x^2/a^2 + 1} dx \\ &= \frac{P}{a^2 - P^2} \left[\frac{1}{P} \arctg \frac{x}{P} - \frac{1}{a} \arctg \frac{x}{a} \right] \Big|_0^\infty \end{aligned}$$

$$= \frac{\pi}{2} \frac{P}{a^2 - P^2} \frac{a - P}{aP} = \frac{\pi}{2a} \frac{1}{a + P},$$

然后求出 $\bar{I}(P)$ 的原函数, 见 § 22 习题 12(1),

$$\therefore I(t) = \frac{\pi}{2a} e^{-at}.$$

$$(2) \quad I(t) = \int_0^\infty \frac{\sin tx}{x} dx.$$

$$\text{解: } \bar{I}(P) = \int_0^\infty \frac{x}{x^2 + P^2} dx = \int_0^\infty \frac{dx}{x^2 + P^2} = \frac{\pi}{2P},$$

然后求出 $\bar{I}(P)$ 的原函数, 见 § 22 习题 12(2), 所以,

$$I(t) = \frac{\pi}{2}.$$

在施以拉普拉斯变换时, 要求 $\sin tx$ 中的 $t > 0$, 从而得 $I = \frac{\pi}{2}$. 如果 $t < 0$, 则

$$\begin{aligned} I(t) &= \int_0^\infty \frac{\sin tx}{x} dx \\ &= - \int_0^\infty \frac{\sin t'x}{x} dx \quad (t' = -t). \end{aligned}$$

再对上式施行拉普拉斯变换得

$$I(P) = -\frac{\pi}{2P}.$$

故

$$I(t) = -\frac{\pi}{2}.$$

于是,

$$I(t) = \int_0^\infty \frac{\sin tx}{x} dx = \begin{cases} \pi/2, & (t > 0), \\ 0, & (t = 0), \\ -\pi/2, & (t < 0). \end{cases}$$

$$(3) \quad I(t) = \int_0^{\infty} \frac{\sin tx}{x(x^2+1)} dx.$$

$$\text{解: } \bar{I}(P) = \int_0^{\infty} \frac{dx}{(x^2+1)(x^2+P^2)} = \frac{\pi}{2P(P+1)},$$

然后求出 $\bar{I}(P)$ 的原函数 (见 § 22 习题12(3))

$$I(t) = \frac{\pi}{2}(1 - e^{-t}).$$

$$\text{当 } t < 0 \text{ 时, } I(t) = \frac{\pi}{2}(e^{-t} - 1).$$

$$\begin{aligned} (4) \quad I(t) &= \int_0^{\infty} \frac{\sin^2 tx}{2x^2} dx \\ &= \int_0^{\infty} \frac{1 - \cos 2xt}{2x^2} dx. \end{aligned}$$

$$\begin{aligned} \text{解: } \bar{I}(P) &= \int_0^{\infty} \frac{1}{P} - \frac{P}{P^2 + (2x)^2} dx \\ &= \int_0^{\infty} \frac{2^2 x^2 dx}{2x^2 P [P^2 + (2x)^2]} \\ &= \frac{1}{P^2} \int_0^{\infty} \frac{d(2x/P)}{\left[1 + \left(\frac{2x}{P}\right)^2\right]} = \frac{\pi}{2P^2}, \end{aligned}$$

然后求出 $\bar{I}(P)$ 的原函数 (见 § 22 习题12(4))

$$I(t) = \frac{\pi}{2}t;$$

$$\text{当 } t < 0 \text{ 时, } I(t) = \int_0^{\infty} \frac{\sin^2 tx}{x} dx = \int_0^{\infty} \frac{\sin^2 |t|x}{x} dx = \frac{\pi}{2}|t|.$$

由上述可知 $I(t) = \frac{\pi}{2}|t|$ (t 为任意实数).

第二篇 傅里叶级数和积分

第六章 傅里叶级数

§24. 周期函数的傅里叶级数

1. 图6-1是硅可控整流电压 $E(t)$ 的图象, 试把它展开为傅里叶级数, 在 $[-\pi/\omega, \pi/\omega]$ 这个周期上, $E(t)$ 可表为

$$E(t) = \begin{cases} 0 & \text{在} [-\pi/\omega, \alpha\pi/\omega] \text{上,} \\ E_0 \sin \omega t & \text{在} [\alpha\pi/\omega, \pi/\omega] \text{上,} \end{cases}$$

其中 α 是触发电路控制的某个参数, 注意直流成分的大小跟 α 有关, 这就是硅可控整流的调压原理。

解: 对任意周期 $2l$ 的傅里叶级数和傅里叶系数表达式为:

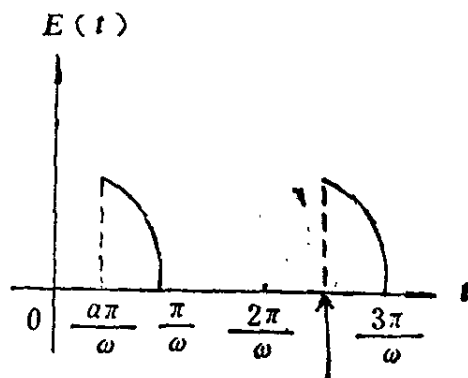


图 6-1 $\frac{(2 \times \alpha)\pi}{\omega}$

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{l} t + b_n \sin \frac{n\pi}{l} t \right),$$

$$a_0 = \frac{1}{2l} \int_{-l}^l f(t) dt,$$

$$a_n = \frac{1}{l} \int_{-l}^l f(t) \cos \frac{n\pi}{l} t dt,$$

$$b_n = \frac{1}{l} \int_{-l}^l f(t) \sin \frac{n\pi}{l} t dt,$$

本题整流电压 $E(t)$ 之周期为 $\frac{2\pi}{\omega}$,

$$\text{令 } 2l = \frac{2\pi}{\omega}, \text{ 得 } \frac{\pi}{l} = \omega,$$

将 l 代入上列公式即可得适合本题傅里叶级数及其系数表达式

$$E(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t),$$

先计算傅里叶系数 a_0

$$\begin{aligned} a_0 &= \frac{\omega}{2\pi} \int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} E(t) dt \\ &= \frac{\omega}{2\pi} \left[\int_{-\frac{\pi}{\omega}}^{\frac{\alpha\pi}{\omega}} 0 dt + \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} E_0 \sin \omega t dt \right] \\ &= \frac{\omega}{2\pi} \cdot \frac{1}{\omega} \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} E_0 \sin \omega t d\omega t \\ &= \frac{E_0}{2\pi} (-\cos \omega t) \Big|_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} \\ &= \frac{E_0}{2\pi} (1 + \cos \alpha\pi), \end{aligned}$$

再计算系数 a_n

$$\begin{aligned} a_n &= \frac{\omega}{\pi} \int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} E(t) \cos n\omega t dt \\ &= \frac{\omega}{\pi} \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} E_0 \sin \omega t \cos n\omega t dt \end{aligned}$$

$$= \frac{\omega E_0}{2\pi} \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} [\sin(1+n)\omega t + \sin(1-n)\omega t] dt.$$

这里要区分两种情况:

(1) $n = 1$ 时

$$\begin{aligned} a_1 &= \frac{\omega E_0}{2\pi} \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} \sin 2\omega t dt \\ &= \frac{E_0}{4\pi} \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} \sin 2\omega t d(2\omega t) \\ &= \frac{E_0}{4\pi} (-\cos 2\omega t) \Big|_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} = \frac{E_0}{4\pi} (\cos 2\alpha\pi - 1), \end{aligned}$$

(2) $n \neq 1$ 时

$$\begin{aligned} a_n &= \frac{\omega E_0}{2\pi} \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} [\sin(1+n)\omega t + \sin(1-n)\omega t] dt \\ &= -\frac{\omega E_0}{2\pi} \left[\frac{\cos(1+n)\omega t}{(1+n)\omega} + \frac{\cos(1-n)\omega t}{(1-n)\omega} \right]_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} \\ &= -\frac{E_0}{2\pi} \left[\frac{\cos(1+n)\omega t - n\cos(1+n)\omega t + \cos(1-n)\omega t + n\cos(1-n)\omega t}{(1+n)(1-n)} \right]_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} \\ &= -\frac{E_0}{2\pi} \left[\frac{2\cos\omega t \cos n\omega t + 2n\sin\omega t \sin n\omega t}{1-n^2} \right]_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} \\ &= \frac{E_0}{\pi} \left[\frac{\cos\alpha\pi \cos n\alpha\pi + n\sin\alpha\pi \sin n\alpha\pi}{1-n^2} \right. \\ &\quad \left. - \frac{\cos\pi \cos n\pi + n\sin\pi \sin n\pi}{1-n^2} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{E_0}{\pi} \left[\frac{\cos \alpha \pi \cos n \alpha \pi + n \sin \alpha \pi \sin n \alpha \pi}{1 - n^2} + \frac{\cos n \pi}{1 - n^2} \right] \\
&= \frac{E_0}{\pi} \left[\frac{\cos \alpha \pi \cos n \alpha \pi + n \sin \alpha \pi \sin n \alpha \pi}{1 - n^2} + \frac{(-1)^n}{1 - n^2} \right],
\end{aligned}$$

用类似的方法可得系数 b_n

$$\begin{aligned}
b_n &= \frac{\omega}{\pi} \int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} E(t) \sin n \omega t dt \\
&= \frac{\omega}{\pi} \int_{-\frac{\pi}{\omega}}^{\frac{\alpha \pi}{\omega}} 0 dt + \frac{\omega}{\pi} \int_{\frac{\alpha \pi}{\omega}}^{\frac{\pi}{\omega}} E_0 \sin \omega t \sin n \omega t dt \\
&= \frac{\omega E_0}{2\pi} \int_{\frac{\alpha \pi}{\omega}}^{\frac{\pi}{\omega}} [\cos(n-1)\omega t - \cos(n+1)\omega t] dt,
\end{aligned}$$

这里也要区分两种情况:

(1) $n = 1$ 时,

$$\begin{aligned}
b_1 &= \frac{\omega E_0}{2\pi} \int_{\frac{\alpha \pi}{\omega}}^{\frac{\pi}{\omega}} [\cos(n-1)\omega t - \cos(n+1)\omega t] dt \\
&= \frac{\omega E_0}{2\pi} \int_{\frac{\alpha \pi}{\omega}}^{\frac{\pi}{\omega}} [1 - \cos 2\omega t] dt \\
&= \frac{E_0}{4\pi} \int_{\frac{\alpha \pi}{\omega}}^{\frac{\pi}{\omega}} [1 - \cos 2\omega t] d 2\omega t \\
&= \frac{E_0}{4\pi} \left[2\omega t - \sin 2\omega t \right]_{\frac{\alpha \pi}{\omega}}^{\frac{\pi}{\omega}} \\
&= \frac{E_0}{4} \left[2(1 - \alpha) + \frac{1}{\pi} \sin 2\alpha \pi \right],
\end{aligned}$$

(2) $n \neq 1$ 时,

$$\begin{aligned}
b_n &= \frac{\omega E_0}{2\pi} \int_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} [\cos(n-1)\omega t - \cos(n+1)\omega t] dt \\
&= \frac{E_0}{2\pi} \left[\frac{\sin(n-1)\omega t}{n-1} - \frac{\sin(n+1)\omega t}{n+1} \right]_{\frac{\alpha\pi}{\omega}}^{\frac{\pi}{\omega}} \\
&= \frac{E_0}{2\pi} \left[\frac{\sin(n+1)\alpha\pi}{n+1} - \frac{\sin(n-1)\alpha\pi}{n-1} \right] \\
&= \frac{E_0}{\pi(1-n^2)} [\cos\alpha\pi \sin n\alpha\pi - n \sin\alpha\pi \cos n\alpha\pi],
\end{aligned}$$

$$\begin{aligned}
\therefore E(t) &= \frac{1}{2\pi} E_0 (1 + \cos\alpha\pi) \\
&\quad + \frac{1}{4\pi} E_0 (\cos 2\alpha\pi - 1) \cos\omega t \\
&\quad + \frac{E_0}{\pi} \sum_{n=2}^{\infty} \frac{1}{1-n^2} [\cos\alpha\pi \cos n\alpha\pi \\
&\quad + n \sin\alpha\pi \sin n\alpha\pi + (-1)^n] \cos n\omega t \\
&\quad + \frac{1}{4} E_0 \left[2(1-\alpha) + \frac{1}{\pi} \sin 2\alpha\pi \right] \sin\omega t \\
&\quad + \frac{E_0}{\pi} \sum_{n=2}^{\infty} \frac{1}{1-n^2} [\cos 2\pi \sin n\alpha\pi \\
&\quad - n \sin\alpha\pi \cos n\alpha\pi] \sin n\omega t.
\end{aligned}$$

计算时，经常用到下列公式，

$$\cos K\pi = (-1)^K, \quad \sin\left(K + \frac{1}{2}\right)\pi = (-1)^K$$

$$\sin\left(K - \frac{1}{2}\right)\pi = (-1)^{K+1}, \quad \cos(K + a)\pi = (-1)^K \cos a\pi$$

$$\sin(K + a)\pi = (-1)^K \sin a\pi, \quad (K \text{ 为整数, } a \text{ 为实数}).$$

2. 试把图6-2的锯齿波展开为傅里叶级数，在 $(0, T)$ 上，这个锯齿波可表为 $f(x) = x/3$ 。

解：锯齿波之周期为 T 。

令

$$2l = T,$$

得

$$l = \frac{T}{2},$$

将 l 代入以 $2l$ 为周期之傅里叶级数和傅里叶系数表达式

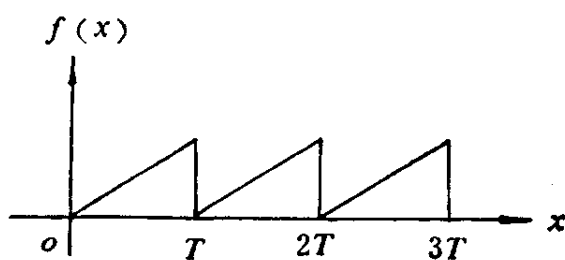


图 6-2

式即可得适合本题傅里叶级数和傅里叶系数表达式：

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2n\pi}{T}t + b_n \sin \frac{2n\pi}{T}t \right).$$

傅里叶系数的计算如下：

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T f(t) dt = \frac{1}{T} \int_0^T \frac{1}{3}x \cdot dx \\ &= \frac{1}{3T} \cdot \frac{1}{2}x^2 \Big|_0^T = \frac{T}{6}, \end{aligned}$$

$$\begin{aligned} a_n &= \frac{2}{T} \int_0^T f(t) \cos \frac{2n\pi}{T}t dt \\ &= \frac{2}{T} \int_0^T \frac{1}{3}x \cos \frac{2n\pi}{T}x dx, \end{aligned}$$

应用积分公式：

$$\int x \cos Px dx = \frac{1}{P^2} \cos Px + \frac{x}{P} \sin Px$$

$$\begin{aligned} \therefore a_n &= \frac{2}{T} \cdot \frac{1}{3} \left[\frac{1}{\left(\frac{2n\pi}{T}\right)^2} \cos \frac{2n\pi}{T}x \right. \\ &\quad \left. + \frac{x}{\frac{2n\pi}{T}} \sin \frac{2n\pi}{T}x \right]_0^T \\ &= \frac{2}{3T} \left(\frac{T}{2n\pi}\right)^2 \left[\cos \frac{2n\pi}{T}x + \frac{2n\pi}{T}x \sin \frac{2n\pi}{T}x \right]_0^T \\ &= 0, \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T f(t) \sin \frac{2n\pi}{T} t dt = \frac{2}{T} \int_0^T \frac{1}{3} x \sin \frac{2n\pi}{T} x dx \\
 &= \frac{2}{T} \cdot \frac{1}{3} \left[\frac{1}{\left(\frac{2n\pi}{T}\right)^2} \sin \frac{2n\pi}{T} x \right. \\
 &\quad \left. - \frac{x}{\frac{2n\pi}{T}} \cos \frac{2n\pi}{T} x \right]_0^T \\
 &= \frac{2}{3T} \left(\frac{T}{2n\pi}\right)^2 \left[\sin \frac{2n\pi}{T} x - \frac{2n\pi}{T} x \cos \frac{2n\pi}{T} x \right]_0^T \\
 &= -\frac{T}{3n\pi},
 \end{aligned}$$

$$\therefore f(x) = \frac{T}{6} - \sum_{n=1}^{\infty} \frac{T}{3n\pi} \sin \frac{2n\pi}{T} x,$$

3. 交流电压 $E_0 \sin \omega t$, 经过全波整流, 成为 $E(t) = E_0 |\sin \omega t|$. 试把它展开为傅里叶级数, 并跟半波整流电压 (课本例) 比较.

解: 交流电压 $E_0 \sin \omega t$ 在区间 $-\pi \leq \omega t \leq \pi$ 上是一周期, 令 $\omega t = x$, 则经过整流后成为:

$E(x) = E(\omega t) = E_0 |\sin x|$,
在周期 $(-\pi, \pi)$ 内均为正值.

其傅里叶级数表为:

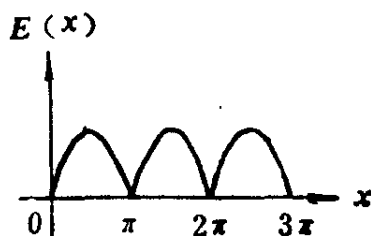


图 6-3

$$E(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx),$$

其中系数

$$\begin{aligned}
 a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{E_0}{\pi} \int_0^{\pi} \sin x dx \\
 &= \frac{E_0}{\pi} (-\cos x) \Big|_0^{\pi} = \frac{2E_0}{\pi}
 \end{aligned}$$

$$\begin{aligned}
a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx dx \\
&= \frac{1}{\pi} \int_{-\pi}^0 E(-\sin x) \cos kx dx + \frac{1}{\pi} \int_0^{\pi} E_0 \sin x \cos kx dx \\
&= \frac{2}{\pi} \int_0^{\pi} E_0 \sin x \cos kx dx \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{E_0}{2} [\sin(kx+x) - \sin(kx-x)] dx \\
&= -\frac{E_0}{\pi} \left[\frac{\cos(k+1)x}{k+1} - \frac{\cos(k-1)x}{k-1} \right]_0^{\pi} \\
&= \begin{cases} 0, & (\text{当 } k \text{ 为奇数时, 但 } k \neq 1) \\ \frac{4E_0}{\pi(1-k^2)}, & (\text{当 } k \text{ 为偶数时}). \end{cases}
\end{aligned}$$

当 $k=1$ 时,

$$a_1 = \frac{2}{\pi} \int_0^{\pi} E_0 \sin x \cos x dx = \frac{E_0}{\pi} \int_0^{\pi} \sin 2x dx = 0,$$

又令 $k=2n$ 时则,

$$a_k = a_{2n} = \frac{4E_0}{\pi(1-4n^2)}, \quad n=1, 2, 3, \dots$$

同理, 可以计算得 b_k

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx dx = \frac{2}{\pi} \int_0^{\pi} E_0 \sin x \sin kx dx = 0,$$

$$\begin{aligned}
\therefore E(t) &= a_0 + \sum_{n=1}^{\infty} a_n \cos 2nx = a_0 + \sum_{n=1}^{\infty} a_n \cos 2n\omega t \\
&= \frac{2E_0}{\pi} + \frac{4E_0}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2n\omega t}{1-4n^2},
\end{aligned}$$

将半波整流和全波整流相比较:

$$E_{\#} = \frac{E_0}{\pi} + \frac{1}{2} E_0 \sin \omega t + \frac{2E_0}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2n\omega t}{1-4n^2}.$$

直流成分：全波整流是 $\frac{2E_0}{\pi}$ ，半波整流是 $\frac{E_0}{\pi}$ 。

基波成分：全波整流中没有和原来频率相同的交流成分，但半波整流中有基波成分，它的数值为 $\frac{E_0}{2}\sin\omega t$ 。

高次谐波：全波整流中，高次谐波部分是半波整流的一倍而高次谐波均为偶次的。

4. 把下列周期函数 $f(x)$ 展开为傅里叶级数。

(1) 在 $(-l, +l)$ 这个周期上， $f(x) = e^{\lambda x}$ 。

解：这是一个周期为 $2l$ 的函数，故可展开为傅里叶级数

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

傅里叶系数计算如下：

$$\begin{aligned} a_0 &= \frac{1}{2l} \int_{-l}^l f(x) dx \\ &= \frac{1}{2l} \int_{-l}^l e^{\lambda x} dx \\ &= \frac{1}{\lambda l} \operatorname{sh} \lambda l \end{aligned}$$

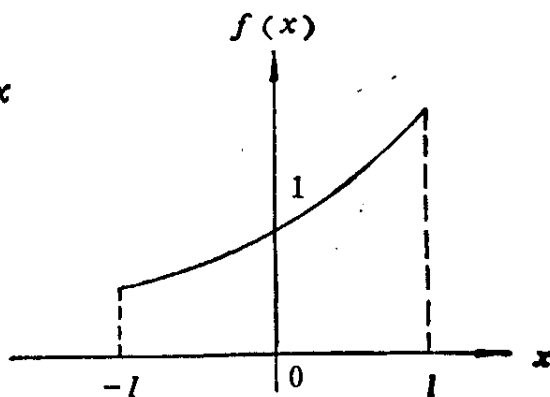


图 6-4

应用已知积分公式

$$\int e^{\lambda x} \cos Px dx = \frac{e^{\lambda x} (\lambda \cos Px + P \sin Px)}{\lambda^2 + P^2}$$

可求得

$$\begin{aligned} a_n &= \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{1}{l} \int_{-l}^l e^{\lambda x} \cos \frac{n\pi x}{l} dx \\ &= \frac{1}{l} \left. \frac{e^{\lambda x} \left(\lambda \cos \frac{n\pi x}{l} + \frac{n\pi}{l} \sin \frac{n\pi x}{l} \right)}{\lambda^2 + \frac{n^2 \pi^2}{l^2}} \right|_{-l}^l \end{aligned}$$

$$\begin{aligned}
&= \frac{l}{\lambda^2 l^2 + n^2 \pi^2} \left[e^{\lambda l} \left(\lambda \cos n\pi + \frac{n\pi}{l} \sin n\pi \right) \right. \\
&\quad \left. - e^{-\lambda l} \left(\lambda \cos(-n\pi) + \frac{n\pi}{l} \sin(-n\pi) \right) \right] \\
&= \frac{\lambda l}{\lambda^2 l^2 + n^2 \pi^2} \cos n\pi (e^{\lambda l} - e^{-\lambda l}) \\
&= (-1)^n \frac{2\lambda l}{\lambda^2 l^2 + n^2 \pi^2} \operatorname{sh} \lambda l,
\end{aligned}$$

再应用积分关系式

$$\int e^{\lambda x} \sin Px dx = \frac{e^{\lambda x} (\lambda \sin Px - P \cos Px)}{\lambda^2 + P^2}$$

可求得:

$$\begin{aligned}
b_n &= \frac{1}{l} \int_{-l}^l e^{\lambda x} \sin \frac{n\pi x}{l} dx \\
&= \frac{1}{l} \left. \frac{e^{\lambda x} \left(\lambda \sin \frac{n\pi x}{l} - \frac{n\pi}{l} \cos \frac{n\pi x}{l} \right)}{\lambda^2 + \frac{n^2 \pi^2}{l^2}} \right|_{-l}^l \\
&= \frac{l}{\lambda^2 l^2 + n^2 \pi^2} \left[e^{\lambda l} \left(\lambda \sin n\pi - \frac{n\pi}{l} \cos n\pi \right) \right. \\
&\quad \left. - e^{-\lambda l} \left(\lambda \sin(-n\pi) - \frac{n\pi}{l} \cos(-n\pi) \right) \right] \\
&= \frac{-2n\pi}{\lambda^2 l^2 + n^2 \pi^2} \cos n\pi (e^{\lambda l} - e^{-\lambda l}) \\
&= (-1)^{n+1} \frac{2n\pi}{\lambda^2 l^2 + n^2 \pi^2} \operatorname{sh} \lambda l.
\end{aligned}$$

将傅里叶系数代入傅里叶级数表达式, 则得

$$f(x) = \frac{1}{\lambda l} \operatorname{sh} \lambda l + \sum_{n=1}^{\infty} \left[(-1)^n \frac{2\lambda l}{\lambda^2 l^2 + n^2 \pi^2} \operatorname{sh} \lambda l \cos \frac{n\pi}{l} x \right]$$

$$+ (-1)^{n+1} \frac{2n\pi}{\lambda^2 l^2 + n^2 \pi^2} \operatorname{sh} \lambda l \sin \frac{n\pi}{l} x \Big] ,$$

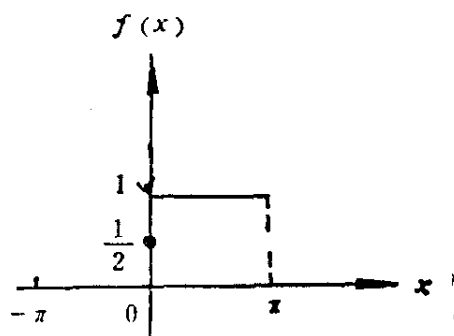
(2) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = H(x)$, 阶跃函数.

解: 根据单位阶跃函数的定义

$$H(x) = \begin{cases} 0, & (x < 0), \\ 1, & (x > 0), \end{cases}$$

可以知道此周期函数之表达式应为

$$f(x) = \begin{cases} 0, & (-\pi < x < 0) \\ 1, & (0 < x < \pi) \end{cases}$$



因此此函数之周期为 2π , 则有

图 6-5

$$2l = 2\pi$$

$$\text{即 } l = \pi$$

将 l 代入以 $2l$ 为周期之傅里叶级数表达式和傅里叶系数公式, 则得

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx),$$

计算傅里叶系数如下:

$$a_0 = \frac{1}{2\pi} \int_0^{\pi} f(x) dx = \frac{1}{2\pi} \int_0^{\pi} dx = \frac{1}{2\pi} x \Big|_0^{\pi} = \frac{1}{2},$$

$$a_n = \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} \cos x dx = \frac{1}{n\pi} \sin nx \Big|_0^{\pi} = 0,$$

$$b_n = \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx = \frac{1}{\pi} \int_0^{\pi} \sin nx dx = -\frac{1}{n\pi} \cos nx \Big|_0^{\pi}$$

$$= \frac{1}{n\pi} [1 - (-1)^n] = \begin{cases} 0, & (n = 2k), \\ \frac{2}{n\pi}, & (n = 2k + 1). \end{cases}$$

$$H(x) = f(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin(2k+1)x.$$

如果给定函数在第一类间断点处的值为左、右极限的算术

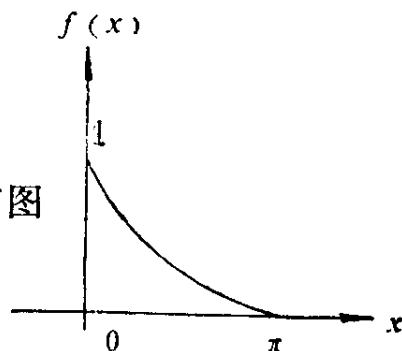
平均值, 则 $H(0) = \frac{1}{2}$, 则上式即为周期是 $(-\pi, \pi)$ 的阶跃函数 $H(x)$ 的傅里叶级数。

(3) 在 $(0, \pi)$ 这个周期上,

$$f(x) = 1 - \sin \frac{x}{2}.$$

解: $f(x) = 1 - \sin \frac{x}{2}$ 的图形如右图

$$\because 2l = \pi, \quad \therefore l = \frac{\pi}{2},$$



所以 $f(x)$ 的傅里叶级数展开式可写成

图 6-6

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos 2nx + b_n \sin 2nx),$$

其中傅里叶系数,

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} \left(1 - \sin \frac{x}{2} \right) dx \\ &= \frac{1}{\pi} \left[x + 2 \cos \frac{x}{2} \right]_0^{\pi} \\ &= \frac{1}{\pi} [\pi - 2] = 1 - \frac{2}{\pi}, \end{aligned}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos 2nx dx \\ &= \frac{2}{\pi} \int_0^{\pi} \left(1 - \sin \frac{x}{2} \right) \cos 2nx dx \\ &= \frac{2}{\pi} \int_0^{\pi} \cos 2nx dx - \frac{2}{\pi} \int_0^{\pi} \sin \frac{x}{2} \cos 2nx dx \\ &= \frac{1}{n\pi} \sin 2nx \Big|_0^{\pi} - \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} \left[\sin \left(\frac{1}{2} + 2n \right) x \right. \end{aligned}$$

$$\begin{aligned}
& + \sin\left(\frac{1}{2} - 2n\right)x \Big] dx \\
& = \frac{1}{\pi\left(2n + \frac{1}{2}\right)} \cos\left(2n + \frac{1}{2}\right)x \Big|_0^{\pi} \\
& \quad - \frac{1}{\pi\left(2n - \frac{1}{2}\right)} \cos\left(2n - \frac{1}{2}\right)x \Big|_0^{\pi} \\
& = \frac{-1}{\pi\left(2n + \frac{1}{2}\right)} + \frac{1}{\pi\left(2n - \frac{1}{2}\right)} = \frac{4}{(16n^2 - 1)\pi}, \\
b_n & = \frac{2}{\pi} \int_0^{\pi} f(x) \sin 2nx dx \\
& = \frac{2}{\pi} \int_0^{\pi} \left(1 - \sin \frac{x}{2}\right) \sin 2nx dx \\
& = \frac{2}{\pi} \int_0^{\pi} \sin 2nx dx - \frac{2}{\pi} \int_0^{\pi} \sin \frac{x}{2} \sin 2nx dx \\
& = -\frac{1}{n\pi} \cos 2nx \Big|_0^{\pi} - \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} \left[\cos\left(2n - \frac{1}{2}\right) \right. \\
& \quad \times x - \cos\left(2n + \frac{1}{2}\right)x \Big] dx \\
& = \frac{-1}{\left(2n - \frac{1}{2}\right)\pi} \sin\left(2n - \frac{1}{2}\right)x \Big|_0^{\pi} + \frac{1}{\left(2n + \frac{1}{2}\right)\pi} \\
& \quad \times \sin\left(2n + \frac{1}{2}\right)x \Big|_0^{\pi} \\
& = \frac{1}{\left(2n - \frac{1}{2}\right)\pi} + \frac{1}{\left(2n + \frac{1}{2}\right)\pi} = \frac{16n}{(16n^2 - 1)\pi},
\end{aligned}$$

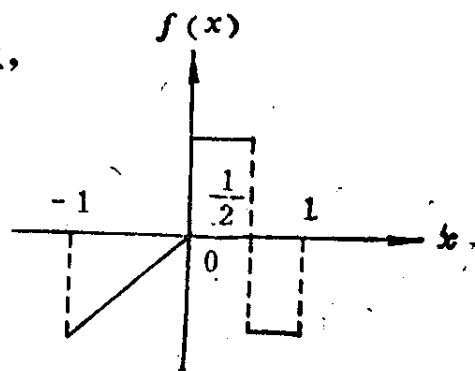
将傅里叶系数代入傅里叶级数表达式则得

$$f(x) = 1 - \frac{2}{\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{4}{16n^2 - 1} \cos 2nx \right)$$

$$+ \frac{16\pi}{16n^2 - 1} \sin 2n\pi x).$$

(4) 在 $(-1, 1)$ 这个周期上,

$$f(x) = \begin{cases} x, & \text{在 } (-1, 0) \text{ 上,} \\ 1, & \text{在 } (0, \frac{1}{2}) \text{ 上,} \\ -1, & \text{在 } (\frac{1}{2}, 1) \text{ 上.} \end{cases}$$



解: $\because 2l = 2, \therefore l = 1,$

所以 $f(x)$ 展开为傅里叶级数的形式是

图 6-7

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos k\pi x + b_k \sin k\pi x)$$

傅里叶系数的计算如下:

$$a_0 = \frac{1}{2} \int_{-1}^1 f(x) dx$$

$$= \frac{1}{2} \left[\int_{-1}^0 x dx + \int_0^{\frac{1}{2}} 1 \cdot dx + \int_{\frac{1}{2}}^1 (-1) dx \right] = -\frac{1}{4}.$$

$$a_k = \int_{-1}^1 f(x) \cos k\pi x dx$$

$$= \int_{-1}^0 x \cos k\pi x dx + \int_0^{\frac{1}{2}} 1 \cdot \cos k\pi x dx \\ + \int_{\frac{1}{2}}^1 (-1) \cos k\pi x dx$$

$$= \left[\frac{1}{k^2 \pi^2} \cos k\pi x + \frac{x}{k\pi} \sin k\pi x \right]_{-1}^0 + \frac{1}{k\pi} \sin k\pi x \Big|_0^{\frac{1}{2}} \\ - \frac{1}{k\pi} \sin k\pi x \Big|_{\frac{1}{2}}^1$$

$$= \frac{1}{k^2 \pi^2} [1 - (-1)^k] + \frac{1}{k\pi} \sin \frac{k\pi}{2} + \frac{1}{k\pi} \sin \frac{k\pi}{2}$$

$$\begin{aligned}
&= \frac{1}{k^2 \pi^2} \left[1 - (-1)^k \right] + \frac{2}{k\pi} \sin \frac{k\pi}{2}, \\
b_k &= \int_{-1}^1 f(x) \sin k\pi x dx \\
&= \int_{-1}^0 x \sin k\pi x dx + \int_0^{\frac{1}{2}} 1 \cdot \sin k\pi x dx \\
&\quad + \int_{\frac{1}{2}}^1 (-1) \sin k\pi x dx \\
&= \left[\frac{1}{k^2 \pi^2} \sin k\pi x - \frac{x}{k\pi} \cos k\pi x \right] \Big|_{-1}^0 - \frac{1}{k\pi} \cos k\pi x \Big|_0^{\frac{1}{2}} \\
&\quad + \frac{1}{k\pi} \cos k\pi x \Big|_{\frac{1}{2}}^1 \\
&= -\frac{1}{k\pi} \cos k\pi - \frac{1}{k\pi} \cos \frac{k\pi}{2} + \frac{1}{k\pi} + \frac{1}{k\pi} \cos k\pi \\
&\quad - \frac{1}{k\pi} \cos \frac{k\pi}{2} \\
&= \frac{1}{k\pi} - \frac{2}{k\pi} \cos \frac{k\pi}{2}, \\
\therefore f(x) &= -\frac{1}{4} + \sum_{k=1}^{\infty} \left\{ \left[\frac{1 - (-1)^k}{k^2 \pi^2} + \frac{2}{k\pi} \sin \frac{k\pi}{2} \right] \right. \\
&\quad \left. \times \cos k\pi x + \frac{1}{k\pi} \left(1 - 2 \cos \frac{k\pi}{2} \right) \sin k\pi x \right\}.
\end{aligned}$$

(5) 在 $(0, l)$ 这个周期上,

$$f(x) = \left(\cos \frac{\pi x}{l} \right) \left[1 - H \left(x - \frac{l}{2} \right) \right].$$

解: 首先分析一下函数 $f(x)$. 函数 $f(x)$ 表达式方括号内之函数 $1 - H \left(x - \frac{l}{2} \right)$ 可以看成是两个单位阶跃函数之叠加, 即

$$1 - H \left(x - \frac{l}{2} \right) = H(x) - H \left(x - \frac{l}{2} \right),$$

单位阶跃函数 $H(x)$ 的定义是

$$H(x) = \begin{cases} 0, & (x < 0), \\ 1, & (x > 0). \end{cases}$$

单位阶跃函数 $H\left(x - \frac{l}{2}\right)$ 的定义则为

$$H\left(x - \frac{l}{2}\right) = \begin{cases} 0, & \left(x < \frac{l}{2}\right), \\ 1, & \left(x > \frac{l}{2}\right), \end{cases}$$

这样，上面二单位阶跃函数之差便表示了一个矩形脉冲，因此有

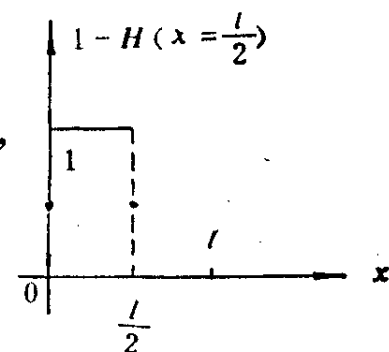
$$1 - H\left(x - \frac{l}{2}\right) = \begin{cases} 0, & (x < 0), \\ 1, & \left(0 < x < \frac{l}{2}\right), \\ 0, & \left(\frac{l}{2} < x\right), \end{cases}$$


图 6-8

从而可以得出

$$f(x) = \cos \frac{\pi x}{l} \left[1 - H\left(x - \frac{l}{2}\right) \right]$$

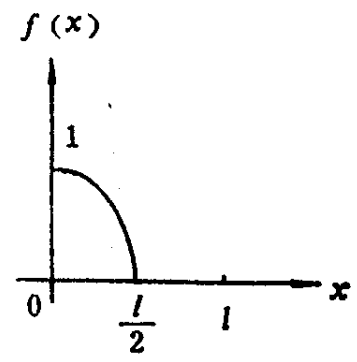
$$= \begin{cases} \cos \frac{\pi x}{l}, & \left(0 < x < \frac{l}{2}\right), \\ 0, & \left(\frac{l}{2} < x < l\right), \end{cases}$$


图 6-9

现将此函数展开成傅里叶级数，因周期为 l ，定义区间为 $(0, l)$ 。故傅里叶级数及其系数表达式为：

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2n\pi}{l} x + b_n \sin \frac{2n\pi}{l} x \right),$$

计算傅里叶系数

$$\begin{aligned}
 a_0 &= \frac{1}{l} \int_0^l f(x) dx = \frac{1}{l} \left[\int_0^{\frac{l}{2}} \cos \frac{\pi x}{l} dx + \int_{\frac{l}{2}}^l 0 \cdot dx \right] \\
 &= \frac{1}{l} \int_0^{\frac{l}{2}} \cos \frac{\pi x}{l} dx = \frac{1}{l} \cdot \frac{l}{\pi} \sin \frac{\pi x}{l} \Big|_0^{\frac{l}{2}} = \frac{1}{\pi}, \\
 a_n &= \frac{2}{l} \left[\int_0^{\frac{l}{2}} \cos \frac{\pi x}{l} \cos \frac{2n\pi}{l} x dx + \int_{\frac{l}{2}}^l 0 \cdot \cos \frac{2n\pi}{l} x dx \right] \\
 &= \frac{2}{l} \int_0^{\frac{l}{2}} \cos \frac{\pi x}{l} \cos \frac{2n\pi}{l} x dx \\
 &= \frac{2}{l} \int_0^{\frac{l}{2}} \frac{1}{2} \left[\cos \left(\frac{\pi x}{l} + \frac{2n\pi}{l} x \right) \right. \\
 &\quad \left. + \cos \left(\frac{\pi x}{l} - \frac{2n\pi}{l} x \right) \right] dx \\
 &= \frac{1}{l} \left[\int_0^{\frac{l}{2}} \cos \frac{2n\pi + \pi}{l} x dx + \int_0^{\frac{l}{2}} \cos \frac{2n\pi - \pi}{l} x dx \right] \\
 &= \frac{1}{l} \frac{l}{2n\pi + \pi} \sin \frac{2n\pi + \pi}{l} x \Big|_0^{\frac{l}{2}} \\
 &\quad + \frac{1}{l} \cdot \frac{l}{2n\pi - \pi} \sin \frac{2n\pi - \pi}{l} x \Big|_0^{\frac{l}{2}} \\
 &= \frac{\cos n\pi}{(2n+1)\pi} - \frac{\cos n\pi}{(2n-1)\pi} = \cos n\pi \frac{-2}{(4n^2-1)\pi} \\
 &= (-1)^{n+1} \frac{2}{(4n^2-1)\pi},
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{2n\pi}{l} x dx \\
 &= \frac{2}{l} \left[\int_0^{\frac{l}{2}} \cos \frac{\pi x}{l} \sin \frac{2n\pi}{l} x dx + \int_{\frac{l}{2}}^l 0 \cdot \sin \frac{2n\pi}{l} x dx \right]
 \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{l} \int_0^{\frac{l}{2}} \cos \frac{\pi}{l} x \sin \frac{2n\pi}{l} x dx \\
&= \frac{2}{l} \int_0^{\frac{l}{2}} \frac{1}{2} \left[\sin \left(\frac{2n\pi}{l} x + \frac{\pi}{l} x \right) \right. \\
&\quad \left. + \sin \left(\frac{2n\pi}{l} x - \frac{\pi}{l} x \right) \right] dx \\
&= \frac{1}{l} \left[\int_0^{\frac{l}{2}} \sin \frac{2n\pi + \pi}{l} x dx + \int_0^{\frac{l}{2}} \sin \frac{2n\pi - \pi}{l} x dx \right] \\
&= -\frac{1}{l} \cdot \frac{l}{2n\pi + \pi} \cos \frac{2n\pi + \pi}{l} x \Big|_0^{\frac{l}{2}} \\
&\quad - \frac{1}{l} \cdot \frac{l}{2n\pi - \pi} \cos \frac{2n\pi - \pi}{l} x \Big|_0^{\frac{l}{2}} \\
&= \frac{1}{(2n+1)\pi} + \frac{1}{(2n-1)\pi} = \frac{4n}{(4n^2-1)\pi},
\end{aligned}$$

将上列傅里叶系数代入傅里叶级数表达式则得

$$\begin{aligned}
f(x) = \frac{1}{\pi} + \sum_{n=1}^{\infty} \left[(-1)^{n+1} \frac{2}{(4n^2-1)\pi} \cos \frac{2n\pi}{l} x \right. \\
\left. + \frac{4n}{(4n^2-1)\pi} \sin \frac{2n\pi}{l} x \right].
\end{aligned}$$

(6) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = x + x^2$, 又在本题答案中, 置 $x = \pi$, 由此验证 $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6}$.

解: $\because 2l = 2\pi, \therefore l = \pi, \quad f(x)$

所以 $f(x) = x^2 + x$ 可以展开为傅里叶级数

$$\begin{aligned}
a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} (x^2 + x) dx
\end{aligned}$$

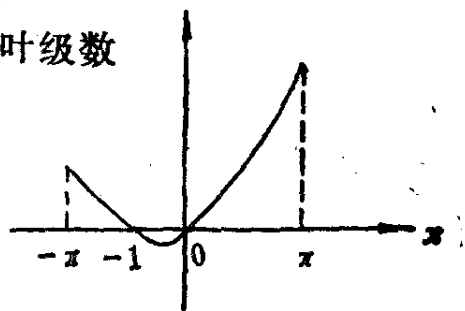


图 6-10

$$= \frac{1}{2\pi} \cdot \frac{x^3}{3} \Big|_{-\pi}^{\pi} + \frac{1}{2\pi} \cdot \frac{x^2}{2} \Big|_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi} \left(\frac{\pi^3}{3} - \frac{(\pi)^3}{3} \right) = \frac{1}{3} \pi^2,$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 + x) \cos nx dx.$$

应用已知积分公式

$$\int x^2 \cos px dx = \frac{2x}{p^2} \cos px + \frac{p^2 x^2 - 2}{p^3} \sin px,$$

$$\int x \cos px dx = \frac{1}{p^2} \cos px + \frac{x}{p} \sin px,$$

得

$$a_n = \frac{1}{\pi} \left[\frac{2x}{n^2} \cos nx + \frac{n^2 x^2 - 2}{n^3} \sin nx \right]_{-\pi}^{\pi}$$

$$+ \frac{1}{\pi} \left[\frac{1}{n^2} \cos nx + \frac{x}{n} \sin nx \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \cdot \frac{4\pi}{n^2} \cos n\pi = \frac{4}{n^2} (-1)^n.$$

应用已知积分公式:

$$\int x^2 \sin nx dx = \frac{2x}{n^2} \sin nx - \frac{n^2 x^2 - 2}{n^3} \cos nx,$$

$$\int x \sin nx dx = \frac{1}{n^2} \sin nx - \frac{x}{n} \cos nx,$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 + x) \sin nx dx$$

$$= \frac{1}{\pi} \left[\frac{2x}{n^2} \sin nx - \frac{n^2 x^2 - 2}{n^3} \cos nx \right]_{-\pi}^{\pi}$$

$$+ \frac{1}{\pi} \left[\frac{1}{n^2} \sin nx - \frac{x}{n} \cos nx \right]_{-\pi}^{\pi}$$

$$= -\frac{2}{n} \cos n\pi = \frac{2}{n} (-1)^{n+1}.$$

将傅里叶系数代入傅里叶级数表达式则得

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left[(-1)^n \frac{4}{n^2} \cos nx + (-1)^{n+1} \times \frac{2}{n} \sin nx \right],$$

在此答案中, 若置 $x = \pi$ 则有,

$$\begin{aligned} f(\pi) &= \frac{\pi^2}{3} + \sum_{n=1}^{\infty} (-1)^n \frac{4}{n^2} \cos n\pi \\ &= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2}. \end{aligned}$$

在 $x = \pi$ 时, 是函数 $f(x)$ 有第一类间断点, 据狄里希里定理知, 此时函数值为

$$f(\pi) = \frac{1}{2} [\pi^2 + \pi + (-\pi)^2 + (-\pi)] = \pi^2,$$

将此结果代入上式则得

$$\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots.$$

§ 25. 奇的和偶的周期函数

把下列函数 $f(x)$ 展开为傅里叶级数

(1) $f(x) = \cos^3 x$

[提示: 可按 (25·4) 和 (25·5) 展开。此外, 还可令 $t = e^{ix}$ 把 $f(x)$ 化为 t 的有理分式, 展开为幂级数, 然后再回到 x].

$$\begin{aligned} \text{解: } f(x) &= \cos^3 x = \left[\frac{e^{ix} + e^{-ix}}{2} \right]^3 \\ &= \frac{1}{8} [e^{i3x} + 3e^{ix} + 3e^{-ix} + e^{-i3x}] \end{aligned}$$

$$\begin{aligned}
&= \frac{3}{4} \cdot \frac{e^{ix} + e^{-ix}}{2} + \frac{1}{4} \cdot \frac{e^{i3x} + e^{-i3x}}{2} \\
&= \frac{3}{4} \cos x + \frac{1}{4} \cos 3x.
\end{aligned}$$

注：本题其实就是三倍角公式：

$$\cos 3x = 4\cos^3 x - 3\cos x,$$

$$\text{则 } f(x) = \cos 3x = \frac{3}{4} \cos x + \frac{1}{4} \cos 3x,$$

$$(2) \quad f(x) = \frac{1-a^2}{1-2a\cos x+a^2}, (|a|<1).$$

$$\text{解：令 } e^{ix}=t, \text{ 则 } \cos x = \frac{e^{ix}+e^{-ix}}{2} = \frac{t+\frac{1}{t}}{2},$$

$$f(x) = \frac{1-a^2}{1-2a\cos x+a^2} = \frac{1-a^2}{1-at-a \cdot \frac{1}{t}+a^2}$$

$$= \frac{1-a^2}{(a-t)(a-\frac{1}{t})} = \frac{t-a^2t}{(t-a)(1-at)}$$

$$= \frac{t-a+a(1-at)}{(t-a)(1-at)} = \frac{1}{1-at} + \frac{\frac{a}{t}}{1-\frac{a}{t}}$$

$$= \sum_{k=0}^{\infty} a^k t^k + \sum_{k=0}^{\infty} \left(\frac{a}{t}\right)^{k+1}$$

$$= 1 + \sum_{k=0}^{\infty} a^k t^k + \sum_{k=1}^{\infty} a^k \frac{1}{t^k},$$

$$\therefore f(x) = 1 + 2 \sum_{k=1}^{\infty} a^k \frac{t^k + t^{-k}}{2}$$

$$= 1 + 2 \sum_{k=1}^{\infty} a^k \cos Kx,$$

$$(3) f(x) = \frac{1 - a \cos x}{1 - 2a \cos x + a^2}, \quad (|a| < 1).$$

解: 令 $t = e^{ix}$, 则 $\cos x = \frac{t + \frac{1}{t}}{2}$.

$$\begin{aligned} f(x) &= \frac{1 - a \left(\frac{t}{2} + \frac{a}{2t} \right)}{(1 - at - a) \cdot \left(\frac{1}{t} + a^2 \right)} = \frac{1}{2} \frac{1 - at + 1 - \frac{a}{t}}{(a - t) \left(a - \frac{1}{t} \right)} \\ &= \frac{1}{2} \left[\frac{-t}{a - t} + \frac{-\frac{1}{t}}{a - \frac{1}{t}} \right] = \frac{1}{2} \left[-\frac{1}{1 - \frac{a}{t}} \right. \\ &\quad \left. + \frac{1}{1 - at} \right] \\ &= \sum_{k=0}^{\infty} a^k \frac{t^k + t^{-k}}{2} = \sum_{k=0}^{\infty} a^k \cos kx. \end{aligned}$$

$$(4) f(x) = \frac{a \sin x}{1 - 2a \cos x + a^2} \quad (|a| < 1).$$

解: 令 $e^{ix} = t$, 则 $\sin x = \frac{e^{ix} - e^{-ix}}{2i} = \frac{1}{2i} \left(t - \frac{1}{t} \right)$,

$$\begin{aligned} f(x) &= \frac{a}{2i} \cdot \frac{t - t^{-1}}{1 - \frac{a}{t} - at + a^2} \\ &= \frac{a}{2i} \cdot \frac{t - \frac{1}{t}}{(a - t) \left(a - \frac{1}{t} \right)} \\ &= \frac{1}{2i} \cdot \frac{1 - a \cdot \frac{1}{t} - (1 - at)}{\left(1 - \frac{a}{t} \right) (1 - at)} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2i} \left[\frac{1}{1-at} - \frac{1}{1-\frac{a}{t}} \right] \\
&= \sum_{k=0}^{\infty} \frac{a^k}{2i} [t^k - t^{-k}] \\
&= \sum_{k=0}^{\infty} a^k \sin Kx \\
&= \sum_{k=1}^{\infty} a^k \sin Kx,
\end{aligned}$$

(5) 在 $[-\pi, \pi]$ 这个周期上, $f(x) = x^2$ 。又在本题答案中, 令 $x=0$, 由此验证: $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}$ 。

解: 由于 $f(x) = x^2$ 是偶函数, 因而 $b_k = 0$, 展开式为如下形式:

$$\begin{aligned}
f(x) &= a_0 + \sum_{k=1}^{\infty} a_k \cos kx \\
a_0 &= \frac{1}{\pi} \int_0^{\pi} f(\xi) d\xi = \frac{1}{\pi} \int_0^{\pi} \xi^2 d\xi = \frac{1}{3\pi} \xi^3 \Big|_0^{\pi} = \frac{\pi^2}{3}.
\end{aligned}$$

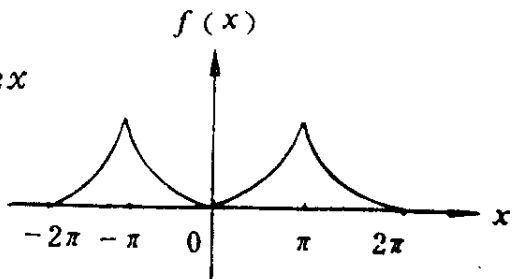


图 6-11

由: $\int x^2 \cos x dx = 2x \cos x + (x^2 - 2) \sin x$

$$\begin{aligned}
a_k &= \frac{2}{\pi} \int_0^{\pi} f(\xi) \cos k\xi d\xi = \frac{2}{\pi} \int_0^{\pi} \xi^2 \cos k\xi d\xi \\
&= \frac{2}{\pi k^3} \int_0^{\pi} (k\xi)^2 \cos k\xi d(k\xi) \\
&= \frac{2}{\pi k^3} \{ 2(k\xi) \cos k\xi + (k^2 \xi^2 - 2) \sin k\xi \} \Big|_0^{\pi} \\
&= \frac{2}{\pi k^3} \{ 2(k\pi) \cos k\pi + (k^2 \pi^2 - 2) \sin k\pi \} = \frac{4}{k^2} (-1)^k,
\end{aligned}$$

$$\therefore f(x) = \frac{\pi^2}{3} + 4 \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2} \cos kx,$$

令 $x = 0$, 得

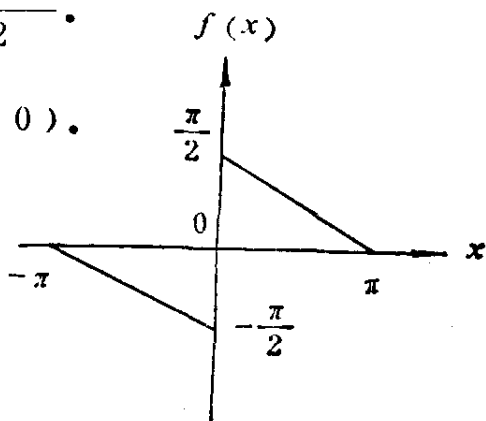
$$0 = \frac{\pi^2}{3} + 4 \left(-1 + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots \right),$$

$$\text{即 } 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}.$$

(6) 在半个周期 $(-\pi, 0)$ 上, $f(x) = -(\pi + x)/2$; 在另外半个周期 $(0, \pi)$ 上, $f(x) = \frac{\pi - x}{2}$.

解:

$$f(x) = \begin{cases} \frac{-(\pi + x)}{2} & (-\pi, 0) \\ \frac{\pi - x}{2} & (0, \pi) \end{cases}$$



因为 $f(x)$ 是奇函数, 可以展开为傅里叶正弦级数.

$$f(x) = \sum_{k=1}^{\infty} b_k \sin kx,$$

图 6-12

$$\text{其中: } b_k = \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi - \xi}{2} \right) \sin k\xi d\xi$$

$$= \frac{2}{\pi} \int_0^{\pi} \frac{\pi}{2} \sin k\xi d\xi - \frac{2}{\pi} \int_0^{\pi} \frac{\xi}{2} \sin k\xi d\xi$$

$$= -\frac{1}{k} \cos k\xi \Big|_0^{\pi} - \frac{1}{\pi k^2} [\sin k\xi - k\xi \cos k\xi] \Big|_0^{\pi}$$

$$= \frac{1}{k},$$

$$\therefore f(x) = \sum_{k=1}^{\infty} \frac{1}{k} \sin kx.$$

(7) 在半个周期 $(-\pi, 0)$ 上, $f(x) = -\cos x$; 在另外半

个周期 $(0, \pi)$ 上, $f(x) = \cos x$.

$$\text{解: } f(x) = \begin{cases} -\cos x, & -\pi < x < 0, \\ \cos x, & 0 < x < \pi, \end{cases}$$

$$\text{又 } 2l = 2\pi, \quad \therefore l = \pi,$$

$\therefore f(x)$ 是奇函数, 所以 $f(x)$ 可以展开为傅里叶正弦级数.

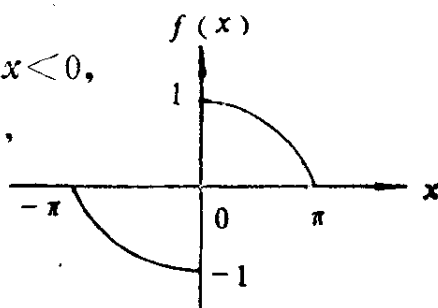


图 6-13

$$f(x) = \sum_{k=1}^{\infty} b_k \sin kx,$$

其中

$$\begin{aligned} b_k &= \frac{2}{\pi} \int_0^{\pi} \cos \xi \sin k\xi d\xi \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} [\sin(k+1)\xi + \sin(k-1)\xi] d\xi \\ &= \frac{1}{\pi} \left[-\frac{\cos(k+1)\xi}{k+1} - \frac{\cos(k-1)\xi}{k-1} \right]_0^{\pi} \\ &= \frac{1}{\pi} \left[-\frac{\cos(k+1)\pi - 1}{k+1} - \frac{\cos(k-1)\pi - 1}{k-1} \right] \\ &= \frac{1}{\pi} \left[\frac{(-1)^{k+2} + 1}{k+1} + \frac{(-1)^k + 1}{k-1} \right] \\ &= \begin{cases} 0, & (k \text{ 为奇数但 } \neq 1), \\ \frac{4k}{\pi(k^2 - 1)}, & (k \text{ 为偶数}), \end{cases} \end{aligned}$$

$$\begin{aligned} b_1 &= \frac{1}{\pi} \int_0^{\pi} \frac{1}{2} \sin 2\xi d\xi = \frac{1}{2\pi} (-\cos 2\xi) \int_0^{\pi} \\ &= \frac{1}{2\pi} [1 - 1] = 0. \end{aligned}$$

$$\therefore f(x) = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{n}{4n^2 - 1} \sin 2nx.$$

(8) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = \cos ax$, (a 非整数).

解: 因为 $f(x)$ 是偶函数 $\therefore b_k = 0$,

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos kx,$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos a\xi d\xi = \frac{1}{2a\pi} \sin a\xi \Big|_{-\pi}^{\pi} \\ = \frac{\sin a\pi}{a\pi},$$

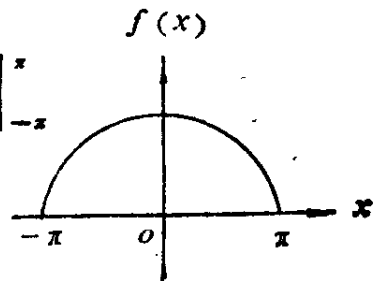


图 6-14

$$a_k = \frac{2}{\pi} \int_0^{\pi} \cos a\xi \cos k\xi d\xi \\ = \frac{1}{\pi} \int_0^{\pi} \cos(k+a)\xi d\xi + \frac{1}{\pi} \int_0^{\pi} \cos(k-a)\xi d\xi \\ = \frac{1}{\pi} \left[\frac{\sin(k+a)\xi}{k+a} + \frac{\sin(k-a)\xi}{k-a} \right]_0^{\pi} \\ = \frac{1}{\pi} \left[\frac{\sin(k+a)\pi}{k+a} + \frac{\sin(k-a)\pi}{k-a} \right] \\ = \frac{1}{\pi} \cdot \frac{1}{k+a} \left[\sin k\pi \cos a\pi + \cos k\pi \sin a\pi \right] \\ + \frac{1}{\pi} \cdot \frac{1}{k-a} \left[\sin k\pi \cos a\pi - \cos k\pi \sin a\pi \right] \\ = \frac{1}{\pi} \cos k\pi \sin a\pi \left[\frac{1}{k+a} - \frac{1}{k-a} \right] \\ = \frac{1}{\pi} (-1)^k \sin a\pi \cdot \frac{-2a}{k^2 - a^2} \\ = \frac{(-1)^{k+1}}{\pi} \sin a\pi \cdot \frac{2a}{k^2 - a^2},$$

$$\therefore f(x) = \frac{2\sin a\pi}{\pi} \left[\frac{1}{2a} + \sum_{k=1}^{\infty} \frac{a(-1)^{k+1}}{k^2 - a^2} \cos kx \right].$$

(9) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = \sin ax$ (a 非整数).

解: $\because f(x)$ 是奇函数, $\therefore a_0 = 0, a_k = 0$,

$$\text{又 } 2l = 2\pi, \therefore l = \pi,$$

$$\begin{aligned} \therefore b_k &= \frac{2}{\pi} \int_0^{\pi} \sin a\xi \sin k\xi d\xi \\ &= -\frac{1}{\pi} \int_0^{\pi} \cos(k+a)\xi d\xi \\ &\quad + \frac{1}{\pi} \int_0^{\pi} \cos(k-a)\xi d\xi \\ &= \frac{(-1)^{k+1} \sin a\pi}{\pi} \frac{2k}{k^2 - a^2}, \end{aligned}$$

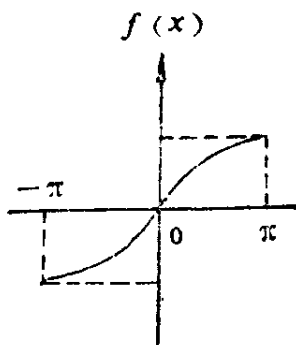


图 6-15

$$\therefore f(x) = \frac{2 \sin a\pi}{\pi} \sum_{k=1}^{\infty} \frac{k}{k^2 - a^2} (-1)^{k+1} \sin kx,$$

(10) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = \operatorname{ch} ax$.

解: $f(x) = \operatorname{ch} ax = \frac{e^{ax} + e^{-ax}}{2}$, 是偶函数

$$\therefore b_k = 0,$$

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{\pi} f(\xi) d\xi = \frac{1}{\pi} \int_0^{\pi} \frac{e^{a\xi} + e^{-a\xi}}{2} d\xi \\ &= \frac{1}{2\pi a} e^{a\xi} \Big|_0^{\pi} - \frac{1}{2\pi a} e^{-a\xi} \Big|_0^{\pi} \\ &= \frac{1}{2\pi a} e^{a\pi} - \frac{1}{2\pi a} e^{-a\pi} \\ &= \frac{1}{\pi a} \operatorname{sh} a\pi. \end{aligned}$$

$$\begin{aligned} a_k &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} (e^{a\xi} + e^{-a\xi}) \cos k\xi d\xi \\ &= \frac{1}{\pi} \int_0^{\pi} e^{a\xi} \cos k\xi d\xi + \frac{1}{\pi} \int_0^{\pi} e^{-a\xi} \cos k\xi d\xi \\ &= \left[\frac{e^{a\xi}}{\pi} \cdot \frac{(a \cos k\xi + k \sin k\xi)}{a^2 + k^2} + \frac{e^{-a\xi}}{\pi} \cdot \frac{(-a \cos k\xi + k \sin k\xi)}{a^2 + k^2} \right]_0^{\pi} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\pi(a^2 + k^2)} \left[e^{a\pi} (a \cos k\pi + k \sin k\pi) \right. \\
&\quad \left. + e^{-a\pi} (-a \cos k\pi + k \sin k\pi) \right] \\
&\quad - \frac{1}{\pi(a^2 + k^2)} \left[(a + 0) + (-a + 0) \right] \\
&= \frac{a}{\pi(a^2 + k^2)} \left[(e^{a\pi} - e^{-a\pi}) \cos k\pi \right] \\
&= \frac{2a \operatorname{sh} a\pi}{\pi(a^2 + k^2)} (-1)^k,
\end{aligned}$$

$$\therefore f(x) = \frac{2 \operatorname{sh} a\pi}{\pi} \left[\frac{1}{2a} + a \sum_{k=1}^{\infty} \frac{(-1)^k \cos kx}{a^2 + k^2} \right].$$

$$\text{注: } \int e^{ax} \cos kx dx = \frac{e^{ax} (a \cos kx + k \sin kx)}{a^2 + k^2}.$$

(11) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = \operatorname{sh} ax$,

解: $f(x) = \operatorname{sh} ax = \frac{e^{ax} - e^{-ax}}{2}$ 是奇函数,

$$\begin{aligned}
b_k &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} (e^{a\xi} - e^{-a\xi}) \sin k\xi d\xi \\
&= \frac{1}{\pi} \int_0^{\pi} e^{a\xi} \sin k\xi d\xi - \frac{1}{\pi} \int_0^{\pi} e^{-a\xi} \sin k\xi d\xi \\
&= \left[\frac{1}{\pi} \frac{e^{a\xi} (a \sin k\xi - k \cos k\xi)}{a^2 + k^2} \right. \\
&\quad \left. - \frac{1}{\pi} \frac{e^{-a\xi} (-a \sin k\xi - k \cos k\xi)}{a^2 + k^2} \right] \\
&= \frac{1}{\pi(a^2 + k^2)} \left[-ke^{a\pi} \cos k\pi + ke^{-a\pi} \cos k\pi \right] \\
&= \frac{2k \operatorname{sh} a\pi}{\pi(a^2 + k^2)} (-1)^{k+1}
\end{aligned}$$

$$\therefore f(x) = \frac{2 \operatorname{sh} a\pi}{\pi} \sum_{k=1}^{\infty} \frac{k(-1)^{k+1}}{a^2 + k^2} \sin kx.$$

注: $\int e^{ax} \sin kx dx = \frac{e^{ax} (a \sin kx - k \cos kx)}{a^2 + k^2}.$

(12) 在半个周期 $(0, \frac{l}{2})$ 上, $f(x) = \sin \frac{\pi x}{l}$; 在另外半个周期 $(\frac{l}{2}, l)$ 上, $f(x) = -\sin \frac{\pi x}{l}.$

解: 在边界上, $f(0) = 0,$
 $f(l) = 0,$ 因此用正弦
 级数展开

$$f(x) = \sum_{k=1}^{\infty} b_k \sin \frac{2k\pi}{l} x,$$

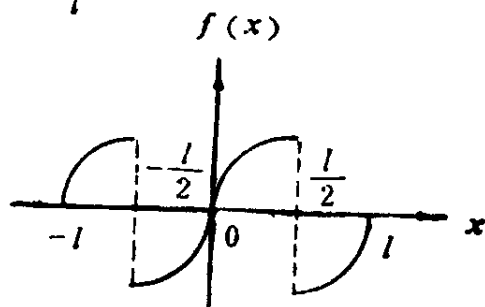


图 6-16

$$\begin{aligned} b_k &= \frac{2}{l} \left[\int_0^{\frac{l}{2}} \sin \frac{\pi \xi}{l} \sin \frac{2k\pi \xi}{l} d\xi - \int_{\frac{l}{2}}^l \sin \frac{\pi \xi}{l} \sin \frac{2k\pi \xi}{l} d\xi \right] \\ &= \frac{1}{l} \left[\int_0^{\frac{l}{2}} \cos \frac{(2k-1)\pi \xi}{l} d\xi - \int_0^{\frac{l}{2}} \cos \frac{(2k+1)\pi \xi}{l} d\xi \right] \\ &\quad - \frac{1}{l} \left[\int_{\frac{l}{2}}^l \cos \frac{(2k-1)\pi \xi}{l} d\xi - \int_{\frac{l}{2}}^l \cos \frac{(2k+1)\pi \xi}{l} d\xi \right] \\ &= \frac{1}{l} \frac{l}{(2k-1)\pi} \sin \frac{(2k-1)\pi}{l} \xi \Big|_0^{\frac{l}{2}} \\ &\quad - \frac{1}{l} \frac{l}{(2k+1)\pi} \sin \frac{(2k+1)\pi}{l} \xi \Big|_0^{\frac{l}{2}} \\ &\quad - \frac{1}{l} \frac{l}{(2k-1)\pi} \sin \frac{(2k+1)\pi}{l} \xi \Big|_{\frac{l}{2}}^l \\ &\quad + \frac{1}{l} \frac{l}{(2k+1)\pi} \sin \frac{(2k+1)\pi}{l} \xi \Big|_{\frac{l}{2}}^l \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{(2k-1)\pi} \sin \frac{(2k-1)}{2} \pi - \frac{1}{(2k+1)\pi} \sin \frac{2k+1}{2} \pi \\
&\quad - \frac{1}{(2k-1)\pi} \sin (2k-1)\pi + \frac{1}{(2k-1)\pi} \sin \frac{(2k-1)}{2} \pi \\
&\quad + \frac{1}{(2k+1)\pi} \sin (2k+1)\pi - \frac{1}{(2k+1)\pi} \sin \frac{(2k+1)}{2} \pi \\
&= \frac{2}{(2k-1)\pi} \sin \frac{2k-1}{2} \pi - \frac{2}{(2k+1)\pi} \sin \frac{2k+1}{2} \pi, \\
b_k &= \frac{2}{(2k-1)\pi} (-1)^{k+1} + \frac{2}{(2k+1)\pi} (-1)^{k+1} \\
&= \frac{2}{\pi} \frac{4k}{4k^2-1} (-1)^{k+1}, \\
\therefore f(x) &= \sum_{k=1}^{\infty} \frac{8}{\pi} \frac{k(-1)^{k+1}}{4k^2-1} \sin \frac{2k\pi x}{l}.
\end{aligned}$$

(13) 在 $(-\pi, \pi)$ 这个区间上,

$$f(x) = \begin{cases} \cos x, & \left(-\frac{\pi}{2} < x < \frac{\pi}{2}\right), \\ 0, & \left(-\pi < x < -\frac{\pi}{2}, \frac{\pi}{2} < x < \pi\right). \end{cases}$$

解: $f(x)$ 在 $(-\pi, \pi)$ 这个区间是偶函数, 因此可展开为傅里叶余弦级数.

$$a_0 = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} \cos \xi d\xi = \frac{1}{\pi} \sin \xi \Big|_0^{\frac{\pi}{2}} = \frac{1}{\pi}.$$

$$\begin{aligned}
a_1 &= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \cos^2 \xi d\xi \\
&= \frac{1}{\pi} \int_0^{\frac{\pi}{2}} (1 + \cos 2\xi) d\xi \\
&= \frac{1}{\pi} \int_0^{\frac{\pi}{2}} d\xi + \frac{1}{\pi} \int_0^{\frac{\pi}{2}} \cos 2\xi d\xi
\end{aligned}$$

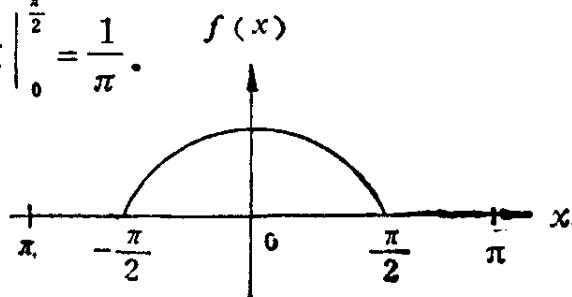


图 6-17

$$= \frac{1}{\pi} \xi \Big|_0^{\frac{\pi}{2}} + \frac{1}{\pi} \frac{1}{2} \sin 2\xi \Big|_0^{\frac{\pi}{2}} = \frac{1}{2}.$$

$$\begin{aligned} a_k &= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \cos \xi \cos k\xi d\xi \\ &= \frac{1}{\pi} \left[\int_0^{\frac{\pi}{2}} \cos (k+1) \xi d\xi + \int_0^{\frac{\pi}{2}} \cos (k-1) \xi d\xi \right] \\ &= \frac{1}{\pi (k+1)} \sin (k+1) \xi \Big|_0^{\frac{\pi}{2}} + \frac{1}{\pi (k-1)} \sin (k-1) \xi \Big|_0^{\frac{\pi}{2}} \\ &= \frac{1}{\pi} \frac{1}{(k+1)} \sin \frac{(k+1)\pi}{2} \\ &\quad + \frac{1}{\pi (k-1)} \sin \frac{(k-1)\pi}{2}. \end{aligned}$$

当 k 为奇数时 $a_k = 0$, 当 k 为偶数时, 则有

$$\begin{aligned} a_n &= a_{2n} = \frac{1}{\pi (2n+1)} \sin \frac{2n+1}{2} \pi \\ &\quad + \frac{1}{\pi (2n-1)} \sin \frac{2n-1}{2} \pi \\ &= \frac{(-1)^n}{\pi (2n+1)} + \frac{(-1)^{n+1}}{\pi (2n-1)} \\ &= \frac{1}{\pi} \left[\frac{-1}{2n+1} + \frac{1}{2n-1} \right] (-1)^{n+1} \\ &= \frac{1}{\pi} \frac{2}{(2n)^2 - 1} (-1)^{n+1}, \\ \therefore f(x) &= \frac{1}{\pi} + \frac{1}{2} \cos x + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{4n^2 - 1} \cos 2nx. \end{aligned}$$

§ 26. 有限区间上的函数的傅里叶级数

1. 要求下列函数 $f(x)$ 在它的定义区间的边界上为零. 试根

据这个要求把 $f(x)$ 展开为傅里叶级数.

(1) $f(x) = \cos ax$, 定义在 $(0, \pi)$ 上.

解: 因为按题意, 在边界 $(0, \pi)$ 上, $f(0) = 0$ 和 $f(\pi) = 0$ 由此可知, 展开式中只有正弦项, 而无余弦项, 即 $a_n = 0$, 因而展开式可表为

$$f(x) = \sum_{k=1}^{\infty} b_k \sin kx,$$

其中
$$b_k = \frac{2}{\pi} \int_0^{\pi} \cos ax \sin kx dx.$$

应用三角公式 $2\sin\alpha\cos\beta = \sin(\alpha+\beta) + \sin(\alpha-\beta)$

可得 $2\cos ax \sin kx = \sin(k+a)x + \sin(k-a)x$

则
$$b_k = \frac{1}{\pi} \int_0^{\pi} [\sin(k+a)x + \sin(k-a)x] dx$$

$$\begin{aligned} &= \frac{1}{\pi} \left[\frac{(-1)}{k+a} \cos(k+a)x \right]_0^{\pi} \\ &\quad + \frac{1}{\pi} \left[\frac{(-1)}{k-a} \cos(k-a)x \right]_0^{\pi} \\ &= \frac{1}{\pi} [1 - \cos(k+a)\pi] \frac{1}{k+a} \\ &\quad + \frac{1}{\pi} \frac{1}{k-a} [1 - \cos(k-a)\pi] \\ &= \frac{2k}{\pi(k^2 - a^2)} [1 + (-1)^{k+1} \cos a\pi], \end{aligned}$$

$$\therefore f(x) = \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{k}{k^2 - a^2} [1 + (-1)^{k+1} \cos a\pi] \sin kx.$$

(2) $f(x) = x^3$, 定义在 $(0, \pi)$ 上.

解:

因为按题意, 在边界上 $f(0) = 0$ 和 $f(\pi) = 0$, 可见展开式中没有余弦项, 即 $a_0 = 0$, $a_k = 0$, 仅有正弦项, 因而展开式可表示

为:

$$f(x) = \sum_{k=1}^{\infty} b_k \sin kx.$$

$$\begin{aligned} \text{其中 } b_k &= \frac{2}{\pi} \int_0^{\pi} f(\xi) \sin k\xi d\xi = \frac{2}{\pi} \int_0^{\pi} \xi^3 \sin k\xi d\xi \\ &= \frac{2}{\pi k^4} \int_0^{\pi} (k\xi)^3 \sin k\xi d(k\xi). \end{aligned}$$

利用公式 $\int x^3 \sin x dx = (3x^2 - 6) \sin - (x^3 - 6x) \cos x$ 代入上式, 则有

$$\begin{aligned} b_k &= \frac{2}{\pi k^4} \left\{ \left[3(k\xi)^2 - 6 \right] \sin k\xi - \left[(k\xi)^3 - 6(k\xi) \right] \cos k\xi \right\}_0^{\pi} \\ &= \frac{2}{\pi k^4} \left\{ - \left[(k\pi)^3 - 6(k\pi) \right] \cos k\pi - 0 \right\} \\ &= (-1)^k \left[\frac{12}{k^3} - \frac{2\pi^2}{k} \right] \end{aligned}$$

$$\text{即 } f(x) = \sum_{k=1}^{\infty} (-1)^k \left[\frac{12}{k^3} - \frac{2\pi^2}{k} \right] \sin kx.$$

请读者将本题和习题 2(2) 比较.

$$(3) \quad f(x) = a \left(1 - \frac{x}{l} \right), \text{ 定义在 } (0, l) \text{ 上.}$$

解: 因为按题意要求, $f(0) = 0$, $f(l) = 0$, 因此应将 $f(x)$ 作奇延拓, 然后展开为傅里叶正弦级数.

$$f(x) = \sum_{k=1}^{\infty} b_k \sin \frac{k\pi x}{l},$$

$$\begin{aligned} \text{其中: } b_k &= \frac{2}{l} \int_0^l f(\xi) \sin \frac{k\pi \xi}{l} d\xi \\ &= \frac{2}{l} \int_0^l a \left(1 - \frac{\xi}{l} \right) \sin \frac{k\pi \xi}{l} d\xi \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{l} \int_0^l a \sin \frac{k\pi \xi}{l} d\xi - \frac{2}{l} \int_0^l \frac{a}{l} \xi \sin \frac{k\pi}{l} \xi d\xi \\
&= \frac{2}{l} \frac{la}{k\pi} \int_0^l \sin \frac{k\pi \xi}{l} d\left(\frac{k\pi \xi}{l}\right) \\
&\quad - \frac{2a}{k^2 \pi^2} \int_0^l \left(\frac{k\pi}{l} \xi\right) \sin \frac{k\pi \xi}{l} d\left(\frac{k\pi \xi}{l}\right) \\
&= \frac{-2a}{k\pi} \cos \frac{k\pi \xi}{l} \Big|_0^l - \frac{2a}{k^2 \pi^2} \left[\sin \frac{k\pi \xi}{l} - \frac{k\pi \xi}{l} \cos \frac{k\pi \xi}{l} \right]_0^l \\
&= \frac{-2a}{k\pi} (\cos k\pi - 1) - \frac{2a}{k^2 \pi^2} [0 - k\pi \cos k\pi - 0 + 0] \\
&= \frac{2a}{k\pi} (1 - \cos k\pi) + \frac{2a}{k\pi} \cos k\pi = \frac{2a}{k\pi}.
\end{aligned}$$

$$\therefore f(x) = \frac{2a}{\pi} \sum_{k=1}^{\infty} \frac{1}{k} \sin \frac{k\pi}{l} x.$$

请将本题与习题 2、(3) 比较。

(4) 在 $(0, \frac{l}{2})$ 上, $f(x) = x$, 在 $(\frac{l}{2}, l)$ 上, $f(x) = l - x$.

解: 按题意要求, 在边界上, $f(0) = 0$ 和 $f(l) = 0$, 因而展开式有下列形式:

$$(fx) = \sum_{k=1}^{\infty} b_k \sin \frac{k\pi x}{l}$$

$$\begin{aligned}
\text{其中 } b_k &= \frac{2}{l} \left[\int_0^{\frac{l}{2}} f(\xi) \sin \frac{k\pi}{l} \xi d\xi + \int_{\frac{l}{2}}^l f(\xi) \sin \frac{k\pi \xi}{l} d\xi \right] \\
&= \frac{2}{l} \left[\int_0^{\frac{l}{2}} \xi \sin \frac{k\pi \xi}{l} d\xi + \int_{\frac{l}{2}}^l (l - \xi) \sin \frac{k\pi \xi}{l} d\xi \right] \\
&= \frac{2l}{k^2 \pi^2} \int_0^{\frac{l}{2}} \left(\frac{k\pi \xi}{l}\right) \sin \frac{k\pi \xi}{l} d\left(\frac{k\pi \xi}{l}\right)
\end{aligned}$$

$$\begin{aligned}
& + \int_{\frac{l}{2}}^l 2 \sin \frac{k\pi\xi}{l} d\xi - \frac{2l}{k^2\pi^2} \\
& \times \int_{\frac{l}{2}}^l \left(\frac{k\pi\xi}{l} \right) \sin \frac{k\pi\xi}{l} d\left(\frac{k\pi\xi}{l} \right) \quad \text{图 6-18} \\
& = \frac{2l}{k^2\pi^2} \left[\sin \frac{k\pi\xi}{l} - \left(\frac{k\pi\xi}{l} \right) \cos \frac{k\pi\xi}{l} \right]_{\frac{l}{2}}^{\frac{l}{2}} \\
& \quad - \frac{2l}{k\pi} \cos \frac{k\pi\xi}{l} \Big|_{\frac{l}{2}}^l - \frac{2l}{k^2\pi^2} \left[\sin \frac{k\pi\xi}{l} \right. \\
& \quad \left. - \left(\frac{k\pi\xi}{l} \right) \cos \frac{k\pi\xi}{l} \right]_{\frac{l}{2}}^l \\
& = \frac{2l}{k^2\pi^2} \sin \frac{k\pi}{2} - \frac{l}{k\pi} \cos \frac{k\pi}{2} - \frac{2l}{k\pi} \cos k\pi \\
& \quad + \frac{2l}{k\pi} \cos \frac{k\pi}{2} + \frac{2l}{k\pi} \cos k\pi \\
& \quad + \frac{2l}{k^2\pi^2} \sin \frac{k\pi}{2} - \frac{l}{k\pi} \cos \frac{k\pi}{2} \\
& = 2 \times \frac{2l}{k^2\pi^2} \sin \frac{k\pi}{2} = \frac{4l}{k^2\pi^2} \sin \frac{k\pi}{2} \\
& = \begin{cases} 0, & (k = 2n), \\ (-1)^n \frac{4l}{(2n+1)^2\pi}, & (k = 2n+1), \end{cases} \\
\therefore f(x) = \sum_{n=0}^{\infty} \frac{4l}{(2n+1)^2\pi^2} (-1)^n \sin \frac{(2n+1)\pi}{l} x.
\end{aligned}$$

请将本题和习题 2(4) 比较

(5) $f(x) = 1$, 定义在 $(0, \pi)$ 上.

解: 因为要满足 $f(0) = 0$ 和 $f(\pi) = 0$, 则展开式中仅有正弦项.

$$\therefore f(x) = \sum_{k=1}^{\infty} b_k \sin kx,$$

$$\begin{aligned} \text{其中: } b_k &= \frac{2}{\pi} \int_0^{\pi} \sin k\zeta d\zeta = -\frac{2}{k\pi} \left[\cos k\zeta \right]_0^{\pi} \\ &= \frac{2}{k\pi} [-\cos k\pi + 1] = \frac{2}{k\pi} [1 - \cos k\pi] \\ &= \begin{cases} 0, & (k=2n), \\ \frac{4}{\pi(2n+1)}, & (k=2n+1), \end{cases} \end{aligned}$$

$$\therefore f(x) = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \sin(2n+1)x.$$

请读者把本题与习题 2(5) 比较.

2. 要求下列函数 $f(x)$ 的导数 $f'(x)$ 在函数定义区间的边界为零. 试根据这个要求把 $f(x)$ 展开为傅里叶级数.

(1) 在 $(0, \frac{l}{2})$ 上, $f(x) = \cos(\frac{\pi x}{l})$; 在 $(\frac{l}{2}, l)$ 上,

$f(x) = 0$.

解: 因为 $f'(0)$ 和 $f'(l) = 0$, 所以应将 $f(x)$ 展开成为傅里叶余弦级数, 其傅里叶系数.

$$\begin{aligned} a_0 &= \frac{1}{l} \int_0^{\frac{l}{2}} \cos \frac{\pi \zeta}{l} d\zeta = \frac{1}{\pi} \sin \frac{k\pi}{l} \zeta \Big|_0^{\frac{l}{2}} \\ &= \frac{1}{\pi} \sin \frac{k\pi}{2} = \frac{1}{\pi}, \\ a_1 &= \frac{2}{l} \int_0^{\frac{l}{2}} \cos^2 \frac{\pi}{l} \zeta d\zeta = \frac{2}{l} \int_0^{\frac{l}{2}} \frac{1}{2} \left(1 + \cos \frac{2\pi}{l} \zeta \right) d\zeta \\ &= \frac{1}{l} \left[\zeta + \frac{1}{\pi} \sin \frac{2\pi}{l} \zeta \right]_0^{\frac{l}{2}} = \frac{1}{l} \left[\frac{l}{2} + \frac{1}{\pi} \sin \pi \right] \end{aligned}$$

$$= \frac{1}{2},$$

$$\begin{aligned} a_k &= \frac{2}{l} \int_0^{\frac{l}{2}} \cos \frac{\pi}{l} \xi \cos \frac{k\pi}{l} \xi d\xi \\ &= \frac{1}{l} \int_0^{\frac{l}{2}} \cos \frac{k+1}{l} \pi \xi d\xi + \frac{1}{l} \int_0^{\frac{l}{2}} \cos \frac{k-1}{l} \pi \xi d\xi \\ &= \left[\frac{1}{(k+1)\pi} \sin \frac{k+1}{l} \pi \xi + \frac{1}{(k-1)\pi} \sin \frac{k-1}{l} \pi \xi \right]_0^{\frac{l}{2}} \\ &= \frac{1}{(k+1)\pi} \sin \frac{k+1}{2} \pi + \frac{1}{(k-1)\pi} \sin \frac{k-1}{2} \pi \\ &= \begin{cases} 0, & (k=2n+1), \\ (-1)^{n+1} \frac{2}{(4n^2-1)\pi} & (k=2n). \end{cases} \end{aligned}$$

$$\begin{aligned} \therefore f(x) &= \frac{1}{\pi} + \frac{1}{2} \cos \frac{\pi x}{l} \\ &\quad + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{4n^2-1} \cos \frac{2n\pi x}{l}. \end{aligned}$$

(2) $f(x) = x^3$, 定义在 $(0, \pi)$ 上.

解: $\because$ 题意要求 $f'(0) = 0$ 和 $f'(\pi) = 0$, 因而应将 $f(x)$ 展开为傅里叶余弦级数, 其傅里叶系数为

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{\pi} \xi^3 d\xi = \frac{1}{\pi} \left. \frac{\xi^4}{4} \right|_0^{\pi} = \frac{\pi^3}{4}, \\ a_k &= \frac{2}{\pi} \int_0^{\pi} \xi^3 \cos k\xi d\xi = \frac{2}{\pi k^4} \int_0^{\pi} (k\xi)^3 \cos k\xi d(k\xi) \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{\pi k^4} \left[(3k^2\xi^2 - 6) \cos k\xi + (k^2\xi^2 - 2) \sin k\xi \right]_0^\pi \\
&= \frac{2}{\pi k^4} \left[(3k^2\pi^2 - 6) \cos k\pi + (k^2\pi^2 - 2) \sin k\pi \right. \\
&\quad \left. - (-6) \cos 0 + (-2) \sin 0 \right] \\
&= \frac{2}{\pi k^4} \left[(3k^2\pi^2 - 6) \cos k\pi + 6 \right], \\
&= \begin{cases} \frac{6\pi}{k^2} (-1)^k, & (k \text{ 为偶数}), \\ \frac{6\pi}{k^2} (-1)^k + \frac{24}{\pi k^4}, & (k \text{ 为奇数}), \end{cases}
\end{aligned}$$

如令 $k = 2n$, 则 $a_k = a_{2n} = \frac{3\pi}{2n^2}$,

$$k = 2n + 1 \text{ 则 } a_k = a_{2n+1} = \frac{24}{\pi(2n+1)^4} - \frac{6\pi}{(2n+1)^4},$$

$$\therefore f(x) = \frac{\pi^3}{4} + \sum_{k=1}^{\infty} a_k \cos kx,$$

请读者将本题和习题 1(2) 比较.

(3) $f(x) = a \left(1 - \frac{x}{l} \right)$, 定义在 $(0, l)$ 上.

解: 因在 $f'(0) = 0$ 和 $f'(l) = 0$, 所以应将 $f(x)$ 展开成余弦级数.

其系数:

$$\begin{aligned}
a_0 &= \frac{1}{l} \int_0^l a \left(1 - \frac{\xi}{l} \right) d\xi = \frac{a}{l} \int_0^l d\xi - \frac{a}{l^2} \int_0^l \xi d\xi \\
&= \frac{a}{2},
\end{aligned}$$

$$a_k = \frac{2}{l} \int_0^l a \left(1 - \frac{\xi}{l} \right) \cos \frac{k\pi}{l} \xi d\xi$$

$$\begin{aligned}
&= \frac{2}{l} \int_0^1 a \cos \frac{k\pi}{l} \xi d\xi - \frac{2a}{l^2} \int_0^1 \xi \cos \frac{k\pi}{l} \xi d\xi \\
&= \frac{2a}{l} \frac{l}{k\pi} \sin \frac{k\pi}{l} \xi \Big|_0^1 - \frac{2a}{k^2 \pi^2} \left[\cos \frac{k\pi}{l} \xi \right. \\
&\quad \left. - \frac{k\pi}{l} \xi \sin \frac{k\pi}{l} \xi \right]_0^1 \\
&= -\frac{2a}{k^2 \pi^2} [\cos k\pi - k\pi \sin k\pi - \cos 0] \\
&= \frac{2a}{k^2 \pi^2} [1 - \cos k\pi] \\
&= \begin{cases} 0 & (k=2n), \\ \frac{4a}{\pi^2 (2n+1)^2} & (k=2n+1). \end{cases}
\end{aligned}$$

$$\therefore f(x) = \frac{a}{2} + \sum_{n=0}^{\infty} \frac{4a}{\pi^2 (2n+1)^2} \cos \frac{(2n+1)\pi}{l} x.$$

请读者将本题和习题 1(3) 比较.

(4) 在 $(0, \frac{l}{2})$ 上, $f(x) = x$; 在 $(\frac{l}{2}, l)$ 上, $f(x) = l - x$.

解: 按题意 $f(x)$ 的展开式为余弦级数:

其系数:

$$\begin{aligned}
a_0 &= \frac{1}{l} \int_0^{\frac{l}{2}} \xi d\xi + \frac{1}{l} \int_{\frac{l}{2}}^l (l - \xi) d\xi \\
&= \frac{1}{2l} \xi^2 \Big|_0^{\frac{l}{2}} + \xi \Big|_{\frac{l}{2}}^l - \frac{1}{2l} \xi^2 \Big|_{\frac{l}{2}}^l = \frac{l}{4}, \\
a_k &= \frac{2}{l} \left[\int_0^{\frac{l}{2}} \xi \cos \frac{k\pi \xi}{l} d\xi + \int_{\frac{l}{2}}^l (l - \xi) \cos \frac{k\pi \xi}{l} d\xi \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{2}{l} \left[\frac{l^2}{k^2 \pi^2} \int_0^{\frac{l}{2}} \left(\frac{k\pi}{l} \xi \right) \cos \frac{k\pi \xi}{l} d \left(\frac{k\pi \xi}{l} \right) \right. \\
&\quad + l \int_{\frac{l}{2}}^l \cos \frac{k\pi \xi}{l} d\xi \\
&\quad \left. - \int_{\frac{l}{2}}^l \xi \cos \frac{k\pi \xi}{l} d\xi \right] \\
&= \frac{2l}{k^2 \pi^2} \left[\cos \frac{k\pi \xi}{l} + \left(\frac{k\pi \xi}{l} \right) \sin \frac{k\pi \xi}{l} \right]_0^{\frac{l}{2}} \\
&\quad + \frac{2l^2}{lk\pi} \sin \frac{k\pi \xi}{l} \Big|_{\frac{l}{2}}^l - \frac{2}{l} \cdot \frac{l^2}{k^2 \pi^2} \left[\cos \frac{k\pi \xi}{l} \right. \\
&\quad \left. + \left(\frac{k\pi \xi}{l} \right) \sin \frac{k\pi \xi}{l} \right]_{\frac{l}{2}}^l \\
&= \frac{2l}{k^2 \pi^2} \left[2 \cos \frac{k\pi}{2} - (1 + (-1)^k) \right] \\
&= \begin{cases} \frac{2l}{k^2 \pi^2} \left(2 \cos \frac{k\pi}{2} - 2 \right) & (k \text{ 为偶数}), \\ 0 & (k \text{ 为奇数}). \end{cases} \\
&= \frac{4l}{\pi^2 (2n)^2} [(-1)^n - 1] \\
&= \begin{cases} 0 & (n \text{ 为偶数}), \\ \frac{-8l}{\pi^2 (2n)^2} & (n \text{ 为奇数}), \end{cases}
\end{aligned}$$

$$\therefore f(x) = \frac{l}{4} - \frac{8l}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{(4n+2)^2} \sin \frac{(2n+2)\pi}{l} x.$$

请读者将本题和习题 1(4) 比较.

(5) $f(x) = 1$, 定义在 $(0, \pi)$ 上.

解: 因 $f'(0) = 0$, $f'(\pi) = 0$, 所以应将 $f(x)$ 展开为余弦

级数.

其系数

$$a_0 = \frac{1}{\pi} \int_0^{\pi} d\xi = \frac{1}{\pi} \xi \Big|_0^{\pi} = 1,$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(\xi) \cos \frac{n\pi}{\pi} \xi d\xi = \frac{2(-1)^n}{\pi n} \sin n\xi \Big|_0^{\pi} = 0,$$

$\therefore f(x) = 1$, 这是只有单项的傅里叶级数.

3. 在区间 $(0, l)$ 上定义了函数 $f(x) = x$. 试根据条件 $f'(0) = 0$, $f(l) = 0$, 把 $f(x)$ 展开为傅里叶级数.

解: 根据边界条件 $f'(0) = 0$ 应将函数 $f(x)$ 对区间 $(0, l)$ 的端点 $x = 0$ 作偶延拓, 又根据边界条件 $f(l) = 0$, 应将函数 $f(x)$ 对区间 $(0, l)$ 的端点 $x = l$ 作

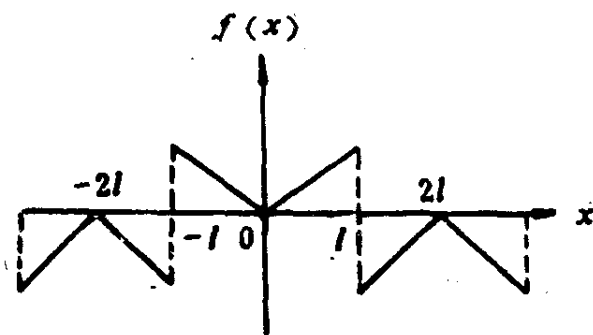


图 6-19

奇延拓, 延拓以后的函数是以 $4l$ 为周期的偶函数. 故展开式为

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos \frac{k\pi x}{2l},$$

现在计算系数

$$\begin{aligned} a_0 &= \frac{1}{2l} \left[\int_0^l x dx + \int_l^{2l} (x-2l) dx \right] \\ &= \frac{1}{2l} \left[\frac{l^2}{2} + \frac{4l^2}{2} - \frac{l^2}{2} - 4l^2 + 2l^2 \right] = 0. \end{aligned}$$

$$\begin{aligned} a_k &= \frac{1}{l} \left[\int_0^l x \cos \frac{k\pi x}{2l} dx + \int_l^{2l} (x-2l) \cos \frac{k\pi x}{2l} dx \right] \\ &= \frac{1}{l} \int_0^l x \cos \frac{k\pi x}{2l} dx + \frac{1}{l} \int_l^{2l} (y-2l) \cos \frac{k\pi y}{2l} dy. \end{aligned}$$

在第二个积分中作代换 $x = 2l - y$ 即 $y = 2l - x$ 则

$$a_k = \frac{1}{l} \int_0^l x \cos \frac{k\pi}{2l} x dx + \frac{1}{l} \int_l^0 x \cos \left(k\pi - \frac{k\pi x}{2l} \right) dx$$

$$= \frac{1}{l} [1 - (-1)^k] \int_0^l x \cos \frac{k\pi x}{2l} dx,$$

而 $1 - (-1)^k = \begin{cases} 0, & (\text{如 } k = \text{偶数}), \\ 2, & (\text{如 } k = \text{奇数}), \end{cases}$

又 $\frac{1}{l} \int_0^l x \cos \frac{k\pi x}{2l} dx = \frac{4l}{k^2 \pi^2} \left[\cos \frac{k\pi x}{2l} + \frac{k\pi x}{2l} \sin \frac{k\pi x}{2l} \right]_0^l$

$$= \frac{4l}{k^2 \pi^2} \left[\cos \frac{k\pi}{2} - 1 + \frac{k\pi}{2} \sin \frac{k\pi}{2} \right],$$

而在 $k = 2n + 1$ 为奇数时, 则有

$$a_k = 2 \cdot \frac{1}{l} \int_0^l x \cos \frac{k\pi x}{2l} dx = \frac{-8l}{(2n+1)^2 \pi^2}$$

$$+ \frac{4l}{(2n+1)\pi} (-1)^n,$$

结果 $f(x) = \sum_{n=0}^{\infty} \left[\frac{(-1)^n 4l}{(2n+1)\pi} - \frac{8l}{(2n+1)^2 \pi^2} \right] \cos \frac{(2n+1)\pi}{2l} x.$

4. 二元函数 $f(x, y) = xy$, 定义在区域 $-\pi < x < \pi, -\pi < y < \pi$ 上. 试根据边界条件 $f|_{x=-\pi} = f|_{x=\pi} = 0$ 把 f 对自变数 x 展为傅里叶级数. 这个级数的“系数”仍然是 y 的函数, 再根据边界条件 $f|_{y=-\pi} = f|_{y=\pi} = 0$ 把这个级数中的“系数”对自变数 y 展为傅里叶级数, 这叫做双重傅里叶级数.

解: 先把 $f(x, y)$ 就自变数 x 展开为傅里叶级数, 根据边界条件, 这傅里叶级数应是正弦级数.

$$f(x, y) = \sum_{k=1}^{\infty} b_k \sin kx = \sum_{k=1}^{\infty} b_k(y) \sin kx,$$

“系数” $b_k(y)$ 的计算如下:

$$\begin{aligned} b_k(y) &= \frac{2}{\pi} \int_0^{\pi} y x \sin kx dx = \frac{2y}{\pi} \left[\frac{1}{k^2} (\sin kx - kx \cos kx) \right]_0^{\pi} \\ &= \frac{2y}{k\pi} [-\pi \cos k\pi] = \frac{2y}{k} (-1)^{k+1}. \end{aligned}$$

再将 $b_k(y)$ 就自变数 y 展开傅里叶级数, 根据边界条件, 这里傅里叶级数应为正弦级数.

$$b_k(y) = \sum_{n=1}^{\infty} b_{kn} \sin ny,$$

系数 b_{kn} 的计算如下:

$$\begin{aligned} b_{kn} &= \frac{2(-1)^{k+1}}{k} \frac{2}{\pi} \int_0^{\pi} y \sin ny dy \\ &= \frac{2(-1)^{k+1}}{k} \frac{2(-1)^{n+1}}{n} = \frac{4(-1)^{k+n}}{kn}, \end{aligned}$$

结果
$$f(x, y) = \sum_{k=1}^{\infty} \sum_{n=1}^{\infty} \frac{4(-1)^{k+n}}{kn} \sin kx \sin ny.$$

§27. 复数形式的傅里叶级数

1. 矩形波 $f(x)$, 在 $\left(-\frac{T}{2}, \frac{T}{2}\right)$ 这个周期上可表为

$$f(x) = \begin{cases} 0, & \text{在 } \left(-\frac{T}{2}, -\frac{\tau}{2}\right) \text{ 上,} \\ H, & \text{在 } \left(-\frac{\tau}{2}, \frac{\tau}{2}\right) \text{ 上,} \\ 0, & \text{在 } \left(\frac{\tau}{2}, \frac{T}{2}\right) \text{ 上,} \end{cases}$$

试将它展开为复数形式的傅里叶级数.

解: $\because l = \frac{T}{2}$, 故

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{i \frac{2k\pi}{T} x},$$

$$\text{其中 } C_0 = \frac{1}{2l} \int_{-l}^l f(x) dx = \frac{1}{2 \cdot \frac{T}{2}} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x) dx$$

$$= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} H dx = \frac{1}{T} H \tau.$$

$$C_k = \frac{1}{2l} \int_{-l}^l f(\xi) \left[e^{i \frac{k\pi \xi}{l}} \right] d\xi$$

$$= \frac{1}{2 \cdot \frac{T}{2}} \int_{-\frac{T}{2}}^{\frac{T}{2}} H e^{-i \frac{2k\pi}{T} x} dx$$

$$= \frac{iH}{2\pi k} \left(-2i \sin \frac{k\pi \tau}{T} \right) = \frac{H}{\pi k} \sin \frac{k\pi \tau}{T} (k \neq 0),$$

$$\therefore f(x) = \frac{H\tau}{T} + \left(\sum_{k=-\infty}^{-1} + \sum_{k=1}^{\infty} \right) \frac{H}{\pi k} \sin \frac{k\pi \tau}{T} \times e^{i \frac{2k\pi}{T} x}.$$

2. 锯齿波 $f(x)$ 在 $(0, T)$ 这个周期上可表为

$$f(x) = \frac{H}{T} x,$$

试把它展开为复数形式的傅里叶级数.

$$\text{解: } f(x) = \sum_{k=-\infty}^{\infty} C_k e^{i \frac{2k\pi}{T} x}, \quad \left(\because l = \frac{T}{2} \right),$$

$$C_0 = \frac{1}{T} \int_0^T f(x) dx = \frac{1}{T} \int_0^T \frac{1}{T} H x dx$$

$$= \frac{1}{T} \left. \frac{H}{2} x^2 \right|_0^T = \frac{H}{2},$$

$$\begin{aligned} C_k &= \frac{1}{T} \int_0^T f(x) \left(e^{i \frac{2k\pi}{T} x} \right)^* dx \\ &= \frac{1}{T} \int_0^T \frac{H}{2} x e^{-i \frac{2k\pi}{T} x} dx \\ &= \frac{H}{T^2} \left(\frac{T}{-i2\pi k} \right)^2 e^{-i \frac{2\pi k}{T} x} \\ &\quad \times \left(-i \frac{2\pi k}{T} x - 1 \right) \Big|_0^T \\ &= \frac{H}{(-i2\pi k)^2} \left[e^{-i2\pi k} (-i2\pi k - 1) - (-1) \right] \\ &= \frac{H}{(-i2\pi k)^2} [(-i2\pi k - 1) + 1] = \frac{H}{-i2\pi k} \\ &= \frac{iH}{2\pi k}, \quad (k \neq 0), \end{aligned}$$

$$\therefore f(x) = \frac{H}{2} + \left(\sum_{k=-\infty}^{-1} + \sum_{k=1}^{\infty} \right) \frac{iH}{2\pi k} e^{i \frac{2\pi k}{T} x}.$$

3. 在实数形式的傅里叶级数24.7式中

$$\left[f(x) = a_0 + \sum_{k=1}^{\infty} \left(a_k \cos \frac{k\pi x}{l} + b_k \sin \frac{k\pi x}{l} \right) \right],$$

把 $\cos \frac{k\pi x}{l}$ 和 $\sin \frac{k\pi x}{l}$ 按欧勒公式用虚指数的指数函数

$e^{i \frac{k\pi x}{l}}$ 和 $e^{-i \frac{k\pi x}{l}}$ 表出, 验证实数形式的傅里叶级数(24.7).

就化为复数形式的傅里叶级数(27.2) [即 $f(x) =$

$$\sum_{k=-\infty}^{\infty} C_k e^{i \frac{k\pi x}{l}}] \text{ 而且 } C_k = \frac{a_k - ib_k}{2}, C_{-k} = \frac{1}{2}(a_k + ib_k). \text{ 其中 } k > 0.$$

$$\begin{aligned}
 \text{解: } f(x) &= a_0 + \sum_{k=1}^{\infty} \left(a_k \cos \frac{k\pi x}{l} + b_k \sin \frac{k\pi x}{l} \right) \\
 &= a_0 + \sum_{k=1}^{\infty} \left[\frac{a_k}{2} \left(e^{i \frac{k\pi x}{l}} + e^{-i \frac{k\pi x}{l}} \right) \right. \\
 &\quad \left. + \frac{b_k}{2i} \left(e^{i \frac{k\pi x}{l}} - e^{-i \frac{k\pi x}{l}} \right) \right] \\
 &= a_0 + \sum_{k=1}^{\infty} \left[\left(\frac{a_k - ib_k}{2} \right) e^{i \frac{k\pi x}{l}} \right. \\
 &\quad \left. + \left(\frac{a_k + ib_k}{2} \right) e^{-i \frac{k\pi x}{l}} \right],
 \end{aligned}$$

$$\text{令 } a_0 = C_0, \quad \frac{a_k - ib_k}{2} = C_k, \quad \frac{a_k + ib_k}{2} = C_{-k},$$

则实数形式的傅里叶级数便化成复数形式:

$$f(x) = C_0 + \sum_{k=1}^{\infty} \left(C_k e^{i \frac{k\pi x}{l}} + C_{-k} e^{-i \frac{k\pi x}{l}} \right).$$

令 $k = 0 \pm 1, \pm 2, \pm 3, \dots$ 则上式可化为统一的复数形式 (即27·2式):

$$f(x) = \sum_{k=-\infty}^{\infty} C_k e^{i \frac{k\pi x}{l}}, \quad \text{其中 } k = |k| = \text{正整数}.$$

从上述讨论可以看出 C_k 和 C_{-k} 的模正好是傅里叶级数展开式中 k 次谐波振幅的一半, 这是因为 k 次谐波

$$\begin{aligned}
 a_k \cos \frac{k\pi x}{l} + b_k \sin \frac{k\pi x}{l} &= \sqrt{a_k^2 + b_k^2} \sin \left(\frac{k\pi x}{l} \right. \\
 &\quad \left. + \arctg \frac{b_k}{a_k} \right) \\
 &= A_k \sin \left(\frac{k\pi x}{l} + \arctg \frac{b_k}{a_k} \right),
 \end{aligned}$$

其中 k 次谐波的振幅 $A_k = \sqrt{a_k^2 + b_k^2}$,

而 $|C_k| = |C_{-k}| = \frac{1}{2} \sqrt{a_k^2 + b_k^2} = \frac{1}{2} A_k$.

第七章 傅里叶积分

§28. 非周期函数的傅里叶积分

1. 把单个锯齿脉冲 $f(t)$ 展开为傅里叶积分.

$$f(t) = \begin{cases} 0, & (t < 0), \\ kt, & (0 < t < T), \\ 0, & (T < t). \end{cases} \quad f(x)$$

解: 因为 $f(t)$ 是无界空间中的非周期函数, 它的周期为 ∞ , 故可展开为傅里叶积分:

$$f(t) = \int_0^{\infty} A(\omega) \cos \omega t d\omega + \int_0^{\infty} B(\omega) \sin \omega t d\omega$$

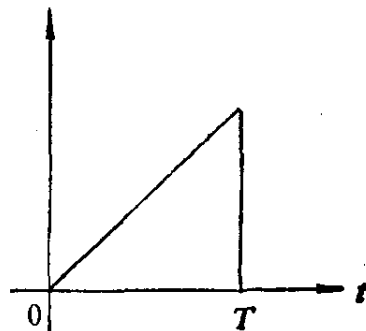


图 7-1

其中傅里叶变换 $A(\omega)$ 和 $B(\omega)$ 为:

$$\begin{aligned} A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x dx = \frac{1}{\pi} \int_0^T kt \cos \omega t dt \\ &= \frac{k}{\pi \omega^2} \int_0^T (\omega t) \cos \omega t d(\omega t) \\ &= \frac{k}{\pi \omega^2} \left[\cos \omega t + \omega t \sin \omega t \right]_0^T \\ &= \frac{k}{\pi \omega^2} [\cos \omega T + \omega T \sin \omega T - 1], \end{aligned}$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x dx = \frac{1}{\pi} \int_0^T kt \sin \omega t dt$$

$$\begin{aligned}
&= \frac{k}{\pi\omega^2} \left[\sin\omega t - \omega t \cos\omega t \right]_0^T \\
&= \frac{k}{\pi\omega^2} [\sin\omega T - \omega T \cos\omega T], \\
\therefore f(t) &= \frac{k}{\pi} \int_0^\infty \frac{1}{\omega^2} (\cos\omega T + \omega T \sin\omega T - 1) \cos\omega t d\omega \\
&\quad + \frac{k}{\pi} \int_0^\infty \frac{1}{\omega^2} (\sin\omega T - \omega T \cos\omega T) \sin\omega t d\omega.
\end{aligned}$$

2. 把振幅按双曲线衰减的振动函数 $f(t)$ 展开为傅里叶积分

$$f(t) = \frac{\sin\Omega t}{t}, \quad (\Omega \text{ 为常数}).$$

试拿本题的频谱跟图(38)比较, 又拿本题的 $f(t)$ 跟图(39)比较. 比较的结果说明什么问题?

解: 因 $\sin\Omega t$ 是奇函数, t 也是奇函数, 所以 $f(t)$ 是偶函数, 应展开为傅里叶余弦积分

$$f(t) = \int_0^\infty A(\omega) \cos\omega t d\omega,$$

其中 $A(\omega)$ 是 $f(t)$ 的傅里叶变换式, 按(28·6)式有

$$\begin{aligned}
A(\omega) &= \frac{1}{\pi} \int_{-\infty}^\infty f(\xi) \cos\omega\xi d\xi = \frac{2}{\pi} \int_0^\infty f(\xi) \cos\omega\xi d\xi \\
&= \frac{2}{\pi} \int_0^\infty \frac{1}{\xi} \sin\Omega\xi \cos\omega\xi d\xi \\
&= \frac{1}{\pi} \left[\int_0^\infty \frac{1}{\xi} \sin(\omega + \Omega)\xi d\xi \right. \\
&\quad \left. - \int_0^\infty \frac{1}{\xi} \sin(\omega - \Omega)\xi d\xi \right].
\end{aligned}$$

应用积分公式

$$\int_0^{\infty} \frac{\sin mx}{x} dx = \begin{cases} \frac{\pi}{2}, & (m > 0), \\ 0, & (m = 0), \\ -\frac{\pi}{2}, & (m < 0), \end{cases}$$

$$\text{得 } A(\omega) = \begin{cases} 0, & (\omega > \Omega), \\ \frac{1}{2}, & (\omega = \Omega), \\ 1, & (\omega < \Omega), \end{cases} = 1 - H(\omega - \Omega),$$

而 $f(t)$ 和 $A(\omega)$ 的图形如图 7-2.

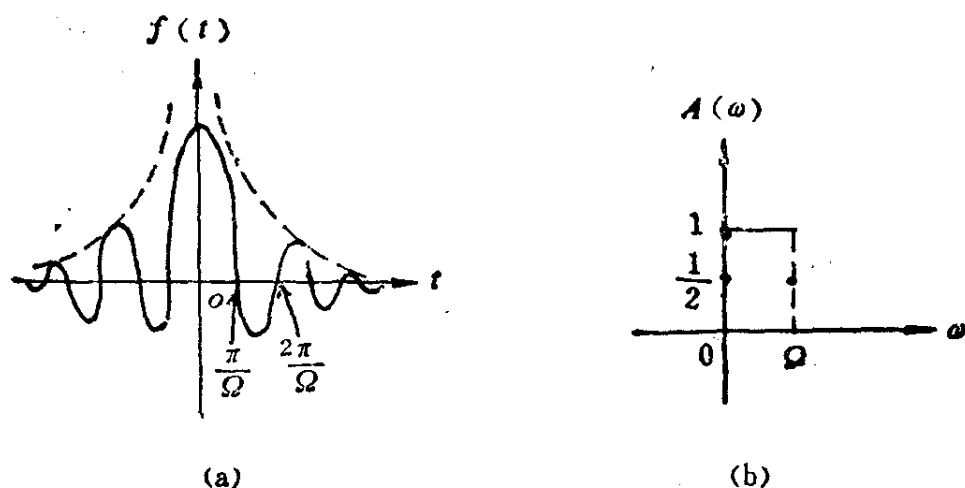
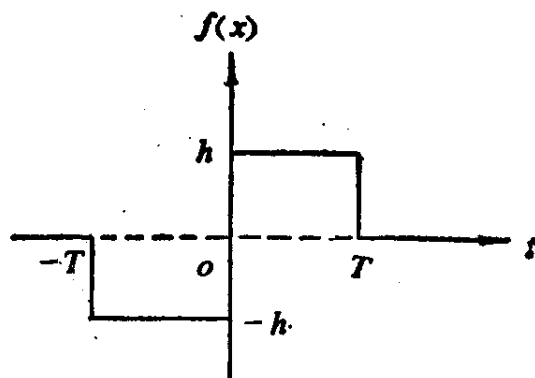


图 7-2

比较知, 本题的 $f(t)$ 的图象同于图 (39) 的 $A(\omega)$, 而本题的频谱 $A(\omega)$ 的图象则同于图 (39) 的 $f(t)$, 这是由于公式 (28·10) 和 (28·11) 对变数 x 和 ω 对称的缘故, 亦即如果不计及常数因子, 其 $f(x)$ 和 $A(\omega)$ 互为傅里叶变换式, 可以说 $A(\omega)$ 是 $f(x)$ 的傅里叶变换式, 也可以说 $f(x)$ 是 $A(\omega)$ 的傅里叶变换式.

3. 把下列脉冲 $f(t)$ 展开为傅里叶积分,

$$f(t) = \begin{cases} 0, & (t < -T), \\ -h, & (-T < t < 0), \\ h, & (0 < t < T), \\ 0, & (T < t). \end{cases}$$



注意在半无界区间 $(0, \infty)$ 上, 本例题的 $f(t)$ 跟例 1 的 $f(t)$ 相同.

图 7-3

解: 因为 $f(t)$ 是奇函数, 所以展开为傅里叶正弦积分:

$$f(t) = \int_0^{\infty} B(\omega) \sin \omega t d\omega$$

其傅里叶变换为:

$$\begin{aligned} B(\omega) &= \frac{2}{\pi} \int_0^T h \sin \omega \xi d\xi = \frac{2}{\pi} \frac{h}{\omega} \int_0^T \sin \omega \xi d(\omega \xi) \\ &= \frac{2h}{\pi \omega} (-\cos \omega \xi) \Big|_0^T = \frac{2h}{\pi \omega} (1 - \cos \omega T). \end{aligned}$$

本题的图7-3和课本中的图38 (第134页例1) 的 $f(t)$ 在区间 $(0, \infty)$ 上, 是相同的, 只是本题属于奇函数, 而第134页的例1为偶函数.

4. $f(t)$ 是定义在半无界区间 $(0, \infty)$ 上的函数,

$$f(t) = \begin{cases} h, & (0 < t < T), \\ 0, & (T < t). \end{cases}$$

(1) 在边界条件 $f'(0) = 0$ 下把 $f(t)$ 展为傅里叶积分;

(2) 在边界条件 $f(0) = 0$ 下把 $f(t)$ 展为傅里叶积分.

解: (1) 要满足边界条件 $f'(0) = 0$, 必须将 $f(t)$ 展开为傅里叶余弦积分.

$$f(t) = \int_0^{\infty} A(\omega) \cos \omega t d\omega,$$

其中

$$A(\omega) = \frac{2}{\pi} \int_0^{\infty} f(\xi) \cos \omega \xi d\xi = \frac{2}{\pi} \int_0^T h \cos \omega \xi d\xi$$

$$= \frac{2h}{\pi \omega} \sin \omega \xi \Big|_0^T = \frac{2h}{\pi \omega} \sin \omega T,$$

$$\therefore f(t) = \int_0^{\infty} \frac{2h}{\pi \omega} \sin \omega T \cos \omega t d\omega$$

$$= \frac{2h}{\pi} \int_0^{\infty} \frac{\sin \omega T \cos \omega t d\omega}{\omega}.$$

(2) 要满足边界条件 $f(0) = 0$, 必须将 $f(t)$ 展开为傅里叶正弦积分:

$$f(t) = \int_0^{\infty} B(\omega) \sin \omega t d\omega,$$

其中

$$B(\omega) = \frac{2}{\pi} \int_0^{\infty} f(\xi) \sin \omega \xi d\xi = \frac{2}{\pi} \int_0^T h \sin \omega \xi d\xi$$

$$= \frac{2}{\pi} \frac{h}{\omega} (-\cos \omega \xi) \Big|_0^T = \frac{2h}{\omega \pi} (1 - \cos \omega T),$$

$$\therefore f(t) = \frac{2h}{\pi} \int_0^{\infty} \frac{(1 - \cos \omega T) \sin \omega t}{\omega} d\omega.$$

5. 在边界条件 $f(0) = 0$ 下, 把定义在 $(0, \infty)$ 上的函数 $f(x) = e^{-\lambda x}$ 展开为傅里叶积分.

解: 要满足边界条件 $f(0) = 0$, 必须将 $f(x)$ 展开为傅里叶正弦积分:

$$f(x) = \int_0^{\infty} B(\omega) \sin \omega x d\omega,$$

其中

$$B(\omega) = \frac{2}{\pi} \int_0^{\infty} f(x) \sin \omega \xi d\xi = \frac{2}{\pi} \int_0^{\infty} e^{-\lambda \xi} \sin \omega \xi d\xi$$

$$= -\frac{2}{\pi \omega} \int_0^{\infty} e^{-\lambda \xi} d(\cos \omega \xi)$$

$$\begin{aligned}
&= -\frac{2}{\pi\omega} e^{-\lambda\xi} \cos\omega\xi \Big|_0^\infty + \frac{2}{\pi\omega} \int_0^\infty \cos\omega\xi d e^{-\lambda\xi} \\
&= \frac{2}{\pi\omega} + \frac{2}{\pi\omega} \int_0^\infty \cos\omega\xi e^{-\lambda\xi} (-\lambda) d\xi \\
&= \frac{2}{\pi\omega} - \frac{2\lambda}{\pi\omega^2} \int_0^\infty e^{-\lambda\xi} d(\sin\omega\xi) \\
&= \frac{2}{\pi\omega} - \frac{2\lambda}{\pi\omega^2} e^{-\lambda\xi} \sin\omega\xi \Big|_0^\infty \\
&\quad + \frac{2\lambda}{\pi\omega^2} \int_0^\infty (-\lambda) e^{-\lambda\xi} \sin\omega\xi d\xi \\
&= \frac{2}{\pi\omega} - \frac{2\lambda^2}{\pi\omega^2} \int_0^\infty e^{-\lambda\xi} \sin\omega\xi d\xi.
\end{aligned}$$

把上式移项整理后得

$$\left(\frac{2}{\pi} + \frac{2\lambda^2}{\pi\omega^2}\right) \int_0^\infty e^{-\lambda\xi} \sin\omega\xi d\xi = \frac{2}{\pi\omega},$$

即
$$\int_0^\infty e^{-\lambda\xi} \sin\omega\xi d\xi = \frac{\frac{2}{\pi\omega}}{\frac{2}{\pi} + \frac{2\lambda^2}{\pi\omega^2}} = \frac{\omega}{\omega^2 + \lambda^2},$$

$$\therefore B(\omega) = \frac{2}{\pi} \int_0^\infty e^{-\lambda\xi} \sin\omega\xi d\xi = \frac{2}{\pi} \cdot \frac{\omega}{\omega^2 + \lambda^2},$$

故 $f(x)$ 的展开式为:

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{\omega}{\omega^2 + \lambda^2} \cos\omega x d\omega.$$

6. 在边界条件 $f'(0) = 0$ 下. 把定义在 $(0, \infty)$ 上的函数 $f(x) = 1 - H(x-a)$ 展为傅里叶积分.

解: 在边界条件 $f'(0) = 0$ 的要求下, $f(x)$ 必须展开为傅里叶余弦积分:

$$f(x) = \int_0^\infty A(\omega) \cos\omega x d\omega,$$

其中

$$\begin{aligned}
 A(\omega) &= \frac{2}{\pi} \int_0^{\infty} [1 - H(x-a)] \cos \omega x dx \\
 &= \frac{2}{\pi} \int_0^{\infty} \cos \omega x dx - \frac{2}{\pi} \int_0^{\infty} H(x-a) \cos \omega x dx \\
 &= \frac{2}{\pi} \int_0^a \cos \omega x dx + \frac{2}{\pi} \int_a^{\infty} \cos \omega x dx \\
 &\quad - \frac{2}{\pi} \int_a^{\infty} 1 \cdot \cos \omega x dx \\
 &= \frac{2}{\pi} \int_0^a \cos \omega x dx \\
 &= \frac{2}{\pi \omega} \sin \omega x \Big|_0^a = \frac{2}{\pi \omega} \sin \omega a,
 \end{aligned}$$

$$\begin{aligned}
 \therefore f(x) &= \int_0^{\infty} A(\omega) \cos \omega x d\omega \\
 &= \frac{2}{\pi} \int_0^{\infty} \frac{\sin \omega a}{\omega} \cos \omega x d\omega.
 \end{aligned}$$

7. 在实数形式的傅里叶积分(28·5)里, 把 $\cos \omega x$ 和 $\sin \omega x$ 按照欧勒公式用虚指数的指数函数 $e^{i\omega x}$ 和 $e^{-i\omega x}$ 表出, 验证实数形式的傅里叶积分(28·5)就化为复数形式的傅里叶积分(28·13)而且

$$C(\omega) = \frac{1}{2} [A(\omega) - iB(\omega)], \quad C(-\omega) = \frac{1}{2} [A(\omega) + iB(\omega)],$$

其中 $\omega > 0$.

证:

$$\begin{aligned}
 f(x) &= \int_0^{\infty} A(\omega) \cos \omega x d\omega + \int_0^{\infty} B(\omega) \sin \omega x d\omega \\
 &= \int_0^{\infty} \left[\frac{A(\omega)}{2} (e^{i\omega x} + e^{-i\omega x}) \right. \\
 &\quad \left. - \frac{i}{2} B(\omega) (e^{i\omega x} - e^{-i\omega x}) \right] d\omega
 \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\infty} \frac{1}{2} [A(\omega) - iB(\omega)] e^{i\omega x} d\omega \\
&\quad + \int_0^{\infty} \frac{1}{2} [A(\omega) + iB(\omega)] e^{-i\omega x} d\omega \\
&= \int_0^{\infty} [C(\omega) e^{i\omega x} + C(-\omega) e^{-i\omega x}] d\omega \\
&= \int_{-\infty}^{\infty} C(\omega) e^{i\omega x} d\omega, \text{ 此即(28.13)式.}
\end{aligned}$$

8. 验证延迟定理、位移定理和卷积定理.

(1) 延迟定理: 如果 $f(x)$ 的傅里叶变换式是 $C(\omega)$ 则 $f(x - x_0)$ 的傅里叶变换式是 $C(\omega) e^{-i\omega x_0}$.

证: $f(x - x_0)$ 的傅里叶变换式是 $\frac{1}{2\pi} \int_{-\infty}^{\infty} f(x - x_0) e^{-i\omega x} dx$,

在上述积分中作代换 $x - x_0 = \xi$ 即 $x = \xi + x_0$,

$$\begin{aligned}
\text{则 } f(x - x_0) \text{ 的傅里叶变换式} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\xi) e^{-i\omega \xi - i\omega x_0} d\xi \\
&= e^{-i\omega x_0} \cdot \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\xi) e^{-i\omega \xi} d\xi \\
&= C(\omega) e^{-i\omega x_0}.
\end{aligned}$$

(2) 位移定理: 如果 $f(x)$ 的傅里叶变换式是 $C(\omega)$ 则 $e^{i\omega_0 x} f(x)$ 的变换式是 $C(\omega - \omega_0)$,

证: $e^{i\omega_0 x} f(x)$ 的傅里叶变换式是

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{i\omega_0 x} e^{-i\omega x} dx &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-i(\omega - \omega_0)x} dx \\
&= C(\omega - \omega_0).
\end{aligned}$$

(3) 卷积定理: 如果 $f_1(x)$ 和 $f_2(x)$ 的傅里叶变换式是 $C_1(\omega)$ 和 $C_2(\omega)$ 则

$$\int_{-\infty}^{\infty} f_1(\xi) f_2(x - \xi) d\xi \text{ 的傅里叶变换式是 } 2\pi C_1(\omega) C_2(\omega)$$

$$\begin{aligned} \text{证: } & \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} dx \int_{-\infty}^{\infty} f_1(\xi) f_2(x-\xi) d\xi \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f_1(\xi) d\xi \int_{-\infty}^{\infty} f_2(x-\xi) e^{-i\omega x} dx. \end{aligned}$$

令 $x - \xi = t$, $dx = dt$, 则上式成为

$$\begin{aligned} & \frac{1}{2\pi} \int_{-\infty}^{\infty} f_1(\xi) d\xi \int_{-\infty}^{\infty} f_2(t) e^{-i(\xi+t)\omega} dt \\ &= 2\pi \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} f_1(\xi) e^{-i\omega\xi} d\xi \cdot \frac{1}{2\pi} \int_{-\infty}^{\infty} f_2(t) e^{-i\omega t} dt \right] \\ &= 2\pi C_1(\omega) \cdot C_2(\omega). \end{aligned}$$

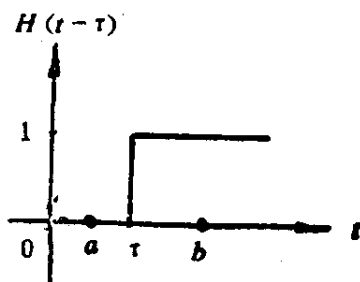
§29. δ 函数和它的傅里叶积分

1. 验证 $H'(t-\tau) = \delta(t-\tau)$, 求 $\delta(t-\tau)$ 的拉普拉斯变换像函数.

解: (1) 验证 $H'(t-\tau) = \delta(t-\tau)$

(i) 按照单位函数的定义

$$H(t-\tau) = \begin{cases} 0, & (t < \tau), \\ 1, & (t > \tau). \end{cases}$$



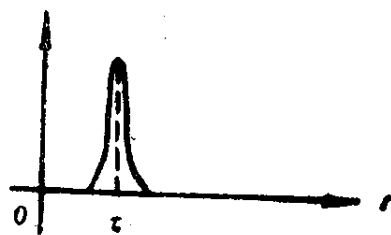
(a)

知当 $t > \tau$ 和 $t < \tau$, $H(t-\tau)$ 为常数, $H'(t-\tau)$

$$\therefore H'(t-\tau) = 0.$$

当 $t = \tau$ 时, $t = \tau$ 是 $H(t-\tau)$ 的第一类间断点.

一般取 $H(0) = \frac{1}{2}$, 则



(b)

图 7-4

$$\lim_{\Delta t \rightarrow +0} \frac{H(\Delta t) - H(0)}{\Delta t} = \lim_{\Delta t \rightarrow +0} \frac{1 - \frac{1}{2}}{\Delta t} = +\infty,$$

$$\lim_{\Delta t \rightarrow -0} \frac{H(\Delta t) - H(0)}{\Delta t} = \lim_{\Delta t \rightarrow -0} \frac{0 - \frac{1}{2}}{\Delta t} = +\infty,$$

$$\therefore H'(t-\tau) \Big|_{t=\tau} = \infty,$$

即

$$H'(t-\tau) = \begin{cases} 0, & (t \neq \tau), \\ \infty, & (t = \tau). \end{cases}$$

$$\begin{aligned} \text{(ii)} \quad \int_a^b H'(t-\tau) dt &= H(t-\tau) \Big|_a^b = H(b-\tau) - H(a-\tau) \\ &= \begin{cases} 0, & (a, b \text{ 都} < \tau \text{ 或都} > \tau), \\ 1, & (a < \tau < b), \end{cases} \end{aligned}$$

由 (i) 和 (ii) 知 $H'(t-\tau) = \delta(t-\tau)$.

(2) 求 $\delta(t-\tau)$ 的拉普拉斯变换象函数.

解: 方法1: 按照拉普拉斯变换的定义

$$\bar{\varphi}(p) = \int_0^{\infty} e^{-pt} \delta(t-\tau) dt = \begin{cases} 0, & (\tau < 0), \\ e^{-p\tau}, & (\tau \geq 0), \end{cases}$$

这是因为 $0 \leq t < \infty$, 当 $\tau < 0$ 时, $t-\tau > 0$,

此时 $\delta(t-\tau) = 0$, 因此 $\bar{\varphi}(p) = 0$.

而当 $\tau > 0$ 时, $0 \leq \tau < \infty$, 则根据 δ 函数的性质

$$\delta(t-\tau) \doteq \int_0^{\infty} e^{-pt} \delta(t-\tau) dt = e^{-pt} \Big|_{t=\tau} = e^{-p\tau}.$$

而当 $\tau = 0$ 时, 则有

$$\delta(t-\tau) = \delta(t) \doteq \int_0^{\infty} e^{-pt} \delta(t) dt = e^{-pt} \Big|_{t=0} = 1.$$

结果

$$\delta(t-\tau) \doteq \begin{cases} 0, & (\tau < 0), \\ e^{-p\tau}, & (\tau \geq 0), \end{cases}$$

$$\delta(t) \doteq 1.$$

$$\text{方法2: } \int_0^{\infty} e^{-pt} \delta(t-\tau) dt = \int_0^{\infty} e^{-pt} H'(t-\tau) dt$$

$$= e^{-pt} H(t-\tau) \Big|_{t=0}^{\infty} - \int_{\tau}^{\infty} -pe^{-pt} H(t-\tau) dt,$$

当 $\tau > 0$ 时, 上式可以写成

$$\begin{aligned} - \int_{\tau}^{\infty} -pe^{-pt} H(t-\tau) dt &= - \int_{\tau}^{\infty} -pe^{-pt} dt \\ &= -e^{-pt} \Big|_{\tau}^{\infty} = e^{-p\tau}. \end{aligned}$$

而当 $\tau < 0$ 时则 $\because H(t-\tau) = 1, H(-\tau) = 1,$

这时上式可写为

$$\begin{aligned} -1 - \int_0^{\infty} -pe^{-pt} dt &= -1 - e^{-pt} \Big|_0^{\infty} = 0, \\ \therefore \delta(t-\tau) &\doteq e^{-p\tau} H(\tau). \end{aligned}$$

2. 验证 § 28 例 2 的频谱 $B(\omega)$ (图 41) 于 $N \rightarrow \infty$ 就成为 $A\delta(\omega - \omega_0) - A\delta(\omega + \omega_0)$, 阐明这结果的物理意义.

$$\begin{aligned} \text{解: } \because B(\omega) &= \frac{2A\omega_0}{\pi(\omega^2 - \omega_0^2)} \sin\left(\frac{\omega}{\omega_0} N 2\pi\right) \\ &= \frac{A}{\pi} \frac{\sin\left(\frac{\omega}{\omega_0} N 2\pi\right)}{\omega - \omega_0} \\ &\quad - \frac{A}{\pi} \frac{\sin\left(\frac{\omega}{\omega_0} N 2\pi\right)}{\omega + \omega_0} \\ &= \frac{A}{\pi} \frac{\sin\left[\frac{2\pi N}{\omega_0} (\omega - \omega_0)\right]}{\omega - \omega_0} \\ &\quad - \frac{A}{\pi} \frac{\sin\left[\frac{2\pi N}{\omega_0} (\omega + \omega_0)\right]}{\omega + \omega_0} \end{aligned}$$

当 $N \rightarrow \infty$ 时, 即 $\frac{2\pi N}{\omega_0} \rightarrow \infty$

这时有限的正弦波列，便成为无限的正弦波列，而

$$\begin{aligned}
 B(\omega) &= A \lim_{N \rightarrow \infty} \frac{1}{\pi} \frac{\sin \frac{N2\pi}{\omega_0} (\omega - \omega_0)}{\omega - \omega_0} \\
 &\quad - A \lim_{N \rightarrow \infty} \frac{1}{\pi} \frac{\sin \frac{N2\pi}{\omega_0} (\omega_0 + \omega)}{\omega + \omega_0} \\
 &= A\delta(\omega - \omega_0) - A\delta(\omega + \omega_0).
 \end{aligned}$$

$$\therefore \lim_{k \rightarrow \infty} \frac{1}{\pi} \frac{\sin kx}{x} = \delta(x),$$

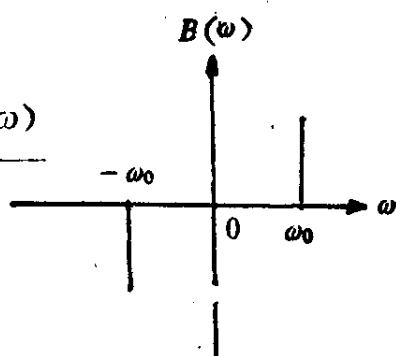


图 7-5

所以对于无限正弦波列，它的频谱成为两条线，一条位于 $\omega = \omega_0$ 处，另一条位于 $\omega = -\omega_0$ 处，振动成为单一圆频率 ω_0 的振动。

3. 把 $\delta(x)$ 展为实数形式的傅里叶积分。

解：

$\therefore \delta(x)$ 是偶函数，它的傅里叶积分可表示为：

$$\delta(x) = \int_0^{\infty} A(\omega) \cos \omega x d\omega,$$

而

$$\begin{aligned}
 A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} \delta(x) \cos \omega x dx \\
 &= \frac{1}{\pi} \cos(\omega \cdot 0) = \frac{1}{\pi},
 \end{aligned}$$

$$\therefore \delta(x) = \frac{1}{\pi} \int_0^{\infty} \cos \omega x d\omega,$$

或

$$\begin{aligned}
 \delta(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega x} d\omega \\
 &= \frac{1}{2\pi} \left[\int_{-\infty}^{\infty} \cos \omega x d\omega + i \int_{-\infty}^{\infty} \sin \omega x d\omega \right] \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos \omega x d\omega = \frac{1}{\pi} \int_0^{\infty} \cos \omega x d\omega.
 \end{aligned}$$

第三篇 数学物理方程

第八章 定解问题

§31. 数学物理方程的导出

1. 拿图51的B段弦作代表, 推导弦振动方程.

解: 取 x 到 $x+dx$ 的B段弦, 这段弦无纵向振动,

$\therefore$ 纵向合力为零

$$T_2 \cos \alpha_2 - T_1 \cos \alpha_1 = 0$$

B段弦的横振动方程为

$$T_1 \sin \alpha_1 - T_2 \sin \alpha_2 = u_{tt} \rho ds,$$

在小振动的情况下, 有 $\alpha_1 \approx \alpha_2 \approx$

$$0, \cos \alpha_1 \approx \cos \alpha_2 \approx 1,$$

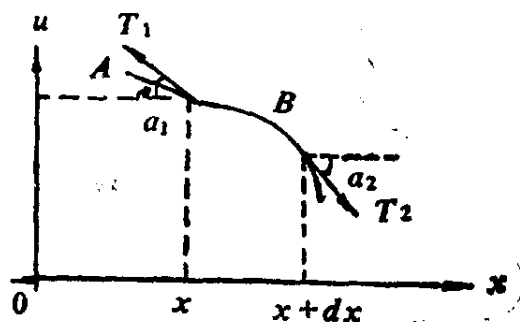


图 8-1

$$du \approx 0, ds = \sqrt{dx^2 + du^2} \approx dx,$$

$$\sin \alpha_1 \approx \tan \alpha_1, \sin \alpha_2 \approx \tan \alpha_2,$$

如图示, $\tan \alpha_1 = -u_x|_x$, $\tan \alpha_2 = -u_x|_{x+dx}$,

故在小振动的情况下, 运动方程为

$$\begin{cases} T_1 = T_2, \\ T_2 u_x|_{x+dx} - T_1 u_x|_x = u_{tt} \rho dx, \end{cases}$$

即

$$\frac{T \frac{\partial u}{\partial x} \Big|_{x+dx} - T \frac{\partial u}{\partial x} \Big|_x}{dx} = \rho u_{tt},$$

上式左边即 $T \frac{\partial u_x}{\partial x} = T \frac{\partial^2 u}{\partial x^2}$ 所以有 $u_{tt} - a^2 u_{xx} = 0$,

其中 $a^2 = \frac{T}{\rho}$.

2. 用均质材料制作细圆锥杆, 试推导它的纵振动方程.

解: 设想在圆锥杆上截取一小段 B , C 段对 B 的拉力是 $Ysu_x|_x$ 合力是

$$Ysu_x|_{x+dx} - Ysu_x|_x = Y \frac{\partial}{\partial x} (su_x) dx,$$

$\therefore B$ 段的质量是 $\rho s dx$,

$\therefore B$ 段的运动方程是

$$(\rho s dx) u_{tt} = Y \frac{\partial}{\partial x} (su_x) dx.$$

其中 $s = \pi r^2 = \pi (x \tan \alpha)^2$,

$\therefore$ 上式又可写为

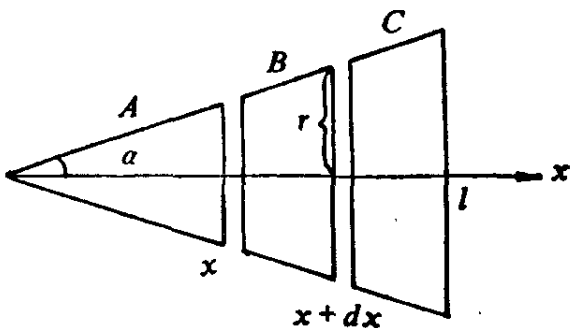


图 8-2

$$\rho (\pi x^2 \tan^2 \alpha) dx u_{tt} = Y \frac{\partial}{\partial x} [\pi x^2 \tan^2 \alpha u_x] dx,$$

$$x^2 u_{tt} = \frac{Y}{\rho} \frac{\partial}{\partial x} (x^2 u_x), \quad \text{令 } a^2 = \frac{Y}{\rho},$$

则得
$$u_{tt} = a^2 \cdot \frac{1}{x^2} \frac{\partial}{\partial x} (x^2 u_x) = \frac{a^2}{x^2} \frac{\partial}{\partial x} \left(x^2 \frac{\partial u}{\partial x} \right).$$

3. 弦在阻尼介质中振动, 单位长度弦所受阻力 $F = -Ru_t$ (比例常数 R 叫做阻力系数), 试推导弦在这阻尼介质中的振动方程.

解: 如 (1) 题图示, B 段弦所受力除了张力 T_1, T_2 外, 还受有阻力 F 的作用, 在小振动的情况下, 其运动方程为

$$T_1 \approx T_2,$$

$$(T_2 u_x|_{x+dx} - T_1 u_x|_x) - Ru_t dx = u_{tt} \rho dx,$$

$$T \frac{u_x|_{x+dx} - u_x|_x}{dx} - Ru_t = \rho u_{tt},$$

$$Tu_{xx} - Ru_t = \rho u_{tt}, u_{tt} - a^2 u_{xx} + \frac{R}{\rho} u_t = 0,$$

其中
$$a^2 = \frac{T}{\rho}.$$

4. 试推导一维和三维的热传导方程 (31、38) 和 (31、39)。

解: (1) 仍采用“隔离物体法”, 任取一小体积 B , 如图8-3所示: 在 Δt 时间内,

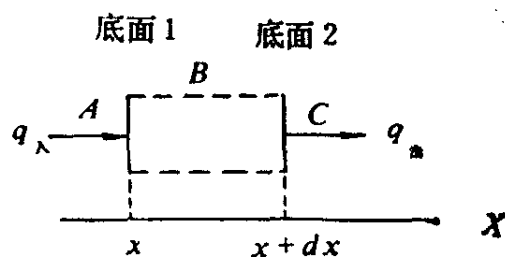


图 8-3

自 A 通过底面 1 流入 B 的热量为 q_λ

$s\Delta t$, 自 B 通过底面 2 流出的热量为 $q_{\text{出}}s\Delta t$, 热量的净流入量

为: $Q = (q_\lambda - q_{\text{出}})s\Delta t$, 但由于 $q_\lambda = -K \frac{\partial u}{\partial x} \Big|_x$, $q_{\text{出}} =$

$-K \frac{\partial u}{\partial x} \Big|_{x+\Delta x}$, 因净流入的热量为 $Q = \left(K \frac{\partial u}{\partial x} \Big|_{x+\Delta x} - K \frac{\partial u}{\partial x} \Big|_x \right) s\Delta t$, 上述净流入热量使 dx 区间内的物质温度升

高 du , 设物质的比热为 C , 密度为 ρ , 则

$$\begin{aligned} C\rho dx du &= (q_\lambda - q_{\text{出}})dt \\ &= \frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) dx dt, \end{aligned}$$

即 $C\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right)$ 这就是 (31~38) 式。

(2) 任取一小体积 dV , 位于 $x, x+dx; y, y+dy; z, z+dz$ 之间, 如图8-4, 先考虑在 $x, x+dx$ 两面与邻域交换热量, 在这两面上热流强度沿正 x 方向的分量为 $q_\lambda = -K u_x|_x$, $q_{\text{出}} = -K u_x|_{x+dx}$, 所以这一小块在 Δt 的时间所流入的热量为 $q_\lambda s\Delta t = -K$

$\times u_x|_x dydz \Delta t$, 所流出的热量为 $q_{\text{出}} S \Delta t = -K u_x|_{x+dx} dydz \Delta t$, 所以它所流入的净热量为

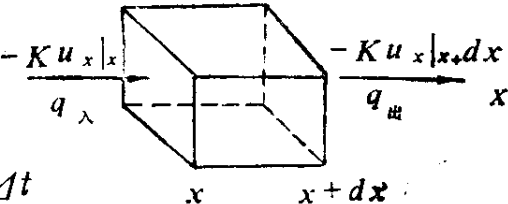
$$Q_1 = \left(K \frac{\partial u}{\partial x} \Big|_{x+dx} - K \frac{\partial u}{\partial x} \Big|_x \right) dydz \Delta t$$


图 8-4

$$= \frac{\partial}{\partial x} \left(K \frac{\partial u}{\partial x} \right) dx dydz \Delta t.$$

同样通过 y_1 $y+dy$, 两面所流入净热量是;

$$Q_2 = \frac{\partial}{\partial y} \left(K \frac{\partial u}{\partial y} \right) dy dx dz \Delta t,$$

通过 z_1 $z+dz$ 两面所流入净热量是;

$$Q_3 = \frac{\partial}{\partial z} \left(K \frac{\partial u}{\partial z} \right) dz dy dx \Delta t,$$

上述流入的净热量 $Q = Q_1 + Q_2 + Q_3$, 使小体积 dV 的温度升高 du , 如仍以 C 表小体积 $dV = dx dy dz$ 内物质的比热, ρ 表密度

$$\text{就有 } C \rho du dx dy dz = \left[\frac{\partial}{\partial x} \left(K \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(K \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(K \frac{\partial u}{\partial z} \right) \right] dx dy dz,$$

$$\text{即 } C \rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(K \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(K \frac{\partial u}{\partial z} \right),$$

此即 (31,39) 式.

5. 混凝土浇灌后逐渐放出“水化热”, 放热速率正比于当时尚储存着水化热密度 Q , 即 $\frac{dQ}{dt} = -\beta Q$, 试推导浇灌后的混凝土内的热传导方程.

解: 设浇灌后的混凝土中在初始时刻储存的水化热密度为

Q_0 , 则在 t 时刻它所储存的水化热密度为 $\int_{Q_0}^Q \frac{dQ}{Q} = \int_0^t -\beta dt$

$$\ln Q = -\beta t + \ln Q_0, \quad Q = Q_0 e^{-\beta t},$$

所以在 t 时刻它的放热速率为 $\frac{dQ}{dt} = -\beta Q_0 e^{-\beta t}$.

其它分析与上题类似. 只是现在还有热源分布于物质之中在单位时间内于单位体积中放出的热量即为 $-\beta Q_0 e^{-\beta t}$, 故上题的热传导方程改为:

$$\rho C u_t = \left[\frac{\partial}{\partial x} (K u_x) + \frac{\partial}{\partial y} (K u_y) + \frac{\partial}{\partial z} (K u_z) \right] + Q_0 \beta e^{-\beta t},$$

$$\text{即 } C \rho u_t - \frac{\partial}{\partial x} (K u_x) + \frac{\partial}{\partial y} (K u_y) + \frac{\partial}{\partial z} (K u_z) = Q_0 \beta e^{-\beta t}.$$

6. 均质导线电阻率为 r , 通过均匀分布的直流电, 电流密度为 j , 试推导导线内的热传导方程.

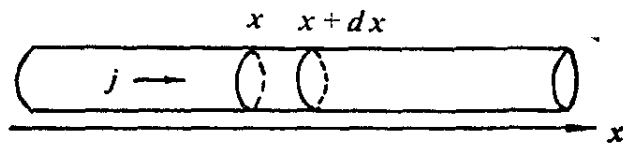


图 8-5

解: 设均质导线面积为 S , 热量沿电流方向传播, 先考虑一维情况, 取 x 方向与电流和热量传播方向相同, 在 x 处和 $x+dx$ 处研究问题.

设 q 为单位时间里通过单位截面的热量, 由热传导定律则有 $q = -K \frac{\partial u}{\partial x}$, 式中 u 是温度, K 为热传导系数. 在 dt 的时间里, 流入体元 $dV = S dx$ 的净热量为

$$\begin{aligned} & \left\{ -K \frac{\partial u}{\partial x} \Big|_x - \left[-K \frac{\partial u}{\partial x} \right]_{x+dx} \right\} S dt \\ &= \frac{\partial}{\partial x} \left(K \frac{\partial u}{\partial x} \right) S dt dx, \end{aligned}$$

由于电流密度为 j ，电阻率为 r 的导体在体元上产生的焦耳热为

$$J^2 R dt = [jS]^2 \left[r \frac{dx}{S} \right] dt = j^2 r S dx dt.$$

小体积 dV 中因温升而需要的热量 = 净流入体元中的热量 + 热源在体元中产生的热量

$$C\rho dV du = \frac{\partial}{\partial x} \left(K \frac{\partial u}{\partial x} \right) S dx dt + j^2 r S dx dt,$$

$$C\rho du/dt = \frac{\partial}{\partial x} \left(K \frac{\partial u}{\partial x} \right) + j^2 r. \quad \text{或} \quad C\rho \frac{\partial u}{\partial t} - K \frac{\partial^2 u}{\partial x^2} = j^2 r.$$

如截面积 S 较大时，应该三维空间的热传导，这时泛定方程为

$$C\rho u_t - K \Delta u = j^2 r.$$

7. 长为 l 的柔软均质绳索，一端固定在以匀速 ω 转动的竖直轴上，由于惯性离心力的作用，这弦的平衡位置应是水平线，试推导此绳相对于水平线的横振动方程。

解：如图示，在小振动的情况下， $\sin \alpha \approx \tan \alpha = \frac{\partial u}{\partial x}$ ，

$\cos \alpha = 1$ ，因而从 x 到 $x+dx$ 这段绳的运动方程为

$$T_2 u_x|_{x+dx} - T_1 u_x|_x = \rho dx \cdot u_{tt},$$

即

$$(T u_x)|_{x+dx} - (T u_x)|_x = u_{tt} \rho dx,$$

为了求出在 x 处的张力 $T(x)$ ，需考虑从 x 到 l 的一段绳上的惯性离心力的作用，设在 x 处的张力为 $T(x)$ ，则

$$\begin{aligned} T(x) &= \int_x^l \omega^2 x \rho dx \\ &= \frac{1}{2} \rho \omega^2 (l^2 - x^2), \end{aligned}$$

$$\therefore \left[\frac{1}{2} \rho \omega^2 (l^2 - x^2) \right]$$

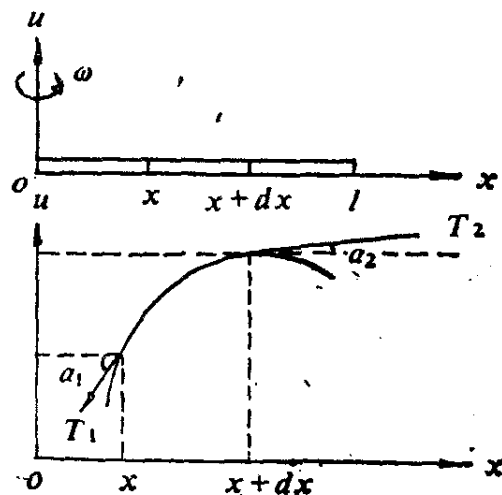


图 8-6

$$\begin{aligned} & \times u_x \Big|_{x+dx} - \left[\frac{1}{2} \rho \omega^2 (l^2 - x^2) \right] u_x \Big|_x \\ & = u_{tt} \rho dx, \\ \text{即 } \rho u_{tt} & = \frac{\left[\frac{1}{2} (l^2 - x^2) \rho \omega^2 u_x \right] \Big|_{x+dx} - \left[\frac{1}{2} (l^2 - x^2) \rho \omega^2 u_x \right] \Big|_x}{dx} \end{aligned}$$

$$\begin{aligned} & = \frac{1}{2} \rho \omega^2 \frac{\partial}{\partial x} [(l^2 - x^2) u_x], \\ u_{tt} - \frac{1}{2} \omega^2 \frac{\partial}{\partial x} [(l^2 - x^2) u_x] & = 0. \end{aligned}$$

8. 长为 l 的柔软均质重绳，上端固定在以匀速 ω 转动的竖直轴上，由于重力作用，绳的平衡位置应是竖直线，试推导此线相对于竖直线的横振动方程。

解：如图示，在小振动的情

况下， $\sin \alpha \approx \tan \alpha = \frac{\partial u}{\partial x}$ ， $ds \approx dx$ ， $\cos \alpha \approx 1$ ，再取 x 到 $x+dx$ 一段绳的运动方程是

$$T_2 u_x \Big|_{x+dx} - T_1 u_x \Big|_x + F = \rho dx \cdot u_{tt},$$

其中 F 是 dx 段弦所受的惯性离心力， $F = \rho dx u \omega^2$ ，在 x 端还受有张力（此处为重力）的作用，张力 T 为 $T = \int_x^l \rho g dx$ ，

以此代入运动方程得：

$$[(l-x) \rho g u_x]_{x+dx} - [(l-x) \rho g u_x]_x + u \omega^2 \rho dx = u_{tt} \rho dx.$$

$$\text{即 } u_{tt} = \frac{[(l-x) g u_x]_{x+dx} - [(l-x) g u_x]_x}{dx} + u \omega^2.$$

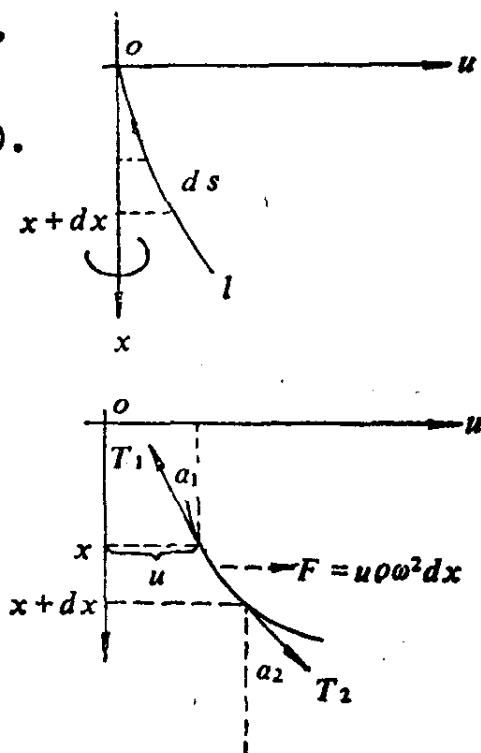


图 8-7

$$u_{tt} - g \frac{\partial}{\partial x} [(l-x)u_x] = u\omega^2$$

或 $u_{tt} - g \frac{\partial}{\partial x} [(l-x)u_x] - u\omega^2 = 0.$

9. 推导均匀圆柱的扭转振动方程,
杆半径为 R , 切变模量为 N

解: 如果沿柱轴的切变是均匀的,
在离轴 r 处的切变角为 $\frac{r\theta}{H}$. 事实上未

必均匀, 所以切变角应改用 $r \frac{\partial \theta}{\partial x}$ 计

算, 从而离轴 r 处的切应力为 $Nr\theta_x$,

从而

$$\begin{aligned} \text{力矩} &= \int_0^{2\pi} d\varphi \int_0^R r (Nr\theta_x) r dr \\ &= 2\pi N\theta_x \int_0^R r^3 dr = \frac{\pi N\theta_x R^4}{2}, \end{aligned}$$

设单位长度对轴的转动惯量为 I , 则 $x-x+dx$ 段的动量矩定理给出

$$I dx \frac{\partial^2 \theta}{\partial t^2} = \frac{\pi N R^4}{2} (\theta_x|_{x+dx} - \theta_x|_x)$$

即
$$\frac{\partial^2 \theta}{\partial t^2} = \frac{\pi N R^4}{2I} \frac{\partial^2 \theta}{\partial x^2}.$$

10. 推导水槽中的重力波方程, 水槽长 l , 截面为矩形、两端由刚性平面封闭, 槽中水在平衡时的深度为 h .

解: 取 x 轴沿水槽的长度方向, 水槽长为 l , 宽为 S , 将水面与静止水面的高度差记作 η , η 随 x 而异, 且随 t 而变.

取 x 处的截面与 $x+dx$ 处的截面之间的水来考虑, 由于这两处的 η 不同, 所以这两部分水前后方所受的压力不等, 其 x 方向

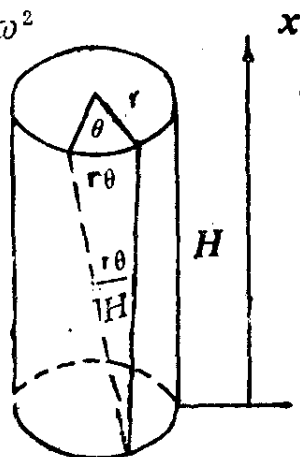


图 8-8

运动方程为;

$$(\rho S dx) u_{tt} = [-\rho g \eta|_{x+dx} + \rho g \eta|_x] S$$

$$= -\rho g S \eta_x dx,$$

即

$$u_{tt} = -g \eta_x. \quad (1)$$

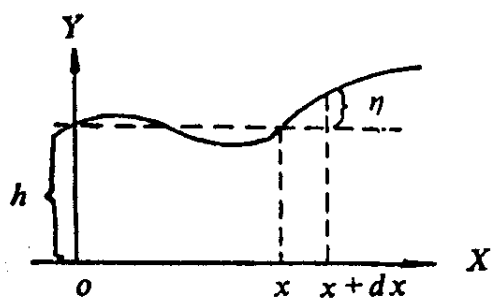


图 8-9(a)

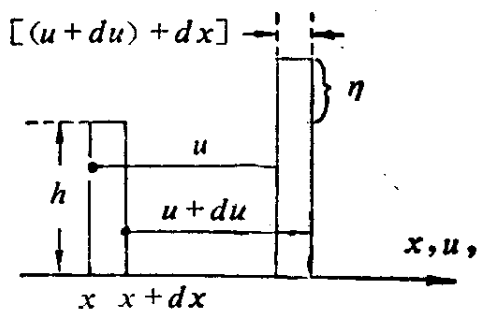


图 8-9(b)

我们还需要找一个方程,以便与(1)消去 η ,事实上,水是不可压缩的,试考察静止时在 $x-x+dx$ 之间的水,其体积为 $Shdx$,在运动中,这部分水的高度变为 $(h+\eta)$,厚度变为

$$[(u+du)+dx-u] = du+dx,$$

从而体积变为 $S(h+\eta)(du+dx)$,

由于水的不可压缩性,所以

$$Shdx = S(h+\eta)(du+dx)$$

$$= S[\eta du + hdu + \eta dx + hdx],$$

$$0 = \eta du + hdu + \eta dx,$$

略去二阶小量 ηdu ,上式给出 $\eta = -hu_x$, (2)

从(1)(2)消去 η ,得到

$$u_{tt} = gh u_{xx}. \quad (3)$$

这就是重力波的纵向运动方程。

再将(2)式微分 $\eta_{tt} = -h \frac{\partial^2}{\partial t^2} u_x = -h \frac{\partial}{\partial x} u_{tt}$, (4)

以(4)代入(1)有 $\eta_{tt} = gh \eta_{xx}$, (5)

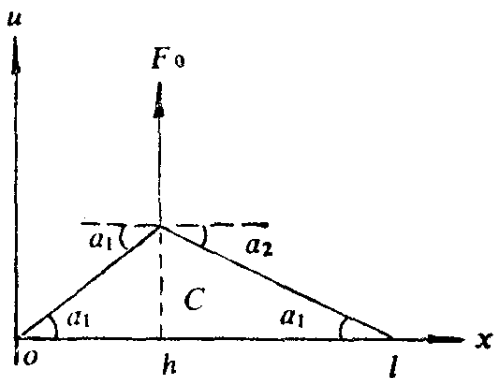
这是重力波的横向运动方程。

§32. 定 解 条 件

1. 长为 l 的均匀弦，两端 $x=0, x=l$ 固定，弦中张力为 T_0 ，在 $x=h$ 点，以横向力 F_0 拉弦，达到稳定后放手任其自由振动，写出初始条件。

解：由点斜式方程，弦的初始位移为

$$u|_{t=0} = \begin{cases} \frac{c}{h}x, & (0 \leq x \leq h), \\ \frac{c}{l-h}(l-x), & (h \leq x \leq l), \end{cases} \quad \text{图 8-10}$$



其中 c 为弦在 $x=h$ 点的初始位移。

如图，因为是小振动，所以 $\sin \alpha_1 \approx \tan \alpha_1 = \frac{c}{h}$,

$$\sin \alpha_2 \approx \tan \alpha_2 = \frac{c}{l-h},$$

$$\cos \alpha_2 \approx \cos \alpha_1 \approx 1, \quad dS \approx dx,$$

然后写出力平衡方程式 $F_0 - T_1 \sin \alpha_1 - T_2 \sin \alpha_2 = 0$ 。

$$T_2 \cos \alpha_2 - T_1 \cos \alpha_1 = 0, \quad T_2 \approx T_1 = T,$$

$$F_0 = T \frac{c}{h} + T \frac{c}{l-h},$$

解得 $C = \frac{F_0 h (l-h)}{Tl}$ ，以 C 代入初始位移中即得：

$$u|_{t=0} = \begin{cases} \frac{F_0 (l-h)}{T_0 l} x, & (0 \leq x \leq h), \\ \frac{F_0 h}{T_0 l} (l-x), & (h \leq x \leq l). \end{cases}$$

初始速度 $u_t|_{t=0} = 0$.

2. 长为 l 的均匀杆两端受拉力 F_0 作用而纵振动, 写出边界条件.

解: 杆的两端所受的拉力 F_0 等于这两端面所受的杨氏弹性力



图 8-11(a)

$$YS \frac{\partial u}{\partial n} \Big|_{x=0} = -YS \frac{\partial u}{\partial x} \Big|_{x=0} = -F_0,$$

$$\therefore YS \frac{\partial u}{\partial x} \Big|_{x=0} = F_0,$$

$$YS \frac{\partial u}{\partial n} \Big|_{x=l} = YS \frac{\partial u}{\partial x} \Big|_{x=l} = F_0.$$

3. 长为 l 的均匀杆, 两端有恒定热流进入, 其强度为 q_0 , 写出这个热传导问题的边界条件.

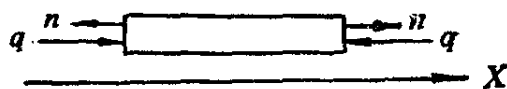


图 8-11(b)

解: 在边界上有 $-K \frac{\partial u}{\partial n} \Big|_{\Sigma} = q_n$,

在 $x=l$ 端,

$$-K \frac{\partial u}{\partial n} \Big|_{x=l} = -K \frac{\partial u}{\partial x} \Big|_{x=l} = q_n = -q,$$

即

$$K \frac{\partial u}{\partial x} \Big|_{x=l} = q.$$

在 $x=0$ 端, $-K \frac{\partial u}{\partial n} \Big|_{x=0} = K \frac{\partial u}{\partial x} \Big|_{x=0} = q_n = -q,$

即

$$K \frac{\partial u}{\partial x} \Big|_{x=0} = -q.$$

4. 半径为 R 而表面熏黑的金属长圆柱体, 受到阳光照射. 阳光方向垂直于柱轴, 热流强度为 M , 写出这个圆柱的热传导

问题的边界条件。

解：设圆柱体周围温度为 θ ，这个圆柱体的表面系熏黑，它的吸收系数为 1，它可以全部吸收垂直照射阳光的热流的法向部分，即 $M \sin \varphi$ ，同时又自然冷却，散热的热流强度为 $H(u - \theta)$ ，因此，

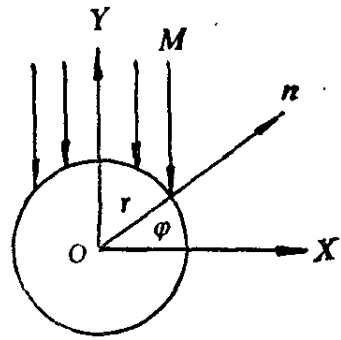


图 8-12

$$-K \frac{\partial u}{\partial n} \Big|_{\rho=R} + M \sin \varphi = H(u - \theta) \Big|_{\rho=R},$$

$$\text{即 } -K \frac{\partial u}{\partial r} \Big|_{\rho=R} + M \sin \varphi = H(u - \theta) \Big|_{\rho=R},$$

$$\text{即 } \left[K \frac{\partial u}{\partial r} + H(u - \theta) \right]_{\rho=R} = \begin{cases} M \sin \varphi, & 0 < \varphi < \pi, \\ 0, & \pi \leq \varphi \leq 2\pi, \end{cases}$$

不妨取圆柱周围环境的温度作为温标的零点，这样作则式中 $\theta = 0$ 。

5. 习题 1 是否需要衔接条件？

解：弦在振动时， F_0 已不作用，所以不需要衔接条件，若弦在振动时，力 F_0 仍然作用着，就要衔接条件。

6. 一根杆由截面相同的两段连接而成。两段的材料不同，杨氏模量分别是 Y^I 、 Y^{II} 密度分别为 ρ_1 、 ρ_2 试写出衔接条件。

解：设二段杆的接点为 $x = 0$ ，在连接处位移 u 是连续的，所以有

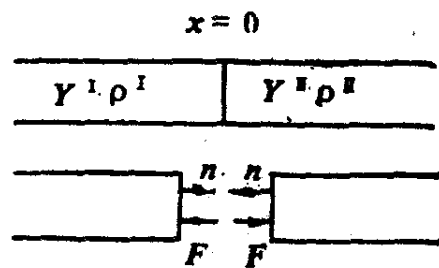


图 8-13

$$u^I|_{x=0} = u^{II}|_{x=0}, \quad (1)$$

又在连接处两方的作用力为

$$Y^I S \frac{\partial u^I}{\partial n} \Big|_{x=0} = Y^{II} S \frac{\partial u^{II}}{\partial n} \Big|_{x=0}$$

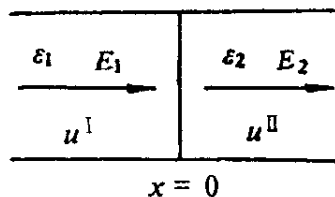
$$\text{与 } Y^{\text{II}} S \frac{\partial u^{\text{II}}}{\partial n} \Big|_{x+0} = -Y^{\text{II}} S \frac{\partial u^{\text{II}}}{\partial x} \Big|_{x+0},$$

这两力是作用力与反作用力所以

$$Y^{\text{I}} S \frac{\partial u^{\text{I}}}{\partial x} \Big|_{x-0} = Y^{\text{II}} S \frac{\partial u^{\text{II}}}{\partial x} \Big|_{x+0}, \quad (2)$$

(1) 和 (2) 就是衔接条件。

7. 写出静电场中电介质表面的衔接条件。



解：在电介质表面，电势是连续的。

$$u^{\text{I}} \Big|_{x-0} = u^{\text{II}} \Big|_{x+0}, \quad (1) \quad \text{图 8-14}$$

又电位移法向分量连续

$$D = \epsilon_1 E_1 = \epsilon_2 E_2 \text{ 即 } \epsilon_1 \frac{\partial u^{\text{I}}}{\partial x} \Big|_{x-0} = \epsilon_2 \frac{\partial u^{\text{II}}}{\partial x} \Big|_{x+0}. \quad (2)$$

8. 一根导热杆由两段构成，二段热传导系数，比热、密度分别是 $K^{\text{I}}, C^{\text{I}}, \rho^{\text{I}}$ 和 $K^{\text{II}}, C^{\text{II}}, \rho^{\text{II}}$ ，初始温度是 u_0 ，然后保持两端温度为零，试把这个热传导问题表为定解问题。

解：



图 8-15

定解问题为

$$\begin{cases} u_t^{\text{I}} - \frac{K^{\text{I}}}{C^{\text{I}} \rho^{\text{I}}} u_{xx}^{\text{I}} = 0, \\ u^{\text{I}}(x_1, t) = 0, \\ u^{\text{I}}(x, 0) = u_0, \end{cases} \quad (x_1 < x < x_2).$$

$$\begin{cases} u_t^{\text{II}} - \frac{K^{\text{II}}}{C^{\text{II}} \rho^{\text{II}}} u_{xx}^{\text{II}} = 0, \\ u^{\text{II}}(x_3, t) = 0, \\ u^{\text{II}}(x, 0) = u_0, \end{cases} \quad (x_2 < x < x_3).$$

衔接条件,

$$u^I|_{x_2=0} = u^{II}|_{x_2=0},$$

$$K^I \frac{\partial u^I}{\partial x} \Big|_{x_2=0} = K^{II} \frac{\partial u^{II}}{\partial x} \Big|_{x_2=0}.$$

§ 33. 二阶线性偏微分方程的分类

1. 把下列方程化为标准形式:

$$(1) \quad au_{xx} + 2au_{xy} + au_{yy} + bu_x + cu_y + u = 0.$$

解: 因为 $a_{12}^2 - a_{11}a_{22} = a^2 - a \cdot a = 0$ 所以该方程是抛物型的

$$\text{特征方程 } \frac{dy}{dx} = \frac{a \pm \sqrt{a^2 - a^2}}{a} = 1,$$

亦即只有一族实的特征线 $y - x = \text{常数}$.

在这种情况下, 我们设 $\xi = y - x$, $\eta = x$ (或令 $\eta = y$, 总之, 此处 η 是与 ξ 无关的任一函数, 当然宜取最简单的函数形式 $\eta = x$ 或 $\eta = y$).

$$\text{解一: 用抛物型方程的标准形式 } \eta_{yy} = -\frac{1}{A_{22}}[B_1u_\xi +$$

$B_2u_\eta + Cu + F]$ 先算出:

$$\left\{ \begin{aligned} A_{22} &= a_{11}\eta_x^2 + 2a_{12}\eta_x\eta_y + a_{22}\eta_y^2 \\ &= a + 2a \cdot 0 + a \cdot 0 = a, \\ B_1 &= a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y \\ &= a \cdot 0 + 2a \cdot 0 + a \cdot 0 + b(-1) + c \cdot 1 = c - b, \\ B_2 &= a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y \\ &= a \cdot 0 + 2a \cdot 0 + a \cdot 0 + b \cdot 1 + c \cdot 0 = b, \\ C &= 1, \quad F = 0, \end{aligned} \right.$$

$$\therefore u_{\eta\eta} = -\frac{1}{a}[(-b+c)u_\xi + bu_\eta + u],$$

即 $u_{\eta\eta} + \frac{c-b}{a} u_{\xi} + \frac{b}{a} u_{\eta} + \frac{1}{a} u = 0.$

解二：应用特征线方程，作自变量变换，求出

$$\begin{cases} u_x = -u_{\xi} + u_{\eta}, & u_y = u_{\xi}, \\ u_{xx} = u_{\xi\xi} - u_{\xi\eta} - u_{\eta\xi} + u_{\eta\eta} = u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}, \\ u_{xy} = -u_{\xi\xi} + u_{\eta\xi}, & u_{yy} = u_{\xi\xi}, \end{cases}$$

代入原方程得， $au_{\eta\eta} + (c-b)u_{\xi} + bu_{\eta} + u = 0.$

$$(2) \quad u_{xx} - 2u_{xy} - 3u_{yy} + 2u_x + 6u_y = 0,$$

解：因为 $a_{12}^2 - a_{11}a_{22} = 4 > 0$ ，所以该方程是双曲型的

其特征方程为 $\frac{dy}{dx} = \frac{-1 \pm \sqrt{1+3}}{1} = \begin{cases} 1, \\ -3, \end{cases}$

特征线为 $x - y = c_1$ 和 $x + 3y = c_2$.

故可令 $\xi = x - y$, $\eta = 3x + y$ ，在双曲型方程的标准型式，

$$u_{\xi\eta} = -\frac{1}{2A_{12}}[B_1u_{\xi} + B_2u_{\eta} + cu + F] \text{ 中，先算出，}$$

$$\begin{cases} A_{12} = a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y \\ \quad = 1 \cdot 1 \cdot 3 + (-1)[1 \cdot 1 + (-1) \cdot 3] + (-3)(-1) \cdot 1 \\ \quad = 3 + 2 + 3 = 8, \\ B_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y = -4, \\ B_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y = 12, \\ C = 0, \quad F = 0, \end{cases}$$

$$\therefore u_{\xi\eta} = -\frac{1}{16}[-4u_{\xi} + 12u_{\eta}], \text{ 即 } 4u_{\xi\eta} - u_{\xi} + 3u_{\eta} = 0,$$

$$(3) \quad u_{xx} + 4u_{xy} + 5u_{yy} + u_x + 2u_y = 0.$$

解：因为 $a_{12}^2 - a_{11}a_{22} = -1 < 0$ ，所以该方程是椭圆型的，

其特征方程为： $\frac{dy}{dx} = \frac{2 \pm \sqrt{2^2 - 5}}{1} = 2 \pm i,$

特征线为： $(2+i)x - y = c_1$ 和 $(2-i)x - y = c_2,$

故可令 $\xi = (2+i)x - y$, $\eta = (2-i)x - y$; 为计算方便, 又令

$$\begin{cases} \alpha = \frac{1}{2}(\xi + \eta) = 2x - y, \\ \beta = \frac{1}{2i}(\xi - \eta) = x, \end{cases}$$

在椭圆型方程的标准形式:

$$u_{\alpha\alpha} + u_{\beta\beta} = -\frac{1}{A_{12}}[(B_1 + B_2)u_\alpha + i(B_2 - B_1)u_\beta + 2cu + F]$$

中, 先算出,

$$\begin{cases} A_{12} = a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y = 2, \\ B_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y = i, \\ B_2 = a_{11}\xi_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y = -i, \\ C = 0, \quad F = 0, \end{cases}$$

$$\therefore u_{\alpha\alpha} + u_{\beta\beta} = -\frac{1}{2}[i(-2i)u_\beta], \text{ 即 } u_{\alpha\alpha} + u_{\beta\beta} + u_\beta = 0.$$

改变自变量 α, β 的记号为 ξ, η , 则 $u_{\xi\xi} + u_{\eta\eta} + u_\eta = 0$.

$$(4) \quad u_{xx} + yu_{yy} = 0.$$

$$\text{解: } a_{12}^2 - a_{11}a_{22} = -y.$$

(i) 如 $y < 0$, 则 $a_{12}^2 - a_{11}a_{22} = -y > 0$, 该方程为双曲型.

其特征方程为: $\frac{dy}{dx} = \sqrt{-y}$, 和 $\frac{dy}{dx} = -\sqrt{-y}$,

其特征线为: $x + 2\sqrt{-y} = c_1$ 和 $x - 2\sqrt{-y} = c_2$,

故可令: $\xi = x + 2\sqrt{-y}$, $\eta = x - 2\sqrt{-y}$.

在双曲型方程的标准形式

$$u_{\xi\eta} = -\frac{1}{2A_{12}}[B_1u_\xi + B_2u_\eta + cu + F] \text{ 中, 先算出}$$

$$\begin{aligned}
 A_{12} &= a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y \\
 &= 1 + 0 + y\left(-\frac{1}{\sqrt{-y}}\right)\left(-\frac{1}{\sqrt{-y}}\right) = 2, \\
 B_1 &= a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y \\
 &= -\frac{1}{2\sqrt{-y}} = \frac{2}{\eta - \xi}, \\
 B_2 &= a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y \\
 &= \frac{1}{2\sqrt{-y}} = \frac{2}{\xi - \eta}, \\
 C &= 0, \quad F = 0.
 \end{aligned}$$

所以原方程化为 $(\xi - \eta)u_{\xi\eta} + \frac{1}{2}(u_\xi - u_\eta) = 0$.

(ii) 如 $y > 0$, 则 $a_{12}^2 - a_{11}a_{22} = -y < 0$, 该方程为椭圆型.

其特征方程为: $\frac{dy}{dx} = \sqrt{y}i$ 和 $\frac{dy}{dx} = -\sqrt{y}i$,

特征线为: $x + 2\sqrt{y}i = c_1$ 和 $x - 2\sqrt{y}i = c_2$.

故可令 $\xi = x + 2\sqrt{y}i$, $\eta = x - 2\sqrt{y}i$, 为方便计, 又令

$$\alpha = \frac{1}{2}(\xi + \eta) = x, \quad \beta = \frac{1}{2i}(\xi - \eta) = 2\sqrt{y} \text{ 或 } y = \frac{\beta^2}{4},$$

则 $u_{xx} = u_{\alpha\alpha}$, $u_y = u_\beta \frac{1}{\sqrt{y}}$, $u_{yy} = -\frac{1}{2y^{3/2}}u_\beta + u_{\beta\beta} \frac{1}{y}$,

原方程为 $u_{\alpha\alpha} + u_{\beta\beta} - \frac{1}{2\sqrt{y}}u_\beta = 0$,

即 $u_{\alpha\alpha} + u_{\beta\beta} - \frac{1}{\beta}u_\beta = 0$.

把符号 α, β 换成 ξ, η , 就有 $u_{\xi\xi} + u_{\eta\eta} - \frac{1}{\eta}u_\eta = 0$.

(5) $u_{xx} + xu_{yy} = 0$.

解: $a_{12}^2 - a_{11}a_{22} = -x$, 所以特征方程为 $\frac{dy}{dx} = -\sqrt{x}$.

(i) 如 $x < 0$, 则 $a_{12}^2 - a_{11}a_{22} = -x > 0$, 所以方程是双曲型的.

特征线: $y + \frac{2}{3}(-x)^{3/2} = C_1$ 和 $y - \frac{2}{3}(-x)^{3/2} = C_2$,

或改写为 $\frac{3}{2}y + (-x)^{3/2} = C_1$ 及 $\frac{3}{2}y - (-x)^{3/2} = C_2$,

$$\text{令 } \xi = \frac{3}{2}y + (-x)^{3/2}, \quad (1)$$

$$\eta = \frac{3}{2}y - (-x)^{3/2}, \quad (2)$$

$$u_{\xi\eta} = -\frac{1}{2A_{12}}[B_1u_\xi + B_2u_\eta + cu + F],$$

$$A_{12} = -\frac{3}{2}\sqrt{-x} \cdot \frac{3}{2}\sqrt{-x} + x \frac{3}{2} \cdot \frac{3}{2} = \frac{9}{2}x,$$

$$B_1 = \frac{3}{4} \cdot \frac{1}{\sqrt{-x}}, \quad B_2 = -\frac{3}{4} \frac{1}{\sqrt{-x}},$$

$$\begin{aligned} u_{\xi\eta} &= -\frac{1}{9x} \frac{3}{4\sqrt{-x}}(u_\xi - u_\eta) \\ &= \frac{1}{12(-x)^{3/2}}(u_\xi - u_\eta). \end{aligned} \quad (3)$$

将(1)减(2)式得

$$\xi - \eta = 2(-x)^{3/2},$$

$$\therefore 12(-x)^{3/2} = 6(\xi - \eta), \text{ 代入 (3)}$$

$$\text{就化成标准形 } u_{\xi\eta} - \frac{1}{6(\xi - \eta)}(u_\xi - u_\eta) = 0.$$

(ii) 如 $x > 0$, 则 $-x < 0$, 则 $a_{12}^2 - a_{11}a_{22} = -x < 0$, 则此方程为椭圆型.

$$\text{特征方程为: } \frac{dy}{dx} = \pm \sqrt{x}i,$$

特征线为: $\frac{3}{2}y + ix^{3/2} = C_1$ 和 $\frac{3}{2}y - ix^{3/2} = C_2$,

令 $\xi = \frac{3}{2}y$, $\eta = -x^{3/2}$,

则 $u_y = u_\xi \xi_y = \frac{3}{2}u_\xi$, $u_{yy} = \frac{9}{4}u_{\xi\xi}$,

$$u_x = u_\eta \left(-\frac{3}{2}x^{\frac{1}{2}} \right) = -\frac{3}{2}u_\eta x^{\frac{1}{2}},$$

$$u_{xx} = u_{\eta\eta} \cdot \frac{1}{4} \cdot x - u_\eta \frac{1}{4\sqrt{x}},$$

方程为 $\frac{9}{4}xu_{\eta\eta} + \frac{9}{4}xu_{\xi\xi} - u_\eta \frac{3}{4\sqrt{x}} = 0$,

$$u_{\xi\xi} + u_{\eta\eta} - \frac{1}{3x^{3/2}}u_\eta = 0 \text{ 即 } u_{\xi\xi} + u_{\eta\eta} + \frac{1}{3\eta}u_\eta = 0.$$

(6) $y^2 u_{xx} + x^2 u_{yy} = 0$.

解: $a_{12}^2 - a_{11}a_{22} = -x^2y^2 < 0$ 故方程是椭圆型,

特征方程: $\frac{dy}{dx} = \frac{\pm \sqrt{-x^2y^2}}{y^2} = \pm i \frac{x}{y}$,

特征线为: $y^2 + ix^2 = C_1$, $y^2 - ix^2 = C_2$,

令 $\xi = y^2$, $\eta = x^2$, 则有

$$u_x = u_\eta \cdot 2x, \quad u_{xx} = 4x^2 u_{\eta\eta} + 2u_\eta,$$

$$u_y = u_\xi \cdot 2y, \quad u_{yy} = 4y^2 u_{\xi\xi} + 2u_\xi,$$

原方程变为 $4x^2y^2u_{\eta\eta} + 2y^2u_\eta + 4y^2x^2u_{\xi\xi} + 2x^2u_\xi = 0$,

$$u_{\xi\xi} + u_{\eta\eta} + \frac{1}{2y^2}u_\xi + \frac{1}{2x^2}u_\eta = 0,$$

$$u_{\xi\xi} + u_{\eta\eta} + \frac{1}{2\xi}u_\xi + \frac{1}{2\eta}u_\eta = 0.$$

(7) $4y^2u_{xx} - e^{2x}u_{yy} - 4y^2u_x = 0$.

解: $a_{12}^2 - a_{11}a_{22} = 4y^2e^{2x} > 0$ 故方程为双曲型,

特征方程: $\frac{dy}{dx} = \frac{\pm \sqrt{4y^2 e^{2x}}}{4y^2} = \pm \frac{e^x}{2y},$

特征线: $y^2 + e^x = C_1, \quad y^2 - e^x = C_2.$

令 $\xi = y^2 + e^x, \quad (1)$

$\eta = y^2 - e^x, \quad (2)$

则 $u_x = u_\xi e^x + u_\eta (-e^x) = e^x (u_\xi - u_\eta),$

$$\begin{aligned} u_{xx} &= e^x (u_\xi - u_\eta) + e^x [u_{\xi\xi} e^x + u_{\xi\eta} (-e^x) - u_{\eta\xi} e^x + u_{\eta\eta} e^x] \\ &= e^x (u_\xi - u_\eta) + e^{2x} (u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}), \end{aligned}$$

$$u_y = u_\xi \cdot 2y + u_\eta \cdot 2y = 2y(u_\xi + u_\eta),$$

$$\begin{aligned} u_{yy} &= 2(u_\xi + u_\eta) + 2y(u_{\xi\xi} \cdot 2y + u_{\xi\eta} \cdot 2y + u_{\eta\xi} \cdot 2y + u_{\eta\eta} \cdot 2y), \\ &= 2(u_\xi + u_\eta) + 4y^2(u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}), \end{aligned}$$

方程成为

$$\begin{aligned} &-16y^2 e^{2x} u_{\xi\eta} + (4ye^x - 2e^{2x}) u_\xi - (4y^2 e^{2x} + 2e^{2x}) u_\eta \\ &- 4y^2 e^x (u_\xi - u_\eta) = 2, \end{aligned}$$

即 $-16y^2 e^x u_{\xi\eta} - 2e^x u_\xi - 2e^x u_\eta = 0,$

$$8y^2 u_{\xi\eta} + u_\xi + u_\eta = 0 \quad (3)$$

(1) + (2) 有 $2y^2 = \xi + \eta, \quad \therefore 8y^2 = 4(\xi + \eta),$

代入 (1) 得 $4(\xi + \eta) u_{\xi\eta} + u_\xi + u_\eta = 0.$

2. 简化下列常系数方程:

(1) $u_{xx} + u_{yy} + \alpha u_x + \beta u_y + \nu u = 0.$

解: 试作函数变换 $u(x, y) = e^{\lambda x + \mu y} v(x, y)$, 其中 λ 和 μ 是待定常数, 于是

$$\begin{cases} u_x = e^{\lambda x + \mu y} (v_x + \lambda v), \\ u_y = e^{\lambda x + \mu y} (v_y + \mu v), \\ u_{xx} = e^{\lambda x + \mu y} (v_{xx} + 2\lambda v_x + \lambda^2 v), \\ u_{xy} = e^{\lambda x + \mu y} (v_{xy} + \lambda v_y + \mu v_x + \lambda \mu v), \\ u_{yy} = e^{\lambda x + \mu y} (v_{yy} + 2\mu v_y + \mu^2 v), \end{cases}$$

以此代入原方程, 约去公共因子 $e^{\lambda x + \mu y}$ 后得:

$$v_{xx} + v_{yy} + (2\lambda + \alpha)v_x + (2\mu + \beta)v_y + (\lambda^2 + \mu^2 + 2\lambda + \beta\mu + \nu)v = 0.$$

令 $\lambda = -\frac{\alpha}{2}$, $\mu = -\frac{\beta}{2}$, 即 $u = e^{-\frac{\alpha}{2}x - \frac{\beta}{2}y}v$, 则一阶偏导数 v_x 和 v_y 的项消去, 方程简化为:

$$v_{xx} + v_{yy} + \left(\gamma - \frac{\alpha^2}{4} - \frac{\beta^2}{4}\right)v = 0.$$

$$(2) \quad u_{xx} = \frac{1}{a^2}u_y + \alpha u + \beta u_x.$$

解: 与 (1) 题一样, 试作函数变换 $u = ve^{\lambda x + \mu y}$, 并以 u_x , u_y , u_{xx} , 及 u 代入原方程, 约去公共因子 $e^{\lambda x + \mu y}$ 后得:

$$v_{xx} + (2\lambda - \beta)v_x - \frac{1}{a^2}v_y + \left(\lambda^2 - \frac{\mu}{a^2} - \alpha - \lambda\beta\right)v = 0.$$

如令 $\lambda = \frac{\beta}{2}$ 则 v_x 项被消去, 如要 v 项也被消去, 则必须

$$\left(\lambda^2 - \frac{\mu}{a^2} - \alpha - \lambda\beta\right) = 0,$$

即 $\mu = -a^2\left(\alpha + \frac{\beta^2}{4}\right)$, 即 $u = ve^{\frac{\beta}{2}x - a^2\left(\alpha + \frac{\beta^2}{4}\right)y}$, 即该常

微分方程简化为 $v_{xx} - \frac{1}{a^2}v_y = 0$.

$$(3) \quad u_{yy} + \frac{c-b}{a}u_x + \frac{b}{a}u_y + u = 0.$$

解: 作函数变换 $u = ve^{\lambda x + \mu y}$, 并以 u_x , u_y , u_{yy} 及 u 代入原方程, 约去公共因子 $e^{\lambda x + \mu y}$ 后得

$$v_{yy} + \frac{c-b}{a}v_x + \left(2\mu + \frac{b}{a}\right)v_y + \left[\mu^2 + \lambda\frac{c-b}{a} + \frac{b}{a}\mu + 1\right]v = 0.$$

如令 $\mu = -\frac{b}{2a}$, 则 v_y 项消失; 如要 v 项也消去, 则必须

$$\mu^2 + \lambda\left(\frac{c-b}{a}\right) + \frac{b}{a}\mu + 1 = 0, \quad \text{即 } \lambda = \frac{4a^2 - b^2}{4a(b-c)} \text{ 才可能.}$$

所以, 作出函数变换 $u = ve^{-\frac{b}{2a}y + \frac{4a^2 - b^2}{4a(b-c)}x}$ 后, 方程简化为

$$v_{xy} + \frac{c-b}{a}v_x = 0.$$

$$(4) \quad u_{xy} + 3u_x + 4u_y + 2u = 0.$$

解: 作函数变换 $u = ve^{\lambda x + \mu y}$, 并以 u_x , u_y , u_{xy} 及 u 代入原方程, 约去公共因子 $e^{\lambda x + \mu y}$ 后得:

$$v_{xy} + (\mu + 3)v_x + (\lambda + 4)v_y + [\lambda\mu + 3\lambda + 4\mu + 2]v = 0.$$

如令 $\lambda = -4$, $\mu = -3$, 即 $u = ve^{-4x - 3y}$ 则方程简化为

$$v_{xy} - 10v = 0.$$

$$(5) \quad 2au_{xx} + 2au_{xy} + au_{yy} + 2bu_x + 2cu_y + u = 0.$$

解: 如直接作函数变换, 该方程不能化简, 所以, 必须先作自变量的变换先消去 u_{xy} 项, 然后再作函数变换, 消去 u_x , u_y 项才行.

(i) 因为 $a_{12}^2 - a_{11}a_{22} = -a^2 < 0$, 该方程为椭圆型

$$\text{其特征方程是: } \frac{dy}{dx} = \frac{a \pm \sqrt{a^2 - 2a^2}}{2a} = \frac{1}{2} \pm i/2,$$

$$\text{即 } 2\frac{dy}{dx} = 1 + i, \quad \text{和 } 2\frac{dy}{dx} = 1 - i,$$

$$\text{特征线为: } y + (1 + i)x = C_1, \quad y - \frac{1}{2}(1 - i)x = C_2,$$

$$\text{令 } \xi = y - \frac{x}{2}, \quad \eta = \frac{x}{2}, \quad y = \xi + \eta,$$

$$\text{则 } u_x = \frac{1}{2}(u_\eta - u_\xi), \quad u_{xx} = u_{\eta\eta} - \frac{1}{4}u_{\xi\xi} - \frac{1}{2}u_{\xi\eta},$$

$$u_{xy} = \frac{1}{2}u_{\eta\xi} - \frac{1}{2}u_{\xi\xi}, \quad u_y = u_\xi, \quad u_{yy} = u_{\xi\xi},$$

方程成为

$$\begin{aligned} & 2a \cdot \frac{1}{4}(u_{\eta\eta} - 2u_{\eta\xi} + u_{\xi\xi}) + 2a \cdot \frac{1}{2}(u_{\eta\xi} - u_{\xi\xi}) + a \cdot u_{\xi\xi} \\ & + 2b \cdot \frac{1}{2}(u_\eta - u_\xi) + 2cu_\xi + u \\ & = 0, \end{aligned}$$

即 $au_{\xi\xi} + au_{\eta\eta} + (4c - 2b)u_\xi + 2bu_\eta + 2u = 0$.

(ii) 对上式作进一步化简, 令 $u = ve^{\lambda\xi + \mu\eta}$,

$$u_\xi = e^{\lambda\xi + \mu\eta} (v_\xi + \lambda v), \quad u_{\xi\xi} = e^{\lambda\xi + \mu\eta} (v_{\xi\xi} + 2\lambda v_\xi + \lambda^2 v),$$

$$u_\eta = e^{\lambda\xi + \mu\eta} (v_\eta + \mu v), \quad u_{\eta\eta} = e^{\lambda\xi + \mu\eta} (v_{\eta\eta} + 2\mu v_\eta + \mu^2 v),$$

$$v_{\xi\xi} + v_{\eta\eta} + \left[\frac{2(2c - b)}{a} + 2\lambda \right] v_\xi + \left[\frac{2b}{a} + 2\mu \right] v_\eta$$

$$+ \left[\lambda^2 + \mu^2 + \frac{2(2c - b)}{a} \lambda + \frac{2b}{a} \mu + \frac{2}{a} \right] v$$

$$= 0.$$

取 $\lambda = -\frac{2c - b}{a}$, $\mu = -\frac{b}{a}$ 代入上式, 则原方程简化为:

$$v_{\xi\xi} + v_{\eta\eta} + \frac{2}{a} \left(\frac{2bc - b^2 - 2c^2}{a} + 1 \right) v = 0,$$

其中 $u = ve^{\frac{b-2c}{a}\xi - \frac{b}{a}\eta}$,

$$\text{代回原来变量 } \frac{b-2c}{a}\xi - \frac{b}{a}\eta = \frac{b-2c}{a} \left(y - \frac{x}{2} \right) - \frac{b}{a} \frac{x}{2}$$

$$= \frac{2c - b - b}{2a} x + \frac{b - 2c}{a} y$$

$$= \frac{c - b}{a} x + \frac{b - 2c}{a} y,$$

$\therefore$

$$u = ve^{\frac{c-b}{a}x + \frac{b-2c}{a}y}.$$

第九章 行波法

§ 34. 行波法

1. 求解无限长弦的自由振动, 设弦的初始位移为 $\varphi(x)$, 初始速度为 $-a\varphi'(x)$.

解:
$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & -\infty < x < +\infty, \\ u|_{t=0} = \varphi(x), \\ u_t|_{t=0} = -a\varphi'(x). \end{cases}$$

这是一个一维的无界空间的问题, 根据达朗伯公式:

$$u(x, t) = \frac{1}{2} \left[\varphi(x+at) + \varphi(x-at) \right] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

将初始位移和初始速度代入上式得:

$$\begin{aligned} u(x, t) &= \frac{1}{2} \left[\varphi(x+at) + \varphi(x-at) \right] \\ &\quad + \frac{1}{2a} \int_{x-at}^{x+at} \left[-a\varphi'(\xi) \right] d\xi \\ &= \frac{1}{2} \left[\varphi(x+at) + \varphi(x-at) \right] - \frac{1}{2} \int_{x-at}^{x+at} \varphi'(\xi) d\xi \\ &= \frac{1}{2} \left[\varphi(x+at) + \varphi(x-at) \right] \\ &\quad - \frac{1}{2} \varphi(x+at) + \frac{1}{2} \varphi(x-at) \\ &= \varphi(x-at). \end{aligned}$$

波只朝一个方向 (x 正方向) 传播, 这是一列行波.

2. 求解无限长理想传输线上电压和电流的传播情况, 设初

始电压分布为 $A \cos kx$, 初始电流分布为 $\sqrt{\frac{C}{L}} A \cos kx$.

解: (1) 电压的传播情况:

传输线方程: $v_{tt} - a^2 v_{xx} = 0$, 式中 $a^2 = \frac{1}{LC}$.

初始条件:

$$\begin{cases} v \Big|_{t=0} = A \cos kx = \varphi(x), \\ v_t \Big|_{t=0} = -\frac{1}{C} j_x \Big|_{t=0} = \left(-\frac{1}{C}\right) \sqrt{\frac{C}{L}} Ak (-\sin kx) \\ = a Ak \sin kx = \psi(x), \end{cases}$$

应用一维无界空间解的达朗伯公式有:

$$\begin{aligned} v(x, t) &= \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \\ &= \frac{1}{2} [A \cos k(x+at) + A \cos k(x-at)] \\ &\quad + \frac{1}{2a} \int_{x-at}^{x+at} a Ak \sin k\xi d\xi \\ &= \frac{1}{2} [A \cos k(x+at) + A \cos k(x-at)] \\ &\quad + \frac{A}{2} [-\cos k(x+at) + \cos k(x-at)] \\ &= A \cos k(x-at). \end{aligned}$$

(2) 电流的传播情况:

传输线方程: $j_{tt} - a^2 j_{xx} = 0$, 式中 $a^2 = \frac{1}{LC}$,

初始条件:

$$\begin{cases} j \Big|_{t=0} = \sqrt{\frac{C}{L}} A \cos kx = \varphi(x), \\ j_t \Big|_{t=0} = -\frac{1}{L} v_x \Big|_{t=0} = \frac{Ak}{L} \sin kx = \psi(x). \end{cases}$$

应用一维无界空间解达朗伯公式:

$$\begin{aligned}
 j(x,t) &= \frac{1}{2} \left[\sqrt{\frac{C}{L}} A \cos k(x+at) + \sqrt{\frac{C}{L}} A \cos k(x- at) \right] + \frac{1}{2a} \int_{x-at}^{x+at} \frac{Ak}{L} \sin k\xi d\xi \\
 &= \frac{A}{2} \sqrt{\frac{C}{L}} [\cos k(x+at) + \cos k(x-at)] \\
 &\quad + \frac{\sqrt{LC}}{2L} A [-\cos k(x+at) + \cos k(x-at)] \\
 &= \sqrt{\frac{C}{L}} A \cos k(x-at).
 \end{aligned}$$

3. 在 $G/C = R/L$ 条件下求无限长传输线上的电报方程的通解.

解: 关于 j 和 v 的电报方程为 (31.13) (31.14):

$$\begin{cases} LC j_{tt} - j_{xx} + (LG + RC) j_t + RG j = 0, \\ LC v_{tt} - v_{xx} + (LG + RC) v_t + RG v = 0. \end{cases}$$

以 j 的方程为代表求其通解. 直接求其通解是比较困难的, 因此要作函数变换, 以消去一阶微分项,

$$\text{令 } j = e^{\lambda x + \mu t} u,$$

$$\text{则 } j_x = e^{\lambda x + \mu t} (u_x + \lambda u), \quad j_t = e^{\lambda x + \mu t} (u_t + \mu u),$$

$$j_{xx} = e^{\lambda x + \mu t} (u_{xx} + 2\lambda u_x + \lambda^2 u),$$

$$j_{tt} = e^{\lambda x + \mu t} (u_{tt} + 2\mu u_t + \mu^2 u).$$

代入关于 j 的方程, 并约去公共因子 $e^{\lambda x + \mu t}$ 后得:

$$\begin{aligned}
 &LC(u_{tt} + 2\mu u_t + \mu^2 u) - (u_{xx} + 2\lambda u_x + \lambda^2 u) \\
 &\quad + (LG + RC)(u_t + \mu u) + RG u \\
 &= 0, \\
 &LC u_{tt} - u_{xx} + [2\mu LC + (LG + RC)] u_t - 2\lambda u_x \\
 &\quad + [LC\mu^2 - \lambda^2 + \mu(LG + RC) + RG] u
 \end{aligned}$$

$$= 0.$$

如果选取 $\lambda = 0$ $\mu = -(LG + RC)/2LC$ 并注意 $G/C = \frac{R}{L}$,

则 $\mu = -\frac{2RC}{2LC} = -\frac{R}{L}$ 代入上式, 方程化简为,

$$LCu_{tt} - u_{xx} + \left[LC \left(-\frac{R}{L} \right)^2 - \frac{R}{L} (LG + RC) + RG \right] u = 0,$$

$$u_{tt} - \frac{1}{LC} u_{xx} = 0, \text{ 即 } u_{tt} - a^2 u_{xx} = 0, \text{ 其中 } a = \sqrt{\frac{1}{LC}},$$

如果初始条件为

$$j|_{t=0} = \varphi(x), \quad j_t|_{t=0} = \psi_1(x),$$

$$\begin{aligned} \text{则 } \begin{cases} u|_{t=0} e^{-(\lambda x + \mu t)} j|_{t=0} = \varphi(x), \\ u_t|_{t=0} = j_t e^{-\lambda x - \mu t}|_{t=0} - \mu u|_{t=0} = j_t|_{t=0} - \mu \varphi(x) \\ \quad = \psi_1(x) - \left(-\frac{R}{L} \right) \varphi(x) = \psi_1(x) + \frac{R}{L} \varphi(x) = \psi(x), \end{cases} \end{aligned}$$

应用达朗伯公式, 可以得到关于 $u_{tt} - a^2 u_{xx} = 0$ 的通解为:

$$u(x, t) = \frac{1}{2} [\varphi(x + at) + \varphi(x - at)]$$

$$+ \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi.$$

$$\therefore j = e^{\mu t} u = e^{-\frac{R}{L}t} \left\{ \frac{1}{2} [\varphi(x + at) + \varphi(x - at)] \right.$$

$$\left. + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi. \right.$$

4. 无限长弦在点 $x = x_0$ 受到初始冲击, 冲量为 I , 试求解弦的振动. [提示: $u_t|_{t=0} = (I/\rho) \delta(x - x_0)$].

解: $u_{tt} - a^2 u_{xx} = 0, \quad (-\infty < x < \infty \text{ 其中 } a^2 = \frac{I}{\rho}),$

$$u|_{t=0} = 0,$$

$$u_t|_{t=0} = \frac{I}{\rho} \delta(\xi - x_0) = \frac{I}{\rho} H'(\xi - x_0),$$

$$\begin{aligned} u(x, t) &= \frac{1}{2a} \int_{x-at}^{x+at} \left(\frac{I}{\rho} \right) \delta(\xi - x_0) d\xi \\ &= \frac{I}{2a\rho} \int_{x-at}^{x+at} H'(\xi - x_0) d(\xi - x_0) \\ &= \frac{I}{2\rho \sqrt{\frac{T}{\rho}}} \left[H(\xi - x_0) \right]_{x-at}^{x+at} \\ &= \frac{I}{2\sqrt{\rho T}} [H(x - x_0 + at) - H(x - x_0 - at)]. \end{aligned}$$

5. 求解细圆锥形均质杆的纵振动 [提示: 泛定方程见 § 31 习题 2, 作变换 $u = v/x$].

解: 细圆锥形的均质杆的纵振动方程已在 § 31 习题中导出,

$$\text{即: } u_{tt} - a^2 \frac{1}{x^2} \frac{\partial}{\partial x} (x^2 u_x) = 0,$$

直接求解该方程是比较困难的, 因此作变换

$$u(x, t) = \frac{v(x, t)}{x},$$

$$\text{则 } \frac{\partial u}{\partial x} = \frac{1}{x} \frac{\partial v}{\partial x} - \frac{v}{x^2},$$

$$\frac{\partial}{\partial x} (x^2 u_x) = \frac{\partial}{\partial x} \left[x \frac{\partial v}{\partial x} - v \right] = x \frac{\partial^2 v}{\partial x^2},$$

$$u_{tt} = \frac{1}{x} v_{tt},$$

代入原方程式即有

$$\frac{1}{x}v_{tt} - a^2 \frac{1}{x^2} \cdot x \frac{\partial^2 v}{\partial x^2} = 0,$$

即 $v_{tt} - a^2 v_{xx} = 0$.

$v(x, t)$ 的通解为 $v(x, t) = f_1(x - at) + f_2(x + at)$,

$$\begin{aligned} \therefore u(x, t) &= \frac{1}{x} v(x, t) \\ &= \frac{1}{x} [f_1(x - at) + f_2(x + at)]. \end{aligned}$$

6. 半无限长杆的端点受到纵向力 $F(t) = A \sin \omega t$ 作用, 求解杆的纵振动.

解: 泛定方程 $u_{tt} - a^2 u_{xx} = 0$,

初始条件:

$$\begin{cases} u|_{t=0} = \varphi(x), & x > 0, \\ u_t|_{t=0} = \psi(x), & x > 0, \end{cases}$$

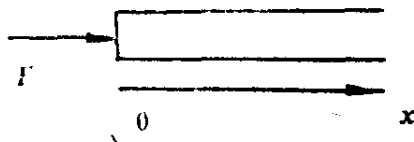


图 9-1

边界条件 $u_x|_{x=0} = \frac{A}{YS} \sin \omega t$,

对 $x > at$ 的地方, 端点的影响未传到, 所以

$$u(x, t) = \frac{1}{2} [\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi.$$

对 $x < at$ 的地方, 需要考虑端点的影响. 对 $x < 0$, $\varphi(x)$ 和 $\psi(x)$ 未定义, 现将它们延拓.

$$\Phi(x) = \begin{cases} \varphi(x), & (x \geq 0), \\ \varphi_1(x), & (x < 0), \end{cases} \quad \Phi(x) = \begin{cases} \psi(x), & (x \geq 0), \\ \psi_1(x), & (x < 0), \end{cases}$$

其中 $\varphi_1(x)$ 和 $\psi_1(x)$ 待定, 应用达朗伯公式:

$$\begin{aligned} u &= \frac{1}{2} [\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_0^{x+at} \psi(\xi) d\xi \\ &\quad + \frac{1}{2a} \int_{x-at}^0 \psi_1(\xi) d\xi, \end{aligned}$$

它应满足边界条件

$$\begin{aligned}
 u_x \Big|_{x=0} &= \frac{1}{2} \varphi'(at) + \frac{1}{2} \varphi_1'(-at) + \frac{1}{2a} \psi(at) \\
 &\quad - \frac{1}{2a} \psi_1(-at) \\
 &= \frac{A}{YS} \sin \omega t,
 \end{aligned}$$

显然, 取 $\varphi_1(x) = \varphi(-x)$, 而 $\psi_1(x) = \psi(-x) + \frac{2aA}{YS} \sin \frac{\omega}{a}x$, 即可满足边界条件.

$$\begin{aligned}
 \therefore u &= \frac{1}{2} [\varphi(x+at) + \varphi(at-x)] \\
 &\quad + \frac{1}{2a} \int_0^{x+at} \psi(\xi) d\xi + \frac{1}{2a} \int_{x-at}^0 \psi(-\xi) d\xi \\
 &\quad + \frac{1}{2a} \int_{x-at}^0 \frac{2aA}{YS} \sin \frac{\omega}{a} \xi d\xi \\
 &= \frac{1}{2} [\varphi(x+at) + \varphi(at-x)] + \frac{1}{2a} \int_0^{x+at} \psi(\xi) d\xi \\
 &\quad + \frac{1}{2a} \int_0^{at-x} \psi(\xi) d\xi \\
 &\quad + \frac{Aa}{YS\omega} \left[\cos \omega \left(t - \frac{x}{a} \right) - 1 \right].
 \end{aligned}$$

7. 求解半无限长理想传输线上电报方程的解, 端点通过电阻 R 而相接, 初始电压分布 $A \cos kx$, 初始电流分布 $\sqrt{\frac{C}{L}} A \cos kx$ 在什么条件下端点没有反射 (这种情况叫作匹配)?

解: $\because$ 是理想传输线 $\therefore G = R = 0$

因此, 定解问题是

$$\begin{cases} j_{tt} - a^2 j_{xx} = 0, \\ j|_{t=0} = \sqrt{\frac{C}{L}} A \cos kx, (x < 0), \\ j_t|_{t=0} = \frac{-1}{L} v_x|_{t=0} = \frac{Ak}{L} \sin kx, \\ v_{tt} - a^2 v_{xx} = 0, \\ v|_{t=0} = A \cos kx, (x < 0), \\ v_t|_{t=0} = \frac{-1}{C} j_x|_{t=0} = \frac{Ak}{\sqrt{LC}} \sin kx, \end{cases}$$

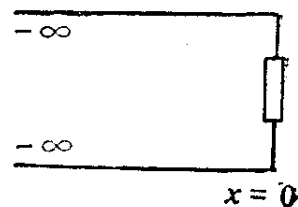


图 9-2

电压 v 和电流 j 在 $x = 0$ 点有.

$$v|_{x=0} = R j|_{x=0},$$

(i) 对于 $t < \frac{|x|}{a}$ 端点的影响尚未到达, 用达朗伯公式:

$$\begin{aligned} j(xt) &= \frac{1}{2} \left[\sqrt{\frac{C}{L}} A \cos k(x+at) + \sqrt{\frac{C}{L}} A \cos k(x-at) \right] \\ &\quad + \frac{1}{2a} \int_{x-at}^{x+at} \frac{Ak}{L} \sin k\xi d\xi. \\ &= \frac{1}{2} \sqrt{\frac{C}{L}} A \cos k(x+at) + \frac{1}{2} \sqrt{\frac{C}{L}} A \cos k(x-at) \\ &\quad + \frac{1}{2} \sqrt{\frac{C}{L}} A \int_{x-at}^{x+at} \sin k\xi d(k\xi). \\ &= \frac{1}{2} \sqrt{\frac{C}{L}} A \cos k(x+at) + \frac{1}{2} \sqrt{\frac{C}{L}} A \cos k(x-at) \\ &\quad + \frac{1}{2} \sqrt{\frac{C}{L}} A [-\cos k(x+at) + \cos k(x-at)] \\ &= \sqrt{\frac{C}{L}} A \cos k(x-at), \end{aligned}$$

同理 $v(xt) = A \cos k(x-at)$,

这就是从 $x < 0$ 的区域沿 x 轴正方向朝着端点 $x = 0$ 行进的入

射波。

(ii) 对于 $t > \frac{|x|}{a}$, 必须考虑到端点的反射, 这里不拟从达朗伯公式(34.6)出发, 而是直接从通解(34.5)出发

$$j(xt) = \sqrt{\frac{C}{L}} A \cos k(x - at) + g_1(x + at), \quad (1)$$

$$v(xt) = A \cos k(x - at) + g_2(x + at), \quad (2)$$

其中 $g_1(x + at)$ 和 $g_2(x + at)$ 是待求的反射波, 因传输是理想的, 故

(1) 和 (2) 应满足 $Lj_t = -v_x$, 和 $cv_t = -j_x$ 。

$$\text{即} \left\{ \begin{aligned} & L \sqrt{\frac{C}{L}} k A \sin k(x - at) + L a g'_1(x + at) \\ &= A k \sin k(x - at) - g'_2(x + at). \\ & C A k \sin k(x - at) + C a g'_2(x + at) \\ &= \sqrt{\frac{C}{L}} A \sin k(x - at) - g'_1(x + at). \end{aligned} \right.$$

由于 $a = \sqrt{\frac{1}{LC}}$, 所以上列两式即

$$\sqrt{\frac{L}{C}} g'_1(x + at) = -g'_2(x + at),$$

$$\sqrt{\frac{C}{L}} g'_2(x + at) = -g'_1(x + at).$$

总之 g_1 和 g_2 两个函数不是独立的, 这样 (1) 和 (2) 应代之以

$$\left\{ \begin{aligned} j(xt) &= \sqrt{\frac{C}{L}} A \cos k(x - at) - \sqrt{\frac{C}{L}} g_2(x + at), \quad (3) \end{aligned} \right.$$

$$\left\{ \begin{aligned} v(xt) &= A \cos k(x - at) + g_2(x + at), \quad (4) \end{aligned} \right.$$

(3) 和 (4) 应满足边界条件 $v|_{x=0} = Rj|_{x=0}$, 即

$$A \cos kat + g_2(at) = R \sqrt{\frac{C}{L}} A \cos kat - R \sqrt{\frac{C}{L}} g_2(at)$$

由此解得

$$g_2(at) = \frac{1 - R\sqrt{\frac{C}{L}}}{1 + R\sqrt{\frac{C}{L}}} A \cos kat = \frac{\sqrt{\frac{L}{C}} - R}{\sqrt{\frac{L}{C}} + R} A \cos kat$$

以此代入(3)和(4)得到解答

$$j(xt) = \sqrt{\frac{C}{L}} A \cos k(x - at) - \sqrt{\frac{C}{L}} \frac{\sqrt{\frac{L}{C}} - R}{\sqrt{\frac{L}{C}} + R} A \cos k(x + at),$$

$$v(xt) = A \cos k(x - at) + \frac{\sqrt{\frac{L}{C}} - R}{\sqrt{\frac{L}{C}} + R} A \cos k(x + at).$$

右边第二项是反射波,要想没有反射波,应令右边第二项的系数为零,即

$$\sqrt{\frac{L}{C}} = R,$$

端点没有反射波,意味着电波的能量全部被电阻吸收,这叫做阻抗匹配,这时负载阻抗 R 等于传输线的特性阻抗

$$\sqrt{\frac{L}{C}}.$$

8. 半无限长的初始位移和速度都是零,端点作微小振动 $u|_{x=0} = A \sin \omega t$, 求解弦的振动.

解: 对于 $x \geq at$, 显然有 $v(x, t) = 0$ 下面研究 $t > \frac{x}{a}$, 将初始条件延拓到 $x < 0$ 的半无界区域,

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & -\infty < x < \infty, \\ u(0, t) = A \sin \omega t, \\ u(x, 0) = \begin{cases} 0, & x \geq 0, \\ \varphi(x), & 0 < x, \end{cases} & u_t(x, 0) = \begin{cases} 0, & x \geq 0, \\ \psi(x), & x < 0, \end{cases} \end{cases}$$

其中 $\varphi(x)$ 和 $\psi(x)$ 尚未确定。

将达朗伯公式应用于延拓后的无界弦。

$$u(x, t) = \frac{1}{2} [\Phi(x + at) + \Phi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \Psi(\xi) d\xi,$$

且令其满足边界条件得到：

$$A \sin \omega t = \frac{1}{2} [0 + \varphi(-at)] + \frac{1}{2a} \int_{-at}^0 \Psi(\xi) d\xi,$$

$$\text{即 } A \sin \omega t = \frac{1}{2} \varphi(-at) + \frac{1}{2a} \int_{-at}^0 \Psi(\xi) d\xi,$$

记 at 为 x 则

$$A \sin \omega \frac{x}{a} = \frac{1}{2} \varphi(-x) + \frac{1}{2a} \int_{-x}^0 \psi(\xi) d\xi,$$

显然若取 $\varphi(x) = 2A \sin\left(-\frac{\omega}{a}x\right)$, $\psi(x) = 0$ 。

$$\text{于是 } u(x, t) = \frac{1}{2} \varphi(x - at) = \frac{1}{2} 2A \sin\left[-\frac{\omega}{a}(x - at)\right]$$

$$= A \sin \omega \left(t - \frac{x}{a}\right). \quad \left(t > \frac{x}{a}\right).$$

9. 在弦的 $x = 0$ 处悬挂着质量为 M 的载荷, 有一行波 $u(x, t) = f\left(t - \frac{x}{a}\right)$ 从 $x < 0$ 的区域向悬挂点行进. 试求反射波和透射波.

解: 设波传到分界点 $x = 0$ 处的时刻为 $t = 0$, 则依题意有:

$$\begin{cases} u_{tt}^I - a^2 u_{xx}^I = 0, & (-\infty < x < 0), \end{cases} \quad (1)$$

$$\begin{cases} u^I|_{t=0} = f\left(t - \frac{x}{a}\right), \end{cases} \quad (2)$$

$$\begin{cases} u_{tt}^{II} - a^2 u_{xx}^{II} = 0, & (0 < x < \infty), \end{cases} \quad (3)$$

$$\begin{cases} u^{II}|_{t=0} = 0, & u_t^{II}|_{t=0} = 0. \end{cases} \quad (4)$$

衔接条件为

$$\begin{cases} u^I|_{x=0} = u^{II}|_{x=0}, \end{cases} \quad (5)$$

$$\begin{cases} u_t^I|_{x=0} = u_t^{II}|_{x=0} = u_t|_{x=0}, \end{cases}$$

$$\begin{cases} u_{tt}^I|_{x=0} = u_{tt}^{II}|_{x=0} = u_{tt}|_{x=0}, \end{cases} \quad (6)$$

$$T(u_x^{II} - u_x^I)|_{x=0} - Mg = Mu_{tt}|_{x=0}.$$

上式中 u 是荷载 Mg 的位移。

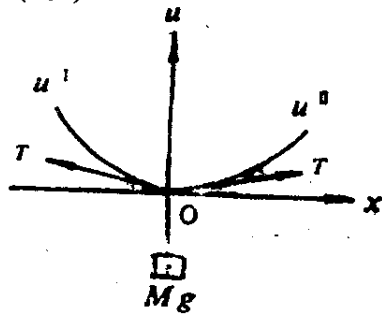


图 9-3

在 $x < 0$ 的区域中, 方程(1)的通解为

$$u^I = f\left(t - \frac{x}{a}\right) + g\left(t + \frac{x}{a}\right),$$

其中 $g\left(t + \frac{x}{a}\right)$ 是待求的反射波。由条件(2)知 $g\left(\frac{x}{a}\right) = 0$, ($x < 0$)。

即 $g(\xi) = 0, \quad (\xi < 0).$

由 u^I 的解知 $u_t^I|_{x=0} = \frac{1}{a}[g'(t) - f'(t)],$

在 $x > 0$ 的区域, 只有透射波, 而没有反射波, 故(3)的解为

$$u^{II}(x,t) = h\left(t - \frac{x}{a}\right), \quad (x > 0).$$

其中 $h\left(t - \frac{x}{a}\right)$ 是待求的反射波, 由条件(4), 可知

$$h\left(-\frac{x}{a}\right) = 0, \quad h'\left(-\frac{x}{a}\right) = 0, \quad (x > 0).$$

即 $h(\xi) = 0, \quad h'(\xi) = 0, \quad (\xi < 0).$

由 $u^{\text{II}}(x|)$ 可得 $u^{\text{II}}|_{x=0} = -\frac{1}{a}h'(t)$.

应用衔接条件(5), (6), 可得

$$\begin{cases} f(t) + g(t) = h(t), & f'(t) + g'(t) = h'(t), \\ f''(t) + g''(t) = h''(t) = u^{\text{II}}|_{x=0}, \\ T\left\{-\frac{1}{a}h'(t) - \frac{1}{a}[g'(t) - f'(t)]\right\} - Mg = Mh''(t), \end{cases}$$

$$\therefore h''(t) + \frac{2T}{Ma}[h'(t) - f'(t)] = -g,$$

将上式对 t 积分, 并利用 $h'|_{t=0} = 0, h|_{t=0} = 0$, 得

$$h'(t) + \frac{2T}{Ma}[h(t) - f(t)] = -gt,$$

$$\text{即 } h'(t) + \frac{2T}{Ma}h(t) = \frac{2T}{Ma}f(t) - gt,$$

$$\begin{aligned} \therefore h(t) &= e^{\frac{-2T}{Ma}t} \left[h(0) + \int_0^t \left[\frac{2T}{Ma}f(\tau) - g\tau \right] e^{\frac{2T}{Ma}\tau} d\tau \right] \\ &= e^{\frac{-2T}{Ma}t} \left[\frac{2T}{Ma} \int_0^t f(\tau) e^{\frac{2T}{Ma}\tau} d\tau - \int_0^t g\tau e^{\frac{2T}{Ma}\tau} d\tau \right] \\ &= e^{\frac{-2T}{Ma}t} \left[\frac{2T}{Ma} \int_0^t f(\tau) e^{\frac{2T}{Ma}\tau} d\tau - g \frac{Ma}{2T} t e^{\frac{2T}{Ma}t} + \frac{gM^2a^2}{4T^2} e^{\frac{2T}{Ma}\tau} \Big|_0^t \right] \\ &= \frac{2T}{Ma} e^{\frac{-2T}{Ma}t} \int_0^t f(\tau) e^{\frac{2T}{Ma}\tau} d\tau - \frac{Ma}{2T} gt + \frac{M^2a^2}{4T^2} g \left[1 - e^{\frac{-2T}{Ma}t} \right]. \end{aligned}$$

而反射波.

$$g(t) = h(t) - f(t).$$

故本题之解为:

透射波

$$h\left(t - \frac{x}{a}\right) = \begin{cases} 0, & \left(t < \frac{x}{a}\right). \\ \frac{2T}{Ma} e^{-\frac{2T}{Ma}\left(t - \frac{x}{a}\right)} \int_0^{t - \frac{x}{a}} f(\tau) e^{\frac{2T}{Ma}\tau} d\tau \\ - \frac{Ma}{2T} g\left(t - \frac{x}{a}\right) + \frac{Ma^2}{4T^2} g \\ \left[1 - e^{-\frac{2T}{Ma}\left(t - \frac{x}{a}\right)}\right] \left(t > \frac{x}{a}, x > 0.\right). \end{cases}$$

反射波

$$g\left(t + \frac{x}{a}\right) = \begin{cases} 0 & \left(t + \frac{x}{a} < 0, x < 0\right) \\ -\frac{2T}{Ma} e^{-\frac{2T}{Ma}\left(t + \frac{x}{a}\right)} \int_0^{t + \frac{x}{a}} f(\tau) e^{\frac{2T}{Ma}\tau} d\tau \\ - \frac{Ma}{2T} g\left(t + \frac{x}{a}\right) + \frac{M^2 a^2}{4T^2} g \\ \left[1 - e^{-\frac{2T}{Ma}\left(t + \frac{x}{a}\right)}\right] - f\left(t + \frac{x}{a}\right). \\ \left(t + \frac{x}{a} > 0, x > 0\right). \end{cases}$$

10. 平面偏振的平面光波沿 x 轴行进而垂直地投射于两种介质的分界面上. 入射光波的电场强度 $E = E_0 \sin \omega\left(t - \frac{n_1}{a}x\right)$,

其中 n_1 是第一种介质的折射率. 求反射光波和透射光波 [提示:

在分界面上, E 连续, $\frac{\partial B}{\partial t}$ (即 $\frac{\partial E}{\partial x}$) 连续].

解：设波传到分界面 $x = 0$ 处的时刻为 $t = 0$ ，得定解问题；

$$\begin{cases} E_{tt}^I - \frac{a^2}{n_1^2} E_{xx}^I = 0, & (-\infty < x < 0), \end{cases} \quad (1)$$

$$\begin{cases} E^I|_{t \leq 0} = E_0 \sin \omega \left(t - \frac{n_1}{a} x \right), \end{cases} \quad (2)$$

$$\begin{cases} E_{tt}^{II} - \frac{a^2}{n_2^2} E_{xx}^{II} = 0, & (0 < x < \infty), \end{cases} \quad (3)$$

$$\begin{cases} E^{II}|_{t \leq 0} = 0, & E_t^{II}|_{t \leq 0} = 0, \end{cases} \quad (4)$$

衔接条件 $E^I|_{x=0} = E^{II}|_{x=0}, \quad (5)$

$$E_x^I|_{x=0} = E_x^{II}|_{x=0}, \quad (6)$$

在 $x < 0$ 的区域中，(1)之解为

$$E^I = E_0 \sin \omega \left(t - \frac{n_1}{a} x \right) + g \left(t + \frac{n_1}{a} x \right), \quad (x < 0),$$

由条件(2)可得 $g(\xi) = 0, (\xi < 0)$ 。

在区域 $x > 0$ 中，没有反射波，只有透射波。因此(3)的解为

$$E^{II} = h \left(t - \frac{n_2}{a} x \right), \quad (x > 0).$$

由条件(4)， $h(\xi) = 0, h'(\xi) = 0, (\xi < 0)$ 。

应用衔接条件(5)(6)，得

$$\begin{cases} E_0 \sin \omega t + g(t) = h(t), \end{cases} \quad (7)$$

$$\begin{cases} -\frac{E_0 n_1 \omega}{a} \cos \omega t + \frac{n_1}{a} g'(t) = -\frac{n_2}{a} h'(t), \end{cases} \quad (8)$$

将(8)对 t 积分，且由于 $g|_{t=0} = 0, h|_{t=0} = 0$ 。

$$-n_1 E_0 \sin \omega t + n_1 g(t) = -n_2 h(t), \quad (9)$$

由(7)(9)消去 $h(t)$ 得

$$g(t) = \frac{(n_1 - n_2)E_0}{n_1 + n_2} \sin \omega t,$$

再得 $h(t) = \frac{2n_1 E_0}{n_1 + n_2} \sin \omega t.$

所以本题的解为:

反射波

$$g\left(t + \frac{n_1 x}{a}\right) = \begin{cases} 0, & \left(t + \frac{n_1 x}{a} < 0, x < 0\right), \\ \frac{n_1 - n_2}{n_1 + n_2} E_0 \sin \omega \left(t + \frac{n_1 x}{a}\right), & \left(t + \frac{n_1 x}{a} > 0, x > 0\right), \end{cases}$$

透射波

$$h\left(t - \frac{n_2 x}{a}\right) = \begin{cases} 0, & \left(t < \frac{n_2 x}{a}, x > 0\right), \\ \frac{2n_1 E_0}{n_1 + n_2} \sin \left(t - \frac{n_2 x}{a}\right), & \left(t > \frac{n_2 x}{a}, x > 0\right). \end{cases}$$

第十章 分离变数 (傅里叶级数) 法

§35. 分离变数法介绍

1. “顾名思义, 分离变数法只能求出分离变数形式的解, 如果一个定解问题的解不是分离变数形式的, 用分离变数法不可能求得这个解。”试对上述说法加以评论.

解: 分离变数法解方程可得到本征解, 本征值说是分离变数形式的, 但定解问题的解一般是本征解的某个叠加, 即由本征解组成的级数, 这种解已不是分离变数形式的了, 事实上, 一个解即使不是分离变数形式的也可展为级数, 所以由分离变数法得到的解, 一般并不一定是分离变数形式的.

2. 演奏琵琶是把弦的某一点向旁拨开一个小距离, 然后放手任其自由振动. 设弦长为 l , 被拨开的点在弦长的 $\frac{1}{n_0}$ (n_0 为正整数) 处, 拨开距离为 h , 试求解弦的振动. 不要套用现成答案, 请按照分离变数法的步骤一步一步求解. [注意: 在解答中, 不存在 n_0 谐音以及 n_0 整倍数次谐音. 因此, 在不同位置拨弦 (n_0 不同), 发出的声音的音色也就不同.]

解: 定解问题为:

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, (0 < x < l), & (1) \\ u|_{x=0} = u|_{x=l} = 0, & (2) \\ u|_{t=0} = \begin{cases} \frac{n_0 h}{l} x, & (0 \leq x \leq \frac{l}{n_0}), \\ \frac{h}{l - \frac{l}{n_0}} (l - x), & (\frac{l}{n_0} \leq x \leq l), \end{cases} & (3) \\ u_t|_{t=0} = 0, (0 \leq x \leq l), & (4) \end{cases}$$



图 10-1

设 $u(x, t) = X(x)T(t)$ 以此代入泛定方程和边界条件:

$$X(x)T''(t) - a^2T(t)X''(x) = 0, \quad (5)$$

$$X(0)T(t) = X(l)T(t) = 0, \quad (6)$$

由 (5) 式得到

$$\frac{T''(t)}{a^2T(t)} = \frac{X''(x)}{X(x)}. \quad (7)$$

只有上式两端均等于同一常数时才有可能成立, 把这个常数记为 $-\lambda$, 代入 (7) 式成为:

$$\frac{T''(t)}{a^2T(t)} = \frac{X''(x)}{X(x)} = -\lambda,$$

即

$$T''(t) + \lambda a^2 T(t) = 0, \quad (8)$$

$$X''(x) + \lambda X(x) = 0, \quad (9)$$

在 (6) 中, 若取 $T(t) = 0$, 得出 $u = X(x)T(t) = 0$, 显然无意义, 只能取 $X(0) = X(l) = 0$ 由此式与 x 的方程 (9) 来求解 $X(x)$, 这要分 $\lambda < 0$, $\lambda = 0$ 和 $\lambda > 0$ 三种情况.

(1) 当 $\lambda = 0$ 时, 由 (9) 式得 $X(x) = C_1x + C_2$ 以此代入 $X(0) = X(l) = 0$ 得 $C_1 = C_2 = 0$ 则 $X(x) \equiv 0$, 无意义, 故得到 $\lambda \neq 0$.

(2) $\lambda < 0$ 时, $-\lambda > 0$, 方程 (9) 的解是 $X(x) = C_1 l^{\sqrt{-\lambda}x} + C_2 l^{-\sqrt{-\lambda}x}$ 以此代入 $X(l) = X(0) = 0$ 得到; $C_1 = C_2 = 0$, $X(x) = 0$, 也是无意义, 可见 $\lambda < 0$ 的情况也要排除.

(3) $\lambda > 0$, 方程 (9) 的解是

$$X(x) = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x, \quad (10)$$

由边界条件 $X(0) = X(l) = 0$ 得到 $C_1 = 0$, $C_2 \sin \sqrt{\lambda} l = 0$, 这里 C_2 不能为 0, 否则得到的解只是零, 无意义. 因此只能取

$$\sin \sqrt{\lambda} l = 0, \quad \lambda = \frac{n^2 \pi^2}{l^2}, \quad (n = 1, 2, 3, \dots)$$

$$\therefore X_n(x) = C_2 \sin \frac{n\pi}{l} x, \quad (11)$$

再解关于 t 的方程 (8), 用 $\lambda = \frac{n^2 \pi^2}{l^2}$, 代入 (8) 式

$$T_n''(t) + \frac{n^2 \pi^2 a^2}{l^2} T_n(t) = 0. \quad (12)$$

(10) 式的解为:

$$T_n(t) = A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l},$$

本征解为

$$u_n(x, t) = \left(A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} \right) \sin \frac{n\pi x}{l},$$

一般解应是它的叠加:

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} \right) \sin \frac{n\pi x}{l}. \quad (13)$$

(13) 式应满足初始条件 (3) 和 (4), 则

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l},$$

$$\begin{aligned} \therefore A_n &= \frac{2}{l} \int_0^{l/n} \frac{n_0 h}{l} \xi \sin \frac{n\pi \xi}{l} d\xi \\ &\quad + \frac{2}{l} \int_{l/n}^l \frac{n_0 h}{l/n(n_0-1)l} (l-\xi) \sin \frac{n\pi \xi}{l} d\xi \\ &= \frac{2n_0^2 h}{n^2 \pi^2 (n_0-1)} \sin \frac{n\pi}{n_0}. \end{aligned}$$

(13) 式还应满足初始条件 (4)

$$u_t(x, 0) = \sum_{n=1}^{\infty} \frac{a\pi n B_n}{l} \sin \frac{n\pi x}{l} = 0,$$

$$\therefore \frac{a\pi n B_n}{l} = 0, \quad \therefore B_n = 0,$$

$$\begin{aligned} \therefore u(x, t) &= \sum_{n=1}^{\infty} A_n \cos \frac{n\pi at}{l} \sin \frac{n\pi x}{l} \\ &= \frac{2n_0^2 h}{\pi^2(n_0 - 1)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{n_0} \cos \frac{n\pi at}{l} \\ &\quad \sin \frac{n\pi x}{l}. \end{aligned}$$

由上式可以得知，不存在 $n = kn_0$ 次谐音，因这时 $\sin \frac{n\pi}{n_0} = 0$ 。

3. 两端固定的弦的长度为 l ，用细棒敲击弦上 $x = x_0$ 点，亦即在 $x = x_0$ 施加冲力，设其冲量为 I ，求解弦的振动。〔注意：上题 n 次谐音的幅度 $\propto \frac{1}{n^2}$ ，本题 n 次谐音幅度 $\propto 1/n$ ，相比之下，细棒敲击弦发出的声音包含较多的高次谐音，比较刺耳。因此，演奏扬琴必须使用锤敲击弦而决不可用细棒〕。

解：定解问题：

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & (0 \leq x \leq l). \\ u|_{x=0} = u|_{x=l} = 0, \\ u|_{t=0} = 0, \quad u_t|_{t=0} = \frac{I}{\rho} \delta(x - x_0). \end{cases}$$

泛定方程的一般解为

$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l} \right) \sin \frac{n\pi x}{l}.$$

代入初始条件：

$$u|_{t=0} = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = 0,$$

$$\therefore A_n = 0,$$

$$\begin{aligned}
u_t|_{t=0} &= \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \sin \frac{n\pi x}{l} = \frac{I}{\rho} \delta(x-x_0), \\
\therefore \frac{n\pi a}{l} B_n &= \frac{2I}{l\rho} \int_0^l \sin \frac{n\pi x}{l} \delta(x-x_0) dx \\
&= \frac{2I}{l\rho} \sin \frac{n\pi x_0}{l}, \\
\therefore B_n &= \frac{2I}{n\pi\rho a} \sin \frac{n\pi x_0}{l}, \\
u &= \sum_{n=1}^{\infty} B_n \sin \frac{n\pi at}{l} \sin \frac{n\pi x}{l} \\
&= \frac{2I}{\pi a\rho} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi at}{l} \sin \frac{n\pi x}{l}.
\end{aligned}$$

§36. 齐次的泛定方程 (傅里叶级数法)

1. 长为 l 的弦, 两端固定, 弦中张力为 T , 在距一端为 x_0 的一点以力 F_0 把弦拉开, 然后突然撤除这力. 求解弦的振动.

解: 定解问题为

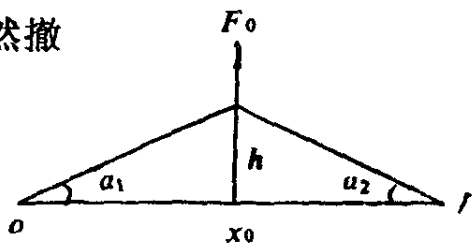


图 10-2

$$\begin{cases}
u_{tt} - a^2 u_{xx} = 0, (0 < x < l). & (1) \\
u(0, t) = u(l, t) = 0, & (2) \\
u(x, 0) = \begin{cases} \frac{F_0}{T} \frac{l-x_0}{l} x, (0 < x < x_0), \\ \frac{F_0}{T} \frac{x_0}{l} (l-x), (x_0 < x < l), \end{cases} & (3) \\
u_t|_{t=0} = 0. & (4)
\end{cases}$$

令 $u(x, t) = X(x)T(t)$ 代入泛定方程 (1) 得

$$\frac{X''}{X} = \frac{T''}{aT} = -\lambda^2.$$

可得
$$\begin{cases} T'' + a^2 \lambda^2 T = 0, \\ X'' + \lambda^2 X = 0, X(0) = X(l) = 0. \end{cases}$$

由此可以得到有关 X 的解是;

$$X(x) = C \sin \lambda x.$$

由 $X(l) = 0$ 可知, $C \sin \lambda l = 0$, C 不能为0, 否则 $x \equiv 0$. 无意义,

$$\therefore \sin \lambda l = 0, \lambda = \frac{n\pi}{l} (n = 1, 2, 3, \dots).$$

以 λ 的数值 $\left(= \frac{n\pi}{l} \right)$ 代入关于 T 的方程得 T 的解:

$$T_n(t) = A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t,$$

$$T(t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t \right),$$

$$\therefore u(x, t) = X(x) T(t)$$

$$= \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t \right) \sin \frac{n\pi}{l} x.$$

将 u 的表达式代入初始条件(4)得:

$$u_t|_{t=0} = \sum_{n=1}^{\infty} \left(-A_n \frac{n\pi a}{l} t \sin \frac{n\pi a}{l} t + B_n \frac{n\pi a}{l} \cos \frac{n\pi a}{l} t \right)$$

$$\sin \frac{n\pi}{l} x \Big|_{t=0}$$

$$= \sum_{n=1}^{\infty} B_n \frac{n\pi a}{l} \cos 0 \sin \frac{n\pi}{l} x = 0, \therefore B_n = 0.$$

$$\text{则 } u(x, t) = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{l} t \sin \frac{n\pi}{l} x,$$

根据初始条件(3)有:

$$u(x, 0) = \sum A_n \sin \frac{n\pi}{l} x = \begin{cases} \frac{F_0}{T} \frac{l-x_0}{l} x, & (0 < x < x_0), \\ \frac{F_0}{T} \frac{x_0}{l} (l-x), & (x_0 < x < l), \end{cases}$$

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^l \varphi(\xi) \sin \frac{n\pi}{l} \xi d\xi \\ &= \frac{2}{l} \int_0^{x_0} \frac{F_0}{T} \frac{l-x_0}{l} \xi \sin \frac{n\pi}{l} \xi d\xi \\ &\quad + \frac{2}{l} \int_{x_0}^l \frac{F_0}{T} \frac{x_0}{l} (l-\xi) \sin \frac{n\pi}{l} \xi d\xi \\ &= \frac{2}{l} \left\{ \frac{F_0}{T} \frac{l-x_0}{l} \left[-\frac{l}{n^2\pi^2} \sin \frac{n\pi\xi}{l} \right. \right. \\ &\quad \left. \left. - \frac{l^2}{n^2\pi^2} \cdot \frac{n\pi\xi}{l} \cos \frac{n\pi}{l} \xi \right]_0^{x_0} \right. \\ &\quad \left. - \frac{F_0}{T} \frac{x_0}{l} \frac{l}{n\pi} \cos \frac{n\pi}{l} \xi \right|_{x_0}^l \\ &\quad \left. - \frac{F_0}{T} \frac{x_0}{l} \frac{l^2}{n^2\pi^2} \left[\sin \frac{n\pi\xi}{l} \right. \right. \\ &\quad \left. \left. - \frac{n\pi\xi}{l} \cos \frac{n\pi\xi}{l} \right]_{x_0}^l \right\} \\ &= \frac{2}{l} \left\{ \frac{F_0}{T} \frac{(l-x_0)l}{n^2\pi^2} \left[\sin \frac{n\pi x_0}{l} \right. \right. \\ &\quad \left. \left. - \frac{n\pi x_0}{l} \cos \frac{n\pi x_0}{l} \right] \right. \\ &\quad \left. - \frac{F_0}{T} \frac{x_0}{n\pi} \left[\cos n\pi - \cos \frac{n\pi x_0}{l} \right] \right. \\ &\quad \left. - \frac{F_0}{T} \frac{x_0 l}{n^2\pi^2} \left[-n\pi \cos n\pi - \sin \frac{n\pi x_0}{l} \right. \right. \\ &\quad \left. \left. + \frac{n\pi x_0}{l} \cos \frac{n\pi x_0}{l} \right] \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{l} \left\{ \left[\frac{F_0 l (1 - x_0)}{T n^2 \pi^2} + \frac{F_0 l x_0}{T n^2 \pi^2} \right] \sin \frac{n \pi x_0}{l} \right. \\
&\quad + \left[\frac{F_0 l x_0}{T n \pi} - \frac{F_0 x_0^2}{T n \pi} - \frac{F_0 (1 - x_0) x_0}{T n \pi} \right] \\
&\quad \cos \frac{n \pi x_0}{l} \\
&\quad \left. - \left[\frac{F_0 l x_0}{T n \pi} - \frac{F_0 x_0 l}{T n \pi} \right] \cos n \pi \right\} \\
&= \frac{2}{l} \frac{F_0}{T} \frac{l^2}{n^2 \pi^2} \sin \frac{n \pi x_0}{l} \\
&= \frac{2 F_0 l}{T \pi^2} \cdot \frac{1}{n^2} \sin \frac{n \pi x_0}{l}, \\
\therefore u(x, t) &= \sum_{n=1}^{\infty} A_n \cos \frac{n \pi a}{l} t \sin \frac{n \pi}{l} x \\
&= \frac{2 F_0 l}{T \pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n \pi x_0}{l} \sin \frac{n \pi}{l} x \cos \frac{n \pi a}{l} t.
\end{aligned}$$

2. 求解细杆导热问题, 杆长 l , 两端保持为零度, 初始温度分布 $u|_{t=0} = bx(l-x)/l^2$.

解: 定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0, \left(a^2 = \frac{k}{C\rho} \right) (0 \leq x \leq l), & (1) \end{cases}$$

$$\begin{cases} u|_{x=0} = u|_{x=l} = 0, & (2) \end{cases}$$

$$\begin{cases} u|_{t=0} = bx(l-x)/l^2. & (3) \end{cases}$$

设 $u(x, t) = X(x)T(t)$ 代入泛定方程:

$$\frac{T'}{a^2 T} = \frac{X''}{X} = -\lambda,$$

由此得到:
$$\begin{cases} T' + a^2 \lambda T = 0. \\ X'' + \lambda X = 0, X(0) = X(l) = 0. \end{cases}$$

解 X 得

$$X(x) = \sum (A'_n \cos \sqrt{\lambda} x + B'_n \sin \sqrt{\lambda} x),$$

由边界条件 (2) 得 $X(0) = 0$.

$$\therefore A'_n = 0,$$

$$X(l) = 0,$$

$$\therefore \sin \sqrt{\lambda} l = 0, \sqrt{\lambda} l = n\pi,$$

$$\therefore \lambda = \frac{n^2 \pi^2}{l^2},$$

$$\therefore X(x) = \sum B'_n \sin \frac{n\pi}{l} x,$$

又根据有关 T 的方程得

$$T'_n + \frac{n^2 \pi^2 a^2}{l^2} T_n = 0,$$

$$T_n = C_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t},$$

$$\begin{aligned} \therefore u(x, t) &= \sum_{n=1}^{\infty} B'_n C_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \sin \frac{n\pi}{l} x \\ &= \sum_{n=1}^{\infty} B_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \sin \frac{n\pi}{l} x. \end{aligned}$$

由初始条件 (3) 得:

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{l} x = \frac{bx(l-x)}{l^2},$$

$$\begin{aligned} \therefore B_n &= \frac{2}{l} \int_0^l \frac{b\xi(l-\xi)}{l^2} \sin \frac{n\pi\xi}{l} d\xi \\ &= \frac{2b}{l^3} \int_0^l \xi(l-\xi) \sin \frac{n\pi\xi}{l} d\xi \\ &= \frac{2b}{l^3} \left\{ l \cdot \frac{l^2}{n^2 \pi^2} \sin \frac{n\pi}{l} \xi \right\}_0^l \end{aligned}$$

$$\begin{aligned}
& -\frac{l^2}{n\pi} \xi \cos \frac{n\pi}{l} \xi \Big|_0^l \\
& -\frac{\xi l^2}{n^2 \pi^2} \left(2 \sin \frac{n\pi}{l} \xi - \frac{n\pi}{l} \xi \cos \frac{n\pi}{l} \xi \right) \Big|_0^l \\
& + \frac{2l}{n^3 \pi^3} \cos \frac{n\pi}{l} \xi \Big|_0^l \Big\} \\
& = \frac{2b}{l^3} \left\{ -\frac{l^3}{n\pi} (-1)^n + \frac{l^3}{n\pi} (-1)^n \right. \\
& \quad \left. + \frac{2l^3}{n^3 \pi^3} [(-1)^n - 1] \right\} \\
& = \begin{cases} \frac{8b}{\pi^3 (2k+1)^3}, & (\text{当 } n \text{ 为奇数 } n=2k+1 \text{ 时}), \\ 0, & (\text{当 } n \text{ 为偶数时}), \end{cases} \\
\therefore u(x,t) &= \sum_{k=0}^{\infty} B_k e^{-\frac{(2k+1)^2 \pi^2 a^2 t}{l^2}} \sin \frac{(2k+1)\pi}{l} x \\
&= \frac{8b}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} e^{-\frac{(2k+1)^2 \pi^2 a^2 t}{l^2}} \\
&\quad \sin \frac{(2k+1)\pi}{l} x.
\end{aligned}$$

3. 两端固定弦, 长为 l . (1) 用宽为 2δ 的平面锤敲击弦的 $x=x_0$ 点. (2) 用宽度为 2δ 的余弦式凸面锤敲击弦的 $x=x_0$, 求解弦的振动.

解: (i) 若锤为平面锤, 定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & (0 < x < l), \\ u|_{x=0} = u|_{x=l} = 0, \\ u|_{t=0} = 0, \\ u_t|_{t=0} = \begin{cases} 0, & (0 < x < x_0 - \delta, x_0 + \delta < x < l), \\ v_0, & (x_0 - \delta < x < x_0 + \delta). \end{cases} \end{cases}$$

根据边界条件, 可知本征函数为 $\sin \frac{n\pi x}{l}$, 故弦的一般振动可表示为:

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t \right) \sin \frac{n\pi x}{l},$$

以此代入初始条件得:

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = 0.$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \sin \frac{n\pi x}{l} = \begin{cases} 0, & (0 < x < x_0 - \delta), \\ x_0 + \delta < x < l \\ v_0 & (x_0 - \delta < x < x_0 + \delta). \end{cases}$$

由此可得傅里叶系数

$$A_n = \frac{2}{l} \int_0^l 0 \cdot \sin \frac{n\pi \xi}{l} d\xi = 0,$$

$$\begin{aligned} B_n &= \frac{2}{n\pi a} \int_{x_0-\delta}^{x_0+\delta} v_0 \sin \frac{n\pi \xi}{l} d\xi = \frac{-2}{n\pi a} \frac{v_0 l}{n\pi} \cos \frac{n\pi \xi}{l} \Big|_{x_0-\delta}^{x_0+\delta} \\ &= \frac{2v_0 l}{n^2 \pi^2 a} \left[\cos \frac{n\pi}{l} (x_0 - \delta) - \cos \frac{n\pi}{l} (x_0 + \delta) \right] \\ &= \frac{4v_0 l}{n^2 \pi^2 a} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi \delta}{l}, \end{aligned}$$

$$\begin{aligned} \therefore u(x, t) &= \frac{4v_0 l}{a\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi \delta}{l} \sin \frac{n\pi a t}{l} \\ &\quad \sin \frac{n\pi x}{l}. \end{aligned}$$

(ii) 若为余弦式锤, 则定解问题为:

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & (0 < x < l), \\ u(0, t) = u(l, t) = 0, \\ u(x, 0) = 0, \\ u_t(x, 0) = \begin{cases} 0, & (0 < x < x_0 - \delta, x_0 + \delta < x < l), \\ v_0 \cos \frac{x - x_0}{2\delta} \pi, & (x_0 - \delta < x < x_0 + \delta). \end{cases} \end{cases}$$

根据边界条件可知其本征函数为 $\sin \frac{n\pi x}{l}$, 因而弦的一般解可表示为:

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l} \right) \sin \frac{n\pi x}{l},$$

代入初始条件得:

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = 0,$$

$$\begin{aligned} u_t(x, 0) &= \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \sin \frac{n\pi x}{l} \\ &= \begin{cases} 0, & (0 < x < x_0 - \delta, x_0 + \delta < x < l), \\ v_0 \cos \frac{x - x_0}{2\delta} \pi, & (x_0 - \delta < x < x_0 + \delta), \end{cases} \end{aligned}$$

从上列二式可得:

$$A_n = 0,$$

$$\begin{aligned} B_n &= \frac{2}{n\pi a} \int_{x_0 - \delta}^{x_0 + \delta} v_0 \cos \frac{x - x_0}{2\delta} \pi \sin \frac{n\pi x}{l} dx \\ &= \frac{2v_0}{n\pi a} \int_{x_0 - \delta}^{x_0 + \delta} \left[\cos \frac{x_0 \pi}{2\delta} \cos \frac{\pi x}{2\delta} + \sin \frac{x_0 \pi}{2\delta} \sin \frac{\pi x}{2\delta} \right] \\ &\quad \sin \frac{n\pi x}{l} dx \\ &= \frac{2v_0}{n\pi a} \int_{x_0 - \delta}^{x_0 + \delta} \left[\cos \frac{x_0 \pi}{2\delta} \cos \frac{\pi x}{2\delta} \sin \frac{n\pi x}{l} + \sin \frac{x_0 \pi}{2\delta} \right. \\ &\quad \left. \sin \frac{\pi x}{2\delta} \sin \frac{n\pi x}{l} \right] dx \\ &= \frac{v_0}{n\pi a} \int \left[\cos \frac{x_0 \pi}{2\delta} \left(\sin \frac{l\pi + 2\delta n\pi}{2\delta l} x \right. \right. \\ &\quad \left. \left. - \sin \frac{l\pi - 2\delta n\pi}{2\delta l} x \right) \right. \end{aligned}$$

$$\begin{aligned}
& + \sin \frac{x_0 \pi}{2\delta} \left(\cos \frac{l\pi - 2\delta n\pi}{2\delta l} x \right. \\
& \left. - \cos \frac{l\pi + 2\delta n\pi}{2\delta l} x \right) dx \\
& = \frac{v_0}{n\pi a} \left[\cos \frac{x_0 \pi}{2\delta} \left(\frac{2l\delta}{l\pi - 2\delta n\pi} \cos \frac{l\pi - 2\delta n\pi}{2l\delta} x \right. \right. \\
& \left. \left. - \frac{2l\delta}{l\pi + 2\delta n\pi} \cos \frac{l\pi + 2\delta n\pi}{2l\delta} x \right) \right]_{x_0 - \delta}^{x_0 + \delta} \\
& + \sin \frac{x_0 \pi}{2\delta} \left(\frac{2l\delta}{l\pi - 2\delta n\pi} \sin \frac{l\pi - 2\delta n\pi}{2l\delta} x \right. \\
& \left. - \frac{2l\delta}{l\pi + 2\delta n\pi} \sin \frac{l\pi + 2\delta n\pi}{2l\delta} x \right) \Big|_{x_0 - \delta}^{x_0 + \delta} \Big] \\
& = \frac{v_0}{n\pi a} \left\{ \frac{2l\delta}{l\pi - 2\delta n\pi} \left[\cos \frac{x_0 \pi}{2\delta} \cos \frac{l\pi - 2\delta n\pi}{2l\delta} \right. \right. \\
& \quad \left. \left. (x_0 + \delta) + \sin \frac{x_0 \pi}{2\delta} \sin \frac{l\pi - 2\delta n\pi}{2l\delta} (x_0 + \delta) \right] \right. \\
& \quad \left. - \frac{2l\delta}{l\pi - 2\delta n\pi} \left[\cos \frac{x_0 \pi}{2\delta} \cos \frac{l\pi - 2\delta n\pi}{2l\delta} (x_0 - \delta) \right. \right. \\
& \quad \left. \left. + \sin \frac{x_0 \pi}{2\delta} \sin \frac{l\pi - 2\delta n\pi}{2l\delta} (x_0 - \delta) \right] \right. \\
& \quad \left. - \frac{2l\delta}{l\pi + 2\delta n\pi} \left[\cos \frac{x_0 \pi}{2\delta} \cos \frac{l\pi + 2\delta n\pi}{2l\delta} (x_0 + \delta) \right. \right. \\
& \quad \left. \left. + \sin \frac{x_0 \pi}{2\delta} \sin \frac{l\pi + 2\delta n\pi}{2l\delta} (x_0 + \delta) \right] \right. \\
& \quad \left. + \frac{2l\delta}{l\pi + 2\delta n\pi} \left[\cos \frac{x_0 \pi}{2\delta} \cos \frac{l\pi + 2\delta n\pi}{2l\delta} (x_0 - \delta) \right. \right. \\
& \quad \left. \left. + \sin \frac{x_0 \pi}{2\delta} \sin \frac{l\pi + 2\delta n\pi}{2l\delta} (x_0 - \delta) \right] \right\} \\
& = \frac{v_0}{n\pi a} \left\{ \frac{2l\delta}{l\pi - 2\delta n\pi} \left[\cos \left(\frac{2n\pi x_0 - l\pi - 2\delta n\pi}{2l} \right) \right. \right.
\end{aligned}$$

$$\begin{aligned}
& - \cos \left(\frac{2n\pi x_0 + l\pi - 2\delta n\pi}{2l} \right) \Bigg] \\
& - \frac{2l\delta}{l\pi - 2\delta n\pi} \left[\cos \left(\frac{-2n\pi x_0 - l\pi - 2\delta n\pi}{2l} \right) \right. \\
& \left. - \cos \left(\frac{-2n\pi x_0 + l\pi + 2\delta n\pi}{2l} \right) \right] \\
& = \frac{4v_0}{n\pi a} \left[\frac{l\delta}{l\pi - 2\delta n\pi} \sin \frac{n\pi x_0}{l} \sin \left(\frac{\pi}{2} - \frac{n\pi\delta}{l} \right) \right. \\
& \quad \left. + \frac{l\delta}{l\pi + 2\delta n\pi} \sin \frac{n\pi x_0}{l} \sin \left(\frac{\pi}{2} + \frac{n\pi\delta}{l} \right) \right] \\
& = \frac{4v_0\delta}{n\pi^2 a} \left[\frac{1}{1 - \frac{2\delta n}{l}} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi\delta}{l} \right. \\
& \quad \left. + \frac{1}{1 + \frac{2\delta n}{l}} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi\delta}{l} \right] \\
& = \frac{8v_0\delta}{n\pi^2 a} \frac{1}{1 - \left(\frac{2\delta n}{l} \right)^2} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi\delta}{l}, \\
\therefore u(x, t) &= \frac{8v_0\delta}{a\pi^2} \sum_{n=1}^{\infty} \frac{1}{n} \cdot \frac{1}{1 - \left(\frac{2\delta n}{l} \right)^2} \sin \frac{n\pi x_0}{l} \\
& \quad \cos \frac{n\pi\delta}{l} \sin \frac{n\pi a t}{l} \sin \frac{n\pi x}{l}.
\end{aligned}$$

4. 长为 l 的均匀杆，两端受压从而长度缩为 $l(1-2\varepsilon)$ ，放手后自由振动，求解杆的这一振动。

解：

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, (0 < x < l), \\ u_x|_{x=0} = u_x|_{x=l} = 0, \\ u|_{t=0} = 2\varepsilon \left(\frac{l}{2} - x \right), \\ u_t|_{t=0} = 0. \end{cases}$$

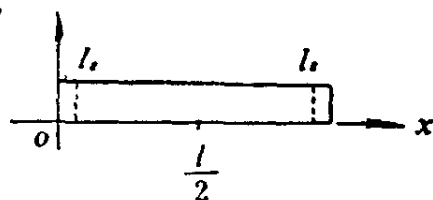


图 10-3

因为是第二类边界条件, 所以要用本征函数 $\cos \frac{n\pi}{l} x$ 展开, 设解为

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l} \right) \cos \frac{n\pi}{l} x,$$

$$\begin{aligned} \therefore \left. \frac{\partial u}{\partial t} \right|_{t=0} &= \sum_{n=1}^{\infty} B_n \frac{n\pi a}{l} \cos \frac{n\pi at}{l} \cos \frac{n\pi}{l} x \Big|_{t=0} \\ &= 0, \end{aligned}$$

$$\therefore B_n = 0.$$

$$u(x, t) = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi at}{l} \cos \frac{n\pi x}{l},$$

$$\therefore u(x, 0) = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l} = 2\varepsilon \left(\frac{l}{2} - x \right),$$

$$\begin{aligned} \therefore A_0 &= \frac{1}{l} \int_0^l 2\varepsilon \left(\frac{l}{2} - x \right) dx = \varepsilon \int_0^l dx - \frac{2\varepsilon}{l} \int_0^l x dx \\ &= \varepsilon x \Big|_0^l - \frac{2\varepsilon}{l} \frac{x^2}{2} \Big|_0^l = \varepsilon l - \varepsilon l = 0. \end{aligned}$$

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^l 2\varepsilon \left(\frac{l}{2} - x \right) \cos \frac{n\pi x}{l} dx \\ &= \frac{4\varepsilon}{l} \int_0^l \frac{l}{2} \cos \frac{n\pi x}{l} dx - \frac{4\varepsilon}{l} \int_0^l x \cos \frac{n\pi x}{l} dx \\ &= 2\varepsilon \left[\frac{l}{n\pi} \sin \frac{n\pi x}{l} \right]_0^l \\ &\quad - \frac{4\varepsilon}{l} \left(\frac{l}{n\pi} \right)^2 \left[\frac{n\pi x}{l} \sin \frac{n\pi x}{l} + \cos \frac{n\pi x}{l} \right]_0^l \\ &= -\frac{4\varepsilon l}{n^2 \pi^2} \left[\cos n\pi - 1 \right] = \frac{4\varepsilon l}{n^2 \pi^2} \left[1 - (-1)^n \right]. \end{aligned}$$

$$= \begin{cases} \frac{8\epsilon l}{\pi^2 (2k+1)^2}, & \text{当 } n \text{ 为奇数 } 2k+1 \text{ 时,} \\ 0, & \text{当 } n \text{ 为偶数时,} \end{cases}$$

$$\therefore u(x, t) = \frac{8\epsilon l}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos \frac{(2k+1)\pi at}{l} \cos \frac{(2k+1)\pi x}{l}.$$

5. 长为 l 的杆, 一端固定, 另一端受力 F_0 而伸长. 求解杆在放手后的振动.

解: 定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & (0 \leq x \leq l), \\ u|_{x=0} = 0, & u_x|_{x=l} = 0, \\ u(x, 0) = \int_0^x \frac{\partial u}{\partial x} dx = \int_0^x \frac{F_0}{YS} dx \\ \quad = \frac{F_0 X}{YS}, & (0 \leq x \leq l), \\ u_t|_{t=0} = 0, \end{cases}$$

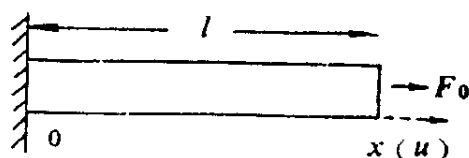


图 10-4

本题是既有第一类边界条件也有第二类边界条件的问题.

令 $u = X(x)T(t)$ 代入泛定方程分离变量得

$$\begin{cases} T'' + \lambda a^2 T = 0, \\ X'' + \lambda X = 0, \\ X(0) = X'(l) = 0. \end{cases}$$

(i) 若 $\lambda < 0$, 则 $X = C_1 l^{\sqrt{-\lambda} x} + C_2 l^{-\sqrt{-\lambda} x}$ 则有 $C_1 + C_2 = 0$ 和 $C_1 l^{+\sqrt{-\lambda} l} + C_2 l^{-\sqrt{-\lambda} l} = 0$.

$\therefore C_1 = C_2 = 0$, $X \equiv 0$ 无意义, 因此, $\lambda < 0$ 的情况应排除.

(ii) $\lambda = 0$, 则方程 $X'' + \lambda X = 0$ 的解为

$$X(x) = C_1 x + C_2,$$

由 X 边界条件便可知, $C_2 = 0$, $C_1 l + C_2 = 0$, $\therefore C_1 = 0$ 从而 $X(x) \equiv 0$ 也没有意义, 也应排除 $\lambda = 0$ 情况.

(iii) 仅当 $\lambda > 0$ 时才有意义的解,

$$X = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x$$

利用边界条件 (8) 可知, $C_1 = 0$,

$$X'(l) = C_2 \sqrt{\lambda} \cos \sqrt{\lambda} l = 0 \quad \therefore \cos \sqrt{\lambda} l = 0$$

$$\sqrt{\lambda} l = n\pi + \frac{1}{2} \pi = \left(n + \frac{1}{2}\right) \pi, \quad n = 0, 1, 2, \dots$$

$$\therefore \lambda = \frac{\left(n + \frac{1}{2}\right)^2 \pi^2}{l^2},$$

$$X_n(x) = C_2 \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x.$$

以本征值 $\lambda = \frac{\left(n + \frac{1}{2}\right)^2 \pi^2}{l^2}$ 代入到关于 T 的方程得解

$$T_n(t) = A'_n \cos \frac{\left(n + \frac{1}{2}\right)}{l} a \pi t + B'_n \sin \frac{\left(n + \frac{1}{2}\right)}{l} a \pi t,$$

$$\begin{aligned} \therefore u(x, t) &= \sum_{n=0}^{\infty} \left(A'_n \cos \frac{\left(n + \frac{1}{2}\right)}{l} a \pi t \right. \\ &\quad \left. + B'_n \sin \frac{\left(n + \frac{1}{2}\right)}{l} a \pi t \right) C_2 \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x \\ &= \sum_{n=0}^{\infty} \left[A_n \cos \frac{\left(n + \frac{1}{2}\right)}{l} a \pi t \right. \\ &\quad \left. + B_n \sin \frac{\left(n + \frac{1}{2}\right)}{l} a \pi t \right] \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x. \end{aligned}$$

利用初始条件, $\left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \therefore B_n = 0,$

再利用初始条件, $u(x, 0) = \frac{F_0 x}{YS},$ 可得

$$\sum_{n=0}^{\infty} A_n \sin \frac{\left(n + \frac{1}{2}\right) \pi x}{l} = \frac{F_0}{YS} x,$$

$$\begin{aligned} \therefore A_n &= \frac{2}{l} \int_0^l \frac{F_0}{YS} \xi \sin \frac{\left(n + \frac{1}{2}\right) \pi}{l} \xi d\xi \\ &= \frac{2l}{\left(n + \frac{1}{2}\right)^2 \pi^2} \frac{F_0}{YS} \int_0^l \left[\frac{\left(n + \frac{1}{2}\right) \pi \xi}{l} \right] \\ &\quad \sin \frac{\left(n + \frac{1}{2}\right) \pi \xi}{l} d \left[\frac{\left(n + \frac{1}{2}\right) \pi \xi}{l} \right] \\ &= \frac{2l}{\left(n + \frac{1}{2}\right)^2 \pi^2} \frac{F_0}{YS} \left[\sin \frac{\left(n + \frac{1}{2}\right) \pi}{l} \xi \right. \\ &\quad \left. - \frac{\left(n + \frac{1}{2}\right) \pi \xi}{l} \cos \frac{\left(n + \frac{1}{2}\right) \pi \xi}{l} \right]_0^l \\ &= \frac{2l}{\left(n + \frac{1}{2}\right)^2 \pi^2} \frac{F_0}{YS} \left[(-1)^n - 0 \right] \\ &= \frac{2l}{\left(n + \frac{1}{2}\right)^2 \pi^2} \frac{F_0}{YS} (-1)^n, \end{aligned}$$

$$\therefore u(x, t) = \frac{8lF_0}{\pi^2 YS} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} \cos \frac{\left(n + \frac{1}{2}\right) \pi a t}{l}$$

$$\sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x.$$

6. 长为 l 的理想传输线, 远端开路, 先把传输线充电到电位差 v_0 , 然后把近端短路. 求解线上的电压 $v(x, t)$.

解: 泛定方程 $v_{tt} - a^2 v_{xx} = 0$, $a^2 = \frac{1}{LC}$, $(0 < x < l)$.

边界条件 $\begin{cases} v(0, t) = 0, \\ v_x(l, t) = -\left(R + L \frac{\partial}{\partial t}\right) j \Big|_{x=l} = 0, \end{cases}$

初始条件 $v(x, 0) = v_0$,

$$v_t(x, 0) = -\frac{1}{C} j_x \Big|_{t=0} = 0,$$

与上题 (第 5 题) 类似, 具有第一类和第二类边界条件, 从而知道其一般解应为:

$$v(x, t) = \sum_{n=0}^{\infty} \left(A_n \cos \frac{n + \frac{1}{2}}{l} a \pi t + B_n \sin \frac{n + \frac{1}{2}}{l} a \pi t \right) \sin \frac{n + \frac{1}{2}}{l} \pi x,$$

由于 $\frac{\partial v}{\partial t} \Big|_{t=0} = 0$,

$$\therefore B_n = 0.$$

$$\therefore v(x, t) \Big|_{t=0} = \sum_{n=0}^{\infty} A_n \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x = v_0.$$

$$\therefore A_n = \frac{2}{l} \int_0^l v_0 \sin \frac{\left(n + \frac{1}{2}\right) \pi}{l} \xi d\xi$$

$$\begin{aligned}
&= \frac{2}{l} \frac{v_0 l}{\left(n + \frac{1}{2}\right)\pi} \left[-\cos \frac{\left(n + \frac{1}{2}\right)\pi \xi}{l} \right]_0^l \\
&= -\frac{4v_0}{(2n+1)\pi} \left[1 - \cos \left(n + \frac{1}{2}\right)\pi \right] \\
&= \frac{4v_0}{\pi(2n+1)}.
\end{aligned}$$

$$\begin{aligned}
\therefore v(x, t) &= \frac{4v_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{(2n+1)} \sin \frac{(2n+1)\pi x}{2l} \\
&\quad \cos \frac{(2n+1)\pi a t}{2l}.
\end{aligned}$$

7. 长为 l 的杆，上端固定在电梯天花板，杆身竖直，下端自由、电梯下降，当速度为 v_0 时突然停止。求解杆的振动。

解：泛定方程 $u_{tt} - a^2 u_{xx} = 0, (0 \leq x \leq l).$ (1)

边界条件
$$\begin{cases} u|_{x=0} = 0, \\ \frac{\partial u}{\partial x} \Big|_{x=l} = 0. \end{cases} \quad (2)$$

初始条件
$$\begin{cases} u|_{t=0} = 0, \\ u_t|_{t=0} = v_0. \end{cases} \quad (3)$$

本题中既有第一类边界条件，也有第二类边界条件，但都是齐次的，可以参阅课本 P.211 的例 2。根据边界条件 (2) 可设

(1) 的解：

$$u = \sum_{n=0}^{\infty} T_n(t) \sin \frac{n + \frac{1}{2}}{l} \pi x, \quad (4)$$

代入泛定方程 (1) 有

$$\sum_{n=0}^{\infty} \left[T_n'' + \frac{\left(n + \frac{1}{2}\right)^2}{l^2} \pi^2 a^2 T_n \right] \sin \frac{n + \frac{1}{2}}{l} \pi x = 0,$$

关于 $T(t)$ 方程的解为:

$$T_n(t) = A_n \cos \frac{\left(n + \frac{1}{2}\right)}{l} \pi a t + B_n \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi a t,$$

$$\text{则: } u(x, t) = \sum_{n=0}^{\infty} \left[A_n \cos \frac{(2n+1)}{2l} \pi a t + B_n \sin \frac{(2n+1)}{2l} \pi a t \right] \sin \frac{(2n+1)}{2l} \pi x.$$

根据初始条件 (3) $u(x, 0) = 0$, 可知 $A_n = 0$.

$$u(x, t) = \sum_{n=0}^{\infty} B_n \sin \frac{(2n+1)}{2l} \pi a t \sin \frac{(2n+1)}{2l} \pi x,$$

$$\text{又从 } \frac{\partial u}{\partial t} \Big|_{t=0} = \sum_{n=0}^{\infty} \frac{(2n+1)}{2l} \pi a B_n \cos \frac{(2n+1)}{2l} \pi x$$

$$\pi a t \sin \frac{(2n+1)}{2l} \pi x \Big|_{t=0} = v_0,$$

$$\text{即 } \sum_{n=0}^{\infty} \frac{(2n+1)}{2l} \pi a B_n \sin \frac{(2n+1)}{2l} \pi x = v_0,$$

$$B_n = \frac{2}{l} \frac{2l}{(2n+1)\pi a} \int_0^l v_0 \sin \frac{(2n+1)}{2l} \pi \xi d\xi$$

$$= \frac{2v_0}{\left(n + \frac{1}{2}\right) \pi a} \cdot \frac{l}{\left(n + \frac{1}{2}\right) \pi}$$

$$\int_0^l \sin \frac{\left(n + \frac{1}{2}\right) \pi}{l} \xi d \left(\frac{n + \frac{1}{2}}{l} \pi \xi \right)$$

$$= \frac{2v_0 l}{\left(n + \frac{1}{2}\right)^2 \pi^2 a} \left[-\cos \frac{\left(n + \frac{1}{2}\right)}{l} \pi \xi \right]_0^l$$

$$= \frac{2v_0 l}{\left(n + \frac{1}{2}\right)^2 \pi^2 a}.$$

$$\therefore u(x, t) = \frac{2v_0 l}{\pi^2 a} \sum_{n=0}^{\infty} \frac{1}{\left(n + \frac{1}{2}\right)^2} \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi a t \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x.$$

8. 在铀块中，除了中子的扩散运动之外，还进行着中子的增殖过程，每秒钟在单位体积中产生的中子数正比于该处的中子浓度 u ，从而可表为 βu ，（ β 是表示增殖快慢的常数）。研究厚度为 l 的层状铀块。求临界厚度（铀块厚度超过临界厚度，则中子浓度将随着时间而增长以致铀块爆炸。原子弹里就是这么回事）。

解：设中子的浓度为 u ，扩散系数为 D 。按题意，由于中子的增殖作用，产生的中子数和该处的中子浓度 u 成正比，设单位体积中产生的中子数为 n ，则 $n = \beta u$ ，则

$$\text{泛定方程 } \frac{\partial u}{\partial t} = D \Delta u + \beta u. \quad (1)$$

$$\text{或 } \frac{\partial u}{\partial t} - a^2 \Delta u - \beta u = 0, \quad (a^2 = D).$$

为了结合边界条件求解方程 (1)

用三种方法来解：

方法 I：在临界厚度时， $\frac{\partial u}{\partial t} = 0$ ，

则 (1) 式成为：

$$D \Delta u + \beta u = 0, \quad (2)$$

如右图所示，将铀块看作一维的，

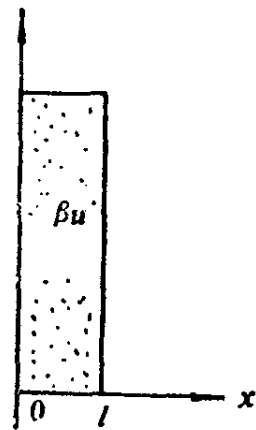


图 10-5

则 (2) 成为:

$$D \frac{\partial^2 u}{\partial x^2} + \beta u = 0 \quad \text{或} \quad \frac{\partial^2 u}{\partial x^2} + \frac{\beta}{D} u = 0,$$

上式的解为:

$$u(x) = A_1 e^{i\sqrt{\frac{\beta}{D}}x} + A_2 e^{-i\sqrt{\frac{\beta}{D}}x}, \quad (3)$$

因为在 $x=0$ 和 $x=l$ 处有相等的浓度, $u(0)=u(l)$, 代入 (3) 式有

$$A_1 + A_2 = A_1 e^{i\sqrt{\frac{\beta}{D}}l} + A_2 e^{-i\sqrt{\frac{\beta}{D}}l},$$

或

$$A_1(1 - e^{i\sqrt{\frac{\beta}{D}}l}) + A_2(1 - e^{-i\sqrt{\frac{\beta}{D}}l}) = 0, \quad (4)$$

$A_1, A_2 \neq 0$, 必然有:

$$\left. \begin{aligned} 1 - e^{i\sqrt{\frac{\beta}{D}}l} &= 1 - \cos \sqrt{\frac{\beta}{D}}l - i \sin \sqrt{\frac{\beta}{D}}l = 0, \\ 1 - e^{-i\sqrt{\frac{\beta}{D}}l} &= 1 - \cos \sqrt{\frac{\beta}{D}}l + i \sin \sqrt{\frac{\beta}{D}}l = 0, \end{aligned} \right\} \quad (5)$$

要 (5) 式成立, 则必须有: $\sqrt{\frac{\beta}{D}}l = \pi$,

$$\therefore \text{临界厚度 } L = \sqrt{\frac{D}{\beta}} \pi = \frac{a\pi}{\sqrt{\beta}} \quad (a = \sqrt{D}).$$

方法 II: (1) 式写成如下形式:

$$u_t - a^2 u_{xx} - \beta u = 0. \quad (6)$$

令 $u = U e^{\beta t}$, $u_t = \beta u e^{\beta t} + v_t e^{\beta t} u_x = U_x e^{\beta t}$, $u_{xx} = v_{xx} e^{\beta t}$,

代入 (6) 式:

$$u \beta e^{\beta t} + v_t e^{\beta t} - a^2 v_{xx} e^{\beta t} - \beta v e^{\beta t} = 0,$$

即

$$v_t - a^2 v_{xx} = 0. \quad (7)$$

由边界条件 $u|_{x=0} = u|_{x=l} = 0$, 即 $v|_{x=0} = v|_{x=l} = 0$,

再令 $v = X(x)T(t)$, 利用边界条件 $v|_{x=0} = v|_{x=l} = 0$ 写出 (7) 的试解为

$$v = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l},$$

代入到 (7) 式有:

$$\sum_{n=1}^{\infty} \left(T'_n + \frac{n^2\pi^2 a^2}{l} T_n \right) \sin \frac{n\pi x}{l} = 0,$$

要上式成立, 只有 $T'_n + \frac{n^2\pi^2 a^2}{l} T_n = 0$ 才行, 于是

$$T_n = C_n e^{-\frac{n^2\pi^2 a^2}{l^2} t}, \quad v_n = C_n e^{-\frac{n^2\pi^2 a^2}{l^2} t} \sin \frac{n\pi x}{l},$$

本征解 $u_n(x, t) = v_n l \beta t = C_n e^{\left(\beta - \frac{n^2\pi^2 a^2}{l^2}\right) t} \sin \frac{n\pi x}{l},$

由指数项可以看出, 当 $\beta > \frac{\pi^2 a^2}{l^2}$ 时, $n=1$ 的解将随时间增长,

设临界厚度为 L , 则 $\beta = \frac{\pi^2 a^2}{L^2},$

$$\therefore L = \frac{\pi a}{\sqrt{\beta}}.$$

方法 II: 令 $u = X(x)T(t)$ 代入泛定方程 (6)

而有 $XT' - a^2 X''T - \beta XT = 0,$

即
$$\frac{T' - \beta T}{a^2 T} = \frac{X''}{X} = -\lambda^2,$$

于是
$$\begin{cases} X'' + \lambda^2 X = 0, & X(0) = X(l) = 0. \end{cases} \quad (8)$$

$$T' + (a^2 \lambda^2 - \beta) T = 0. \quad (9)$$

由 (8) 可得 $X_n = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{l} x, \quad \lambda^2 = \frac{n^2 \pi^2}{l^2}.$

由 (9) 可得 $T_n = C_n e^{-\left(\frac{a^2 \pi^2 n^2}{l^2} - \beta\right) t},$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} C_n e^{-\left(\frac{a^2 n^2 \pi^2}{l^2} - \beta\right)t} \sin \frac{n\pi}{l} x. \quad (10)$$

(10) 的指数项当 $n=1$ 时有三种情况:

若 $\beta - \frac{a^2 \pi^2}{l^2} > 0$, 则浓度 u 将随时间而增长, 便可能产生爆炸。

$\beta - \frac{a^2 \pi^2}{l^2} < 0$, 则浓度将随时间增长而减小, 反应堆可能熄灭。

$\beta - \frac{a^2 \pi^2}{l^2} = 0$, 则浓度 u 不随时间而变化, 这时的 l 就是临界厚度, 写作 L , 则有

$$L = \frac{a\pi}{\sqrt{\beta}}.$$

9. 求解薄膜的恒定表面浓度扩散问题。薄膜厚度为 l 。杂质从两面进入薄膜。由于薄膜周围气氛中含有充分的杂质, 薄膜表面上的杂质浓度得以保持为恒定的 N_0 。对于较大的 t , 把所得答案简化。

$$\text{解: } u_t - a^2 u_{xx} = 0, \quad \begin{cases} u(0, t) = N_0, & u(x, 0) = 0. \\ u(l, t) = N_0, \end{cases}$$

$$\text{令 } u = W + N_0,$$

则 $W = u - N_0$. 代入泛定方程:

$$W_t - a^2 W_{xx} = 0, \quad \begin{cases} W(0, t) = 0, & W(x, 0) = -N_0. \\ W(l, t) = 0, \end{cases}$$

经代换后的 W 的方程组中, 有齐次的边界条件。

$$\text{令 } W(x, t) = X(x)T(t), \quad \frac{T'}{a^2 T} = \frac{X''}{X} = -\lambda,$$

$$\text{即 } \begin{cases} T' + a^2 \lambda T = 0, & (1) \\ X'' + \lambda X = 0, & X(0) = X(l) = 0, \end{cases} \quad (2)$$

由 (2) 式得 $X_n(x) = A_n \cos \sqrt{\lambda} x + B_n \sin \sqrt{\lambda} x$.

$$\because X(0) = 0,$$

$$\therefore A_n = 0.$$

$$\text{又 } X(l) = 0, B_n \neq 0, \therefore \sin \sqrt{\lambda} x = 0,$$

$$\lambda = \frac{n^2 \pi^2}{l^2},$$

$$\therefore X(x) = \sum B_n \sin \frac{n\pi}{l} x.$$

由关于 T 的方程 (1) 式得

$$T_n = C_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t},$$

$$\begin{aligned} \therefore W(x, t) &= \sum_{n=1}^{\infty} X_n T_n \\ &= \sum_{n=1}^{\infty} B_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \sin \frac{n\pi}{l} x. \end{aligned}$$

为了确定系数 B_n , 可以利用初始条件:

$$W(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} = -N_0,$$

$$\begin{aligned} \therefore B_n &= \frac{2}{l} \int_0^l (-N_0) \sin \frac{n\pi}{l} \xi d\xi \\ &= -\frac{2N_0}{l} \cdot \frac{l}{n\pi} \left[-\cos \frac{n\pi}{l} \xi \right]_0^l \\ &= \frac{2N_0}{n\pi} [\cos n\pi - 1] = \frac{2N_0}{n\pi} [(-1)^n - 1], \end{aligned}$$

$$= \begin{cases} -\frac{4N_0}{\pi(2k+1)}, & (\text{当 } n=2k+1 \text{ 奇数时, } k=0, 1, 2, \dots), \\ 0, & (\text{当 } n \text{ 为偶数时}), \end{cases}$$

$$\therefore W(x, t) = \sum_{k=0}^{\infty} \left[-\frac{4N_0}{\pi(2k+1)} \right] e^{-\frac{(2k+1)^2 \pi^2 a^2}{l^2} t}$$

$$\begin{aligned}
 u(x,t) &= N_0 + W \sin \frac{(2k+1)\pi x}{l}, \\
 &= N_0 - \frac{4N_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} e^{-\frac{(2k+1)^2 \pi^2 a^2}{l^2} t} \sin \frac{(2k+1)\pi}{l} x.
 \end{aligned}$$

对于较大的 t , 考虑指数因子 (当 $t > 0$) :

$$e^{-\frac{(2k+1)^2 \pi^2 a^2}{l^2} t},$$

它随时间 t 的增大而急剧减小 (参阅课本 P217), u 的级数解将收敛得很快, t 越大, 级数收敛得越快. 当 $t > 0.18 \frac{l^2}{a^2}$ 时, 可以只保留 $k=0$ 的一项, 略去 $k > 0$ 的项, 其误差 $< 1\%$, 故 t 很大时,

$$u(x,t) = N_0 - \frac{4N_0}{\pi} e^{-\frac{\pi^2 a^2}{l^2} t} \sin \frac{\pi x}{l}.$$

10. 把上题改为限定源扩散. 这是说, 薄膜两面的表层已含有一定的杂质, 比方说, 每单位表面积下杂质总量 Φ_0 , 但此外不再有杂质进入薄膜.

解: 泛定方程: $u_t - a^2 u_{xx} = 0$,

对于限定源扩散问题, 薄膜两面的表面上含有一定的杂质浓度, 设每单位表面积下杂质总量 Φ_0 , 随着扩散时间的增长, 杂质浓度具有趋向均匀的趋势, 这是因为表面上不再有杂质粒子流 $j(x)$ 进入界面, 这条件可写为:

$$j(x) \Big|_{x=0} = -D \frac{\partial u(x,t)}{\partial x} \Big|_{x=0} = 0,$$

因为扩散系数 $D \neq 0$

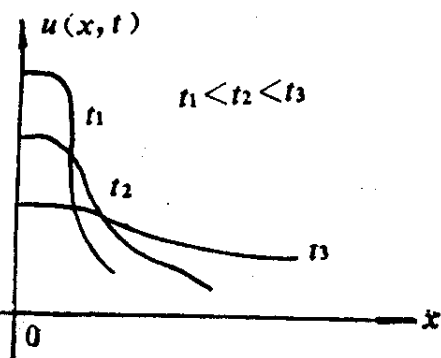


图 10-6

∴ 可以写出边界条件 $\frac{\partial u}{\partial x} \Big|_{x=0} = 0$,

同理, 对薄片的另一面 $x=l$ 处, 同样有 $\frac{\partial u}{\partial x} \Big|_{x=l} = 0$,

在初始时刻 $t=0$ 时的表面杂质浓度可表示为:

$$u(x,0) = \begin{cases} \Phi_0 \delta(x-0), & (0 < x < \epsilon), \\ 0, & (\epsilon < x < l-\epsilon), \\ \Phi_0 \delta[x-(l-0)], & (l-\epsilon < x < l), \end{cases}$$

以 $u(x,t) = X(x)T(t)$ 代入泛定方程和边界条件得:

$$\begin{cases} T' + a^2 \lambda T = 0, \\ X'' + \lambda X = 0, \quad X'(0) = X'(l) = 0. \end{cases}$$

对于 X 组成的本征问题, 解得本征值为 $\lambda = \frac{k^2 \pi^2}{l^2}$,

本征函数 $X(x) = C_1 \cos \frac{k\pi x}{l}$, 代入泛定方程

$$u(x,t) = T_0(t) + \sum_{k=1}^{\infty} T_k(t) \cos \frac{k\pi x}{l}.$$

$$T'_0(t) + \sum_{k=1}^{\infty} \left[T'_k(t) + \frac{k^2 \pi^2 a^2}{l^2} T_k(t) \right] \cos \frac{k\pi x}{l} = 0.$$

$$\because T'_0(t) = 0,$$

$$\therefore T_0 = a_0.$$

由 $T'_k(t) + \frac{k^2 \pi^2 a^2}{l^2} T_k(t) = 0$. 解得

$$T_k(t) = a_k e^{-\frac{k^2 \pi^2 a^2}{l^2} t},$$

于是解 $u(x,t)$ 可表为:

$$u(x,t) = a_0 + \sum_{k=1}^{\infty} a_k e^{-\frac{k^2 \pi^2 a^2}{l^2} t} \cos \frac{k\pi x}{l}.$$

应用初始条件以决定傅里叶系数, 有:

$$\begin{aligned}
u(x, 0) &= a_0 + \sum_{k=1}^{\infty} a_k \cos \frac{k\pi x}{l} \\
&= \begin{cases} \Phi_0 \delta(x-0), & (0 < x < \varepsilon), \\ 0, & (\varepsilon < x < l - \varepsilon), \\ \Phi_0 \delta[x - (l-0)], & (l - \varepsilon < x < l). \end{cases} \\
\therefore a_0 &= \frac{1}{l} \int_0^l \left\{ \Phi_0 \delta(x-0) + \Phi_0 \delta[x - (l-0)] \right\} dx \\
&= \frac{1}{l} \int_0^{\varepsilon} \Phi_0 \delta(x-0) dx + \frac{1}{l} \int_{l-\varepsilon}^l \Phi_0 \delta[x - (l-0)] dx \\
&= 2 \frac{\Phi_0}{l} \\
a_k &= \frac{2}{l} \int_0^l \left\{ \Phi_0 \delta(x-0) + \Phi_0 \delta[x - (l-0)] \right\} \cos \frac{k\pi x}{l} dx \\
&= \frac{2\Phi_0}{l} [1 + (-1)^k] = \begin{cases} \frac{4\Phi_0}{l}, & (\text{当 } k \text{ 为偶数时}), \\ 0, & (\text{当 } k \text{ 为奇数时}), \end{cases} \\
\therefore u(x, t) &= \frac{2\Phi_0}{l} + \frac{4\Phi_0}{l} \sum_{n=1}^{\infty} e^{-\frac{4n^2\pi^2 a^2}{l^2} t} \cos \frac{2n\pi x}{l}.
\end{aligned}$$

对于较大的 t , 只取 $n = 1$ 的一项时, 上式成为:

$$u(x, t) = \frac{2\Phi_0}{l} + \frac{4\Phi_0}{l} e^{-\frac{4\pi^2 a^2}{l^2} t} \cos \frac{2\pi x}{l}.$$

11. 求解细杆导热问题. 杆长 l , 初始温度均匀为 u_0 , 两端分别保持温度为 u_1 和 u_2 .

解: 定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u|_{x=0} = u_1, \quad u|_{x=l} = u_2, \\ u|_{t=0} = u_0, \end{cases}$$

首先设法化去非齐次边界条件,

令
$$u = w + u_1 + \frac{x}{l}(u_2 - u_1),$$

则 w 的定解问题是

$$\begin{cases} \omega_t - a^2 \omega_{xx} = 0, \\ w|_{x=0} = w|_{x=l} = 0, \\ w|_{t=0} = u_0 - u_1 - \frac{x}{l}(u_2 - u_1). \end{cases}$$

$\therefore w$ 有第一类齐次边界条件, 它的解可写为

$$w(x, t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{l} x e^{-\frac{n^2 \pi^2 a^2}{l^2} t}.$$

利用初始条件来确定系数 B_n :

$$w(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{l} x = u_0 - u_1 - \frac{x}{l}(u_2 - u_1),$$

$$\begin{aligned} \therefore B_n &= \frac{2}{l} \int_0^l \left[u_0 - u_1 - \frac{x}{l}(u_2 - u_1) \right] \sin \frac{n\pi}{l} x dx \\ &= \frac{2}{l} \left[(u_0 - u_1) \frac{l}{n\pi} \int_0^l \sin \frac{n\pi}{l} x dx \left(\frac{n\pi}{l} x \right) \right. \\ &\quad \left. - (u_2 - u_1) \int_0^l \frac{x}{l} \sin \frac{n\pi}{l} x dx \right] \\ &= \frac{2}{l} \left[(u_0 - u_1) \frac{-l}{n\pi} \left(\cos \frac{n\pi x}{l} \right) \right]_0^l - \frac{2}{l} (u_2 - u_1) \\ &\quad \frac{l}{n^2 \pi^2} \int_0^l \left(\frac{n\pi x}{l} \right) \sin \frac{n\pi}{l} x dx \left(\frac{n\pi x}{l} \right) \\ &= \frac{2(u_0 - u_1)}{n\pi} [1 - (-1)^n] - \frac{2(u_2 - u_1)}{n^2 \pi^2} \\ &\quad \left[\sin \frac{n\pi x}{l} - \frac{n\pi x}{l} \cos \frac{n\pi x}{l} \right] \\ &= \frac{2(u_0 - u_1)}{n\pi} [1 - (-1)^n] \end{aligned}$$

$$\begin{aligned}
& - \frac{2(u_2 - u_1)}{n^2 \pi^2} [0 - n\pi \cos n\pi - 0 + 0] \\
& = \frac{2(u_0 - u_1)}{n\pi} [1 - (-1)^n] + \frac{2(u_2 - u_1)}{n\pi} (-1)^n. \\
\therefore w(x, t) &= \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left\{ (u_0 - u_1) [1 - (-1)^n] \right. \\
& \quad \left. + (u_2 - u_1)(-1)^n \right\} \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2 a^2}{l^2} t}. \\
u(x, t) &= u_1 + \frac{x}{l} (u_2 - u_1) + w(x, t) \\
&= u_1 + \frac{(u_2 - u_1)x}{l} \\
& \quad + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left\{ (u_0 - u_1) [1 - (-1)^n] + (u_2 - u_1) \right. \\
& \quad \left. (-1)^n \right\} \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2 a^2}{l^2} t}.
\end{aligned}$$

12. 求解细杆导热问题, 初始温度为零, 一端 $x = l$ 保持零度另一端 $x = 0$ 的温度为 At 度 (A 是常数, t 代表时间).

解:

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (1) \end{cases}$$

$$\begin{cases} u|_{x=0} = At, \quad u|_{x=l} = 0, & (2) \end{cases}$$

$$\begin{cases} u|_{t=0} = 0, & (3) \end{cases}$$

解本题时, 希望将边界条件化为齐次时, 仍保持泛定方程为齐次, 令

$$u = Atf(x) + g(x) + v, \quad (4)$$

代入泛定方程(1), 整理后有:

$$v_t - a^2 v_{xx} + Af(x) - a^2 Atf''(x) - a^2 g''(x) = 0, \quad (5)$$

为了得到关于 v 的齐次泛定方程, 就要选择(4)中的 $f(x)$ 和 $g(x)$, 使(5)式中的后面几项消去. 显然, 若 $f(x)$ 取 x 的一次

式, 则 $f''(x) = 0$, 因此选 $f(x) = 1 - \frac{x}{l}$ 即可。从而在 (5) 式

中, 因 $f'(x) = -\frac{1}{l}$, $f''(x) = 0$, 所以 $a^2 A t f''(x) = 0$ 。

同时又为了得到齐次边界条件, 可按下列方程决定 $g(x)$:

$$\begin{cases} a^2 g''(x) = A f(x) = A \left(1 - \frac{x}{l}\right), & (6) \end{cases}$$

$$\begin{cases} g(0) = g(l) = 0, & (7) \end{cases}$$

对 (6) 积分两次:

$$g'(x) = -\frac{A}{a^2} \left(x - \frac{x^2}{2l}\right) + C_1,$$

$$g(x) = -\frac{A}{a^2} \left(\frac{x^2}{2} - \frac{x^3}{6l}\right) + C_1 x + C_2,$$

由边界条件 (7): $\begin{cases} g(0) = C_2 = 0. \\ g(l) = -\frac{A}{a^2} \left(\frac{l^2}{2} - \frac{l^3}{6l}\right) + C_1 l = 0, \end{cases}$

$$\therefore C_1 = -\frac{Al}{3a^2},$$

$$\therefore g(x) = -\frac{l^2 A}{6a^2} \left[\left(\frac{x}{l}\right)^3 - 3 \left(\frac{x}{l}\right)^2 + 2 \left(\frac{x}{l}\right) \right], \quad (8)$$

$$\begin{aligned} u(x, t) = A \left(1 - \frac{x}{l}\right) t - \frac{l^2 A}{6a^2} & \left[\left(\frac{x}{l}\right)^3 - 3 \left(\frac{x}{l}\right)^2 \right. \\ & \left. + 2 \left(\frac{x}{l}\right) \right] + v, \end{aligned} \quad (9)$$

由 (5) 式得齐次泛定方程:

$$v_t - a^2 v_{xx} = 0, \quad (10)$$

由 (2) 和 (9) 得齐次边界条件:

$$v|_{x=0} = v|_{x=l} = 0, \quad (11)$$

由 (3) 和 (9) 得初始条件:

$$v|_{t=0} = \frac{l^2 A}{6a^2} \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 + 2 \left(\frac{x}{l} \right) \right] \quad (12)$$

对于(10)(11)和(12)的定解问题, 其解为:

$$v = \sum_{k=1}^{\infty} C_k \sin \frac{k\pi x}{l} e^{-\frac{k^2 \pi^2 a^2}{l^2} t},$$

$$v|_{t=0} = \sum_{k=1}^{\infty} C_k \sin \frac{k\pi x}{l} = \frac{Al^2}{6a^2} \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 + 2 \left(\frac{x}{l} \right) \right],$$

$$\begin{aligned} C_k &= \frac{2}{l} \int_0^l \frac{Al^2}{6a^2} \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 + 2 \left(\frac{x}{l} \right) \right] \sin \frac{k\pi x}{l} dx \\ &= \frac{Al^2}{3k\pi a^2} \left\{ - \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 + 2 \left(\frac{x}{l} \right) \right] \cos \frac{k\pi x}{l} \Big|_0^l \right. \\ &\quad \left. + \int_0^l \cos \frac{k\pi x}{l} d \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 + 2 \left(\frac{x}{l} \right) \right] \right\} \\ &= \frac{Al^2}{3k\pi a^2} \left[\int_0^l \frac{2}{l} \cos \frac{k\pi x}{l} dx + \int_0^l \frac{6}{l^2} x \cos \frac{k\pi x}{l} dx \right. \\ &\quad \left. + \int_0^l \frac{3}{l^3} x^2 \cos \frac{k\pi x}{l} dx \right] \\ &= \frac{Al^2}{3k^2 \pi^2 a^2} \left\{ 2 \sin \frac{k\pi x}{l} \Big|_0^l + \frac{6}{l} \int_0^l \sin \frac{k\pi x}{l} dx \right. \\ &\quad \left. - \frac{6}{l} x \sin \frac{k\pi x}{l} \Big|_0^l \right. \\ &\quad \left. + \frac{3}{l^2} \left[x^2 \sin \frac{k\pi x}{l} \Big|_0^l - 2 \int_0^l x \sin \frac{k\pi x}{l} dx \right] \right\}. \\ &= \frac{Al^2}{3k^2 \pi^2 a^2} \left\{ - \frac{6}{k\pi} \int_0^l d \left(\cos \frac{k\pi x}{l} \right) + \frac{6}{l^2} \right. \\ &\quad \left. \left[\frac{l}{k\pi} x \cos \frac{k\pi x}{l} \Big|_0^l - \int_0^l \cos \frac{k\pi x}{l} dx \right] \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{Al^2}{3k^2\pi^2a^2} \left\{ -\frac{6}{k\pi} \cos \frac{k\pi x}{l} \Big|_0^l + \frac{6}{l^2} \left[\frac{l^2}{k\pi} \cos k\pi \right. \right. \\
&\quad \left. \left. - \frac{l}{k\pi} \sin \frac{k\pi}{l} x \Big|_0^l \right] \right\} \\
&= \frac{Al^2}{3k^2\pi^2a^2} \left\{ -\frac{6}{k\pi} \cos k\pi + \frac{6}{k\pi} + \frac{6}{k\pi} \cos k\pi \right\} \\
&= \frac{2Al^2}{k^3\pi^3a^2}.
\end{aligned}$$

$$\therefore v = \sum_{k=1}^{\infty} \frac{2Al^2}{\pi^3a^2k^3} \sin \frac{k\pi x}{l} e^{-\frac{k^2\pi^2a^2}{l^2}t},$$

$$\begin{aligned}
u(x,t) &= Af(x)t + g(x) + v(x,t) \\
&= At \left(1 - \frac{x}{l} \right) - \frac{Al^2}{6a^2} \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 \right. \\
&\quad \left. + 2 \left(\frac{x}{l} \right) \right] \\
&\quad + \frac{2Al^2}{\pi^3a^2} \sum_{k=1}^{\infty} \frac{1}{k^3} \sin \frac{k\pi x}{l} e^{-\frac{k^2\pi^2a^2}{l^2}t}.
\end{aligned}$$

13. 求解均匀杆的纵振动, 杆长 l , 一端固定, 另一端受纵向力 $F(t)$ 作用, 初始位移和速度分别是 $\varphi(x)$ 和 $\psi(x)$.

$$F(t) = F_0 \sin \omega t$$

解: 与上题解法同样, 是希望将边界条件化为齐次, 并且又保持泛定方程为齐次. 设:

$$u = f(x) \frac{F_0}{YS} \sin \omega t + v(x,t), \quad (1)$$

代入到关于 $u(x,t)$ 的定解问题:

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, \end{cases} \quad (2)$$

$$\begin{cases} u|_{x=0} = 0, u_x|_{x=l} = \frac{F_0}{YS} \sin \omega t, \end{cases} \quad (3)$$

$$\begin{cases} u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x), \end{cases} \quad (4)$$

由(1)式: $u_t = f(x) \frac{F_0 \omega}{YS} \cos \omega t + v_t,$

$$u_{tt} = -f(x) \frac{F_0 \omega^2}{YS} \sin \omega t + v_{tt},$$

$$u_x = f_x \frac{F_0}{YS} \sin \omega t + v_x(x),$$

$$u_{xx} = f_{xx} \frac{F_0}{YS} \sin \omega t + v_{xx},$$

代入泛定方程(2):

$$\left\{ \begin{array}{l} v_{tt} - f(x) \frac{F_0 \omega^2}{YS} \sin \omega t - a^2 f_{xx} \frac{F_0}{YS} \sin \omega t - a^2 v_{xx} = 0, \end{array} \right. \quad (5)$$

$$\left\{ \begin{array}{l} u(0, t) = f(0) \frac{F_0}{YS} \sin \omega t + v(0, t) = 0 \end{array} \right. \quad (6)$$

$$\left\{ \begin{array}{l} u_x(l, t) = f_x(l) \frac{F_0}{YS} \sin \omega t + v_x(l, t) = \frac{F_0}{YS} \sin \omega t, \end{array} \right. \quad (7)$$

为了得到 $\left\{ \begin{array}{l} u_{xx} - a^2 u_{tt} = 0, \end{array} \right. \quad (8)$

$$\left\{ \begin{array}{l} v(0, t) = 0, v_x(l, t) = 0, \end{array} \right. \quad (9)$$

因此(5)(6)和(7)式必须是

$$\left\{ \begin{array}{l} -f(x) \frac{F_0 \omega^2}{YS} \sin \omega t - a^2 f_{xx} \frac{F_0}{YS} \sin \omega t = 0, \end{array} \right.$$

$$\left\{ \begin{array}{l} f(0) \frac{F_0}{YS} \sin \omega t = 0, \end{array} \right.$$

$$\left\{ \begin{array}{l} f_x(l) \frac{F_0}{YS} \sin \omega t = \frac{F_0}{YS} \sin \omega t, \end{array} \right.$$

即 $\left\{ \begin{array}{l} f_{xx} + \frac{\omega^2}{a^2} f(x) = 0, \end{array} \right. \quad (10)$

$$\left\{ \begin{array}{l} f(0) = 0 \text{ 和 } f_x(l) = 1, \end{array} \right. \quad (11)$$

(10)在(11)条件下的解为

$$f(x) = \frac{a}{\omega \cos\left(\frac{\omega l}{a}\right)} \sin \frac{\omega x}{a}, \quad (12)$$

$$\therefore u(x, t) = \frac{aF_0}{\omega Y S \cos\left(\frac{\omega l}{a}\right)} \sin \frac{\omega x}{a} \sin \omega t + v, \quad (13)$$

由(4)、(1)和(12)式得:

$$v|_{t=0} = \varphi(x), \quad v_t|_{t=0} = \frac{-F_0 a \sin \frac{\omega x}{a}}{Y S \cos \frac{\omega l}{a}} + \psi(x), \quad (14)$$

从而问题转为求解(8)(9)和(13)的定解问题:

$$\text{即} \quad \begin{cases} v_{tt} - a^2 v_{xx} = 0, \\ v|_{x=0} = 0, v_x|_{x=l} = 0, \\ v|_{t=0} = \varphi(x), v_t|_{t=0} = \psi(x) - \frac{F_0 a \sin \frac{\omega x}{a}}{Y S \cos \frac{\omega l}{a}}. \end{cases}$$

参照课本第211页例2, 可以写出定解问题(8)(9)(13)的解:

$$v = \sum_{k=0}^{\infty} \left[A_k \cos \frac{\left(k + \frac{1}{2}\right) a \pi t}{l} + B_k \sin \frac{\left(k + \frac{1}{2}\right) a \pi t}{l} \right] \sin \frac{\left(k + \frac{1}{2}\right) \pi x}{l}. \quad (15)$$

利用初始条件(14), 求出系数 A_k 和 B_k .

$$A_k = \frac{2}{l} \int_0^l \varphi(\xi) \sin \frac{\left(k + \frac{1}{2}\right) \pi \xi}{l} d\xi, \quad (16)$$

$$B_k = \frac{2l}{\left(k + \frac{1}{2}\right)\pi al} \int_0^l \left[\psi(\xi) - \frac{F_0 a \sin \frac{\omega \xi}{a}}{Y S \cos \frac{\omega l}{a}} \right] \sin \frac{\left(k + \frac{1}{2}\right)\pi}{l} \xi d\xi, \quad (17)$$

将(14)代入(13)式

$$u(x,t) = \frac{aF_0 \sin \frac{\omega}{a} x \sin \omega t}{Y \omega S \cos \frac{\omega l}{a}} + \sum_{k=0}^{\infty} \left[A_k \cos \frac{\left(k + \frac{1}{2}\right)a\pi t}{l} + B_k \sin \frac{\left(k + \frac{1}{2}\right)a\pi t}{l} \right] \cdot \sin \frac{\left(k + \frac{1}{2}\right)\pi}{l} x,$$

其中的系数 A_k 和 B_k 由(16)式与(17)式确定。

14. 把弹簧上端 $x = 0$ 加以固定,在静止弹簧下端 $x = l$ 轻轻地挂上质量为 m 的物体,求解弹簧的纵振动.弹簧本身的重量可以忽略不计。

解: 以重力作用下的平衡状态作为基准来计算位移 u , 则泛定方程是齐次的, 定解问题为:

$$u_{tt} - a^2 u_{xx} = 0, \quad (a^2 = Y/\rho), \quad (1)$$

$$u|_{x=0} = 0, \quad (2)$$

$$\left. \begin{aligned} mg - Y S u_x|_{x=l} &= m u_{tt}|_{x=l}, \text{即} \left(u_x + \frac{m}{Y S} u_{tt} \right) \Big|_{x=l} \\ &= \frac{mg}{Y S}, \end{aligned} \right\} \quad (3)$$

$$u|_{t=0} = 0, \quad u_t|_{t=0} = 0, \quad (4)$$

$$\text{令 } v = \frac{mg}{Y S} x, \quad u = v + \omega,$$

$$\text{则有} \begin{cases} w_{tt} - a^2 w_{xx} = 0, & (5) \end{cases}$$

$$\begin{cases} w|_{x=0} = 0, & (6) \end{cases}$$

$$\begin{cases} \left(w_x + \frac{m}{YS} w_{tt} \right) \Big|_{x=l} = 0, & (7) \end{cases}$$

$$\begin{cases} w|_{t=0} = -\frac{mg}{YS} x, & (8) \end{cases}$$

$$\begin{cases} w_t|_{t=0} = 0. & (9) \end{cases}$$

令 $w = X(x)T(t)$, 代入 (5)、(6)、(7), 则有

$$T'' + \lambda a^2 T = 0, \quad (10)$$

$$X'' + \lambda X = 0, \quad (11)$$

$$X(0) = 0, \quad (12)$$

$$X'(l)T(t) + \frac{m}{YS}X(l)T''(t) = 0. \quad (13)$$

由 (13) 并利用 (10) 式, 则有

$$\frac{\frac{X'(l)}{-\frac{m}{YS}X(l)}}{\frac{T''(t)}{T(t)}} = -\lambda a^2 = -\lambda \frac{Y}{\rho}.$$

$$\therefore X'(l) - \frac{\lambda m}{\rho S} X(l) = 0.$$

现由 (11), (12), (14) 求解 X ,

当 $\lambda > 0$ 时, $X = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x$,

由 (12), $C_1 = 0$, $X = C_2 \sin \sqrt{\lambda} x$,

$$\text{由 (14)} \quad \sqrt{\lambda} \cos \sqrt{\lambda} l - \frac{\lambda m}{\rho S} \sin \sqrt{\lambda} l = 0, \quad (15)$$

$$\text{或} \quad \sqrt{\lambda} \operatorname{tg} \sqrt{\lambda} l = \frac{\rho S}{m},$$

记 $\xi = \sqrt{\lambda} l$, 则方程 (15) 可写成

$$\operatorname{tg} \xi = \frac{\rho S l}{m} \frac{1}{\xi}.$$

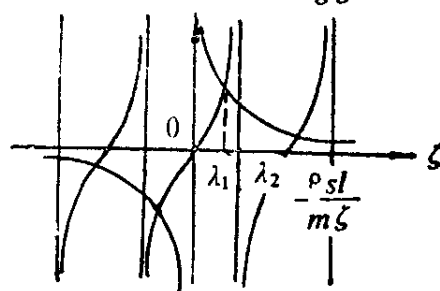


图 10-7

这个超越方程不能用初等方法求解, 可用图解法求其根.

设 λ_n 是方程 (15) 的第 n 个正根, 则

本征函数为 $X_n(x) = C_n \sin \sqrt{\lambda_n} x$, ($n = 1, 2, \dots$),

$w(x, t)$ 的一般解为

$$w(x, t) = \sum_{n=1}^{\infty} (A_n \cos \sqrt{\lambda_n} at + B_n \sin \sqrt{\lambda_n} at) \sin \sqrt{\lambda_n} x, \quad (16)$$

由 (9) 式, 知 $B_n = 0$,

$$\text{由 (8) 式, 得 } \sum_{n=1}^{\infty} A_n \sin \sqrt{\lambda_n} x = -\frac{mg}{YS} x = \varphi(x). \quad (17)$$

为求 A_n , 注意到本征函数族 $\{\sin \sqrt{\lambda_n} x\}$ 在 $(0, l)$ 上并不正交, 事实上, $K \neq n$ 时,

$$\begin{aligned} & \int_0^l \sin \sqrt{\lambda_k} x \sin \sqrt{\lambda_n} x dx \\ &= \frac{1}{2} \left[\int_0^l \cos (\sqrt{\lambda_k} - \sqrt{\lambda_n}) x dx - \int_0^l (\cos \sqrt{\lambda_k} + \sqrt{\lambda_n}) x dx \right] \\ &= \frac{1}{2} \left[\frac{\sin (\sqrt{\lambda_k} - \sqrt{\lambda_n}) l}{\sqrt{\lambda_k} - \sqrt{\lambda_n}} - \frac{\sin (\sqrt{\lambda_k} + \sqrt{\lambda_n}) l}{\sqrt{\lambda_k} + \sqrt{\lambda_n}} \right] \\ &= \frac{\sqrt{\lambda_n} \sin \sqrt{\lambda_k} l \cos \sqrt{\lambda_n} l - \sqrt{\lambda_k} \cos \sqrt{\lambda_k} l \sin \sqrt{\lambda_n} l}{\lambda_k - \lambda_n} \\ &= -\frac{m}{\rho S} \sin \sqrt{\lambda_k} l \sin \sqrt{\lambda_n} l \quad (\text{用 (15) 式, 得到}), \end{aligned} \quad (18)$$

$$\begin{aligned} & \int_0^l \sin^2 \sqrt{\lambda_n} x dx \\ &= \frac{1}{2} \int_0^l (1 - \cos 2\sqrt{\lambda_n} x) dx \\ &= \frac{l}{2} - \frac{1}{4\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} x \Big|_0^l \\ &= \frac{l}{2} - \frac{m}{2\rho S} \sin^2 \sqrt{\lambda_n} l = \frac{l}{2} - \frac{1}{2\sqrt{\lambda_n}} \sin \sqrt{\lambda_n} l \cos \sqrt{\lambda_n} l. \end{aligned} \quad (19)$$

由(17), (18), (19), 有:

$$\begin{aligned}\int_0^l \varphi(x) \sin \sqrt{\lambda_n} x dx &= \sum_{k=1}^{\infty} \int_0^l A_k \sin \sqrt{\lambda_k} x \sin \sqrt{\lambda_n} x dx \\ &= A_n \left[\frac{l}{2} - \frac{m}{2\rho S} \sin^2 \sqrt{\lambda_n} l \right] + \sum_{k \neq n} \\ &\quad - \frac{m}{\rho S} A_k \sin \sqrt{\lambda_k} l \sin \sqrt{\lambda_n} l, \quad (20)\end{aligned}$$

$$\text{而 } \frac{m}{\rho S} \varphi(l) \sin \sqrt{\lambda_n} l = \sum_{k=1}^{\infty} \frac{m}{\rho S} A_k \sin \sqrt{\lambda_k} l \sin \sqrt{\lambda_n} l, \quad (21)$$

将(20)与(21)两式逐项相加, 即得,

$$\begin{aligned}&\int_0^l \varphi(x) \sin \sqrt{\lambda_n} x dx + \frac{m}{\rho S} \varphi(l) \sin \sqrt{\lambda_n} l \\ &= A_n \left[\frac{l}{2} - \frac{m}{2\rho S} \sin^2 \sqrt{\lambda_n} l \right] + \frac{m}{\rho S} A_n \sin^2 \sqrt{\lambda_n} l \\ &= A_n \left[\frac{l}{2} + \frac{m}{2\rho S} \sin^2 \sqrt{\lambda_n} l \right], \quad (22)\end{aligned}$$

$$\begin{aligned}\therefore \int_0^l \varphi(x) \sin \sqrt{\lambda_n} x dx &= \int_0^l -\frac{mg}{YS} x \sin \sqrt{\lambda_n} x dx \\ &= \frac{mg}{YS} \frac{1}{\lambda_n} \left[\sqrt{\lambda_n} l \cos \sqrt{\lambda_n} l \right. \\ &\quad \left. - \sin \sqrt{\lambda_n} l \right] \\ &= \frac{mg}{YS} \frac{1}{\lambda_n} \left(\frac{\lambda_n m}{\rho S} l - 1 \right) \sin \sqrt{\lambda_n} l, \\ &\quad - \frac{m}{\rho S} \varphi(l) \sin \sqrt{\lambda_n} l \\ &= \frac{m}{\rho S} \left(-\frac{mg}{YS} l \right) \sin \sqrt{\lambda_n} l.\end{aligned}$$

$$\therefore (22) \text{ 式的左端} = -\frac{mg}{YS\lambda_n} \sin \sqrt{\lambda_n} l,$$

$$\text{从而 } A_n = \frac{-2 \frac{mg}{YS\lambda_n} \sin \sqrt{\lambda_n} l}{1 + \frac{m}{\rho S} \sin^2 \sqrt{\lambda_n} l},$$

最后得本问题的解为:

$$\begin{aligned} u(x,t) &= v + w \\ &= \frac{mg}{YS} x + \sum_{n=1}^{\infty} \frac{-2mg \sin \sqrt{\lambda_n} l}{YS\lambda_n \left(1 + \frac{m}{\rho S} \sin^2 \sqrt{\lambda_n} l \right)} \\ &\quad \cos a \sqrt{\lambda_n} t \sin \sqrt{\lambda_n} x. \end{aligned}$$

又 $\because$ 由(15)式有:

$$\begin{aligned} \sin^2 \sqrt{\lambda_n} l &= \frac{1}{1 + \operatorname{ctg}^2 \sqrt{\lambda_n} l} = \frac{1}{1 + \frac{m^2 \lambda_n}{\rho^2 S^2}} \\ &= \frac{\rho^2 S^2}{\rho^2 S^2 + m^2 \lambda_n}, \end{aligned}$$

$\therefore$ 本问题的解也可写成

$$\begin{aligned} u(x,t) &= \frac{mg}{YS} x + \sum_{n=1}^{\infty} \frac{-2mg\rho}{Y\lambda_n \left(1 + \frac{\rho Sm}{\rho^2 S^2 + m^2 \lambda_n} \right) \sqrt{\rho^2 S^2 + m^2 \lambda_n}} \\ &\quad \cos a \sqrt{\lambda_n} t \sin \sqrt{\lambda_n} x. \end{aligned}$$

15. 长为 l 的柱形管, 一端封闭, 另一端开放. 管外空气中含有某种气体, 其浓度为 u_0 , 向管内扩散, 求解该气体在管内的浓度 $u(x,t)$.

$$\text{解: } \begin{cases} u_t - a^2 u_{xx} = 0, \\ u|_{x=0} = u_0, \\ u_x|_{x=l} = 0, \\ u|_{t=0} = 0, \end{cases}$$

令 $u = v + u_0$,

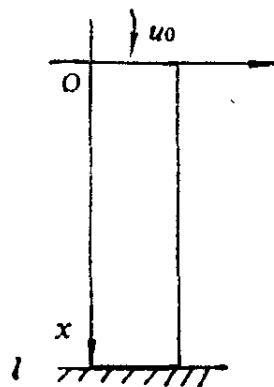


图 10-8

则 $v_t - a^2 v_{xx} = 0$, $v|_{x=0} = 0$, $v_x|_{x=l} = 0$, $v|_{t=0} = -u_0$.

这样 v 的边界条件变成齐次的了. 显然 v 具有第一类和第二类的边界条件, 它的本征值问题决定了它的本征函数是

$\sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x$, 即 v 的解为

$$v(x, t) = \sum_{n=0}^{\infty} C_n e^{-\frac{\left(n + \frac{1}{2}\right)^2}{l^2} \pi^2 a^2 t} \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x,$$

$$v(x, 0) = \sum_{n=0}^{\infty} C_n \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x = -u_0.$$

$$\therefore C_n = \frac{2}{l} \int_0^l -u_0 \sin \frac{\left(n + \frac{1}{2}\right)}{l} \pi x dx$$

$$= -\frac{2u_0}{l} \left[\frac{l}{\left(n + \frac{1}{2}\right)\pi} \cos \frac{\left(n + \frac{1}{2}\right)}{l} \pi x \right]_0^l$$

$$= \frac{-2u_0}{\left(n + \frac{1}{2}\right)\pi} = \frac{-4u_0}{\pi} \frac{1}{2n+1},$$

$$v = -\frac{4u_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} e^{-\frac{(2n+1)^2}{4l^2} \pi^2 a^2 t} \sin \frac{(2n+1)}{2l} \pi x$$

$$u(x, t) = u_0 + v$$

$$= u_0 - \frac{4u_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} e^{-\frac{(2n+1)^2}{4l^2} \pi^2 a^2 t} \sin \frac{(2n+1)}{2l} \pi x$$

16. 在矩形区域 $0 < x < a$, $0 < y < b$ 上求解拉氏方程 $\Delta u = 0$. 使满足边界条件

$$u|_{x=0} = Ay(b-y), \quad u|_{x=a} = 0.$$

$$u|_{y=0} = B \sin \frac{\pi x}{a}, \quad u|_{y=b} = 0.$$

解: 令 $u = v + w$, 可使 v 和 w 分别满足:

$$v_{xx} + v_{yy} = 0, \quad \begin{cases} v(0, y) = 0, \\ v(a, y) = 0, \end{cases} \quad \begin{cases} v(x, 0) = B \sin \frac{\pi x}{a} \\ v(x, b) = 0. \end{cases} \quad (1)$$

$$w_{xx} + w_{yy} = 0, \quad \begin{cases} w(0, y) = Ay(b-y), \\ w(a, y) = 0, \end{cases} \quad \begin{cases} w(x, 0) = 0 \\ w(x, b) = 0. \end{cases} \quad (2)$$

先解 $v(x, y)$, 设 $v(x, y) = X(x)Y(y)$, 代入 $\Delta u = 0$ 得,

$$X''Y + XY'' = 0, \quad \frac{X''}{X} = -\frac{Y''}{Y} = -\lambda^2,$$

$$X = A'_n \cos \lambda x + B'_n \sin \lambda x,$$

$$\because X(0)Y(y) = 0,$$

$$\therefore A'_n = 0,$$

$$X(a)Y(y) = 0, \quad 0 = B'_n \sin \lambda a, \quad B'_n \neq 0, \therefore \sin \lambda a = 0,$$

$$\lambda a = n\pi, \quad \lambda = \frac{n\pi}{a}.$$

$$\therefore X = B'_n \sin \frac{n\pi}{a} x.$$

$$Y'' - \frac{n^2 \pi^2}{a^2} Y = 0, \quad Y = A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}}.$$

$$\therefore v(x, y) = XY$$

$$= \sum_{n=1}^{\infty} \left(A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}} \right) \sin \frac{n\pi}{a} x.$$

由条件 $v(x, 0) = B \sin \frac{\pi x}{a}$ 得.

$$\sum_{n=1}^{\infty} (A_n + B_n) \sin \frac{n\pi x}{a} = B \sin \frac{\pi x}{a}$$

(式中 B 为常数 A_n, B_n 为系数),

$$\therefore A_1 + B_1 = \frac{2B}{a} \int_0^a \sin^2 \frac{\pi x}{a} dx = B, (n=1),$$

$$\begin{aligned} A_n + B_n &= \frac{2B}{a} \int_0^a \sin \frac{\pi x}{a} \sin \frac{n\pi x}{a} dx \\ &= \frac{B}{a} \int_0^a \left[\cos \frac{(n-1)\pi}{a} x - \cos \frac{(n+1)\pi}{a} x \right] dx \\ &= 0, (n \neq 1). \end{aligned}$$

从条件 $v(x, b) = 0$ 得

$$\sum_{n=0}^{\infty} \left(A_n e^{\frac{n\pi b}{a}} + B_n e^{-\frac{n\pi b}{a}} \right) \sin \frac{n\pi x}{a} = 0,$$

$$\therefore A_n e^{\frac{n\pi b}{a}} + B_n e^{-\frac{n\pi b}{a}} = 0.$$

因有:
$$\begin{cases} A_1 + B_1 = B, \\ A_1 e^{\frac{\pi b}{a}} + B_1 e^{-\frac{\pi b}{a}} = 0, \end{cases}$$

$$\begin{cases} A_n + B_n = 0, (n \neq 1), \\ A_n e^{\frac{n\pi b}{a}} + B_n e^{-\frac{n\pi b}{a}} = 0. \end{cases}$$

可解得:
$$A_1 = \frac{-Be^{-\frac{\pi b}{a}}}{\left(e^{\frac{\pi b}{a}} - e^{-\frac{\pi b}{a}} \right)} = \frac{-Be^{-\frac{\pi b}{a}}}{2\operatorname{sh} \frac{\pi b}{a}},$$

$$B_1 = B - A_1 = \frac{Be^{-\frac{\pi b}{a}}}{2\operatorname{sh} \frac{\pi b}{a}},$$

$$A_n = B_n = 0, (n \neq 1),$$

$$\begin{aligned} \therefore v(x, y) &= \frac{B}{2 \operatorname{sh} \frac{\pi b}{a}} \left[e^{\frac{\pi(b-y)}{a}} - e^{-\frac{\pi(b-y)}{a}} \right] \sin \frac{\pi x}{a} \\ &= \frac{B \operatorname{sh} \frac{\pi(b-y)}{a}}{\operatorname{sh} \frac{\pi b}{a}} \sin \frac{\pi x}{a}, \end{aligned}$$

同样可以得到 $W(x, y)$ 的一般解:

$$W(x, y) = \sum_{n=1}^{\infty} \left(C_n e^{\frac{n\pi x}{b}} + D_n e^{-\frac{n\pi x}{b}} \right) \sin \frac{n\pi y}{b}.$$

从条件 $W(0, y) = 0$ 得:

$$\sum_{n=1}^{\infty} (C_n + D_n) \sin \frac{n\pi y}{b} = Ay(b-y),$$

$$\begin{aligned} \therefore C_n + D_n &= \frac{2A}{b} \int_0^b \xi(b-\xi) \sin \frac{n\pi \xi}{b} d\xi \\ &= \frac{4Ab^2}{(n\pi)^3} [1 - (-1)^n]. \end{aligned}$$

从条件 $W(a, y) = 0$ 得:

$$\sum_{n=1}^{\infty} \left(C_n e^{\frac{n\pi a}{b}} + D_n e^{-\frac{n\pi a}{b}} \right) \sin \frac{n\pi y}{b} = 0,$$

$$\therefore C_n e^{\frac{n\pi a}{b}} + D_n e^{-\frac{n\pi a}{b}} = 0,$$

当 $n = \text{偶数}$ 时:

当 $n = \text{奇数}$ 时:

$$\begin{aligned} C_n + D_n &= 0, \\ C_n e^{\frac{n\pi a}{b}} + D_n e^{-\frac{n\pi a}{b}} &= 0, \end{aligned} \quad \left\{ \begin{aligned} C_n + D_n &= \frac{8Ab^2}{(n\pi)^3}, \\ C_n e^{\frac{n\pi a}{b}} + D_n e^{-\frac{n\pi a}{b}} &= 0. \end{aligned} \right.$$

由此可得: $n = \text{偶数}$ 时, $C_n = D_n = 0$,

$n = \text{奇数}$ 时,

$$C_n = \frac{-8Ab^2 e^{-\frac{n\pi a}{b}}}{(n\pi)^3 \left(e^{\frac{n\pi a}{b}} - e^{-\frac{n\pi a}{b}} \right)} = \frac{-4Ab^2 e^{-\frac{n\pi a}{b}}}{(n\pi)^3 \operatorname{sh} \frac{n\pi a}{b}},$$

$$D_n = \frac{8Ab^2}{(n\pi)^3} - C_n = \frac{8Ab^2 e^{\frac{n\pi a}{b}}}{(n\pi)^3 \left(e^{\frac{n\pi a}{b}} - e^{-\frac{n\pi a}{b}} \right)}$$

$$= \frac{4Ab^2 e^{\frac{n\pi a}{b}}}{(n\pi)^3 \operatorname{sh} \frac{n\pi a}{b}},$$

$$\begin{aligned} \therefore W(x, y) &= \sum_{k=0}^{\infty} \frac{4Ab^2}{(2k+1)^3 \pi^3 \operatorname{sh} \frac{(2k+1)\pi a}{b}} \\ &\quad \left[e^{\frac{(2k+1)\pi(a-x)}{b}} - e^{-\frac{(2k+1)\pi(a-x)}{b}} \right] \\ &\quad \cdot \sin \frac{(2k+1)\pi y}{b} \\ &= \sum_{k=0}^{\infty} \frac{8Ab^2 \operatorname{sh} \frac{(2k+1)\pi(a-x)}{b}}{(2k+1)^3 \pi^3 \operatorname{sh} \frac{(2k+1)\pi a}{b}} \sin \frac{(2k+1)\pi y}{b}, \end{aligned}$$

$$\therefore u(x, y) = v(x, y) + W(x, y)$$

$$\begin{aligned} &= \frac{B \operatorname{sh} \frac{\pi(b-y)}{a}}{\operatorname{sh} \frac{\pi b}{a}} \sin \frac{\pi x}{a} \\ &\quad + \sum_{k=0}^{\infty} \frac{8Ab^2 \operatorname{sh} \frac{(2k+1)\pi(a-x)}{b}}{(2k+1)^3 \pi^3 \operatorname{sh} \frac{(2k+1)\pi a}{b}} \\ &\quad \sin \frac{(2k+1)\pi y}{b}. \end{aligned}$$

17. 均匀的薄板占据区域 $0 < x < a, 0 < y < \infty$, 边界上温度

$$u|_{x=0} = 0, u|_{x=a} = 0, u|_{y=0} = u_0, \lim_{y \rightarrow \infty} u = 0.$$

求解板的稳定温度分布.

解: 泛定方程 $u_{xx} + u_{yy} = 0$. (1)

边界条件已由题目中明确给出, 令 $u = X(x)Y(y)$ 代入 (1):

$$X''Y + XY'' = 0, \text{ 即 } \frac{X''}{X} = -\frac{Y''}{Y} = -\lambda^2.$$

$$X'' + \lambda^2 X = 0. \quad (2)$$

$$Y'' - \lambda^2 Y = 0. \quad (3)$$

先解本征值问题的 (2):

$$X = \sum A_n \cos \lambda x + B_n \sin \lambda x,$$

$$\because X(0) = 0,$$

$$\therefore A_n = 0.$$

$$\because X(a) = 0,$$

$$\therefore \sum B_n \sin \lambda a = 0.$$

$$\sin \lambda a = 0, \lambda a = n\pi.$$

$$\therefore \lambda = \frac{n\pi}{a}, (n = 1, 2, 3 \cdots).$$

$$\therefore X(x) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{a} x.$$

再求方程 (3) 的解, 将 (3) 写成

$$Y'' - \frac{n^2 \pi^2}{a^2} Y = 0,$$

$$Y_n = A'_n e^{\frac{n\pi y}{a}} + B'_n e^{-\frac{n\pi y}{a}}.$$

$$u(x, y) = \sum_{n=1}^{\infty} \left(C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}} \right) \sin \frac{n\pi x}{a}.$$

利用 y 的边界条件:

$$u(x, 0) = X(x)Y(0) = \sum_{n=1}^{\infty} (C_n + D_n) \sin \frac{n\pi x}{a}$$

$$= u_0,$$

$$u(x, \infty) = X(x)Y(\infty) = \sum_{n=1}^{\infty} (C_n e^{\infty} + D_n e^{-\infty}) \sin \frac{n\pi x}{a} = 0,$$

$$\therefore C_n = 0.$$

$$u(x, 0) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{a} = u_0,$$

$$\begin{aligned} D_n &= \frac{2}{l} \int_0^l u_0 \sin \frac{n\pi x}{a} dx = \frac{2}{a} \int_0^a u_0 \sin \frac{n\pi x}{a} dx \\ &= \frac{2u_0}{a} \frac{a}{n\pi} \left[-\cos \frac{n\pi x}{a} \right]_0^a = \frac{2u_0}{n\pi} [1 - (-1)^n] \end{aligned}$$

$$= \begin{cases} \frac{4u_0}{(2k+1)\pi}, & (\text{当 } n \text{ 为奇数 } 2k+1), \\ 0, & (\text{当 } n \text{ 为偶数}), \end{cases}$$

$$\therefore u = \frac{4u_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} e^{-\frac{(2k+1)\pi}{a}y} \sin \frac{(2k+1)\pi}{a}x.$$

18. 在带状区域 $0 < x < a, 0 < y < \infty$ 上求解 $\Delta u = 0$ 使

$$u|_{x=0} = 0, u|_{x=a} = 0, u|_{y=0} = A\left(1 - \frac{x}{a}\right), \lim_{y \rightarrow \infty} u = 0.$$

解: 如上题, 令 $u = X(x)Y(y)$, 则可得

$$u(x, y) = \sum_{n=1}^{\infty} \left(A_n e^{\frac{n\pi}{a}y} + B_n e^{-\frac{n\pi}{a}y} \right) \sin \frac{n\pi}{a}x.$$

由 $\lim_{y \rightarrow \infty} u = 0$, 得 $A_n = 0$,

由 $u|_{y=0} = A\left(1 - \frac{x}{a}\right)$, 得

$$\sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{a}x = A\left(1 - \frac{x}{a}\right),$$

$$\therefore B_n = \frac{2}{a} \int_0^a A \left(1 - \frac{x}{a} \right) \sin \frac{n\pi}{a} x dx = \frac{2A}{n\pi}.$$

$$\therefore \text{原问题的解为 } u(x, y) = \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} e^{-\frac{n\pi}{a}y} \sin \frac{n\pi}{a} x.$$

19. 矩形膜, 边长为 l_1 和 l_2 , 边缘固定, 求它的本征振动模式.

解: 设膜上各点的横向位移为 $u(x, y, t)$, 则有

$$\begin{cases} u_{tt} - a^2(u_{xx} + u_{yy}) = 0, (0 < x < l_1, 0 < y < l_2), & (1) \\ u|_{x=0} = 0, u|_{x=l_1} = 0, & (2) \\ u|_{y=0} = 0, u|_{y=l_2} = 0, & (3) \end{cases}$$

令 $u(x, y, t) = X(x)Y(y)T(t)$, 代入泛定方程及边界条件得:

$$T''XY - a^2(X''Y + XY'')T = 0,$$

$$\frac{T''}{a^2T} = \frac{X''Y + XY''}{XY} = -\lambda.$$

$$T'' + \lambda a^2 T = 0, \quad (4)$$

$$X''Y + XY'' + \lambda XY = 0, XY'' + (X'' + \lambda X)Y = 0,$$

$$Y'' + \left(\frac{X''}{X} + \lambda \right) Y = 0, \quad (5)$$

为了利用 (2), (3) 求得 X 与 Y 的本征函数, 可以记 $\lambda = \lambda_1$

$$+ \lambda_2, \lambda_1 \text{ 是 } \begin{cases} X'' + \lambda X = 0, \\ X(0) = X(l_1) = 0, \end{cases} \text{ 决定的本征值}$$

$$\text{当 } \lambda_1 = \left(\frac{n\pi}{l_1} \right)^2 \text{ 时, } X = C_1 \sin \frac{n\pi}{l_1} x, \text{ 可使 } \frac{X''}{X} + \lambda_1 = 0,$$

这时 (5) 成为 $Y'' + \lambda_2 Y = 0$.

注意到边界条件 (3), 应取 $\lambda_2 = \left(\frac{m\pi}{l_2} \right)^2$, 亦即关于 y 的本征函数为

$$Y = C_2 \sin \frac{m\pi}{l_2} y,$$

将 $\lambda = \lambda_1 + \lambda_2 = \left(\frac{n\pi}{l_1}\right)^2 + \left(\frac{m\pi}{l_2}\right)^2$ 代入 (4)

可得 $T = A \cos \sqrt{\lambda} at + B \sin \sqrt{\lambda} at$,

如此可得本征振动模式为

$$u_{m,n}(x, y, t) = (A_{m,n} \cos \sqrt{\lambda} at + B_{m,n} \sin \sqrt{\lambda} at) \cdot \sin \frac{n\pi x}{l_1} \sin \frac{m\pi y}{l_2}.$$

$$\text{其中 } \sqrt{\lambda} = \pi \sqrt{\left(\frac{n}{l_1}\right)^2 + \left(\frac{m}{l_2}\right)^2}.$$

20. 长为 l 的均匀杆两端被支承, 求解它的横振动.

解: 泛定方程 $\begin{cases} u_{tt} + a^2 u_{xxxx} = 0, (0 < x < l), \end{cases} \quad (1)$

因两端被支承 $\begin{cases} u|_{x=0} = u_{xx}|_{x=0} = 0, \\ u|_{x=l} = u_{xx}|_{x=l} = 0, \end{cases} \quad (2)$

$u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x). \quad (3)$

假设

以 $u(x, t) = X(x)T(t)$, 代入 (1)、(2) 得

$$T''X + a^2 X^{(4)}T = 0, \quad \frac{X^{(4)}}{X} = -\frac{T''}{a^2 T} = \lambda^2,$$

$$T'' + a^2 \lambda^2 T = 0, \quad (4)$$

$$\begin{cases} X^{(4)} - \lambda^2 X = 0, \end{cases} \quad (5)$$

$$\begin{cases} X(0) = X''(0) = X(l) = X''(l) = 0. \end{cases} \quad (6)$$

若 $\lambda = 0$, 则 $X = C_1 + C_2 x + C_3 x^2 + C_4 x^3$.

由 $X(0) = X''(0) = 0$, 显有 $C_1 = C_3 = 0$,

又由 $X(l) = X''(l) = 0$, 得 $\begin{cases} C_2 + C_4 l^2 = 0, \\ 6C_4 l = 0, \end{cases}$

$\therefore C_4 = 0$.

$$C_2 = 0.$$

从而 $\lambda = 0$ 时只有零解,故 $\lambda \neq 0$, 这时不妨设 $\lambda > 0$, (5)有通解

$$X = C_1 \operatorname{ch} \sqrt{\lambda} x + C_2 \operatorname{sh} \sqrt{\lambda} x + C_3 \cos \sqrt{\lambda} x + C_4 \sin \sqrt{\lambda} x.$$

$$\text{由(6)} \quad \begin{cases} C_1 + C_3 = 0, \end{cases} \quad (7)$$

$$\begin{cases} C_1 \lambda - C_3 \lambda = 0, \end{cases} \quad (8)$$

$$\begin{cases} C_1 \operatorname{ch} \sqrt{\lambda} l + C_2 \operatorname{sh} \sqrt{\lambda} l + C_3 \cos \sqrt{\lambda} l \\ + C_4 \sin \sqrt{\lambda} l = 0, \end{cases} \quad (9)$$

$$\begin{cases} C_1 \operatorname{ch} \sqrt{\lambda} l + C_2 \operatorname{sh} \sqrt{\lambda} l - C_3 \cos \sqrt{\lambda} l \\ - C_4 \sin \sqrt{\lambda} l = 0. \end{cases} \quad (10)$$

由(7), (8), $C_1 = C_3 = 0$, (9) + (10), (9) - (10) 分别得

$$C_2 \operatorname{sh} \sqrt{\lambda} l = 0, \quad C_4 \sin \sqrt{\lambda} l = 0.$$

$$\because \operatorname{sh} \sqrt{\lambda} l \neq 0, \therefore C_2 = 0,$$

$$\text{又} \because C_4 \neq 0 \text{ (否则得零解)}, \therefore \sin \sqrt{\lambda} l = 0.$$

$$\sqrt{\lambda} l = n\pi, \text{ 本征值 } \lambda = \frac{n^2 \pi^2}{l^2}, \quad X_n = C_n \sin \frac{n\pi}{l} x.$$

将 λ 代入(4), 得

$$T'' + \frac{a^2 n^4 \pi^4}{l^4} T = 0,$$

$$T_n = A_n \cos \frac{n^2 \pi^2 a}{l^2} t + B_n \sin \frac{n^2 \pi^2 a}{l^2} t,$$

因而有

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n^2 \pi^2 a}{l^2} t + B_n \sin \frac{n^2 \pi^2 a}{l^2} t \right) \sin \frac{n\pi}{l} x.$$

由(3)知, 其中

$$\begin{cases} A_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} dx, \\ B_n = \frac{2l}{n^2 \pi^2 a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} dx. \end{cases}$$

21. 细圆环, 半径为 R , 初始温度分布已知为 $f(\varphi)$, φ 是以环心为极点的极角. 环的表面是绝热的. 求解环内温度变化情况.

解: 设圆环的截面积为 S , 比热为 C , 密度为 ρ , 热传导系数是 k , 因是细环, 内外半径近似相等为 R . 位于 φ 和 $\varphi + d\varphi$ 之间的体元 B 的体积为

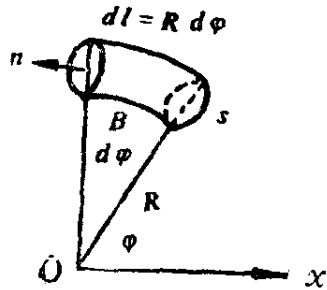


图 10-9

$$Sdl = SRd\varphi, \quad \frac{\partial u}{\partial n} = \frac{\partial u}{\partial l} = \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial l} = \frac{1}{R} u_{\varphi}.$$

于是热平衡方程可以写成为.

$$\rho CSRd\varphi du = -k \frac{1}{R} u_{\varphi}|_{\varphi} + k \frac{1}{R} u_{\varphi}|_{\varphi+d\varphi} S dt,$$

$$\therefore u_t - \frac{a^2}{R^2} u_{\varphi\varphi} = 0, \quad \left(a^2 = \frac{k}{\rho c} \right).$$

于是可以写出细圆环的定解问题:

$$\begin{cases} u_t - a^2 u_{\varphi\varphi} = 0, 0 \leq \varphi < 2\pi, & (1) \end{cases}$$

$$\begin{cases} u_{t=0} = f(\varphi), & (2) \end{cases}$$

$$\begin{cases} u(\varphi + 2\pi) = u(\varphi). & (3) \end{cases}$$

由于 u 有自然周期条件, 设 u 有分离变量形式的本征解 $u_n = T_n(t) \Phi_n(\varphi)$, 则 $\Phi_n(\varphi)$ 以 2π 为周期, 从而可将 u 展开为以 2π 为周期的傅里叶级数为,

$$u(\varphi, t) = \sum_{n=0}^{\infty} T_n(t) (A_n \cos n\varphi + B_n \sin n\varphi),$$

代入泛定方程(1)得

$$T'_n + \frac{a^2 n^2}{R^2} T_n = 0, \quad T_n(t) = C_n e^{-\frac{n^2 a^2}{R^2} t},$$

$$\therefore u(\varphi, t) = \sum_{n=0}^{\infty} (A_n \cos n\varphi + B_n \sin n\varphi) e^{-\frac{n^2 a^2}{R^2} t},$$

其中 A_n 和 B_n 是由初始条件所决定的傅里叶系数,

$$A_n = \frac{1}{2\pi} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi,$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi.$$

22. 在圆形域内求解 $\Delta u = 0$ 使满足边界条件

$$(1) u|_{\rho=a} = A \cos \varphi, (2) u|_{\rho=a} = A + B \sin \varphi.$$

解: 在极坐标系下, 泛定方程为

$$u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\varphi\varphi} = 0, \quad \begin{pmatrix} 0 < \rho < a \\ 0 < \varphi < 2\pi \end{pmatrix}, \quad (1)$$

设 $u(\rho, \varphi) = R(\rho)\Phi(\varphi)$, 有

$$R''\Phi + \frac{1}{\rho} R'\Phi + \frac{1}{\rho^2} R\Phi'' = 0,$$

$$(\rho^2 R'' + \rho R')\Phi + R\Phi'' = 0,$$

$$\Phi'' + \lambda\Phi = 0, \quad (2)$$

$$\rho^2 R'' + \rho R' - \lambda R = 0. \quad (3)$$

利用自然的周期条件, 得本征值 $\lambda = m^2 (m = 0, 1, 2, \dots)$,

本征函数 $\Phi_m(\varphi) = A_m \cos m\varphi + B_m \sin m\varphi$,

将 λ 值代入 (3), 得 $\rho^2 R'' + \rho R' - m^2 R = 0$,

$m = 0$ 时, 解为 $R_0 = C_0 + D_0 \ln \rho$,

$m > 0$ 时, 解为 $R_m = C_m \rho^m + D_m \rho^{-m}$,

从而得 (1) 之本征解:

$$u_0(\rho, \varphi) = C_0 + D_0 \ln \rho,$$

$$u_m(\rho, \varphi) = (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho^m + D_m \rho^{-m}).$$

$$\therefore \text{一般解为 } u(\rho, \varphi) = (C_0 + D_0 \ln \rho) + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho^m + D_m \rho^{-m}),$$

为使 $u(\rho, \varphi)$ 在圆形域内有界, 应有 $D_0 = D_m = 0$,

(1) 由 $u|_{\rho=a} = A \cos \varphi$, 有

$$C_0 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) C_m a^m = A \cos \varphi,$$

$$\therefore C_0 = 0, A_1 C_1 a = A, A_1 C_1 = \frac{A}{a}, \text{其余系数均为 } 0,$$

$$\therefore u(\rho, \varphi) = \frac{A}{a} \rho \cos \varphi,$$

(2) 在一般解 $u(\rho, \varphi) = C_0 + \sum_{m=1}^{\infty} (a_m \cos m\varphi + b_m \sin m\varphi) \rho^m$ 中,

由 $u|_{\rho=a} = A + B \sin \varphi$, 有:

$$C_0 + \sum_{m=1}^{\infty} (a_m \cos m\varphi + b_m \sin m\varphi) a^m = A + B \sin \varphi,$$

$$\therefore C_0 = A, b_1 a = B, \text{其余系数均为 } 0.$$

$$\therefore u(\rho, \varphi) = A + \frac{B}{a} \rho \sin \varphi.$$

23. 半圆形薄板, 板面绝热, 边界直径上温度保持零度, 圆周上保持 u_0 , 求稳定状态下的板上温度分布.

解: 板面绝热, 方程为齐次的, 稳定状态下 $u_t = 0$, 所以在极坐标系下定解问题为:



$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\varphi\varphi} = 0, (0 < \rho < R, 0 < \varphi < \pi), & (1) \\ u|_{\varphi=0} = 0, & \\ u|_{\varphi=\pi} = 0, & (0 < \rho < R), & (2) \\ u|_{\rho=R} = u_0, & (0 < \varphi < \pi). & (3) \end{cases}$$

设 $u(\rho, \varphi) = R(\rho) \Phi(\varphi)$, 代入 (1), 有

$$\Phi'' + \lambda \Phi = 0, \quad (4)$$

$$\rho^2 R'' + \rho R' - \lambda R = 0. \quad (5)$$

又, 由 (2), 有 $\Phi(0) = 0, \Phi(\pi) = 0. \quad (6)$

解 (4), (6) (i) $\lambda < 0, \Phi = C_1 e^{\sqrt{-\lambda}\varphi} + C_2 e^{-\sqrt{-\lambda}\varphi}$, 由 (6) $C_1 +$

$$C_2 = 0,$$

$C_1 e^{\sqrt{-\lambda}\pi} + C_2 e^{-\sqrt{-\lambda}\pi} = 0$, 得 $C_1 = C_2 = 0$, $\Phi \equiv 0$, 应排除 $\lambda < 0$.

(i) $\lambda = 0$. $\Phi = C_1 \varphi + C_2$ 由 (6) $C_1 = C_2 = 0$, 应排除 $\lambda = 0$,

(ii) $\lambda > 0$ $\Phi = C_1 \cos \sqrt{\lambda} \varphi + C_2 \sin \sqrt{\lambda} \varphi$ 由 (6) $C_1 = 0$, $C_2 \sin \sqrt{\lambda} \pi = 0$.

$\therefore$ 本征值为 $\lambda = n^2, (n = 1, 2, \dots)$,

从而本征函数为 $\Phi_n(\varphi) = A_n \sin n\varphi$,

以 λ 值代入 (5) $\rho^2 R'' + \rho R' - n^2 R = 0$,

$$R = C_n \rho^n + D_n \rho^{-n}. (n = 1, 2, \dots),$$

$\therefore$ 一般解为 $u(\rho, \varphi) = \sum_{n=1}^{\infty} A_n \sin n\varphi (C_n \rho^n + D_n \rho^{-n})$,

$\therefore$ 在半圆形薄板内 u 有界, $\rho \rightarrow \infty$ 时有界, 所以 $D_n = 0$,

$$u(\rho, \varphi) = \sum_{n=1}^{\infty} a_n (\sin n\varphi) \rho^n.$$

由边界条件 (3), $\sum_{n=1}^{\infty} a_n R^n \sin n\varphi = u_0$.

$a_n R^n$ 为 u_0 的傅里叶系数,

$$\begin{aligned} a_n R^n &= \frac{2}{\pi} \int_0^{\pi} u_0 \sin n\varphi d\varphi = -\frac{2u_0}{\pi n} \cos n\varphi \Big|_0^{\pi} \\ &= \frac{2u_0}{n\pi} (1 - (-1)^n), \end{aligned}$$

$$\therefore a_n = \begin{cases} 0, & n = \text{偶数}. \\ \frac{2}{R^n} \cdot \frac{2u_0}{n\pi}, & n = \text{奇数}. \end{cases}$$

$$\therefore u(\rho, \varphi) = \frac{4u_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{R^{2k+1} (2k+1)} \rho^{2k+1} \sin (2k+1)\varphi.$$

24. 把例 6 的导体圆柱换为介质圆柱, 介质的介电常数为

ε . 求解柱内外的电场。(提示: 柱内电势必须有限. 在柱面上, 电势连续, 电位移的法向分量连续.)

解: 取长圆柱的轴为 Z 轴, 可设场强、电势与 Z 无关, 只需研究图示 xy 平面上的截面.

x 轴方向为场强 E 的方向, 设圆柱半径为 a .

柱内外的电势均应满足拉普拉斯方程. 取极坐标系, 且设柱内电势为 u^I , 柱外电势为 u^{II} , 则有:

$$u_{\rho\rho}^I + \frac{1}{\rho} u_{\rho}^I + \frac{1}{\rho^2} u_{\varphi\varphi}^I = 0, (0 \leq \rho < a), \quad (1)$$

$$u_{\rho\rho}^{II} + \frac{1}{\rho} u_{\rho}^{II} + \frac{1}{\rho^2} u_{\varphi\varphi}^{II} = 0, (\rho > a), \quad (2)$$

$$u^I(\varphi + 2\pi) = u^I(\varphi), u^{II}(\varphi + 2\pi) = u^{II}(\varphi). \quad (3)$$

在自然周期条件(3)下, 方程(1), (2)的一般解分别如下:

$$u^I(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho^m + D_m \rho^{-m}),$$

$\because$ 在柱内电势有限, $D_0 = D_m = 0$,

$$\therefore u^I(\rho, \varphi) = C_0 + \sum_{m=1}^{\infty} (a_m \cos m\varphi + b_m \sin m\varphi) \rho^m, \quad (4)$$

$$u^{II}(\rho, \varphi) = C'_0 + D'_0 \ln \rho + \sum_{m=1}^{\infty} (A'_m \cos m\varphi + B'_m \sin m\varphi) (C'_m \rho^m + D'_m \rho^{-m}),$$

$\because$ 在无限远处的静电场保持为匀强的场强 E_0 , 在给定坐标系下,

$$\text{则} \quad \lim_{\rho \rightarrow \infty} \frac{\partial u^{II}}{\partial x} = -E_0, \quad \lim_{\rho \rightarrow \infty} u^{II} = -E_0 x = -E_0 \rho \cos \varphi,$$

从而 $C'_0 = D'_0 = C'_m = 0, m \neq 1$.

$$A'_1 C'_1 = -E_0 \rho, \text{ 记 } a'_m = A'_m D'_m, b'_m = B'_m D'_m.$$

$$\text{有: } u^{\text{II}}(\rho, \varphi) = -E_0 \rho \cos \varphi + \sum_{m=1}^{\infty} (a'_m \cos m\varphi + b'_m \sin m\varphi) \rho^{-m}. \quad (5)$$

$$\text{现以衔接条件 } u^{\text{I}}|_{\rho=a} = u^{\text{II}}|_{\rho=a}, \quad (6)$$

$$\varepsilon \frac{\partial u^{\text{I}}}{\partial \rho} \Big|_{\rho=a} = \frac{\partial u^{\text{II}}}{\partial \rho} \Big|_{\rho=a}, \quad (7)$$

决定(4)与(5)的系数

由(6)得

$$\begin{cases} C_0 = 0, \\ a_1 a = -E_0 a + a'_1 a^{-1}, \end{cases} \quad (8)$$

$$\begin{cases} a_m a^m = a'_m a^{-m}, \\ b_m a^m = b'_m a^{-m}, \end{cases} \quad (9)$$

由(7)得

$$\varepsilon a_1 = -E_0 - a'_1 a^{-2}, \quad (10)$$

$$\varepsilon a_m \cdot m a^{m-1} = -m a'_m a^{-m-1}, \quad (11)$$

$$\varepsilon b_m \cdot m a^{m-1} = -m b'_m a^{-m-1},$$

$$\text{由(8), (10) 两式, 解得 } a_1 = \frac{-2E_0}{1+\varepsilon}, \quad a'_1 = \frac{(\varepsilon-1)}{\varepsilon+1} a^2 E_0.$$

由(9), (11)两式, 得:

$$\varepsilon m \cdot a'_m a^{-m} = -m a'_m a^{-m},$$

$$(\varepsilon+1) m a^{-m} a'_m = 0,$$

$$\therefore a'_m = 0, \text{ 从而 } a_m = 0, \quad (m=2, 3, \dots),$$

$$\text{同理 } b_m = b'_m = 0, \quad (m=1, 2, \dots),$$

$\therefore$ 本问题的解为:

$$\text{柱内电势分布 } u^{\text{I}}(\rho, \varphi) = \frac{-2E_0}{1+\varepsilon} \rho \cos \varphi,$$

$$\text{柱外电势分布 } u^{\text{II}}(\rho, \varphi) = -E_0 \rho \cos \varphi + \frac{\varepsilon - 1}{\varepsilon + 1} a^2 E_0 \rho^{-1} \cos \varphi$$

$$= -\left(\rho - \frac{\varepsilon - 1}{\varepsilon + 1} \cdot \frac{a^2}{\rho}\right) E_0 \cos \varphi,$$

$$\begin{aligned} \text{柱内场强 } E^{\text{I}}(\rho, \varphi) &= -\frac{\partial u^{\text{I}}}{\partial x} = -\frac{\partial}{\partial x} \left(\frac{-2E_0}{1+\varepsilon} x \right) \\ &= \frac{2E_0}{1+\varepsilon}. \end{aligned}$$

$$\text{柱内极化强度 } P = (\varepsilon - 1) \varepsilon_0 E = 2\varepsilon_0 \frac{\varepsilon - 1}{\varepsilon + 1} E_0.$$

$$\text{柱面束缚电荷面密度} = P \text{ 的法向分量} = 2\varepsilon_0 \frac{\varepsilon - 1}{\varepsilon + 1} E_0 \cos \varphi.$$

在 $u^{\text{II}}(\rho, \varphi)$ 的第二项, 表示介质柱体极化电荷对柱体周围电势的影响。

25. 半径为 a , 表面熏黑了的均匀长圆柱, 在温度为零度的空气中受着阳光照射。阳光垂直于柱轴, 热流强度为 q , 试求柱内稳定温度分布。〔提示: 泛定方程为 $\Delta u = 0$, 边界条件为 $(ku_p + Hu)|_{\rho=a} = f(\varphi)$, $f(\varphi)$ 是热流强度的法向分量。如取极轴垂直于阳光, 则

$$f(\varphi) \equiv \begin{cases} q \sin \varphi, & (0 < \varphi < \pi), \\ 0, & (\pi < \varphi < 2\pi). \end{cases}$$

解: 如图, 取极轴方向垂直于阳光, 对于“无限长”圆柱, 温度分布与轴向无关, 故可取圆柱的一个截面考虑。

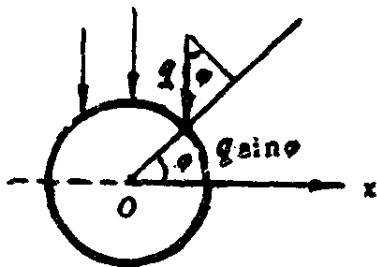


图 10-10

以轴心为极点, 与阳光垂直的方向为极轴方向, 则对于稳定温度而言, $u(\rho, \varphi)$ 满足拉普拉斯方程 $\Delta u = 0$,

$$\text{即 } u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\varphi\varphi} = 0, \quad \begin{pmatrix} 0 < \rho < a \\ 0 < \varphi < 2\pi \end{pmatrix}, \quad (1)$$

在柱面处, 一方面有热流流入, 另一方面与周围空气进行热交

换, 由热传导的定律, 可得边界条件如下:

$$[ku_{\rho} + H(u - 0)]|_{\rho=a} = \begin{cases} q \sin \varphi, & 0 < \varphi < \pi, \\ 0, & \pi < \varphi < 2\pi, \end{cases} \quad (2)$$

自然周期条件 $u(\rho, \varphi) = u(\rho, \varphi + 2\pi)$,

设 $u(\rho, \varphi) = R(\rho)\Phi(\varphi)$, 代入 (1) 有

$$\Phi'' + \lambda\Phi = 0, \quad 0 < \varphi < 2\pi, \quad (4)$$

$$\rho^2 R'' + \rho R' - \lambda R = 0, \quad 0 < \rho < a, \quad (5)$$

$$\text{由 (3) } \Phi(\varphi) = \Phi(\varphi + 2\pi), \quad (6)$$

解本征问题 (4), (6), 得本征值 $\lambda = n^2$, ($n = 0, 1, 2, \dots$),

本征函数为 $\Phi_n(\varphi) = A_n \cos n\varphi + B_n \sin n\varphi$, ($n = 0, 1, 2, \dots$),

以 λ 值代入 (5) 得 $R_0 = C_0 + D_0 \ln \rho$,

$$R_n = C_n \rho^n + D_n \rho^{-n}, \quad n \neq 0,$$

$$\therefore u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n})$$

$$(A_n \cos n\varphi + B_n \sin n\varphi),$$

$\because$ 在柱内 u 有界,

$\therefore D_n = 0$, ($n = 0, 1, 2, \dots$),

$$u = C_0 + \sum_{n=1}^{\infty} \rho^n (a_n \cos n\varphi + b_n \sin n\varphi),$$

由条件 (2),

$$HC_0 + \sum_{n=1}^{\infty} (kna^{n-1} + Ha^n) (a_n \cos n\varphi + b_n \sin n\varphi)$$

$$= \begin{cases} q \sin \varphi, & (0 < \varphi < \pi) \\ 0, & (\pi < \varphi < 2\pi) \end{cases}$$

在 $[0, 2\pi]$ 上, 利用三角函数族之正交性. 得

$$C_0 = \frac{1}{H} \cdot \frac{1}{2\pi} \int_0^{2\pi} q \sin \varphi d\varphi = \frac{q}{\pi H},$$

$$b_1 = \frac{1}{(k+Ha)\pi} \int_0^\pi q \sin^2 \varphi d\varphi = \frac{q}{2(k+Ha)},$$

当 $n \geq 2$ 时, $b_n = 0$,

$$\begin{aligned} a_n &= \frac{1}{(k\pi a^{n-1} + Ha^n)\pi} \int_0^\pi q \sin \varphi \cdot \cos n \varphi d\varphi \\ &= \frac{q}{(kna^{n-1} + Ha^n)\pi} \frac{1}{2} \left[\frac{-\cos(1+n)\varphi}{1+n} \right. \\ &\quad \left. + \frac{-\cos(1-n)\varphi}{1-n} \right]_0^\pi \\ &= \frac{q}{2\pi(kna^{n-1} + Ha^n)} \left[\frac{1 - (-1)^{n+1}}{1+n} + \frac{1 - (-1)^{n-1}}{1-n} \right]. \\ &= \begin{cases} 0, & n = 2m+1, \\ \frac{q}{\pi(2kma^{2m-1} + Ha^{2m})} \cdot \frac{2}{1-4m^2}, & n = 2m, \end{cases} \end{aligned}$$

$\therefore$ 本问题的解为:

$$\begin{aligned} u(\rho, \varphi) &= \frac{q}{\pi H} + \frac{q}{2(k+Ha)} \sin \varphi \\ &\quad + \frac{2q}{\pi} \sum_{m=1}^{\infty} \frac{\rho^{2m} \cos^2 m \varphi}{a^{2m-1} (2km + Ha) (1-4m^2)}. \end{aligned}$$

26. 在以原点为心, 以 R_1 和 R_2 为半径的两个同心圆所围成的环域上求解 $\Delta u = 0$, 使满足边界条件 $u|_{\rho=R_1} = f_1(\varphi)$, $u|_{\rho=R_2} = f_2(\varphi)$.

解: 取极坐标系, 定解问题为

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_\rho + \frac{1}{\rho^2} u_{\varphi\varphi} = 0, & (R_1 < \rho < R_2), \\ u|_{\rho=R_1} = f_1(\varphi), \\ u|_{\rho=R_2} = f_2(\varphi), \end{cases}$$

与上题类似, 利用自然的周期条件得本征值 $\lambda = n^2$, ($n = 0, 1, 2, \dots$),

本征函数为 $\Phi_n(\varphi) = A_n \cos n\varphi + B_n \sin n\varphi$,

一般解为 $u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} (A_n \cos n\varphi + B_n \sin n\varphi) (C_n \rho^n + D_n \rho^{-n})$,

$$\text{由边界条件} \begin{cases} C_0 + D_0 \ln R_1 + \sum_{n=1}^{\infty} (A_n \cos n\varphi + B_n \sin n\varphi) (C_n R_1^n + D_n R_1^{-n}) = f_1(\varphi), \\ C_0 + D_0 \ln R_2 + \sum_{n=1}^{\infty} (A_n \cos n\varphi + B_n \sin n\varphi) (C_n R_2^n + D_n R_2^{-n}) = f_2(\varphi), \end{cases}$$

$$\therefore \text{ i) } \begin{cases} C_0 + D_0 \ln R_1 = \frac{1}{2\pi} \int_0^{2\pi} f_1(\varphi) d\varphi \equiv \beta_0^{(1)}, & (1) \\ C_0 + D_0 \ln R_2 = \frac{1}{2\pi} \int_0^{2\pi} f_2(\varphi) d\varphi \equiv \beta_0^{(2)}, & (2) \end{cases}$$

$$C_0 = \frac{\beta_0^{(1)} \ln R_2 - \beta_0^{(2)} \ln R_1}{\ln R_2 - \ln R_1}, \quad D_0 = \frac{\beta_0^{(1)} - \beta_0^{(2)}}{\ln R_1 - \ln R_2},$$

$$\text{ii) } (C_n R_1^n + D_n R_1^{-n}) A_n = \frac{1}{\pi} \int_0^{2\pi} f_1(\varphi) \cos n\varphi d\varphi \equiv \beta_n^{(1)}, \quad (3)$$

$$(C_n R_2^n + D_n R_2^{-n}) A_n = \frac{1}{\pi} \int_0^{2\pi} f_2(\varphi) \cos n\varphi d\varphi \equiv \beta_n^{(2)}, \quad (4)$$

$$(C_n R_1^n + D_n R_1^{-n}) B_n = \frac{1}{\pi} \int_0^{2\pi} f_1(\varphi) \sin n\varphi d\varphi \equiv \alpha_n^{(1)}, \quad (5)$$

$$(C_n R_2^n + D_n R_2^{-n}) B_n = \frac{1}{\pi} \int_0^{2\pi} f_2(\varphi) \sin n\varphi d\varphi \equiv \alpha_n^{(2)}. \quad (6)$$

记 $K = R_2^{-n} R_1^n - R_1^{-n} R_2^n$,

解 (1) , (2) , 得

$$C_n A_n = \frac{1}{K} (R_2^{-n} \beta_n^{(1)} - R_1^{-n} \beta_n^{(2)});$$

$$D_n A_n = -\frac{1}{K} (R_2^n \beta_n^{(1)} - R_1^n \beta_n^{(2)}).$$

解 (3) , (4) , 得

$$C_n B_n = \frac{1}{K} (R_2^{-n} \alpha_n^{(1)} - R_1^{-n} \alpha_n^{(2)}),$$

$$D_n B_n = -\frac{1}{K} (R_2^n \alpha_n^{(1)} - R_1^n \alpha_n^{(2)}),$$

则原问题的解为:

$$\begin{aligned} u(\rho, \varphi) = & \frac{\beta_0^{(1)} \ln R_2 - \beta_0^{(2)} \ln R_1}{\ln R_2 / R_1} + \frac{\beta_0^{(2)} - \beta_0^{(1)}}{\ln R_2 / R_1} \ln \rho \\ & + \sum_{n=1}^{\infty} \frac{1}{R_2^{-n} R_1^n - R_1^{-n} R_2^n} \left\{ \left[(R_2^{-n} \beta_n^{(1)} \right. \right. \\ & \left. \left. - R_1^{-n} \beta_n^{(2)}) \rho^n - (R_2^n \beta_n^{(1)} - R_1^n \beta_n^{(2)}) \rho^{-n} \right] \cos n\varphi \right. \\ & \left. + \left[(R_2^{-n} \alpha_n^{(1)} - R_1^{-n} \alpha_n^{(2)}) \rho^n - (R_2^n \alpha_n^{(1)} \right. \right. \\ & \left. \left. - R_1^n \alpha_n^{(2)}) \rho^{-n} \right] \sin n\varphi \right\}. \end{aligned}$$

其中 $\beta_0^{(1)}, \beta_0^{(2)}, \beta_n^{(1)}, \beta_n^{(2)}, \alpha_n^{(1)}, \alpha_n^{(2)}$ 分别由 (1) — (6) 式决定.

27. 求解绕圆柱的水流问题. 在远离圆柱因而未受圆柱干扰处的水流是均匀的, 流速为 v_0 圆柱半径为 a .

解: 对于水的无旋流动, 有速度 u , 满足下列关系:

$$\begin{cases} \Delta u = 0, & (\vec{v} = \nabla u, \rho > a). \\ \left. \frac{\partial u}{\partial \rho} \right|_{\rho=a} = 0, & u|_{\rho \text{ 很大}} \approx v_0 \rho \cos \varphi. \end{cases}$$

由于流体流动有自然周期条件, 所以方程的解是

$$u(\rho, \varphi) = C_0 + D_0 \ln \rho + A_0 \varphi + \sum_{n=1}^{\infty} \left[\rho^n (A_n \cos n\varphi + B_n \sin n\varphi) \right]$$

$$+ \frac{1}{\rho^n} (A'_n \cos n\varphi + B'_n \sin n\varphi) \Big\},$$

由边界条件 $u|_{\rho \rightarrow \infty} \approx v_0 \cos \varphi$ 得

$$u|_{\rho \rightarrow \infty} = \sum_{n=1}^{\infty} \rho^n (A_n \cos n\varphi + B_n \sin n\varphi) \approx v_0 \rho \cos \varphi.$$

$$A_n = 0, \quad (n \neq 0, 1), \quad A_1 = v_0, \quad B_n = 0.$$

又由

$$\begin{aligned} \frac{\partial u}{\partial \rho} \Big|_{\rho=a} &= \frac{\partial}{\partial \rho} \Big|_{\rho=a} \left\{ C_0 + D_0 \ln \rho + v_0 \rho \cos \varphi \right. \\ &\quad \left. + \sum_{n=1}^{\infty} \frac{1}{\rho^n} [A'_n \cos n\varphi + B'_n \sin n\varphi] \right\} \\ &= \frac{D_0}{a} + \sum_{n=1}^{\infty} \frac{-n}{a^{n+1}} [A'_n \cos n\varphi + B'_n \sin n\varphi] \\ &\quad + v_0 \cos \varphi = 0, \end{aligned}$$

$$\therefore D_0 = 0, \quad A'_n = 0, \quad (n \neq 1), \quad -\frac{1}{a^2} A'_1 + v_0 = 0,$$

$$\therefore A'_1 = a^2 v_0.$$

$$\therefore u = C_0 + v_0 \rho \cos \varphi + \frac{v_0 a^2}{\rho} \cos \varphi + A_0 \varphi.$$

$$\text{记 } A_0 = \frac{\Gamma}{2\pi}, \quad \Gamma \text{ 称为环流,}$$

$$\text{则 } u(\rho, \varphi) = C_0 + v_0 \rho \cos \varphi + \frac{v_0 a^2}{\rho} \cos \varphi + \frac{\Gamma}{2\pi} \varphi.$$

28. 长为 l 的理想传输线, 一端接于电动势为 $v_0 \sin \omega t$ 的交流电源. 另一端短路, 求解线上的稳恒电振荡, 并计算输入阻抗.

解: 本题为没有初始条件的问题. 其定解问题为:

$$\text{泛定方程: } v_{tt} - a^2 v_{xx} = 0, \quad \left(a = \frac{1}{\sqrt{LC}} \right), \quad (1)$$

$$\text{边界条件: } \begin{cases} v|_{x=0} = v_0 \sin \omega t, & (2) \\ v|_{x=l} = 0, & (3) \end{cases}$$

为了计算方便, 将电动势 $v_0 \sin \omega t$ 写成 $v_0 e^{i\omega t}$, 这就需要约定在计算结果中取其虚部, 设 $v(x, t) = X(x) e^{i\omega t}$, 代入泛定方程 (1)

$$X(i\omega)^2 e^{i\omega t} - a^2 X'' e^{i\omega t} = 0,$$

$$\text{即 } X'' + \left(\frac{\omega}{a}\right)^2 X = 0. \quad (4)$$

(4) 式的解为:

$$X = A e^{i \frac{\omega}{a} x} + B e^{-i \frac{\omega}{a} x},$$

$$\therefore v(x, t) = X e^{i\omega t} = \left[A e^{i \frac{\omega}{a} x} + B e^{-i \frac{\omega}{a} x} \right] e^{i\omega t}, \quad (5)$$

利用 (2) 式 $v|_{x=0} = v_0 e^{i\omega t}$, 代入 (5) 式:

$$[A + B] e^{i\omega t} = v_0 e^{i\omega t}, \text{ 即 } A + B = v_0, \quad (6)$$

利用 (3) 式 $v|_{x=l} = 0$, 代入 (5) 式有:

$$\left[A e^{i \frac{\omega}{a} l} + B e^{-i \frac{\omega}{a} l} \right] = 0, \quad (7)$$

解 (6) 式和 (7) 式得

$$A = \frac{v_0}{1 - e^{2i \frac{\omega}{a} l}} = \frac{i v_0 e^{-i \frac{\omega}{a} l}}{2 \sin\left(\frac{\omega}{a} l\right)},$$

$$B = \frac{v_0}{1 - e^{-2i \frac{\omega}{a} l}} = \frac{-i v_0 e^{i \frac{\omega}{a} l}}{2 \sin\left(\frac{\omega}{a} l\right)},$$

以 A, B 代入 (5) 式得

$$v(x, t) = \frac{v_0 [e^{i \frac{\omega}{a} (x-l)} - e^{-i \frac{\omega}{a} (x-l)}] e^{i\omega t}}{-2i \sin\left(\frac{\omega}{a} l\right)}$$

$$= \frac{v_0 [e^{i \frac{\omega}{a} (l-x)} - e^{-i \frac{\omega}{a} (l-x)}] e^{i \omega t}}{2i \sin\left(\frac{\omega l}{a}\right)}$$

$$= \frac{\sin \frac{\omega}{a} (l-x)}{\sin\left(\frac{\omega l}{a}\right)} e^{i \omega t},$$

取 $v(x, t)$ 的虚部, 并以 $a = \sqrt{\frac{1}{LC}}$ 代入, 可得

$$v(x, t) = \frac{v_0 \sin \omega \sqrt{LC} (l-x)}{\sin \omega l \sqrt{LC}} \sin \omega t,$$

而 $\frac{\partial v}{\partial x} = \frac{-v_0 \omega \sqrt{LC} \cos \omega \sqrt{LC} (l-x) \sin \omega t}{\sin \omega l \sqrt{LC}},$

对于理想传输线, $R = G = 0$, 由 (31, 12) 式得

$$\frac{\partial j}{\partial t} = -\frac{1}{L} v_x = v_0 \omega \sqrt{\frac{C}{L}} \frac{\cos \omega \sqrt{LC} (l-x)}{\sin \omega l \sqrt{LC}} \sin \omega t,$$

$$\therefore j(x, t) = \int \frac{\partial j}{\partial t} dt$$

$$= v_0 \sqrt{\frac{C}{L}} \frac{\cos \omega \sqrt{LC} (l-x)}{\sin \omega l \sqrt{LC}} \int \omega \sin \omega t dt$$

$$= -v_0 \sqrt{\frac{C}{L}} \frac{\cos \omega \sqrt{LC} (l-x)}{\sin \omega l \sqrt{LC}} \cos \omega t + f(t).$$

(8)

由于 (8) 对 t 求导后为 $-\frac{1}{L} v_x$, 可知 $f'(t) = 0$, 故可取

$$f(t) = 0.$$

$$\text{输入阻抗 } Z_1 = \frac{v_{\max}|_{x=0}}{j_{\max}|_{x=0}} = \sqrt{\frac{L}{C}} \operatorname{tg} \omega l \sqrt{LC}.$$

若以 $l = \frac{\lambda}{4}$, 且 $\omega = \frac{2\pi a}{\lambda}$ 代入上式

$$\begin{aligned} \text{得 } Z_{\lambda} &= \sqrt{\frac{L}{C}} \operatorname{tg} \frac{2\pi a}{\lambda} \cdot \frac{\lambda}{4} \cdot \frac{1}{a} \\ &= \sqrt{\frac{L}{C}} \operatorname{tg} \frac{\pi}{2} = \infty. \end{aligned}$$

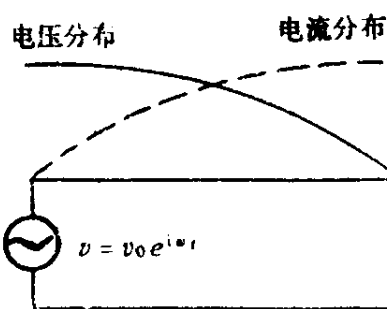


图 10-11

29. 长为 l 的非理想传输线, 一端接于电源 $v_0 \sin \omega t$, 另一端通过阻抗元件 R_0 , L_0 和 C_0 而相接。求解线上的稳恒电振荡。在什么情况下不存在反射波 (这叫作匹配) ?

解: 传输线上电压方程为 (即31、14式):

$$LCv_{xx} - v_{xx} + (LG + RC)v_t + RGv = 0. \quad (1)$$

边界上, $x = 0$ 处将电压 $v|_{x=0} = v_0 \sin \omega t$ 写成复数形式, 并约定其结果中取虚部:

$$v|_{x=0} = v_0 e^{i\omega t}, \quad (2)$$

$x = l$ 处串接阻抗元件, 电压为:

$$v|_{x=l} = \left(jR_0 + L_0 \frac{dj}{dt} + \frac{1}{C_0} \int_{t_0}^t j dt \right) \Big|_{x=l}.$$

由于是稳定电振荡, 可以认为振荡周期和电源周期相同, 因此可将 v 表示成 $v = X(x) e^{i\omega t}$, (3)

则 $v_{xx} = X''(x) e^{i\omega t}$, $v_t = i\omega X e^{i\omega t}$,

$$v_{tt} = -\omega^2 X(x) e^{i\omega t},$$

将上列各式代入方程 (1):

$$\begin{aligned} & -\omega^2 X(x) e^{i\omega t} LC - X''(x) e^{i\omega t} \\ & + (LG + RC) i\omega X(x) e^{i\omega t} \\ & + RG X(x) e^{i\omega t} \\ & = 0, \end{aligned}$$

整理后有:

$$\frac{X''(x)}{X(x)} = -\omega^2 LC + i\omega (LG + RC) + RG$$

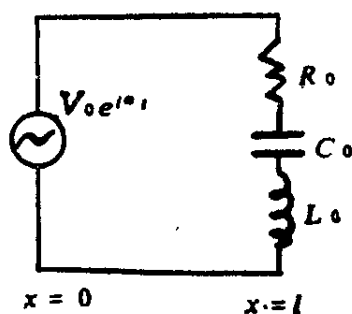


图 10-12

$$= (R + i\omega L)(G + i\omega C),$$

上面微分方程的解为:

$$X(x) = Ae^{\alpha x} + Be^{-\alpha x}, \quad (4)$$

$$\text{式中 } \alpha = \sqrt{(R + i\omega L)(G + i\omega C)}, \quad (5)$$

由 (31.12) 式得:

$$\begin{aligned} j_x &= -Gv - Cv_t = -GXe^{i\omega t} - C(i\omega)Xe^{i\omega t}, \\ &= -(G + i\omega C)Xe^{i\omega t} \\ &= -(G + i\omega C)(Ae^{\alpha x} + Be^{-\alpha x})e^{i\omega t} \end{aligned}$$

$$\begin{aligned} j &= -(G + i\omega C) \left[\frac{A}{\alpha} \int e^{\alpha x} d(\alpha x) - \frac{B}{\alpha} \right. \\ &\quad \left. \int e^{-\alpha x} d(\alpha x) \right] e^{i\omega t} + f(t) \\ &= -\frac{(G + i\omega C)}{\alpha} [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} + f(t), \quad (6) \end{aligned}$$

$$j_t = -\frac{(G + i\omega C)}{\alpha} [Ae^{\alpha x} - Be^{-\alpha x}] (i\omega) e^{i\omega t} + f'(t).$$

代入 (31.12) 式中关于 v_x 的方程.

$$\begin{aligned} v_x &= -Rj - Lj_t = -[Rj + Lj_t] \\ &= \frac{R(G + i\omega C)}{\alpha} [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} \\ &\quad + \frac{i\omega L(G + i\omega C)}{\alpha} [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} \\ &\quad - (Rf + Lf') \\ &= \frac{R(G + i\omega C)}{\alpha} [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} \\ &\quad + \frac{i\omega L(G + i\omega C)}{\alpha} [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} \\ &\quad - (Rf + Lf') \end{aligned}$$

$$\begin{aligned}
&= \frac{(G + i\omega C)}{\alpha} [R + i\omega L] [Ae^{\alpha x} - Be^{-\alpha x}] \\
&\quad e^{i\omega t} - (Rf - Lf') \\
&= \alpha [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} - (Rf + Lf'). \quad (7)
\end{aligned}$$

另一方面, 从 (3) 式和 (4) 式中得到

$$\begin{aligned}
v_x &= X'(x) e^{i\omega t} = [\alpha Ae^{\alpha x} - \alpha Be^{-\alpha x}] e^{i\omega t} \\
&= \alpha [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t}, \quad \text{代入 (7) 式的左边有} \\
&\quad \alpha [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} \\
&= \alpha [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} - (Rf + Lf').
\end{aligned}$$

即 $Rf(t) + Lf'(t) = 0,$

$$\frac{df(t)}{dt} = -\frac{R}{L}f(t), \quad \frac{df}{f} = -\frac{R}{L}dt.$$

$\therefore f(t) = f_0 e^{-\frac{R}{L}t},$ 代入 (6) 式:

$$j = -\sqrt{\frac{G + i\omega C}{R + i\omega L}} [Ae^{\alpha x} - Be^{-\alpha x}] e^{i\omega t} + f_0 e^{-\frac{R}{L}t}. \quad (8)$$

$$v = [Ae^{\alpha x} + Be^{-\alpha x}] e^{i\omega t}, \quad (9)$$

代入边界条件: $\begin{cases} v|_{x=0} = v_0 e^{i\omega t}, \\ v|_{x=l} = \left(R_0 + i\omega L_0 - \frac{i}{C_0\omega}\right) j|_{x=l}, \end{cases}$

得 $f_0 = 0,$ 及

$$\begin{cases} A + B = v_0, \\ Ae^{\alpha l} + Be^{-\alpha l} = \left(R_0 + i\omega L_0 - \frac{i}{\omega C_0}\right) \sqrt{\frac{G + i\omega C}{R + i\omega L}} (Ae^{\alpha l} - Be^{-\alpha l}). \end{cases}$$

解之得:

$$\begin{cases} A = \frac{R_0 + i\omega L_0 - \frac{i}{C_0\omega} - \sqrt{\frac{R + i\omega L}{G + i\omega C}}}{M}, \\ B = \frac{R_0 + i\omega L_0 - \frac{i}{C_0\omega} + \sqrt{\frac{R + i\omega L}{G + i\omega C}}}{M} \cdot e^{2\alpha l}, \end{cases}$$

式中, $M = \left(R_0 + i\omega L_0 - \frac{i}{C_0\omega} + \sqrt{\frac{R + i\omega L}{G + i\omega C}} \right) e^{\alpha l} + \left(R_0 + i\omega L_0 - \frac{i}{C_0\omega} - \sqrt{\frac{R + i\omega L}{G + i\omega C}} \right) e^{-\alpha l}.$

从方程(9)中可以看出, $Be^{-\alpha x + i\omega t}$ 表示前进波,

$Ae^{\alpha x + i\omega t}$ 表示反射波.若 $A = 0$ 表示不存在反射波

即必须 $R_0 + i\omega L_0 - \frac{i}{C_0\omega} = \sqrt{\frac{R + i\omega L}{G + i\omega C}}$, 等号右边称为传输线的特征阻抗.

30. 长为 l 的均匀杆, 一端固定, 另一端在纵向力 $F(t) = F_0 \sin \omega t$ 长期作用下, 求解杆的稳恒振动.

解: 这个杆的稳恒振动完全是由力 $F(t)$ 长期作用下形成的, 因而是个没有初始条件的问题. 定解问题是:

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & (1) \\ u|_{x=0} = 0, & (2) \\ u_x|_{x=l} = \frac{F_0}{YS} e^{i\omega t}, & (3) \end{cases}$$

因为是稳定振动, 因此可设振动周期和外力有相同的周期,

$$u(x, t) = X(x) \sin \omega t,$$

代入(1)得 $-\omega^2 X \sin \omega t - a^2 X'' \sin \omega t = 0,$

$$X'' + \frac{\omega^2}{a^2} X = 0, \therefore X = C_1 \cos \frac{\omega}{a} x + C_2 \sin \frac{\omega}{a} x,$$

由边界条件(2)得 $C_1 = 0.$

由条件 (3) 得: $C_2 \frac{\omega}{a} \cos \frac{\omega}{a} x \Big|_{x=l} = \frac{F_0}{YS}$.

$$\therefore C_2 = \frac{F_0 a}{YS \omega} \frac{1}{\cos \frac{\omega}{a} l},$$

$$\therefore \text{所求解是 } u(x, t) = \frac{F_0 a}{YS \omega} \frac{\sin \frac{\omega}{a} x}{\cos \frac{\omega}{a} l} \sin \omega t.$$

§37. 非齐次的泛定方程 (傅里叶级数法)

1. 两端固定的弦在线密度为 $\rho f(x, t) = \rho \Phi(x) \sin \omega t$ 的横向力作用下振动, 求解其振动情况, 研究共振的可能性, 并求共振时的解.

解: (i) 冲量定理法

$$\begin{cases} u_{tt} - a^2 u_{xx} = \Phi(x) \sin \omega t, \\ u|_{x=0} = 0, \quad u|_{x=l} = 0, \\ u|_{t=0} = 0, \quad u_t|_{t=0} = 0, \end{cases}$$

应用冲量定理法, 先求解

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0, & (1) \end{cases}$$

$$\begin{cases} v|_{x=0} = 0, \quad v|_{x=l} = 0, & (2) \end{cases}$$

$$\begin{cases} v|_{t=\tau+0} = 0, \quad v_t|_{t=\tau+0} = \Phi(x) \sin \omega \tau, & (3) \end{cases}$$

$\therefore$ 是第一类齐次边界条件, 因此可设

$$v(x, t, \tau) = \sum_{n=1}^{\infty} T_n(t, \tau) \sin \frac{n\pi x}{l},$$

代入 (1), 得

$$\sum_{n=1}^{\infty} \left(T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n \right) \sin \frac{n\pi x}{l} = 0,$$

解
$$T'_n + \frac{n^2 \pi^2 a^2}{l^2} T_n = 0,$$

得
$$T_n(t, \tau) = A_n(\tau) \cos \frac{n\pi a(t-\tau)}{l} + B_n(\tau) \sin \frac{n\pi a(t-\tau)}{l}.$$

$$\therefore v(x, t, \tau) = \sum_{n=1}^{\infty} \left[A_n(\tau) \cos \frac{n\pi a(t-\tau)}{l} + B_n(\tau) \sin \frac{n\pi a(t-\tau)}{l} \right] \sin \frac{n\pi x}{l},$$

由(3)
$$\sum_{n=1}^{\infty} A_n(\tau) \sin \frac{n\pi x}{l} = 0, \quad A_n(\tau) = 0,$$

$$\sum_{n=1}^{\infty} B_n(\tau) \frac{n\pi a}{l} \sin \frac{n\pi x}{l} = \Phi(x) \sin \omega \tau,$$

$$\therefore B_n(\tau) = \frac{2}{n\pi a} \sin \omega \tau \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi.$$

$$\therefore v(x, t, \tau) = \sum_{n=1}^{\infty} \left[\frac{2}{n\pi a} \sin \omega \tau \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \right] \sin \frac{n\pi a(t-\tau)}{l} \cdot \sin \frac{n\pi x}{l}.$$

从而原问题的解为

$$\begin{aligned} u(x, t) &= \int_0^t v(x, t, \tau) d\tau \\ &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \left[\frac{1}{n} \left(\int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \right) \sin \frac{n\pi x}{l} \right] \int_0^t \sin \omega \tau \sin \frac{n\pi a(t-\tau)}{l} d\tau \\ &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \left[\frac{\omega \sin \frac{n\pi a t}{l} - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right] \sin \frac{n\pi x}{l}. \end{aligned}$$

下面研究共振的可能性，并求共振时的解。

如外力的频率等于基音或谐音的频率，亦即当

$$\omega = \frac{n\pi a}{l}, \quad (n = 1, 2, \dots \text{中的某一个值}) \text{ 时, 解 } u(x, t)$$

的表达式中 [] 内为 $\frac{0}{0}$ 型，由洛必达法则

$$\begin{aligned} & \lim_{\omega \rightarrow \frac{n\pi a}{l}} \frac{\omega \sin \frac{n\pi a t}{l} - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \\ &= \lim_{\omega \rightarrow \frac{n\pi a}{l}} \frac{\sin \frac{n\pi a t}{l} - t \frac{n\pi a}{l} \cos \omega t}{2\omega} \\ &= \frac{1}{2\omega} \sin \omega t - \frac{1}{2} t \cos \omega t, \end{aligned}$$

这时 $\frac{1}{2} t \cos \omega t$ 项的振幅 $\frac{1}{2} t$ 随时间 t 而增长，亦即发生共振。

(ii) 格林函数法，先求格林函数 G 。

$$\begin{cases} G_{tt} - a^2 G_{xx} = \delta(x - \xi) \delta(t - \tau); \\ G|_{x=0} = 0, \quad G|_{x=l} = 0, \\ G|_{t=0} = 0, \quad G_t|_{t=0} = 0, \end{cases}$$

$$\text{即 } \begin{cases} G_{tt} - a^2 G_{xx} = 0, \\ G|_{x=0} = 0, \quad G|_{x=l} = 0, \\ G|_{t=\tau+0} = 0, \quad G_t|_{t=\tau+0} = \delta(x - \xi), \end{cases}$$

$$\begin{aligned} G(x, t; \xi, \tau) = \sum_{n=1}^{\infty} & \left[A_n(\xi, \tau) \cos \frac{n\pi a(t - \tau)}{l} \right. \\ & \left. + B_n(\xi, \tau) \sin \frac{n\pi a(t - \tau)}{l} \right] \sin \frac{n\pi x}{l}, \end{aligned}$$

由初始条件， $A_n = 0$ ，

$$\begin{aligned}
 B_n(\xi, \tau) &= \frac{2}{n\pi a} \int_0^l \delta(x - \xi) \sin \frac{n\pi x}{l} dx \\
 &= \frac{2}{n\pi a} \sin \frac{n\pi \xi}{l}. \\
 \therefore G(x, t; \xi, \tau) &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi a(t - \tau)}{l} \\
 &\quad \sin \frac{n\pi x}{l},
 \end{aligned}$$

从而原问题的解为:

$$\begin{aligned}
 u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l f(\xi, \tau) G(x, t; \xi, \tau) d\xi d\tau \\
 &= \int_{\tau=0}^t \int_{\xi=0}^l \left(\Phi(\xi) \sin \omega \tau \cdot \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi a(t - \tau)}{l} \sin \frac{n\pi x}{l} \right) d\xi d\tau \\
 &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \left[\left(\int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \right) \sin \frac{n\pi x}{l} \right] \\
 &\quad \int_0^t \sin \omega \tau \sin \frac{n\pi a(t - \tau)}{l} d\tau \\
 &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \\
 &\quad \left[\frac{\omega \sin \frac{n\pi a t}{l} - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right] \sin \frac{n\pi x}{l}.
 \end{aligned}$$

(iii) 傅里叶级数法

$$\text{设 } u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l},$$

为定 $T_n(t)$, 将 $f(x, t) = \Phi(x) \sin \omega t$ 展成以 $\left\{ \sin \frac{n\pi x}{l} \right\}$

为基函数族的傅里叶级数，其傅里叶系数为

$$f_n(t) = \frac{2}{l} \int_0^l \Phi(x) \sin \omega t \sin \frac{n\pi x}{l} dx,$$

将 $u(x,t)$ 及 $f(x,t)$ 的傅氏级数代入泛定方程，比较系数得

$$T'_n + \frac{n^2 \pi^2 a^2}{l^2} T_n = f_n(t), \quad \left. \vphantom{\frac{n^2 \pi^2 a^2}{l^2}} \right\}$$

由零初始条件，得 $T_{11}(0) = 0, T'_n(0) = 0$,

用拉氏变换解此常微分方程，得

$$p^2 T_n(p) + \frac{n^2 \pi^2 a^2}{l^2} T_n(p) = \bar{f}_n(p),$$

$$T_n(p) = \frac{1}{p^2 + \frac{n^2 \pi^2 a^2}{l^2}} \bar{f}_n(p),$$

利用卷积定理反演，得

$$T_n(t) = \frac{l}{n\pi a} \int_0^t f_n(\tau) \sin \frac{n\pi a(t-\tau)}{l} d\tau$$

$$= \frac{2}{n\pi a} \int_0^t \int_0^l f(x, \tau) \sin \frac{n\pi x}{l} \sin$$

$$\frac{n\pi a(t-\tau)}{l} dx d\tau,$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$$

$$= \int_0^t \int_0^l \sum_{n=1}^{\infty} \frac{2}{n\pi a} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi a(t-\tau)}{l} \sin$$

$$\frac{n\pi x}{l} f(\xi, \tau) d\xi d\tau.$$

这就是弦振动方程在第一类齐次边界条件与零初始条件下解的公式。对于本题而言， $f(x, t) = \Phi(x) \sin \omega t$ ，以此代入后，则有

$$\int_0^t \sin \omega \tau \sin \frac{n\pi a(t-\tau)}{l} d\tau$$

$$= \frac{\omega \sin \frac{n\pi a t}{l} - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2}.$$

所得结果与前两法相同。

在上述一般公式中，积分号下与 $f(\xi, \tau)$ 相乘的级数，正是解法(ii)中所求得的格林函数。

2. 两端固定弦在点 x_0 受谐变力 $\rho f(t) = \rho f_0 \sin \omega t$ 作用而振动，求解振动情况。〔提示：外加力的线密度可表为 $\rho f(x, t) = \rho f_0 \sin \omega t \delta(x - x_0)$ 。〕

解：(i) 用冲量定理法求解

$$\begin{cases} u_{tt} - a^2 u_{xx} = f_0 \sin \omega t \delta(x - x_0), \\ u|_{x=0} = 0, \quad u|_{x=l} = 0, \\ u|_{t=0} = 0, \quad u_t|_{t=0} = 0, \end{cases}$$

应用冲量定理法，先解

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0, \\ v|_{x=0} = 0, \quad v|_{x=l} = 0, \\ v|_{t=t_0+0} = 0, \quad v_t|_{t=t_0+0} = f_0 \delta(x - x_0) \sin \omega \tau. \end{cases}$$

与上一题一样，可得

$$v(x, t, \tau) = \sum_{n=1}^{\infty} \left[A_n(\tau) \cos \frac{n\pi a(t-\tau)}{l} + B_n(\tau) \sin \frac{n\pi a(t-\tau)}{l} \right] \sin \frac{n\pi x}{l}.$$

由初始条件，诸 $A'_n(\tau) = 0$ ，

$$B_n(\tau) = \frac{2}{n\pi a} f_0 \sin \omega \tau \int_0^l \delta(x - x_0) \sin \frac{n\pi x}{l} dx$$

$$= \frac{2f_0}{n\pi a} \sin \omega \tau \sin \frac{n\pi x_0}{l},$$

$$\therefore v(x, t; \tau) = \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \sin \omega \tau \sin \frac{n\pi a(t-\tau)}{l} \sin \frac{n\pi x}{l},$$

从而原问题的解为

$$\begin{aligned} u(x, t) &= \int_0^t v(x, t; \tau) d\tau \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi x}{l} \int_0^t \sin \omega \tau \sin \frac{n\pi a(t-\tau)}{l} d\tau \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi x}{l} \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right]. \end{aligned}$$

(ii) 用格林函数法求解, 如上题, 对弦振动方程第一类齐次边界条件, 零初始条件的格林函数是:

$$G(x, t; \xi, \tau) = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a(t-\tau)}{l} \sin \frac{n\pi x}{l} \sin \frac{n\pi \xi}{l},$$

所以原问题的解为

$$\begin{aligned} u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l f(\xi, \tau) G(x, t; \xi, \tau) d\xi d\tau \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \left[\left(\int_0^l \delta(\xi - x_0) \sin \frac{n\pi \xi}{l} d\xi \right) \right. \\ &\quad \left. \sin \frac{n\pi x}{l} \int_0^t \sin \omega \tau \right] \end{aligned}$$

$$\begin{aligned}
& \cdot \sin \frac{n\pi a(t-\tau)}{l} d\tau \Big] \\
& = \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi x}{l} \\
& \quad \left[\frac{\omega \sin \frac{n\pi at}{l} - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right].
\end{aligned}$$

3. 均匀细导线，每单位长的电阻为 R ，通以恒定电流 I ，导线表面跟周围温度为零度的介质进行热量交换，试解线上温度变化，设初始温度和两端温度都是零度。

解：泛定方程是 $c\rho u_t d_t - ku_{xx} dt = -kudt + RI^2 dt$ ，即

$$\begin{cases} u_t - a^2 u_{xx} + \frac{h}{c\rho} u = \frac{1}{c\rho} I^2 R, & (1) \end{cases}$$

$$\begin{cases} u(0t) = u(lt) = 0, u|_{t=0} = 0, & (2) \end{cases}$$

从泛定方程和齐次边界条件得知 (1) 的特征函数是 $\sin \frac{n\pi}{l} x$

将 $\frac{1}{c\rho} I^2 R$ 展开为和特征函数相对应的傅立叶级数

$$\begin{aligned}
C_n &= \frac{2}{l} \int_0^l \frac{1}{c\rho} I^2 R \sin \frac{n\pi}{l} \xi d\xi \\
&= \frac{2}{n\pi} I^2 R \frac{1}{c\rho} \int_0^l \sin \frac{n\pi}{l} \xi d \left(\frac{n\pi}{l} \xi \right) \\
&= \frac{2}{n\pi} I^2 R \frac{1}{c\rho} \left[-\cos \frac{n\pi}{l} \xi \right]_0^l \\
&= \frac{2}{n\pi} \frac{I^2 R}{c\rho} [1 - (-1)^n],
\end{aligned}$$

取 $n = 2k + 1 = 2n + 1$ 为奇数， $c_n = \frac{4}{(2n+1)\pi} \frac{I^2 R}{c\rho}$ ，

$$\therefore \frac{1}{c\rho} I^2 R = \frac{4I^2 R}{c\rho} \sum_{n=1}^{\infty} \frac{1}{(2n+1)\pi} \sin \frac{(2n+1)\pi}{l} x.$$

(3)

$$\text{令 } u(x,t) = \sum T_n(t) X(x) = \sum T_n \sin \frac{(2n+1)\pi}{l} x, \quad (4)$$

将(3)(4)代入(1)中

$$\begin{aligned} & \sum_{n=1}^{\infty} \left(T'_{2n+1} + \frac{(2n+1)^2 \pi^2 a^2}{l^2} T_{2n+1} + \frac{h}{c\rho} T_{2n+1} \right) \\ & \times \sin \frac{(2n+1)\pi}{l} x \\ & = \sum_{n=1}^{\infty} \frac{4I^2 R}{c\rho(2n+1)\pi} \sin \frac{(2n+1)\pi}{l} x, \end{aligned}$$

比较 $\sin \frac{(2n+1)\pi}{l} x$ 的系数得到

$$T' + \left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] T = \frac{4I^2 R}{c\rho(2n+1)\pi}. \quad (5)$$

微分方程(5)的相应齐次方程的解是

$$T = C_1 e^{-\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] t} \quad \text{应用系数变更法方程}$$

(5)的全解是,

$$T = e^{-\int p dt} \left[c_1 + \int Q(t) e^{\int p dt} dt \right], \quad (6)$$

$$(6) \text{ 式中的 } p = \frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}, \quad Q = \frac{4I^2 R}{c\rho(2n+1)\pi},$$

p 、 Q , 都是常数

$$\therefore T = e^{-pt} \left[c_1 + \int Q e^{pt} dt \right] = e^{-pt} \left[c_1 + \frac{Q}{p} e^{pt} \right].$$

$$\text{应用初始条件决定 } T(0) = 0, \quad \therefore c_1 = -\frac{Q}{p}.$$

$$\therefore \text{ 因此 } T = \frac{Q}{p} (1 - e^{-pt}) = \frac{4I^2 R}{c\rho(2n+1)\pi} \cdot \frac{1}{p} (1 - e^{-pt}).$$

$$\therefore u(x,t) = \frac{4I^2 R}{c\rho} \sum_{n=1}^{\infty} \frac{1}{(2n+1)\pi} \frac{1}{\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}} \left[1 - e^{-\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] t} \right] \cdot \sin \frac{(2n+1)\pi}{l} x.$$

4. 在圆域 $\rho < a$ 上求解 $\Delta u = -4$ 边界条件是 $u|_{\rho=a} = 0$.

解: 在圆柱坐标中的拉氏方程是

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = -4, \quad u|_{\rho=a} = 0. \quad (1)$$

由试探法找一个特解, 使泛定方程化为齐次的.

设 $u = v - \rho^2$, 则定解问题成为

$$\frac{\partial^2 v}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial v}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 v}{\partial \varphi^2} = 0, \quad (2)$$

$$v|_{\rho=a} = a^2, \quad (\because u|_{\rho=a} = v|_{\rho=a} - a^2 = 0), \quad (3)$$

泛定方程(2)的解是

$$v(\rho, \varphi) = C_0 + D_0 \ln \rho + \left(C_n \rho^n + D_n \frac{1}{\rho^n} \right) (A_m \cos m\varphi + B_m \sin m\varphi),$$

$$\text{代入边界条件 } v|_{\rho=a} = C_0 + D_0 \ln a + \left(C_n a^n + D_n \frac{1}{a^n} \right) (A_m \cos m\varphi + B_m \sin m\varphi) = a^2.$$

$$(A_m \cos m\varphi + B_m \sin m\varphi) = a^2.$$

$$\begin{aligned} \text{因为边界条件与 } \varphi \text{ 无关 } v|_{\rho=a} &= C_0 + D_0 \ln a + \left(C_n a^n + D_n \frac{1}{a^n} \right) \\ &= a^2, \end{aligned}$$

比较系数 $C_0 = 0, D_0 = 0, C_2 = 1, D_2 = 0$, 所以 $v = a^2$.

所以所求的解 $u = v - \rho^2 = a^2 - \rho^2$.

5. 在圆域 $\rho < a$ 上求解 $\Delta u = -xy$, 边界条件是 $u|_{\rho=a} = 0$.

解: (i) 用傅里叶级数法求解.

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\varphi\varphi} = -\rho^2 \sin\varphi \cos\varphi = -\frac{1}{2}\rho^2 \sin 2\varphi, \\ u|_{\rho=a} = 0, \end{cases} \quad \begin{matrix} (1) \\ (2) \end{matrix}$$

$$\text{设 } u = \sum_{m=0}^{\infty} R_m(\rho) (A_m \cos m\varphi + B_m \sin m\varphi),$$

$$\text{代入(2)有} \quad R_m(a) = 0, m = 0, 1, 2, \dots, \quad (2')$$

$$\begin{aligned} \text{代入(1)有} \quad \sum_{m=0}^{\infty} \left(R_m'' + \frac{1}{\rho} R_m' - \frac{m^2}{\rho^2} R_m \right) (A_m \cos m\varphi + B_m \sin m\varphi) \\ = -\frac{1}{2}\rho^2 \sin\varphi, \end{aligned}$$

$\therefore$ 诸 $A_m = 0$, 且 $B_m = 0$, ($m \neq 2$). 取 $B_2 = 1$, 有

$$R_2'' + \frac{1}{\rho} R_2' - \frac{2^2}{\rho^2} R_2 = -\frac{1}{2}\rho^2,$$

$$\text{令 } \rho = e^t, \text{ 有 } \ddot{R}_2 - 4R_2 = -\frac{1}{2}e^{4t},$$

$$R_2 = c_1 e^{2t} + c_2 e^{-2t} - \frac{1}{24} e^{4t} = c_1 \rho^2 + c_2 \rho^{-2} - \frac{1}{24} \rho^4,$$

由于 u 在圆内有界,

$$\therefore c_2 = 0, \text{ 又由 (2) } c_1 = -\frac{1}{24} a^2,$$

$$\therefore \text{ 解为 } u(\rho, \varphi) = \frac{1}{24} \rho^2 (a^2 - \rho^2) \sin 2\varphi.$$

(ii) 用特解法求定解问题. 注意到

$$\Delta(x^3 y) = 6xy, \Delta(xy^3) = 6xy,$$

为便于化为极坐标, 取 $v = -\frac{1}{12}(x^3 y + xy^3)$, $\Delta v = -xy$,

令 $u = v + w$, 在极坐标下 $v = -\frac{1}{12} xy(x^2 + y^2)$

$$= -\frac{1}{12}\rho^4 \sin\varphi \cos\varphi,$$

则 $\Delta w = 0$,

$$w|_{\rho=a} = u|_{\rho=a} - v|_{\rho=a} = \frac{1}{12}\rho^4 \sin\varphi \cos\varphi|_{\rho=a} = \frac{1}{24}a^4 \sin 2\varphi,$$

解上述定解问题

$$w = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho^m + D_m \rho^{-m}),$$

由于 w 在圆内有界, $D_0 = D_m = 0$, 记 $A_m C_m = a_m, B_m C_m = b_m$, 由边界条件

$$C_0 + \sum_{m=1}^{\infty} (a_m \cos m\varphi + b_m \sin m\varphi) a^m = \frac{1}{24}a^4 \sin 2\varphi,$$

知 $C_0 = 0, a_m = 0, b_2 = \frac{1}{24}a^2, m \neq 2$ 时, $b_m = 0$,

$$\therefore w = \frac{1}{24}a^2 \rho^2 \sin 2\varphi,$$

$$\begin{aligned} \therefore u(\rho, \varphi) = v + w &= -\frac{1}{24}\rho^4 \sin 2\varphi + \frac{1}{24}a^2 \rho^2 \sin 2\varphi \\ &= \frac{1}{24}\rho^2 (a^2 - \rho^2) \sin 2\varphi, \end{aligned}$$

6. 在矩形域 $0 < x < a, -\frac{b}{2} < y < \frac{b}{2}$ 上 求解 $\Delta u = -2$ 且 u 在边界上的值为零.

$$\text{解: } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -2 \quad (1)$$

$$u(0, y) = u(a, y) = 0, \quad u\left(x, \frac{b}{2}\right) = u\left(x, -\frac{b}{2}\right) = 0 \quad (2)$$

取试探解 $V = A + BX + CX^2$ 使 $\Delta_2 V = -2$ 且 $V|_{x=0} = V|_{x=a}$

$$= 0,$$

(由于 $\Delta_2 V = -2$, $V|_{x=0} = V|_{x=a} = 0$ 有三个条件, 所以可以定出三个系数 A 、 B 、 C).

由 $\Delta_2 V = -2$, 得出 $C = -1$, 由 $V|_{x=0} = 0$, $A = 0$.

$$V|_{x=a} = 0, B = a.$$

$$\therefore V = ax - x^2 = x(x-a),$$

$$\text{令 } u = v + w, \Delta_2 w = 0, \quad (3)$$

$$w|_{x=0} = u - v|_{x=0} = 0, w|_{x=a} = u - v|_{x=a} = 0,$$

$$w \Big|_{y=\frac{b}{2}} = w \Big|_{y=-\frac{b}{2}} = u - v \Big|_{y=\frac{b}{2}} = -x(x-a),$$

泛定方程 (3) 的 x 有齐次边界条件 $X = \sum C_n \sin \frac{n\pi}{a} x$, 对于 y 没有齐次边界条件,

$$\therefore W = \sum_{n=1}^{\infty} \left(A_n \operatorname{Ch} \frac{n\pi y}{a} + B_n \operatorname{Sh} \frac{n\pi y}{a} \right) \sin \frac{n\pi}{a} x,$$

应用边界条件 $W \Big|_{y=\frac{b}{2}} = -x(x-a)$, 及 $W \Big|_{y=-\frac{b}{2}} = -x(x-a)$, 有

$$\sum_{n=1}^{\infty} \left(A_n \operatorname{Ch} \frac{n\pi}{a} \cdot \frac{b}{2} + B_n \operatorname{Sh} \frac{n\pi b}{2a} \right) \sin \frac{n\pi}{e} x$$

$$= -x(x-a),$$

$$\sum_{n=1}^{\infty} \left(A_n \operatorname{Ch} \frac{n\pi b}{2a} - B_n \operatorname{Sh} \frac{n\pi b}{2a} \right) \sin \frac{n\pi}{e} x = -x(x-a).$$

$$A_n \operatorname{Ch} \frac{n\pi b}{2a} + B_n \operatorname{Sh} \frac{n\pi b}{2a}$$

$$= \frac{2}{a} \int_0^a -x(a-x) \sin \frac{n\pi}{e} x dx,$$

$$A_n \operatorname{Ch} \frac{n\pi b}{2a} - B_n \operatorname{Sh} \frac{n\pi b}{2a} \\ = \frac{2}{a} \int_0^a -x(a-x) \sin \frac{n\pi}{e} x dx.$$

$$\therefore B_n = 0$$

$$\begin{aligned} A_n &= \frac{2}{n\pi \operatorname{Ch} \frac{n\pi b}{2a}} \left\{ x(a-x) \cos \frac{n\pi x}{a} \Big|_0^a \right. \\ &\quad \left. - \int_0^a \cos \frac{n\pi}{e} x d[x(a-x)] \right\} \\ &= \frac{2}{n\pi \operatorname{Ch} \frac{n\pi b}{2a}} \left\{ \int_0^a x \cos \frac{n\pi}{a} x dx - \int_0^a (a-x) \cos \frac{n\pi}{e} x dx \right\} \\ &= \frac{2a}{n^2 \pi^2 \operatorname{Ch} \frac{n\pi b}{2a}} \left\{ x \sin \frac{n\pi}{e} x \Big|_0^a - \int_0^a \sin \frac{n\pi}{a} x dx \right. \\ &\quad \left. - (a-x) \sin \frac{n\pi}{a} x \Big|_0^a - \int_0^a \sin \frac{n\pi}{a} x dx \right\} \\ &= \frac{4a^2}{n^3 \pi^3 \operatorname{Ch} \frac{n\pi b}{2a}} [(-1)^n - 1] \\ &= \begin{cases} \frac{-8a^2}{\pi^3 (2k+1)^3 \operatorname{Ch} \frac{(2k+1)\pi b}{2a}}, & [n \text{ 为奇数 } 2k+1], \\ 0, & n \text{ 为偶数.} \end{cases} \end{aligned}$$

n 为偶数.

$$\therefore w(xy) = -\frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{\operatorname{Ch} \frac{(2k+1)\pi y}{a} \sin \frac{(2k+1)\pi x}{a}}{(2k+1)^3 \operatorname{Ch} \frac{(2k+1)\pi b}{2a}},$$

$$u(xy) = x(a-x) - \frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{\operatorname{Ch} \frac{(2k+1)\pi y}{a} \sin \frac{(2k+1)\pi x}{a}}{(2k+1)^3 \operatorname{Ch} \frac{(2k+1)\pi b}{2a}},$$

7. 在矩形域 $0 < x < a - \frac{b}{2} < y < \frac{b}{2}$ 上求解 $\Delta u = -x^2 y$, 且

u 在边界上的值为零:

解: 找一个特解 v 使泛定方程为齐次, 且仍保持齐次的边界条件 $v|_{x=0} = v|_{x=a} = 0$.

设 $v = Axy + Bx^4y$, 由 $\Delta_2 v = 12Bx^2y = -x^2y$,

$$\therefore B = -\frac{1}{12},$$

$$\text{由 } v|_{x=0} = 0 \text{ 得 } A = \frac{a^3}{12},$$

$$\therefore v = \frac{1}{12} (a^3xy - x^4y) = \frac{xy}{12} (a^3 - x^3),$$

$u = v + w$ 得出

$$\Delta w = 0 \quad w|_{x=0} = w|_{x=a} = 0,$$

$$w|_{y=\frac{b}{2}} = +\frac{xb}{24} (a^3 - x^3),$$

$$w|_{y=-\frac{b}{2}} = -\frac{bx}{24} (a^3 - x^3).$$

$$w = \sum_{n=1}^{\infty} \left[A_n \operatorname{Ch} \frac{n\pi y}{a} + B_n \operatorname{Sh} \frac{n\pi y}{a} \right] \sin \frac{n\pi x}{a},$$

应用关于 y 的边界条件来确定 A_n 和 B_n , 先求积分

$$I_n = \frac{2}{a} \int_0^a \frac{bx}{24} (a^3 - x^3) \sin \frac{n\pi x}{a} dx$$

$$= \frac{-b}{12} \cdot \frac{12a^3}{n^3\pi^3} \left\{ a^2(-1)^n + \frac{2a^2}{n^2\pi^2} \left[1 - (-1)^n \right] \right\}$$

$$= \frac{-ba^3}{n^5\pi^5} [n^2\pi^2(-1)^n + 2 - 2(-1)^n],$$

$$\begin{cases} A_n \operatorname{Ch} \frac{n\pi b}{2a} - B_n \operatorname{Sh} \frac{n\pi b}{2a} = I_n, \text{ 得 } A_n = 0. \\ A_n \operatorname{Ch} \frac{n\pi b}{2a} + B_n \operatorname{Sh} \frac{n\pi b}{2a} = -I_n, \quad B_n = -\frac{I_n}{\operatorname{Sh} \frac{n\pi b}{2a}}, \end{cases}$$

$$\begin{aligned} \therefore u(xy) &= -\frac{xy}{12}(a^3 - x^3) + \frac{a^4b}{\pi^5} \sum_{n=1}^{\infty} \frac{n^2\pi^2(-1)^n + 2 - 2(-1)^n}{n^5 \operatorname{Sh} \frac{n\pi b}{2a}} \\ &\quad \operatorname{Sh} \frac{n\pi y}{a} \sin \frac{n\pi}{a} x. \end{aligned}$$

第十一章 分离变数 (傅里叶积分) 法

§38. 齐次的泛定方程 (傅里叶积分法)

1. 求解无限长传输线上的电振荡传播问题. $G:C = R:L$ 的情况跟 $G:C \neq R:L$ 的情况有什么不同?

解: 在电报方程式中令 $j = u$, $v = u$, 则有同一方程式

$$LCu_{tt} - u_{xx} + (LG + RC)u_t + RG u = 0, \quad (1)$$

因此对于电压或电流都只须求同一个方程式 (1) 的解.

以分离变数的试探解, $u = X(x)T(t)$ 代入 (1) 式有

$$LCT''X - TX'' + (LG + RC)T'X + RGT X = 0,$$

$$LC \frac{T''}{T} + (LG + RC) \frac{T'}{T} + RG = \frac{X''}{X} = -k^2, \quad (2)$$

无穷空间 X 的解是 $X = \int e^{ikx} dk, \quad (3)$

(3) 式中的 $k = \frac{\omega}{a} = \frac{2\pi}{\lambda}, \quad (4)$

k 称为波矢, λ 是波长.

关于 t 的方程为

$$LCT'' + (LG + RC)T' + (RG + k^2)T = 0, \text{ 以 } T = e^{mt} \text{ 代入得}$$

$$m^2 LC + m(LG + RC) + RG + k^2 = 0,$$

$$\therefore m = \frac{-(LG + RC) \pm \sqrt{(LG + RC)^2 - 4LC(RG + k^2)}}{2LC},$$

$$\therefore m_1, m_2 = -\frac{G}{2C} - \frac{R}{2L} \pm \frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C}\right)^2 - \frac{4k^2}{LC}},$$

$$\therefore T(t) = A(k)e^{m_1 t} + B(k)e^{m_2 t}.$$

可以得 $u(x, t)$ 的一般解为:

$$\begin{aligned} u(x, t) = & \int TX dk = e^{-\frac{G}{2C}t - \frac{R}{2L}t} \\ & \times \int_{-\infty}^{\infty} \left[A(k) e^{\frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C}\right)^2 - \frac{4k^2}{LC}} t} \right. \\ & \left. + B(k) e^{-\frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C}\right)^2 - \frac{4k^2}{LC}} t} \right] dk \end{aligned} \quad (5)$$

(i) 在 (5) 式如有 $\frac{G}{C} = \frac{R}{L}$ 时, 这时 (5) 式成为

$$\begin{aligned} u(x, t) = & e^{-\frac{G}{2C}t - \frac{R}{2L}t} \int_{-\infty}^{\infty} \left(A(k) e^{\frac{ik}{\sqrt{LC}}t} \right. \\ & \left. + B(k) e^{-\frac{ik}{\sqrt{LC}}t} \right) dk, \end{aligned}$$

即对任意波矢都有相同的衰减振荡, 波没有色散现象. 这说明传输线对通过频率没有限制.

(ii) 在 $\frac{G}{C} \neq \frac{R}{L}$ 时, 若 $\left[\left(\frac{R}{L} - \frac{G}{C}\right)^2 - \frac{4k^2}{LC}\right] < 0$ 即当 $k > \frac{\sqrt{LC}}{2} \left(\frac{R}{L} - \frac{G}{C}\right)$ 时, 这时有衰减振荡, 若以 $\omega \equiv \frac{1}{2} \sqrt{\frac{4k^2}{LC} - \left(\frac{R}{L} - \frac{G}{C}\right)^2}$,

则可得振荡传播速度

$$a = v \times \lambda = \frac{\lambda}{2\pi} \cdot 2\pi v = \frac{\omega}{k} = \frac{1}{2} \sqrt{\frac{4}{LC} - \frac{1}{k^2} \left(\frac{R}{L} - \frac{G}{C}\right)^2},$$

即波速 a 随 k 的不同而有差异, 这时波有色散现象.

(iii) 若 $\frac{G}{C} \neq \frac{R}{L}$ 且 $k < \frac{\sqrt{LC}}{2} \left(\frac{R}{L} - \frac{G}{C}\right)$, 则 (5) 式中不出现虚数项, 可以看出这时只有指数式的衰减而无振荡.

2. 研究半无限长细杆的导热问题, 杆端 $x = 0$ 温度保持为零度, 初始温度分布为 $K(e^{-\lambda x} - 1)$.

解: $u_t - a^2 u_{xx} = 0$, 设 $u = TX$,

$$\frac{T'}{a^2 T} = \frac{X''}{X} = -\lambda, \lambda \text{ 须为正实数记 } \lambda = \omega^2, \omega \text{ 为实数,}$$

则有 $X = C_1 e^{i\omega x}, T = C_2 e^{-\omega^2 a^2 t}$.

本征解为 $C(\omega) e^{-\omega^2 a^2 t} e^{i\omega x}$, 一般解 $u = \int_{-\infty}^{\infty} C(\omega) e^{-\omega^2 a^2 t}$

$\cdot e^{i\omega x} d\omega$. 若在无界空间求解 $u|_{t=0} = \int_{-\infty}^{\infty} C(\omega) e^{i\omega x} d\omega$, 其中 $C(\omega)$

即 $u|_{t=0}$ 的傅里叶变换式, 现在初始条件 $u|_{t=0} = \varphi(x)$ 定义在半无界空间, 根据边界条件应将 u 作奇延拓

$$\text{即 } u|_{t=0} = \begin{cases} -K(e^{\lambda x} - 1), (x < 0), \\ K(e^{-\lambda x} - 1), (x > 0). \end{cases} \text{ 于是}$$

$$C(\omega) = \frac{1}{2\pi} \int_0^{\infty} K(e^{-\lambda \xi} - 1) e^{-i\omega \xi} d\xi - \frac{1}{2\pi} \int_{-\infty}^0 K(e^{\lambda \xi} - 1) \times e^{-i\omega \xi} d\xi,$$

$$\begin{aligned} u(x,t) &= \int_{-\infty}^{\infty} C(\omega) e^{i\omega x} e^{-\omega^2 a^2 t} d\omega \\ &= \frac{K}{2\pi} \int_0^{\infty} (e^{-\lambda \xi} - 1) \int_{-\infty}^{\infty} e^{i\omega(x-\xi) - \omega^2 a^2 t} d\omega d\xi \\ &\quad - \frac{K}{2\pi} \int_{-\infty}^0 (e^{\lambda \xi} - 1) \left[\int_{-\infty}^{\infty} e^{i\omega(x-\xi) - \omega^2 a^2 t} d\omega d\xi \right] \\ &= \frac{K}{2\pi} \frac{\sqrt{\pi}}{a\sqrt{t}} \left[\int_0^{\infty} (e^{-\lambda \xi} - 1) e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \right. \\ &\quad \left. - \int_{-\infty}^0 (e^{\lambda \xi} - 1) e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \right] \\ &= \frac{K}{2a\sqrt{\pi t}} \left[-2a\sqrt{\pi t} \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right) \right] \end{aligned}$$

$$\begin{aligned}
& + \int_0^\infty e^{-\left(\lambda\xi + \frac{(x-\xi)^2}{4a^2t}\right)} d\xi \\
& - \int_{-\infty}^0 e^{-\left(\lambda\xi - \frac{(x-\xi)^2}{4a^2t}\right)} d\xi \Big] \\
& = -K \operatorname{erf} \frac{x}{2a\sqrt{t}} + \frac{K}{2a\sqrt{\pi t}} (I_1 + I_2).
\end{aligned}$$

在 I_1 的积分中令 $z = \frac{\xi - (x - 2a^2\lambda t)}{2a\sqrt{t}}$,

在 I_2 的积分中令 $Z = \frac{(x + 2a^2\lambda t) - \xi}{2a\sqrt{t}}$,

$$\begin{aligned}
\text{则 } I_1 &= \int_0^\infty e^{-\left(\lambda\xi + \frac{(x-\xi)^2}{4a^2t}\right)} d\xi \\
&= \int_0^\infty e^{-\{[\xi - (x - 2a^2\lambda t)]^2 - 4a^2\lambda^2 t^2 + 4a^2\lambda t x\}} d\xi \\
&= e^{a^2\lambda^2 t - \lambda x} \int_0^\infty e^{-[\xi - (x - 2a^2\lambda t)]^2 / 4a^2 t} d\xi \\
&= 2a\sqrt{t} e^{a^2\lambda^2 t - \lambda x} \int_{\frac{2\lambda a^2 t - x}{2a\sqrt{t}}}^\infty e^{-z^2} dz. \\
&= 2a\sqrt{t} e^{a^2\lambda^2 t - \lambda x} \left[\int_0^\infty e^{-z^2} dz - \int_0^{\frac{2\lambda a^2 t - x}{2a\sqrt{t}}} e^{-z^2} dz \right] \\
&= a\sqrt{\pi t} e^{a^2\lambda^2 t - \lambda x} \operatorname{erfc} \left(\frac{2\lambda a^2 t - x}{2a\sqrt{t}} \right).
\end{aligned}$$

$$\begin{aligned}
I_2 &= - \int_{-\infty}^0 e^{-\left(\lambda\xi - \frac{(x-\xi)^2}{4a^2t}\right)} d\xi \\
&= \int_0^\infty e^{-\frac{-\{[\xi - (x + 2a^2\lambda t)]^2 - 4a^4 t^2 \lambda^2 - 4xa^2\lambda t\}}{4a^2 t}} d\xi \\
&= e^{a^2\lambda^2 t + \lambda x} \int_0^\infty e^{-\left(\frac{\xi - (x + 2a^2\lambda t)}{2a\sqrt{t}}\right)^2} d\xi
\end{aligned}$$

$$\begin{aligned}
&= -2a\sqrt{t} e^{a^2\lambda^2 t + \lambda x} \int_{\frac{x+2a^2\lambda t}{2a\sqrt{t}}}^{\infty} e^{-z^2} dz \\
&= -2a\sqrt{t} e^{a^2\lambda^2 t + \lambda x} \left[\int_0^{\infty} e^{-z^2} dz - \int_0^{\frac{x+2a^2\lambda t}{2a\sqrt{t}}} e^{-z^2} dz \right] \\
&= -a\sqrt{\pi t} e^{a^2\lambda^2 t + \lambda x} \operatorname{erfc} \left[\frac{x+2a^2\lambda t}{2a\sqrt{t}} \right].
\end{aligned}$$

$$\begin{aligned}
\therefore u(xt) &= -K \operatorname{erf} \left(\frac{x}{2a\sqrt{t}} \right) + \frac{K}{2} e^{a^2\lambda^2 t} \left[e^{-\lambda x} \right. \\
&\quad \left. \times \operatorname{erfc} \left(\frac{2\lambda a^2 t - x}{2a\sqrt{t}} \right) - e^{\lambda x} \operatorname{erfc} \left(\frac{x+2a^2\lambda t}{2a\sqrt{t}} \right) \right].
\end{aligned}$$

在做上面的积分时，应用到余误差积分：

$$\begin{aligned}
\frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-z^2} dz &= \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-z^2} dz - \int_0^x e^{-z^2} dz \\
&= \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} - \operatorname{erf} x \\
&= 1 - \operatorname{erf} x = \operatorname{erfc} x.
\end{aligned}$$

3. 半无界杆，杆端 $x = 0$ 有谐变热流 $B \sin \omega t$ 进入，求长时间后杆上温度分布 $u(xt)$ 。

解： $u_t - a^2 u_{xx} = 0, x > 0,$

$u_x|_{x=0} = -\frac{B}{k} \sin \omega t$ ，将边界条件改写为 $u_x|_{x=0} = -\frac{B}{k}$

$e^{i\omega t}$ ，在解答中取虚部，可以预计各点温度都以同一频率作周期变化，所以 u 可表为 $u = X(x) e^{i\omega t}$ ，代入泛定方程后得

$$i\omega X - a^2 X'' = 0, \text{ 以 } X = e^{mx}, m = \pm \sqrt{\frac{\omega}{a^2}} \sqrt{i},$$

由于 $\sqrt{i} = \sqrt{e^{\frac{\pi}{2}i}} = e^{\frac{\pi}{4}i} = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$ 从而得到

$$X_1(x) = e^{\sqrt{\frac{\omega}{a^2}} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) x},$$

$$X_2(x) = e^{-\sqrt{\frac{\omega}{a^2}}\left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right)x}.$$

应用自然边界条件当 $x \rightarrow \infty$ u 有限, 舍去 $X_1(x)$, 因而有

$$u = Ce^{-\sqrt{\frac{\omega}{2a^2}}(1+i)x} e^{i\omega t}, \text{ 将 } u \text{ 代入边界条件, 得}$$

$$-\sqrt{\frac{\omega}{2a^2}}(1+i)Ce^{i\omega t} = -\frac{B}{k}e^{i\omega t},$$

$$C = \frac{aB}{k}\sqrt{\frac{1}{2\omega}}(1-i), \quad (1-i) = \sqrt{2}e^{-\frac{\pi}{4}i}.$$

$$\begin{aligned} u &= \frac{aB}{k} \frac{1}{\sqrt{2\omega}}(1-i)e^{-\sqrt{\frac{\omega}{2}}\frac{x}{a}} e^{i\left(\omega t - \sqrt{\frac{\omega}{2}}\frac{x}{a}\right)} \\ &= \frac{aB}{k\sqrt{\omega}} e^{-\sqrt{\frac{\omega}{2}}\frac{x}{a}} e^{i\left(\omega t - \sqrt{\frac{\omega}{2}}\frac{x}{a} - \frac{\pi}{4}\right)}. \end{aligned}$$

$$\text{取虚部得 } u(x,t) = \frac{aB}{k\sqrt{\omega}} e^{-\sqrt{\frac{\omega}{2}}\frac{x}{a}} \sin\left(\omega t - \sqrt{\frac{\omega}{2}}\frac{x}{a} - \frac{\pi}{4}\right).$$

4. 应用泊松公式计算下述定解问题的解. $u_{tt} - a^2 \Delta u = 0$, 初始速度为零, 初始位移在某个单位球内为 1, 在球外为零.

解: 取单位球的球心为坐标原点, 则定解问题为:

$$\begin{cases} u_{tt} - a^2 \Delta u = 0, \\ u|_{t=0} = \varphi(\vec{r}) = \begin{cases} 1, \gamma < 1, \\ 0, \gamma > 1, \end{cases} \\ u_t|_{t=0} = \psi(\vec{r}) = 0, \end{cases}$$

由泊松公式

$$u(\vec{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{a,t}'} \frac{\varphi(\vec{r}')}{at} ds' + \frac{1}{4\pi a} \iint_{S_{a,t}'} \frac{\psi(\vec{r}')}{at} ds'$$

$$= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S'_{at}} \frac{\varphi(\vec{r}')}{at} ds'.$$

(i) 当点 $\vec{r}$ (以 $\vec{r}$ 为矢径的点简称为点 $\vec{r}$, 下同) 在单位球内时:

a. 若 $r + at < 1$, 球面 S'_{at} 完全在单位球内. 从而 $\varphi(\vec{r}') =$

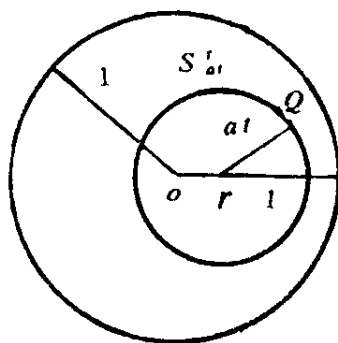


图 11-1

$$\begin{aligned} u(\vec{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \left(\frac{1}{at} \iint_{S'_{at}} ds' \right) \\ &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \left(\frac{1}{at} \cdot 4\pi (at)^2 \right) = 1, \end{aligned}$$

b. 若 $at > r + 1$, 单位球将在球面 S'_{at} 内, 这时 $\varphi(\vec{r}') = 0$, 从而 $u(\vec{r}, t) = 0$.

c. 若 $1 - r < at < 1 + r$, 则 S'_{at} 与单位球相交, 设它在球内的部分为 $\bar{S}'_{at}$, 因在球外 $\varphi(\vec{r}') = 0$ 而在 $\bar{S}'_{at}$ 上, $\varphi(\vec{r}') = 1$.

$$\therefore u(\vec{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{\bar{S}'_{at}} \frac{1}{at} ds'.$$

$$\therefore \iint_{\bar{S}'_{at}} ds' = \int_0^{2\pi} \int_0^{\theta_0} (at)^2 \sin\theta d\theta d\varphi$$

$$= 2\pi (at)^2 (-\cos\theta) \Big|_0^{\theta_0}$$

$$= 2\pi (at)^2 (1 - \cos\theta_0)$$

$$= 2\pi (at)^2 \left(1 - \frac{r^2 + a^2 t^2 - 1}{2rat} \right)$$

$$= \frac{\pi at}{r} [1 - (r - at)^2],$$

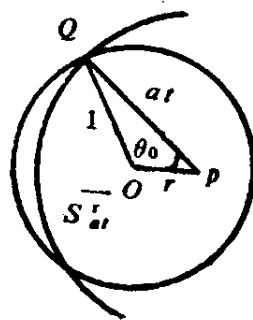


图 11-2

$$\begin{aligned}\therefore u(\vec{r}, t) &= \frac{1}{4\pi a} - \frac{\partial}{\partial t} \left[-\frac{\pi}{r} (1 - (r - at)^2) \right] \\ &= \frac{1}{2r} (r - at).\end{aligned}$$

(ii) 当点 $\vec{r}$ 在单位球外时.

a. 若 $at + 1 < r$, S'_{at} 与单位球分离, 在 S'_{at} 上 $\varphi(\vec{r}') = 0$,

$$\therefore u(\vec{r}, t) = 0.$$

b. 若 $at > 1 + r$, S'_{at} 将单位球包含于内, 在 S'_{at} 上 $\varphi(\vec{r}') = 0$,

$$\therefore u(\vec{r}, t) = 0.$$

c. 若 $r - 1 < at < 1 + r$, S'_{at} 与单位球相交, 设它在球内的部

分为 $\bar{S}'_{at}$, 与(i)之c相同的计算,

$$\text{得 } u(\vec{r}, t) = \frac{1}{2r} (r - at)$$

综上所述,

在球内

$$u(\vec{r}, t) = \begin{cases} 1, & \left(t < \frac{1-r}{a} \right), \\ 0, & \left(t > \frac{1+r}{a} \right), \\ \frac{1}{2r} (r - at), & \left(\frac{1-r}{a} < t < \frac{1+r}{a} \right), \end{cases}$$

在球外

$$u(\vec{r}, t) = \begin{cases} 0, & \left(t < \frac{r-1}{a} \right), \\ 0, & \left(t > \frac{1+r}{a} \right), \\ \frac{1}{2r} (r - at), & \left(\frac{r-1}{a} < t < \frac{1+r}{a} \right). \end{cases}$$

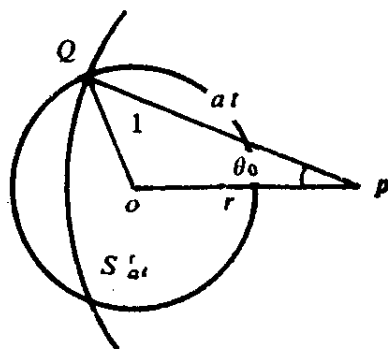


图 11-3

5. 应用泊松公式计算下述定解问题的解, $u_{tt} - a^2 \Delta u = 0$, 初始速度为零, 初始位移在球 $r = r_0$ 以内为 $A \cos(\pi r / 2r_0)$, 在球外为零.

解:

$$\begin{cases} u_{tt} - a^2 \Delta u = 0, \\ u_t|_{t=0} = \varphi(\vec{r}) = 0, \\ u|_{t=0} = \varphi(\vec{r}) = \begin{cases} A \cos\left(\frac{\pi r}{2r_0}\right), & (r < r_0), \\ 0, & (r > r_0). \end{cases} \end{cases}$$

由泊松公式 $u(\vec{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^{\vec{r}}} \frac{\varphi(\vec{r}')}{at} ds',$

(i) 当点 $\vec{r}$ 在球 $r = r_0$ 内时,

a. 若 $r + at < r_0$, $S_{at}^{\vec{r}}$ 在球内,

$$u(\vec{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^{\vec{r}}} \frac{A \cos\left(\frac{\pi r'}{2r_0}\right)}{at} ds'.$$

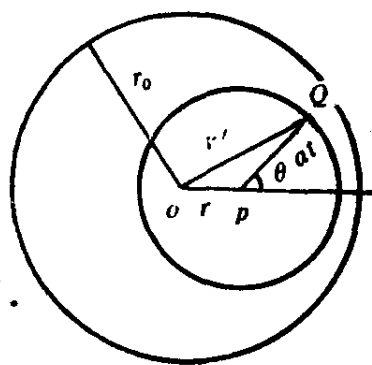


图 11-4

为计算上式右端积分, 如右图所示, 以 $p(\vec{r})$ 为原点, $o\vec{p}$ 的方向为 Z 轴方向建立球坐标系, 设 $S_{at}^{\vec{r}}$ 上的点 $Q(\vec{r}')$ 在球坐标系内的坐标为 $Q'(at, \theta, \varphi)$.

则 $\angle QPZ = \theta, r'^2 = r^2 + (at)^2 + 2rat \cos \theta$.

注意到 $d \cos \theta = \frac{dr'^2}{2rat} = \frac{r'}{rat} dr',$

有

$$u(\vec{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} d\varphi \int_0^\pi \frac{A \cos\left(\frac{\pi r'}{2r_0}\right)}{at} \cdot (at)^2 \sin \theta d\theta$$

$$\begin{aligned}
&= \frac{1}{2a} \frac{\partial}{\partial t} \int_0^x A \cos \left(\frac{\pi r'}{2r_0} \right) \cdot (-at) d \cos \theta \\
&= \frac{1}{2a} \frac{\partial}{\partial t} \int_{r+at}^{|r-at|} - A \cos \left(\frac{\pi r'}{2r_0} \right) \frac{r'}{r} dr' \\
&= \frac{A}{2ar} \frac{\partial}{\partial t} \int_{|r-at|}^{r+at} r' \cos \left(\frac{\pi r'}{2r_0} \right) dr'. \\
\therefore \frac{d}{dx} \int_0^x f(\xi) d\xi &= f(x), \\
\text{又 } \frac{\partial}{\partial t} &= \frac{\partial}{\partial(r+at)} \cdot \frac{\partial(r+at)}{\partial t} = a \frac{\partial}{\partial(r+at)} \\
\therefore \frac{\partial}{\partial t} \int_{r-at}^{r+at} f(r') dr' &= af(r+at) - (-a)f(r-at) \\
&= a[f(r+at) + f(r-at)]. \\
\frac{\partial}{\partial t} \int_{at-r}^{r+at} f(r') dr' &= af(r+at) - af(at-r).
\end{aligned}$$

从而

$$\begin{aligned}
u(\vec{r}, t) &= \begin{cases} \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (r-at) \cos \frac{\pi(r-at)}{2r_0} \right], & (r > at), \\ \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (at-r) \cos \frac{\pi(at-r)}{2r_0} \right], & (r < at). \end{cases} \\
&= \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (r-at) \cos \frac{\pi(r-at)}{2r_0} \right],
\end{aligned}$$

b. 若 $at > r + r_0$, 球 $r = r_0$ 将在 S'_{at} 内部, 这时 $\varphi(\vec{r}') = 0$, 从而 $u(\vec{r}, t) = 0$,

c. 若 $r_0 - r < at < r_0 + r$, 则 S'_{at} 与球 $r = r_0$ 相交.

与情形 i) 一样建立坐标系, 一样讨论, 只不过应在 S'_{at} 在球内的部分积分 ($\because$ 在球外 $\varphi(\vec{r}') = 0$), 即为右图所示, θ 应从 θ_0 积到 π , 而 θ_0 所对应之点 $A, r' = r_0$, 从而

$$\begin{aligned}
 u(\vec{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} d\varphi \\
 &\quad \times \int_{\theta_0}^{\pi} \frac{A \cos\left(\frac{\pi r'}{2r_0}\right)}{at} \cdot (at)^2 \sin\theta d\theta \\
 &= \frac{A}{2ar} \frac{\partial}{\partial t} \int_{|r-at|}^{r_0} r' \cos\left(\frac{\pi r'}{2r_0}\right) dr' \\
 &= \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0}.
 \end{aligned}$$

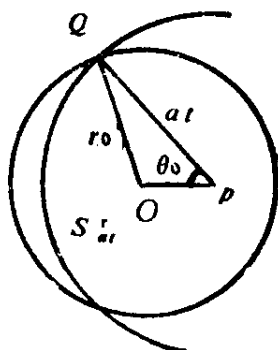


图 11-5

(ii) 当点 $\vec{r}$ 在球外时.

a. 若 $at + r_0 < r$, S'_{at} 与球分离, $\varphi(\vec{r}') = 0, u(\vec{r}, t) = 0$,

b. 若 $at > r_0 + r$, S'_{at} 将球 $r = r_0$ 包含于内, 仍有 $\varphi(\vec{r}') = 0$, $u(\vec{r}, t) = 0$.

c. 若 $r - r_0 < at < r_0 + r$, 球面 S'_{at} 与球 $r = r_0$ 相交.

完全类似于 (i) 之 c 的讨论得到

$$u(\vec{r}, t) = \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0},$$

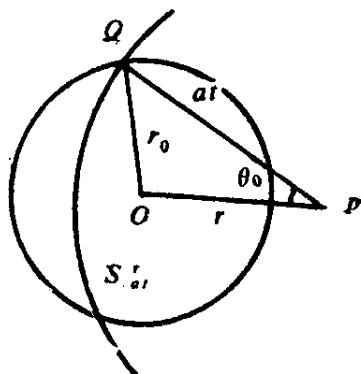


图 11-6

综上述, 本问题的解为

在球内 $u(\vec{r}, t) =$

$$\begin{aligned}
 &\frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (r-at) \cos \frac{\pi(r-at)}{2r_0} \right], \\
 &\quad \left(t < \frac{r_0-r}{a} \right), \\
 &= \begin{cases} \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0}, & \left(\frac{r_0-r}{a} < t < \frac{r_0+r}{a} \right), \\ 0, & \left(t > \frac{r_0+r}{a} \right). \end{cases}
 \end{aligned}$$

在球外 $u(\vec{r}, t) =$

$$= \begin{cases} 0, & \left(t < \frac{r-r_0}{a}\right), \\ \frac{A}{2r}(r-at) \cos \frac{\pi(r-at)}{2r_0}, & \left(\frac{r-r_0}{a} < t < \frac{r+r_0}{a}\right), \\ 0, & \left(t > \frac{r+r_0}{a}\right). \end{cases}$$

6. 二维波动, 初始速度为零, 初始位移在圆 $\rho=1$ 以内为 1, 在圆外为零, 试求 $u|_{t=0}$.

解: 应用二维泊松公式

$$u(\rho, \theta, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \iint \Sigma_{at}^{xy} \frac{\varphi(\rho', \theta')}{\sqrt{a^2 t^2 - \rho'^2}} \rho' d\rho' d\theta',$$

(i) 当 $at > 1$ 时,

$$\begin{aligned} u &= \frac{1}{2\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} d\theta \int_0^1 \frac{\rho d\rho}{\sqrt{a^2 t^2 - \rho^2}} \\ &= \frac{1}{a} \frac{\partial}{\partial t} \sqrt{a^2 t^2 - \rho^2} \Big|_0^1 \\ &= \frac{1}{a} \frac{\partial}{\partial t} (at - \sqrt{a^2 t^2 - 1}) \\ &= 1 - \frac{at}{\sqrt{a^2 t^2 - 1}}, \end{aligned}$$

(ii) 当 $at < 1$,

$$\begin{aligned} u &= \frac{1}{2\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} d\theta \int_0^{at} \frac{\rho d\rho}{\sqrt{a^2 t^2 - \rho^2}} \\ &= -\frac{1}{a} \frac{\partial}{\partial t} \left(\sqrt{a^2 t^2 - \rho^2} \Big|_0^{at} \right) = 1. \end{aligned}$$

7. 求解三维无界空间的输送问题, $u_t - a^2 \Delta u = 0$, $u|_{t=0} = \varphi(x, y, z)$.

解: 将 u 展开为三重傅里叶积分 $\vec{k} = k_1 \vec{i}_1 + k_2 \vec{i}_2 + k_3 \vec{i}_3$

$u(r, t) = \iiint_{-\infty}^{\infty} T(\vec{k}) e^{i\vec{k}\cdot\vec{r}} d k_1 d k_2 d k_3$ 代入泛定方程, 得

$$\iiint_{-\infty}^{\infty} (T' + k^2 a^2 T) e^{i\vec{k}\cdot\vec{r}} d k_1 d k_2 d k_3 = 0,$$

得关于 T 的方程为 $T' + k^2 a^2 T = 0$, 即 $T = C(k) e^{-k^2 a^2 t}$.

$u(\vec{r}, t) = \iiint_{-\infty}^{\infty} c(\vec{k}) e^{-k^2 a^2 t} e^{i\vec{k}\cdot\vec{r}} d k_1 d k_2 d k_3$, 代入初始条件

$$u|_{t=0} = \varphi(\vec{r}'),$$

即 $\iiint_{-\infty}^{\infty} c(\vec{k}) e^{i\vec{k}\cdot\vec{r}} d k_1 d k_2 d k_3 = \varphi(\vec{r})$, $c(k)$ 是 $\varphi(\vec{r})$ 的三重傅里叶

变换式:

$$c(\vec{k}) = \left(\frac{1}{2\pi}\right)^3 \iiint_{-\infty}^{\infty} \varphi(\vec{r}') e^{-i\vec{k}\cdot\vec{r}'} d x' d y' d z'.$$

把 $c(k)$ 代入 $u(r, t)$ 的式子得:

$$\begin{aligned} u(r, t) &= \iiint_{-\infty}^{\infty} \left[\left(\frac{1}{2\pi}\right)^3 \iiint_{-\infty}^{\infty} \varphi(\vec{r}') e^{-i\vec{k}\cdot\vec{r}'} d x' d y' d z' \right] \\ &\quad \times e^{-k^2 a^2 t} e^{i\vec{k}\cdot\vec{r}} d k_1 d k_2 d k_3 \\ &= \iiint_{-\infty}^{\infty} \varphi(\vec{r}') \left[\left(\frac{1}{2\pi}\right)^3 \iiint_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{i\vec{k}\cdot(\vec{r}-\vec{r}')} \right. \\ &\quad \left. \times d k_1 d k_2 d k_3 \right] d x' d y' d z' \end{aligned}$$

$$\text{其中 } \left(\frac{1}{2\pi}\right)^3 \iiint_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{i\vec{k}\cdot(\vec{r}-\vec{r}')} d k_1 d k_2 d k_3$$

$$= \left(\frac{1}{2\pi}\right)^3 \int_{-\infty}^{\infty} e^{-k_1^2 a^2 t} e^{i k_1 (x-x')} d k_1$$

$$\times \int_{-\infty}^{\infty} e^{-k_2^2 a^2 t} e^{i k_2 (y-y')} d k_2 \int_{-\infty}^{\infty} e^{-k_3^2 a^2 t} e^{i k_3 (z-z')} d k_3$$

$$\begin{aligned}
&= \left(\frac{1}{2\pi} \right)^3 \frac{\sqrt{\pi}}{a\sqrt{t}} e^{-\frac{(x-x')^2}{4a^2t}} \frac{\sqrt{\pi}}{a\sqrt{t}} e^{-\frac{(y-y')^2}{4a^2t}} \frac{\sqrt{\pi}}{a\sqrt{t}} \\
&\quad \times e^{-\frac{(z-z')^2}{4a^2t}} \\
&= \frac{1}{(2a\sqrt{\pi t})^3} e^{-\frac{|\vec{r}-\vec{r}'|^2}{4a^2t}}. \\
\therefore u &= \iiint_{-\infty}^{\infty} \varphi(\vec{r}') \frac{1}{(2a\sqrt{\pi t})^3} e^{-\frac{|\vec{r}-\vec{r}'|^2}{4a^2t}} dx' dy' dz'.
\end{aligned}$$

8. 例6 研究三维无界空间中的自由振动是从初始 ($t = 0$) 状况推算以后 ($t > 0$) 的状况. 试重新求解例6, 从初始状况反推以前 ($t < 0$) 的状况.

解: $u_{tt} - a^2 \Delta_3 u = 0$, $u|_{t=0} = \varphi(\vec{r})$, $u_t|_{t=0} = \psi(\vec{r})$,

将 u 展开为三重傅里叶积分

$u(\vec{r}) = \iiint_{-\infty}^{\infty} T(t, \vec{k}) e^{i\vec{k}\cdot\vec{r}} dk_1 dk_2 dk_3$ 代入泛定方程, 分离出关于 T 的方程.

$$T'' + a^2 k^2 T = 0, T = A(\vec{k}) e^{i\vec{k}\cdot\vec{r}} + B(\vec{k}) e^{-i\vec{k}\cdot\vec{r}},$$

$$u = \iiint_{-\infty}^{\infty} \left[A(k) e^{i\vec{k}\cdot\vec{r}} + B(k) e^{-i\vec{k}\cdot\vec{r}} \right] e^{i\vec{k}\cdot\vec{r}} dk_1 dk_2 dk_3, \text{ 代}$$

入初始条件得到

$$\begin{cases}
\iiint_{-\infty}^{\infty} [A(k) + B(k)] e^{i\vec{k}\cdot\vec{r}} dk_1 dk_2 dk_3 = \varphi(\vec{r}), \\
\iiint_{-\infty}^{\infty} ika [A(\vec{k}) - B(\vec{k})] e^{i\vec{k}\cdot\vec{r}} dk_1 dk_2 dk_3 = \psi(\vec{r}), \\
A(\vec{k}) + B(\vec{k}) = \left(\frac{1}{2\pi} \right)^3 \iiint_{-\infty}^{\infty} \varphi(\vec{r}') e^{-i\vec{k}\cdot\vec{r}'} dx' dy' dz', \\
ika[A(\vec{k}) - B(\vec{k})] = \left(\frac{1}{2\pi} \right)^3 \iiint_{-\infty}^{\infty} \psi(\vec{r}') e^{-i\vec{k}\cdot\vec{r}'} dx' dy' dz'
\end{cases}$$

解之得

$$A(\vec{k}) = \left(\frac{1}{2\pi}\right)^3 \iiint \frac{1}{2} \left[\varphi(\vec{r}') + \frac{1}{ika} \psi(\vec{r}') \right] e^{-i\vec{k}\vec{r}'} \\ \times dx' dy' dz',$$

$$B(\vec{k}) = \left(\frac{1}{2\pi}\right)^3 \iiint \frac{1}{2} \left[\varphi(\vec{r}') - \frac{1}{ika} \psi(\vec{r}') \right] e^{-i\vec{k}\vec{r}'} \\ \times dx' dy' dz',$$

$$\begin{aligned} u(\vec{r}, t) &= \iiint \left[A(\vec{k}) e^{i\vec{k}\vec{r}} + B(\vec{k}) e^{-i\vec{k}\vec{r}} \right] e^{i\vec{k}\vec{r}} \\ &\quad \times dk_1 dk_2 dk_3 \\ &= \frac{1}{4\pi a} \iiint \varphi(\vec{r}') \left[\frac{a}{4\pi^2} \iiint (e^{i\vec{k}\vec{r}} + e^{-i\vec{k}\vec{r}}) \right. \\ &\quad \times e^{i\vec{k}(\vec{r}-\vec{r}')} dk_1 dk_2 dk_3 \left. \right] dx' dy' dz' \\ &\quad + \frac{1}{4\pi a} \iiint \varphi(\vec{r}') \left[\frac{a}{4\pi^2} \iiint \frac{1}{ik} (e^{i\vec{k}\vec{r}} - e^{-i\vec{k}\vec{r}}) \right. \\ &\quad \times e^{i\vec{k}(\vec{r}-\vec{r}')} dk_1 dk_2 dk_3 \left. \right] dx' dy' dz' \\ &= \frac{1}{4\pi a} \iiint \varphi(\vec{r}') \frac{\partial}{\partial t} \left\{ \frac{1}{|\vec{r}-\vec{r}'|} \left[\delta(|\vec{r}-\vec{r}'|-at) \right. \right. \\ &\quad \left. \left. - \delta(|\vec{r}-\vec{r}'|+at) \right] \right\} dx' dy' dz' \\ &\quad + \frac{1}{4\pi a} \iiint \psi(\vec{r}') \frac{1}{|\vec{r}-\vec{r}'|} \left[\delta(|\vec{r}-\vec{r}'|-at) \right. \\ &\quad \left. - \delta(|\vec{r}-\vec{r}'|+at) \right] dx' dy' dz'. \end{aligned}$$

对 $t < 0$, $|\vec{r}-\vec{r}'|-at \neq 0$, 舍去 $\delta(|\vec{r}-\vec{r}'|-at)$, 所以

$$u = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iiint -\frac{\varphi(\vec{r}')}{|\vec{r}-\vec{r}'|} \delta(|\vec{r}-\vec{r}'|+at) dx' dy' dz'$$

$$+ \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} - \frac{\varphi(r')}{|\vec{r} - \vec{r}'|} \delta(|\vec{r} - \vec{r}'| + at) dx' dy' dz',$$

积分只需在球面上进行, 这个球面使 $|\vec{r} - \vec{r}'| + at = 0$,

$|\vec{r} - \vec{r}'| = -at$, $\vec{r}'$ 为球面上点的矢端, 球心在以 $\vec{r}$ 为矢径的点

(即要求解的点, 半径为 $-at$ 或写作 $a|t|$, 这个球面记为 $S_{a|t|}^r$,

因而所求的解

$$u = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iiint_{S_{a|t|}^r} \frac{\varphi(\vec{r}')}{at} ds' + \frac{1}{4\pi a} \iiint_{S_{a|t|}^r} \frac{\psi(\vec{r}')}{at} ds'.$$

§39. 非齐次的泛定方程(傅里叶积分法)

1. 求解一维半无界空间的输送问题

$$u_t - a^2 u_{xx} = 0, \quad u|_{x=0} = At, \quad u|_{t=0} = 0.$$

解: (i) 令 $u = W + At$, 以消去非齐次边界条件, 得 W 的定解问题:

$$\begin{cases} W_t - a^2 W_{xx} = -A, \\ W|_{x=0} = u|_{x=0} - At = 0, \\ W|_{t=0} = 0. \end{cases}$$

再找定解问题的格林函数:

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x_0 - \xi) \delta(t - \tau), \\ G|_{x=0} = 0, \\ G|_{t=0} = 0, \end{cases} \quad \begin{cases} G_t - a^2 G_{xx} = 0, \\ G|_{x=0} = 0, \\ G|_{t=\tau} = \delta(x_0 - \xi), \end{cases}$$

于是得到格林函数为

$$G(x, t; \xi, \tau) = \int_{-\infty}^{\infty} \delta(x_0 - \xi) \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{\frac{-(x-\xi)^2}{4a^2(t-\tau)}} d\xi$$

$$= \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-x_0)^2}{4a^2(t-\tau)}}$$

$$\begin{aligned} \text{得 } u(x, t) &= At - \int AG(x, t; \xi, \tau) d\tau \\ &= At - \int_0^t A \frac{1}{2a\sqrt{t-\tau}} e^{-\frac{(x-x_0)^2}{4a^2(t-\tau)}} d\tau \\ &= A \int_0^t \left(1 - \operatorname{erf} \frac{x}{2a\sqrt{t-\tau}} \right) \\ &= A \int_0^t \operatorname{erfc} \frac{x}{2a\sqrt{t-\tau}} d\tau. \end{aligned}$$

(ii) 参照边界条件作奇延拓

$$W_t - a^2 W_{xx} = \begin{cases} -A, & x > 0, \\ A, & x < 0. \end{cases} \quad W|_{t=0} = 0,$$

$$\begin{aligned} W &= \int_0^t \left[\int_0^\infty -A \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi \right. \\ &\quad \left. + \int_{-\infty}^0 \frac{A e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}}}{2a\sqrt{\pi(t-\tau)}} d\xi \right] d\tau \\ &= A \int_0^t \left[-\frac{1}{\sqrt{\pi}} \int_0^\infty e^{-\left(\frac{x-\xi}{2a\sqrt{t-\tau}}\right)^2} d\left(\frac{\xi-x}{2a\sqrt{t-\tau}}\right) \right. \\ &\quad \left. - \frac{1}{\sqrt{\pi}} \int_{-\infty}^0 e^{-\left(\frac{x-\xi}{2a\sqrt{t-\tau}}\right)^2} d\left(\frac{x-\xi}{2a\sqrt{t-\tau}}\right) \right] d\tau, \end{aligned}$$

在 w 的第一个积分中, 作替换 $z = \frac{\xi-x}{2a\sqrt{t-\tau}}$, 第二个积分中,

$$z = \frac{x-\xi}{2a\sqrt{t-\tau}}, \text{ 则}$$

$$W = -A \int_0^t \left[\frac{1}{\sqrt{\pi}} \int_0^\infty \frac{-x}{2a\sqrt{t-\tau}} e^{-z^2} dz + \frac{1}{\sqrt{\pi}} \int_{-\infty}^0 \frac{x}{2a\sqrt{t-\tau}} \right]$$

$$\begin{aligned} & \times e^{-z^2} dz \Big] d\tau \\ &= -A \int_0^t \left[-\frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{2a\sqrt{t-\tau}}} e^{-z^2} dz \right] d\tau = -A \\ & \int_0^t \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau, \end{aligned}$$

$$\begin{aligned} u &= At - \int_0^t \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \\ &= A \int_0^t \left[1 - \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) \right] d\tau \\ &= A \int_0^t \operatorname{erfc} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau. \end{aligned}$$

2. 在一维半无界空间求解 $u_t - a^2 u_{xx} = 0$, $u|_{x=0} = f(t)$, $u|_{t=0} = \varphi(x)$.

解: $u = f(t) + v + w$ 定解问题成为

$$\begin{cases} v_t - a^2 v_{xx} = -f'(t), \\ v|_{x=0} = 0, v|_{t=0} = 0, \end{cases} \quad \begin{cases} W_t - a^2 W_{xx} = 0, \\ W|_{x=0} = 0, W|_{t=0} = \varphi(x) - f(0), \end{cases}$$

v 的方程参照边界条件, 作奇延拓

$$v_t - a^2 v_{xx} = \begin{cases} -f'(t), & x > 0, \\ f'(t), & x < 0, \end{cases} \quad v|_{t=0} = 0,$$

$$\begin{aligned} v(x, t) &= \int_0^t \left\{ \int_{-\infty}^0 f'(\tau) \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi \right. \\ & \quad \left. + \int_0^{\infty} -f'(\tau) \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi \right\} d\tau \\ &= - \int_0^t f'(\tau) \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \end{aligned}$$

$$\begin{aligned}
&= \int_0^t -f'(\tau) \left[1 - \operatorname{erfc} \left(\frac{x}{2a\sqrt{t-\tau}} \right) \right] d\tau \\
&= f(0) - f(t) + \int_0^t f'(\tau) \operatorname{erfc} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau,
\end{aligned}$$

w 的方程, 参照边界条件作奇延拓.

$$W_t - a^2 W_{xx} = 0, W|_{t=0} = \begin{cases} \varphi(x) - \varphi(0), & x > 0, \\ -[\varphi(x) - f(0)], & x < 0, \end{cases}$$

引用书上第257页的结果, 得出 W 的解

$$\begin{aligned}
W &= \int_0^\infty \left[\varphi(\xi) - f(0) \right] \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \\
&\quad - \int_{-\infty}^0 [\varphi(-\xi) - f(0)] \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi,
\end{aligned}$$

在 W 的第一个积分中令 $z = \frac{\xi - x}{2a\sqrt{t}}$, 第二个积分中令

$$z = \frac{x - \xi}{2a\sqrt{t}}.$$

$$\begin{aligned}
\text{则 } W &= \int_{\frac{-x}{2a\sqrt{t}}}^\infty \varphi(x + z2a\sqrt{t}) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{\frac{x}{2a\sqrt{t}}}^\infty \varphi \\
&\quad \times (z2a\sqrt{t} - x) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - f(0) \operatorname{erf} \left(\frac{x}{2a\sqrt{t}} \right),
\end{aligned}$$

$$\begin{aligned}
u &= f(t) + v + W = -f(0) \operatorname{erf} \left(\frac{x}{2a\sqrt{t}} \right) + \int_0^t f'(\tau) d\tau \\
&\quad + f(0) - \int_0^t f'(\tau) \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \\
&\quad + \int_{\frac{-x}{2a\sqrt{t}}}^\infty \varphi(x + z2a\sqrt{t}) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{\frac{x}{2a\sqrt{t}}}^\infty \varphi \\
&\quad \times (z2a\sqrt{t} - x) \frac{1}{\sqrt{\pi}} e^{-z^2} dz
\end{aligned}$$

$$\begin{aligned}
&= f(0) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t}}\right) + \int_0^t f'(\tau) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau \\
&\quad + \int_{\frac{-x}{2a\sqrt{t}}}^{\infty} \varphi(x+z2a\sqrt{t}) \frac{1}{\sqrt{\pi}} e^{-z^2} dz \\
&\quad - \int_{\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(z2a\sqrt{t}-x) \frac{1}{\sqrt{\pi}} e^{-z^2} dz.
\end{aligned}$$

3. 在一维半无界空间求解 $u_t - a^2 u_{xx} = 0$, $u_x|_{x=0} = q(t)$, $u|_{t=0} = 0$.

解: 本题实际还存在自然边界条件 $u|_{x \rightarrow \infty}$ 应有限, 所以应先选择一个函数以消除非齐次边界条件. 函数 $-e^{-x}q(t)$ 是适合这个条件的, 为此

令 $u = -e^{-x}q(t) + v + w$, 而 v, w 满足

$$\begin{cases} v_t - a^2 v_{xx} = e^{-x}q'(t) - a^2 e^{-x}q(t), \\ v_x|_{x=0} = 0, v|_{t=0} = 0, \end{cases}$$

$$\begin{cases} W_t - a^2 W_{xx} = 0, \\ W_x|_{x=0} = 0, w|_{t=0} = e^{-x}q(0). \end{cases}$$

参照边界条件, 将 v, w 偶延拓.

$$v_t - a^2 v_{xx} = \begin{cases} e^{-x}[q'(t) - a^2 q(t)]x > 0, \\ e^x[q'(t) - a^2 q(t)]x < 0, \end{cases}$$

$$\begin{cases} W_t - a^2 W_{xx} = 0, \\ W|_{t=0} = \begin{cases} e^{-x}q(0), x > 0, \\ e^x q(0), x < 0, \end{cases} \end{cases}$$

$$\begin{aligned}
v &= \int_{\tau=0}^t \left[\int_{-\infty}^0 e^{\xi} \frac{q(\tau) - a^2 q(\tau)}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi \right. \\
&\quad \left. + \int_0^{\infty} e^{-\xi} \frac{q'(\tau) - a^2 q(\tau)}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi \right] d\tau,
\end{aligned}$$

在 v 的第一个积分中, 令 $z = \frac{[x - 2a^2(t - \tau)] - \xi}{2a\sqrt{t - \tau}}$,

第二个积分中令 $z = \frac{\xi - [x + 2a^2(t - \tau)]}{2a\sqrt{t - \tau}}$

$$\begin{aligned} \therefore v &= \int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] \left[-e^{\frac{4a^4(t-\tau)^2 - 4a^2 x(t-\tau)}{4a^2(t-\tau)}} \right. \\ &\quad \left. \operatorname{erfc}\left(\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}}\right) \right. \\ &\quad \left. + e^{\frac{4a^4(t-\tau)^2 + 4a^2 x(t-\tau)}{4a^2(t-\tau)}} \operatorname{erfc}\left(\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}}\right) \right] d\tau \\ &= \frac{1}{2} \int_0^t [a^2 q(\tau) - q'(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc}\left[\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}}\right] \right. \\ &\quad \left. - e^{-x} \operatorname{erfc}\left[\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}}\right] \right\} d\tau, \end{aligned}$$

$$\begin{aligned} W &= \int_{-\infty}^0 q(0) e^{-x} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \int_0^{\infty} q(0) \\ &\quad e^x \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \\ &= \frac{1}{2} q(0) e^{a^2 t + x} \operatorname{erfc}\left[\frac{2a^2 t + x}{2a\sqrt{t}}\right] - \frac{1}{2} q(0) e^{a^2 t - x} \\ &\quad \times \operatorname{erfc}\left[\frac{2a^2 t - x}{2a\sqrt{t}}\right] \\ &= \frac{1}{2} q(0) \left[e^x \operatorname{erfc}\left(\frac{2a^2 t + x}{2a\sqrt{t}}\right) - e^{-x} \right. \\ &\quad \left. \times \operatorname{erfc}\left(\frac{2a^2 t - x}{2a\sqrt{t}}\right) \right] e^{a^2 t}, \end{aligned}$$

$$u = -e^{-x} q(t) + v + W$$

$$\begin{aligned}
&= -e^{-x} q(t) + \frac{1}{2} q(0) \left[e^x \operatorname{erfc} \left(\frac{2a^2 t + x}{2a \sqrt{t}} \right) - e^{-x} \right. \\
&\quad \times \operatorname{erfc} \left(\frac{2a^2 t - x}{2a \sqrt{t}} \right) \Big] e^{a^2 t} \\
&\quad + \frac{1}{2} \int_0^t [a^2 q(\tau) - q'(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc} \left[\frac{2a^2(t-\tau) + x}{2a \sqrt{t-\tau}} \right] - e^{-x} \operatorname{erfc} \left[\frac{2a^2(t-\tau) - x}{2a \sqrt{t-\tau}} \right] \right\} d\tau.
\end{aligned}$$

4. 用拉普拉斯变换法求解例 8 的常微分方程 $T'' + \omega^2 a^2 T = \bar{f}(t)$,

$$T(0) = 0, T'(0) = 0,$$

解: 考虑初始条件, 对方程进行拉普拉斯变换

$p^2 \bar{T}(p) + \omega^2 a^2 \bar{T}(p) = \bar{\bar{f}}(p)$, 其中 $\bar{\bar{f}}(p)$ 是 $\bar{f}(t)$ 的拉氏变换象函数.

$$\bar{T}(p) = \bar{\bar{f}}(p) \frac{1}{p^2 + \omega^2 a^2} \cdot \bar{f}(p) \doteq f(t).$$

$$\frac{1}{p^2 + \omega^2 a^2} \doteq \frac{e^{i\omega a t} - e^{-i\omega a t}}{2ai\omega}$$

应用卷积定理

$$\bar{T} p \doteq T(t) = \frac{1}{2ai\omega} \int_0^t \bar{f}(\tau) \left[e^{i\omega a(t-\tau)} - e^{-i\omega a(t-\tau)} \right] d\tau.$$

5. 用拉普拉斯变换求解例 9 的常微分方程 $T' + \omega^2 a^2 T = \bar{f}(t)$, $T(0) = 0$.

解: 对方程进行拉氏变换得 $p \bar{T}(p) + \omega^2 a^2 \bar{T}(p) = \bar{\bar{f}}(p)$,

$\bar{\bar{f}}(p)$ 是 $\bar{f}(t)$ 的拉氏变换的像函数,

$$\bar{T}(p) = \bar{\bar{f}}(p) \frac{1}{p + \omega^2 a^2}, \quad \bar{f}(p) \doteq \bar{f}(t), \quad \frac{1}{p + \omega^2 a^2} \doteq e^{-\omega^2 a^2 t},$$

由卷积定理, 即得到

$$\bar{T}(p) = T(t) \int_0^t \bar{f}(\tau) e^{-\omega^2 a^2 (t-\tau)} d\tau.$$

6. 例10研究三维无界空间中的受迫振动, 从初始($t = 0$)状况推算以后($t > 0$)的状况, 试重新求解例10, 从初始($t = 0$)状况反推以前($t < 0$)的状况.

解: $u_{tt} - a^2 \Delta u = f(\vec{r}t)$, $u|_{t=0} = 0$, $u_t|_{t=0} = 0$, ($t < 0$).

将 u 展开为三重傅里叶积分

$$u = \iiint T e^{i\vec{k} \cdot \vec{r}} dk_1 dk_2 dk_3, \text{ 代入方程及初始条件分离出}$$

$$T'' + k^2 a^2 T = \left(\frac{1}{2\pi}\right)^3 \iiint f(\vec{r}'t) e^{-i\vec{k} \cdot \vec{r}'} dx' dy' dz',$$

$$T|_{t=0} = 0, T'|_{t=0} = 0.$$

$$\text{解得: } T = \frac{1}{2ai k} \int_0^t \left(\frac{1}{2\pi}\right)^3 \iiint f(\vec{r}'\tau) e^{-i\vec{k} \cdot \vec{r}'} dx' dy' dz' \\ \times \left[e^{ika(t-\tau)} - e^{-ika(t-\tau)} \right] d\tau,$$

$$u = \iiint T e^{i\vec{k} \cdot \vec{r}} dk_1 dk_2 dk_3 \\ = \iiint \int_{-\infty}^{\infty} \frac{1}{2ai k} \int_0^t \left(\frac{1}{2\pi}\right)^3 \iiint f(\vec{r}'\tau) \left[e^{ika(t-\tau)} - e^{-ika(t-\tau)} \right] \\ \times dx' dy' dz' \cdot d\tau e^{i\vec{k} \cdot \vec{r}} dk_1 dk_2 dk_3 \\ = \frac{1}{4\pi a} \iiint \int_0^t f(\vec{r}'\tau) \frac{1}{4\pi^2} \iiint \frac{1}{ik} \left[e^{ika(t-\tau)} - e^{-ika(t-\tau)} \right] \\ \times e^{i\vec{k} \cdot (\vec{r} - \vec{r}')} dk_1 dk_2 dk_3 d\tau dx' dy' dz',$$

$$\text{在上式中 } \frac{1}{4\pi^2} \iiint \frac{1}{ik} \left[e^{ika(t-\tau)} - e^{-ika(t-\tau)} \right] e^{i\vec{k} \cdot (\vec{r} - \vec{r}')} dk_1 dk_2 dk_3$$

$$dk_1 dk_2 dk_3 \\ = \frac{1}{R} \left\{ \delta[a(t-\tau) - R] - \delta[a(t-\tau) + R] \right\},$$

$$R = |\vec{r} - \vec{r}'|,$$

从初始情况推算以前情况, 所以 $t < 0$, 因而 $a(t-\tau) - R \neq 0$, 所以舍弃 $\delta[a(t-\tau) - R]$ 项, 而只保留 $\delta[a(t-\tau) + R]$ 项,

$$u = \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} f(\vec{r}', \tau) \frac{-1}{R} \delta[a(t-\tau) + R] d\tau dx' dy' dz' \\ = \frac{1}{4\pi a^2} \iiint_{-\infty}^{\infty} \frac{1}{R} \int_0^t f(\vec{r}', \tau) \delta[a(t-\tau) + R] d[a(t-\tau) + R] \\ dx' dy' dz'$$

$$= \frac{1}{4\pi a^2} \iiint_{-\infty}^{\infty} \frac{f(\vec{r}', t + \frac{R}{a})}{R} dx' dy' dz',$$

$$\because t + \frac{R}{a} < 0, R \equiv |\vec{r} - \vec{r}'| < -at = a|t|,$$

$|\vec{r} - \vec{r}'| < a|t|$ 表示 $(x'y'z')$ 在这样一个球内, 圆心的矢径为 $\vec{r}$, 半径为 $a|t|$, 记作 $T_{a|\vec{r}|}$.

$$\therefore u = \frac{1}{4\pi a^2} \iiint_{T_{a|\vec{r}|}} \frac{f(\vec{r}', t + \frac{R}{a})}{R} dx' dy' dz', \text{ 其中 } R \equiv |\vec{r} - \vec{r}'|.$$

第十二章 二阶常微分方程级数解法 本征值问题

§ 40. 特殊函数常微分方程

1. 试用平面极坐标系把二维波动方程分离变数.

解: $u_{tt} - a^2 \Delta_2 u = 0,$ (1)

先把时间变数 t 分离出来.

令 $u(\rho, \varphi; t) = U(\rho, \varphi) \cdot T(t)$, 代入方程 (1)

$$U(\rho, \varphi) \cdot T''(t) - a^2 \Delta_2 U(\rho, \varphi) \cdot T(t) = 0,$$

各项遍乘以 $\frac{1}{a^2 U T}$ 并移项, 得

$$\frac{T''}{a^2 T} = \frac{\Delta_2 U}{U}.$$

上式左边仅是 t 的函数; 右边是 ρ 和 φ 的函数. 若要等式成立, 两边应为同一常数, 记为 $-k^2$, 即有

$$T'' + a^2 k^2 T = 0, \quad (2)$$

$$\Delta_2 U + k^2 U = 0, \quad (3)$$

(3) 式为二维亥姆霍兹方程, 它在平面极坐标系下的表达式为:

$$U_{\varphi\varphi} + \frac{1}{\rho} U_{\rho} + \frac{1}{\rho^2} U_{\varphi\varphi} + k^2 U = 0.$$

进一步分离变数,

令 $U(\rho, \varphi) = R(\rho) \Phi(\varphi)$, 代入上式

$$R''(\rho) \Phi(\varphi) + \frac{1}{\rho} R'(\rho) \Phi(\varphi) + \frac{1}{\rho^2} R(\rho) \Phi''(\varphi) + k^2 R(\rho) \Phi(\varphi) = 0$$

$$+ k^2 R(\rho) \Phi(\varphi) = 0.$$

各项遍乘以 $\frac{\rho^2}{R\Phi}$, 并适当移项, 得

$$\frac{\rho^2 R''}{R} + \frac{\rho R'}{R} + k^2 \rho^2 = - \frac{\Phi''}{\Phi}.$$

同上讨论, 等式两边应为同一常数, 记为 m^2 ,

$$\text{即得: } \Phi'' + m^2 \Phi = 0, \quad (4)$$

$$\rho^2 R'' + \rho R' + (k^2 \rho^2 - m^2) R = 0. \quad (5)$$

对方程 (5) 作变数代换 $x = k\rho$ 后变为贝塞尔方程

$$x^2 R'' + x R' + (x^2 - m^2) R = 0. \quad (6)$$

由周期性条件, 方程 (4) 的解为:

$$\Phi_m = A_m \cos m\varphi + B_m \sin m\varphi,$$

由波动问题及解在 $\rho \rightarrow 0$ 有限的条件, 方程 (3) 的解为:

$$T_n = C_n \cos k_n a t + D_n \sin k_n a t.$$

2. 试用平面极坐标系把二维输运过程方程分离变数.

$$\text{解: } u_t - a^2 \Delta_2 u = 0, \quad (1)$$

在平面极坐标系下方程 (1) 为:

$$u_t - a^2 \left(u_{\rho\rho} + \frac{1}{\rho} u_\rho + \frac{1}{\rho^2} u_{\varphi\varphi} \right) = 0,$$

令 $u(\rho, \varphi; t) = R(\rho) \Phi(\varphi) T(t)$, 代入方程 (1)

$$T' R \Phi - a^2 \left(R u'' \Phi T + \frac{1}{\rho} R' \Phi T + \frac{1}{\rho^2} \Phi'' R T \right) = 0,$$

各项遍乘以 $\frac{1}{a^2 R \Phi T}$, 并适当移项, 得

$$\frac{T'}{a^2 T} = \frac{R''}{R} + \frac{1}{\rho} \cdot \frac{R'}{R} + \frac{1}{\rho^2} \cdot \frac{\Phi''}{\Phi}.$$

同上题讨论, 等式两边应为同一常数, 记为 $-K^2$. 得

$$T' + a^2 K^2 T = 0, \quad (2)$$

$$\frac{R''}{R} + \frac{1}{\rho} \cdot \frac{R'}{R} + \frac{1}{\rho^2} \frac{\Phi''}{\Phi} = -K^2. \quad (3)$$

方程 (3) 各项遍乘以 ρ^2 , 并适当移项, 得

$$\rho^2 \frac{R''}{R} + \rho \frac{R'}{R} + K^2 \rho^2 = -\frac{\Phi''}{\Phi} = m^2,$$

$$\text{即} \quad \Phi'' + m^2 \Phi = 0, \quad (4)$$

$$\rho^2 R'' + \rho R' + (k^2 \rho^2 - m^2) R = 0. \quad (5)$$

方程 (2) 和 (4) 的解为

$$T_n = A_n e^{-a^2 k_n^2 t},$$

$$\Phi_m = B_m \cos m\varphi + C_m \sin m\varphi.$$

方程 (5) 作变数代换 $x = k\rho$ 变成贝塞尔方程

$$x^2 R'' + x R' + (x^2 - m^2) R = 0.$$

3. 氢原子定态问题的量子力学薛定谔方程是 $-\frac{\hbar^2}{8\pi^2\mu} \Delta u$

$-\frac{Ze^2}{r} u = Eu$, 其中 $\hbar, \mu, Z, e, E$ 都是常数. 试用球坐标系把这个方程分离变数.

解: 先令 $A = \frac{\hbar^2}{8\pi\mu}$, $B = Ze^2$.

定态薛定谔方程在球坐标系下表达式是:

$$\begin{aligned} & A \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) \right. \\ & \left. + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \varphi^2} \right] + B \frac{u}{r} + Eu \\ & = 0, \end{aligned}$$

令 $u(r, \theta, \varphi) = R(r)Y(\theta, \varphi)$, 代入上式

$$\begin{aligned} \text{得到} & \frac{AY}{r^2} \cdot \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{AR}{r^2 \sin^2 \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{AR}{r^2 \sin^2 \theta} \frac{\partial^2 Y}{\partial \varphi^2} \\ & + \left(\frac{B}{r} + E \right) RY = 0, \end{aligned}$$

各项遍乘以 $-\frac{r^2}{ARY}$, 则有

$$\begin{aligned} & \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{r^2}{A} \left(\frac{B}{r} + E \right) \\ &= -\frac{1}{Y \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) - \frac{1}{Y \sin^2 \theta} \frac{\partial^2 Y}{\partial \varphi^2}, \end{aligned}$$

上式不可能成立, 除非两边等于同一个常数, 把这个数记作 $l(l+1)$, 则得:

$$\begin{aligned} & \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{B}{A} r + \frac{E}{A} r^2 - l(l+1) \right] R = 0, \\ \text{即 } & \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{8\pi^2 \mu}{h^2} \left(\frac{Ze^2}{r} + E \right) - \frac{l(l+1)}{r^2} \right] R = 0, \end{aligned} \quad (1)$$

至于 Y 则满足球函数方程

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \varphi^2} + l(l+1)Y = 0, \quad (2)$$

球函数方程 (2) 的进一步分离变数见课本第284—285页, 如令 $Y(\theta, \varphi) = \Theta(\theta)\Phi(\varphi)$, 则 Φ 满足

$$\Phi'' + m^2 \Phi = 0, \quad (3)$$

它的解是 $\Phi_m(\varphi) = A_m \cos m\varphi + B_m \sin m\varphi$.

Θ 则满足缔合勒让德方程

$$(1+x^2) \frac{d^2 \Theta}{d\varrho^2} - 2x \frac{d\Theta}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2} \right] \Theta = 0, \quad (4)$$

其中 $x = \cos \theta$.

4. 研究电磁波在矩形波导中的传播, 取波导管的管轴为 z 轴, 并设电磁波以谐波形式传播, 通常令:

$$\begin{aligned} E_z(x, y, z, t) &= e_z(x, y) e^{i(hz - kCt)}, \\ H_z(x, y, z, t) &= \mathcal{H}_z(x, y) e^{i(hz - kCt)}, \end{aligned}$$

其中 h 应为实数, (由 z 的上下底齐次边界条件确定 h 为实数)。

代入方程 $\Delta E_z + k^2 E_z = 0$ 和 $\Delta H_z + k^2 H_z = 0$ 得

$$\Delta_z \mathcal{E}_z + (k^2 - h^2) \mathcal{E}_z = 0 \text{ 和 } \Delta_z \mathcal{H}_z + (k^2 - h^2) \mathcal{H}_z = 0.$$

设波导的 x 和 y 的边宽度为 a 和 b . 试在边界条件:

$$(1) \quad \mathcal{E}_z|_{x=0} = \mathcal{E}_z|_{x=a} = \mathcal{E}_z|_{y=0} = \mathcal{E}_z|_{y=b} = 0$$

$$(2) \quad \left. \frac{\partial \mathcal{H}_z}{\partial x} \right|_{x=0} = \left. \frac{\partial \mathcal{H}_z}{\partial x} \right|_{x=a} = \left. \frac{\partial \mathcal{H}_z}{\partial y} \right|_{y=0} = \left. \frac{\partial \mathcal{H}_z}{\partial y} \right|_{y=b} = 0$$

下求解 $\mathcal{E}_z$ 和 $\mathcal{H}_z$. 在何种条件下得不到谐波形式的解?

解: (1) $\mathcal{E}_z$ 的定解问题为:

$$\Delta_z \mathcal{E}_z + (k^2 - h^2) \mathcal{E}_z = 0, \quad (1)$$

$$\begin{cases} \mathcal{E}_z|_{x=0} = \mathcal{E}_z|_{x=a} = 0, \\ \mathcal{E}_z|_{y=0} = \mathcal{E}_z|_{y=b} = 0, \end{cases} \quad (2)$$

令 $\mathcal{E}_z(x, y) = X(x)Y(y)$, 代入方程(1)和条件(2)进行分离变数, 得:

$$\frac{X''}{X} = -\frac{Y''}{Y} - (k^2 - h^2) = -\lambda,$$

$$\begin{cases} X|_{x=0}Y = X|_{x=a}Y = 0, \\ XY|_{y=0} = XY|_{y=b} = 0, \end{cases}$$

即

$$\begin{cases} X'' + \lambda X = 0, \\ X|_{x=0} = X|_{x=a} = 0, \end{cases} \quad (3)$$

和

$$\begin{cases} Y'' + (k^2 - h^2 - \lambda)Y = 0, \\ Y|_{y=0} = Y|_{y=b} = 0. \end{cases} \quad (4)$$

由定解问题(3)解得:

$$\lambda_m = \left(\frac{m\pi}{a} \right)^2,$$

$$X_m = A \sin \frac{m\pi x}{a}, \quad (m = 1, 2, \dots),$$

由定解问题 (4) 解得:

$$k^2 - h^2 - \left(\frac{m\pi}{a}\right)^2 = \left(\frac{n\pi}{b}\right)^2,$$

$$Y_n = B \sin \frac{n\pi y}{b}, \quad (n=1, 2, \dots),$$

$$\therefore \mathcal{G}_z = C \cdot \sin \frac{m\pi x}{a} \cdot \sin \frac{n\pi y}{b}. \quad (5)$$

(2) $\mathcal{H}_z$ 的定解问题为:

$$\Delta_z \mathcal{H}_z + (k^2 - h^2) \mathcal{H}_z = 0, \quad (6)$$

$$\begin{cases} \frac{\partial \mathcal{H}_z}{\partial x} \Big|_{x=0} = \frac{\partial \mathcal{H}_z}{\partial x} \Big|_{x=a} = 0, \\ \frac{\partial \mathcal{H}_z}{\partial y} \Big|_{y=0} = \frac{\partial \mathcal{H}_z}{\partial y} \Big|_{y=b} = 0, \end{cases} \quad (7)$$

令 $\mathcal{H}_z(x, y) = X(x)Y(y)$ 代入方程 (6) 和条件 (7) 得

$$\begin{cases} X'' + \lambda X = 0, \\ X' \Big|_{x=0} = X' \Big|_{x=a} = 0, \end{cases} \quad (8)$$

及

$$\begin{cases} Y'' + (k^2 - h^2 - \lambda) Y = 0, \\ Y' \Big|_{y=0} = Y' \Big|_{y=b} = 0. \end{cases} \quad (9)$$

由定解问题 (8) 解得:

$$\lambda_m = \left(\frac{m\pi}{a}\right)^2,$$

$$X_m = A \cos \frac{m\pi x}{a}, \quad (m=0, 1, \dots).$$

由定解问题 (9) 解得:

$$k^2 - h^2 - \left(\frac{m\pi}{a}\right)^2 = \left(\frac{n\pi}{b}\right)^2,$$

$$Y_n = B \cos \frac{n\pi y}{b}, \quad (n=0, 1, \dots),$$

$$\therefore \mathcal{H}_z = C \cdot \cos \frac{m\pi x}{a} \cdot \cos \frac{n\pi y}{b}, \quad (10)$$

故

$$\begin{cases} E_z = \mathcal{E}_z e^{i(hz - kCt)} = A \cdot \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{i(hz - kCt)}, \\ H_z = \mathcal{H}_z e^{i(hz - kCt)} = B \cdot \cos \frac{m\pi x}{a} \cdot \cos \frac{n\pi y}{b} e^{i(hz - kCt)}. \end{cases} \quad (11)$$

(3) 从解 (11) 知, 当 h 为虚数 $h = i\beta$ 时, 因子

$$e^{ihz} = e^{-\beta z},$$

这表示电磁波沿 z 轴正方向衰减而不能传播, 亦即没有谐波形式的解.

$$\because k^2 - \left[\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right] = h^2, \text{ 当 } h \text{ 为虚数时,}$$

$$\text{有} \quad k^2 < \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2.$$

设 $a > b$, 当 m, n 不同时为零时, 则凡 $k < \frac{\pi}{a}$ 的波以任何模式都通不过. 这时

$$k_c = \frac{\pi}{a} \text{ 称为截止波矢,}$$

$$\omega_c = ck_c = \frac{c\pi}{a} \text{ 称为截止圆频率,}$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{c}{2a} \text{ 称为截止频率,}$$

$$\lambda_c = \frac{c}{f} = 2a \text{ 称为截止波长.}$$

设 $a < b$, 当 m, n 不同时为零时, 则 $nk < \frac{\pi}{b}$ 的波以任何模式都通不过.

§41. 常点邻域上的级数解法

1. 在 $x_0 = 0$ 的邻域上求解 $y'' - xy = 0$.

$$\text{解: } y'' - xy = 0, \quad (1)$$

这里 $p(x) = 0$, $q(x) = -x$, $\therefore x_0 = 0$ 是方程 (1) 的常点.

$$\text{设 } y = \sum_{n=0}^{\infty} a_n x^n, \quad (2)$$

$$\text{则 } y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k, \quad (3)$$

$$xy = \sum_{n=0}^{\infty} a_n x^{n+1} = \sum_{k=1}^{\infty} a_{k-1} x^k, \quad (4)$$

把 (3) 和 (4) 式代入方程 (1), 合并用幂次项, 令各幂次的系数为零, 得系数递推公式:

$$a_{k+2} = \frac{1}{(k+1)(k+2)} a_{k-1}, \quad (5)$$

由 (5) 可推得

$$a_2 = a_{-1} = 0 \quad (\because a_{-1} = 0), \quad a_5 = 0, \quad \dots a_{3k+2} = 0,$$

$$a_3 = \frac{1}{2 \cdot 3} a_0 \quad (a_0 \neq 0, \text{ 待定}),$$

$$a_6 = \frac{1}{5 \cdot 6} a_3 = \frac{1 \cdot 4}{6!} a_0, \quad \dots a_{3k} = \frac{1 \cdot 4 \cdots (3k-2)}{(3k)!} a_0,$$

$$a_4 = \frac{1}{3 \cdot 4} a_1 \quad (a_1 \neq 0, \text{ 待定}),$$

$$a_7 = \frac{1}{6 \cdot 7} a_4 = \frac{2 \cdot 5}{7!} a_1, \quad \dots a_{3k+1} = \frac{2 \cdot 5 \cdots (3k-1)}{(3k+1)!} a_1.$$

故

$$y_0(x) = a_0 \left[1 + \frac{1}{2 \cdot 3} x^3 + \dots + \frac{1 \cdot 4 \cdots (3k-2)}{(3k)!} x^{3k} + \dots \right], \quad (6)$$

$$y_1(x) = a_1 \left[x + \frac{1}{3 \cdot 4} x^4 + \dots + \frac{2 \cdot 5 \cdots (3k-1)}{(3k+1)!} \times x^{3k+1} + \dots \right], \quad (7)$$

其级数解的收敛半径为:

$$\begin{aligned} R_0 &= \lim_{k \rightarrow \infty} \left| \frac{1 \cdot 4 \cdots (3k-2)}{(3k)!} \cdot \frac{[3(k+1)]!}{1 \cdot 4 \cdots [3(k+1)-2]} \right| \\ &= \lim_{k \rightarrow \infty} \left| \frac{1 \cdot 4 \cdots (3k-2)}{(3k)!} \cdot \frac{(3k+3)!}{1 \cdot 4 \cdots (3k+1)} \right| \\ &= \lim_{k \rightarrow \infty} |(3k+2)(3k+3)| = \infty. \end{aligned}$$

同样可得

$$R_1 = \lim_{k \rightarrow \infty} |(3k+3)(3k+4)| = \infty.$$

2. 在 $x_0 = 0$ 的邻域上求解厄密方程

$$y'' - 2xy' + (\lambda - 1)y = 0.$$

λ 取什么数值可使级数解退化为多项式? 这些多项式乘以适当常数使最高幂项成为 $(2x)^n$ 形式, 叫做厄密多项式, 记作 $H_n(x)$. 写出前几个 $H_n(x)$.

$$\text{解: } y'' - 2xy' + (\lambda - 1)y = 0, \quad (1)$$

这里 $p(x) = -2x$, $q(x) = \lambda - 1$. 知 $x_0 = 0$ 是方程 (1) 的常点.

$$\text{设 } y = \sum_{n=0}^{\infty} a_n x^n, \quad (2)$$

$$\text{则 } (\lambda - 1)y = \sum_{k=0}^{\infty} (\lambda - 1)a_k x^k, \quad (3)$$

$$-2xy' = \sum_{n=1}^{\infty} -2na_n x^n = \sum_{k=1}^{\infty} -2ka_k x^k, \quad (4)$$

$$y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} = \sum_{k=0}^{\infty} (k+1) \times (k+2)a_{k+2} x^k, \quad (5)$$

把 (3), (4), (5) 式代入方程 (1), 可得系数递推公式

$$a_{k+2} = \frac{2k+1-\lambda}{(k+1)(k+2)} a_k, \quad (6)$$

由公式 (6) 可推得

$$a_2 = \frac{1-\lambda}{1 \cdot 2} a_0 (a_0 \neq 0, \text{ 待定}),$$

$$a_3 = \frac{3-\lambda}{2 \cdot 3} a_1 (a_1 \neq 0, \text{ 待定}),$$

$$a_4 = \frac{5-\lambda}{3 \cdot 4} a_2 = \frac{(1-\lambda)(5-\lambda)}{4!} a_0,$$

$$a_5 = \frac{7-\lambda}{4 \cdot 5} a_3 = \frac{(3-\lambda)(7-\lambda)}{5!} a_1,$$

.....

$$a_{2k} = \frac{(1-\lambda)(5-\lambda) \cdots (4k-3-\lambda)}{(2k)!} a_0,$$

$$a_{2k+1} = \frac{(3-\lambda)(7-\lambda) \cdots (4k-1-\lambda)}{(2k+1)!} a_1,$$

$$\text{故 } y(x) = a_0 y_0(x) + a_1 y_1(x), \quad (7)$$

$$\begin{aligned} \text{其中 } y_0(x) = & 1 + \frac{(1-\lambda)}{2!} x^2 + \frac{(1-\lambda)(5-\lambda)}{4!} x^4 + \cdots \\ & + \frac{(1-\lambda)(5-\lambda) \cdots (4k-3-\lambda)}{(2k)!} x^{2k} + \cdots, \end{aligned} \quad (8)$$

$$y_1(x) = x + \frac{(3-\lambda)}{3!} x^3 + \frac{(3-\lambda)(7-\lambda)}{5!} x^5 + \cdots$$

$$+ \frac{(3-\lambda)(7-\lambda)\cdots(4k-1-\lambda)}{(2k+1)!} x^{2k+1} + \dots, \quad (9)$$

收敛半径均是无限大。

从级数解 (8) 和 (9) 知:

当 $\lambda = 4k - 3$ ($k = 1, 2, \dots$) 时, $y_0(x)$ 退化成多项式,

当 $\lambda = 4k - 1$ ($k = 1, 2, \dots$) 时, $y_1(x)$ 退化成多项式。

使多项式最高幂项为 $(2x)^n$ 形式, 称为厄密多项式, 记为 $H_n(x)$ 。其前几个 $H_n(x)$ 是:

取 $k = 1$, 有

$$\lambda = 4k - 3 = 1, \quad y_0(x) = 1, \quad \text{记为 } H_0(x) = 1,$$

$$\lambda = 4k - 1 = 3, \quad y_1(x) = x, \quad \text{记为 } H_1(x) = 2y_1(x) = 2x,$$

取 $k = 2$, 有

$$\lambda = 4k - 3 = 5, \quad y_0(x) = 1 - 2x^2,$$

$$\text{记为 } H_2(x) = -2y_0(x) = (2x)^2 - 2.$$

$$\lambda = 4k - 1 = 7, \quad y_1(x) = x - \frac{2}{3}x^3,$$

$$\text{记为 } H_3(x) = -(2^2) \cdot 3y_1(x) = (2x)^3 - 12x.$$

3. 在 $x_0 = 0$ 的邻域上求解 $(1-x^2)y'' - 6xy' + 6y = 0$ 即

$$(1-x^2)y'' - 2(2+1)xy' + [3(3+1) - 2(2+1)]y = 0,$$

在例 2 的 l 阶勒让德方程 $(1-x^2)R'' - 2xy' + l(l+1)R = 0$ 的级数解

$$\begin{aligned} R_0(x) &= 1 + \frac{(-l)(l+1)}{2!}x^2 + \dots \\ &+ \frac{(2k-2-l)(2k-4-l)\cdots(-l)(l+1)(l+3)\cdots(l+2k-1)}{(2k)!} \\ &\times x^{2k} + \dots, \end{aligned}$$

$$R_1(x) = x + \frac{(1-l)(l+2)}{3!}x^3 + \dots$$

$$+ \frac{(2k-1-l)(2k-3-l)\cdots(1-l)(l+2)(l+4)\cdots(l+2k)}{(2k+1)!} \\ \times x^{2k+1} + \cdots$$

之中以 $l = 3$ 代入, 并求它的二阶导数, 然后与本题的答案比较一下.

解: $\because x_0 = 0$ 是方程的常点,

$$\therefore \text{令 } y = \sum_{n=0}^{\infty} a_n x^n.$$

$$\text{则: } 6y = \sum_{k=0}^{\infty} 6a_k x^k, \quad (1)$$

$$-6xy' = \sum_{n=1}^{\infty} -6na_n x^n = \sum_{k=1}^{\infty} -6ka_k x^k, \quad (2)$$

$$y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} \\ = \sum_{k=0}^{\infty} (k+1)(k+2)a_{k+2} x^k, \quad (3)$$

$$-x^2 y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^n = \sum_{k=2}^{\infty} k(k-1)a_k x^k. \quad (4)$$

把 (1) — (4) 式代入方程得系数递推公式

$$(k+1)(k+2)a_{k+2} - k(k-1)a_k - 6ka_k + 6a_k = 0,$$

$$\text{即 } a_{k+2} = \frac{(k-1)(k+6)}{(k+1)(k+2)} a_k = \frac{[(k+2)-3][(k+2)+4]}{(k+2)(k+1)} a_k.$$

由系数递推公式知:

$$a_2 = -\frac{6}{1 \cdot 2} a_0, \quad (a_0 \neq 0),$$

$$a_3 = 0 \cdot a_1 = 0, \quad (\text{设 } a_1 \neq 0),$$

$$a_4 = \frac{8}{3 \cdot 4} a_2 = \frac{6 \cdot 8}{4!} a_0,$$

$$a_5 = \frac{2 \cdot 9}{4 \cdot 5} a_3 = 0,$$

.....

$$a_{2n} = \frac{(2n+4)(2n+2)\cdots 8\cdot 6\cdot 1\cdot (-1)\cdots (2n-5)(2n-3)}{(2n)!} a_0,$$

$$a_{2n+1} = 0,$$

$$\therefore y(x) = a_0 y_0(x) + a_1 y_1(x), \quad (5)$$

$$\begin{aligned} \text{其中 } y_0(x) = & 1 + \frac{(-1)\cdot 6}{2!} x^2 + \frac{(-1)\cdot 1\cdot 6\cdot 8}{4!} x^4 + \cdots \\ & + \frac{(2n-3)\cdots (-1)\cdot 1\cdots (2n+4)}{(2n)!} x^{2n} + \cdots, \end{aligned} \quad (6)$$

$$y_1(x) = x, \quad (7)$$

又, $l = 3$ 阶勒让德方程的解为:

$$\begin{aligned} R_0(x) = & 1 + \frac{(-3)\cdot 4}{2!} x^2 + \cdots \\ & + \frac{(2n-5)(2n-7)\cdots (-3)(4)(6)\cdots (2n+2)}{(2n)!} \\ & \times x^{2n} + \cdots, \end{aligned}$$

$$R_1(x) = x + \frac{(-2)\cdot 5}{3!} x^3.$$

对 $R_0(x)$ 和 $R_1(x)$ 求二阶导数

$$\begin{aligned} \frac{d^2 R_0}{dx^2} = & (-3)\cdot (4) + \frac{(-1)(-3)\cdot 4\cdot 6\cdot 3}{3!} x^2 \\ & + \frac{(2n-5)\cdots (-3)\cdot 4\cdot 6\cdots (2n+2)(2n-1)}{(2n-1)!} x^{2n-2} + \cdots \\ = & (-3)\cdot 4 \left[1 + \frac{(-1)\cdot 6}{2!} x^2 + \cdots \right. \\ & \left. + \frac{(2n-3)\cdots (-1)\cdot 1\cdot 6\cdot 8\cdots (2n+4)}{(2n)!} x^{2n} + \cdots \right] \\ = & (-3)\cdot 4\cdot y_0(x), \end{aligned}$$

$$\frac{d^2 R_1}{dx^2} = (-2) \cdot 5 \cdot y_1(x).$$

因此,可以说本题的解正是3阶勒让德方程解的二阶导数.

4. 在 $x_0 = 0$ 的邻域上求解雅可俾方程

$$(1-x^2)y'' + [\beta - \alpha - (\alpha + \beta + 2)x]y' + \lambda(\alpha + \beta + \lambda + 1)y = 0.$$

解: $x_0 = 0$ 是方程的常点.

$$\text{设 } y = \sum_{n=0}^{\infty} a_n x^n, \quad (1)$$

$$\text{则 } \lambda(\alpha + \beta + \lambda + 1)y = \sum_{k=0}^{\infty} \lambda(\alpha + \beta + \lambda + 1)a_k x^k, \quad (2)$$

$$\begin{aligned} (\beta - \alpha)y' &= \sum_{n=1}^{\infty} (\beta - \alpha) n a_n x^{n-1} \\ &= \sum_{k=0}^{\infty} (\beta - \alpha) \cdot (k+1) a_{k+1} x^k, \end{aligned} \quad (3)$$

$$\begin{aligned} -(\alpha + \beta + 2)x \cdot y' &= \sum_{n=1}^{\infty} -(\alpha + \beta + 2) \cdot a_n n x^n \\ &= \sum_{k=1}^{\infty} -(\alpha + \beta + 2) k a_k x^k, \end{aligned} \quad (4)$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{k=0}^{\infty} (k+1)(k+2) a_{k+2} x^k, \quad (5)$$

$$-x^2 y'' = \sum_{n=2}^{\infty} -n(n-1) a_n x^n = \sum_{k=2}^{\infty} -(k-1)k a_k x^k, \quad (6)$$

把 (2) — (6) 代入雅可俾方程可得系数递推公式:

$$\begin{aligned} (k+1)(k+2)a_{k+2} + (\beta - \alpha)(k+1)a_{k+1} + [\lambda(\alpha + \beta + \lambda + 1) \\ - (\alpha + \beta + 2)k - k(k-1)]a_k = 0, \end{aligned}$$

$$a_{k+2} = \frac{\alpha - \beta}{k+2} a_{k+1}$$

$$-\frac{\lambda(\alpha+\beta+\lambda+1)+(\alpha+\beta+2)k-(k-1)k}{(k+1)(k+2)}a_k,$$

$$\text{即 } a_{k+2} = \frac{\alpha-\beta}{k+2}a_{k+1} + \frac{(k-\lambda)(k+\alpha+\beta+\lambda+1)}{(k+1)(k+2)}a_k.$$

可以写出前面几个系数, 但难以写出一般的系数公式.

§ 42. 正则奇点邻域上的级数解法

1. 在 $x_0 = 0$ 的邻域上求解 $x^2 y'' + 2xy' - l(l+1)y = 0$.

解: $x_0 = 0$ 是 $p(x) = \frac{2}{x}$ 的一阶极点, 是 $q(x) = \frac{l(l+1)}{x^2}$

的二阶极点, 所以是方程的正则奇点.

设 $y = \sum_{k=s}^{\infty} a_k x^k$, 代入方程, 各同幂次项分别集合如下

表.

	x^s	x^{s+1}	...	x^{s+k}	...
$x^2 y'' =$	$s(s-1)a_s$	$(s+1)s a_{s+1}$	...	$(s+k)(s+k-1)a_{s+k}$	...
$2xy' =$	$2sa_s$	$2(s+1)a_{s+1}$	...	$2(s+k)a_{s+k}$	...
$-l(l+1)y =$	$-l(l+1)a_s$	$-l(l+1)a_{s+1}$	...	$-l(l+1)a_{s+k}$	...

令最低幂项系数为零, 得判定方程:

$$[s(s-1) + 2s - l(l+1)]a_s = 0.$$

$$\because a_s \neq 0,$$

$$\therefore s^2 + s - l(l+1) = 0.$$

$$\text{解得 } s_1 = l, \quad s_2 = -(l+1). \quad (1)$$

令 x^{s+k} 项系数为零, 得系数递推公式:

$$[(s+k)(s+k-1) + 2(s+k) - l(l+1)]a_{s+k} = 0. \quad (2)$$

当 $s_1 = l$ 时, 由 (2) 式

$$[(l+k)(l+k-1) + 2(l+k) - l(l+1)]a_{s_1+k} \\ = k(k+2l+1)a_{s_1+k} = 0.$$

若 $k \neq 0$, l 取整数, 则有 $a_{s_1+k} = 0$,

若 $k = 0$, 则 $a_{s_1} \neq 0$, 于是得一特解

$$y_0(x) = a_0 x^l. \quad (3)$$

当 $s_2 = -(l+1)$ 时, 由 (1) 式得

$$[(-l-1+k) + (-l-1+k-1) + 2(-l-1+k) \\ - l(l+1)]a_{s_2+k} \\ = k(k-2l-1)a_{s_2+k} = 0,$$

同上讨论, 有 $a_{s_2+k} = 0$, $a_{s_2} \neq 0$, 得另一特解

$$y_1(x) = a_1 x^{-(l+1)}, \quad (4)$$

$$\therefore y(x) = a_0 x^l + a_1 x^{-(l+1)}. \quad (5)$$

2. 在 $x_0 = 0$ 的邻域上求拉盖尔方程 $xy'' + (1-x)y' + \lambda y = 0$ 的有限解. λ 取什么数值可使级数退化为多项式? 这些多项式乘以适当常数使最高幂项成为 $(-x)^n$ 形式就叫作拉盖尔多项式, 记作 $L_n(x)$. 写出前几个 $L_n(x)$.

解: $x_0 = 0$ 是 $p(x) = \frac{1-x}{x}$ 和 $q(x) = \frac{\lambda}{x}$ 的一阶极点, 所以是方程的正则奇点.

$$\text{设 } y = \sum_{k=s}^{\infty} a_k x^k,$$

代入方程, 把各同幂次项分别集合如下表.

	$x^s - 1$	x^s	...	x^{s+k}	...
$xy'' =$	$s(s-1)a_s$	$(s+1)sa_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
$y' =$	sa_s	$(s+1)a_{s+1}$	...	$(s+k+1)a_{s+k+1}$	...
$-xy' =$		$-sa_s$	...	$-(s+k)a_{s+k}$	...
$\lambda y =$		λa_s	...	λa_{s+k}	...

令最低次幂系数为零, 得判定方程

$$(s^2 - s + s)a_s = 0,$$

$$\because a_s \neq 0,$$

$$\therefore s_{1,2} = 0. \quad (1)$$

令 x^{s+k} 项系数为零, 得系数递推公式

$$a_{s+k+1} = \frac{s+k-\lambda}{(s+k+1)^2} a_{s+k}, \quad (2)$$

当 $s = 0$ 时, 由 (2) 式推算出:

$$a_1 = -\lambda a_0,$$

$$a_2 = \frac{1-\lambda}{2^2} a_1 = \frac{(-\lambda)(1-\lambda)}{1^2 \cdot 2^2} a_0 = \frac{(-\lambda)(1-\lambda)}{(2!)^2} a_0,$$

.....

$$a_n = \frac{(-\lambda)(1-\lambda) \cdots (k-1-\lambda)}{(k!)^2} a_0, \dots$$

$$\therefore y(x) = a_0 \left[1 + \frac{-\lambda}{(1!)^2} x + \frac{(-\lambda)(1-\lambda)}{(2!)^2} x^2 + \dots + \frac{(-\lambda)(1-\lambda) \cdots (k-1-\lambda)}{(k!)^2} x^k + \dots \right], \quad (3)$$

如 $\lambda = 0, 1, 2 \dots$ 时, 级数解 (2) 退化为多项式.

使多项式最高幂项为 $(-x)^n$ 形式, 称为拉盖尔多项式, 记为 $L_n(x)$. 其前几个 $L_n(x)$ 是:

$$\lambda = 0, \quad y_0(x) = 1,$$

$$\text{记为 } L_0(x) = 1,$$

$$\lambda = 1, \quad y_1(x) = 1 - x,$$

$$\text{记为 } L_1(x) = -x + 1,$$

$$\lambda = 2, \quad y_2(x) = 1 - 2x + \frac{1}{2}x^2,$$

$$\text{记为 } L_2(x) = 2y_2 = (-x)^2 - 4x + 2,$$

$$\lambda = 3, \quad y_3(x) = 1 - 3x + \frac{3}{2}x^2 - \frac{1}{6}x^3,$$

$$\text{记为 } L_3(x) = 6y_3 = (-x)^3 + 9x^2 - 18x + 6.$$

方程的另一特解在 $x = 0$ 处为无限大, 舍去不讨论.

3. 在 $x_0 = 0$ 的邻域上求

$$y'' - 2\lambda y' + \left[\frac{2z}{x} - \frac{l(l+1)}{x^2} \right] y = 0 \text{ 的有限解. } \lambda \text{ 取什}$$

么数值可使级数退化为多项式?

解: 因 $x_0 = 0$ 是 $p(x) = -2\lambda$ 的常点, 是 $q(x) = \frac{2z}{x} -$

$\frac{l(l+1)}{x^2}$ 的二阶极点, 所以是方程的正则奇点.

$$\text{设 } y(x) = \sum_{k=s}^{\infty} a_k x^k,$$

代入方程, 则有:

	x^s	x^{s+1}	...	x^{s+k}	...
$x^2 y''$	$s(s-1)a_s$	$(s+1)sa_{s+1}$	...	$(s+k)(s+k-1)a_{s+k}$	...
$-2\lambda xy'$		$-2\lambda sa_{s+1}$	...	$-2\lambda(s+k-1)a_{s+k-1}$	...
$2zxy$		$2za_s$	...	$2za_{s+k-1}$	...
$-l(l+1)y$	$-l(l+1)a_s$	$-l(l+1)a_{s+1}$	...	$-l(l+1)a_{s+k}$	...

令最低幂项系数为零, 得判定方程

$$[s(s-1) - l(l+1)]a_s = 0,$$

$$\because a_s \neq 0,$$

$$\therefore s^2 - s - l(l+1) = 0,$$

$$\text{得 } s_1 = l+1, \quad s_2 = -l, \quad (1)$$

令 x^{s+k} 项系数为零, 得系数递推公式

$$[(s+k)(s+k-1) - l(l+1)]a_{s+k}$$

$$+ [2z - 2\lambda(s+k-1)]a_{s+k-1} = 0,$$

$$\text{即} \quad a_{s+k} = \frac{2\lambda(s+k-1) - 2z}{(s+k)(s+k-1) - l(l+1)} a_{s+k-1}. \quad (2)$$

当 $s_1 = l+1$ 时, 由 (2) 式得

$$\begin{aligned} a_{l+1+k} &= \frac{2\lambda(l+k) - 2z}{(l+1+k)(l+k) - l(l+1)} a_{l+k} \\ &= \frac{2[\lambda(l+k) - z]}{k(k+2l+1)} a_{l+k} \\ &= \frac{2^2[\lambda(l+k) - z][\lambda(l+k-1) - z]}{k(k-1)(k+2l+1)(k+2l)} a_{l+k-1} \\ &= \dots \\ &= \frac{2^k[\lambda(l+k) - z][\lambda(l+k-1) - z] \dots [\lambda(l+1) - z]}{k!(k+2l+1)(k+2l) \dots (2+2l)} a_{l+1}, \end{aligned}$$

$$\begin{aligned} \therefore y(x) &= a_0 x^{l+1} \left[1 + \frac{(l+1-z/\lambda)}{1!(2l+2)} (2\lambda x) \right. \\ &\quad + \frac{(l+1-z/\lambda)(l+2-z/\lambda)}{2!(2l+2)(2l+3)} (2\lambda x)^2 \\ &\quad \left. + \dots \right]. \quad (3) \end{aligned}$$

如果 $\frac{z}{\lambda} = \text{整数 } n$, 则级数退化成 $n-l-1$ 次多项式.

当 $s_2 = -l$ 时, 由上面讨论知这级数解中含有负幂项 x^{-l} , (因 $l > 0$), 在 $x_0 = 0$ 处发散, 不为有限解, 故应舍去, 不加讨论.

4. 在 $x_0 = 0$ 的邻域上求解 m 阶虚宗量贝塞耳方程 $x^2 y'' + xy' - (x^2 + m^2)y = 0$, 暂且认为 m 非整数, 象例 2 那样选取 $a \pm m = 1/2 \pm m\Gamma(\pm m + 1)$, 所得的两个解分别叫作 m 阶和 $-m$ 阶虚宗量贝塞耳函数, 分别记作 $I_{\pm m}(x)$.

验证 $I_{\pm m}(x)$ 没有实的零点.

比较 $J_{\pm m}(ix)$ 和 $I_{\pm m}(x)$.

解: $x_0 = 0$ 是 $p(x) = \frac{1}{x}$ 的一阶极点, 是 $q(x) = -\frac{(x^2 + m^2)}{x^2}$ 的二阶极点, 故为方程的正则奇点.

设
$$y(x) = \sum_{k=s}^{\infty} a_k x^k,$$

代入方程, 则有:

	x^s	x^{s+1}	x^{s+2}	...	x^{s+k}	...
$x^2 y'' =$	$s(s-1)a_s$	$(s+1)s a_{s+1}$	$(s+2)(s+1)a_{s+2}$	...	$(s+k)(s+k-1)a_{s+k}$	...
$xy' =$	$s a_s$	$(s+1)a_{s+1}$	$(s+2)a_{s+2}$	...	$(s+k)a_{s+k}$	...
$-x^2 y =$			$-a_s$	...	$-a_{s+k-2}$	...
$-m^2 y =$	$-m^2 a_s$	$-m^2 a_{s+1}$	$-m^2 a_{s+2}$	...	$-m^2 a_{s+k}$	...

令各同次幂项系数为零, 得一系列方程

$$[s(s-1) + s - m^2]a_s = 0, \quad (1)$$

$$[(s+1)s + (s+1) - m^2]a_{s+1} = 0, \quad (2)$$

.....

$$[(s+k)(s+k-1) + (s+k) - m^2]a_{s+k} - a_{s+k-2} = 0, \quad (3)$$

.....

$$\text{由(1)式, } \because a_s \neq 0, \therefore s_{1,2} = \pm m. \quad (4)$$

$$\text{由(2)式} [(s+1)^2 - m^2]a_{s+1} = 0,$$

$$\text{当 } s_{1,2} = \pm m \text{ 时, } [(\pm m+1)^2 - m^2]a_{s+1} = 0, \therefore a_{s+1} = 0. \quad (5)$$

由(3)式得系数递推公式:

$$a_{s+k} = \frac{1}{(s+k+m)(s+k-m)} a_{s+k-2}, \quad (6)$$

取 $s_1 = m$, 由 (6) 式有:

$$a_{m+2} = \frac{1}{(2m+2) \cdot 2} a_m = \frac{1}{2^2 \cdot (m+1) \cdot 1} a_m,$$

$$a_{m+3} = \frac{1}{(2m+3) \cdot 3} a_{m+1} = 0, \quad (\because a_{m+1} = a_{s+1} = 0),$$

$$\begin{aligned} a_{m+4} &= \frac{1}{(2m+4) \cdot 4} a_{m+2} \\ &= \frac{1}{2^4 \cdot (m+1)(m+2)1 \cdot 2} a_m, \end{aligned}$$

类推有:

$$\begin{aligned} a_{m+2k} &= \frac{1}{2^{2k} \cdot k! (m+1)(m+2) \cdots (m+k)} a_m, \\ a_{m+2k+1} &= 0. \end{aligned}$$

于是得方程一个特解:

$$\begin{aligned} y_1(x) = a_m x^m &\left[1 + \frac{1}{1! (m+1)} \left(\frac{x}{2}\right)^2 + \cdots \right. \\ &\left. + \frac{1}{k! (m+1) \cdots (m+k)} \left(\frac{x}{2}\right)^{2k} + \cdots \right], \end{aligned}$$

选取 $a_m = 1/2^m \Gamma(m+1)$,

$$\text{则 } I_m(x) \equiv y_1(x) = \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(m+k+1)} \left(\frac{x}{2}\right)^{m+2k} \quad (7)$$

取 $s_2 = -m$ 依照上面同样讨论, 并选取 $a_{-m} = 1/2^{-m} \Gamma(-m+1)$ 可得方程另一特解:

$$I_{-m}(x) \equiv y_2(x) = \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(-m+k+1)} \left(\frac{x}{2}\right)^{-m+2k}, \quad (8)$$

因 (7) 和 (8) 式是对 k 求和, 所以 $\left(\frac{x}{2}\right)^{\pm m}$ 可提出放在求和号外

面, 这样求和号内只出现 x 的偶次幂, 且系数全为正值 (当 $k+1 \leq m$ 时, $\Gamma(-m+k+1) = \infty$, 这部分系数为零), 因而 $I_{\pm m}(x)$ 没有实的零点.

$$\text{又} \because J_{\pm m}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\pm m + k + 1)} \left(\frac{x}{2}\right)^{\pm m + 2k},$$

$$\begin{aligned} \therefore J_{\pm m}(ix) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\pm m + k + 1)} \left(\frac{ix}{2}\right)^{\pm m + 2k} \\ &= (i)^{\pm m} \sum_{k=0}^{\infty} \frac{(-1)^k (i)^{2k}}{k! \Gamma(\pm m + k + 1)} \\ &\quad \times \left(\frac{x}{2}\right)^{\pm m + 2k} \\ &= (i)^{\pm m} I_{\pm m}(x). \end{aligned}$$

5. 在 $x_0 = 1$ 的邻域上, 求勒让德方程 $(1-x^2)y'' - 2xy' + l(l+1)y = 0$ 的有限解.

$$\text{解: 令 } t = \frac{1-x}{2}, \quad (1)$$

$$\text{则 } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{-1}{2} \frac{dy}{dx};$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx} = \frac{1}{4} \frac{d^2y}{dt^2}.$$

原方程化为:

$$(t-t^2)y'' + (1-2t)y' + l(l+1)y = 0, \quad (\text{且 } t_0 = 0), \quad (2)$$

$\because t_0 = 0$ 是 $p(t) = (1-2t)/t(1-t)$ 和 $q(t) = l(l+1)/t(1-t)$ 的一阶极点,

$\therefore t_0 = 0$ 是方程的正则奇点.

设 $y(t) = \sum_{k=0}^{\infty} a_k t^k$ 代入方程 (2), 则有:

	t^{s-1}	t^s	...	t^{s+k}	...
$-t^2 y'' =$		$-s(s-1)a_s$	...	$-(s+k)(s+k-1)a_{s+k}$	...
$ty'' =$	$s(s-1)a_s$	$(s+1)sa_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
$y' =$	sa_s	$(s+1)a_{s+1}$	...	$(s+k+1)a_{s+k+1}$	...
$-2ty' =$		$-2sa_s$	...	$-2(s+k)a_{s+k}$	...
$l(l+1)y =$		$l(l+1)a_s$	...	$l(l+1)a_{s+k}$	...

令上表各个幂次合并后的系数为零, 得

$$[s(s-1) + s]a_s = 0,$$

.....

$$\begin{aligned} & [(s+k+1)(s+k) + (s+k+1)]a_{s+k+1} \\ & = [(s+k)(s+k-1) + 2(s+k) - l(l+1)]a_{s+k}, \end{aligned}$$

.....

$$\because a_s \neq 0, \quad \therefore s_{1,2} = 0, \quad (3)$$

$$\text{又} \quad a_{s+k+1} = \frac{(s+k)(s+k+1) - l(l+1)}{(s+k+1)^2} a_{s+k}, \quad (4)$$

取 $s = 0$, $a_{k+1} = \frac{(k-l)(l+k+1)}{(k+1)^2} a_k$, 其前几个系为:

$$a_1 = \frac{-l(l+1)}{1^2} a_0 = (-1)^1 \frac{(l+1)l \cdot \Gamma(l)}{\Gamma(l)} \cdot \frac{1}{1^2} a_0$$

$$= (-1)^1 \frac{\Gamma(l+2)}{\Gamma(l)} \frac{1}{1^2} a_0,$$

$$a_2 = \frac{(1-l)l+2}{2^2} a_1$$

$$= (-1)^2 \frac{(l-1)(l+2)}{2^2} \cdot (-1)^1 \frac{\Gamma(l+2)}{1^2 \cdot \Gamma(l)} a_0$$

$$= (-1)^2 \frac{(l-1) \cdot (l+2) \Gamma(l+2)}{(2!)^2 (l-1) \Gamma(l-1)} a_0$$

$$= (-1) \frac{\Gamma(l+3)}{\Gamma(l-1)} \cdot \frac{a_0}{(2!)^2},$$

类推有:

$$a_k = (-1)^k \frac{\Gamma(l+k+1)}{\Gamma(l-k+1)} \frac{1}{(k!)^2} a_0,$$

$$\begin{aligned} \therefore y(x) = a_0 & \left[1 + (-1)^1 \frac{\Gamma(l+2)}{\Gamma(l)} \frac{1}{(1!)^2} \left(\frac{1-x}{2}\right) \right. \\ & + (-1)^2 \frac{\Gamma(l+3)}{\Gamma(l-1)} \cdot \frac{1}{(2!)^2} \left(\frac{1-x}{2}\right)^2 + \dots \\ & + (-1)^k \frac{\Gamma(l+k+1)}{\Gamma(l-k+1)} \frac{1}{(k!)^2} \\ & \left. \left(\frac{1-x}{2}\right)^k + \dots \right], \quad \left(\text{因 } t = \frac{1-x}{2} \right), \end{aligned} \quad (5)$$

另一特解在 $x_0 = 1$ (即 $t_0 = 0$) 为无限大, 舍去不讨论.

6. 在 $x_0 = 0$ 的邻域上求解 $xy'' - xy' + y = 0$.

解: 这里 $x_0 = 0$ 是方程正则奇点.

$$\text{令 } y(x) = \sum_{k=s}^{\infty} a_k x^k,$$

代入方程, 把各幂次项集合如下表.

	x^{s-1}	x^s	...	x^{s+k}	...
$xy'' =$	$s(s-1)a_s$	$(s+1)sa_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
$-xy' =$		$-sa_s$	...	$-(s+k)a_{s+k}$	...
$y =$		a_s	...	a_{s+k}	...

由最低幂项系数为零, 得

$$s(s-1)a_s = 0,$$

$$\because a_s \neq 0, \quad \therefore s_1 = 1, \quad s_2 = 0. \quad (1)$$

取 $s = s_1 = 1$, 这时系数递推公式是:

$$a_{k+2} = \frac{k}{(k+1)(k+2)} a_{k+1}, \quad (2)$$

当 $k = 0$ 时, $a_2 = 0 \cdot a_1$, $\because a_1 \neq 0, \therefore a_2 = 0$,

推知 $a_k = 0, (k \neq 1)$,

从而得到方程的一个特解:

$$y_1(x) = a_1 x. \quad (3)$$

由于 $s_1 - s_2 = 1 - 0 = 1$ 为整数, 所以对于判定方程较小的根 s_2 对应的另一个特解的形式是:

$$y_2(x) = Ax \ln x + \sum_{k=0}^{\infty} b_k x^k,$$

且
$$y_2'(x) = A + A \ln x + \sum_{k=1}^{\infty} k b_k x^{k-1},$$

$$y_2''(x) = \frac{A}{x} + \sum_{k=2}^{\infty} k(k-1) b_k x^{k-2},$$

代入原方程, 集合如下表.

	x^0	$x \ln x$	x	...	x^k	...
$xy_2'' =$	A		$2b_2$	...	$(k+1)kb_{k+1}$	...
$-xy_2' =$		$-A$	$-A - b_1$	...	$-kb_k$	...
$y_2 =$	b_0	A	b_1	...	b_k	...

令上表各幂次项系数为零, 有

$$b_0 = -A,$$

$$-A + A = 0, \text{ 知 } A \text{ 为任意,}$$

$$2b_2 - A - b_1 + b_1 = 0, \text{ 知 } b_2 = \frac{A}{2}, b_1 \text{ 为任意,}$$

.....

$$b_{k+1} = \frac{k-1}{k(k+1)} b_k, (k \geq 2),$$

推之有: $b_3 = \frac{1}{2 \cdot 3} b_2 = \frac{1}{2 \cdot 3} \cdot \frac{1}{2} A = \frac{1}{2!} \cdot \frac{1}{3!} A,$

$$b_4 = \frac{2}{3 \cdot 4} b_3 = \frac{2!}{3!4!} A,$$

...

$$b_k = \frac{(k-2)!}{(k-1)!k!} A,$$

$$\begin{aligned} \therefore y_2(x) &= Ax \ln x + \left[-A + b_1 x \right. \\ &\quad \left. + \sum_{k=2}^{\infty} \frac{(k-2)!}{(k-1)!k!} Ax^k \right] \\ &= A \left[x \ln x - 1 + x + \sum_{k=2}^{\infty} \frac{(k-2)!}{(k-1)!k!} x^k \right], \quad (4) \end{aligned}$$

故

$$\begin{aligned} y(x) &= y_1(x) + y_2(x) \\ &= a_0 x + a_1 \left[x \ln x - 1 + \sum_{k=2}^{\infty} \frac{(k-2)!}{(k-1)!k!} x^k \right]. \end{aligned}$$

7. 在 $x_0 = 0$ 的邻域上, 求解 $xy'' + y = 0$.

解: 易知 $x_0 = 0$ 是方程的正则奇点.

设 $y = \sum_{k=s}^{\infty} a_k x^k,$

代入方程, 则有:

	x^{s-1}	x^s	...	x^{s+k}	...
$xy'' =$	$s(s-1)a_s$	$(s+1)sa_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
$y =$		a_s	...	a_{s+k}	...

这里判定方程为: $s(s-1) = 0$, $\therefore s_1 = 1, s_2 = 0$. (1)

其系数递推公式为:

$$a_{s+k+1} = \frac{-1}{(k+s+1)(s+k)} a_{s+k}, \quad (2)$$

取 $s_1 = 1$, 系数递推公式是: $a_{k+2} = \frac{-1}{(k+1)(k+2)} a_{k+1}$,

$$\text{即有 } a_2 = \frac{(-1)}{1 \cdot 2} a_1 (a_1 \text{ 任意}), \quad a_3 = \frac{-1}{2 \cdot 3} a_2 = \frac{(-1)^2}{2!3!} a_1,$$

$$\dots\dots a_k = \frac{(-1)^{k-1}}{(k-1)!k!} a_1,$$

于是得方程的一个特解:

$$y_1(x) = a_1 \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(k-1)!k!} x^k, \quad (3)$$

由于 $s_1 - s_2 = 1$ 为整数, 所以第二个特解形式为

$$y_2(x) = A y_1(x) \ln x + \sum_{k=0}^{\infty} b_k x^k,$$

$$\text{且 } y_2'(x) = \frac{A}{x} y_1(x) + A \ln x \cdot y_1'(x) + \sum_{k=1}^{\infty} k b_k x^{k-1}.$$

$$y_2''(x) = -\frac{A}{x^2} y_1(x) + \frac{2A}{x} y_1'(x) + A y_1'(x) \ln x + \sum_{k=2}^{\infty} k(k-1) b_k x^{k-2},$$

代入方程并注意到 $x y_1' + y_1 = 0$, 于是有

$$2A y_1' - \frac{A}{x} y_1 + \sum_{k=2}^{\infty} [b_k k(k-1) + b_{k-1}] x^{k-1} + b_0 = 0.$$

$$\begin{aligned} \because -\frac{A}{x} y_1 &= -A a_1 \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(k-1)!k!} x^{k-1} \\ &= -A a_1 - A a_1 \sum_{k=2}^{\infty} \frac{(-1)^{k-1}}{(k-1)!k!} x^{k-1}, \end{aligned}$$

$$2A y_1' = 2A a_1 \sum_{k=1}^{\infty} \frac{k(-1)^{k-1}}{(k-1)!k!} x^{k-1},$$

$$= 2Aa_1 + 2Aa_1 \sum_{k=2}^{\infty} \frac{k(-1)^k}{(k-1)!k!} x^{k-1},$$

代入上式, 并令同次幂系数为零, 有

$$2Aa_1 - Aa_1 + b_0 = 0, \quad \therefore b_0 = -Aa_1 = -A_1,$$

$$(2A_1k - A_1) \frac{(-1)^{k-1}}{(k-1)!k!} + k(k-1)b_k + b_{k-1} = 0,$$

$$\therefore b_k = \left[\frac{A_1(2k-1)(-1)^k}{(k-1)!k!} - b_{k-1} \right] \frac{1}{k(k-1)},$$

对 b_1 无特殊要求, 可令 $b_1 = 0$,

$$b_2 = \frac{3}{1!2!} \cdot A_1 \cdot \frac{1}{2},$$

$$\begin{aligned} b_3 &= \left(\frac{-5}{2!3!} A_1 - b_2 \right) \cdot \frac{1}{3 \cdot 2} = \left(\frac{-5A_1}{2!3!} - \frac{3A_1}{1!2!} \cdot \frac{1}{2} \right) \frac{1}{2 \cdot 3} \\ &= \frac{-A_1}{2!3!} \left(\frac{3}{1 \cdot 2} + \frac{5}{2 \cdot 3} \right), \end{aligned}$$

$$\text{推之得: } b_k = \frac{(-1)^k A_1}{(k-1)!k!} \left[\frac{3}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \cdots + \frac{(2k-1)}{(k-1)k} \right].$$

$$\begin{aligned} \therefore y_2(x) &= A_1 \ln x \cdot y_1(x) - A_1 \\ &\quad + A_1 \sum_{k=2}^{\infty} \frac{(-1)^k}{(k-1)!k!} \left[\frac{3}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \cdots \right. \\ &\quad \left. + \frac{(2k-1)}{(k-1)k} \right] x^k. \end{aligned} \quad (4)$$

8. 在 $x_0 = 0$ 的邻域上求解高斯方程 (超几何级数微分方程)

$$x(x-1)y'' + [(1+\alpha+\beta)x - \gamma]y' + \alpha\beta y = 0.$$

解: $\because x_0 = 0$ 是方程的 E' 则奇点.

设 $y(x) = \sum_{k=0}^{\infty} a_k x^k$, 则有 (如下页表).

其判定方程为

$$[-s(s-1) - \gamma_s]a_s = 0.$$

	x^{s-1}	x^s	...	x^{s+k}	...
$x^2 y'' =$		$s(s-1)a_s$	...	$(s+k)(s+k-1)a_{s+k}$	...
$-xy' =$	$-s(s-1)a_s$	$-(s-1)sa_{s+1}$	...	$-(s+k+1)(s+k)a_{s+k+1}$	...
$(1+\alpha+\beta)xy' =$		$(1+\alpha+\beta)sa_s$	...	$(1+\alpha+\beta)(s+k)a_{s+k}$	...
$-\gamma y' =$	$-\gamma sa_s$	$-\gamma(s+1)a_{s+1}$	...	$-\gamma(s+k+1)a_{s+k+1}$	...
$\alpha\beta y =$		$\alpha\beta a_s$	...	$\alpha\beta a_{s+k}$	...

$$\because a_s \neq 0,$$

$$\therefore s_1 = 0, s_2 = 1 - \gamma. \quad (1)$$

其系数递推公式为

$$\begin{aligned} & [(s+k-1)(s+k) + (s+k)(1+\alpha+\beta) + \alpha\beta]a_{s+k} \\ & = [(s+k)(s+k+1) + \gamma(s+k+1)]a_{s+k+1}. \end{aligned}$$

$$\text{即} \quad a_{k+s+1} = \frac{(s+k+\alpha)(s+k+\beta)}{(s+k+\gamma)(s+k+1)} a_{s+k}. \quad (2)$$

取 $s_1 = 0$,

$$\text{则} \quad a_{k+1} = \frac{(k+\alpha)(k+\beta)}{(k+\gamma)(k+1)} a_k, \quad (3)$$

由 (3) 式推算出:

$$\begin{aligned} a_1 &= \frac{\alpha\beta}{1!\gamma} a_0 \quad (a_0 \text{ 为任意}), \\ a_2 &= \frac{(\alpha+1)(\beta+1)}{2(\gamma+1)} a_1 = \frac{\alpha(\alpha+1)\beta(\beta+1)}{2!\gamma(\gamma+1)} a_0, \\ \therefore y_1(x) &= a_0 \left[1 + \frac{\alpha\beta}{1!\gamma} x + \frac{\alpha(\alpha+1)\beta(\beta+1)}{2!\gamma(\gamma+1)} x^2 + \dots \right] \\ &\equiv a_0 F(\alpha, \beta, \gamma; x), \end{aligned} \quad (4)$$

其中 $F(\alpha, \beta, \gamma; x)$ 是超几何函数.

再取 $s_2 = 1 - \gamma$ (非整数),

则有系数递推公式:

$$a_{k+1} = \frac{(k+\alpha+1-\gamma)(k+\beta+1-\gamma)}{(k+1)(k+2-\gamma)} a_k, \quad (5)$$

公式(5)是公式(3)中以 $(\alpha+1-\gamma)$ 、 $(\beta+1-\gamma)$ 、 $(2-\gamma)$ 代替 α 、 β 、 γ 所得的结果,故知另一特解

$$y_2(x) = a_{1-\gamma} x^{1-\gamma} F(\alpha+1-\gamma, \beta+1-\gamma, 2-\gamma; x). \quad (6)$$

9. 在 $x_0 = 0$ 的邻域上求解合流超几何级数微分方程
 $xy'' + (\gamma - x)y' - \alpha y = 0.$

解: $x_0 = 0$ 是方程的正则奇点,

设 $y(x) = \sum_{k=0}^{\infty} a_k x^k$, 则有

	x^{s-1}	x^s	...	x^{s+k}	...
xy''	$s(s-1)a_s$	$(s+1)sa_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
$\gamma y'$	γsa_s	$\gamma(s+1)a_{s+1}$	...	$\gamma(s+k+1)a_{s+k+1}$	...
$-xy'$		$-sa_s$	...	$-(s+k)a_{s+k}$	...
$-\alpha y$		$-\alpha a_s$	...	$-\alpha a_{s+k}$	...

其判定方程为:

$$[s(s-1) + \gamma s]a_s = 0.$$

$$\because a_s \neq 0,$$

$$\therefore s_1 = 0, s_2 = 1 - \gamma. \quad (1)$$

其系数递推公式为:

$$[(s+k)(s+k+1) + \gamma(s+k+1)]a_{s+k+1} - (s+k+\alpha)a_{s+k} = 0,$$

$$\text{即} \quad a_{s+k+1} = \frac{(s+k+\alpha)}{(s+k+1)(s+k+\gamma)} a_{s+k}, \quad (2)$$

取 $s_1 = 0$, 其系数递推公式为:

$$a_{k+1} = \frac{(k+\alpha)}{(k+1)(k+\gamma)} a_k, \quad (3)$$

与上题 (3) 式比较知, 上式仅不含 β 因子, 故其解为

$$y_1(x) = a_0 \left[1 + \frac{\alpha}{1! \gamma} x + \frac{\alpha(\alpha+1)}{2! \gamma(\gamma+1)} x^2 + \dots \right] \\ \equiv a_0 F(\alpha, \gamma, x), \quad (4)$$

再取 $s_1 = 1 - \gamma$, 其系数递推公式为:

$$a_{k+1} = \frac{(k + \alpha + 1 - \gamma)}{(k + 1)(k + 2 - \gamma)} a_k, \quad (5)$$

完全类似的讨论有:

$$y_2(x) = a_{1-\gamma} x^{1-\gamma} F(\alpha + 1 - \gamma, 2 - \gamma, x). \quad (6)$$

第十三章 球函数

§44. 轴对称球函数

1. 计算 $\frac{2l+1}{2} \int_{-1}^1 x^n p_l(x) dx$.

$$\begin{aligned} \text{解: } & \frac{2l+1}{2} \int_{-1}^{+1} x^n p_l(x) dx \\ &= \frac{2l+1}{2} \cdot \frac{1}{2^l l!} \int_{-1}^{+1} \frac{d^l (x^2-1)^l}{dx^l} \cdot x^n dx \\ &= \frac{2l+1}{2} \cdot \frac{1}{2^l l!} \left\{ x^n \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l \Big|_{-1}^{+1} - n \int_{-1}^{+1} x^{n-1} \right. \\ & \quad \times \frac{d^{l-1} (x^2-1)^l}{dx^{l-1}} dx \Big\}, \\ \therefore & \frac{d^{l-1} (x^2-1)^l}{dx^{l-1}} \Big|_{-1}^1 = 0, \end{aligned}$$

$$\text{原积分} = -\frac{2l+1}{2} \cdot \frac{1}{2^l l!} \left\{ n \int_{-1}^{+1} x^{n-1} \frac{d^{l-1} (x^2-1)^l}{dx^{l-1}} \cdot dx \right\},$$

分部积分 n 次, 由 $(x^n)^{(n)} = n!$,

$$\text{原积分} = (-1)^n \frac{2l+1}{2} \cdot \frac{n!}{2^l l!} \int_{-1}^{+1} \frac{d^{l-n} (x^2-1)^l}{dx^{l-n}} \cdot dx.$$

(1) 如 $n < l$,

$$\therefore \int_{-1}^{+1} \frac{d^{l-n} (x^2-1)^l}{dx^{l-n}} \cdot dx = \frac{d^{l-n-1}}{dx^{l-n-1}} (x^2-1)^l \Big|_{-1}^{+1} = 0,$$

$\therefore$ 原积分为零.

(2) 如 $n > l$,

(i) 若 $n-l$ 为奇数,

$$\begin{aligned} \text{原积分} &= (-1)^l \cdot \frac{2l+1}{2} \cdot \frac{1}{2^l \cdot l!} n(n-1) \cdots (n-l+1) \\ &\quad \times \int_{-1}^{+1} x^{n-l} (x^2-1)^l dx, \end{aligned}$$

$\because x^{n-l} (x^2-1)^l$ 为奇函数,

$$\therefore \int_{-1}^{+1} x^{n-l} (x^2-1)^l dx = 0,$$

因此原积分等于零.

(ii) 若 $n-l$ 为偶数

$\because x^{n-l} (x^2-1)^l$ 为偶函数,

$$\begin{aligned} &\because \int_{-1}^{+1} x^{n-l} (x^2-1)^l dx \\ &= 2 \int_0^1 x^{n-l} (x^2-1)^l dx \\ &= 2 \left[\frac{x^{n-l+1}}{n-l+1} (x^2-1)^l \right]_0^1 - \frac{2 \cdot l}{n-l+1} \int_0^1 x^{n-l+1} \\ &\quad \times (x^2-1)^{l-1} x \cdot dx \\ &= 2(-1)^l \frac{2^l \cdot l}{n-l+1} \int_0^1 x^{n-l+2} (x^2-1)^{l-1} \cdot dx \\ &= 2(-1)^l \frac{2^l}{(n-l+1)} \left[\frac{x^{n-l+3}}{n-l+3} (x^2-1)^{l-1} \right]_0^1 \\ &\quad - \frac{2(l-1)}{n-l+3} \int_0^1 x^{n-l+4} (x^2-1)^{l-2} dx \Big] \\ &= 2(-1)^l \frac{2^2 \cdot l(l-1)}{(n-l+1)(n-l+3)} \int_0^1 x^{n-l+4} \\ &\quad \times (x^2-1)^{l-2} dx \\ &= \cdots \\ &= 2(-1)^l \frac{2^l \cdot l!}{(n-l+1) \cdots [n-l+(2l-1)]} \end{aligned}$$

$$\begin{aligned}
& \int_0^1 x^{n-l+2l} dx \\
&= 2(-1)^l \frac{2^l \cdot l!}{(n-l+1) \cdots (n+l-1)} \cdot \frac{1}{n+l+1} \\
&= 2(-1)^l \frac{(2l)!!(n-l-1)!!}{(n+l+1)!!},
\end{aligned}$$

$$\begin{aligned}
\text{因此原积分} &= (-1)^l \frac{2l+1}{2} \cdot \frac{1}{2^l l!} \cdot \frac{n!}{(n-l)!} \\
&\quad \cdot \frac{2(-1)^l \cdot (2l)!!(n-l-1)!!}{(n+l+1)!!} \\
&= \frac{2l+1}{2^l} \cdot \frac{n!}{l!(n-l)!} \cdot \frac{(2l)!!(n-l-1)!!}{(n+l+1)!!} \\
&= \frac{(2l+1)n!(n-l-1)!!}{(n-l)!(n+l+1)!!} \\
&= \frac{(2l+1)n!}{(n-l)!!(n+l+1)!!}, \\
&\quad [\text{因 } (n-l)! = (n-l)!!(n-l-1)!!],
\end{aligned}$$

故

$$\frac{2l+1}{2} \int_{-1}^1 x^n p_l(x) dx = \begin{cases} 0, & (n < l \text{ 及 } n-l = \text{奇数}), \\ \frac{(2l+1)n!}{(n-l)!!(n+l+1)!!}, & \end{cases}$$

($n-l = \text{非负偶数}$).

(记号 $k!! = k(k-2)(k-4) \cdots$ 直到 1 或 2 为止.)

2. 以勒让德多项式为基本函数族, 在区间 $[-1, +1]$ 上把下列函数展为傅里叶级数.

$$(1) f(x) = x^3.$$

$$\text{解: } x^3 = \sum_{l=0}^{\infty} f_l p_l(x),$$

$$\text{其中 } f_l = \frac{2l+1}{2} \int_{-1}^1 x^3 p_l(x) dx,$$

这里 $n=3$, 由第1题结果知:

$l>3$ 及 $3-l=$ 奇数 (即 $l=0, 2$) 时积分为零, 只有:

$$f_1 = \frac{3 \cdot 3!}{2!1!5!!} = \frac{3}{5}, \quad f_3 = \frac{7 \cdot 3!}{0!1!7!!} = \frac{2}{5},$$

$$\therefore x^3 = \frac{3}{5} p_1(x) + \frac{2}{5} p_3(x).$$

$$(2) f(x) = x^4.$$

解: 这里 $n=4$, 由第1题结果知:

$l>4$ 及 $4-l=$ 奇数 (即 $l=1, 3$) 时, 积分为零. 只有:

$$\begin{aligned} f_0 &= \frac{1}{2} \int_{-1}^1 x^4 p_0(x) dx \\ &= \frac{1}{2} \int_{-1}^1 x^4 dx = \frac{1}{2} \cdot \frac{x^5}{5} \Big|_{-1}^1 = \frac{1}{5}, \end{aligned}$$

$$\begin{aligned} f_2 &= \frac{3}{2} \int_{-1}^1 x^4 p_2(x) dx \\ &= \frac{3}{2} \int_{-1}^1 x^4 \left(\frac{3x^2-1}{2} \right) dx = \frac{4}{7}, \end{aligned}$$

$$\begin{aligned} f_4 &= \frac{9}{2} \int_{-1}^1 x^4 p_4(x) dx \\ &= \frac{9}{2} \int_{-1}^1 x^4 (35x^4 - 3x^2 + 3) \frac{1}{8} dx = \frac{8}{35}. \end{aligned}$$

$$\therefore x^4 = \frac{1}{5} p_0(x) + \frac{4}{7} p_2(x) + \frac{8}{35} p_4(x).$$

$$(3) f(x) = |x| = \begin{cases} x, & (0 \leq x \leq 1), \\ -x, & (-1 \leq x < 0). \end{cases}$$

解: $\because f(x) = |x|$ 在区间 $[-1, 1]$ 上是偶函数,

$\therefore \underline{f(x)p_{2n}(x)}$ 是偶函数, $f(x)p_{2n+1}(x)$ 是奇函数.

$$\therefore f_l = \begin{cases} 0, & (l=2n+1), \\ (2l+1) \int_0^1 x p_l(x) dx, & (l=2n), \end{cases}$$

$$\begin{aligned}
f_{2n} &= \frac{4n+1}{2^{2n}(2n)!} \int_0^1 x \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} dx \\
&= \frac{4n+1}{2^{2n}(2n)!} \left[x \frac{d^{2n-1}}{dx^{2n-1}} (x^2-1)^{2n} \Big|_0^1 \right. \\
&\quad \left. - \int_0^1 \frac{d^{2n-1}}{dx^{2n-1}} (x^2-1)^{2n} dx \right] \\
&= \frac{4n+1}{2^{2n}(2n)!} \left[- \frac{d^{2n-2}}{dx^{2n-2}} (x^2-1)^{2n} \Big|_0^1 \right] \\
&= \frac{4n+1}{2^{2n}(2n)!} C_{2n}^{n-1} (2n-2)! (-1)^{n+1} \\
&= (-1)^{n+1} \frac{(4n+1)(2n-2)!}{2^{2n}(n-1)!(n+1)!} \\
&= (-1)^{n+1} \frac{(4n+1)(2n-2)!}{(2n-2)!!(2n+2)!!}.
\end{aligned}$$

$$\begin{aligned}
\text{其中 } C_{2n}^{n-1} &= \frac{(2n)(2n-1)\cdots[2n-(n-1-1)]}{(n-1)!} \\
&= \frac{(2n)!}{(n-1)!(n+1)!}.
\end{aligned}$$

对于 $n=0$, 由于系数计算公式中出现 $(-2)!$ 和 $(-2)!!$, 而它们都无定义, 故对于 $n=0$ 的系数 f_0 应另外算出

$$f_0 = \int_0^1 x dx = \frac{1}{2}.$$

$$\begin{aligned}
\therefore |x| &= \frac{1}{2} p_0(x) + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(4n+1)(2n-2)!}{(2n-2)!!(2n+2)!!} \\
&\quad \times p_{2n}(x)
\end{aligned}$$

$$\text{或 } |x| = \frac{1}{2} p_0(x) + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(4n+1)(2n-2)!}{2^{2n}(n-1)!(n+1)!} p_{2n}(x).$$

$$(4) \quad f(x) = \begin{cases} x, & (0 < x < 1). \\ 0, & (-1 < x < 0). \end{cases}$$

解:

$$f(x) = \sum_{l=0}^{\infty} f_l p_l(x),$$

其中

$$f_l = \frac{2l+1}{2} \int_0^1 x p_l(x) dx.$$

由前题结果知:

$$f_{2n} = (-1)^{n+1} \frac{(4n+1)(2n-2)!}{2(2n-2)!!(2n+2)!!},$$

及

$$f_0 = \frac{1}{2} \int_0^1 x dx = \frac{1}{4}.$$

$$\begin{aligned} f_{2n+1} &= \frac{4n+3}{2^{2n+2}(2n+1)!} \int_0^1 x \frac{d^{2n+1}}{dx^{2n+1}} (x^2-1)^{2n+1} dx \\ &= \frac{4n+3}{2^{2n+2}(2n+1)!} \left[x \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n+1} \Big|_0^1 \right. \\ &\quad \left. - \int_0^1 \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n+1} dx \right] \\ &= \frac{-(4n+3)}{2^{2n+2}(2n+1)!} \frac{d^{2n-1}}{dx^{2n-1}} (x^2-1)^{2n+1} \Big|_0^1 = 0. \end{aligned}$$

$\because$ 二项式 $(x^2-1)^{2n+1}$ 的展开式中只有偶次幂, 求导次数为 $(2n-1)$ 奇次数, 其积分值总为零,

$$\therefore f_{2n+1} = 0,$$

$\because$ 上式不适用 $n=0$ 即 $l=1$ 的情况, (因为其中有 $\frac{d^{2n-1}}{dx^{2n-1}}$),

$\therefore f_1$ 应另算出:

$$f_1 = \frac{3}{2} \int_0^1 x p_1(x) dx = \frac{3}{2} \int_0^1 x^2 dx = \frac{1}{2}.$$

$$\begin{aligned} \therefore f(x) &= \frac{1}{4} p_0(x) + \frac{1}{2} p_1(x) + \sum_{n=1}^{\infty} (-1)^{n+1} \\ &\quad \times \frac{(4n+1)(2n-2)!}{2(2n-2)!!(2n+2)!!} p_{2n}(x). \end{aligned}$$

3. 在本来是匀强的静电场 E_0 中 放置导体球, 球的半径为 a , 试研究导体球怎样改变了匀强静电场.

解: 取球坐标系, 以球心为极点, 过球心而平行于 E_0 的直线为极轴.

因是导体球, 所以球内电势处处相等, 设为 C . 由于球外不存在电荷, 所以球外电势满足拉普拉斯方程, 又因球面上感应电荷在无限远处产生的电场为零, 所以在无限远处仍是原来电场 E .

于是可写出定解问题:

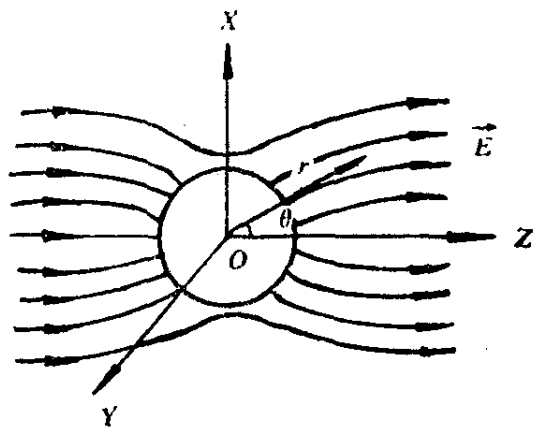


图 13-1

$$\Delta_3 u = 0, \quad (1)$$

$$\begin{cases} u|_{r \rightarrow \infty} \approx -E_0 Z = -E_0 r \cos \theta, & (\text{设导体放入前, } u|_{r=0} = 0), \\ u|_{r=a} = C, \end{cases} \quad (2)$$

$$(3)$$

由于电势具有轴对称性, 方程 (1) 的有限解

$$u = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-(l+1)}) p_l(\cos \theta),$$

由条件 (2):

$$\begin{aligned} u|_{r \rightarrow \infty} &= \sum_{l=0}^{\infty} A_l r^l p_l(\cos \theta) \Big|_{r \rightarrow \infty} \\ &\approx -E_0 r \cos \theta = -E_0 r p_1(\cos \theta). \end{aligned}$$

得

$$A_1 = -E_0, A_l = 0, (l \neq 1),$$

$$\text{于是 } u = -E_0 r p_1(\cos \theta) + \sum_{l=0}^{\infty} B_l r^{-(l+1)} p_l(\cos \theta),$$

$$\text{由条件 (3): } u|_{r=a} = -E_0 a p_1 + \sum_{l=0}^{\infty} B_l a^{-(l+1)} p_l = C,$$

得 $-E_0 a + B_1 a^{-2} = 0, B_0 = C a, B_1 = 0, (l \neq 0, 1),$
 即 $B_1 = E_0 a^3,$

$$\therefore u = -E_0 r p_1(\cos\theta) + C a \frac{1}{r} p_0(\cos\theta) \\ + E_0 a^3 \frac{1}{r^2} p_1(\cos\theta).$$

或 $u = \left(-E_0 r + E_0 a^3 \frac{1}{r} \right) \cos\theta + C a \frac{1}{r}, (r \geq a), \quad (4)$

若导体接地, 则 $C = 0$, 这时

$$u = \left(-E_0 r + E_0 a^3 \frac{1}{r} \right) \cos\theta, (r \geq a). \quad (5)$$

4. 在点电荷 $4\pi\epsilon_0 q$ 的电场中放置导体球, 球的半径为 a , 球心与点电荷相距 d ($d > a$), 求解这个静电场.

解: 选择球心为极点, 极轴通过点电荷 q , 则问题与 φ 无关;

又设导体球接地, 所以导体球内电势为 0;

这样就剩下求解球外电势的问题.

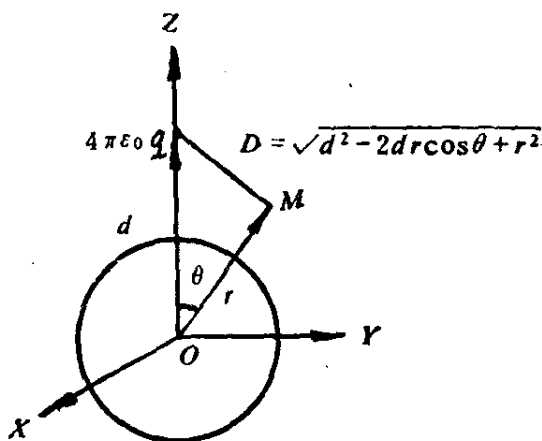


图 13-2

在球外 (除点电荷处) 任意点 M 的电势是点电荷 $4\pi\epsilon_0 q$ 产生的电势 $\frac{q}{D}$ 和导体球感应电荷产生的电势 v 的叠加. 而 v 满足拉普拉斯方程. 于是定解问题为

$$\begin{cases} u = \frac{q}{D} + v, \end{cases} \quad (1)$$

$$\begin{cases} v|_{r=a} = 0, \end{cases} \quad (2)$$

$$\begin{cases} \Delta v = 0, & (3) \\ v|_{r \rightarrow \infty} = 0, & (4) \end{cases}$$

定解问题(3)、(4)，由于轴对称，其有限解为：

$$v = \sum_{l=0}^{\infty} C_l r^{-(l+1)} P_l(\cos\theta), \quad (5)$$

在球附近， $d > r$

$$\begin{aligned} \frac{q}{D} &= \frac{q}{\sqrt{d^2 + r^2 - 2dr\cos\theta}} = \frac{q}{d} \cdot \frac{1}{\sqrt{1 - 2\left(\frac{r}{d}\right)\cos\theta + \left(\frac{r}{d}\right)^2}} \\ &= \frac{q}{d} \sum_{l=0}^{\infty} \left(\frac{r}{d}\right)^l P_l(\cos\theta), \end{aligned}$$

由条件(2)：

$$q \sum_{l=0}^{\infty} \frac{a^l}{d^{l+1}} P_l(\cos\theta) + \sum_{l=0}^{\infty} \frac{C_l}{a^{l+1}} P_l(\cos\theta) = 0,$$

得
$$\frac{qa^l}{d^{l+1}} + \frac{C_l}{a^{l+1}} = 0,$$

$$\therefore C_l = -qa^{2l+1}d^{-(l+1)},$$

故
$$\begin{aligned} u &= \frac{q}{\sqrt{d^2 - 2dr\cos\theta + r^2}} - q \sum_{l=0}^{\infty} \frac{a^{2l+1}}{d^{l+1}} \cdot \\ &\quad \frac{1}{r^{l+1}} P_l(\cos\theta) \\ &= \frac{q}{\sqrt{d^2 - 2dr\cos\theta + r^2}} - \frac{a}{d} \\ &\quad \times \frac{q}{\sqrt{\left(\frac{a^2}{d}\right)^2 - 2\left(\frac{a^2}{d}\right)r\cos\theta + r^2}}. \end{aligned} \quad (6)$$

在解(6)中，第二项为象电荷产生的电势，这象电荷处在

球内极轴 $d_0 = \frac{a^2}{d}$ 上，带电量为 $-4\pi\epsilon_0 q \cdot \frac{a}{d}$ 。

5. 求解

$$\begin{cases} \Delta_3 u = 0, (r < a), \\ u|_{r=a} = \cos^2 \theta, \end{cases}$$

解：因为是球内问题，所求的有限解为

$$u = \sum_{l=0}^{\infty} \sum_{m=0}^l C_{l,m} r^l P_l^m(\cos \theta) \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix},$$

又因边界条件与 φ 无关，所以知其解也应与 φ 无关，

即
$$u(r, \theta) = \sum_{l=0}^{\infty} C_l \cdot r^l \cdot P_l(\cos \theta).$$

代入边界条件：

$$u(a, \theta) = \sum_{l=0}^{\infty} C_l a^l P_l(\cos \theta) = \cos^2 \theta$$

$$\text{或 } u(a, x) = \sum_{l=0}^{\infty} C_l a^l P_l(x) = x^2,$$

$$\therefore C_l = \frac{2l+1}{2a^l} \int_{-1}^1 x^2 P_l(x) dx,$$

$$C_0 = \frac{1}{2} \int_{-1}^1 x^2 \cdot 1 dx = \frac{1}{2} \left. \frac{x^3}{3} \right|_{-1}^1$$

$$= \frac{1}{6} [1 - (-1)^3] = \frac{1}{3},$$

$$C_2 = \frac{5}{2a^2} \int_{-1}^1 x^2 \frac{1}{2} (3x^2 - 1) dx$$

$$= -\frac{5}{2a^2} \left[\frac{3}{2} \frac{x^5}{5} - \frac{1}{2} \frac{x^3}{3} \right]_{-1}^1$$

$$= -\frac{5}{2a^2} \left[\frac{3}{5} - \frac{1}{3} \right] = -\frac{2}{3a^2},$$

$$C_l = 0, (l \neq 0, 2),$$

$$\therefore u(r, \theta) = \frac{1}{3} P_0 + \frac{2}{3a^2} r^2 P_2$$

$$= \frac{1}{3} + \frac{2}{3} \left(\frac{r}{a} \right)^2 P_2(\cos\theta).$$

6. 用一层不导电的物质把半径为 a 的导体球壳分隔为两个半球壳, 使半球各充电到电势为 v_1 和 v_2 , 试解电场中的电势分布.

解: 取球坐标系, 以球心为极点, 以过 O 且垂直于介质平面的直线为极轴, 则电势与 φ 无关.

(1) 球壳内电势 u_i 的定解问题为:

$$\Delta u_i = 0, (r < a), \quad (1)$$

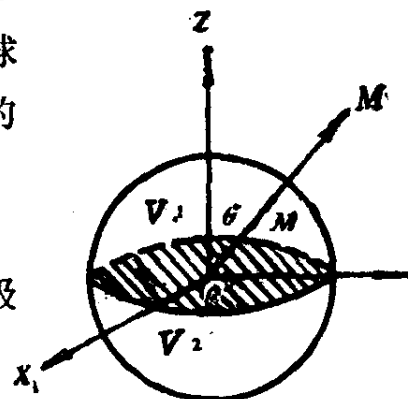


图 13-3

$$\begin{cases} u_i|_{r=0} \text{ 有限, } & (\text{自然边界条件}) \end{cases} \quad (2)$$

$$\begin{cases} u_i|_{r=a} = \begin{cases} v_1, & (0 \leq \theta < \frac{\pi}{2}) \text{ 或 } (0 \leq x < 1), \\ v_2, & (\frac{\pi}{2} < \theta \leq \pi) \text{ 或 } (-1 < x \leq 0), \end{cases} \end{cases} \quad (3)$$

其有限解为: $u_i = \sum_{l=0}^{\infty} C_l r^l P_l(x), (x = \cos\theta),$

由边值(3): $u_i|_{r=a} = \sum_{l=0}^{\infty} C_l a^l P_l(x) = \begin{cases} v_1, \\ v_2, \end{cases}$

$$\begin{aligned} \therefore C_l &= \frac{2l+1}{2a^l} \left[\int_{-1}^0 v_2 P_l(x) dx + \int_0^1 v_1 P_l(x) dx \right] \\ &= \frac{(2l+1)v_2}{2a^l} \int_{-1}^0 P_l dx + \frac{(2l+1)v_1}{2a^l} \int_0^1 P_l dx, \end{aligned}$$

$$\begin{aligned} \therefore \int_{-1}^0 P_l(x) dx &= \int_1^0 P_l(-x) (-dx) \\ &= \int_0^1 P_l(-x) dx = (-1)^l \int_0^1 P_l(x) dx, \end{aligned}$$

$$\begin{aligned}
\text{而} \int_0^1 P_l(x) dx &= \frac{1}{2^l l!} \int_0^1 \frac{d^l}{dx^l} (x^2 - 1)^l dx \\
&= \frac{1}{2^l l!} \left[\frac{d^{l-1}}{dx^{l-1}} (x^2 - 1)^l \right] \Big|_{x=0}^{x=1} \\
&= \frac{-1}{2^l l!} \left[\frac{d^{l-1}}{dx^{l-1}} \sum_{k=0}^l C_l^k x^{2k} (-1)^{l-k} \right]_{x=0} \\
&= \begin{cases} 0, & (l = \text{偶数且} l \neq 0), \\ \frac{(-1)^n (2n)!}{2^{2n+1} (n+1)! n!}, & (l = 2n+1 \text{ 时}), \end{cases}
\end{aligned}$$

$$\begin{aligned}
\therefore C_l &= \frac{v_1 + (-1)^l v_2}{a^l} \cdot \frac{2l+1}{2} \int_0^1 P_l(x) dx \\
&= \begin{cases} 0, & (l = 2k, \text{ 且 } k \neq 0), \\ \frac{v_1 - v_2}{2} (-1)^k \frac{(4k+3)(2k)!}{(2k+2)!!(2k)!!} \cdot \frac{1}{a^{2k+1}}, & (l = 2k+1), \end{cases}
\end{aligned}$$

$$C_0 = \frac{v_1 - v_2}{2} \int_0^1 P_0(x) dx = \frac{v_1 - v_2}{2},$$

故球壳内电势为:

$$\begin{aligned}
u_i(r, \theta) &= \frac{v_1 + v_2}{2} + \frac{v_1 - v_2}{2} \sum_{k=0}^{\infty} (-1)^k \\
&\quad \times \frac{(4k+3)(2k)!}{(2k+2)!!(2k)!!} \left(\frac{r}{a}\right)^{2k+1} P_{2k+1}(\cos \theta). \quad (4)
\end{aligned}$$

(2) 球壳外电势 u_e 的定解问题为:

$$\Delta u_e = 0, \quad (r > a), \quad (1)$$

$$\begin{cases} u_e|_{r \rightarrow \infty} = 0, \end{cases} \quad (2)$$

$$\begin{cases} u_e|_{r=a} = \begin{cases} v_1, & (0 \leq \theta < \frac{\pi}{2}), \\ v_2, & (\frac{\pi}{2} < \theta \leq \pi), \end{cases} \end{cases} \quad (3)$$

其有限解是:

$$u_e = \sum_{l=0}^{\infty} d_l \frac{1}{r^{l+1}} P_l(x), \quad (x = \cos\theta),$$

代入条件 (3):

$$u_e|_{r=a} = \sum_{l=0}^{\infty} d_l \frac{1}{a^{l+1}} P_l(x) = \begin{cases} v_1, \\ v_2, \end{cases}$$

$$\text{同上有 } d_0 = \frac{a}{2} (v_1 + v_2) \int_0^1 P_0(x) dx = \frac{a}{2} (v_1 + v_2),$$

$$\begin{aligned} d_{2k+1} &= \frac{v_1 - v_2}{2} (-1)^k \cdot (4k+3) \frac{(2k-1)!!}{(2k+2)!!} a^{2k+2} \\ &= \frac{v_1 - v_2}{2} (-1)^k \cdot (4k+3) \frac{(2k)!}{(2k+2)!! (2k)!!}, \end{aligned}$$

$$d_{2k} = 0, \quad (k \neq 0),$$

∴ 球壳外的电势为:

$$\begin{aligned} u_e(r, \theta) &= \frac{v_1 + v_2}{2} \cdot \frac{a}{r} + \frac{v_1 - v_2}{2} \sum_{k=0}^{\infty} (-1)^k \\ &\quad \times \frac{(4k+3)(2k)!}{(2k+2)!! (2k)!!} \left(\frac{a}{r}\right)^{2k+2} P_{2k+1}(\cos\theta). \end{aligned} \quad (4)$$

7. 半球的球面保持一定温度 u_0 , 半球底面①保持 0°C , ②绝热, 试求这个半球里的稳定温度分布.

解: (1) 半球底面保持 0°C , 取球坐标系如图 (13-4), 则温度 u 与 φ 无关, 其定解问题为:

$$\begin{aligned} \Delta u &= 0, \quad (r < a), \\ \begin{cases} u|_{r=0} \text{ 有限,} \\ u|_{r=a} = u_0, \end{cases} \\ u|_{\theta = \frac{\pi}{2}} &= 0, \end{aligned}$$

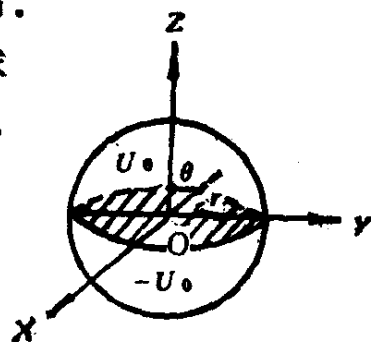


图 13-4

将半球问题化成全球问题, 需将 θ 延拓至 $\left[\frac{\pi}{2}, \pi\right]$, 根据

$u|_{\theta=\frac{\pi}{2}} = 0$ 知需作奇延拓, 于是定解问题为:

$$\Delta u = 0, \quad (r < a), \quad (1)$$

$$\begin{cases} u|_{r=0} \text{ 有限,} \end{cases} \quad (2)$$

$$u|_{r=a} = \begin{cases} u_0, & (0 \leq \theta < \frac{\pi}{2}), \\ -u_0, & (\frac{\pi}{2} < \theta \leq \pi), \end{cases} \quad (3)$$

其有限解是:

$$u = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta),$$

代入条件 (3):

$$u|_{r=a} = \sum_{l=0}^{\infty} A_l a^l P_l(\cos \theta) = \begin{cases} u_0, \\ -u_0, \end{cases}$$

由第 6 题 (1) 的结果知:

$$A_0 = \frac{1}{2}(u_0 - u_0) = 0, \quad A_{2k} = 0,$$

$$A_{2k+1} = \frac{(-1)^k}{a^{2k+1}} \frac{(4k+3)(2k)!}{(2k+2)!!(2k)!!} u_0,$$

$$\therefore u = u_0 \sum_{k=0}^{\infty} (-1)^k \frac{(4k+3)(2k)!}{(2k+2)!!(2k)!!}$$

$$\times \left(\frac{r}{a}\right)^{2k+1} P_{2k+1}(\cos \theta). \quad (4)$$

(2) 半球底面绝热.

定解问题为:

$$\Delta u = 0, \quad (r < a),$$

$$\begin{cases} u|_{r=a} \text{有限}, \\ u|_{r=a} = u_0, \\ \frac{\partial u}{\partial \theta} \Big|_{\theta=\frac{\pi}{2}} = 0, \end{cases}$$

由于 $\frac{\partial u}{\partial \theta} \Big|_{\theta=\frac{\pi}{2}} = 0$, 需作偶延拓, 于是定解问题为:

$$\Delta u = 0, \quad (r < a), \quad (1)$$

$$\begin{cases} u|_{r=a} \text{有限}, \end{cases} \quad (2)$$

$$u|_{r=a} = \begin{cases} u_0, & (0 \leq \theta < \frac{\pi}{2}), \\ u_0, & (\frac{\pi}{2} < \theta \leq \pi), \end{cases} \quad (3)$$

其有限解是:

$$u = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta),$$

由条件(3):

$$u|_{r=a} = \sum_{l=0}^{\infty} A_l a^l P_l(\cos \theta) = u_0 \equiv u_0 P_0(\cos \theta),$$

比较等式两边得:

$$A_0 = u_0, \quad A_l = 0, \quad (l \neq 0),$$

$$\therefore u = u_0. \quad (4)$$

8. 半径为 a , 表面熏黑的均匀球, 在温度为 0° 的空气中, 受着阳光的照射, 阳光的热流强度为 q_0 , 求解小球里的稳定温度分布.

解: 取坐标系如图(13-5), 则温度与 φ 无关, 球仅有上半部

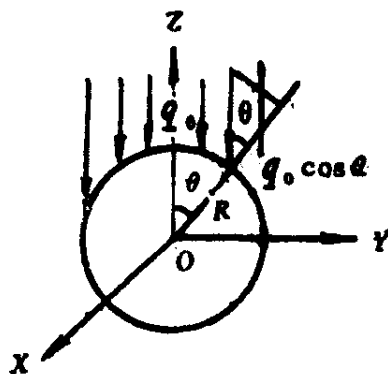


图 13-5

分球面受阳光照射, 进行热量交换, 而下半球面则不受阳光照射. 于是定解问题为:

$$\Delta u = 0, \quad (r < a), \quad (1)$$

$$\begin{cases} |u|_{r=0} \text{ 有限,} \\ \left(\frac{\partial u}{\partial r} + Hu \right) \Big|_{r=a} = \begin{cases} q_0 \cos \theta, & (0 \leq \theta < \frac{\pi}{2}), \\ 0, & (\frac{\pi}{2} < \theta \leq \pi), \end{cases} \end{cases} \quad (2)$$

$$\begin{cases} \left(\frac{\partial u}{\partial r} + Hu \right) \Big|_{r=a} = \begin{cases} q_0 \cos \theta, & (0 \leq \theta < \frac{\pi}{2}), \\ 0, & (\frac{\pi}{2} < \theta \leq \pi), \end{cases} \end{cases} \quad (3)$$

这里 $H = h/k$, h 为热交换系数, k 为热传导系数.

其有限解为:

$$u = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta),$$

由边界条件(3), 得

$$\begin{aligned} \sum_{l=1}^{\infty} A_l \cdot l \cdot a^{l-1} \cdot P_l(\cos \theta) + \sum_{l=0}^{\infty} H A_l a^l P_l(\cos \theta) = \\ \begin{cases} q_0 \cos \theta, & (0 \leq \theta < \frac{\pi}{2}), \\ 0, & (\frac{\pi}{2} < \theta \leq \pi), \end{cases} \end{aligned}$$

$$\text{即} \quad H A_0 + \sum_{l=1}^{\infty} A_l (l + H a) a^{l-1} P_l(\cos \theta) = \begin{cases} q_0 \cos \theta, \\ 0, \end{cases}$$

利用本节第2题(4)的结果, 有

$$\left. \begin{aligned} A_0 &= \frac{1}{2H} \int_0^1 q_0 x dx = \frac{q_0}{4H}, \\ A_1 &= \frac{1}{Ha+1} \cdot \frac{3}{2} q_0 \int_0^1 x \cdot x dx = \frac{1}{Ha+1} \cdot \frac{q_0}{2}, \\ A_{2k+1} &= 0, \\ A_{2k} &= \frac{(-1)^{k+1} q_0}{Ha-2k} \cdot \frac{4k+1}{2} \cdot \frac{(2k-2)!}{(2k-2)!!(2k+2)!!} \cdot \frac{1}{a^{2k-1}}, \end{aligned} \right\} \begin{aligned} (x = \cos \theta), \\ (k \neq 0), \end{aligned}$$

$$\begin{aligned} \therefore u(r, \theta) = & \frac{q_0}{4H} + \frac{q_0}{2(Ha+1)} \cdot r \cdot P_1(\cos\theta) \\ & + \sum_{k=1}^{\infty} \frac{(-1)^{k+1} q_0}{(Ha+2k) a^{2k-1}} \cdot \\ & \frac{(4k+1)(2k-2)!}{2(2k-2)!!(2k+2)!!} r^{2k} P_{2k}(\cos\theta). \quad (4) \end{aligned}$$

9. 求解 § 31 习题 7 的轻绳的振动. 初位移 $\varphi(x)$, 初速 $\psi(x)$.

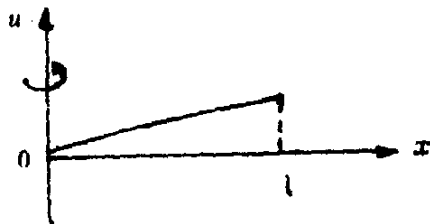


图 13-6

解: 轻绳的横振动方程为:

$$u_{tt} - a^2 \frac{\partial}{\partial x} \left[(l^2 - x^2) \frac{\partial u}{\partial x} \right] = 0, \quad \left(a^2 = \frac{\omega^2}{2} \right), \quad (1)$$

定解条件为:

$$\begin{cases} u|_{x=0} = 0, & (\text{端点固定}), \\ u|_{x=l} = \text{有限}, & (\text{端点自由}), \end{cases} \quad (2)$$

$$\begin{cases} u|_{t=0} = \varphi(x), \\ u_t|_{t=0} = \psi(x), \end{cases} \quad (3)$$

令 $u(x, t) = X(x) \cdot T(t)$ 代入泛定方程及边界条件:

$$T'' X - a^2 \cdot T \cdot \frac{d}{dx} [(l^2 - x^2) X'] = 0,$$

$$\text{即} \quad \frac{(l^2 - x^2) X'' - 2x X'}{X} = \frac{T''}{a^2 T} = -k(k+1),$$

由此分解为两个常微分方程

$$(l^2 - x^2) X'' - 2x X' + k(k+1) X = 0, \quad (4)$$

$$T'' + a^2 k(k+1) T = 0, \quad (5)$$

$$\text{及} \begin{cases} TX|_{x=0} = 0, \\ TX|_{x=l} = \text{有限}. \end{cases}$$

令 $\xi = \frac{x}{l}$ 代入方程(4), 于是得本征值问题:

$$(1 - \xi^2) \frac{d^2 X}{d\xi^2} - 2\xi \frac{dX}{d\xi} + k(k+1)X = 0, \quad (6)$$

$$\begin{cases} X|_{x=0} = 0, \\ X|_{x=l} = \text{有限}, \end{cases} \quad (7)$$

$$(8)$$

方程(6)为 k 阶勒让德方程, 其解为

$$X_k(\xi) = X_k\left(\frac{x}{l}\right) = P_k\left(\frac{x}{l}\right),$$

由边界条件(8) $X|_{x=l} = \text{有限}$, 知 $k = 0, 1, 2, \dots$,

由边界条件(7) $X|_{x=0} = P_k(0) = 0$,

知 $k = 2n-1$, ($n = 1, 2, \dots$),

方程(5)的解为

$$T_n = A_n \cos \sqrt{2n(2n-1)} at + B_n \sin \sqrt{2n(2n-1)} at,$$

$$\begin{aligned} \therefore u(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos \sqrt{2n(2n-1)} at \right. \\ \left. + B_n \sin \sqrt{2n(2n-1)} at \right] P_{2n-1} \left(\frac{x}{l} \right). \end{aligned} \quad (9)$$

由初始条件(3)有

$$\begin{cases} u(x, t)|_{t=0} = \sum_{n=1}^{\infty} A_n P_{2n-1} \left(\frac{x}{l} \right) = \varphi(x), \\ u_t(x, t)|_{t=0} = \sum_{n=1}^{\infty} B_n \sqrt{2n(2n-1)} a P_{2n-1} \left(\frac{x}{l} \right) = \psi(x), \end{cases}$$

$$\therefore \begin{cases} A_{2n-1} = \frac{4n-1}{l} \int_0^l \varphi(x) P_{2n-1} \left(\frac{x}{l} \right) dx, \\ B_{2n-1} = \frac{4n-1}{\sqrt{2n(2n-1)} al} \int_0^l \psi(x) P_{2n-1} \left(\frac{x}{l} \right) dx. \end{cases} \quad (10)$$

故本题的解为(9), 其中系数由(10)决定.

10. 半径为 a 的圆形铁环, 充有 $4\pi\epsilon_0 q$ 单位电荷, 求铁环周围电场中的电势. [在初等电学课程中已知圆环轴上距环心 r 处的电势为 $q/\sqrt{a^2+r^2}$.]

解: 除圆铁环外, 均无电荷, 所以电势 u_i, u_e 均满足拉氏方程. 为求定解条件, 考察在圆环上的一电荷元 dq , 它在环轴上距环心 r 远点 P 处所产生的电势为:

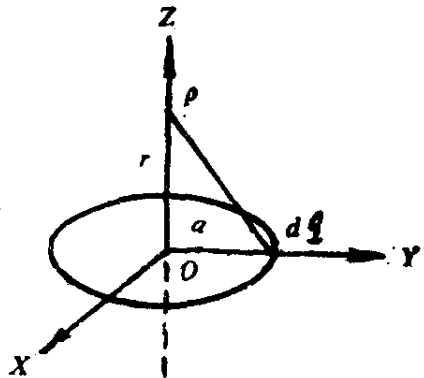


图 13-7

$$\frac{dq}{4\pi\epsilon_0\sqrt{r^2+a^2}},$$

$\therefore$ 圆环上全部电荷在点 P 处的合电势为:

$$\int \frac{dq}{4\pi\epsilon_0\sqrt{r^2+a^2}} = \frac{4\pi\epsilon_0 q}{4\pi\epsilon_0\sqrt{r^2+a^2}} = \frac{q}{\sqrt{r^2+a^2}},$$

同理可知, 在圆环中心 O 处的合电势为 $\frac{q}{a}$.

以环心为极点, 环轴为极轴建立球坐标系, 电场分布具对称性, 极轴即为对称轴, 从而定解问题为:

$$(1) r < a \text{ 时, } \Delta_3 u_i = 0, \quad (1)$$

$$u_i|_{\theta=0, \pi} = \frac{q}{\sqrt{r^2+a^2}}, \quad (2)$$

方程(1)在球内有限解为 $u_i = \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)$,

由条件(2): $\sum_{l=0}^{\infty} A_l r^l P_l(1) = \frac{q}{\sqrt{r^2+a^2}},$

$$\text{即: } \sum_{l=0}^{\infty} A_l r^l = \frac{q}{a} \frac{1}{\sqrt{1 + \left(\frac{r}{a}\right)^2}}$$

$$= \frac{q}{a} \left\{ 1 - \frac{1}{2} \left(\frac{r}{a}\right)^2 + \frac{1.3}{2.4} \left(\frac{r}{a}\right)^4 + \dots + (-1)^n \right\}$$

$$\times \frac{(2n-1)!!}{(2n)!!} \left(\frac{r}{a}\right)^{2n} + \dots \},$$

$$\left(\text{因} \left|\frac{r}{a}\right| < 1\right)$$

$$\therefore A_{2n+1} = 0, \quad A_0 = \frac{q}{a},$$

$$A_{2n} = (-1)^n \frac{(2n-1)!!}{(2n)!!} \frac{q}{a^{2n+1}},$$

$$\therefore u_i(r, \theta) = \frac{q}{a} + \frac{q}{a} \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!} \times \left(\frac{r}{a}\right)^{2n} P_{2n}(\cos \theta), \quad (r < a). \quad (3)$$

如果细铁环，不计其高度，占据一个平面： $\left\{ \theta = \frac{\pi}{2}, \right.$

$r < a \left\{ \right.$ ，这时定解问题成为 $\Delta u = 0$ ， $u \Big|_{\theta = \frac{\pi}{2}} = \frac{q}{a}$ ，解为 $u \equiv \frac{q}{a}$ ，这也可以由上面所得解 $u_i(r, \theta)$ 中令 $\theta = \pi/2$ 而得到，

$$\therefore P_{2n} \text{ 是 } \cos \theta \text{ 的偶次幂, } \cos \frac{\pi}{2} = 0,$$

$$\therefore u_i \left(r, \frac{\pi}{2} \right) = \frac{q}{a}.$$

(2) $r > a$ 时,

$$\Delta u_e = 0, \quad (4)$$

$$u_e \Big|_{r=0, \pi} = q / \sqrt{r^2 + a^2}, \quad (5)$$

方程(4)在球外的有限解为

$$u_e = \sum_{l=0}^{\infty} B_l \frac{1}{r^{l+1}} P_l(\cos \theta),$$

由条件(5)得:

$$\begin{aligned}\sum_{l=0}^{\infty} B_l \frac{1}{r^{l+1}} P_l(1) &= q/\sqrt{r^2+a^2} = \frac{q}{r} \cdot \frac{1}{\sqrt{1+\left(\frac{a}{r}\right)^2}} \\ &= \frac{q}{r} \left[1 + \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!} \right. \\ &\quad \left. \times \left(\frac{a}{r}\right)^{2n} \right],\end{aligned}$$

$$\therefore B_0 = q, \quad B_{2n+1} = 0,$$

$$B_{2n} = (-1)^n \frac{(2n-1)!!}{(2n)!!} q a^{2n},$$

$$\begin{aligned}\therefore u(r, \theta) &= \frac{q}{r} + \frac{q}{a} \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!} \\ &\quad \times \left(\frac{q}{r}\right)^{2n+1} P_{2n}(\cos \theta), \quad (r > a) \quad (6)\end{aligned}$$

11. 求证 $P_l(x) = P'_{l+1}(x) - 2xP'_l(x) + P'_{l-1}(x)$. [提示: 拿(44.20)对 x 求导.]

$$\text{证: } \because \frac{1}{\sqrt{1-2rx+r^2}} = \sum_{l=0}^{\infty} r^l P_l(x),$$

将上式对 x 求导, 得:

$$\frac{r}{(1-2rx+r^2)^{3/2}} = \sum_{l=0}^{\infty} r^l P'_l(x),$$

$$\frac{r}{\sqrt{1-2rx+r^2}} = (1-2rx+r^2) \sum_{l=0}^{\infty} r^l P'_l(x),$$

$$r \sum_{l=0}^{\infty} r^l P_l(x) = (1-2rx+r^2) \sum_{l=0}^{\infty} r^l P'_l(x),$$

比较等式两边 r^{l+1} 的系数, 得:

$$P_l(x) = P'_{l+1}(x) - 2xP'_l(x) + P'_{l-1}(x).$$

12. 利用上题和(44.21)求证 $(2l+1)P_l(x) = P'_{l+1}(x) - P'_{l-1}(x)$.

证：由递推公式

$$(l+1)P_{l+1}(x) - (2l+1)xP_l(x) + lP_{l-1}(x) = 0,$$

对 x 求导，得：

$$(l+1)P'_{l+1}(x) - (2l+1)P_l(x) - (2l+1)xP'_l(x) + lP'_{l-1}(x) = 0,$$

整理得：

$$(2l+1)P_l(x) = \frac{2l+1}{2} [P'_{l+1}(x) - 2xP'_l(x) + P'_{l-1}(x)] + \frac{1}{2} [P'_{l+1}(x) - P'_{l-1}(x)],$$

应用上题结果得：

$$(2l+1)P_l(x) = \frac{2l+1}{2} P_l(x) + \frac{1}{2} [P'_{l+1}(x) - P'_{l-1}(x)],$$

$$\therefore (2l+1)P_l(x) = P'_{l+1}(x) - P'_{l-1}(x).$$

§45. 一般的球函数

1. 用球函数把下列函数展开

$$(1) \sin^2\theta \cdot \cos^2\varphi.$$

解一：利用三角公式。

$$\begin{aligned} \sin^2\theta \cdot \cos^2\varphi &= \frac{1}{4}(1 - \cos 2\theta)(1 + \cos 2\varphi) \\ &= \frac{1}{6} \left[\frac{3}{2}(1 - \cos 2\theta)\cos 2\varphi \right] \\ &\quad + \frac{1}{2}(1 - \cos^2\theta) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{6} \left[\frac{3}{2} (1 - \cos 2\theta) \cos 2\varphi \right] \\
&\quad - \frac{1}{3} \left(\frac{3}{2} \cos^2 \theta - \frac{1}{2} \right) + \frac{1}{3} \\
&= \frac{1}{3} - \frac{1}{3} p_2(\cos \theta) + \frac{1}{6} p_2^2(\cos \theta) \cos 2\varphi.
\end{aligned}$$

解二：利用展开公式。

$$f(\theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=0}^l [A_l^m \cos m\varphi + B_l^m \sin m\varphi] p_l^m(\cos \theta),$$

其中

$$\begin{cases} A_l^m = \frac{2l+1}{2\pi\delta_m} \cdot \frac{(l-m)!}{(l+m)!} \int_0^{2\pi} \int_0^\pi f(\theta, \varphi) p_l^m(\cos \theta) \\ \quad \cdot \cos m\varphi \sin \theta d\theta d\varphi, \\ B_l^m = \frac{2l+1}{2\pi} \cdot \frac{(l-m)!}{(l+m)!} \int_0^{2\pi} \int_0^\pi f(\theta, \varphi) p_l^m(\cos \theta) \\ \quad \cdot \sin m\varphi \sin \theta d\theta d\varphi, \end{cases}$$

先对 φ 积分：

$\because \cos^2 \varphi$ 是偶函数，

$\therefore B_l^m = 0$,

$$\text{而 } \int_0^{2\pi} \cos^2 \varphi \cdot \cos m\varphi d\varphi = \frac{1}{2} \int_0^{2\pi} (1 - \cos 2\varphi) \cos m\varphi d\varphi$$

$$= \begin{cases} \pi, & (m=0), \\ \pi/2, & (m=2), \\ 0, & (m \neq 0, 2). \end{cases}$$

再对 θ 积分：

$$\begin{aligned}
\int_0^\pi \sin^2 \theta \cdot p_l^m(\cos \theta) \sin \theta d\theta &= - \int_0^\pi (1 - \cos^2 \theta) p_l^m(\cos \theta) d(\cos \theta) \\
&= \int_{-1}^1 (1-x^2) p_l^m(x) dx, \quad (\text{令 } x = \cos \theta),
\end{aligned}$$

当 $m=0$ 时，

$$\therefore \int_{-1}^1 p_l(x) dx = \begin{cases} 2, & (l=0), \\ 0, & (l \neq 0), \end{cases}$$

$$- \int_{-1}^1 x^2 p_l(x) dx = \begin{cases} -\frac{2}{3}, & (l=0), \\ -\frac{4}{15}, & (l=2), \\ 0, & (l \neq 0, 2), \end{cases}$$

$$\therefore \int_{-1}^1 (1-x^2) p_l(x) dx = \begin{cases} \frac{4}{3}, & (l=0), \\ -\frac{4}{15}, & (l=2), \\ 0, & (l \neq 0, 2), \end{cases}$$

当 $m = 2$ 时,

$$\therefore \int_{-1}^1 p_l^2(x) dx = \begin{cases} 4, & (l=2), \\ 0, & (l \neq 2), \end{cases}$$

$$- \int_{-1}^1 x^2 p_l^2(x) dx = \begin{cases} -\frac{4}{5}, & (l=2), \\ 0, & (l \neq 2), \end{cases}$$

$$\therefore \int_{-1}^1 (1-x^2) p_l^2(x) dx = \begin{cases} \frac{16}{5}, & (l=2), \\ 0, & (l \neq 2), \end{cases}$$

于是得: $A_0^0 = -\frac{1}{4\pi} \pi \cdot \frac{4}{3} = \frac{1}{3}$, $A_2^0 = -\frac{5}{4\pi} \pi \left(-\frac{4}{15}\right) = -\frac{1}{3}$,

$$A_2^2 = -\frac{5}{2\pi} \cdot \frac{1}{4!} \cdot \frac{\pi}{2} \cdot \frac{16}{5} = \frac{1}{6},$$

$$\therefore \sin^2\theta \cdot \cos^2\varphi = \frac{1}{3} - \frac{1}{3} p_2(\cos\theta) + \frac{1}{6} p_2^2(\cos\theta) \cos 2\varphi.$$

$$(2) (1+3\cos\theta) \sin\theta \cdot \cos\varphi.$$

解: $(1+3\cos\theta) \sin\theta \cdot \cos\varphi = \sin\theta \cdot \cos\varphi + 3\cos\theta \cdot \sin\theta \cdot$

$$\begin{aligned}
 & \cdot \cos \varphi \\
 &= \sin \theta \cdot \cos \varphi + \frac{3}{2} \sin 2\theta \cdot \cos \varphi \\
 &= p_1^1(\cos \theta) \cos \varphi + p_2^1(\cos \theta) \\
 & \quad \cdot \cos \varphi.
 \end{aligned}$$

$$(3) (1 - |\cos \theta|)(1 + \cos 2\varphi).$$

$$\begin{aligned}
 \text{解: } (1 - |\cos \theta|)(1 + \cos 2\varphi) &= (1 - |\cos \theta|) + (1 \\
 & \quad - |\cos \theta|) \cos 2\varphi,
 \end{aligned}$$

其中 $(1 - |\cos \theta|)$ 可利用 §.44 习题 2(3) 的展开结果,

$$\begin{aligned}
 (1 - |\cos \theta|) &= \frac{1}{2} + \sum_{n=1}^{\infty} (-1)^n \frac{(4n+1)(2n-2)!}{2^{2n}(n+1)!(n-1)!} \\
 & \quad \times p_{2n}(\cos \theta),
 \end{aligned}$$

我们还需要把 $(1 - |\cos \theta|) \cos 2\varphi$ 展开,

$$\text{令 } (1 - |\cos \theta|) \cos 2\varphi = \sum_k A_k p_k^2(\cos \theta) \cos 2\varphi,$$

$$\begin{aligned}
 \text{系数 } A_k &= \frac{2k+1}{2\pi} \frac{(k-2)!}{(k+2)!} \int_{-1}^1 (1 - |x|) p_k^2(x) dx \\
 & \quad \times \int_0^{2\pi} \cos 2\varphi \cos 2\varphi d\varphi,
 \end{aligned}$$

对 x 迭次分部积分得

$$\begin{aligned}
 A_k &= \frac{2k+1}{2} \frac{(k-2)!}{(k+2)!} \frac{1}{2^k k!} \int_{-1}^1 (1 - |x|)(1 - x^2) \\
 & \quad \times \frac{d^{k+2}}{dx^{k+2}} (x^2 - 1)^k dx,
 \end{aligned}$$

如 k 为奇数, 则被积函数为奇函数, 积分为零, 即

$$A_k = 0, (k \text{ 为奇数}),$$

至于 $k = \text{偶数 } 2n$,

$$\text{则 } A_{2n} = \frac{(4n+1)}{2} \frac{(2n-2)!}{(2n+2)!} \frac{2}{2^{2n}(2n)!} \int_0^1 (1-x)(1-x^2)$$

$$\begin{aligned}
& \times \frac{d^{2n+2}}{dx^{2n+2}} (x^2 - 1)^{2n} dx \\
& = \frac{(4n+1)(2n-2)!}{(2n+2)! 2^{2n} (2n)!} \left\{ \left[(1-x)(1-x^2) \frac{d^{2n+1}}{dx^{2n+1}} \right. \right. \\
& \quad \left. \left. (x^2 - 1)^{2n} \right]_0^1 \right. \\
& \quad \left. - \int_0^1 \frac{d^{2n+1}}{dx^{2n+1}} (x^2 - 1)^{2n} d[(1-x)(1-x^2)] \right\}.
\end{aligned}$$

$\frac{d^{2n+1}}{dx^{2n+1}} (x^2 - 1)^{2n}$ 只有奇次幂, 以 $x = 0$ 代入必为零,

$$\begin{aligned}
\therefore A_{2n} &= \frac{(4n+1)(2n-2)!}{(2n+2)! 2^{2n} (2n)!} \int_0^1 (1-x)(3x+1) \frac{d^{2n+1}}{dx^{2n+1}} \\
& \quad \times (x^2 - 1)^{2n} dx \\
&= \frac{(4n+1)(2n-2)!}{(2n+2)! 2^{2n} (2n)!} \left\{ \left[(1-x)(3x+1) \frac{d^{2n}}{dx^{2n}} \right. \right. \\
& \quad \left. \left. (x^2 - 1)^{2n} \right]_0^1 \right. \\
& \quad \left. + 2 \int_0^1 (3x-1) \frac{d^{2n}}{dx^{2n}} (x^2 - 1)^{2n} dx \right\} \\
&= \frac{(4n+1)(2n-2)!}{(2n+2)! 2^{2n} (2n)!} \left[-(-1)^n 2^{2n} (2n)! \frac{(2n-1)!!}{(2n)!!} \right. \\
& \quad \left. + 6 \frac{(2n-2)!(2n)!(-1)^{n+1}}{(n+1)!(n-1)!} \right] \\
&= -(-1)^n \frac{(4n+1)(2n-2)!(2n-1)!!}{(2n+2)!(2n)!!} \\
& \quad \times \left[1 + \frac{6}{(2n-1)(2n+2)} \right],
\end{aligned}$$

$$\therefore (1 - |\cos \theta|)(1 + \cos 2\varphi) = \frac{1}{2} + \sum_{n=1}^{\infty} (-1)^n$$

$$\begin{aligned} & \times \frac{(4n+1)(2n-2)!}{(2n+2)!!(2n-2)!!} p_{2n}(\cos\theta) \\ & - \sum_{n=1}^{\infty} (-1)^n (4n+1) \left[1 + \frac{6}{(2n-1)(2n+2)} \right] \\ & \frac{(2n-2)!(2n-1)!!}{(2n+2)!(2n)!!} p_{2n}^2(\cos\theta) \cos 2\varphi. \end{aligned}$$

2. 在半径为 a 的球外 ($r > a$) 求解

$$\begin{cases} \Delta_3 u = 0, \\ u|_{r=a} = f(\theta, \varphi). \end{cases}$$

解: 球外问题的解为:

$$\begin{aligned} u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=0}^l \left(\frac{a}{r}\right)^{l+1} [A_l^m \cos m\varphi \\ + B_l^m \sin m\varphi] p_l^m(\cos\theta), \end{aligned} \quad (1)$$

由边界条件得:

$$\begin{aligned} u(a, \theta, \varphi) &= \sum_{l=0}^{\infty} \sum_{m=0}^l \left(\frac{a}{a}\right)^{l+1} [A_l^m \cos m\varphi \\ &+ B_l^m \sin m\varphi] p_l^m(\cos\theta) \\ &= f(\theta, \varphi), \end{aligned}$$

$$\therefore \begin{cases} A_l^m = \frac{(2l+1)}{2\pi\delta_m} \frac{(l-m)!}{(l+m)!} \iint f(\theta, \varphi) p_l^m(\cos\theta) \\ \quad \times \cos m\varphi \cdot \sin\theta d\theta d\varphi, \\ B_l^m = \frac{(2l+1)}{2\pi} \frac{(l-m)!}{(l+m)!} \iint f(\theta, \varphi) \\ \quad \times p_l^m(\cos\theta) \sin m\varphi \cdot \sin\theta d\theta d\varphi. \end{cases} \quad (2)$$

3. 在半径为 a 的球的 (1) 内部, (2) 外部, 求解:

$$\begin{cases} \Delta_3 u = 0, \\ \left. \frac{\partial u}{\partial r} \right|_{r=a} = f(\theta, \varphi). \end{cases}$$

研究一个特例 $f(\theta, \varphi) = A \cos\theta$.

解: (1) 半径为 a 的球的内部有限解为:

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=0}^l \left(\frac{r}{a}\right)^l [A_l^m \cos m\varphi + B_l^m \sin m\varphi] P_l^m(\cos\theta), \quad (1)$$

代入边界条件:

$$\begin{aligned} \left. \frac{\partial u}{\partial r} \right|_{r=a} &= \sum_{l=1}^{\infty} \sum_{m=0}^l \frac{l}{a} \cdot [A_l^m \cos m\varphi + B_l^m \sin m\varphi] p_l^m(\cos\theta). \\ \therefore \begin{cases} A_l^m = \frac{a}{l} \frac{(2l+1)}{2\pi\delta_m} \frac{(l-m)!}{(l+m)!} \iint f(\theta, \varphi) p_l^m(\cos\theta) \\ \quad \times \cos m\varphi \cdot \sin\theta d\theta d\varphi, \\ B_l^m = \frac{a}{l} \frac{(2l+1)}{2\pi} \frac{(l-m)!}{(l+m)!} \iint f(\theta, \varphi) p_l^m(\cos\theta) \\ \quad \times \sin m\varphi \cdot \sin\theta d\theta d\varphi, \end{cases} \quad (2) \end{aligned}$$

或者写为:

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=0}^l \frac{r^l}{l a^{l-1}} [A_l^m \cos m\varphi + B_l^m \sin m\varphi] p_l^m(\cos\theta), \quad (3)$$

其中系数 A_l^m 和 B_l^m 同上题,

当 $f(\theta, \varphi) = A \cdot \cos\theta$ 时, 知问题与 φ 无关, 即 $m=0$, 这时

$$u(r, \theta) = \sum_{l=0}^{\infty} A_l \left(\frac{r}{a}\right)^l p_l(\cos\theta),$$

代入边界条件:

$$\begin{aligned} \left. \frac{\partial u}{\partial r} \right|_{r=a} &= \sum_{l=0}^{\infty} A_l \cdot l \cdot a^{-1} p_l(\cos\theta) = A \cos\theta, \\ \therefore A_l &= \frac{a}{l} \cdot \frac{2l+1}{2} \int_{-1}^1 A p_l(x) \cdot x \cdot dx \\ &= \frac{Aa}{l} \frac{2l+1}{2} \int_{-1}^1 x p_l dx. \end{aligned}$$

$$\therefore \int_{-1}^1 x p_l(x) dx = \begin{cases} \int_{-1}^1 x p(x) dx = \int_{-1}^1 x^2 dx \\ = \frac{1}{3} (x^3) \Big|_{-1}^1 = \frac{2}{3}, (l=1), \\ 0, (l \neq 1), \end{cases}$$

$$\therefore A_1 = \frac{3Aa}{2} \cdot \frac{2}{3} = A, \quad A_l = 0, (l \neq 1),$$

故: $u(r, \theta) = Ar p_1(\cos \theta) = A r \cos \theta$.

(2) 外部的解:

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=0}^l \left(\frac{a}{r}\right)^{l+1} [A_l^m \cos m\varphi + B_l^m \sin m\varphi] p_l^m(\cos \theta), \quad (1)$$

代入边界条件:

$$\left. \frac{\partial u}{\partial r} \right|_{r=a} = \sum_{l=0}^{\infty} \sum_{m=0}^l \frac{-(l+1)}{a^{l+2}} [A_l^m \cos m\varphi + B_l^m \sin m\varphi] p_l^m(\cos \theta),$$

$$\therefore \left\{ \begin{aligned} A_l^m &= -\frac{a^{l+2}}{l+1} \cdot \frac{2l+1}{2\pi \delta_m} \cdot \frac{(l-m)!}{(l+m)!} \\ &\quad \times \iint f(\theta, \varphi) \cdot p_l^m(\cos \theta) \cdot \cos m\varphi \cdot \sin \theta d\theta d\varphi, \\ B_l^m &= -\frac{a^{l+2}}{l+1} \cdot \frac{2l+1}{2\pi} \cdot \frac{(l-m)!}{(l+m)!} \\ &\quad \times \iint f(\theta, \varphi) p_l^m(\cos \theta) \sin m\varphi \cdot \sin \theta \cdot d\theta d\varphi, \end{aligned} \right. \quad (2)$$

或者写为:

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=0}^l \frac{-a^{l+2}}{(l+1)r^{l+1}} [A_l^m \cos m\varphi + B_l^m \sin m\varphi] p_l^m(\cos \theta), \quad (3)$$

其中系数 A_l^m 和 B_l^m 见上题.

当 $f(\theta, \varphi) = A \cos \theta$ 时, 知问题与 φ 无关, 即有 $m=0$, 这时,

$$u(r, \theta) = \sum_{l=0}^{\infty} B_l \frac{1}{r^{l+1}} p_l(\cos \theta),$$

代入边界条件:

$$\left. \frac{\partial u}{\partial r} \right|_{r=a} = \sum_{l=0}^{\infty} \frac{-(l+1)B_l}{a^{l+2}} p_l = A \cos \theta.$$

$$\begin{aligned} \therefore B_l &= \frac{-a^{l+2}}{(l+1)} \cdot \frac{2l+1}{2} A \int_0^\pi p_l(\cos \theta) \cos \theta d\theta \\ &= \begin{cases} -\frac{a^3 A}{2}, (l=1), \\ 0, (l \neq 1), \end{cases} \end{aligned}$$

$$\begin{aligned} \therefore u(r, \theta) &= -A \frac{a^3}{2r^2} p_1(\cos \theta) \\ &= -\frac{a^3 A}{2} \frac{1}{r^2} \cos \theta. \end{aligned}$$

第十四章 柱函数

§46. 贝塞耳函数

1. 计算下列积分

$$(1) \int x^3 J_0(x) dx.$$

$$\text{解一: } \because \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x),$$

$$\text{即有 } xJ_0(x) = [xJ_1(x)]', x^2J_1(x) = [x^2J_2(x)]',$$

$$\begin{aligned} \therefore \int x^3 J_0(x) dx &= \int x^2 [xJ_0(x)] dx \\ &= \int x^2 [xJ_1(x)]' dx \\ &= x^3 J_1(x) - \int 2x^2 J_1(x) dx \\ &= x^3 J_1(x) - 2x^2 J_2(x) + C. \end{aligned}$$

$$\text{解二: } \because J_1(x) = -J_0'(x),$$

$$\begin{aligned} \therefore \int x^3 J_0(x) dx &= x^3 J_1(x) - \int 2x^2 J_1(x) dx \\ &= x^3 J_1(x) + \int 2x^2 J_0'(x) dx \\ &= x^3 J_1(x) + 2x^2 J_0(x) - \int 4x J_0(x) dx \\ &= x^3 J_1(x) + 2x^2 J_0(x) - 4 \int [xJ_1(x)]' dx \\ &= x^3 J_1(x) + 2x^2 J_0(x) - 4xJ_1(x) + C, \end{aligned}$$

由 $J_2(x) = \frac{2J_1(x)}{x} - J_0(x)$,

知解二与解法一结果相同.

(2) $\int x^4 J_1(x) dx$.

解一: 利用公式 $\frac{d}{dx}[x^n J_n] = x^n J_{n-1}$,

$$\begin{aligned}\int x^4 J_1 dx &= \int x^2 [x^2 J_1] dx \\ &= x^4 J_2 - 2 \int x^3 J_2 dx \\ &= x^4 J_2 - 2x^3 J_3 + C.\end{aligned}$$

解二: 利用公式 $J_1 = -J'_0$,

$$\begin{aligned}\int x^4 J_1 dx &= -\int x^4 J'_0 dx \\ &= -x^4 J_0 + 4 \int x^3 J_0 dx = -x^4 J_0 + 4 \int x^2 [x J_0] dx \\ &= -x^4 J_0 + 4x^3 J_1 - 4 \int 2xx J_1 dx \\ &= -x^4 J_0 + 4x^3 J_1 - 8 \int x^2 J_1 dx = -x^4 J_0 + 4x^3 J_1 \\ &\quad - 8x^2 J_2 + C,\end{aligned}$$

解三: 对上式 $-8 \int x^2 J_1 dx$ 再利用公式 $J_1 = -J'_0$,

$$\begin{aligned}\therefore -8 \int x^2 J_1 dx &= 8 \int x^2 J'_0 dx \\ &= 8x^2 J_0 - 16 \int x J_0 dx \\ &= 8x^2 J_0 - 16x J_1 + C,\end{aligned}$$

$$\begin{aligned}\therefore \int x^4 J_1 dx &= -x^4 J_0 + 4x^3 J_1 + 8x^2 J_0 - 16x J_1 + C \\ &= (8x^2 - x^4) J_0 + (4x^3 - 16x) J_1 + C.\end{aligned}$$

$$(3) \int J_3(x) dx$$

解: 利用公式 $\frac{d}{dx} \left[\frac{J_m}{x^m} \right] = -\frac{J_{m+1}}{x^m}$,

$$\begin{aligned} \int J_3 dx &= \int x^2 [x^{-2} J_3] dx = \int x^2 [x^{-2} J_2]' dx \\ &= -J_2 + \int 2x^{-1} J_2 dx \\ &= -J_2 - 2x^{-1} J_1 + C, \end{aligned}$$

由递推公式 $J_{n+1} - \frac{2nJ_n}{x} + J_{n-1} = 0$, 有 $J_2 = \frac{2J_1}{x} - J_0$,

$$\therefore \int J_3 dx = J_0 - \frac{2J_1}{x} - 2 \frac{J_1}{x} + C = J_0 - 4 \frac{J_1}{x} + C.$$

2. 在区间 $(0, 1)$ 上, 第一类齐次边界条件下, 用零阶贝塞耳函数把 $f(x) = 1$ 展开为傅里叶—贝塞耳级数.

解: 展开公式是:

$$\begin{cases} f(\rho) = \sum_{n=1}^{\infty} f_n J_n \left(\frac{x_n^{(m)}}{\rho_0} \rho \right), \\ \text{系数 } f_n = \frac{1}{[N_n^{(m)}]^2} \int_0^{\rho_0} f(\rho) J_n \left(\frac{x_n^{(m)}}{\rho_0} \rho \right) \rho d\rho, \end{cases}$$

这里可设 $x = \rho$, 则 $f(x) = f(\rho) = 1$, 同时 $\rho_0 = 1$, 又第一类齐次边界条件, $[N_n^{(0)}]^2 = \frac{1}{2} \cdot 1 \cdot [J_1(x_n^{(0)})]^2$,

$$\begin{aligned} \therefore f_n &= \frac{2}{[J_1(x_n^{(0)})]^2} \int_0^1 J_0(x_n^{(0)} \rho) \rho d\rho \\ &= \frac{2}{[J_1(x_n^{(0)})]^2} \cdot \frac{1}{(x_n^{(0)})^2} \int_0^1 (x_n^{(0)} \rho) \\ &\quad \times J_0(x_n^{(0)} \rho) d(x_n^{(0)} \rho) \\ &= \frac{2}{[x_n^{(0)} J_1(x_n^{(0)})]^2} \cdot (x_n^{(0)} \rho) J_1(x_n^{(0)} \rho) \Big|_0^1 \end{aligned}$$

$$= \frac{2}{x_n^{(0)} J_1(x_n^{(0)})},$$

故

$$\begin{aligned} 1 &= 2 \sum_{n=1}^{\infty} \frac{J_0(x_n^{(0)} \rho)}{x_n^{(0)} J_1(x_n^{(0)})} \\ &= 2 \sum_{n=1}^{\infty} \frac{J_0(x_n^{(0)} x)}{x_n^{(0)} J_1(x_n^{(0)})}, \end{aligned}$$

其中 $x_n^{(0)}$ 是 $J_0(x)$ 的第 n 个零点.

3. 求解半径为 R 的圆形膜的稳恒振动. 每单位面积上作用的周期力为

$$(1) \quad f = A \sin \omega t,$$

$$(2) \quad f = A(1 - \rho^2/R^2) \sin \omega t.$$

解: 设薄膜均匀, 单位面积的质量为 p , 则均匀圆形膜的受迫横振动方程为:

$$u_{tt} - a^2 u = f/p, \quad (a^2 = T/p),$$

取极坐标系, 以圆形膜中心为极点, 则振动与 φ 无关, 从而方程成为:

$$u_{tt} - a^2 \left(u_{\rho\rho} + \frac{1}{\rho} u_{\rho} \right) = f/p,$$

求解在周期力作用下的稳恒振动, 即是“没有初始条件”的问题, 振动的周期与外力周期相同. 可设

$$u(\rho, t) = v(\rho) \sin \omega t,$$

代入方程得:

$$\left[\omega^2 v + a^2 \left(v_{\rho\rho} + \frac{1}{\rho} v_{\rho} \right) \right] \sin \omega t = -f/p. \quad (1)$$

$$(1) \quad f = A \sin \omega t.$$

这时有:

$$\begin{cases} v'' + \frac{1}{\rho} v' + \frac{\omega^2}{a^2} v = -\frac{A}{a^2 p}, & (2) \end{cases}$$

$$\begin{cases} v|_{\rho=R} = 0, \text{ (圆膜边缘固定)}. & (3) \end{cases}$$

上面是非齐次方程，不能直接求解。但非齐次项是常数，故可

$$\text{令 } W = v + \frac{A}{a^2 p}, \text{ 及 } x = \frac{\omega}{a} \rho,$$

则定解问题(2)和(3)变成

$$\begin{cases} W'' + \frac{1}{x}W' + W = 0, & (4) \end{cases}$$

$$\begin{cases} W \Big|_{x = \frac{\omega}{a} R} = \frac{A}{a^2 p}, & (5) \end{cases}$$

方程(4)是零阶贝塞尔方程，在圆内的有限解为：

$$W(x) = C J_0(x),$$

代入边界条件(5)：

$$C J_0\left(\frac{\omega}{a} R\right) = \frac{A}{a^2 p},$$

$$\therefore C = A/a^2 p J_0\left(\frac{\omega}{a} R\right),$$

$$\text{于是 } W = \left[A/a^2 p J_0\left(\frac{\omega}{a} R\right) \right] J_0(x),$$

$$\text{即 } v = \frac{A}{a^2 p} \left[\frac{J_0\left(\frac{\omega}{a} \rho\right)}{J_0\left(\frac{\omega}{a} R\right)} - 1 \right],$$

$$\therefore u = \frac{A}{a^2 p} \left[\frac{J_0\left(\frac{\omega}{a} \rho\right)}{J_0\left(\frac{\omega}{a} R\right)} - 1 \right] \sin \omega t. \quad (6)$$

$$(2) \quad f = A \left(1 - \frac{\rho^2}{R^2} \right) \sin \omega t,$$

这时有

$$\begin{cases} v'' + \frac{1}{\rho} v' + \frac{\omega^2}{a^2} v = -\frac{A}{a^2 p} \left(1 - \frac{\rho^2}{R^2} \right), & (7) \end{cases}$$

$$\begin{cases} v|_{\rho=R} = 0. & (8) \end{cases}$$

非齐次方程(7)有形如 $\lambda\rho^2 + \mu$ 的特解, λ, μ 为待定系数. 以特解代入方程(7)得:

$$2\lambda + 2\lambda + \frac{\omega^2}{a^2} \lambda \rho + \frac{\omega^2}{a^2} \mu + \frac{A}{a^2 p} \left(1 - \frac{\rho^2}{R} \right) = 0,$$

即有

$$\begin{cases} 4\lambda + \frac{\omega^2}{a^2} \mu + \frac{A}{a^2 p} = 0, \\ \frac{\omega^2}{a^2} \lambda - \frac{A}{a^2 p R} = 0, (\because \rho \text{ 不恒为零}), \end{cases}$$

解得:

$$\begin{cases} \lambda = \frac{A}{p R^2 \omega^2}, \end{cases} \quad (9)$$

$$\begin{cases} \mu = -\frac{A}{p \omega^2} - \frac{4 A a^2}{p R^2 \omega^4}, \end{cases} \quad (10)$$

于是可令 $v = W + \lambda\rho^2 + \mu$, $x = \frac{\omega}{a} \rho$,

则方程(7)成为:

$$W'' + \frac{1}{x} W' + W = 0, \quad (11)$$

方程(11)是零阶贝塞尔方程, 在圆内的有限解为:

$$W = C J_0(x),$$

则

$$v = C J_0\left(\frac{\omega}{a} \rho\right) + \lambda\rho^2 + \mu,$$

代入边界条件(8):

$$C J_0\left(\frac{\omega}{a} R\right) + \frac{A}{p R^2 \omega^2} R^2 - \frac{A}{p \omega^2} - \frac{4 A a^2}{p R^2 \omega^4} = 0,$$

即

$$C = \frac{4 A a^2}{p R^2 \omega^4 J_0\left(\frac{\omega}{a} R\right)},$$

$$\therefore v = \frac{4Aa^2}{pR^2\omega^4} \left(\frac{J_0\left(\frac{\omega}{a}\rho\right)}{J_0\left(\frac{\omega}{a}R\right)} - 1 \right) + \frac{A}{p\omega^2} \left(\frac{\rho^2}{R^2} - 1 \right), \quad (12)$$

故 $u(\rho, t) = v(\rho) \sin \omega t,$ (13)

其中 v 由 (12) 式表出.

4. 半径为 R 的圆形膜, 边缘固定. 初始形状是旋转抛物面. $u|_{t=0} = (1 - \rho^2/R^2)H$, 初速为零. 求解膜的振动情况.

解: 取以圆形膜中心为极点的极坐标系. 由于定解条件与 φ 无关, 因而问题亦与 φ 无关. 于是定解问题是:

$$u_{tt} - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho}) = 0, \quad (1)$$

$$\begin{cases} u|_{\rho=0} \text{有限}, \\ u|_{\rho=R} = 0. \end{cases} \quad (2)$$

$$\begin{cases} u|_{t=0} = H\left(1 - \frac{\rho^2}{R^2}\right), \\ u_t|_{t=0} = 0. \end{cases} \quad (3)$$

令 $u(\rho, t) = U(\rho)T(t)$ 代入泛定方程 (1) 和边界条件 (2) 分离变数得: $T'' + \lambda^2 a^2 T = 0,$ (4)

$$U'' + \frac{1}{\rho}U' + \lambda^2 U = 0, \quad (5)$$

$$\begin{cases} U|_{\rho=0} \text{有限}, \\ U|_{\rho=R} = 0, \end{cases} \quad (6)$$

方程 (5) 是零阶贝塞耳方程, 在圆内的有限解是:

$$U = C \cdot J_0(\lambda\rho),$$

代入边界条件 (6) 的第二条:

$$J_0(\lambda R) = 0, \text{ 令 } x^{(0)}_n \text{ 为 } J_0(x) \text{ 的第 } n \text{ 个零点, } (n=1, 2, \dots),$$

则

$$\lambda_n^{(0)} = \frac{x_n^{(0)}}{R},$$

于是

$$v_n = J_0\left(\frac{x_n^{(0)}}{R}\rho\right),$$

方程 (4) 的解为:

$$\begin{aligned} T_n &= A_n \cos \frac{x_n^{(0)}}{R} at + B_n \sin \frac{x_n^{(0)}}{R} at, \\ \therefore u &= \sum_{n=1}^{\infty} \left(A_n \cos \frac{x_n^{(0)}}{R} at + B_n \sin \frac{x_n^{(0)}}{R} at \right) J_0 \\ &\quad \times \left(\frac{x_n^{(0)}}{R} \rho \right), \end{aligned} \quad (7)$$

由初始条件 (3), $u_t|_{t=0} = 0$, 知 $B_n = 0$ 及

$$\sum_{n=1}^{\infty} A_n J_0\left(\frac{x_n^{(0)}}{R}\rho\right) = H\left(1 - \frac{\rho^2}{R^2}\right),$$

$$\therefore A_n = \frac{2H}{R^2 [J_1(x_n^{(0)})]^2} \int_0^R \left(1 - \frac{\rho^2}{R^2}\right) J_0\left(\frac{x_n^{(0)}}{R}\rho\right) \rho d\rho,$$

$$\begin{aligned} \therefore \int_0^R J_0\left(\frac{x_n^{(0)}}{R}\rho\right) \rho d\rho &= \frac{R^2}{[x_n^{(0)}]^2} \int_0^{x_n^{(0)}} J_0(x) x dx \\ &= \frac{R^2}{x_n^{(0)}} J_1(x_n^{(0)}), \end{aligned}$$

$$\begin{aligned} \int_0^R \rho^2 J_0\left(\frac{x_n^{(0)}}{R}\rho\right) \rho d\rho &= \frac{R^4}{[x_n^{(0)}]^4} \int_0^{x_n^{(0)}} x^3 J_0(x) dx \\ &= \frac{R^4}{[x_n^{(0)}]^4} [x^3 J_1(x) \\ &\quad + 2x^2 J_0(x) - 4x J_1(x)] \Big|_0^{x_n^{(0)}} \\ &= \frac{R^4}{[x_n^{(0)}]^4} J_1(x_n^{(0)}) \\ &\quad - 4 \frac{R^4}{[x_n^{(0)}]^3} J_1(x_n^{(0)}), \end{aligned}$$

$$\begin{aligned} \therefore A_n &= \frac{2H}{R^2 [J_1(x_n^{(0)})]^2} \left[\frac{R^2}{x_n^{(0)}} J_1(x_n^{(0)}) \right. \\ &\quad \left. - \frac{R^2}{x_n^{(0)}} J_1(x_n^{(0)}) + \frac{4R^2}{[x_n^{(0)}]^3} J_1(x_n^{(0)}) \right] \\ &= \frac{8H}{[x_n^{(0)}]^3 J_1(x_n^{(0)})}, \end{aligned} \quad (8)$$

$$\begin{aligned} \text{故 } u(\rho, t) &= 8H \sum_{n=1}^{\infty} \frac{1}{[x_n^{(0)}]^3 J_1(x_n^{(0)})} J_0\left(\frac{x_n^{(0)}}{R} \rho\right) \\ &\quad \times \cos \frac{x_n^{(0)}}{R} at. \end{aligned} \quad (9)$$

5. 半径为 R 的圆形膜, 在 ρ_0, φ_0 受到冲量 K 作用, 求解其后的振动.

解: 膜在 (ρ_0, φ_0) 点受冲量 K 作用, 可用 δ -函数来表示, 即 $K \cdot \delta(\rho - \rho_0) \cdot \delta(\varphi - \varphi_0)$, 其冲量密度为

$$K \cdot \delta(\rho - \rho_0) \cdot \delta(\varphi - \varphi_0) / \rho_0,$$

由于冲量仅作用在起始时刻 $t = 0$, 因此也就是 $t = 0$ 时的动量. 设单位面积质量为 p , 则膜受到的初始速度为:

$$u_t|_{t=0} = \frac{k}{p\rho_0} \delta(\rho - \rho_0) \cdot \delta(\varphi - \varphi_0). \text{ 于是定解问题为:}$$

$$u_{tt} - a^2 \Delta_2 u = 0, \quad (1)$$

$$\begin{cases} u|_{\rho=R} = 0, \text{ (膜边缘固定),} \\ u|_{\rho=0} \text{ 有限,} \end{cases} \quad (2)$$

$$\begin{cases} u|_{t=0} = 0, \text{ (初位移为零),} \\ u_t|_{t=0} = \frac{k}{p\rho_0} \delta(\rho - \rho_0) \cdot \delta(\varphi - \varphi_0), \end{cases} \quad (3)$$

方程 (1) 在圆内的有限解为:

$$u(\rho, \varphi, t) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \left(A_n \cos \frac{x_n^{(m)}}{R} at + B_n \sin \frac{x_n^{(m)}}{R} at \right)$$

$$\times (C_m \cos m\varphi + D_m \sin m\varphi) J_m \left(\frac{x_n^{(m)}}{R} \rho \right),$$

其中 $x_n^{(m)}$ 是 m 阶贝塞尔函数的第 n 个零点。

由 $u|_{t=0} = 0$, 知 $A_n = 0$,

由 $u_t|_{t=0} = \frac{k}{p\rho_0} \delta(\rho - \rho_0) \delta(\varphi - \varphi_0)$,

$$\begin{aligned} \text{有 } \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{x_n^{(m)}}{R} a (C_{m,n} \cos m\varphi + D_{m,n} \sin m\varphi) J_m \left(\frac{x_n^{(m)}}{R} \rho \right) \\ = \frac{k}{p\rho_0} \delta(\rho - \rho_0) \delta(\varphi - \varphi_0), \end{aligned}$$

$$\begin{aligned} \therefore C_{m,n} &= \frac{k}{p\rho_0} \cdot \frac{R}{x_n^{(m)} a} \cdot \frac{1}{\pi \delta_m} \cdot \frac{2}{R^2 [J'_m(x_n^{(m)})]^2} \\ &\quad \times \int_0^{\rho_0} \delta(\rho - \rho_0) \cdot J_m \left(\frac{x_n^{(m)}}{R} \rho \right) \rho d\rho \\ &\quad \times \int_0^{\varphi_0} \delta(\varphi - \varphi_0) \cos m\varphi d\varphi \\ &= \frac{2k}{pa\pi R \delta_m x_n^{(m)}} \cdot \frac{J_m \left(\frac{x_n^{(m)}}{R} \rho_0 \right)}{[J'_m(x_n^{(m)})]^2} \cos m\varphi_0. \end{aligned}$$

同理

$$D_{m,n} = \frac{2k}{pa\pi R x_n^{(m)}} \cdot \frac{J_m \left(\frac{x_n^{(m)}}{R} \rho_0 \right)}{[J'_m(x_n^{(m)})]^2} \sin m\varphi_0,$$

故:

$$\begin{aligned} u &= \frac{2k}{pa\pi R} \cdot \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{J_m \left(\frac{x_n^{(m)}}{R} \rho_0 \right)}{\delta_m x_n^{(m)} [J'_m(x_n^{(m)})]^2} \\ &\quad \cdot (\cos m\varphi \cdot \cos m\varphi_0 + \sin m\varphi \cdot \sin m\varphi_0) J_m \\ &\quad \times \left(\frac{x_n^{(m)}}{R} \rho \right) \sin \frac{x_n^{(m)}}{R} at, \end{aligned}$$

$$\text{即 } u = \frac{2k}{pa\pi R} \cdot \sum_{n=0}^{\infty} \sum_{m=1}^{\infty} \frac{J_m\left(\frac{x_n^{(m)}}{R}\rho_0\right)}{\delta_m x_n^{(m)} [J'_m(x_n^{(m)})]^2} \\ \cdot \cos(\varphi - \varphi_0) J_m\left(\frac{x_n^{(m)}}{R}\rho\right) \sin \frac{x_n^{(m)}}{R} at. \quad (4)$$

6. 半径为 R 的半圆形膜, 边缘固定, 求其本征频率和本征振动.

解: 采用极坐标, 定解问题为:

$$u_{tt} - a^2 \Delta_2 u = 0, \quad (0 \leq \varphi \leq \pi, \quad 0 \leq \rho \leq R), \quad (1)$$

$$\begin{cases} u|_{\rho=0} \text{ 有限,} \\ u|_{\rho=R} = 0, \end{cases} \quad (2)$$

$$u|_{\varphi=0} = u|_{\varphi=\pi} = 0. \quad (3)$$

令 $u(\rho, \varphi, t) = v(\rho)\Phi(\varphi)T(t)$ 代入定解问题, 进行分离变数得:

$$\begin{cases} \Phi'' + m^2\Phi = 0, \\ \Phi|_{\varphi=0} = \Phi|_{\varphi=\pi} = 0, \end{cases} \quad (4)$$

$$\begin{cases} v'' + \frac{1}{\rho}v' + \left(\lambda^2 - \frac{m^2}{\rho^2}\right)v = 0, \\ v|_{\rho=0} \text{ 有限,} \\ v|_{\rho=R} = 0, \end{cases} \quad (5)$$

$$T'' + \lambda^2 a^2 T = 0, \quad (6)$$

本征问题 (4) 的解

$$\Phi = A_1 \cos m\varphi + A_2 \sin m\varphi,$$

由 $\Phi|_{\varphi=0} = 0$ 知 $A_1 = 0$,

由 $\Phi|_{\varphi=\pi} = 0$ 有 $A_2 \sin m\varphi = 0$,

$\therefore A_2 \neq 0$ (否则得零解),

$\therefore m = 0, 1, 2, \dots$,

则 $\Phi_m = A_m \sin m\varphi$,

本征问题 (5) 的在圆内的有限解:

$$v = B_1 J_m(\lambda \rho),$$

由 $v|_{\rho=R} = 0$, 有 $J_m(\lambda R) = 0$,

令 $x_n^{(m)} = \lambda_n^{(m)} R$ 为 $J_m(\lambda \rho)$ 的第几个零点,

$$\therefore \lambda_n^{(m)} = \frac{x_n^{(m)}}{R}, \quad (n = 1, 2, \dots),$$

则
$$v_n = B_n J_m\left(\frac{x_n^{(m)}}{R} \rho\right),$$

方程 (6) 的解为

$$T_{n,m} = C \cdot \cos \frac{x_n^{(m)}}{R} at + D \cdot \sin \frac{x_n^{(m)}}{R} at,$$

$\therefore$ 本征振动为:

$$\begin{aligned} u_{n,m} = & \left[A_{n,m} \cos \frac{x_n^{(m)}}{R} at + B_{n,m} \sin \frac{x_n^{(m)}}{R} at \right] J_m \\ & \times \left(\frac{x_n^{(m)}}{R} \rho \right) \sin m\varphi, \end{aligned} \quad (7)$$

本征圆频率为: $\omega_{n,m} = \frac{a}{R} x_n^{(m)}. \quad (8)$

7. 半径为 R 而高为 H 的圆柱体下底和侧面保持零度, 上底温度分布为 $f(\rho) = \rho^2$. 求柱体内各点的稳恒温度.

解: 定解问题为:

$$\Delta u = 0 \quad (1)$$

$$\begin{cases} u|_{z=0} = 0, \\ u|_{z=H} = \rho^2, \end{cases} \quad (2)$$

$$\begin{cases} u|_{\rho=0} \text{ 有限}, \\ u|_{\rho=R} = 0, \end{cases} \quad (3)$$

取柱坐标系, 以柱下底面为 $z = 0$ 的坐标面, 以柱轴为 z 轴. 由边界条件知问题与 φ 无关 (即 $m = 0$),

因侧面是第一类齐次边界条件, 故其本征值 λ 由 $J_0(\lambda R) = 0$ 决定.

即 $\lambda_n^{(0)} = \frac{x_n^{(0)}}{R}$, ($n=1, 2, \dots$),

其中 $x_n^{(0)}$ 为 $J_0(x)$ 的第 n 个零点。

柱内问题的有限解为:

$$u = \sum_{n=1}^{\infty} \left(A_n e^{\frac{x_n^{(0)}}{R} Z} + B_n e^{-\frac{x_n^{(0)}}{R} Z} \right) J_0(\lambda_n^{(0)} \rho),$$

由边界条件 (2), 得:

$$\begin{cases} \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(0)}}{R} \rho\right) = 0, \\ \sum_{n=1}^{\infty} \left(A_n e^{\frac{x_n^{(0)}}{R} H} + B_n e^{-\frac{x_n^{(0)}}{R} H} \right) J_0\left(\frac{x_n^{(0)}}{R} \rho\right) = \rho^2. \end{cases} \quad (4)$$

由 (4) 得: $B_n = -A_n$,

$$\begin{aligned} \text{由 (5) 得: } A_n &= \frac{1}{R^2 \cdot [J_1(x_n^{(0)})]^2 \cdot \text{sh}\left(\frac{x_n^{(0)}}{R} H\right)} \\ &\quad \times \int_0^R \rho^2 J_0\left(\frac{x_n^{(0)}}{R} \rho\right) \rho d\rho, \end{aligned}$$

$$\begin{aligned} \therefore \text{积分 } \int_0^R \rho^3 J_0\left(\frac{x_n^{(0)}}{R} \rho\right) d\rho \\ &= \frac{R^4}{[x_n^{(0)}]^4} \left[x^3 J_1(x) + 2x^2 J_0(x) - 4x J_1(x) \right]_0^{x_n^{(0)}} \\ &= \frac{R^4}{x_n^{(0)}} J_1(x_n^{(0)}) - \frac{4R^4}{[x_n^{(0)}]^3} J_1(x_n^{(0)}), \end{aligned}$$

$$\begin{aligned} \therefore A_n &= \frac{R^2}{x_n^{(0)} \cdot J_1(x_n^{(0)}) \cdot \text{sh}(x_n^{(0)} H/R)} \\ &\quad \cdot \left[1 - \frac{4}{(x_n^{(0)})^2} \right], \end{aligned}$$

$$\text{故 } u(\rho, z) = 2R^2 \sum_{n=1}^{\infty} \frac{J_0(x_n^{(0)} \rho/R) \cdot \text{sh}(x_n^{(0)} Z/R)}{x_n^{(0)} J_1(x_n^{(0)}) \cdot \text{sh}(x_n^{(0)} H/R)}$$

$$\times \left[1 - \frac{4}{(x^{(0)})^2} \right]. \quad (6)$$

8. 圆柱体半径为 R , 高为 H , 上底保持温度 u_1 , 下底保持温度 u_2 , 侧面温度分布为

$$f(z) = \frac{2u_1}{H^2} \left(Z - \frac{H}{2} \right) Z + \frac{u_2}{H} (H - Z). \quad \text{求柱内各点的稳定温度.}$$

度.

$$\text{解: 定解问题为: } \Delta u = 0, \quad (1)$$

$$\begin{cases} u|_{\rho=R} = \frac{2u_1}{H^2} \left(Z - \frac{H}{2} \right) Z + \frac{u_2}{H} (H - Z), \end{cases} \quad (2)$$

$$\begin{cases} u|_{z=0} = u_2, \end{cases} \quad (3)$$

$$\begin{cases} u|_{z=H} = u_1. \end{cases} \quad (4)$$

解一: 这里边界条件全是非齐次, 不能直接求解. 考虑到计算简单, 我们化上下底为齐次边界.

$$\text{令 } u = u_2 + \frac{u_1 - u_2}{H} z + v, \quad (5)$$

则 v 的定解问题为:

$$\Delta v = 0, \quad (6)$$

$$\begin{cases} v|_{z=0} = 0, \end{cases} \quad (7)$$

$$\begin{cases} v|_{z=H} = 0, \end{cases} \quad (8)$$

$$\begin{cases} v|_{\rho=R} = \frac{2u_1}{H^2} z^2 - \frac{2u_1}{H} z, \end{cases} \quad (9)$$

因为上下底是齐次边界, 所以本征问题是:

$$\begin{cases} Z = A_1 \cosh z + A_2 \sinh z, \\ Z|_{z=0} = 0, \\ Z|_{z=H} = 0, \end{cases}$$

$$\text{即有 } A_1 = 0. \quad h_n = \frac{n\pi}{H}, \quad (n=1, 2, \dots), \quad Z_n = A_n \sin \frac{n\pi}{H} Z,$$

又因问题与 φ 无关 (即 $m=0$), 所以柱内的有限解为:

$$v = \sum_{n=1}^{\infty} A_n I_0\left(\frac{n\pi}{H} \rho\right) \sin \frac{n\pi}{H} Z,$$

代入条件 (9), 得:

$$\begin{aligned} \sum_{n=1}^{\infty} A_n I_0\left(\frac{n\pi R}{H}\right) \sin \frac{n\pi}{H} Z &= \frac{2u_1}{H^2} Z^2 - \frac{2u_1}{H} Z, \\ \therefore A_n &= \frac{2}{H} \cdot \frac{1}{I_0(n\pi R/H)} \int_0^H \left(\frac{2u_1}{H^2} Z^2 - \frac{2u_1}{H} Z \right) \\ &\quad \times \sin \frac{n\pi}{H} Z dz, \end{aligned}$$

$$\begin{aligned} \therefore \int_0^H z^2 \cdot \sin \frac{n\pi z}{H} \cdot dz &= \left\{ \left(\frac{H}{n\pi} \right)^2 z \left[2 \sin \frac{n\pi z}{H} \right. \right. \\ &\quad \left. \left. - \frac{n\pi}{H} z \cos \frac{n\pi z}{H} \right] + \left(\frac{H}{n\pi} \right)^3 \cdot 2 \cos \frac{n\pi z}{H} \right\} \Big|_0^H \\ &= (-1)^{n+1} \frac{H^3}{n\pi} + \frac{2H^3}{(n\pi)^3} [(-1)^n - 1], \\ &\quad - \int_0^H z \sin \frac{n\pi z}{H} dz \\ &= - \left[\left(\frac{H}{n\pi} \right)^2 \sin \frac{n\pi z}{H} - \left(\frac{H}{n\pi} \right) \cos \frac{n\pi z}{H} \right] \Big|_0^H \\ &= (-1)^n \frac{H^2}{n\pi}, \end{aligned}$$

$$\begin{aligned} \therefore A_n &= \frac{4u_1}{H I_0(n\pi R/H)} \left[(-1)^{n+1} \frac{H}{n\pi} + \frac{2H}{(n\pi)^3} \right. \\ &\quad \left. \times \left((-1)^n - 1 \right) + (-1)^n \frac{H}{n\pi} \right] \\ &= \begin{cases} \frac{-16u_1}{(n\pi)^3 I_0(n\pi R/H)}, & (n = 2k+1, k = 0, 1, 2, \dots), \\ 0, & (n = 2k, k = 0, 1, 2, \dots), \end{cases} \end{aligned}$$

故: $u = u_2 + \frac{u_1 - u_2}{H} z$

$$- \sum_{k=0}^{\infty} \frac{16u_1}{(2k+1)^3 \pi^3 \cdot I_0[(2k+1)\pi R/H]} I_0 \times \left(\frac{2k+1}{H} \pi \rho \right) \sin \frac{2k+1}{H} \pi z. \quad (10)$$

解二：令 $u = v_1 + v_2$ ，将定解问题化为：

$$\text{I} \begin{cases} \Delta v_1 = 0, \\ v_1|_{z=0} = 0, \\ v_1|_{z=H} = 0, \\ v_1|_{\rho=R} = f(z), \end{cases} \quad \text{及 II} \begin{cases} \Delta v_2 = 0, \\ v_2|_{z=0} = u_2, \\ v_2|_{z=H} = u_1, \\ v_2|_{\rho=R} = 0, \end{cases}$$

定解问题 I 的解为：

$$v_1 = \sum_{n=1}^{\infty} C_n I_0 \left(\frac{n\pi}{H} \rho \right) \sin \frac{n\pi}{H} z, \quad (1)$$

由边界条件， $v_1|_{\rho=R} = f(\rho)$ ，有

$$\begin{aligned} \sum_{n=1}^{\infty} C_n I_0 \left(\frac{n\pi R}{H} \right) \cdot \sin \frac{n\pi}{H} z &= \frac{2u_1}{H^2} \left(z - \frac{H}{2} \right) + \frac{u_2}{H} (H - z), \\ \therefore C_n &= \frac{2}{H I_0(n\pi R/H)} \int_0^H \left[\frac{2u_1}{H^2} \left(z - \frac{H}{2} \right) + \frac{u_2}{H} (H - z) \right] \sin \frac{n\pi}{H} z \cdot dz \\ &= \frac{2}{H I_0(n\pi R/H)} \left\{ \frac{2u_1 H}{n\pi} \left[(-1)^{n+1} + 2 \frac{(-1)^n - 1}{(n\pi)^2} \right] + \frac{H(u_1 + u_2)}{n\pi} (-1)^n - \frac{H u_2}{n\pi} \left[(-1)^n - 1 \right] \right\} \\ &= \frac{1}{I_0(n\pi R/H)} \begin{cases} \frac{u_2 - u_1}{k\pi}, & (n = 2k), \\ \frac{2}{(2k+1)\pi} \left[\left(1 - \frac{8}{(2k+1)^2 \pi^2} \right) u_1 + u_2 \right], & (n = 2k+1). \end{cases} \end{aligned} \quad (2)$$

定解问题 II 的解为：

$$v_2 = \sum_{n=1}^{\infty} \left(A_n e^{\frac{x_n^{(0)}}{R} Z} + B_n e^{-\frac{x_n^{(0)}}{R} Z} \right) J_0 \left(\frac{x_n^{(0)}}{R} \rho \right), \quad (3)$$

由上、下底的边界条件, 有

$$\begin{cases} \sum_{n=1}^{\infty} (A_n + B_n) J_0 \left(\frac{x_n^{(0)}}{R} \rho \right) = u_2, \\ \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} H/R} + B_n e^{-x_n^{(0)} H/R}) J_0 \left(\frac{x_n^{(0)}}{R} \rho \right) = u_1, \end{cases}$$

$$\begin{aligned} \therefore A_n + B_n &= \frac{2}{R^2 [J_1(x_n^{(0)})]^2} \int_0^R u_2 J_0 \left(\frac{x_n^{(0)}}{R} \rho \right) \rho d\rho \\ &= \frac{2u_2}{[x_n^{(0)} J_0(x_n^{(0)})]^2} \int_0^{x_n^{(0)}} x J_1(x) dx \\ &= \frac{2u_2}{x_n^{(0)} J_1(x_n^{(0)})}. \end{aligned}$$

$$\text{同理有 } A_n e^{x_n^{(0)} H/R} + B_n e^{-x_n^{(0)} H/R} = \frac{2u_1}{x_n^{(0)} J_1(x_n^{(0)})},$$

$$\text{解得: } \begin{cases} A_n = \frac{u_1 - u_2 e^{-x_n^{(0)} H/R}}{x_n^{(0)} J_1(x_n^{(0)}) \cdot \text{sh}(x_n^{(0)} H/R)}, \\ B_n = \frac{u_1 - u_2 e^{x_n^{(0)} H/R}}{x_n^{(0)} J_1(x_n^{(0)}) \cdot \text{sh}(x_n^{(0)} H/R)}, \end{cases} \quad (4)$$

$$\text{故: } u = v_1 + v_2, \quad (5)$$

其中 v_1 和 v_2 分别为(1)和(3)式, 而系数分别由(2)和(4)表出.

9. 圆柱体半径为 R , 高为 H , 上底有均匀分布的强度为 q_0 的热流流入, 下底有同样热流流出, 柱侧保持为 0°C , 求柱内的稳恒温度.

解法一: 定解问题为:

$$\Delta_2 u = 0, \quad (1)$$

$$\left. \frac{\partial u}{\partial z} \right|_{z=0} = \frac{q}{R} \quad (\text{热流方向与 } z \text{ 轴反向}), \quad (2)$$

$$\left. \frac{\partial u}{\partial z} \right|_{z=H} = \frac{q_0}{R} \quad (\text{热流方向与 } z \text{ 轴同向}), \quad (3)$$

$$u|_{\rho=R} = 0, \quad (4)$$

$\therefore$ 侧面是第一类齐次边界条件, 且问题与 φ 无关,
则有 $J_0(\omega R) = 0$,

$$\therefore \omega_n^{(0)} = \frac{x_n^{(0)}}{R}, \quad (n=1, 2, \dots),$$

$x_n^{(0)}$ 为 $J_0(x)$ 的第 n 个零点.

$$u(\rho, z) = \sum_{n=1}^{\infty} (A_n e^{\omega_n^{(0)} z} + B_n e^{-\omega_n^{(0)} z}) + J_0(\omega_n^{(0)} \rho), \quad (5)$$

由非齐次边界条件定出系数 A_n, B_n ,

$$\left. \frac{\partial u}{\partial z} \right|_{z=0} = \sum_{n=1}^{\infty} (A_n - B_n) \omega_n^{(0)} J_0(\omega_n^{(0)} \rho) = \frac{q_0}{k},$$

$$\begin{aligned} \therefore A_n - B_n &= \frac{2q_0}{R^2 \omega_n^{(0)} [J_1(\omega_n^{(0)} R)]^2 k} \int_0^R J_0(\omega_n^{(0)} \rho) \rho d\rho \\ &= \frac{2q_0}{k R^2 (\omega_n^{(0)})^3 [J_1(\omega_n^{(0)} R)]^2} \\ &\quad \times \int_0^{\omega_n^{(0)} R} J_0(x) x dx \\ &= \frac{2q_0}{k R^2 (\omega_n^{(0)})^3 [J_1(\omega_n^{(0)} R)]^2} \\ &\quad \times x J_1(x) \Big|_0^{\omega_n^{(0)} R}, \end{aligned}$$

$$A_n - B_n = \frac{2q_0}{k R (\omega_n^{(0)})^2 J_1(\omega_n^{(0)} R)}, \quad (6)$$

$$\begin{aligned}\left. \frac{\partial u}{\partial z} \right|_{z=H} &= \sum_{n=1}^{\infty} (A_n e^{\omega_n^{(0)} H} - B_n e^{-\omega_n^{(0)} H}) \omega_n^{(0)} J_0(\omega_n^{(0)} \rho) \\ &= \frac{q_0}{k},\end{aligned}$$

$$\therefore A_n e^{\omega_n^{(0)} H} - B_n e^{-\omega_n^{(0)} H} = \frac{2q_0}{kR(\omega_n^{(0)})^2 J_1(\omega_n^{(0)} R)}, \quad (7)$$

由(5), (7)解出:

$$\begin{cases} A_n = \frac{-Rq_0(-1 + e^{-x_n^{(0)} H/R})}{k[x_n^{(0)}]^2 J_1(x_n^{(0)}) \cdot \text{sh}(x_n^{(0)} H/R)}, \\ B_n = \frac{-Rq_0(e^{x_n^{(0)} H/R} - 1)}{k[x_n^{(0)}]^2 J_1(x_n^{(0)}) \cdot \text{sh}(x_n^{(0)} H/R)}. \end{cases} \quad (8)$$

(5)式是定解问题的解, 其中系数 A_n 和 B_n 由(8)式表出.

解二:

因为上下底非齐次边界的非齐次项是常数, 故可较易化成齐次边界. 这样本征值问题就变成傅里叶级数本征问题, 而不是贝塞尔函数本征问题, 同时求系数亦为简单.

$$\text{选 } v_1 = \frac{q_0}{k} z, \quad \text{令 } u = \frac{q_0}{k} z + v,$$

则 v 的定解问题为:

$$\Delta_2 v = 0, \quad (1)$$

$$\begin{cases} v_z|_{z=0} = 0, \end{cases} \quad (2)$$

$$\begin{cases} v_z|_{z=H} = 0, \end{cases} \quad (3)$$

$$\begin{cases} v|_{\rho=R} = -\frac{q_0}{k} z, \end{cases} \quad (4)$$

分离变数得到的本征值问题为:

$$Z'' + h^2 Z = 0,$$

$$\begin{cases} Z'|_{z=0} = 0, \\ Z'|_{z=H} = 0, \end{cases}$$

解得 $h_n = \frac{n\pi}{H}, (n = 0, 1, 2, \dots),$

$$Z_n = A_n \cos \frac{n\pi}{H} z,$$

问题在柱内的有限解为:

$$v(\rho, z) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi}{H} z \cdot I_0\left(\frac{n\pi}{H} \rho\right),$$

由条件(4), 有

$$\sum_{n=0}^{\infty} A_n I_0\left(\frac{n\pi R}{H}\right) \cdot \cos \frac{n\pi}{H} z = -\frac{q_0}{k} z,$$

将上式右端展为傅里叶余弦级数, 则有

$$\begin{aligned} A_n &= \frac{2}{H I_0(n\pi R/H)} \int_0^H -\frac{q_0}{k} z \cdot \cos \frac{n\pi z}{H} dz \\ &= -\frac{2q_0}{H k I_0(n\pi R/H)} \left[\left(\frac{H}{n\pi}\right)^2 \cos \frac{n\pi z}{H} \right. \\ &\quad \left. + \frac{H}{n\pi} z \cdot \sin \frac{n\pi z}{H} \right]_0^H \\ &= -\frac{2q_0}{H k I_0(n\pi R/H)} \cdot \left(\frac{H}{n\pi}\right)^2 [(-1)^n - 1] \\ &= \begin{cases} 0, & (n = 2R \text{ 但不为零}), \\ \frac{4Hq_0}{k\pi^2 n^2 I_0(n\pi R/H)}, & (n = 2R + 1), \end{cases} \end{aligned}$$

$$A_0 = \frac{1}{H I_0(0)} \int_0^H -\frac{q_0}{k} z dz = -\frac{q_0 H}{2k}, \quad (\because I_0(0) = 1),$$

$$\begin{aligned} \therefore v &= -\frac{q_0 H}{2k} + \frac{4q_0 H}{k\pi^2} \sum_{l=0}^{\infty} \\ &\quad \times \frac{I_0\left(\frac{2l+1}{H} \pi \rho\right)}{(2l+1)^2 I_0\left(\frac{2l+1}{H} \pi R\right)} \cdot \cos \frac{2l+1}{H} \pi z, \end{aligned}$$

(5)

故
$$u = \frac{q_0}{k} z + v, \quad (6)$$

其中 v 由(5)式表出.

10. 研究横电波 (指 $E_z = 0$ 的情况, z 为波导的管轴方向, 横电波通常记作 TE 波) 在半径为 R 的圆形波导中传播. [提示: 在管壁上可以认为 $\mathcal{E}_\varphi = 0$, 参看(40.53), 对于 TE 波, $\mathcal{E}_\varphi = 0$, 意味 $\frac{\partial \mathcal{H}_z}{\partial \rho} = 0$, 这就是 $\mathcal{H}_z$ 的边界条件.]

解: (40.48) 式指出: 如果电磁波沿管轴以谐波形式传播,

则
$$\begin{cases} \vec{E}(\rho, \varphi, z, t) = \mathcal{E}(\rho, \varphi) e^{i(hz - kCt)}, \\ \vec{H}(\rho, \varphi, z, t) = \mathcal{H}(\rho, \varphi) e^{i(hz - kCt)}, \end{cases}$$

其中 h 应为实数, 否则意味着 $\vec{E}$ 和 $\vec{H}$ 沿管轴衰减而通不过波导. 问题归结为求解 $\mathcal{E}$ 和 $\mathcal{H}$. 为此, 又只需求解 $\mathcal{E}_z$ 和 $\mathcal{H}_z$, 因为由此可按(40.53)式将其它分量 $\mathcal{E}_\rho, \mathcal{E}_\varphi, \mathcal{H}_\rho, \mathcal{H}_\varphi$ 算出.

$$\because \begin{cases} \Delta_2 \mathcal{E}_z + (k^2 - h^2) \mathcal{E}_z = 0, \\ \Delta_2 \mathcal{H}_z + (k^2 - h^2) \mathcal{H}_z = 0, \end{cases} \quad (1)$$

$$(2)$$

又 TE 波之 $\mathcal{E}_z = 0$, 故只需解 $\mathcal{H}_z$.

在管壁上可认为 $\mathcal{E}_\varphi = 0$, 又由(40.53)知

$$\mathcal{E}_\varphi = \frac{i}{k^2 - h^2} \left(h \frac{1}{\rho} \frac{\partial \mathcal{E}_z}{\partial \varphi} - k \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{\partial \mathcal{H}_z}{\partial \rho} \right),$$

由于 $\mathcal{E}_z = 0$, $\mathcal{E}_\varphi = 0$, 知 $\frac{\partial \mathcal{H}_z}{\partial \rho} \Big|_{\rho=R} = 0$, (3)

亥姆霍兹方程(2)的分离变数形式的解是

$$\mathcal{H}_z = J_m(\sqrt{k^2 - h^2} \rho) \begin{cases} \cos m\varphi \\ \sin m\varphi \end{cases},$$

由边界条件(3), 知

$$J'_m(\sqrt{k^2 - h^2} R) = 0,$$

从而 $\sqrt{k^2 - h^2} R = x_n^{(m)}$ ($x_n^{(m)}$ 是 $J'_m(x)$ 的第 n 个零点), (4)

$$\therefore \mathcal{H}_z = J_m\left(\frac{x_n^{(m)}}{R}\rho\right) \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix}, \quad (5)$$

将 $\mathcal{E}_z = 0$ 与 (4) 代入 (40.53), 得到 $\mathcal{E}$ 、 $\mathcal{H}$ 的各个分量为:

$$\left\{ \begin{aligned} \mathcal{E}_\rho &= \frac{i}{k^2 - h^2} \frac{k}{\rho} \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{\partial \mathcal{H}_z}{\partial \varphi} = \frac{imkR^2}{\rho [x_n^{(m)}]^2} \sqrt{\frac{\mu_0}{\epsilon_0}} J_m \\ &\quad \times \left(\frac{x_n^{(m)}}{R}\rho\right) \begin{Bmatrix} -\sin m\varphi \\ \cos m\varphi \end{Bmatrix}, \\ \mathcal{E}_\varphi &= \frac{-ik}{k^2 - h^2} \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{\partial \mathcal{H}_z}{\partial \rho} = \frac{-ikR}{x_n^{(m)}} \sqrt{\frac{\mu_0}{\epsilon_0}} J'_m \\ &\quad \times \left(\frac{x_n^{(m)}}{R}\rho\right) \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix}, \\ \mathcal{E}_z &= 0, \\ \mathcal{H}_\rho &= \frac{ih}{k^2 - h^2} \frac{\partial \mathcal{H}_z}{\partial \rho} = \frac{ihR}{x_n^{(m)}} J'_m\left(\frac{x_n^{(m)}}{R}\rho\right) \\ &\quad \times \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix}, \\ \mathcal{H}_\varphi &= \frac{ih}{(k^2 - h^2)\rho} \frac{\partial \mathcal{H}_z}{\partial \varphi} = \frac{ihmR^2}{[x_n^{(m)}]^2 \rho} J_m\left(\frac{x_n^{(m)}}{R}\rho\right) \\ &\quad \times \begin{Bmatrix} -\sin m\varphi \\ \cos m\varphi \end{Bmatrix}, \\ \mathcal{H}_z &= J_m\left(\frac{x_n^{(m)}}{R}\rho\right) \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix}, \end{aligned} \right.$$

其中 $x_n^{(m)}$ 是 $J'_m(x)$ 的第 n 个零点.

由 (4) 式有 $h = \sqrt{k^2 - (x_n^{(m)}/R)^2}$,

要使电磁波能通过波导, h 必须为实数, 则有 $k \geq \frac{x_n^{(m)}}{R}$,

$$\therefore k = \frac{2\pi}{\lambda},$$

$$\therefore x_n^{(m)} \leq \frac{2\pi R}{\lambda}, \quad (6)$$

由贝塞尔函数知, n 一定, m 越大, 则 $x_n^{(m)}$ 越大; m 一定, n 越大, 则 $x_n^{(m)}$ 也越大.

又由(6)式知, 对一定的电磁波(λ 一定), 波导越粗(R 越大), 符合(6)式的 $x_n^{(m)}$ 个数越多, 即能通过的电磁波的模式越多, 称为多模传播.

对于 TE 波, $x_n^{(m)}$ 是 $J'_n(x)$ 的零点, 其绝对值最小的零点是 $J'_1(x)$ 的第一个零点, $\therefore J'_0(x) = -J'_1(x)$, $x_1^{(1)} = 3.832$, 其次轮到 $J'_0(x)$ 的第二个零点 $x_2^{(0)} = 5.5201$. 于是波导半径 R

$$\frac{\lambda}{2\pi} x_1^{(1)} < R < \frac{\lambda}{2\pi} x_2^{(0)},$$

$$\text{即} \quad \frac{2\pi R}{x_1^{(1)}} > \lambda > \frac{2\pi R}{x_2^{(0)}}.$$

则只有 $m = 1$ 和 $n = 1$ 的模式通过波导, 称单模传播.

如果 $R < \frac{\lambda}{2\pi} x_1^{(1)}$, 即 $\lambda > \frac{2\pi R}{x_1^{(1)}}$, 则什么模式也不能通过波导.

11. 求长圆柱形铀块的临界半径. [“临界”一词参看 § 36 习题 8].

解: 取柱坐标系. 因为增殖反应相当于存在着扩散源, 而这些源与扩散浓度成正比, 因此扩散方程为:

$$u_t - a^2 \Delta u = \beta u,$$

其中 a^2 是扩散系数, β 是增殖常数.

长圆柱铀块的侧面给予防护, 设圆柱半径为 R , 因而有条件

$$u|_{\rho=R} = 0,$$

因为是长圆柱，可取为平面问题，即问题与 z 无关；又 z 轴是对称轴，问题又与 φ 无关。所以在柱坐标系下定解问题为

$$u_t - a^2 \left(u_{\rho\rho} + \frac{1}{\rho} u_\rho \right) = \beta u, \quad (1)$$

$$u|_{\rho=R} = 0, \quad (2)$$

分离变数(1)和(2)，得

$$T = Ae^{\lambda t}, \quad (3)$$

$$\begin{cases} v'' + \frac{1}{\rho} v' + \frac{(\beta - \lambda)}{a^2} v = 0, \\ v|_{\rho=R} = 0, \end{cases} \quad (4)$$

$$(5)$$

方程(4)是零阶贝塞尔方程，其柱内有限解是

$$v(\rho) = BJ_0\left(\frac{\sqrt{\beta - \lambda}}{a}\rho\right), \quad (6)$$

由边界条件(5)，有

$$J_0\left(\frac{\sqrt{\beta - \lambda}}{a}R\right) = 0,$$

$$\therefore x_n^{(0)} = \frac{\sqrt{\beta - \lambda_n^{(0)}}}{a}R, (x_n^{(0)} \text{ 是 } J_0(x) \text{ 的第 } n \text{ 个零点}),$$

$$\text{即 } \lambda_n^{(0)} = \beta - \left(\frac{ax_n^{(0)}}{R}\right)^2,$$

由解(3)知，对于 $\beta > 0$ (增殖扩散)，即使指数中只有一个 $\lambda_n^{(0)} > 0$ ，即 $\beta > \left(\frac{ax_n^{(0)}}{R}\right)^2$ ，则随时间 t 的增长，中子浓度将按指数而增大，铀块将爆炸，当 β 一定时， $\beta - (ax_n^{(0)}/R)^2 = 0$ ，即

$$R = \frac{ax_n^{(0)}}{\sqrt{\beta}}, \text{ 称为“临界尺寸”。}$$

$x_n^{(0)}$ 中最小的一个零点是 $x_1^{(0)} = 2.4048$ ，代入上式便得长圆柱铀

块的临界半径

$$R_{kp} = \frac{2.4048a}{\sqrt{\beta}}. \quad (7)$$

12. 样品放入烘炉之前的温度同于室温即 $u_0^\circ\text{C}$, 把它放入温度为 $u_1^\circ\text{C}$ 的烘炉进行保温. 但是, 样品内的温度不可能立即变为 $u_1^\circ\text{C}$, 它与 u_1 的差随时间作指数衰减. 今约定把差值降到 $1/e$, 才算作保温开始, 试计算圆柱形样品放入烘炉内多少时间才可开始计算保温时间.

解: 定解问题为:

$$\begin{aligned} u_t - a^2 \Delta_2 u &= 0, \\ \left. \begin{aligned} u|_{\rho=R} &= u_1, \\ u|_{z=0} &= u_1, \\ u|_{z=H} &= u_1, \end{aligned} \right\} & \text{(同于烘炉温度),} \\ u|_{t=0} &= u_0, \text{(同于室温).} \end{aligned}$$

解: 改取温度 u_1 作为温标的零点, 即作变换:

$$v = u - u_1, \quad (1)$$

v 的定解问题:

$$\begin{cases} v_t - a^2 \Delta_3 v = 0, (\rho \leq R, 0 \leq z \leq H), & (2) \\ v|_{t=0} = u_0 - u_1, & (3) \\ v|_{\rho=R} = v|_{z=0} = v|_{z=H} = 0, & (4) \end{cases}$$

分离变数形式的解为:

$$e^{-k^2 a^2 t} \begin{Bmatrix} \cos m\rho \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} \cosh kz \\ \sinh kz \end{Bmatrix} J_m(\sqrt{k^2 - h^2} \rho),$$

既然问题与 φ 无关, 所以 $m=0$, 又根据上下底的第一类齐次边

界条件舍弃 $\cosh kz$ 而取 $\sinh kz$, 且 $h = \frac{m\pi}{H}$, ($m=1, 2, \dots$).

由于侧面的第一类齐次边界条件:

$\sqrt{k^2 - h^2} = \frac{x_n^{(0)}}{R}$, $x_n^{(0)}$ 是 $J_0(x)$ 的第 n 个零点.

$$\text{于是 } v = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} C_{m,n} e^{-\left[\left(\frac{x_n^{(0)}}{R}\right)^2 + \left(\frac{m\pi}{H}\right)^2\right] a^2 t} \cdot \sin \frac{m\pi}{H} z \cdot J_0\left(\frac{x_n^{(0)}}{R} \rho\right),$$

其中 $n=1$ 和 $m=1$ 的一项随时间减小最慢, 可把此项降到 $t=0$ 时的值的 $\frac{1}{e}$ 作为保温开始的时间即

$$e^{-\left[\left(\frac{x_1^{(0)}}{R}\right)^2 + \left(\frac{\pi}{H}\right)^2\right] a^2 t} = e^{-1},$$

$$\therefore t = \frac{1}{a^2 \left[\left(\frac{2.4048}{R} \right)^2 + \left(\frac{\pi}{H} \right)^2 \right]}. \quad (5)$$

13. 电子光学透镜的某一部件由两个中空圆柱筒组成, 其电势分别为 $+v_0$ 和 $-v_0$, 在圆柱中间隙缝的边缘处电势可近似表为 $v = v_0 \sin \frac{\pi z}{2\delta}$, 求圆柱筒内的电势分布, 圆柱两端边界条件可近似表为 $v|_{z=\pm l} = \pm v_0$.

解: 取柱坐标系. 由于电势为 z 的奇函数, 可取通过隙缝中间的平面为柱坐标系底面, 这样可在 $[0, l]$ 上求解.

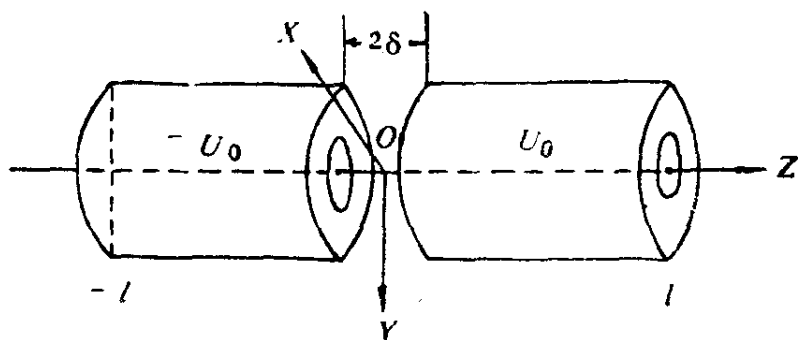


图 14-1

又因圆柱筒是等势体, 筒内无电荷, 所以筒内电势分布的

定解问题为:

$$\begin{aligned} \Delta u &= 0, \\ \begin{cases} u|_{z=0} = v_0 \sin \frac{\pi z}{2\delta} \Big|_{z=0} = 0, \\ u|_{z=l} = v_0, \\ u|_{\rho=R} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta}, & (0 \leq z \leq \delta), \\ v_0, & (\delta \leq z \leq l), \end{cases} \end{cases} \end{aligned}$$

将定解问题化为两个定解问题.

$$\text{令 } u = u^I + u^{II}, \quad (1)$$

$$\begin{cases} \Delta u^I = 0, \\ u^I|_{z=0} = 0, \\ u^I|_{z=l} = 0, \\ u^I|_{\rho=R} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta}, & (0 \leq z \leq \delta), \\ v_0, & (\delta \leq z \leq l), \end{cases} \end{cases} \quad \text{及} \quad \begin{cases} \Delta u^{II} = 0, \\ u^{II}|_{z=0} = 0, \\ u^{II}|_{z=l} = v_0, \\ u^{II}|_{\rho=R} = 0, \end{cases}$$

先解 u^I : 由于上、下底都是第一类齐次边界, 且问题与 φ 无关, 可知本征值和本征函数为:

$$h_n = \frac{n\pi}{l}, \quad Z_n = C_n \sin \frac{n\pi}{l} z, \quad (n = 1, 2, \dots),$$

柱内有限解为:

$$u^I = \sum_{n=1}^{\infty} C_n I_0\left(\frac{n\pi}{l} \rho\right) \cdot \sin \frac{n\pi}{l} z,$$

由柱侧边界, 得:

$$\sum_{n=1}^{\infty} C_n I_0\left(\frac{n\pi R}{l}\right) \sin \frac{n\pi}{l} z = \begin{cases} v_0 \sin \frac{\pi z}{2\delta}, & (0 \leq z \leq \delta), \\ v_0, & (\delta \leq z \leq l), \end{cases}$$

$$C_n = \frac{2}{l} \cdot \frac{1}{I_0(n\pi R/l)} \left[\int_0^\delta v_0 \sin\left(\frac{\pi}{2\delta} z\right) \cdot \sin\left(\frac{n\pi}{l} z\right) dz + \int_\delta^l v_0 \sin\left(\frac{n\pi}{l} z\right) dz \right],$$

$$\begin{aligned} \therefore \int_0^\delta \sin\left(\frac{\pi}{2\delta} z\right) \cdot \sin\left(\frac{n\pi}{l} z\right) dz &= \left[-\frac{\sin\left(\frac{\pi}{2\delta} + \frac{n\pi}{l}\right)z}{2\left(\frac{\pi}{2\delta} + \frac{n\pi}{l}\right)} + \frac{\sin\left(\frac{\pi}{2\delta} - \frac{n\pi}{l}\right)z}{2\left(\frac{\pi}{2\delta} - \frac{n\pi}{l}\right)} \right]_0^\delta \\ &= -\frac{\sin\left(\frac{\pi}{2} + \frac{n\pi\delta}{l}\right)}{2(\pi/2\delta + n\pi/l)} + \frac{\sin\left(\frac{\pi}{2} - \frac{n\pi\delta}{l}\right)}{2(\pi/2\delta - n\pi/l)} \\ &= \frac{\cos\frac{n\pi\delta}{l}}{2} \left[\frac{\pi/2\delta + n\pi/l - \pi/2\delta + n\pi/l}{\left(\frac{\pi}{2\delta}\right)^2 - \left(\frac{n\pi}{l}\right)^2} \right] \\ &= \frac{\cos\left(\frac{n\pi\delta}{l}\right)}{2} \cdot \frac{2\frac{n\pi}{l}}{\left(\frac{\pi}{2\delta}\right)^2 - \left(\frac{n\pi}{l}\right)^2} \\ &= \cos\left(\frac{n\pi\delta}{l}\right) \cdot \frac{n\pi}{l} \cdot \frac{4\delta^2 l^2}{l^2 \pi^2 - 4\delta^2 n^2 \pi^2}, \end{aligned}$$

$$\begin{aligned} \int_\delta^l \sin\left(\frac{n\pi}{l} z\right) dz &= \frac{l}{n\pi} \left[-\cos\left(\frac{n\pi}{l} z\right) \right]_\delta^l \\ &= \frac{l}{n\pi} \left[(-1)^{n+1} + \cos\left(\frac{n\pi\delta}{l}\right) \right] \end{aligned}$$

$$\begin{aligned} \therefore C_n &= \frac{U_0}{I_0(n\pi R/l)} \cdot \frac{2}{l} \left\{ \frac{n\pi}{l} \cdot \frac{4\delta^2 l^2}{l^2 \pi^2 - 4\delta^2 n^2 \pi^2} \right. \\ &\quad \left. \times \cos\left(\frac{n\pi\delta}{l}\right) + \frac{l}{n\pi} \left[(-1)^{n+1} + \cos\left(\frac{n\pi\delta}{l}\right) \right] \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{2v_0}{I_0(n\pi R/l)} \cdot \frac{1}{n\pi} \left[(-1)^{n+1} + \cos\left(\frac{n\pi\delta}{l}\right) \right. \\
&\quad \left. + \frac{n^2 4\delta^2}{l^2 - 4n^2\delta^2} \cos\left(\frac{n\pi\delta}{l}\right) \right] \\
&= \frac{2v_0}{n\pi} \cdot \frac{1}{I_0(n\pi R/l)} \left[(-1)^{n+1} \right. \\
&\quad \left. + \left(1 + \frac{4\delta^2 n^2}{l^2 - 4n^2\delta^2} \right) \cos\left(\frac{n\pi\delta}{l}\right) \right] \\
&= \frac{2v_0}{\pi} \cdot \frac{1}{I_0(n\pi R/l)} \cdot \frac{1}{n} \left[(-1)^{n+1} \right. \\
&\quad \left. + \frac{l^2}{l^2 - 4\delta^2 n^2} \cos\left(\frac{n\pi\delta}{l}\right) \right],
\end{aligned}$$

故:
$$u^I(\rho, z) = \frac{2v_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left[(-1)^{n+1} + \frac{l^2}{l^2 - 4\delta^2 n^2} \right. \\ \left. \times \cos\left(\frac{n\pi\delta}{l}\right) \right] \frac{I_0\left(\frac{n\pi}{l}\rho\right)}{I_0\left(\frac{n\pi}{l}R\right)} \sin\left(\frac{n\pi}{l}z\right) \quad (2)$$

再解 u^{II} : 由于侧面是齐次边界, 且问题与 φ 无关($m=0$), 可知本征值和本征函数为:

$$\lambda_n^{(0)} = \frac{x_n^{(0)}}{R}, \quad (n=1, 2, \dots), \quad x_n^{(0)} \text{ 是 } J_0(x) \text{ 的第 } n \text{ 个零}$$

点,

$$v_n(\rho) = J_0\left(\frac{x_n^{(0)}}{R}\rho\right),$$

则柱内有限解为

$$\begin{aligned}
u^{II} &= \sum_{n=1}^{\infty} \left(A_n e^{x_n^{(0)} z/R} + B_n e^{-x_n^{(0)} z/R} \right) \\
&\quad \times J_0\left(\frac{x_n^{(0)}}{R}\rho\right),
\end{aligned}$$

由下底边界条件:

$$\sum_{n=1}^{\infty} (A_n + B_n) J_0 \left(\frac{x_n^{(0)}}{R} \rho \right) = 0,$$

得 $A_n = -B_n$,

由上底边界条件

$$\sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} l/R} + B_n e^{-x_n^{(0)} l/R}) J_0 \left(\frac{x_n^{(0)}}{R} \rho \right) = v_0,$$

得

$$\begin{aligned} A_n &= \frac{2}{R^2 [J_1(x_n^{(0)})]^2} \cdot \frac{1}{2 \cdot \text{sh}(x_n^{(0)} l/R)} \\ &\quad \times \int_0^R v_0 J_0 \left(\frac{x_n^{(0)}}{R} \rho \right) \rho d\rho \\ &= \frac{v_0 R^2}{R^2 [J_1(x_n^{(0)})]^2 \cdot \text{sh}(x_n^{(0)} l/R) (x_n^{(0)})^2} \\ &\quad \times \left[\frac{x_n^{(0)}}{R} \rho J_1 \left(\frac{x_n^{(0)}}{R} \rho \right) \right]_0^R \\ &= \frac{v_0}{x_n^{(0)} \cdot J_1(x_n^{(0)}) \cdot \text{sh}(x_n^{(0)} l/R)}. \end{aligned}$$

$$\therefore u^{\text{II}}(\rho, z) = 2v_0 \sum_{n=1}^{\infty} \frac{J_0(x_n^{(0)} \rho/R)}{x_n^{(0)} \cdot J_1(x_n^{(0)})} \cdot \frac{\text{sh}(x_n^{(0)} z/R)}{\text{sh}(x_n^{(0)} l/R)}, \quad (3)$$

故: $u = u^{\text{I}} + u^{\text{II}}, \quad (4)$

其中 u^{I} 和 u^{II} 分别为 (2) 和 (3) 表出.

§47. 球贝塞耳方程

1. 确定球形铀块的临界半径. [“临界”一词参看 §36 习题 8].

解: 取球坐标系, 以铀块中心为极点. 同上节第11题讨论相同, 有定解问题:

$$u_t - a^2 \Delta u = \beta u, \quad (1)$$

$$u|_{r=R} = 0, \quad (2)$$

其中 a^2 为扩散系数, β 为增殖常数, R 为铀球半径.

解一: 作函数变换, 消去方程 (1) 中函数项 βu ,

$$\text{令 } u(r, \theta, \varphi, t) = v(r, \theta, \varphi, t) e^{\beta t},$$

$$\text{则 } v_t - a^2 \Delta v = 0, \quad (3)$$

$$v|_{r=R} = 0, \quad (4)$$

$\therefore$ 问题与 θ, φ 无关 (中子浓度 u 的变化只与 r 有关).

$$\text{即 } l = 0, m = 0,$$

$\therefore$ 球内问题的分离变数形式解为

$$j_0(kr) e^{-k^2 a^2 t},$$

由边界条件 (4), 有

$$j_0(kr)|_{r=R} = \frac{\sin kr}{kr} \Big|_{r=R} = 0,$$

$$\therefore k_n = \frac{n\pi}{R}, (n = 1, 2, \dots),$$

则 v 的本征解为:

$$v_n = A_n \cdot j_0\left(\frac{n\pi}{R} r\right) \cdot e^{-n^2 \pi^2 a^2 t / R^2},$$

u 的本征解为:

$$u_n = A_n \cdot j_0\left(\frac{n\pi}{R} r\right) e^{(\beta - n^2 \pi^2 a^2 / R^2) t},$$

根据“临界尺寸”定义, 临界半径

$$R_{kp} = \frac{\pi a}{\sqrt{\beta}}, (\text{取 } n = 1). \quad (5)$$

解二: 直接进行分离变量. 由于问题与 θ, φ 无关, 可令

$u(r, t) = v(r) T(t)$ 代入方程 (1), 得

$$T' - KT = 0, \quad (6)$$

$$r^2 u'' + 2r u' + \frac{r^2}{a^2}(\beta - k)v = 0, \quad (7)$$

方程(7)是零阶球贝塞尔方程, 在球内有限解为:

$$v = j_0\left(\frac{\sqrt{\beta - k}}{a} r\right),$$

由边界条件(2),

$$j_0\left(\frac{\sqrt{\beta - k}}{a} r\right)\Big|_{r=R} = \frac{\sin\frac{\sqrt{\beta - k}}{a} r}{\sqrt{\beta - k} r/a}\Big|_{r=R} = 0,$$

即
$$\sqrt{\beta - k_n} \cdot \frac{R}{a} = n\pi, (n = 1, 2, \dots),$$

$$\therefore k_n = \beta - n^2 \pi^2 a^2 / R^2,$$

根据“临界尺寸”定义, 临界半径

$$R_{kp} = \frac{\pi a}{\sqrt{\beta}}, (\text{取 } n = 1). \quad (8)$$

2. 均质球, 半径为 r_0 , 初始温度分布为 $f(r)$, 把球面温度保持为零度而使它冷却, 求解球内各处温度变化情况.

解: 定解问题为:

$$u_t - a^2 \Delta u = 0, \quad (1)$$

$$\begin{cases} u|_{r=r_0} = 0, & (2) \\ u|_{t=0} = f(r), & (3) \end{cases}$$

取球坐标系, 以球心为极点, 由定解条件知, 问题与 θ, φ 无关,

($l = 0, m = 0$) 则球内问题的分离变数形式解为:

$$j_0(kr) e^{-a^2 k^2 t} = \frac{\sin kr}{kr} e^{-a^2 k^2 t},$$

其本征值由 $R(kr)\Big|_{r=r_0} = 0$ 求得:

$$\frac{\sin kr_0}{kr_0} = 0, \text{ 即 } \sin kr_0 = 0,$$

$$\therefore k_n = \frac{n\pi}{r_0}, (n = 1, 2, \dots),$$

$$\therefore u(r, t) = \sum_{n=1}^{\infty} C_n \frac{\sin(n\pi r/r_0)}{n\pi r/r_0} e^{-\left(\frac{a^2 n^2 \pi^2}{r_0^2}\right)t}.$$

由初始条件(3),

$$u|_{t=0} = \sum_{n=1}^{\infty} C_n j_0\left(\frac{n\pi}{r_0} r\right) = f(r),$$

$$\begin{aligned} \therefore C_n &= \frac{\int_0^{r_0} f(r) j_0\left(\frac{n\pi}{r_0} r\right) r^2 dr}{\int_0^{r_0} \left[j_0\left(\frac{n\pi}{r_0} r\right)\right]^2 r^2 dr} \\ &= \int_0^{r_0} f(r) r^2 \cdot \frac{\sin(n\pi r/r_0)}{\frac{n\pi r}{r_0}} \\ &\quad \cdot dr / \int_0^{r_0} \frac{\sin^2(n\pi r/r_0)}{\left(\frac{n\pi r}{r_0}\right)^2} r^2 dr \\ &= \left(\frac{n\pi}{r_0}\right)^2 \cdot \frac{2}{r_0} \cdot \frac{r_0}{n\pi} \int_0^{r_0} f(r) \cdot r \cdot \sin(n\pi r/r_0) dr \\ &= \frac{2\pi n}{r_0^2} \int_0^{r_0} f(r) r \cdot \sin(n\pi r/r_0) dr, \end{aligned}$$

故:

$$\begin{aligned} u(r, t) &= \sum_{n=1}^{\infty} \frac{2\pi n}{r_0^2} j_0\left(\frac{n\pi r}{r_0}\right) e^{-n^2 \pi^2 a^2 t / r_0^2} \\ &\quad \cdot \int_0^{r_0} f(r) r \cdot \sin(n\pi r/r_0) dr \\ &= \frac{2}{r_0 r} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi r}{r_0}\right) \cdot e^{-n^2 \pi^2 a^2 t / r_0^2} \\ &\quad \cdot \int_0^{r_0} f(r) \cdot r \cdot \sin\left(\frac{n\pi r}{r_0}\right) dr. \quad (4) \end{aligned}$$

3. 均质球, 半径为 r_0 , 初始温度分布为 $f(r) \cos \theta$, 把球面

温度保持为零度而使它冷却. 求解球内各处温度变化情况.

解: 定解问题为:

$$\begin{cases} u_t - a^2 \Delta u = 0, & (1) \\ u|_{t=0} = f(r) \cos \theta, & (2) \\ u|_{r=r_0} = 0, & (3) \end{cases}$$

从初始条件知问题与 φ 无关, 即 $m = 0$, 又因初始条件中因子 $\cos \theta = P_1(\cos \theta)$, 即 $l = 1$, 于是问题的一般解为:

$$P_1(\cos \theta) \cdot j_1(kr) \cdot e^{-a^2 k^2 t},$$

由边界条件(3), 有 $j_1(kr)|_{r=r_0} = \frac{\sin kr_0 - kr_0 \cos kr_0}{kr_0} = 0$,

即 $\operatorname{tg} kr_0 = kr_0$, 令其第 n 个根为 $x_n = k_n r_0$, ($n = 1, 2, \dots$),

$$\therefore k_n = \frac{x_n}{r_0},$$

于是解为:

$$u(r, \theta, t) = \sum_{n=1}^{\infty} A_{1,n} P_1(\cos \theta) j_1(k_n r) e^{-a^2 k_n^2 t},$$

由初始条件(2),

$$\begin{aligned} u|_{t=0} &= \sum_{n=1}^{\infty} A_{1,n} P_1(\cos \theta) j_1(k_n r) = f(r) \cos \theta \\ &= f(r) P_1(\cos \theta), \end{aligned}$$

$$\therefore A_{1,n} = \int_0^{r_0} f(r) j_1(k_n r) r^2 dr / \int_0^{r_0} j_1^2(k_n r) r^2 dr,$$

$$\begin{aligned} \because \int_0^{r_0} [j_1(k_n r)]^2 r^2 dr &= \frac{\pi}{2k_n} \int_0^{r_0} \left[J_{\frac{3}{2}}(k_n r) \right]^2 r dr \\ &= \frac{\pi}{2k_n} \cdot \frac{1}{2} r_0^2 \left[J_{\frac{3}{2}}(k_n r_0) \right]^2 \\ &= \frac{\pi}{2k_n} \cdot \frac{r_0}{2} \left[J_{\frac{1}{2}}(k_n r_0) \right]^2 \end{aligned}$$

$$\begin{aligned}
& -\frac{3}{2} \frac{J_{\frac{3}{2}}(k_n r_0)}{k_n r_0} \Bigg]^2 \\
& = \frac{\pi}{2k_n} \cdot \frac{r_0^2}{2} [J_2(k_n r_0)]^2 \\
& = \frac{1}{2} r_0^3 \left[\sqrt{\frac{\pi}{2k_n r_0}} J_{\frac{1}{2}}(k_n r_0) \right]^2 \\
& = \frac{r_0^3}{2} [j_0(k_n r_0)]^2,
\end{aligned}$$

$$\begin{aligned}
\text{故: } u(r, \theta, t) &= \frac{2}{r_0^3} \sum_{n=1}^{\infty} \frac{1}{[j_0(k_n r_0)]^2} j_1(k_n r) P_1(\cos \theta) e^{-a^2 k_n^2 t} \cdot \\
& \int_0^{r_0} f(r) \cdot r^2 \cdot j_1(k_n r) dr. \quad (4)
\end{aligned}$$

4. 半径为 $2r_0$ 的均质球. 初始温度 = $\begin{cases} u_0 (0 < r < r_0), \\ 0 (r_0 < r < 2r_0), \end{cases}$ 把球面保持为零度而使它冷却, 求解球内温度变化情况.

解: 定解问题为:

$$\begin{cases} u_t - a^2 \Delta u = 0, & (1) \end{cases}$$

$$\begin{cases} u|_{r=2r_0} = 0, & (2) \end{cases}$$

$$\begin{cases} u|_{t=0} = \begin{cases} u_0, (0 < r < r_0), \\ 0, (r_0 < r < 2r_0), \end{cases} & (3) \end{cases}$$

由定解条件知问题与 θ, φ 无关, 即 $l = 0, m = 0$, 于是球内的分离变数形式解为:

$$j_0(kr) e^{-a^2 k^2 t},$$

其中本征值 k 由 $j_0(2kr_0) = \frac{\sin 2kr_0}{2kr_0} = 0$ 来定:

$$2kr_0 = n\pi, \therefore k_n = \frac{n\pi}{2r_0}, (n = 1, 2, \dots),$$

$$\therefore u(r, t) = \sum_{n=1}^{\infty} C_n j_0\left(\frac{n\pi}{2r_0} r\right) e^{-\left(\frac{an\pi}{2r_0}\right)^2 t},$$

由初始条件(3),

$$u|_{t=0} = \sum_{n=1}^{\infty} C_n j_0\left(\frac{n\pi}{2r_0} r\right) = \begin{cases} u_0, (0 < r < r_0), \\ 0, (r_0 < r < 2r_0), \end{cases}$$

$$\begin{aligned} C_n &= \frac{\int_0^{r_0} u_0 j_0(k_n r) \cdot r^2 dr}{\int_0^{2r_0} [j_0(k_n r)]^2 r^2 dr} \\ &= \int_0^{r_0} u_0 \frac{\sin(k_n r)}{k_n r} r^2 dr / \int_0^{2r_0} \frac{\sin^2(k_n r)}{(k_n r)^2} r^2 dr \\ &= \frac{u_0}{k_n} \int_0^{r_0} r_0 \sin(k_n r) dr / \frac{1}{k_n^2} \int_0^{2r_0} \sin^2(k_n r) dr, \end{aligned}$$

$$\begin{aligned} \therefore \frac{1}{k_n^2} \int_0^{2r_0} \sin^2 k_n r dr \\ = \frac{1}{k_n^2} \cdot \frac{1}{k_n} \left[\frac{1}{2} k_n r - \frac{1}{4} \sin(2k_n r) \right]_0^{2r_0} = \frac{r_0}{k_n^2}, \end{aligned}$$

$$\begin{aligned} \text{及 } \frac{u_0}{k_n} \int_0^{r_0} r \cdot \sin(k_n r) dr &= \frac{u_0}{k_n} \left[-\frac{1}{k_n^2} \sin(k_n r) \right. \\ &\quad \left. - \frac{1}{k_n} r \cos(k_n r) \right]_0^{r_0} \\ &= \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin(k_n r_0) - \frac{r_0}{k_n} \cos(k_n r_0) \right] \\ &= \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin\left(\frac{n\pi}{2}\right) - \frac{r_0}{k_n} \cos\left(\frac{n\pi}{2}\right) \right] \\ &= \frac{u_0}{k_n^2} \left[\frac{1}{k_n} \sin\left(\frac{n\pi}{2}\right) - r_0 \cos\left(\frac{n\pi}{2}\right) \right], \end{aligned}$$

$$\therefore C_n = \frac{u_0}{k_n^2} \left[\frac{1}{k_n} \sin\left(\frac{n\pi}{2}\right) - r_0 \cos\left(\frac{n\pi}{2}\right) \right] / \frac{r_0}{k_n^2}$$

$$\begin{aligned}
&= \frac{u_0}{r_0} \left[\frac{1}{k_n} \sin\left(\frac{n\pi}{2}\right) - r_0 \cos\left(\frac{n\pi}{2}\right) \right] \\
&= \frac{u_0}{r_0 k_n} \sin\left(\frac{n\pi}{2}\right) - u_0 \cos\left(\frac{n\pi}{2}\right), \\
\text{故: } u &= \sum_{n=1}^{\infty} \left(\frac{u_0}{r_0 k_n} \sin\frac{n\pi}{2} - u_0 \cos\frac{n\pi}{2} \right) \frac{\sin k_n r}{k_n r} e^{-k_n^2 a^2 t} \\
&= \sum_{n=1}^{\infty} \left(\frac{4r_0 u_0}{n^2 \pi^2} \sin\frac{n\pi}{2} - \frac{2r_0 u_0}{n\pi} \cos\frac{n\pi}{2} \right) \\
&\quad \times \frac{1}{r} \sin(k_n r) e^{-a^2 k_n^2 t} \\
&= \sum_{n=1}^{\infty} A_n \frac{1}{r} \sin(k_n r) e^{-a^2 k_n^2 t}, \tag{4}
\end{aligned}$$

其中 $A_{2k} = (-1)^{k+1} \frac{u_0 r_0}{k\pi},$

$$A_{2k+1} = (-1)^k \frac{4u_0 r_0}{(2k+1)^2 \pi^2},$$

$$\therefore \cos\frac{n\pi}{2} = \begin{cases} 0, & (n = 2k+1), \\ (-1)^k, & (n = 2k), \end{cases}$$

$$\sin\frac{n\pi}{2} = \begin{cases} 0, & (n = 2k), \\ (-1)^k, & (n = 2k+1). \end{cases}$$

5. 均质球，半径为 r_0 ，初始温度为 v_0 ，放在温度为 u_0 的空气中自由冷却，（按牛顿冷却定律跟空气交换热量），求解球内各处温度变化情况。

解：定解问题为：

$$\begin{cases} u_t - a^2 \Delta u = 0, \\ (u + H u_r) |_{r=r_0} = u_0, (H = k/h), \\ u |_{t=0} = v_0, \end{cases}$$

边界条件为非齐次，应化为齐次

令 $u = u_0 + W$ ，则 W 的定解问题为：

$$W_t - a^2 \Delta W = 0, \quad (1)$$

$$\begin{cases} (W + HWr) |_{r=r_0} = 0, & (2) \\ W |_{t=0} = v_0 - u_0, & (3) \end{cases}$$

由定解条件知问题与 θ, φ 无关, 即 $l=0, m=0$, 球内问题的分离变数形式解是:

$$j_0(kr) \cdot e^{-a^2 k^2 t},$$

本征值 k 由 $(W + HWr) |_{r=r_0} = 0$ 决定.

$$\left[\frac{\sin kr}{kr} + H \frac{k^2 r \cos kr - k \sin kr}{(kr)^2} \right]_{r=r_0} = 0,$$

即 $kr_0 \sin kr_0 + H k^2 r_0 \cos kr_0 - H k \sin kr_0 = 0.$

$$\therefore \operatorname{tg} kr_0 = \frac{H kr_0}{H - r_0},$$

将这个方程的第 n 个根记作 $k_n, (n=1, 2, \dots)$,

$$\therefore W(r, t) = \sum_{n=1}^{\infty} C_n \frac{\sin(k_n r)}{k_n r} e^{-a^2 k_n^2 t},$$

由初始条件(3),

$$\sum_{n=1}^{\infty} C_n \frac{\sin k_n r}{k_n r} = v_0 - u_0,$$

$$\therefore C_n = (v_0 - u_0) \int_0^{r_0} \frac{\sin(k_n r)}{k_n r} r^2 dr \bigg/ \int_0^{r_0} \frac{\sin^2(k_n r)}{k_n^2 r^2} r^2 dr$$

$$= k_n (v_0 - u_0) \int_0^{r_0} r \cdot \sin(k_n r) dr \bigg/ \int_0^{r_0} \sin^2(k_n r) dr,$$

$$\begin{aligned} \therefore \int_0^{r_0} \sin^2(k_n r) dr &= \frac{1}{k_n} \left[\frac{k_n r}{2} - \frac{1}{4} \sin(2k_n r) \right]_0^{r_0} \\ &= \frac{1}{k_n} \left[\frac{k_n r_0}{2} - \frac{1}{4} \sin(2k_n r_0) \right] \\ &= \frac{r_0}{2} - \frac{1}{k_n^2} \sin(k_n r_0) \cos(k_n r_0), \end{aligned}$$

$$\begin{aligned}
& \text{及} \quad k_n(v_0 - u_0) \int_0^{r_0} r \cdot \sin(k_n r) dr \\
& = (v_0 - u_0) k_n \left[\frac{1}{k_n^2} \sin(k_n r) - \frac{1}{k_n} r \cos(k_n r) \right]_0^{r_0} \\
& = (v_0 - u_0) \left[\frac{\sin(k_n r_0)}{k_n} - r_0 \cos(k_n r_0) \right], \\
& \therefore C_n = (v_0 - u_0) \frac{1}{k_n} [\sin(k_n r_0) - r_0 k_n \cos(k_n r_0)] \bigg/ \frac{1}{2k_n} \\
& \quad \times [r_0 k_n - \sin(k_n r_0) \cos(k_n r_0)] \\
& = 2(v_0 - u_0) \frac{\sin(k_n r_0) - r_0 k_n \cos(k_n r_0)}{r_0 k_n - \sin(k_n r_0) \cos(k_n r_0)} \\
& = \frac{2r_0(v_0 - u_0)}{H} \cdot \frac{\sin(k_n r_0)}{k_n r_0 - \sin(k_n r_0) \cdot \cos(k_n r_0)},
\end{aligned}$$

故:

$$\begin{aligned}
u(r, t) = u_0 + \sum_{n=1}^{\infty} \frac{2(v_0 - u_0)r_0}{H} \cdot \frac{\sin k_n r_0}{k_n r_0 - \sin k_n r_0 \cdot \cosh k_n r_0} \\
\cdot \frac{\sin k_n r}{k_n r} \cdot e^{-k_n^2 a^2 t}. \quad (4)
\end{aligned}$$

6. 半径为 r_0 的球面径向速度分布为 $v = v_0 \frac{1}{4} (3 \cos 2\theta + 1) \times \cos \omega t$, 试求解这个球在空气中辐射出去的声场中的速度势, 设 $r_0 \ll \lambda$ (波长). 本题径向速度对空间中的方向的依赖性由因子 $\frac{1}{4} (3 \cos 2\theta + 1)$ 即 $P_2(\cos \theta)$ 描写, 因而是轴对称四极声源.

解: 速度势满足三维波动方程, 即

$$v_{rr} - a^2 \Delta v = 0, \quad (1)$$

其中 $a^2 = \frac{p_0 r}{\rho_0}$, p_0 是初始压强, ρ_0 是初始密度, γ 是定压比热与定容比热的比值.

又声波传播的速度与速度势的关系为 $\vec{v} = \nabla v$, 所以在球面上有条件

$$\left. \frac{\partial v}{\partial r} \right|_{r=r_0} = v_0 \frac{1}{4} (3 \cos 2\theta + 1) \cdot \cos \omega t = v_0 P_2(\cos \theta) e^{-i\omega t}, \quad (2)$$

这里时间因子取指数形式是为了便利计算, 在最后结果中取其
实部。

三维波动方程(1)的分离变数形式解是

$$\begin{Bmatrix} h_l^{(1)}(kr) \\ h_l^{(2)}(kr) \end{Bmatrix} P_l^m(\cos \theta) \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} e^{ikat} \\ e^{-ikat} \end{Bmatrix},$$

与边界条件(2)比较, 有 $l = 2$, $m = 0$, 时间因子取 e^{-ikat} , 且 $ka = \omega$, 即 $k = \omega/a$,

又 $h_l^{(1)}(kr)$ 含因子 e^{ikr} , 与 e^{-ikat} 结合给出 $e^{ik(r-at)}$ 的辐射波, 符合题意, 应保留. 而 $h_l^{(2)}(kr)$ 含因子 e^{-ikr} , 与 e^{-ikat} 结合给出 $e^{-ik(r+at)}$ 的收缩波, 不符题意, 应舍去。

综上所述, 其解应为:

$$\begin{aligned} U(r, \theta, t) &= A h_2^{(1)}(kr) \cdot P_2(\cos \theta) e^{-ikat} \\ &= A \left[-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right] e^{ikr} \\ &\quad \times P_2(\cos \theta) e^{-ikat}, \end{aligned}$$

代入边界条件(2), 有

$$A \left[\frac{9i}{(kr_0)^4} + \frac{9}{(kr_0)^3} - \frac{2i}{(kr_0)^2} + \frac{1}{kr_0} \right] k e^{ikr_0} = v_0,$$

$\because r_0 \ll \lambda, \therefore kr_0 \ll 2\pi$, 即 kr_0 很小, 而 $e^{ikr_0} \approx 1$, 于是上式可用第一项来表示,

$$\frac{9i}{(kr_0)^4} Ak = v_0,$$

$$\therefore A = -i \frac{v_0 k^3 r_0^4}{9},$$

故
$$U = -i \frac{U_0 k^3 r_0^4}{9} h_{\frac{1}{2}}^{(1)}(kr) P_2(\cos\theta) e^{-ikat},$$

取实部
$$U = \operatorname{Re} \left[-i \frac{U_0 k^3 r_0^4}{9} h_{\frac{1}{2}}^{(1)}(kr) \cdot P_2(\cos\theta) \cdot e^{-ikat} \right],$$

(3)

在远场区 ($r \gg 0$):

$$U = -i \frac{U_0 k^3 r_0^4}{9} \left[-\frac{3i}{k^3 r^3} - \frac{3}{k^2 r^2} + \frac{i}{kr} \right] e^{ikr} \cdot P_2(\cos\theta) \times e^{-ikat},$$

$\therefore r \gg 0$, $\therefore$ 可只取第三项

$$U_{r \text{ 很大}} = \frac{U_0 k^2 r_0^4}{9r} e^{ik(r-at)} \cdot P_2(\cos\theta),$$

取实部

$$U_{r \text{ 很大}} = \frac{U_0 k^2 r_0^4}{9r} P_2(\cos\theta) \cdot \cos k(r-at). \quad (4)$$

§48. 路积分表示式和渐近公式

1. 半径为 ρ_0 的长圆柱面上一条母线作谐振动, 即柱面径向速度为 $U = v_0 \delta(\varphi - \varphi_0) \cos \omega t$, 试求解这个长圆柱在空气中辐射出去的声场中的速度势, 设 $\rho_0 \ll \lambda$.

解: $\therefore$ 是长圆柱, $\therefore$ 问题与 z 无关, 其速度势满足二维波动方程:

$$U_{rr} - a^2 \Delta_2 U = 0, \quad (1)$$

$$\left. \frac{\partial U}{\partial \rho} \right|_{\rho=\rho_0} = v_0 \delta(\varphi - \varphi_0) \cos \omega t = v_0 \delta(\varphi - \varphi_0) e^{-i\omega t}, \quad (2)$$

注意在最后结果中取实部。

方程(1)的分离变数形式解

$$\left\{ \begin{array}{l} H_m^{(1)}(k\rho) \\ H_m^{(2)}(k\rho) \end{array} \right\} \left\{ \begin{array}{l} \cos m\varphi \\ \sin m\varphi \end{array} \right\} \left\{ \begin{array}{l} e^{ikat} \\ e^{-ikat} \end{array} \right\},$$

考虑到边界条件,时间因子应取 e^{-ikat} ,且有

$$ka = \omega, \text{ 即 } k = \omega/a,$$

又 $\because H_m^{(1)}(k\rho)$ 含有因子 $e^{ik\rho}$,与 e^{-ikat} 结合给出 $e^{ik(\rho-at)}$ 的发散波,符合题意,应保留,

而 $H_m^{(2)}(k\rho)$ 中含有因子 $e^{-ik\rho}$,与 e^{-ikat} 结合给出 $e^{-ik(\rho+at)}$ 的收敛波,不符题意,应舍去。

所求的解

$$v(r, \varphi, t) = \sum_{m=0}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) H_m^{(1)}(k\rho) e^{-ikat}, \quad (3)$$

为确定系数 A_m, B_m ,把上式代入边界条件(2),有

$$\sum_{m=0}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \left. \frac{d}{d\rho} H_m^{(1)}(k\rho) \right|_{\rho=\rho_0} = v_0 \delta(\varphi - \varphi_0), \quad (4)$$

$\because \rho_0 \ll \lambda, \therefore k\rho_0 \ll 2\pi$, 即 $k\rho_0$ 很小,

$$\begin{aligned} \text{这时有 } H_m^{(1)} &= J_m + iN_m \approx \frac{1}{m!} \left(\frac{k\rho}{2} \right)^m - i \frac{(m-1)!}{\pi} \left(\frac{2}{k\rho} \right)^m \\ &\approx -i \frac{(m-1)!}{\pi} \left(\frac{2}{k\rho} \right)^m, \end{aligned}$$

$$H_0^{(1)} = J_0 + iN_0 \approx 1 - \frac{1}{4}(k\rho)^2 + i \frac{2}{\pi} \left[\ln \left(\frac{k\rho}{2} \right) + C \right]$$

$$\approx i \frac{2}{\pi} \left[\ln \left(-\frac{k\rho}{2} \right) + C \right],$$

$$\begin{aligned} \therefore \left. \frac{d}{d\rho} H_m^{(1)}(k\rho) \right|_{\rho=\rho_0} &\approx \frac{d}{d\rho} \left[-i \frac{(m-1)!}{\pi} \left(-\frac{2}{k\rho} \right)^m \right]_{\rho=\rho_0} \\ &= i \frac{m!}{\pi} \left(\frac{2}{k} \right)^m \frac{1}{\rho_0^{m+1}}, \end{aligned} \quad (5)$$

$$\begin{aligned} \left. \frac{d}{d\rho} H_m^{(1)}(k\rho) \right|_{\rho=\rho_0} &\approx \frac{d}{d\rho} \left[i \frac{2}{\pi} \ln \left(-\frac{k\rho}{2} + C \right) \right]_{\rho=\rho_0} \\ &= i \frac{2}{\pi \rho_0}, \end{aligned} \quad (6)$$

将 $\delta(\varphi - \varphi_0)$ 在 $[0, 2\pi]$ 上展成傅里叶级数,

$$\begin{aligned} \delta(\varphi - \varphi_0) &= \frac{1}{2\pi} + \sum_{m=1}^{\infty} (\cos m\varphi_0 \cos m\varphi \\ &\quad + \sin m\varphi_0 \sin m\varphi), \end{aligned} \quad (7)$$

把 (5)、(6)、(7) 代入 (4) 式,

$$\begin{aligned} &A_0 \left(i \frac{2}{\pi \rho_0} \right) + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \left[i \frac{m!}{\pi} \left(\frac{2}{k} \right)^m \frac{1}{\rho_0^{m+1}} \right] \\ &= v_0 \left[\frac{1}{2\pi} + \sum_{m=1}^{\infty} (\cos m\varphi_0 \cos m\varphi + \sin m\varphi_0 \sin m\varphi) \right], \end{aligned}$$

比较两边系得:

$$\begin{aligned} A_0 &= -i \frac{v_0 \rho_0}{4}, \\ B_0 &= 0, \\ \left. \begin{aligned} A_m &= -i \frac{v_0 \rho_0^{m+1} k^m}{2^m \cdot m!} \cos m\varphi_0, \\ B_m &= -i \frac{v_0 \rho_0^{m+1} k^m}{2^m \cdot m!} \sin m\varphi_0, \end{aligned} \right\} (m \neq 0), \end{aligned}$$

故
$$U = \operatorname{Re} \left\{ \left[\frac{v_0 \rho_0}{4} H_0^{(1)}(k\rho) \right. \right.$$

$$+ \sum_{m=1}^{\infty} \frac{U_0 \rho_0^{m+1} k^m}{2^m \cdot m!} H_m^{(1)}(k\rho) \cdot \cos m(\varphi - \varphi_0) \Big\} e^{-i\frac{\pi}{2}} \cdot e^{-i\omega t} \Big\}. \quad (8)$$

$$(\because -i = e^{-i\pi/2}),$$

2. 半径为 r_0 的球面径向速度为 $v = v_0 \frac{3}{2} (1 - \cos 2\theta) \cdot \sin 2\varphi \cdot \cos \omega t$, 试求解这个球在空气中辐射出去的声场中的速度势, 设 $r_0 \ll \lambda$. 本题是非轴对称的四极声源.

解: 速度势满足三维波动方程, 定解问题为

$$U_{rr} - a^2 \Delta U = 0, \quad (1)$$

$$\begin{aligned} \left. \frac{\partial U}{\partial r} \right|_{r=r_0} &= U_0 \frac{3}{2} (1 - \cos 2\theta) \sin 2\varphi \cdot \cos \omega t \\ &= U_0 P_2^2(\cos \theta) \cdot \sin 2\varphi \cdot e^{-i\omega t}, \end{aligned} \quad (2)$$

注意最后取实部.

方程(1)的分离变数形式解

$$\left\{ \begin{matrix} h_l^{(1)}(kr) \\ h_l^{(2)}(kr) \end{matrix} \right\} P_l^m(\cos \theta) \left\{ \begin{matrix} \cos m\varphi \\ \sin m\varphi \end{matrix} \right\} \left\{ \begin{matrix} e^{ikat} \\ e^{-ikat} \end{matrix} \right\},$$

由题意和边界条件(2), 应取:

$$l = 2, m = 2, e^{-ikat},$$

$$(ka = \omega), h_l^{(1)}(kr),$$

$$\therefore U = A h_2^{(1)}(kr) \cdot P_2^2(\cos \theta) \cdot \sin 2\varphi e^{-ikat},$$

$$\text{同时有} \quad A \frac{d}{dr} [h_2^{(1)}(kr)] \Big|_{r=r_0} = U_0,$$

$$\because r_0 \ll \lambda,$$

$$\therefore kr_0 \ll 2\pi, \text{即 } kr_0 \text{ 很小},$$

$$\therefore e^{ikr_0} \approx 1,$$

$$\text{因而 } h_{\frac{1}{2}}^{(1)}(kr) = \left[-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right] \approx -i \frac{3}{(kr)^3},$$

$$\therefore A \frac{d}{dr} \left[-\frac{3i}{(kr)^3} \right]_{r=0} = v_0,$$

$$\text{即} \quad A = -i \frac{v_0 r_0^4 k^3}{9},$$

$$\text{故 } U = \text{Re} \left[-i \frac{U_0 r_0^4 k^3}{9} \cdot h_{\frac{1}{2}}^{(1)}(kr) P_2^2(\cos\theta) \cdot \sin 2\varphi \cdot e^{-i\omega t} \right], \quad (3)$$

对于远场区 ($r \gg 0$):

$$h_{\frac{1}{2}}^{(1)}(kr) \approx i \frac{1}{kr} e^{ikr},$$

$$\therefore U = \text{Re} \left[-i \frac{U_0 r_0^4 k^3}{9} \cdot i \frac{1}{kr} e^{ikr} P_2^2(\cos\theta) \cdot \sin 2\varphi \cdot e^{-i\omega t} \right],$$

$$\text{即} \quad U = \frac{U_0 r_0^4 k^2}{9r} P_2^2(\cos\theta) \cdot \sin 2\varphi \cdot \cos k(r - at). \quad (4)$$

第十五章 数学物理方程的 解的积分公式

§ 50. 格林公式应用于拉普拉斯方程和泊松方程

1. 在圆 $\rho = a$ 内求解拉普拉斯方程的第一边值问题

$$\begin{cases} \Delta_2 u = 0, & (\rho \leq a), \\ u|_{\rho=a} = f(\varphi). \end{cases}$$

解: 圆的第一边值问题的格林函数已在 § 37 例 8 中求出 (37·37):

$$G(M, M_0)$$

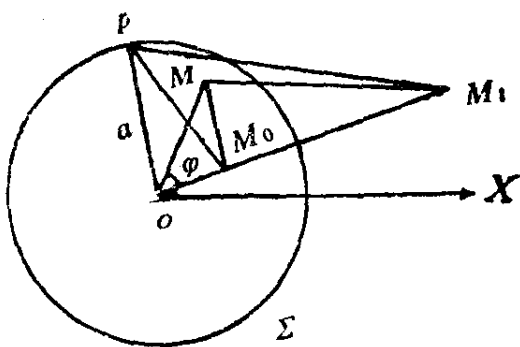


图 15-1

$$= \frac{1}{2\pi} \left[-\ln \frac{1}{|\vec{r} - \vec{r}_0|} + \ln \frac{1}{|\vec{r} - \vec{r}_1|} + \ln \frac{a}{\rho_0} \right],$$

其中 $\vec{r}$, $\vec{r}_0$ 和 $\vec{r}_1$ 分别是点 $M(\rho, \varphi)$, $M_0(\rho_0, \varphi_0)$ 和 $M_1(\rho_1, \varphi_1)$ 的矢径, 且 M_1 是 M_0 关于圆 O 的“电像”, 因而有

$$\rho_1 = a^2 / \rho_0,$$

又由图 (15-1) 知: (设 $\phi = \varphi - \varphi_0$),

$$|\vec{r} - \vec{r}_0| = \sqrt{\rho^2 - 2\rho\rho_0\cos\phi + \rho_0^2},$$

$$|\vec{r} - \vec{r}_1| = \sqrt{\rho^2 - 2\rho\rho_1\cos\phi + \rho_1^2},$$

$$\begin{aligned} \text{于是 } G = \frac{1}{2\pi} & \left[\ln \sqrt{\rho^2 - 2\rho\rho_0\cos\phi + \rho_0^2} \right. \\ & \left. - \ln \sqrt{\rho^2 - 2\rho\rho_1\cos\phi + \rho_1^2} + \ln \frac{a}{\rho_0} \right]. \end{aligned}$$

$$\begin{aligned}
\therefore \frac{\partial G}{\partial n} \Big|_Z &= \frac{\partial G}{\partial \rho} \Big|_{\rho=a} = \frac{1}{2\pi} \left[\frac{\rho - \rho_0 \cos \phi}{\rho^2 - 2\rho\rho_0 \cos \phi + \rho_0^2} \right. \\
&\quad \left. - \frac{\rho - \rho_1 \cos \phi}{\rho^2 - 2\rho\rho_1 \cos \phi + \rho_1^2} \right]_{\rho=a} \\
&= \frac{1}{2\pi} \left[\frac{a - \rho_0 \cos \phi}{a^2 - 2a\rho_0 \cos \phi + \rho_0^2} \right. \\
&\quad \left. - \frac{a - a^2 \cos \phi / \rho_0}{a^2 - 2a \cos \phi \cdot a^2 / \rho_0 + a^4 / \rho_0^2} \right] \\
&= \frac{(a - \rho_0 \cos \phi) - (\rho_0/a)^2 (a - a^2 \cos \phi / \rho_0)}{a^2 - 2a\rho_0 \cos \phi + \rho_0^2} \\
&= \frac{1}{2\pi a} \cdot \frac{a^2 - \rho_0^2}{a^2 - 2a\rho_0 \cos \phi + \rho_0^2},
\end{aligned}$$

又 $\because G(M, M_0) \Big|_Z = 0, ds = ad\varphi,$

故代入解的积分公式得:

$$\begin{aligned}
u(\rho_0, \varphi_0) &= \frac{1}{2\pi a} \int_0^{2\pi} \frac{(a^2 - \rho_0^2) f(\varphi)}{a^2 - 2a\rho_0 \cos \phi + \rho_0^2} ad\varphi \\
&= \frac{a^2 - \rho_0^2}{2\pi} \int_0^{2\pi} \frac{f(\varphi)}{a^2 - 2a\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2} d\varphi.
\end{aligned}$$

2. 在半平面 $y > 0$ 内求解拉普拉斯方程的第一边值问题

$$\begin{cases} \Delta_2 u = 0, & (y > 0), \\ u|_{y=0} = f(x). \end{cases}$$

解: 平面第一边值问题的格林函数满足定解问题

$$\begin{cases} \Delta_2 G(M, M_0) = \delta(x - x_0) \cdot \delta(y - y_0), \\ G|_{y=0} = 0, \end{cases}$$

其解为:

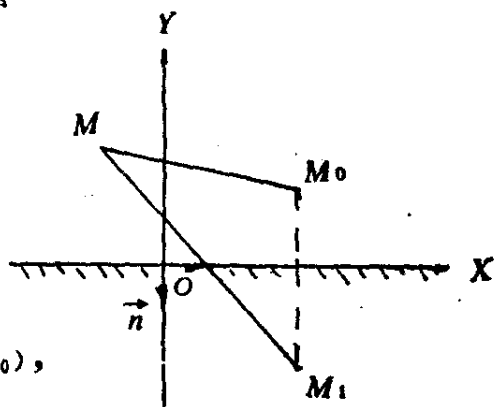


图 15-2

$$G(M, M_0) = \frac{1}{2\pi} \left[-\ln \frac{1}{|\vec{r} - \vec{r}_0|} + \ln \frac{1}{|\vec{r} - \vec{r}_1|} \right],$$

式中 $\vec{r}$, $\vec{r}_0$ 和 $\vec{r}_1$ 是点 $M(x, y)$, $M_0(x_0, y_0)$ 和 $M_1(x_0, -y_0)$ 的矢径, 且 M_1 是 M_0 关于平面 $y=0$ 的“电像”。

由图 (15-2) 知:

$$\begin{aligned} |\vec{r} - \vec{r}_0| &= \sqrt{(x-x_0)^2 + (y-y_0)^2}, \\ |\vec{r} - \vec{r}_1| &= \sqrt{(x-x_0)^2 + (y+y_0)^2}, \\ \therefore \frac{\partial G}{\partial n} \Big|_z &= - \frac{\partial G}{\partial y} \Big|_{y=0} = - \frac{1}{2\pi} \left[\frac{y-y_0}{(x-x_0)^2 + (y-y_0)^2} \right. \\ &\quad \left. - \frac{y+y_0}{(x-x_0)^2 + (y+y_0)^2} \right]_{y=0} \\ &= \frac{1}{\pi} \cdot \frac{y_0}{(x-x_0)^2 + y_0^2}, \end{aligned}$$

故代入解的积分公式, 得

$$u(x_0, y_0) = \frac{y_0}{\pi} \int_{-\infty}^{\infty} \frac{f(x)}{(x-x_0)^2 + y_0^2} dx,$$

3. 在圆形域 $\rho \leq a$ 上求解, $\Delta u = 0$, 使满足边界条件

$$(1) \quad u|_{\rho=a} = A \cdot \cos \varphi, \quad (2) \quad u|_{\rho=a} = A + B \sin \varphi.$$

$$\text{解一: } \begin{cases} \Delta_2 u = 0, \\ u|_{\rho=a} = A \cdot \cos \varphi, \end{cases}$$

由习题 1 的结果, 有

$$u(\rho_0, \varphi_0) = \frac{a^2 - \rho_0^2}{2\pi} \int_0^{2\pi} \frac{A \cdot \cos \varphi \cdot d\varphi}{a^2 - 2a\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2},$$

$\therefore$ 是圆内问题, 有 $a > \rho_0$, 上式分子、分母同除以 a^2 ,

并令 $\frac{\rho_0}{a} = \varepsilon$, ($0 < \varepsilon < 1$), 有:

$$u(\rho_0, \varphi_0) = \frac{(1 - \varepsilon^2)A}{2\pi} \int_0^{2\pi} \frac{\cos \varphi d\varphi}{1 - 2\varepsilon \cos(\varphi - \varphi_0) + \varepsilon^2},$$

$$\therefore \cos \varphi = \frac{1}{2} (e^{i\varphi} + e^{-i\varphi}) = \frac{1}{2} \left(z + \frac{1}{z} \right), (\text{令 } z = e^{i\varphi}),$$

$$\cos(\varphi - \varphi_0) = \frac{1}{2}(e^{i\varphi}e^{-i\varphi_0} + e^{-i\varphi}e^{i\varphi_0})$$

$$= \frac{1}{2}\left(e^{-i\varphi_0}z + e^{i\varphi_0}\frac{1}{z}\right),$$

$$d\varphi = \frac{dz}{iz},$$

$$\begin{aligned}\therefore I &= \int_0^{2\pi} \frac{\cos\varphi d\varphi}{1 - 2\varepsilon\cos(\varphi - \varphi_0) + \varepsilon^2} \\ &= \frac{-1}{2i} \oint_{|z|=1} \frac{(1+z^2)dz}{z[e^{-i\varphi_0}\varepsilon z^2 - (1+\varepsilon)^2z + \varepsilon e^{i\varphi_0}]},\end{aligned}$$

被积函数在 $|z|=1$ 内有两个单极点 0 和 $\varepsilon e^{i\varphi_0}$, 其留数为

$$\begin{aligned}&\frac{1}{\varepsilon}e^{-i\varphi_0} \text{ 和 } \frac{1+\varepsilon^2 e^{i2\varphi_0}}{\varepsilon e^{i\varphi_0}(\varepsilon^2-1)}, \text{ 留数之和为 } \frac{(\varepsilon^2-1)+1+\varepsilon^2 e^{i2\varphi_0}}{\varepsilon e^{i\varphi_0}(\varepsilon^2-1)} \\ &= \frac{2\varepsilon}{\varepsilon^2-1} \cos\varphi_0,\end{aligned}$$

$$\therefore I = \frac{-1}{2i} 2\pi i \cdot \frac{2\varepsilon}{\varepsilon^2-1} \cos\varphi_0 = \frac{2\pi\varepsilon}{1-\varepsilon^2} \cos\varphi_0,$$

$$\begin{aligned}\text{故 } u(\rho_0, \varphi_0) &= \frac{1 - (\rho_0/a)^2}{2\pi} A \cdot \frac{2\pi\rho_0/a}{1 - (\rho_0/a)^2} \cos\varphi_0 \\ &= \frac{A\rho_0}{a} \cos\varphi_0.\end{aligned}$$

$$\text{解二: } \begin{cases} \Delta u = 0, \\ u|_{\rho=a} = A + B\sin\varphi, \end{cases}$$

$$\begin{aligned}u(\rho, \varphi_0) &= \frac{a^2 - \rho_0^2}{2\pi} \int_0^{2\pi} \frac{A + B\sin\varphi}{a^2 - 2a\rho_0\cos(\varphi_0 - \varphi_0) + \rho_0^2} d\varphi \\ &= \frac{1 - \varepsilon^2}{2\pi} \int_0^{2\pi} \frac{A + B\sin\varphi}{1 - 2\varepsilon\cos(\varphi - \varphi_0) + \varepsilon^2} d\varphi,\end{aligned}$$

式中 $\varepsilon = \rho_0/a$,

$$\text{计算积分 } I = \int_0^{2\pi} \frac{A + B\sin\varphi}{1 - 2\varepsilon\cos(\varphi - \varphi_0) + \varepsilon^2} d\varphi.$$

令 $z = e^{i\varphi}$, 则

$$I = \frac{1}{2} \oint_{|z|=1} \frac{2Aiz + Bz^2 - B}{z[ee^{-i\varphi_0}z^2 - (1+\varepsilon^2)z + \varepsilon e^{i\varphi_0}]} dz,$$

被积函数在 $|z|=1$ 内有两个单极点 0 与 $\varepsilon e^{i\varphi_0}$, 相应的留数为

$$\frac{-B}{\varepsilon e^{i\varphi_0}} \text{ 与 } \frac{2Aie^{i\varphi_0} + B(\varepsilon e^{i2\varphi_0} - 1)}{\varepsilon e^{i\varphi_0}(\varepsilon^2 - 1)},$$

$$\text{留数之和} = \frac{-B\varepsilon^2 + B + 2Aie^{i\varphi_0} + B(\varepsilon^2 e^{i2\varphi_0} - 1)}{\varepsilon e^{i\varphi_0}(\varepsilon^2 - 1)}$$

$$= -\frac{2Ai}{1-\varepsilon^2} - \frac{B\varepsilon(e^{i\varphi_0} - e^{-i\varphi_0})}{1-\varepsilon^2}$$

$$= -\frac{2i(A + B\varepsilon \sin \varphi_0)}{1-\varepsilon^2},$$

$$\therefore I = 2\pi i \cdot \frac{1}{2} \cdot \left[-\frac{2i(A + B\varepsilon \sin \varphi_0)}{1-\varepsilon^2} \right]$$

$$= -\frac{2\pi(A + B\varepsilon \sin \varphi_0)}{1-\varepsilon^2}.$$

应用这个结果

$$\begin{aligned} u(\rho_0, \varphi_0) &= \frac{1 - \left(\frac{\rho_0}{a}\right)^2}{2\pi} \cdot \frac{2\pi[A + B(\rho_0/a)\sin \rho_0]}{1 - \left(\frac{\rho_0}{a}\right)^2} \\ &= A + B\frac{\rho_0}{a}\sin \varphi_0. \end{aligned}$$

4. 圆内拉氏方程第一边值问题中, 对于一般的 $f(\varphi)$ 积分不易计算. 试把 $1/[a^2 - 2a\rho_0\cos(\varphi - \varphi_0) + \rho_0^2]$ 展开为傅里叶级数, 然后逐项积分.

作为对照, 再用分离变量法求解圆的第一边值问题.

解: 圆内拉氏方程第一边值问题解的积分公式是:

$$u(\rho_0, \varphi_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho_0^2}{a^2 - 2a\rho_0\cos(\varphi - \varphi_0) + \rho_0^2} f(\varphi) d\varphi$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - \varepsilon^2}{1 - 2\varepsilon \cos(\varphi - \varphi_0) + \varepsilon^2} f(\varphi) d\varphi, \\ (\varepsilon = \rho_0/a),$$

利用 § 25 习题 2 的答案

$$\frac{1 - \varepsilon^2}{1 - 2\varepsilon \cos(\varphi - \varphi_0) + \varepsilon^2} = 1 + 2 \sum_{n=1}^{\infty} \varepsilon^n \cos n(\varphi - \varphi_0),$$

将上式逐项积分, 即得

$$\begin{aligned} u(\rho_0, \varphi_0) &= \frac{1}{2\pi} \int_0^{2\pi} f(\varphi) \left[1 + 2 \sum_{n=1}^{\infty} \varepsilon^n \cos n(\varphi - \varphi_0) \right] d\varphi \\ &= \sum_{n=0}^{\infty} \frac{1}{\delta_n} \left(\frac{\rho_0}{a} \right)^n \left[(\cos n\varphi_0) \cdot \frac{1}{\pi} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi \right. \\ &\quad \left. + (\sin n\varphi_0) \cdot \frac{1}{\pi} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi \right] \\ &= \sum_{n=0}^{\infty} [A_n \cos n\varphi_0 + B_n \sin n\varphi_0] \rho_0^n, \end{aligned}$$

其中 $A_n = \frac{1}{\delta_n \pi a^n} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi,$

$$B_n = \frac{1}{\pi a^n} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi,$$

作为对照, 用分离变数法重新求解, 圆内拉普拉斯方程的一般解是

$$u(\rho, \varphi) = \sum_{n=0}^{\infty} (A_n \cos n\varphi + B_n \sin n\varphi) \rho^n,$$

为确定系数 A_n 和 B_n , 以这式代入边界条件得

$$\sum_{n=0}^{\infty} (A_n a^n \cos n\varphi + B_n a^n \sin n\varphi) = f(\varphi),$$

这样 $A_n a^n$, $B_n a^n$, 是 $f(\varphi)$ 的傅里叶系数, 所以

$$A_n = \frac{1}{\delta_n \pi a^n} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi,$$

$$B_n = \frac{1}{\pi a^n} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi,$$

两种解法结果相同。

5. 试求层状空间 $0 < z < H$ 第一边值问题的格林函数。

解：格林函数的定解问题为：

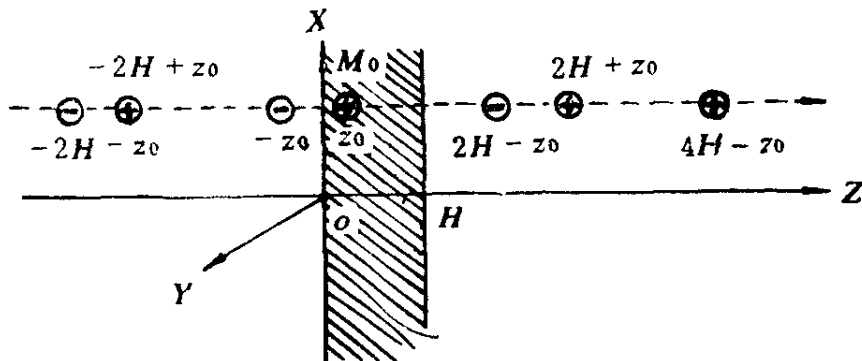


图 15-3

$$\begin{cases} \Delta G = \delta(\vec{r} - \vec{r}_0), \\ G|_{z=0} = 0, \\ G|_{z=H} = 0. \end{cases}$$

平面 $z=0$ 和 $z=H$ 相当于两面平面镜，电荷放在中间，出现反复反射，造成无限多个“电像”。对于放在 $M_0(x_0, y_0, z_0)$ 点的正点电荷（负电荷亦可），所有位于 $(x_0, y_0, 2nH + z_0)$ 处“电像”带正电荷，而所有位于 $(x_0, y_0, 2nH - z_0)$ 处“电像”带负电荷。

空间第一边值问题的格林函数已在 §50 例 2 中求出为 (50.22) 式

$$G(\vec{r}; \vec{r}_0) = -\frac{1}{4\pi} \sum_{n=-\infty}^{+\infty} \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-2nH-z_0)^2}} \\ + \frac{1}{4\pi} \sum_{n=-\infty}^{\infty} \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-2nH+z_0)^2}},$$

这里的格林函数是无穷级数。

§ 51. 推广的格林公式及其应用

$$1. \text{ 求解 } \begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), \\ u|_{t=0} = \varphi(x), \\ u_t|_{t=0} = \psi(x). \end{cases}$$

解: (1) 作变换, 化方程为标准型. 因为上面方程的特征线方程是 $x \pm at = C$.

$$\text{令 } y = at, \quad (1)$$

则定解问题成为:

$$\begin{aligned} u_{xx} - u_{yy} &= -\frac{1}{a^2} f\left(x, \frac{y}{a}\right) \\ &= f_1(x, y), \quad (2) \end{aligned}$$

$$\begin{cases} u|_{y=0} = \varphi(x), \\ u_y|_{y=0} = \psi(x). \end{cases} \quad (3)$$

(2) 求里曼函数 v .

$\because$ 方程(2)的系数 $a_{11} = 1, a_{22} = -1, a_{12} = b_1 = b_2 = c = 0$,

$\therefore$ 算符 $L = M = \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$ 是自伴的, 其里曼函数 v 应

是方程 $MU = 0$ 的解, 且满足条件

$$\begin{cases} U = e^{\int_{s_0}^s \frac{b_2 + b_1}{2\sqrt{2}} ds} = e^0 = 1, & (\text{在 } M_2 M_0 \text{ 上}), \\ U = e^{\int_{s_0}^s \frac{b_2 - b_1}{2\sqrt{2}} ds} = e^0 = 1, & (\text{在 } M_0 M_1 \text{ 上}), \end{cases}$$

如取 $v(M, M_0) = 1$, (4)

则上述条件均被满足.

(3) 定解问题的解. 由§51中(51·33)式, 解为:

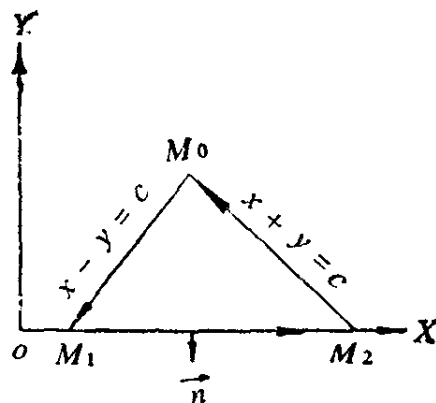


图 15-4

$$\begin{aligned}
u(x, y) - v(x, y) = & \frac{1}{2} [u(x_1, y_1)v(x_1, y_1) \\
& + u(x_2, y_2)v(x_2, y_2)] \\
& + \frac{1}{2} \int_{M_1 M_2} [X \cdot \cos(\vec{n}, \vec{x}) + Y \cdot \cos(\vec{n}, \vec{y})] \\
& dl - \frac{1}{2} \iint_{M_0 M_1 M_2} v f_1 ds, \quad (5)
\end{aligned}$$

其中 X 和 Y 由 §51 中 (51.30) 式给出:

$$\begin{cases} X = -(uv)_x + (2u_x + b_1 u)v, \\ Y = (uv)_y - (2u_y + b_2 u)v, \end{cases} \quad (6)$$

下面分别算出 (5) 式各项.

由图 (15.4) 知: $M_1(x_1, y_1)$ 的坐标为:

$$x_1 = x - y = x - at, \quad y_1 = 0,$$

$M_2(x_2, y_2)$ 的坐标为:

$$x_2 = x + y = x + at, \quad y_2 = 0,$$

而 $v(M_1, M_0) = v(M_2, M_0) = 1$,

$$u|_{y_1=0} = \varphi(x_1) = \varphi(x - at),$$

$$u|_{y_2=0} = \varphi(x_2) = \varphi(x + at),$$

$\therefore$ (5) 式中第一项为:

$$\frac{1}{2} [\varphi(x - at) + \varphi(x + at)],$$

由 (6) 知: $X = u_x$,

$$Y = -u_y,$$

对于直线段 $M_1 M_2$, 有

$$y = 0, \quad \cos(\vec{n}, \vec{x}) = \cos\left(\frac{\pi}{2}\right) = 0, \quad \cos(\vec{n}, \vec{y}) = -1,$$

$\therefore$ (5) 式中第二项积分为

$$\frac{1}{2} \int_{M_1}^{M_2} [u_y]|_{y=0} dx = \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi,$$

$\therefore$ 在区域 $M_0M_1M_2$ 内, $v \equiv 1$, 又 $dy = a dt$,

$\therefore$ (5) 式中最后一项积分为:

$$-\frac{1}{2} \int \int_{M_0M_1M_2} f_1(x, y) dx dy$$

$$= \frac{1}{2a} \int_0^{at} \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau,$$

故 $u(x, y) = \frac{1}{2} [\varphi(x-at) + \varphi(x+at)]$

$$+ \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

$$+ \frac{1}{2a} \int_0^{at} \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi \cdot d\tau. \quad (7)$$

2. 求解 $\begin{cases} x^2 u_{xx} - t^2 u_{tt} = 0, \\ u|_{t=1} = \varphi(x), \\ u_t|_{t=1} = \psi(x). \end{cases}$

解: (1) 作变换, 化方程为标准型, 上面方程的特征方程为

$$x^2 - t^2 \left(\frac{dx}{dt} \right)^2 = 0,$$

则特征线为 $\ln x \pm \ln t = C$.

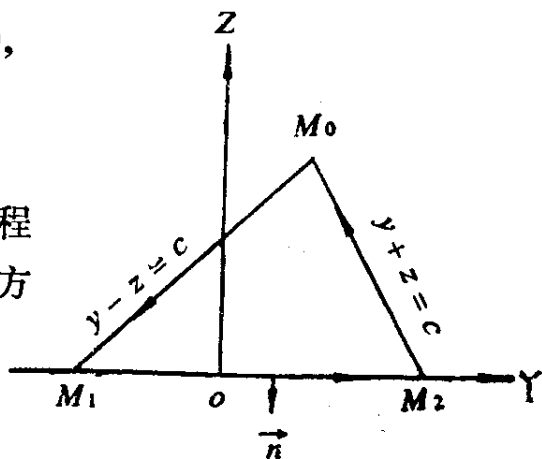


图 15-5

令 $y = \ln x, z = \ln t$,

则定解问题变为:

$$u_{yy} - u_{zz} - u_y + u_z = 0,$$

$$\begin{cases} u|_{z=0} = \varphi(e^y), \\ u_z|_{z=0} = u_t \cdot \frac{dt}{dz} \Big|_{t=1} = \psi(e^y) \end{cases} \quad (3)$$

(2) 求里曼函数 v

$\therefore a_{11} = 1, a_{22} = -1, b_1 = -1,$

$b_2 = 1, a_{12} = c = 0,$

$$\therefore M = \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} + \frac{\partial}{\partial y} - \frac{\partial}{\partial z},$$

于是有 $Mv = 0$,

$$v = e^{\int_{s_0}^s \frac{b_2 - b_1}{2\sqrt{2}} ds} = e^0 = 1, \quad (\text{在 } M_2M_0 \text{ 上}),$$

$$v = e^{\int_{s_0}^s \frac{b_2 + b_1}{2\sqrt{2}} ds} = e^{-(z - z_0)}, \quad (\text{在 } M_0M_1 \text{ 上}),$$

$\therefore$ 可取

$$v = e^{-\frac{1}{2}(y - y_0 + z - z_0)}. \quad (4)$$

(3) 定解问题的解, 其解式同上题公式(5). 下面分别计算(5)式中各项.

由图(15·5)知 $M_1(x_1, y_1)$ 的坐标为 $y_1 = y_0 - z_0, z_1 = 0$,

$M_2(x_2, y_2)$ 的坐标为 $y_2 = y_0 + z_0, z_2 = 0$,

则 $v(y_1, z_1) = e^{-\frac{1}{2}(y_0 - z_0 - y_0 + 0 - z_0)} = e^{z_0} = t_0$,

$$v(y_2, z_2) = e^{-\frac{1}{2}(y_0 + z_0 - z_0 + 0 - z_0)} = e^0 = 1,$$

$$u(y_1, z_1) \big|_{z_1=0} = \varphi(e y_1) = \varphi(e y_0 - z_0) = \varphi\left(\frac{x_0}{t_0}\right),$$

$$u(y_2, z_2) \big|_{z_2=0} = \varphi(e y_2) = \varphi(e^{y_0 + z_0}) = \varphi(x_0 t_0),$$

$\therefore$ (5) 式中第一项为:

$$\frac{1}{2} [\varphi(e^{y_0 - z_0}) e^{z_0} + \varphi(e^{y_0 + z_0})] = \frac{1}{2} t_0 \varphi\left(\frac{x_0}{t_0}\right) + \frac{1}{2} \varphi(x_0 t_0),$$

又在 M_1M_2 上, $y = 0, \cos(\vec{n}, \vec{y}) = 0, \cos(\vec{n}, \vec{z}) = -1$, 所以只要算出上题公式(6)中 Z 的表达式.

$$\therefore Z = uv_z - u_z v + b_2 uv$$

$$= \frac{1}{2} uv - vu_z, \quad (\because v_z = -\frac{1}{2}v),$$

$$\begin{aligned}
\therefore Z|_{z=0} &= \frac{1}{2} e^{-\frac{1}{2}(y-y_0-z_0)} u|_{z=0} \\
&\quad - e^{-\frac{1}{2}(y-y_0-z_0)} u_z|_{z=0} \\
&= \frac{1}{2} \varphi(e^y) e^{-\frac{y}{2}} \cdot e^{\frac{1}{2}(y_0+z_0)} \\
&\quad - \psi(e^y) e^{-\frac{1}{2}y} \cdot e^{\frac{1}{2}(y_0+z_0)},
\end{aligned}$$

$$\therefore e^y = x,$$

$$\therefore e^{-\frac{1}{2}y} = x^{-1/2}, \quad e^{\frac{1}{2}(y_0+z_0)} = (x_0 t_0)^{1/2},$$

$$\therefore Z|_{z=0} = \frac{1}{2} (x_0 t_0)^{1/2} \cdot x^{-1/2} \cdot \varphi(x) - (x_0 t_0)^{1/2} x^{-1/2} \psi(x).$$

$$\text{又 } dy = \frac{1}{x} dx,$$

$\therefore$ (5) 式中第二项积分为

$$\begin{aligned}
\frac{1}{2} \int_{M_1 M_2} (-Z) dl &= -\frac{1}{2} \int_{y_1}^{y_2} Z|_{z=0} dy \\
&= -\frac{\sqrt{x_0 t_0}}{4} \int_{x_0/t_0}^{x_0 t_0} \varphi(x) x^{-3/2} dx \\
&\quad + \frac{\sqrt{x_0 t_0}}{2} \int_{x_0/t_0}^{x_0 t_0} \psi(x) x^{-3/2} dx,
\end{aligned}$$

$$\text{又 } \therefore f=0,$$

$\therefore$ (5) 式中最后一项积分为零.

故

$$\begin{aligned}
u(x_0, t_0) &= \frac{1}{2} t_0 \varphi\left(\frac{x_0}{t_0}\right) + \frac{1}{2} \varphi(x_0, t_0) \\
&\quad - \frac{\sqrt{x_0 t_0}}{4} \int_{x_0/t_0}^{x_0 t_0} \varphi(x) x^{-3/2} dx \\
&\quad + \frac{\sqrt{x_0 t_0}}{2} \int_{x_0/t_0}^{x_0 t_0} \psi(x) x^{-3/2} dx. \quad (5)
\end{aligned}$$

第十六章 拉普拉斯变换法

§52. 拉普拉斯变换法

1. 求解一维无界空间的扩散问题, 即 $u_t - a^2 u_{xx} = 0$,

$u|_{t=0} = \varphi(x)$, [本题即 §38 例2, 可对照].

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (-\infty < x < \infty), \\ u|_{t=0} = \varphi(x), \end{cases} \quad (1)$$

对定解问题施行拉普拉斯变换得

$$P\bar{u} - \varphi(x) - a^2 \bar{u}_{xx} = 0, \text{ 即 } a^2 \bar{u}_{xx} - P\bar{u} = -\varphi(x), \quad (2)$$

现在解方程 $a^2 y'' - Py = -\varphi(x)$ 即 $y'' - \frac{P}{a^2} y = -\frac{1}{a^2} \varphi(x)$,

$$(3) \text{ 对应的齐次方程的解是 } y = Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x} \quad (3)$$

(4)

(4) 式的 A 和 B 是积分常数. 为了求得非齐次方程 (3) 的特解, 可以把 (4) 式中的参数 A 和 B 看作是 x 的函数, 而用参数变易法求 (3) 的特解. 具体步骤如下:

$$\begin{aligned} y' = & \frac{\sqrt{p}}{a} Ae^{\frac{\sqrt{p}}{a}x} - \frac{\sqrt{p}}{a} Be^{-\frac{\sqrt{p}}{a}x} + A'e^{\frac{\sqrt{p}}{a}x} \\ & + B'e^{-\frac{\sqrt{p}}{a}x}, \end{aligned} \quad (5)$$

$$\text{引入辅助条件} \quad A'e^{\frac{\sqrt{p}}{a}x} + B'e^{-\frac{\sqrt{p}}{a}x} = 0, \quad (6)$$

则(5)成为 $y' = \frac{\sqrt{p}}{a} A e^{\frac{\sqrt{p}}{a}x} - \frac{\sqrt{p}}{a} B e^{-\frac{\sqrt{p}}{a}x}, \quad (7)$

$$\begin{aligned} y'' &= \frac{p}{a^2} \left(A e^{\frac{\sqrt{p}}{a}x} + B e^{-\frac{\sqrt{p}}{a}x} \right) \\ &\quad + \frac{\sqrt{p}}{a} \left(A' e^{\frac{\sqrt{p}}{a}x} - B' e^{-\frac{\sqrt{p}}{a}x} \right) \\ &= \frac{p}{a^2} \left(A e^{\frac{\sqrt{p}}{a}x} + B e^{-\frac{\sqrt{p}}{a}x} \right), \end{aligned} \quad (8)$$

以(8)式代入(3)式中得

$$\frac{\sqrt{p}}{a} \left(A' e^{\frac{\sqrt{p}}{a}x} - B' e^{-\frac{\sqrt{p}}{a}x} \right) = -\frac{1}{a^2} \varphi(x),$$

或 $A' e^{\frac{\sqrt{p}}{a}x} - B' e^{-\frac{\sqrt{p}}{a}x} = -\frac{1}{a\sqrt{p}} \varphi(x), \quad (9)$

(9)+(6)得 $A' = \frac{-1}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \varphi(x),$

积分得 $A = -\frac{1}{2a\sqrt{p}} \int e^{-\frac{\sqrt{p}}{a}\xi} \varphi(\xi) d\xi, \quad (10)$

以(6)-(9) $B' = \frac{1}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} \varphi(x),$

$$B = \frac{1}{2a\sqrt{p}} \int e^{\frac{\sqrt{p}}{a}\xi} \varphi(\xi) d\xi, \quad (11)$$

以(10)(11)代入(4)中即得方程(3)的特解为:

$$\begin{aligned} y_{\text{特}} &= -\frac{1}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} \int e^{-\frac{\sqrt{p}}{a}\xi} \varphi(\xi) d\xi \\ &\quad + \frac{1}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \int e^{\frac{\sqrt{p}}{a}\xi} \varphi(\xi) d\xi, \end{aligned} \quad (12)$$

从而得到方程(3)的通解为:

$$y = A e^{\frac{\sqrt{p}}{a}x} + B e^{-\frac{\sqrt{p}}{a}x} - \frac{1}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} \int e^{-\frac{\sqrt{p}}{a}\xi} \varphi(\xi) d\xi + \frac{1}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \int e^{\frac{\sqrt{p}}{a}\xi} \varphi(\xi) d\xi$$

$$\begin{aligned}
& \times \int e^{-\frac{\sqrt{P}}{a}\xi} \varphi(\xi) d\xi, \\
& + \frac{1}{2a\sqrt{p}} e^{-\frac{\sqrt{P}}{a}x} \int e^{\frac{\sqrt{P}}{a}\xi} \varphi(\xi) d\xi. \\
& = Ae^{\frac{\sqrt{P}}{a}x} + Be^{-\frac{\sqrt{P}}{a}x} - \frac{1}{2a} e^{\frac{\sqrt{P}}{a}x} \\
& \times \int \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{P}}{a}\xi} \varphi(\xi) d\xi \\
& + \frac{1}{2a} e^{-\frac{\sqrt{P}}{a}x} \int \frac{1}{\sqrt{p}} e^{\frac{\sqrt{P}}{a}\xi} \varphi(\xi) d\xi.
\end{aligned}$$

为了使解在 $x \rightarrow \pm \infty$ 时为有限, 必须取 $A = B = 0$,

$$\begin{aligned}
\therefore \bar{u}(P) = y &= \frac{1}{2a} \int_x^{+\infty} \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{P}}{a}(\xi-x)} \varphi(\xi) d\xi \\
&+ \frac{1}{2a} \int_{-\infty}^x \frac{1}{\sqrt{p}} e^{\frac{\sqrt{P}}{a}(\xi-x)} \varphi(\xi) d\xi,
\end{aligned}$$

应用第496页公式(13) $\frac{e^{-a\sqrt{P}}}{\sqrt{p}} \doteq \frac{1}{\sqrt{\pi t}} e^{-a^2/4t}$, 进行反演.

$$\therefore \frac{e^{\pm \frac{(x-\xi)}{a}\sqrt{P}}}{\sqrt{p}} \doteq \frac{1}{\sqrt{\pi t}} e^{-(x-\xi)^2/4a^2t},$$

$$\begin{aligned}
\therefore u &= \frac{1}{2a} \int_x^{\infty} \frac{\varphi(\xi)}{\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2t}} d\xi + \frac{1}{2a} \\
&\times \int_{-\infty}^x \frac{\varphi(\xi)}{\sqrt{\pi t}} e^{-\frac{(\xi-x)^2}{4a^2t}} d\xi \\
&= \int_{-\infty}^{\infty} \frac{\varphi(\xi)}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2t}} d\xi.
\end{aligned}$$

2. 求解硅片的限定源扩散问题, 把硅片的厚度作为无限大, 这是半无界空间的定解问题 $u_t - a^2 u_{xx} = 0, u_x|_{x=0} = 0$,

$u|_{t=0} = \Phi_0 \delta(x-0)$ [本题即 §38 例 3 可对照].

解一: 对泛定方程和边界条件施行拉普拉斯变换, 得:

$$\begin{cases} p\bar{u} - \Phi_0\delta(x-0) - a^2\bar{u}_{xx} = 0, & (x>0), \\ \bar{u}_x|_{x=0} = 0, \end{cases} \quad (1)$$

由上题, 可知(1)的通解为

$$\begin{aligned} \bar{u}(x, p) = & Ae^{\sqrt{p}x/a} + Be^{-\sqrt{p}x/a} - \frac{1}{2a} \\ & \times \int^{(x)} \frac{e^{\sqrt{p}\left(\frac{x-\xi}{a}\right)}}{\sqrt{p}} \Phi_0\delta(\xi-0) d\xi \\ & + \frac{1}{2a} \int^{(x)} \frac{e^{\sqrt{p}\left(\frac{x-\xi}{a}\right)}}{\sqrt{p}} \Phi_0\delta(\xi-0) d\xi, \end{aligned}$$

$\because x \rightarrow +\infty$ 时, $\bar{u}$ 有界, $\therefore A=0$,

$$\begin{aligned} \text{又 } \bar{u}_x = & -\frac{\sqrt{p}}{a} Be^{-\sqrt{p}x/a} - \frac{1}{2a} \frac{e^0}{\sqrt{p}} \Phi_0\delta(x-0) \\ & + \frac{1}{2a} \frac{e^0}{\sqrt{p}} \Phi_0\delta(x-0) \\ = & -\frac{\sqrt{p}}{a} Be^{-\sqrt{p}x/a}, \end{aligned}$$

$$0 = \bar{u}_x|_{x=0} = -\frac{\sqrt{p}}{a} B,$$

$\therefore B=0$,

$$\begin{aligned} \bar{u}(x, p) = & -\frac{\Phi_0}{2a} \int^{(x)} \frac{e^{\sqrt{p}\left(\frac{x-\xi}{a}\right)}}{\sqrt{p}} \delta(\xi-0) d\xi \\ & + \frac{\Phi_0}{2a} \int^{(x)} \frac{e^{\sqrt{p}\left(\frac{x-\xi}{a}\right)}}{\sqrt{p}} \delta(\xi-0) d\xi \\ = & -\frac{\Phi_0}{2a} \int_{-\infty}^x \frac{\delta(\xi-0)}{\sqrt{p}} e^{\sqrt{p} \cdot \frac{x-\xi}{a}} d\xi \\ & + \frac{\Phi_0}{2a} \int_{-\infty}^x \frac{\delta(\xi-0)}{\sqrt{p}} e^{\sqrt{p}\left(\frac{\xi-x}{a}\right)} d\xi \end{aligned}$$

$$\begin{aligned}
&= \frac{\Phi_0}{2a} \frac{1}{\sqrt{p}} e^{\sqrt{P} \frac{x}{a}} + \frac{\Phi_0}{2a} \frac{1}{\sqrt{p}} e^{\sqrt{P} \cdot \frac{-x}{a}}, \\
u(x, t) &= \frac{\Phi_0}{2a} \frac{1}{\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}} + \frac{\Phi_0}{2a} \frac{1}{\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}} \\
&= \frac{\Phi_0}{a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}}. \quad (2)
\end{aligned}$$

解二：由 $u_x|_{x=0} = 0$ ，作偶延拓，得无界空间的定解问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u|_{t=0} = \begin{cases} \Phi_0 \delta(x-0), & (x > 0), \\ \Phi_0 \delta(x+0), & (x < 0), \end{cases} \end{cases}$$

由上题结果，有

$$\begin{aligned}
\therefore u &= \frac{1}{2a} \int_{-\infty}^{\infty} \frac{\Phi_0 \delta(x-0) + \Phi_0 \delta(x+0)}{\sqrt{\pi t}} e^{-\frac{(\xi-x)^2}{4a^2 t}} d\xi \\
&= \frac{\Phi_0}{2a\sqrt{\pi t}} \left[e^{-\frac{(0-x)^2}{4a^2 t}} + e^{-\frac{(-0-x)^2}{4a^2 t}} \right] \\
&= \frac{\Phi_0}{a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}}.
\end{aligned}$$

3. 求解一维无界空间的有源输运问题， $u_t - a^2 u_{xx} = f(x, t)$ ， $u|_{t=0} = 0$ ，（本题即 § 39 例 3 可对照）。

解：将方程进行拉普拉斯变换， $P\bar{u} - a^2 \bar{u}_{xx} = \bar{f}(xP)$ ，应用第一题的结果

$$\begin{aligned}
\bar{u} &= Ae^{\frac{\sqrt{P}}{a}x} + Be^{-\frac{\sqrt{P}}{a}x} + \frac{1}{2a} \int_{-\infty}^{\infty} \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{P}}{a}(\xi-x)} \\
&\quad \times \bar{f}(\xi p) d\xi + \frac{1}{2a} \int_{-\infty}^{\infty} \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{P}}{a}(\xi-x)} \bar{f}(\xi p) d\xi,
\end{aligned}$$

为使 $x \rightarrow \pm \infty$ 时有限，必须取 $A = B = 0$ 。所以

$$\begin{aligned}
\bar{u} &= \frac{1}{2a} \int_{-\infty}^{\infty} \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{P}}{a}(\xi-x)} \bar{f}(\xi, p) d\xi \\
&\quad + \frac{1}{2a} \int_{-\infty}^x \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{P}}{a}(x-\xi)} \bar{f}(\xi, P) d\xi,
\end{aligned}$$

$$\begin{aligned}\text{进行反演 } \bar{f}(\xi, P) &\doteq f(x, t), \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{P}}{a}(x-\xi)} \\ &\doteq \frac{1}{\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2t}},\end{aligned}$$

再应用卷积定理

$$u = \int_{-\infty}^{\infty} \int_0^t \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} f(\xi\tau) d\tau d\xi.$$

4. 求解一维半界空间的输运问题, $u_t - a^2 u_{xx} = 0$,
 $u|_{t=0} = 0$, 边界条件是 $u|_{x=0} = f(t)$, [本题为 § 39 习题2的一部分
 可对照].

解: 进行拉普拉斯变换得

$$\begin{cases} P\bar{u} - a^2 \bar{u}_{xx} = 0, & (1) \end{cases}$$

$$\begin{cases} u|_{x=0} = \bar{f}(P), & (2) \end{cases}$$

$$\text{微分方程(1)的通解为 } \bar{u} = Ae^{\frac{\sqrt{P}}{a}x} + Be^{-\frac{\sqrt{P}}{a}x}, \quad (3)$$

由于条件 $x \rightarrow \infty$, $\bar{u}$ 应有限, 知 $A = 0$,

$$\therefore \bar{u} = Be^{-\frac{\sqrt{P}}{a}x}, \text{ 根据边界条件, 可得到 } B = \bar{f}(P),$$

$$\therefore \bar{u} = \bar{f}(P) e^{-\frac{\sqrt{P}}{a}x} = P \left[\bar{f}(P) \frac{1}{P} e^{-\frac{\sqrt{P}}{a}x} \right],$$

$$\text{进行反演 } \bar{f}(P) \doteq f(t); \frac{1}{P} e^{-\frac{\sqrt{P}}{a}x} \doteq \text{erfc} \frac{x}{a\sqrt{t}},$$

$$\text{应用卷积定理 } \bar{f}(P) \frac{1}{P} e^{-\frac{\sqrt{P}}{a}x}$$

$$\doteq \int_0^t f(t-\tau) \times \text{erfc} \frac{x}{2a\sqrt{t}} d\tau, \quad (4)$$

(4) 式积分在 $t = 0$ 时为零, 于是按 (21.14)

$$u(x, t) = \frac{\partial}{\partial t} \int_0^t f(t-\tau) \text{erfc} \frac{x}{2a\sqrt{t}} d\tau.$$

5. 求解无界弦的受迫振动 $u_{tt} - a^2 u_{xx} = f(x, t)$, $u|_{t=0} = \varphi(x)$, $u_t|_{t=0} = \psi(x)$.

$$\text{解: } \begin{cases} u_{tt} - a^2 u_{xx} = f(x, t) \\ u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x), \end{cases} \quad (1)$$

进行拉普拉斯变换

$$P^2 \bar{u} - P\varphi - \psi - a^2 \bar{u}_{xx} = \bar{f}(x, P),$$

$$\text{即 } \bar{u}_{xx} - \frac{P^2}{a^2} \bar{u} = \frac{1}{a^2} [P\varphi + \psi - \bar{f}(x, P)]. \quad (2)$$

仿照第 1 题得到方程②的解是:

$$\begin{aligned} \bar{u} = & A e^{\frac{P}{a}x} + B e^{-\frac{P}{a}x} - \frac{1}{2} e^{\frac{Px}{a}} \int \frac{1}{P} e^{-\frac{P\xi}{a}} \\ & \times [\psi(\xi) + P\varphi(\xi) + \bar{f}(\xi, P)] d\xi + \frac{1}{2a} e^{-\frac{Px}{a}} \int \frac{1}{P} \\ & \times e^{\frac{P\xi}{a}} [\psi(\xi) + P\varphi(\xi) + \bar{f}(\xi, P)] d\xi, \end{aligned}$$

$\bar{u}$ 在 $\pm\infty$ 应有限, 必须 $A = B = 0$,

$$\begin{aligned} \bar{u} = & -\frac{1}{2a} \int_{-\infty}^x \frac{1}{P} e^{-P(\xi-x)/a} [\psi(\xi) + P\varphi(\xi) + \bar{f}(\xi, P)] d\xi \\ & + \frac{1}{2a} \int_{-\infty}^x \frac{1}{P} e^{-P(x-\xi)/a} [\psi(\xi) + P\varphi(\xi) + \bar{f}(\xi, P)] d\xi, \\ = & \left[-\frac{1}{2a} \int_x^\infty \frac{1}{P} e^{-P\frac{(\xi-x)}{a}} \psi(\xi) d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{1}{P} \right. \\ & \times e^{-P\frac{(x-\xi)}{a}} \psi(\xi) d\xi \left. \right] + \left[\frac{1}{2a} \int_x^\infty \frac{1}{P} e^{-P\frac{(\xi-x)}{a}} P\varphi(\xi) d\xi \right. \\ & + \frac{1}{2a} \int_{-\infty}^x \frac{1}{P} e^{-P\frac{(x-\xi)}{a}} P\varphi(\xi) d\xi \left. \right] + \left[\frac{1}{2a} \int_x^\infty \frac{1}{P} \right. \\ & \times e^{-P\frac{(\xi-x)}{a}} \bar{f}(\xi, P) d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{1}{P} \\ & \times e^{-P\frac{(x-\xi)}{a}} \bar{f}(\xi, P) d\xi \left. \right]. \end{aligned}$$

进行反演:

$$\begin{aligned}
 \frac{1}{P} e^{-P \frac{(\xi-x)}{a}} &\doteq H\left(t - \frac{\xi-x}{a}\right) = \begin{cases} 1, & \xi < x+at, \\ 0, & \xi > x+at, \end{cases} \\
 \frac{1}{P} e^{-P \frac{(x-\xi)}{a}} &\doteq H\left(t - \frac{x-\xi}{a}\right) = \begin{cases} 1, & \xi > x-at, \\ 0, & \xi < x-at, \end{cases} \\
 \int_x^\infty \frac{1}{P} e^{-P \frac{(\xi-x)}{a}} \psi(\xi) d\xi + \int_{-\infty}^x \frac{1}{P} e^{-P \frac{(x-\xi)}{a}} \psi(\xi) d\xi \\
 &\doteq \int_{x-at}^{x+at} \psi(\xi) d\xi, \\
 P \left[\int_x^\infty \frac{1}{P} e^{-P \frac{(\xi-x)}{a}} \varphi(\xi) d\xi + \int_{-\infty}^x \frac{1}{P} e^{-P \frac{(x-\xi)}{a}} \varphi(\xi) d\xi \right] \\
 &\doteq \frac{\partial}{\partial t} \int_{x-at}^{x+at} \varphi(\xi) d\xi \\
 &= a[\varphi(x+at) + \varphi(x-at)], \\
 \left[\int_x^\infty \frac{1}{P} e^{-P \frac{(\xi-x)}{a}} f(\xi, P) d\xi + \int_{-\infty}^x \frac{1}{P} e^{-P \frac{(x-\xi)}{a}} \right. \\
 &\quad \left. \times \bar{f}(\xi, P) d\xi \right] \\
 &\doteq \int_0^t \int_{x-a(t-\tau)}^{x+a(t+\tau)} f(\xi, \tau) \varphi \tau d\xi,
 \end{aligned}$$

所以总的解是:

$$\begin{aligned}
 u(x, t) &= \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \\
 &\quad + \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^{x+a(t+\tau)} f(\xi, \tau) d\tau d\xi.
 \end{aligned}$$

第十七章 保角变换法

§54. 某些常用的保角变换

1. 例1的二面角(60°)的二等分面上有一带电细导线, 平行于二面角的顶角线, 相距为 a , 导线每单位长度带电量为 Q , 试求电势分布.

解: 作变换 $\zeta = Z^3$

这样就把 Z 平面变换到 ζ 平面上, 在虚轴 η 上距原点 a^3 处有一细导线, 下半平面为导体, 用电像法求得 ζ 平面电场的复势是

$$-\frac{Q}{2\pi\epsilon_0} \ln \frac{\zeta - a^3 i}{\zeta + a^3 i},$$

回到 Z 平面复势为 $-\frac{Q}{2\pi\epsilon_0} \ln \frac{Z^3 - a^3 i}{Z^3 + a^3 i},$

电势为实数部分 $\text{Re} \left[-\frac{Q}{2\pi\epsilon_0} \ln \frac{Z^3 - a^3 i}{Z^3 + a^3 i} \right].$

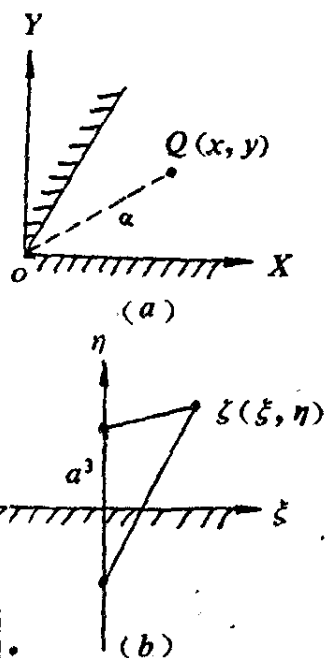


图 17-1

2. 接地甚长空心金属圆柱半径为 a , 柱内有细导线平行于柱轴, 与柱轴相距为 b , 导线每单位长度带电量为 Q , 试求柱内电势分布?

解: 试作变换使 b 点变到圆心, 考虑到 $x=b$ 的对称点是 $\frac{a^2}{b}$, 作分式线性变换

$$\zeta = \frac{Z - b}{Z - \frac{a^2}{b}}, \quad (1)$$

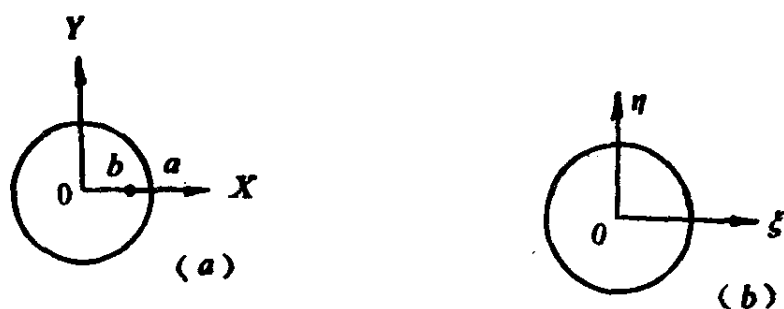


图 17-2

经过这个变换, Z 平面的 b 变成 ζ 平面上的原点, $x=b$ 的对称点 $x = \frac{a^2}{b}$ 变到 ζ 平面上的无限远点. 圆仍变为圆, ζ 平面上的原点 $\zeta=0$ 和无限远点对于变换后的圆是对称的, 可见原点 $\zeta=0$ 是圆心.

为了确定 ζ 平面上圆的半径, 以 $Z=a$ 代入 (1) 式得出

$$R = \left| \frac{a-b}{a-a^2/b} \right| = \frac{b(a-b)}{a|b-a|} = \frac{b}{a},$$

在 ζ 平面上圆内的电势是容易求出的 (设圆周上电势为零),

$$\text{它应是 } -\frac{Q}{2\pi\epsilon_0} \ln \frac{\zeta}{\frac{b}{a}} = -\frac{Q}{2\pi\epsilon_0} \ln \frac{a\zeta}{b}.$$

$$\text{回到 } Z \text{ 平面上, } Z \text{ 平面的复势是 } -\frac{Q}{2\pi\epsilon_0} \ln \frac{a(Z-b)}{b(Z-a^2/b)},$$

$$\begin{aligned} \text{电势分布为 } & \operatorname{Re} \left[-\frac{Q}{2\pi\epsilon_0} \ln \frac{a(Z-b)}{b(Z-a^2/b)} \right] \\ &= \frac{-Q}{2\pi\epsilon_0} \ln \left| \frac{a^2(x-b)^2 + a^2y^2}{(bx-a^2)^2 + b^2y^2} \right|^{1/2}. \end{aligned}$$

3. 甚长金属圆柱的轴平行于甚大金属平板, 两者相距为 b , 平板接地, 圆柱半径为 a , 试求每单位长度的电容量.

解: 先找 A 和 B 两点, 使它们对于圆和实轴对称, 设 A 在 y_1i , B 在 $-y_2i$, ($y_1 > 0, y_2 > 0$), 根据对称的定义,

$$\begin{cases} (b-y_1)(b+y_2) = a^2, \\ y_1 = y_2, \end{cases}$$

解得 $y_1 = y_2 = \sqrt{b^2 - a^2}$,

于是分式线性变换 $\zeta = \frac{Z - iy_1}{Z + iy_2}$,

把原来Z平面上的圆周和实轴变为在 ζ 平面上的同心圆，圆心在原点半径分别为

$$\begin{aligned} R_1 &= \left| \frac{(b-a-y_1)i}{(b-a+y_2)i} \right| \\ &= \left| \frac{b-a-\sqrt{b^2-a^2}}{b-a+\sqrt{b^2-a^2}} \right| \\ &= \left| \frac{a^2+b^2-2ab-b^2+a^2}{(b-a+\sqrt{b^2-a^2})^2} \right| \end{aligned}$$

$$= \left| \frac{-2a(b-a)}{(b-a)(\sqrt{b-a}+\sqrt{b+a})^2} \right|$$

$$= \frac{2a}{b-a+b+a+a\sqrt{b^2-a^2}} = \frac{a}{b+\sqrt{b^2-a^2}},$$

$$R_2 = \left| \frac{0-iy_1}{0+iy_1} \right| = 1,$$

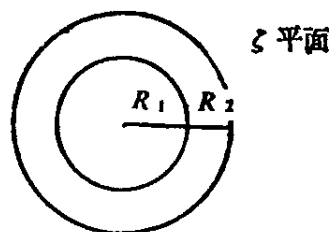
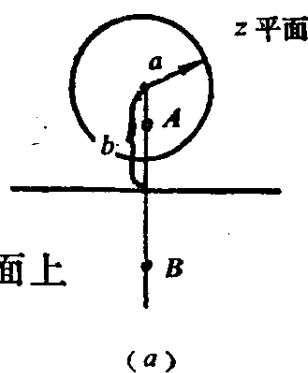


图 17-3

从而电容量 $c = \frac{2\pi\epsilon_0}{\ln \frac{R_2}{R_1}} = \frac{2\pi\epsilon_0}{\ln \left(\frac{b}{a} + \sqrt{\frac{b^2}{a^2} - 1} \right)}$, 由于保角变换不改

变电容量，回到Z平面，原来的电容量也是这数值。

4. 甚大金属平面有柱形隆起，其横截面为弓形，弓形在 c 和 a 之间，弓形弧内半径为 a ，求解带电后的静电场。

解：作分式线性变换

$$Z_1 = \frac{-Z}{Z-a},$$

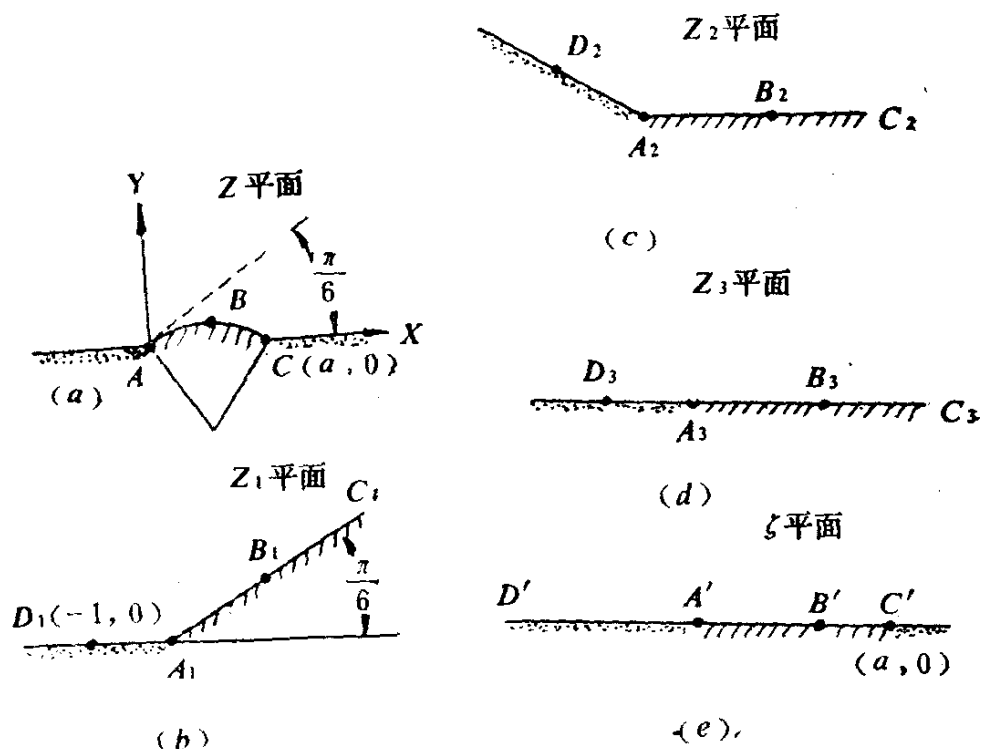


图 17-4

这个变换使 C 点变为 Z_1 平面的 ∞ , A, B, C 变为射线 $A_1B_1C_1$. 圆弧和 AC' 的交角 $\pi/6$ 不变, 无限远点 D 则变为 $Z_1 = -1$ 即 D_1 点.

再作变换

$$Z_2 = e^{-\frac{\pi}{6}i} Z_1,$$

这相应于 Z_1 顺时针转 $\frac{\pi}{6}$ 再作变换

$$Z_3 = Z_2^{\frac{6}{5}},$$

在这个变换下正实轴这一段不变, A_2, D_2, C_2 , 则变成负实轴 A_3, D_3, C_3 .

再作分式线性变换

$$\xi = \frac{aZ_3}{Z_3 + 1},$$

使 D_3 变为 ∞ 点 D' , Z_3 上的 ∞ 点 C_3 变成 ξ 平面上的 C' 点(a 点), ξ 平面上的电势为 $C_1\eta + C_2$, 即 $C_1I_m\xi + C_2$, 回到原来的变数, 即得解.

注: 如果将 Z_3 作变换 $\xi' = \sqrt{Z_3}$, 则 Z_3 的幅角为 0° 时不变, 幅角为 2π 时变为 π 度, 则有



图 17-5

这时好象可以得出 ξ' 平面上的电势为 $C_1\eta' + C_2$ 或 $C_1I_m\xi' + C_2$, 但是仔细考虑一下, 在 ξ' 的等势线相交于 ∞ 点, 即 C' 点, 而在 Z 平面上却不相交于 C' 点, 而是相交于 ∞ 的 D 点, 因此作 $\xi' = \sqrt{Z_3}$ 变换而得出的等势是不对的.

注: 保角变换, 解题的一般方法:

上半平面及平面上, 一点 Z_0	令 $\zeta = \frac{Z - Z_0}{Z + Z_0}$	Z_0 变换成 ζ 平面上的圆心, 圆半径是1
相离二圆或一圆及圆外一直线	求出公共对称点 Z_1, Z_2 , 再令 $\zeta = \frac{Z - Z_1}{Z - Z_2}$	变换成同心圆
二圆相切、或一直线与圆相切	设切点为 Z_0 令 $\zeta = \frac{1}{Z - Z_0}$	变换成带形域
带形域	令 $\zeta = e^Z$	变成角域或环域
角域	令 $\zeta = Z^m$	化成半平面或全平面

二相交圆弧	设交点为 Z_1, Z_2 令 $\zeta = \frac{Z - Z_1}{Z - Z_2}$	变换成角域
上半平面有半圆突起	令 $\zeta = \frac{1}{2} \left(\frac{Z}{R} + \frac{R}{Z} \right)$	化成上半平面
椭圆	$Z = \frac{C}{2} \left(\zeta - \frac{1}{\zeta} \right)$	变成圆, 半径 $R = \frac{a+b}{c}$

5. 长金属柱, 其截面由两段圆弧围成, 这两段圆弧是相等的, 其半径为 a , 交点在 b 和 a , 求解金属柱带电后的静电场。

解: (i) 分式线性变换

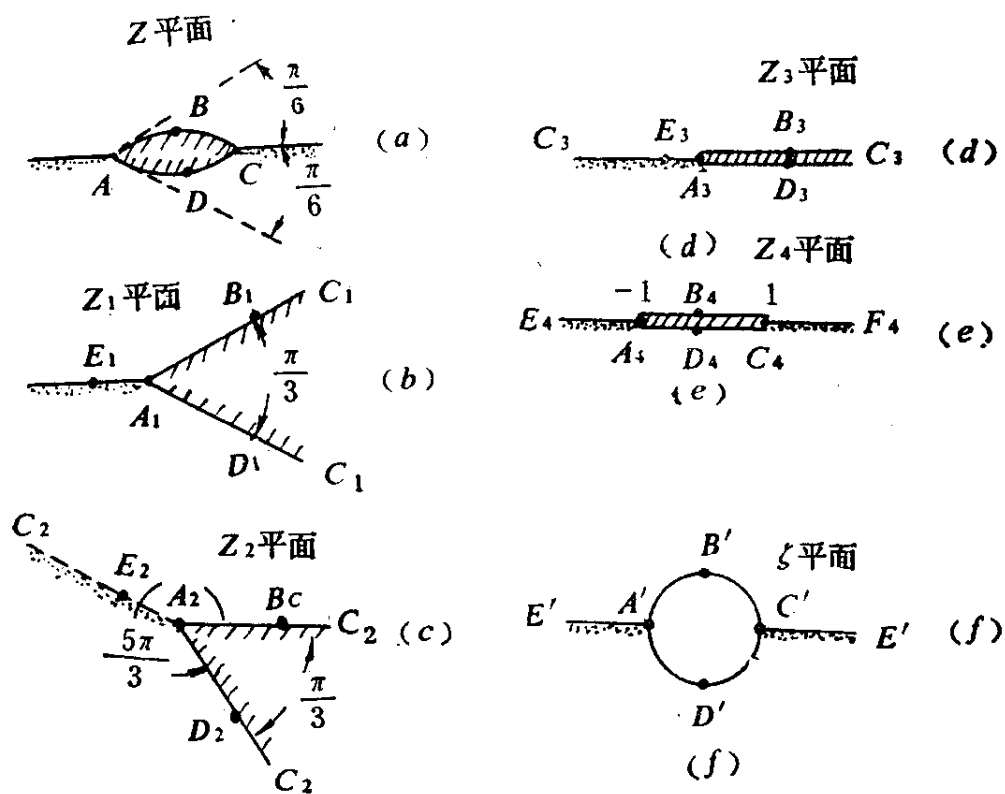


图 17-6

$$z_1 = \frac{-z}{z-a},$$

将圆弧 ABC 变为射线 $A_1B_1C_1$,

将圆弧 ADC 变为射线 $A_1D_1C_1$,

二者交角 $\frac{\pi}{3}$ 不变.

再作变换 $z_2 = e^{-\frac{\pi}{6}i} z_1$, 把图形顺时针转 $\frac{\pi}{6}$, 射线 $A_1B_1C_1$ 变为

正轴 $A_2B_2C_2$, 射线 $A_2B_2C_2$ 与 $A_2D_2C_2$ 之间夹角为 $\frac{5\pi}{3}$,

再作变换 $z_3 = z_2^{6/5}$,

将夹角放大 $\frac{6}{5}$ 倍, 从而射线 $A_3B_3C_3$ 与 $A_3D_3C_3$ 成为正实轴割线两岸,

再作变换 $z_4 = \frac{z_3 - 1}{z_3 + 1}$,

将正实轴割线两岸, 变为从 -1 到 $+1$ 的割线两岸. 引用儒阔夫斯基变换

$$z_4 = \frac{1}{2} \left(\xi + \frac{1}{\xi} \right),$$

将 $-1, +1$, 割线两岸变为单位圆, 于是问题变为长金属圆柱带电后的静电场, ξ 平面的电势为 $C_1 \ln |\xi| + C_2$ 回到 z 平面, 即得原来那种长金属柱带电后的静电场.

解: (ii) 如(i)一样先作变换

$z_1 = \frac{-z}{z-a}$ 再作变换 $z_2 = e^{-\frac{\pi}{6}i} z_1$, 然后

再作变换

$z_3 = z_2^{\frac{3}{5}}$ 这相当于 z_2 的模缩小原

来的 $\frac{3}{5}$ 倍, 幅角成为原来的 $\frac{3}{5}$ 倍即 $\frac{3}{5}$

$\times \frac{5}{3}\pi = \pi$, 这就是说 $A_5B_5C_5$ 和

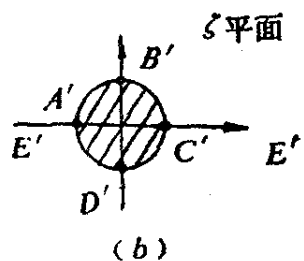
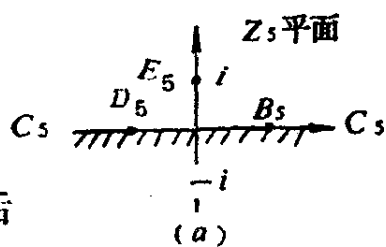


图 17-7

$A_5 D_5 C_5$ 成一直线. 再作变换

$$z_6 = \frac{z_5 + i}{z_5 - i},$$

$-i$ 变成 ζ 平面上的原点, 它是圆心, 圆半径是 $R = |z_6| =$

$$\left| \frac{0+i}{0-i} \right| = 1, \text{电势} = C_1 \ln |z_6| + C_2,$$

可以验证: 解(ii)与解(i)等价, 事实上

$$\begin{aligned} z_4 &= \frac{z_3 - 1}{z_3 + 1} = \frac{z_5^2 - 1}{z_5^2 + 1} = \frac{[i(z_6 + 1)/(z_6 - 1)]^2 - 1}{[i(z_6 + 1)/(z_6 - 1)]^2 + 1} \\ &= \frac{-(z_6 + 1)^2 - (z_6 - 1)^2}{-(z_6 + 1)^2 + (z_6 - 1)^2} = \frac{-2(z_6^2 + 1)}{-4z_6} = \frac{1}{2} \left(z_6 + \frac{1}{z_6} \right), \end{aligned}$$

可见 $\zeta = z_6$,

不过, 在解(ii)之中, 各点的对应关系不如解(i)那样明显.

6. 把下列区域保角变换为圆.

(1) 弓形 $I_m z \geq 1, |z| \leq 2$.

这是以原点为圆心, 以2为半径的弧以及经过 $-\sqrt{3} + i$ 和 $\sqrt{3} + i$ 的直线, 两者所围的图形. 作变换 z_1

$$= \frac{z + \sqrt{3} - i}{-z + \sqrt{3} + i}$$

A 点变成 z_1 平面上的原点, B 点成为 z_1 平面上的 ∞ 点, 再作变换 $z_2 = z_1^3$,

z_1 的 $\frac{\pi}{3}$ 角域变成 z_2 的上半平面, 再作变换

$$\zeta = \frac{z_2 - i}{z_2 + i},$$

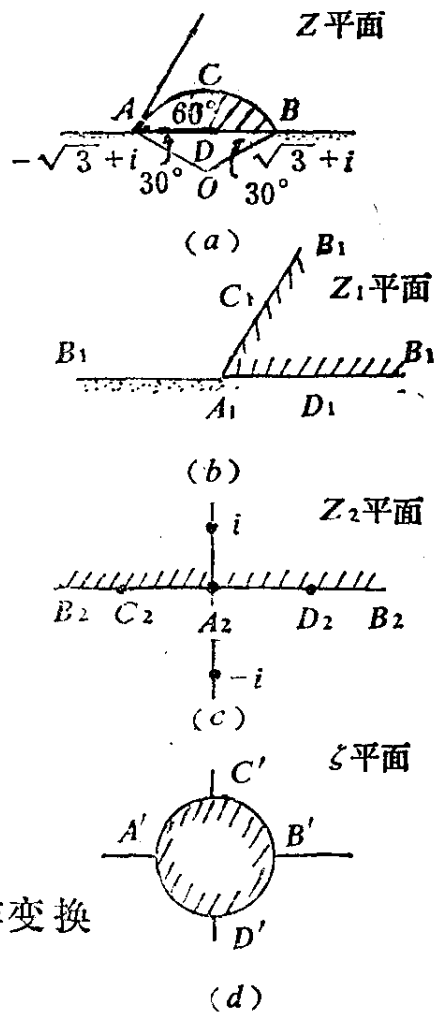


图 17-8

圆的半径 $R = |\xi| = \left| \frac{0-i}{0+i} \right| = 1$,

(2) 圆 $|z| = 2$ 外, 除去第一象限作变换

$$z_1 = e^{-i\frac{\pi}{2}z},$$

再作变换

$$z_2 = z_1^{\frac{2}{3}},$$

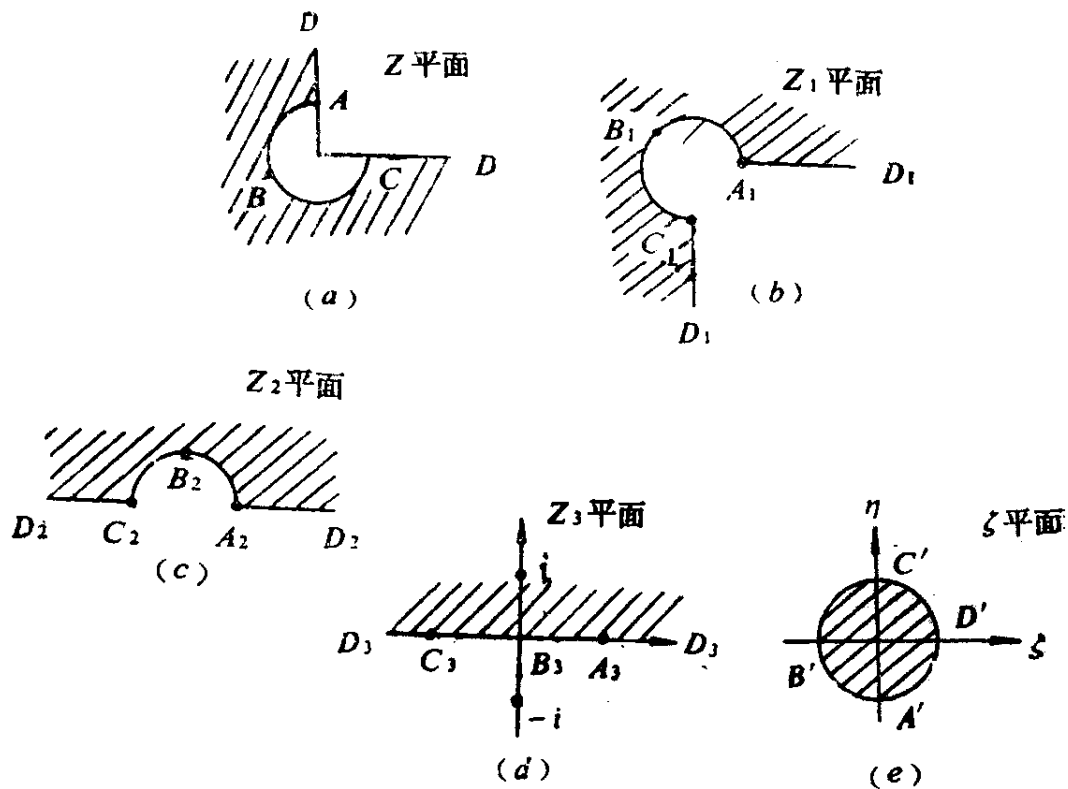


图 17-9

引用§54, 例7的结果 $z_3 = \frac{1}{2} \left(\frac{z_2}{2^{3/2}} + \frac{2^{2/3}}{z_2} \right)$ 将 z_2 平面的划线区域变为 z_3 的上半平面.

再作变换

$$\xi = \frac{z_3 - i}{z_3 + i},$$

变成 ξ 平面上的一个圆,

$$\text{半径 } R = |\xi| = \left| \frac{0-i}{0+i} \right| = 1.$$

(3) 二个相切的圆 $|z| \leq 2$, 和 $|z-3| \leq 1$ 以外的区域.

$$\text{作变换 } z_1 = \frac{1}{z-2},$$

B 点变成 ∞ 点, A 变为 $-\frac{1}{4}$, C 为 $+\frac{1}{2}$, 两圆和实轴的交角不变, AB 的上半圆弧变成 z_1 的实轴下部, 以 $z=0$ 代入 z_1 得 $z_1 = -\frac{1}{2}$.

由此得知 z 平面上给定的区域变为 z_1 平面的条形区域, 再作变换 $z_2 = e^{iz_1}$,

$$\text{变换 } z_3 = z_2 e^{\frac{i}{4}},$$

$$\text{变换 } z_4 = z_3 e^{\frac{1}{3}\pi},$$

最后作变换

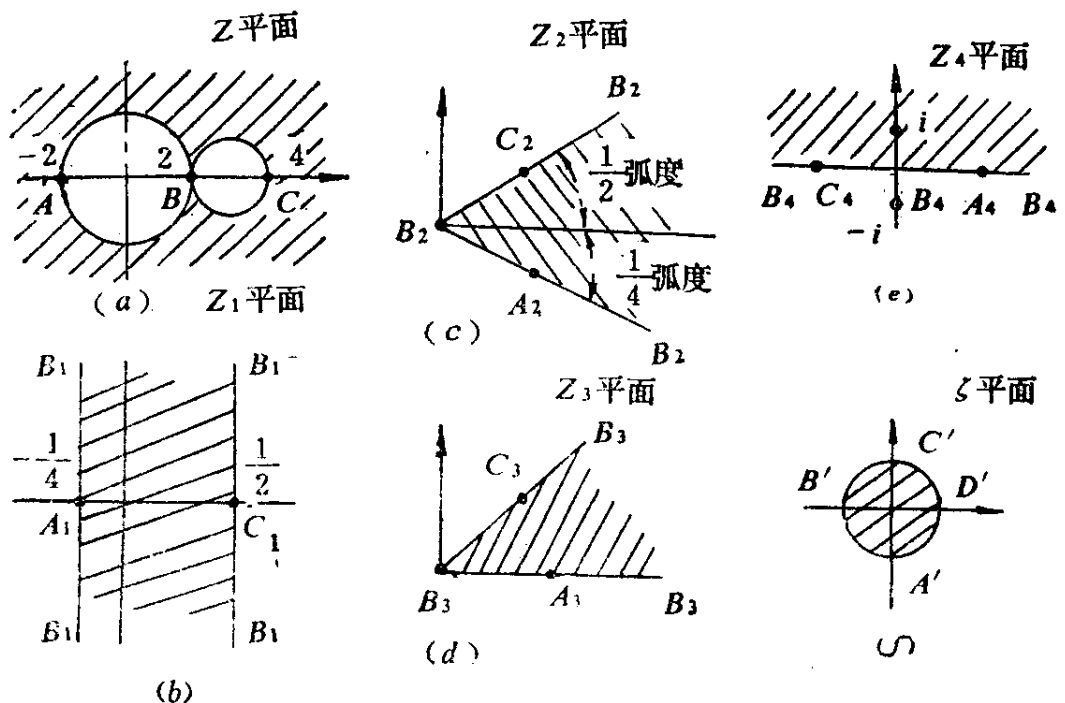
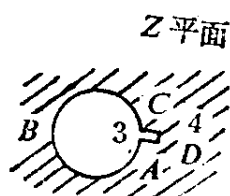


图 17-10

$$\xi = \frac{z_3 - 1}{z_4 + i},$$

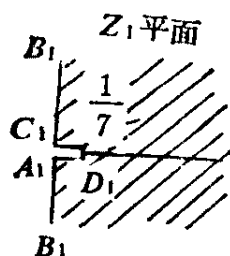
即将给定的区域变为圆。

$$(4) \text{ 圆 } |z| \leq 3 \text{ 除去突起 } \begin{cases} \operatorname{Im} z = 0, \\ 3 \leq \operatorname{Re} z \leq 4, \end{cases}$$



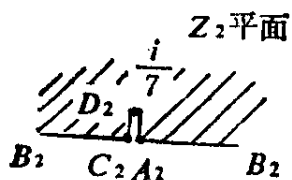
(a)

$$\text{作 } z_1 = \frac{z-3}{z+3}$$

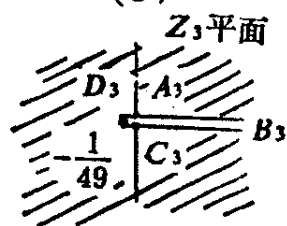


(b)

$$\text{变换 } z_2 = iz_1$$



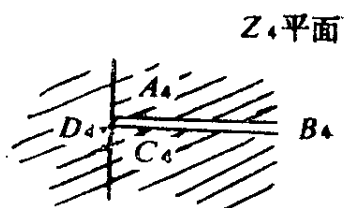
(c)



(d)

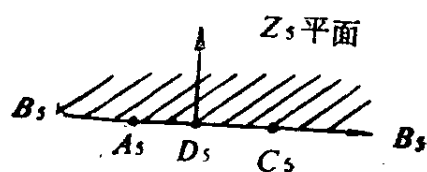
$$z_4 = z_3 + \frac{1}{49}$$

$$\text{再作变换 } z_5 = z_4^2$$

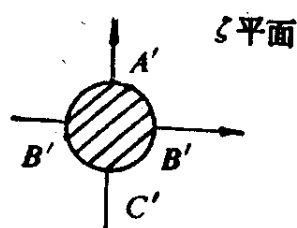


(e)

$$\text{作变换 } z_5 = \sqrt{z_4}$$



(f)



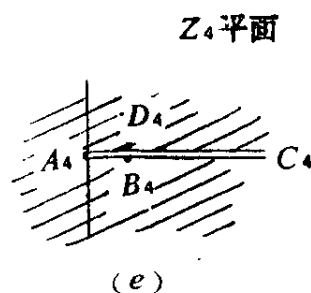
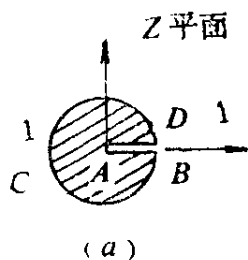
(g)

图 17-11

从 z_2 平面到 z_5 平面的变换就是课本中§54例2的变换,最后作变换

$$\xi = \frac{z_5 - i}{z_5 + i}$$

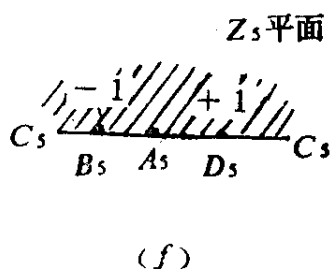
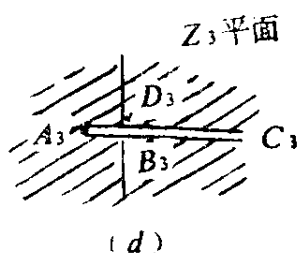
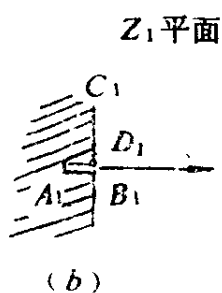
(5) 在圆形区域有割线 $\begin{cases} \text{Im} z = 0, \\ 0 \leq \text{Re} z \leq 1, \end{cases}$



用分式线性变换, 将圆变成直线

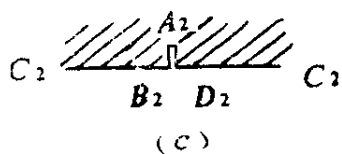
$$z_1 = \frac{z-1}{z+1},$$

再作变换 $z_5 = \sqrt{z_4}$,



再变换 $z_4 = z_3 + 1$,

再作变换 $z_2 = -iz_1$,



从 z_2 到 z_5 的变换, 即课本
§54 例 2 的变换, 最后作

变换 $\xi = \frac{z-i}{z+i}$, ξ 平面

再作变换 $z_3 = z_2^2$,

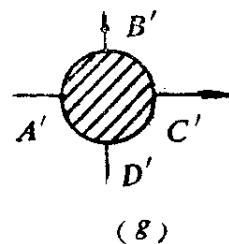


图 17-12

(6) 心脏线 $|z| \leq \cos^2 \left[\frac{1}{2} \arg z \right]$ 的内部

令 $\xi = \sqrt{z}$ 即 $z = \xi^2$, 心脏内部区域变为 $|\xi^2| = \cos^2 \left[\frac{1}{2} \arg \xi^2 \right]$

的内部.

即 $|\xi|^2 = \cos^2 [\arg \xi]$,

再即 $|\xi| = 1 \cdot \cos \arg \xi$, 用极坐标表出则是 $\rho = \cos \psi$, 这正是圆.

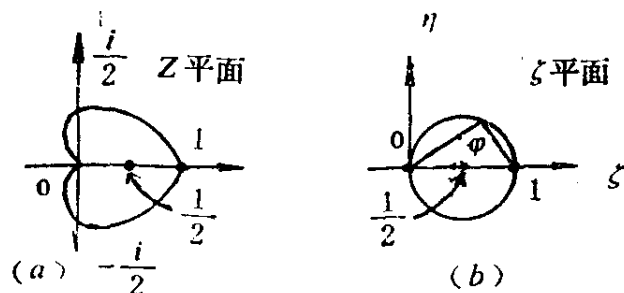


图 17-13

(7) 双纽线一支 $|z| \leq \sqrt{\cos[2\arg z]}$,

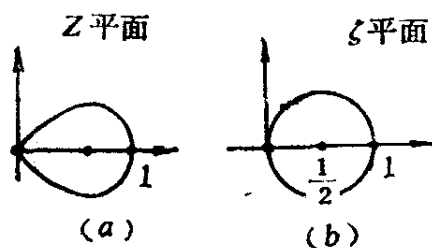


图 17-14

令 $\xi = z^2$, 即 $z = \sqrt{\xi}$,

z 平面的双纽线在 ξ 平面上变为 $|\xi| = 1, \cos \arg \xi$, 这正是圆.

7. 研究甚长带电导体周围的静电场, 带宽为 $2a$.

解: 带电导体可以看作是焦点在 $-a, a$, 半长轴为 a , 半短

轴为 0 的椭圆。作儒阔夫斯基变换 $\frac{z}{a} =$

$\frac{1}{2}\left(\xi + \frac{1}{\xi}\right)$, 带电导体变成在 ξ 平面上

半径为 1 的圆柱, 圆柱外的电势可表示为 $C_1 \ln|\xi| + C_2$, 将这个表达式中的 ξ 变换到 z 即得原来那个静电场中的电势。

8. 研究甚长带电椭圆导体柱周围的静电场, 椭圆半长轴为 a , 半短轴为 b 。

解: 用儒阔夫斯基变换

$$\frac{z}{\sqrt{a^2 - b^2}} = \frac{1}{2}\left(\xi + \frac{1}{\xi}\right),$$

把 z 平面的椭圆变成 ξ 平面的圆。

圆的半径为 $\frac{a + b}{\sqrt{a^2 - b^2}}$,

电势为 $C_1 \ln|\xi| + C_2$ 。

9. 二个椭圆柱构成柱形电容器, 横截面是两个共焦点椭圆, 半长轴分别为 a_1 和 a_2 , 半短轴分别为 b_1 和 b_2 试求每单位长度的电容。

解: 两焦点距离的一半

$$C = \sqrt{a_1^2 - b_1^2} = \sqrt{a_2^2 - b_2^2},$$

通过变换 $\frac{Z}{C} = \frac{1}{2}\left(\xi + \frac{1}{\xi}\right)$,

把共焦点椭圆变成 ξ 平面上的同心圆, 半径分别为 $R_1 =$

$\frac{a_1 + b_1}{C}$ 和 $R_2 = \frac{a_2 + b_2}{C}$ 所以

所求的电容量是

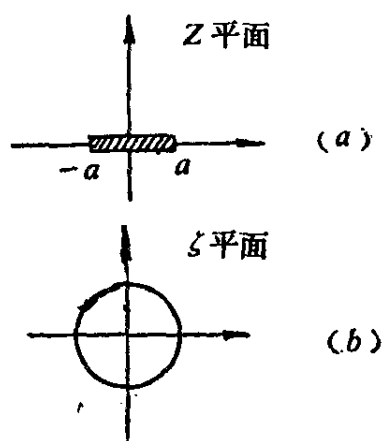


图 17-15

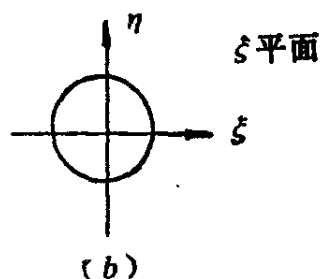
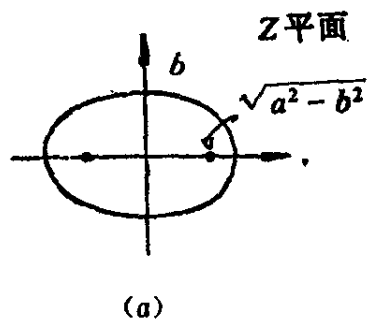


图 17-16

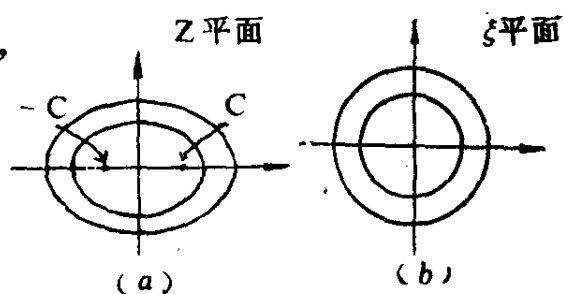


图 17-17

$$C = 2\pi\epsilon_0 / \ln \frac{R_2}{R_1} = 2\pi\epsilon_0 / \ln \frac{a_2 + b_2}{a_1 + b_1}.$$

10. 求解二维稳恒水流通过宽度为 $2a$ 的闸门的情形.

解, 闸门处即 $\begin{cases} l_m z = 0, \\ -a < R_c z < a, \end{cases}$

由于水没有横向流动, 因而速度势相等. 通过变换

$$\frac{z}{a} = \frac{1}{2} \left(\xi + \frac{1}{\xi} \right), \text{ 闸门变为半}$$

径为 1 的圆柱, 而圆柱表面上的速度势相等. 与静电势相对比易知速度势 $= C_1 \ln |\xi| + C_2$.

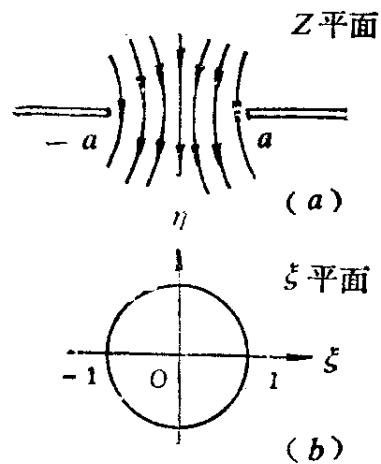


图 17-18

11. 右图是六角“星”的六个臂, 彼此相隔 60° , 各臂的长度为 1, 试把星的外部变为 ξ 平面单位圆的外部 (提示 $z_1 = z^3$, $\xi_1 = \xi^3$ 再找出 z_1 和 ξ_1 之间的关系).

解: 作变换 $z_1 = z^3$,
和 $\xi_1 = \xi^3$,
则 z 平面变为 z_1 的三叶交叉平面, 原六角星成为三叶上沿实轴的一段 (从 -1 到 $+1$) 割线, ξ 平面单位圆的外部变为 ξ_1 的三叶交叉平面的单位圆外部.

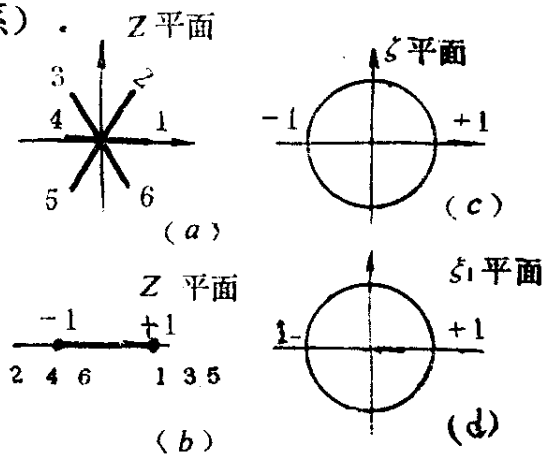


图 17-19

儒阔夫斯基变换 $z_1 = \frac{1}{2} \left(\xi_1 + \frac{1}{\xi_1} \right)$ 将 z_1 平面的图形变为 ξ_1 平面上的图形, 因此, 所求变换为

$$z^3 = \frac{1}{2} \left(\xi^3 + \frac{1}{\xi^3} \right),$$

12. 研究电机的转子和定子之间 (图17-20) 的磁场, 求最大磁场和最小磁场之比.

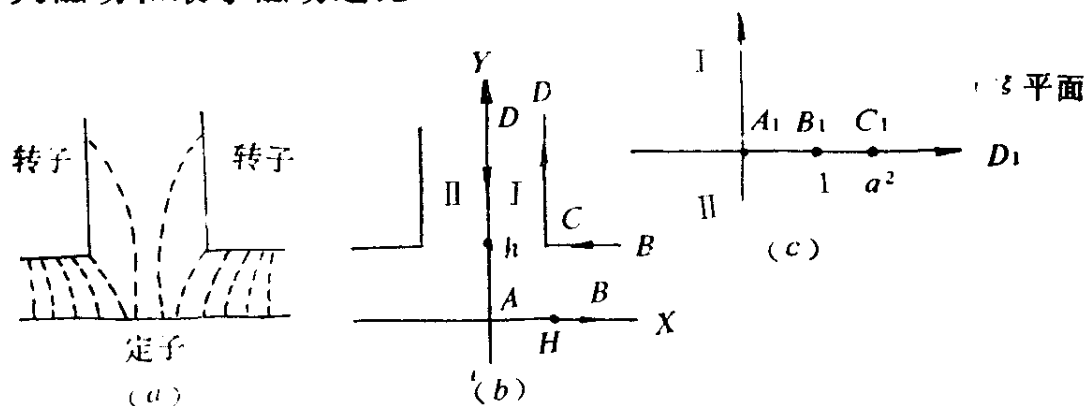


图 17-20

解: “磁势” u 在转子和定子之间的区域中满足拉普拉斯方程, 定子和转子的表面为等势面. 找变换, 使 z 空间变为 ζ_1 的全平面, 区域 I 和 II 分别变为 ζ_1 平面的上半和下半平面, 由于对称性, 我们只考虑区域 I 和 ζ_1 上半平面. 将区域 I 的 A, B, C, D 变成 ζ_1 平面的 A_1, B_1, C_1, D_1 (其对应坐标见附表) 的许瓦兹-克利斯多菲变换为:

附表

z 平面	θ	$-\frac{\theta}{\pi}$	ζ_1 平面
$A(0, 0)$	$\frac{\pi}{2}$	$-\frac{1}{2}$	$A_1(0, 0)$
$B(\infty, \frac{0}{h})$	π	-1	$B_1(\frac{1-0}{1+0}, 0)$
$C(H, h)$	$-\frac{\pi}{2}$	$+\frac{1}{2}$	$C_1(a^2, 0)$
$D(\frac{H}{0}, \infty)$	π	-1	$D_1(\frac{+\infty}{-\infty}, 0)$

$$z = z_0 + A \int \sqrt{\frac{\zeta_1 - a^2}{\zeta_1}} \frac{1}{\zeta_1 - 1} d\zeta_1$$

作代换

$$\sqrt{\frac{\xi_1 - a^2}{\xi_1}} = u,$$

即

$$\xi_1 = \frac{a^2}{1 - u^2},$$

$$\frac{1}{\xi_1 - 1} = \frac{1 - u^2}{a^2 - 1 + u^2},$$

$$d\xi_1 = \frac{2a^2 u du}{(1 - u^2)^2},$$

变换公式成为

$$\begin{aligned} z &= z_0 + A \int u \frac{1 - u^2}{a^2 - 1 + u^2} \frac{2a^2 u}{(1 - u^2)^2} du \\ &= z_0 + 2A \int \frac{a^2 u^2 du}{(a^2 - 1 + u^2)(1 - u^2)} \\ &= z_0 - 2A \int \left[\frac{a^2 - 1}{a^2 - 1 + u^2} - \frac{1}{1 - u^2} \right] du \\ &= z_0 - 2A \left[\sqrt{a^2 - 1} \operatorname{arctg} \frac{u}{\sqrt{a^2 - 1}} - \operatorname{arcth} u \right], \end{aligned}$$

我们注意到

$$\operatorname{arctg} x = -i \ln \sqrt{\frac{1 + ix}{1 - ix}},$$

$$\operatorname{arcth} x = \ln \sqrt{\frac{1 + x}{1 - x}},$$

由此读者可以验证.

(1) 如 $x < 1$, 则 $\operatorname{arcth} x$ 为实数, 如 $x > 1$ 则

$$\begin{aligned} \operatorname{arcth} x &= \ln \sqrt{\frac{1 + x}{1 - x}} = \ln \sqrt{\frac{1 + x}{-(x - 1)}} = \ln \left(-i \sqrt{\frac{x + 1}{x - 1}} \right) \\ &= \ln(-i) + \ln \sqrt{\frac{x + 1}{x - 1}} = -\frac{\pi}{2} i + \ln \sqrt{\frac{x + 1}{x - 1}} \end{aligned}$$

即 $\operatorname{arcth} x$ 是复数.

(2) 如 $x \rightarrow 1-0$ 则 $\operatorname{arcth} x = \ln \sqrt{\frac{1+x}{1-x}} = \infty$ 为实数,

如 $x \rightarrow 1+0$ 则 $\operatorname{arcth} x = \ln \sqrt{\frac{1+x}{1-x}} = \ln \sqrt{\frac{x+1}{-(x-1)}} = -\frac{\pi}{2}i + \infty$

两者虚数部分不同, 相差 $-\frac{\pi}{2}$.

(3) 如 $x \rightarrow \pm \infty$ 时,

$$\begin{aligned} \text{则 } \operatorname{arcth} x &= \ln \sqrt{\frac{1+x}{1-x}} = \ln \sqrt{\frac{\frac{1}{x} + 1}{\frac{1}{x} - 1}} = \ln \frac{1}{\sqrt{-1}} \\ &= \ln(-i) = -\frac{\pi}{2}i. \end{aligned}$$

$$\begin{aligned} (4) \operatorname{arctg} \left(\frac{1}{x} \right) &= -i \ln \sqrt{\frac{1 + i \frac{1}{x}}{1 - i \frac{1}{x}}} = -i \ln \sqrt{\frac{x + i}{x - i}} \\ &= -i \ln \sqrt{\frac{-x_i + 1}{-x_i - 1}} = -i \ln \sqrt{\frac{1 + (-x_i)}{-[1 - (-x_i)]}} \\ &= -i \ln(-i) \sqrt{\frac{1 + (-x_i)}{1 - (-x_i)}} \\ &= -i [\ln(-i) + \ln \sqrt{\frac{1 + (-x_i)}{1 - (-x_i)}}] \\ &= +\frac{\pi}{2} - i \ln \sqrt{\frac{1 + (-x_i)}{1 - (-x_i)}} \\ &= +\frac{\pi}{2} + \operatorname{arctg}(-x) = +\frac{\pi}{2} - \operatorname{arctg} x. \end{aligned}$$

$$(5) \operatorname{arcth} x = \ln \sqrt{\frac{1+x}{1-x}} = \ln \sqrt{\frac{\frac{1}{x} + 1}{\frac{1}{x} - 1}} = \ln \sqrt{\frac{1 + \frac{1}{x}}{-\left(1 - \frac{1}{x}\right)}}$$

$$\begin{aligned}
&= \ln(-i) + \ln \sqrt{\frac{1 + \frac{1}{x}}{1 - \frac{1}{x}}} = -\frac{\pi}{2}i + \operatorname{arcth} \frac{1}{x} \\
&= \ln(-i) + \operatorname{arcth} \frac{1}{x}.
\end{aligned}$$

$$(6) \operatorname{arctg} ix = i \operatorname{arcth} x, \operatorname{arcth} ix = i \operatorname{arctg} x.$$

$$(7) \lim_{x \rightarrow a^2} \operatorname{arctg} \sqrt{a^2 - 1} \sqrt{\frac{x}{x - a^2}} = \operatorname{arctg} \infty = \frac{\pi}{2},$$

$$\begin{aligned}
\text{另一方面 } \lim_{x \rightarrow a^2} \operatorname{arctg} \sqrt{a^2 - 1} \sqrt{\frac{x}{x - a^2}} &= \lim_{x \rightarrow a^2} i \operatorname{arcth} \sqrt{a^2 - 1} \sqrt{\frac{x}{a^2 - x}} \\
&= i \left(-\frac{\pi}{2} \right) i = \frac{\pi}{2}.
\end{aligned}$$

现在根据点之间的对应关系确定变换公式中的 z_0 和 A , 首先由 $z = 0 + 0$ 和 $\zeta_1 = 0 + 0$ 的对应知 $u = i\infty$ 而 $0 = z_0 - 2A[\sqrt{a^2 - 1}(-\frac{\pi}{2}) - \ln(-i)]$, 因而 $z_0 = 2A[\sqrt{a^2 - 1}(-\frac{\pi}{2}) - \ln(-i)]$, 而变换公式成为

$$\begin{aligned}
z &= 2A[\sqrt{a^2 - 1}(-\frac{\pi}{2}) - \ln(-i)] \\
&\quad - 2A[\sqrt{a^2 - 1} \operatorname{arctg} \frac{u}{\sqrt{a^2 - 1}} - \operatorname{arcth} u] \\
&= 2A\left[\sqrt{a^2 - 1} \operatorname{arctg} \sqrt{\frac{a^2 - 1}{u}} + \operatorname{arcth} \frac{1}{u}\right] \\
&= 2A\left[\sqrt{a^2 - 1} \operatorname{arctg} \sqrt{a^2 - 1} \sqrt{\frac{\zeta_1}{\zeta_1 - a^2}} + \operatorname{arcth} \sqrt{\frac{\zeta_1}{\zeta_1 - a^2}}\right],
\end{aligned} \tag{1}$$

其次由 $\zeta_1 = 1 - 0$ 和 $z = \infty$ 的对应知

$$\infty = 2A \left\{ \sqrt{a^2-1} i \left[\operatorname{arcth} \left(\sqrt{a^2-1} \sqrt{\frac{\xi_1}{a^2-\xi_1}} \right) \right]_{\xi_1 \rightarrow 1-0} + i \operatorname{arctg} \sqrt{\frac{1}{a^2-1}} \right\},$$

$$\begin{aligned} \text{即 } \infty &= 2A \left\{ \sqrt{a^2-1} \cdot i \cdot \infty + i \operatorname{arctg} \sqrt{\frac{1}{a^2-1}} \right\} \\ &= i 2A \left\{ \sqrt{a^2-1} \cdot \infty + \operatorname{arctg} \sqrt{\frac{1}{a^2-1}} \right\}, \end{aligned}$$

由此可见 $2A$ 应为纯虚数, 把它记作 iC (C 为实数) 变换公式改写为:

$$\begin{aligned} z &= iC \left[\sqrt{a^2-1} \operatorname{arctg} \left(\sqrt{a^2-1} \sqrt{\frac{\xi_1}{\xi_1-a^2}} \right) \right. \\ &\quad \left. + \operatorname{arcth} \sqrt{\frac{\xi_1}{\xi_1-a^2}} \right], \end{aligned} \quad (2)$$

同时还有 $\xi_1 = 1+0$ 对 $z = \infty + ih$ 对应即

$$\begin{aligned} \infty + ih &= iC \left\{ \sqrt{a^2-1} i \left[\operatorname{arcth} \left(\sqrt{a^2-1} \sqrt{\frac{\xi_1}{a^2-\xi_1}} \right) \right]_{\xi_1 \rightarrow 1+0} \right. \\ &\quad \left. + i \operatorname{arctg} \sqrt{\frac{1}{a^2-1}} \right\} \\ &= iC \left\{ \sqrt{a^2-1} i \left(\infty - i \frac{\pi}{2} \right) + i \operatorname{arctg} \sqrt{\frac{1}{a^2-1}} \right\} \\ &= i \frac{\pi}{2} C \sqrt{a^2-1} - C \left\{ \sqrt{a^2-1} \cdot \infty + \operatorname{arctg} \sqrt{\frac{1}{a^2-1}} \right\}, \end{aligned}$$

$$\text{由此可见 } C = \frac{2}{\pi} \frac{h}{\sqrt{a^2-1}}, \quad (3)$$

又其次由于 $\xi_1 = +\infty$ 和 $z = H + i\infty$ 的对应知

$$H + i\infty = iC \left\{ \sqrt{a^2-1} \operatorname{arctg} \sqrt{a^2-1} + \left(\infty - \frac{\pi}{2} i \right) \right\},$$

由 $\xi_1 = -\infty$ 和 $z = i\infty$ 对应知

$$i\infty = iC \left\{ \sqrt{a^2-1} \operatorname{arctg} \sqrt{a^2-1} + \infty \right\}, \text{ 由此两式可见}$$

$$C = +\frac{2}{\pi}H, \quad (4)$$

(3)和(4)式应不矛盾, 这表明 $\frac{h}{\sqrt{a^2-1}} = H$, 即

$$a^2 = 1 + \frac{h^2}{H^2}, \quad (5)$$

还有 $\zeta_1 = a^2$ 和 $z = H + ih$ 对应, 即

$$H + ih = iC \left[\sqrt{a^2-1} \cdot \frac{\pi}{2} - \frac{\pi}{2} i \right],$$

这就是说 $C\sqrt{a^2-1}\frac{\pi}{2} = h$, $C = \frac{2}{\pi}H$, 可得 $a^2 = 1 + \frac{h^2}{H^2}$,

这正是(3)和(4)式这样从 z 平面到 ζ_1 平面的变换为

$$z = +i\frac{2}{\pi} \left[h \operatorname{arctg} \frac{h}{H\sqrt{\zeta_1-a^2}} + H \operatorname{arcth} \sqrt{\frac{\zeta_1}{\zeta_1-a^2}} \right].$$

接着作变换 $\zeta_2 = \sqrt{\zeta_1}$ 区域 I 变成 ζ_2 平面的第一象限, 又作变换

$$\zeta_3 = \frac{1+\zeta_2}{1-\zeta_2},$$

$$\zeta = \frac{V}{\pi} \ln \zeta_3 = \frac{V}{\pi} \ln \frac{1+\sqrt{\zeta_1}}{1-\sqrt{\zeta_1}},$$

在 ζ 平面的磁势为 η_1 , 复势为 ζ , 而磁场

$$\begin{aligned} B &= \left| \frac{d\text{复势}}{dz} \right| = \left| \frac{d\zeta}{dz} \right| \\ &= \frac{d\zeta}{d\zeta_1} \bigg/ \frac{dz}{d\zeta_1} = \frac{V}{H} \frac{1}{\sqrt{\zeta_1^2 - a^2}}, \end{aligned} \quad (6)$$

在点 A, 磁场最小, A 点为 $z = 0$, 它对应于 $\zeta_1 = 0$,

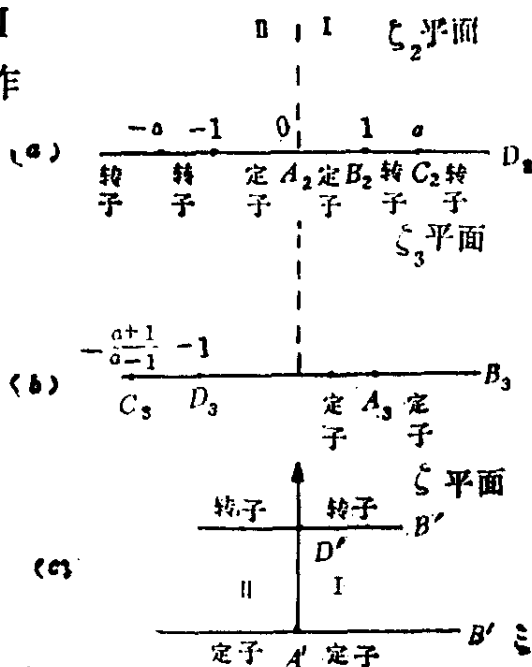


图 17-21

$$\therefore B_{\min} = \frac{V}{H} \cdot \frac{1}{a},$$

在 B 点磁场最大, B 点为 $z = \infty$, 它对应于 $\xi_1 = 1$,

$$\therefore B_{\max} = \frac{V}{H} \frac{1}{\sqrt{a^2 - 1}},$$

$$\therefore B_{\max} : B_{\min} = a : \sqrt{a^2 - 1} = \sqrt{H^2 + h^2} : h.$$

在得到最后结果时应用到 $a^2 = 1 + \frac{h^2}{H^2}$ 以及

$$\begin{aligned} \left| \frac{dz}{d\xi_1} \right| &= \frac{2}{\pi} \left\{ \frac{h \cdot \frac{h}{H}}{1 + \frac{h^2}{H^2} \frac{\xi_1}{\xi_1 - a^2}} + \frac{H}{1 - \frac{\xi_1}{\xi_1 - a^2}} \right\} \frac{d}{d\xi_1} \sqrt{\frac{\xi_1}{\xi_1 - a^2}} \\ &= \frac{2}{\pi} \left\{ \frac{h^2 H (\xi_1 - a^2)}{H^2 (\xi_1 - a^2) + h^2 \xi_1} + \frac{H (\xi_1 - a^2)}{-a^2} \right\} \\ &\quad \times \frac{1}{2} \sqrt{\frac{\xi_1 - a^2}{\xi_1}} \cdot \frac{-a^2}{(\xi_1 - a^2)^2} \\ &= \frac{H}{\pi} \frac{(h^2 + H^2) (\xi_1 - a^2)}{[(h^2 + H^2) \xi_1 - H^2 a^2] \cdot (-a^2)} \sqrt{\frac{1}{\xi_1 (\xi_1 - a^2)}} (-a^2) \\ &= \frac{H}{\pi} \frac{h^2 + H^2}{(h^2 + H^2) \xi_1 - h^2 a^2} \sqrt{\frac{\xi_1 - a^2}{\xi_1}} \\ &= \frac{H}{\pi} \frac{\sqrt{\xi_1 - a^2}}{(\xi_1 - 1) \sqrt{\xi_1}}, \end{aligned}$$

$$\left| \frac{d\xi}{d\xi_1} \right| = \frac{V}{\pi} \frac{1 - \sqrt{\xi_1}}{1 + \sqrt{\xi_1}} \cdot \frac{1}{\sqrt{\xi_1} (1 - \sqrt{\xi_1})^2}.$$

13. 求 z 平面半无界长条 $0 < R_c z < a$, $I_m z > 0$ 上的调和函数, 边界条件为 $u|_{x=0} = 0$, $u|_{x=a} = 0$, $u|_{y=0} = u_0$.

解: 要把 z 平面上的 $A(0, 0)$, $B(a, 0)$, $C(0 \text{ 或 } a, \infty)$ 点变到 ξ 平面的 $A'(-1, 0)$, $B'(1, 0)$, $C(\pm \infty, 0)$ 作施瓦兹-克利斯多菲变换,

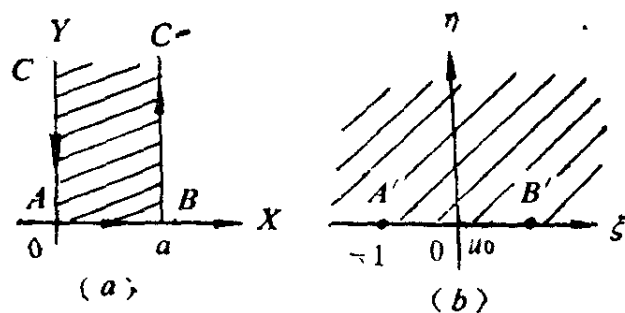


图 17-21

z 平面	θ	$-\frac{\theta}{\pi}$	ζ 平面
$A(0, 0)$	$\frac{\pi}{2}$	$-\frac{1}{2}$	-1
$B(a, 0)$	$\frac{\pi}{2}$	$-\frac{1}{2}$	1
$C\left(\begin{smallmatrix} 0 \\ a \end{smallmatrix}, \infty\right)$	π	-1	$\pm\infty$

$$\begin{aligned}
 z &= z_0 + A \int (\zeta + 1)^{-\frac{1}{2}} (\zeta - 1)^{-\frac{1}{2}} d\zeta \\
 &= \zeta_0 + A_i \int \frac{d\zeta}{\sqrt{1 - \zeta^2}} \\
 &= z_0 - A_i \arccos \zeta,
 \end{aligned} \tag{1}$$

应用点的对应关系，代入 (1) 式， $A(0, 0)$ 对应 $A'(-1, 0)$ 得， $0 = z_0 - A_i \pi$ 、 $B(a, 0)$ 对应 $B'(1, 0)$ 得 $a = z_0$ ，

解得 $z_0 = a$ ， $A = -\frac{ai}{\pi}$ ，

于是所求的变换是

$$z = a + \frac{a}{\pi} \arccos \zeta \text{ 即 } \left(\frac{z - a}{a} \right) \pi = \arccos \zeta,$$

$$\text{亦即 } \zeta = \cos \left(\frac{\pi z}{a} - \pi \right) = -\cos \frac{\pi z}{a}, \tag{2}$$

应用(50.25), 求 ξ 平面上的解.

$$\begin{aligned}
 u &= \frac{u_0}{\pi} \int_{-1}^1 \frac{d\xi_0}{(\xi - \xi_0)^2 + \eta^2} = \frac{u_0}{\pi} \operatorname{arctg} \frac{\xi - \xi_0}{\eta} \Big|_{-1}^1 \\
 &= \frac{u_0}{\pi} \left[\operatorname{arctg} \frac{\xi + 1}{\eta} - \operatorname{arctg} \frac{\xi - 1}{\eta} \right] \\
 &= \frac{u_0}{\pi} \operatorname{arctg} \frac{2\eta}{\xi^2 + \eta^2 - 1}, \tag{3}
 \end{aligned}$$

求得 ξ 平面上的 u , 再回到 z 平面上来即为所求的解, 利用

$$\begin{aligned}
 \eta &= I_m \xi = I_m \left(-\cos \frac{\pi z}{a} \right) \\
 &= \operatorname{sh} \left(\frac{\pi y}{a} \right) \sin \left(\frac{\pi x}{a} \right). \\
 \xi^2 + \eta^2 &= |\xi|^2 = \left| \cos \frac{\pi z}{a} \right|^2 \\
 &= \operatorname{ch}^2 \frac{\pi y}{a} \cos^2 \frac{\pi x}{a} + \operatorname{sh}^2 \frac{\pi y}{a} \sin^2 \frac{\pi x}{a} \\
 &= \operatorname{sh}^2 \frac{\pi y}{a} \left(\cos^2 \frac{\pi x}{a} + \sin^2 \frac{\pi x}{a} \right) \\
 &\quad + \left(\operatorname{ch}^2 \frac{\pi y}{a} - \operatorname{sh}^2 \frac{\pi y}{a} \right) \cos \frac{2\pi x}{a} \\
 &= \operatorname{sh}^2 \frac{\pi y}{a} + \cos^2 \frac{\pi x}{a} \\
 &= \operatorname{sh}^2 \frac{\pi y}{a} + 1 - \sin^2 \frac{\pi x}{a},
 \end{aligned}$$

将求得的 η 和 $\xi^2 + \eta^2$ 代入(3)就得解

$$\therefore u = \frac{u_0}{\pi} \operatorname{arctg} \frac{2 \operatorname{sh} \frac{\pi y}{a} \sin \frac{\pi x}{a}}{\operatorname{sh}^2 \frac{\pi y}{a} - \sin^2 \frac{\pi x}{a}}.$$

14. 把可变电容器中的静电场所占的空间 (图17-22) 变为

上半面。

解：把 z 平面上的 $ABCD$ 点，变换到 ζ 平面上的 $A'B'C'D'$ 点，

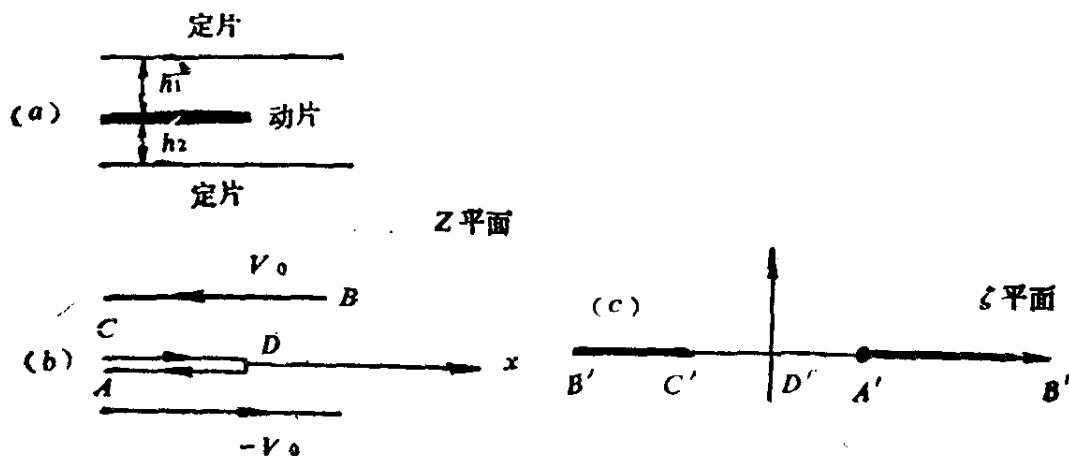


图 17-22

z 平面	θ	$-\frac{\theta}{\pi}$	ζ 平面
$A(-\infty, 0 \text{ 或 } -ih_2)$	π	-1	1
$B(\infty, -ih_2 \text{ 或 } 0)$	π	-1	$+\infty$
$C(-\infty, ih_1 \text{ 或 } 0)$	π	-1	$-a$
$D(0, 0)$	$-\pi$	$+1$	0

$$\begin{aligned}
 z &= z_0 + A \int (\zeta - 1)^{-1} (\zeta + a)^{-1} \zeta d\zeta \\
 &= z_0 + \frac{A}{1+a} \int \frac{d(-\zeta)}{1-\zeta} + \frac{Aa}{1+a} \int \frac{d\zeta}{\zeta+a} \\
 &= z_0 + \frac{A}{1+a} \left[\ln\left(\frac{\zeta}{a} + 1\right) + \ln(1-\zeta) \right]. \quad (1)
 \end{aligned}$$

应用对应关系，由 D 点 $(0, 0)$ 变到 $D'(0, 0)$ 所以 $z_0 = 0$ 。
由于 A 与 A' 点对应

$$\begin{cases} -\infty - ih_2 = \frac{A}{1+a} (-\infty - i\pi), \\ -\infty + 0 = \frac{A}{1+a} (-\infty + 0), \end{cases}$$

$$\therefore \frac{A}{1+a} = \frac{h_2}{\pi},$$

由C点 $(-\infty, ih_2, \text{或 } 0)$ 对应于 $C'(-a, 0)$ 有

$$\begin{cases} -\infty + ih_1 = \frac{h_1}{\pi} a, & (-\infty + i\pi), \\ -\infty = \frac{h_1}{\pi} a (-\infty), \end{cases}$$

$$\therefore \frac{h_1}{\pi} = \frac{h_2}{\pi} a, \quad a = \frac{h_1}{h_2}, \quad (2)$$

代回(1)式便有

$$z = \frac{h_2}{\pi} \ln(1 - \xi) + \frac{h_1}{\pi} \ln\left(1 + \frac{h_2}{h_1} \xi\right), \quad (3)$$

应用(50, 25)求 ξ 平面上的解

$$\begin{aligned} u &= \frac{1}{\pi} \left(\int_{-\infty}^{-a} + \int_1^{\infty} \right) \frac{V_0}{(\xi_0 - \xi)^2 + \eta^2} d\xi_0 \\ &= V_0 - \frac{V_0}{\pi} \left[\operatorname{arctg} \frac{a + \xi}{\eta} + \operatorname{arctg} \frac{1 - \xi}{\eta} \right] \\ &= V_0 - \frac{V_0}{\pi} \operatorname{arctg} \frac{\eta \left(1 + \frac{h_1}{h_2} \right)}{\xi^2 + \eta^2 + \left(\frac{h_1}{h_2} - 1 \right) \xi - \frac{h_1}{h_2}}, \end{aligned}$$

回到原变数，即为所求的解。

15. 研究回旋加速器D形盒(图17-23)的静电场，可把D形盒的侧壁当作在无限远处。

解：要把1 $(\pm\infty, H)$, 2 $(-h, H)$, 3 $(-\infty, \pm H)$, 4 $(-h, -H)$, 5 $(\pm\infty, -H)$, 6 $(h, -H)$, 7 $(\infty, \pm H)$, 8 (h, H) 分别变

为 $1'(\pm\infty, 0)$, $2'(-b, 0)$, $3'(-a, 0)$, $4'(-1, 0)$, $5'(0, 0)$, $6'(1, 0)$, $7'(a, 0)$, $8'(b, 0)$.

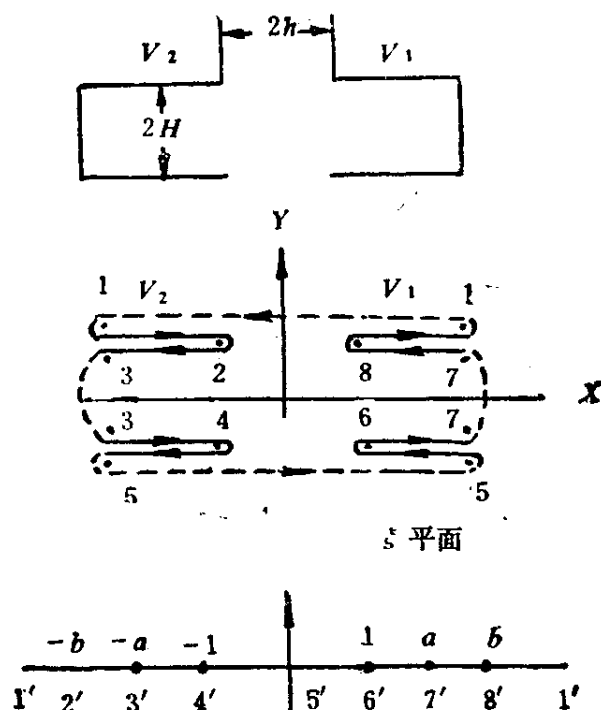


图 17-23

须作变换

$$\begin{aligned}
 z &= z_1 + A \int (\xi_1 + b)(\xi_1 + a)^{-1}(\xi_1 + 1)\xi_1^{-2}(\xi_1 - 1) \\
 &\quad \times (\xi_1 - a)^{-1}(\xi_1 - b)d\xi_1 \\
 &= z_1 + A \int \frac{(\xi_1^2 - 1)(\xi_1^2 - b^2)}{\xi_1^2(\xi_1^2 - a^2)} d\xi_1 \\
 &= z_1 + A \left\{ \xi_1 + \frac{b^2}{a^2} \frac{1}{\xi_1} + \frac{a^2 + \frac{a^2}{b^2} - (1+b)^2}{2a} \ln \frac{\xi_1 - a}{\xi_1 + a} \right\},
 \end{aligned} \tag{1}$$

利用点的对应关系

$$(\pm\infty, H) \text{ 变为 } (\pm\infty, 0), \pm\infty + iH = z_1 + A(\pm\infty + 0 + 0)$$

$$\therefore z_1 = iH, \tag{2}$$

$$(\pm \infty, -H) \text{ 变为 } (0, 0), \pm \infty - iH = iH + A \frac{b^2}{a^2} (\pm \infty) + A \\ \times \frac{a^2 + \frac{b^2}{a^2} - (1 + b^2)}{2a} \pi i,$$

$$\therefore A \frac{a^2 + \frac{b^2}{a^2} - (1 + b^2)}{2a} = \frac{-2H}{\pi}, \quad (3)$$

(h, H) 变为 $(b, 0)$, $(h, -H)$ 变为 $(1, 0)$,

$$\begin{cases} h + iH = iH + A \left(b + \frac{b}{a^2} \right) - \frac{2H}{\pi} \ln \left(\frac{b-a}{b+a} \right), \\ h - iH = iH + A \left(1 + \frac{b^2}{a^2} \right) - \frac{2H}{\pi} \ln \left(\frac{1-a}{1+a} \right), \end{cases}$$

$$\text{即} \begin{cases} h = A \left(b + \frac{b}{a^2} \right) - \frac{2H}{\pi} \ln \frac{\frac{b}{a} - 1}{\frac{b}{a} + 1}, \end{cases} \quad (4)$$

$$h - iH = iH + A \left(\frac{b^2}{a^2} + 1 \right) - \frac{2H}{\pi} i\pi - \frac{2H}{\pi} \ln \frac{a-1}{a+1}, \quad (5)$$

$$\text{亦即} \begin{cases} h = A \left(b + \frac{b}{a^2} \right) - \frac{2H}{\pi} \ln \frac{\frac{b}{a} - 1}{\frac{b}{a} + 1}, \\ h = A \left(\frac{b^2}{a^2} + 1 \right) - \frac{2H}{\pi} \ln \frac{a-1}{a+1}, \end{cases} \quad \text{两式比较知 } b = a^2, \quad (6)$$

以(6)代入(3)得

$$A = \frac{4Ha}{(a^2 - 1)^2 \pi}, \text{ 而 } a \text{ 须满足(4),}$$

$$\text{即} \quad h = \frac{4Ha}{(a^2 - 1)^2 \pi} (1 + a^2) - \frac{2H}{\pi} \ln \frac{a-1}{a+1}, \quad (7)$$

把(2)(3)(6)(7)代入(1)得到所作的变换是

$$z = iH + \frac{4Ha}{(a^2 - 1)^2 \pi} \left(\zeta_1 + \frac{a^2}{\zeta_1} \right) - \frac{2H}{\pi} \ln \frac{\zeta_1 - a}{\zeta_1 + a}, \quad (8)$$

再作变换

$\zeta = \ln \zeta_1$ 如图 17-24.

$$\begin{aligned} \text{电势} &= V_1 + \frac{V_2 - V_1}{\pi} I_m \zeta \\ &= V_1 + \frac{V_2 - V_1}{\pi} \arg \zeta_1, \quad (9) \end{aligned}$$

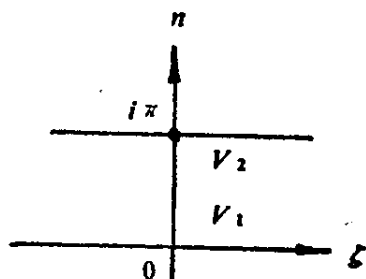


图 17-24

回到原变数, 即得解.

而(9)式中的 ζ_1 由(8)式决定, (8)式中的 a 由(7)式决定.

编 后 记

梁昆森教授所编《数学物理方法》于一九六〇年初版，一九七八年七月修订后出第二版，到一九七九年二月已第十次印刷。可见本书流传广泛，甚受欢迎。本书将数学方法与物理内容紧密结合，既着重物理概念的叙述，又适当地保持了数学的严谨性，为综合大学及师范院校的物理专业广泛采用，不少工科院校的有关系科也选为专业数学的教学用书。

本书再版时增补了许多与物理专业密切有关的例题和习题，求解这些习题有助于加深对正文的理解，且丰富和补充了正文的内容。为了启发读者思考，能在比较短的时间内取得较好的学习效果，同时便于教师教学参考，我们收集有关资料解出了该书的全部习题、编成此书。

由于数学物理方法习题的求解往往需要许多时间，把书中全部习题解出来亦非易事，本书的出版是相互协作的结果。梁昆森教授对本书的出版给予了很大的支持，审阅了原稿，并作了若干订正，我们表示衷心地感谢。杭州大学许健民同志，徐州师范学院周明儒、苏跃中同志，江汉石油学院姜书时同志为本书编写提供过资料及改进意见，谨向他们表示谢意。还有袁士和同志和其他关心支持本书出版的同志，我们均表示谢意。

解题的方法往往是多样的，考虑到篇幅，不可能一一列出。由于我们水平所限，虽经反复审查修改，仍可能有不妥和错误之处，敬希同志们批评指正。

编 者

一九八一年三月

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